

 MET.29291084		Task Card Title: INTERIOR PLACARDS CHECK					
Task Card NO: 11-11-00-01-0-SVA		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP			
A/C Type: 321		Block Taskcard		Main Skill: A-PAINT,ACFT_TECH		PROD Final Acceptance:	
Project No: 01BASPC72608		Related Task Card:		A/C Area: 05			
Internal Check:		Check Type:		Work Order NO: 31909295		Tally Sequence NO: 3	
ENCLOSURE: ☐ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐							
Type Code: CHK		Special Code: CAB		MSG Code: 7, 4		Request:	
Phase: PRE INSP							
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRIED OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.							
Zones: 200							
DOCUMENT REFERENCES:							
Doc Typ		Reference					
CAP		S05-C10003 APPROVED.pdf					
DO		sb:53-20-15					
DO		IPC 11-31/11-32/11-42/11-52					
ACCESS PANELS:							
MATERIAL REQUIREMENTS:							
Part Number		Description					Qty.
No parts assigned to this Task Item							
TOOLING REQUIREMENTS:							
Part Number		Description					Qty.
No tools assigned to this Task Item							
AIRCRAFT INTERIOR PLACARDS CHECK							
Job Description							
A. CHECK CABIN INTERIOR PLACARDS FOR FOLLOWING: NOTE: FOR REFERENCE SEE (IPC: 11-31, 11-32, 11-42, 11-52). (1) PRESENCE, READABILITY AND ADHESION. (2) TEARS, DIRT, STAIN AND WRITINGS THAT ARE NOT PART OF THE PLACARDS.							
STEP COMPLETION SIGN OFF:							
	Qualification / Skill	Number of employees	Duration		Number./Signature	Date	
						Day	Month
Mechanic/s	A-PAINT/ACFT_TECH H			HRS			
Insp.				HRS			

 MET.29291084	Task Card Title: INTERIOR PLACARDS CHECK			
Task Card NO: 11-11-00-01-0-SVA	Work Package: ASP/H-26 C7-CHK	A/C Registration: HZ-ASP		
- Aircraft Maintenance Manual used: Revision No. _____ Dated. _____. Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below Work order No: _____ _____ Note: Make sure to create the finding Work Order on task card level. Remark's: _____ _____				
(1) If Task card interrupted or performed by different persons updates the system and update the following accordingly (Tick Applicable Box) - Interruption No. 1 [] Task card [] AMM [] Other specify. _____ Performed by: _____ Page No. _____ Step From _____ To _____ - Interruption No. 2 [] Task card [] AMM [] Other specify. _____ Performed by: _____ Page No. _____ Step From _____ To _____ - Interruption No. 3 [] Task card [] AMM [] Other specify. _____ Performed by: _____ Page No. _____ Step From _____ To _____ - Interruption No. 4 [] Task card [] AMM [] Other specify. _____ Performed by: _____ Page No. _____ Step From _____ To _____ - Interruption No. 5 [] Task card [] AMM [] Other specify. _____ Performed by: _____ Page No. _____ Step From _____ To _____				
(2) Record any dimension or torque through the performed steps and sign-off as applicable: - Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____ Remark's: _____ _____ - Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____ Remark's: _____ _____ - Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____ Remark's: _____ _____ - Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____ Remark's: _____ _____ - Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____ Remark's: _____ _____				
Task Completion Close Up And System Update		Station	Date	Signature

 MET.25495527		Task Card Title: COCKPIT VOICE RECORDER OPC					
Task Card NO: 237100-01-1		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP			
A/C Type: 321		Block Taskcard		Main Skill: AV,ACFT_TECH		PROD Final Acceptance:	
Project No: 01BASPC72608		Related Task Card:		A/C Area: 06			
Internal Check:		Check Type:		Work Order NO: 31909295		Tally Sequence NO: 9	
ENCLOSURE: ☐ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐							
Type Code: OPC		Special Code: AVN		MSG Code: GACA*, MRB, 9		Request: 	
Phase: PRE HGR							
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRIED OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.							
Zones: 210							
DOCUMENT REFERENCES:							
Doc Typ		Reference					
AMM		23-71-00-710-001-A					
AMM		237100-710-001					
GACA		GACAR Part 91.452					
ACCESS PANELS:							
MATERIAL REQUIREMENTS:							
Part Number		Description				Qty.	
No parts assigned to this Task Item							
TOOLING REQUIREMENTS:							
Part Number		Description				Qty.	
No tools assigned to this Task Item							
TASK INSTRUCTIONS							
Job Description							
A. PERFORM AN OPERATIONAL TEST OF THE CVR AND CVR CHANNEL RECORDING WITH HOT MIKE FUNCTION AS PER AMM TASK 23-71-00-710-001-A.							
STEP COMPLETION SIGN OFF:							
	Qualification / Skill	Number of employees	Duration		Number./Signature	Date	
				HRS		Day	Month
Mechanic/s	AV/ACFT_TECH			HRS			
Insp.				HRS			



MET.25495527

Task Card Title:

COCKPIT VOICE RECORDER OPC

Task Card NO:

237100-01-1

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below

Work order No: _____

Note: Make sure to create the finding Work Order on task card level.

Remark's: _____

(1) If Task card interrupted or performed by different persons updates the system and update the following accordingly (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 2 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 3 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 4 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 5 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

**Task Completion Close Up And
System Update****Station****Date****Signature**

TF-N
AIRCRAFT MAINTENANCE MANUAL
COCKPIT VOICE RECORDER - ADJUSTMENT/TEST

TASK 23-71-00-710-001 -A

On A/C 003-250

Operational Check of CVR and CVR-Channel Recording

WARNING: **MAKE SURE THAT ALL THE CIRCUITS IN MAINTENANCE ARE ISOLATED BEFORE YOU SUPPLY ELECTRICAL POWER TO THE AIRCRAFT. UNWANTED ELECTRICAL POWER CAN CAUSE INJURY TO PERSONS AND/OR DAMAGE TO EQUIPMENT.**

NOTE : This procedure is applicable to the Cockpit Voice Recorder (CVR) and the Solid State Cockpit Voice Recorder (SSCVR).

NOTE : For this procedure, you must make sure that the PARK BRK control handle is in the ON position (accumulator pressure pointer on the triple indicator is in the green range).

1. Reason for the Job

Refer to the MPD TASK: 237100-01

OPERATIONAL CHECK OF CVR AND CVR-CHANNEL RECORDING

To do an operational check of:

- The CVR
- The recording of the communications from the CAPT, F/O and third occupant stations, and the CVR microphone.

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	AR	BOOMSET 600 OHMS
No specific	AR	HEADSET 600 OHMS
No specific	AR	MICROPHONE 150 OHMS - HAND

B. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
210	CKPT,FWD COMPT BHD TO FLT COMPT BULKHEAD

C. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 24-41-00-861-002-A	Energize the Aircraft Electrical Circuits from the External Power
Ref. AMM 24-41-00-861-002-A-01	Energize the Aircraft Electrical Circuits from the APU
Ref. AMM 24-41-00-861-002-A-02	Energize the Aircraft Electrical Circuits from Engine 1(2)
Ref. AMM 24-41-00-862-002-A	De-energize the Aircraft Electrical Circuits Supplied from the External Power
Ref. AMM 24-41-00-862-002-A-01	De-energize the Aircraft Electrical Circuits Supplied from the APU
Ref. AMM 24-41-00-862-002-A-02	De-energize the Aircraft Electrical Circuits Supplied from the Engine 1(2)

3. Job Set-up

Subtask 23-71-00-860-051-A

On A/C 003-226

A. Aircraft Maintenance Configuration

- (1) Energize the aircraft electrical circuits

[Ref. AMM 24-41-00-861-002-A](#) [Ref. AMM 24-41-00-861-002-A-01](#)

[Ref. AMM 24-41-00-861-002-A-02.](#)

- (2) In the cockpit, on the Audio Control Panels (ACP) of the CAPT, F/O and third occupant stations, make sure that:

- The INT/RAD selector switch is in the neutral position
- The green lines of all the transmission keys are off
- All the reception pushbutton switches are pushed in (no reception).

TF-N
AIRCRAFT MAINTENANCE MANUAL
COCKPIT VOICE RECORDER - ADJUSTMENT/TEST

- (3) Make sure that all the BOOMSET 600 OHMS are disconnected from the crew stations.
- (4) On the center pedestal, on panel 110VU, make sure that the PARK BRK control switch is in the ON position.
- (5) Make sure that the aircraft is in the ground configuration.
- (6) Make sure that the engine shutdown occurred five minutes or more before you do this procedure.
- (7) On the overhead maintenance panel 50VU, connect the HEADSET 600 OHMS to the CVR/ HEADSET jack (22RK).
- (8) Connect the MICROPHONE 150 OHMS - HAND to the hand microphone connectors of the CAPT, F/O and third occupant stations.

Subtask 23-71-00-865-050-A

On A/C 003-250

- B. Make sure that this (these) circuit breaker(s) is (are) closed:

PANEL	DESIGNATION	FIN	LOCATION
On A/C 003-250 EMB SB 23-1449 for A/C 003-004			
49VU	COM/CVR/SPLY	23RK	E14
On A/C 003-250 EMB SB 23-1449 for A/C 003-004			
49VU	COM/CVR/CTL	26RK	E13

4. Procedure

Subtask 23-71-00-710-050-D

On A/C 054-100; 103-226

- A. Test of the CVR

NOTE : You can use the HEADSET 600 OHMS with the jack (22RK) as an alternative to the cockpit loudspeakers.

NOTE : Before you do this test, you can hear the frequency shift keying-signal that gives an audio signal at intervals of four seconds. The Flight Data Interface Unit (FDIU) or the Flight Data Interface and Management Unit (FDIMU) gives the frequency shift keying-signal the synchronization of the SSCVR clock with the Solid State Flight Data Recorder (SSFDR). This audio signal operates independently from the test of the CVR.

NOTE : If the HEADSET 600 OHMS is not available on the aircraft, you can keep your hand on the cockpit microphone to decrease the Larsen effect.

NOTE : During the test, you can hear your voice from the HEADSET 600 OHMS or from the cockpit loudspeakers.

NOTE : During the test, if there is a sound (Larsen effect), Airbus recommends that you do the test of the CVR with the jack (22RK).

WARNING: BEFORE YOU SET THE PARKING BRAKE TO THE OFF POSITION, MAKE SURE THAT THE WHEEL CHOCKS ARE INSTALLED ON THE MLG WHEELS. MOVEMENT OF THE AIRCRAFT CAN CAUSE INJURY TO PERSONS AND/OR DAMAGE TO THE AIRCRAFT.

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AIRCRAFT MAINTENANCE MANUAL
COCKPIT VOICE RECORDER - ADJUSTMENT/TEST

- (1) If you do the test of the CVR with the jack (22RK), on RUD TRIM/PARKING BRK panel 110VU, set the PARK BRK control switch to the OFF position.
- (2) Do the steps that follow:

ACTION	RESULT
1. In the cockpit, on overhead control and indicating panel 21VU: • Push the RCDR/GND CTL pushbutton switch.	On the RCDR/GND CTL pushbutton switch: • The blue ON legend comes on.
2. Speak in the cockpit.	You can hear your voice from the HEADSET 600 OHMS, which is connected to the jack (22RK).
3. On overhead control and indicating panel 21VU: • Push and hold the RCDR/CVR TEST pushbutton switch.	At the CAPT (F/O or third occupant) station: • You can hear a test tone in the cockpit loudspeakers or the HEADSET 600 OHMS. NOTE : At the F/O station only, during the test, it is possible that there is a sound interference. It is not necessary to remove the CVR in this condition.
4. On overhead control and indicating panel 21VU: • Release the RCDR/CVR TEST pushbutton switch.	• You cannot hear the test tone.
5. On overhead control and indicating panel 21VU: • Release the RCDR/GND CTL pushbutton switch.	On the RCDR/GND CTL pushbutton switch: • The blue ON legend goes off.
6. On RUD TRIM/PARKING BRK panel 110VU: • If the PARK BRK control switch is in the OFF position, set it to the ON position.	

Subtask 23-71-00-710-055-B

On A/C 003-250

B. Test of the CVR Channel Recording with the Hot-Mike Function

This procedure is applicable to the CVR supplied with 115VAC or 28VDC.

For this procedure:

- Make sure that all the boomsets are disconnected.
- Use only the HEADSET 600 OHMS on maintenance panel 50VU.
- Make sure that the INT/RAD selector switch is set to the neutral position.

NOTE : This test is applicable to the CAPT station. To do this test on the other stations, you can use the indications between the parentheses.

ACTION	RESULT
1. On overhead control and indicating panel 21VU: • Push the RCDR/GND CTL pushbutton switch.	On the RCDR/GND CTL pushbutton switch: • The blue ON legend comes on.
2. Speak in the cockpit.	You can hear your voice from the CVR microphone in the CVR headset.
3. Put a cover on the CVR microphone. • Speak in the cockpit.	You cannot hear your voice from the CVR microphone in the CVR headset.
4. On the CAPT (F/O or third occupant) ACP: • Set the INT/RAD selector switch to the neutral position.	
5. On the CAPT (F/O or third occupant) station: • Connect the MICROPHONE 150 OHMS - HAND to panel 17VU (16VU or 62VU).	

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AIRCRAFT MAINTENANCE MANUAL
COCKPIT VOICE RECORDER - ADJUSTMENT/TEST

ACTION	RESULT
• Push the Push-to-Talk (PTT) switch of the MICROPHONE 150 OHMS - HAND.	
• Speak in the MICROPHONE 150 OHMS - HAND.	You can hear your voice from the hand microphone in the CVR headset.
6. On the CAPT (F/O or third occupant) station: • Release the PTT switch of the MICROPHONE 150 OHMS - HAND.	
• Disconnect the MICROPHONE 150 OHMS - HAND from panel 17VU (16VU or 62VU).	
7. Remove the cover from the CVR microphone.	
8. On overhead control and indicating panel 21VU: • Release the RCDR/GND CTL pushbutton switch.	On the RCDR/GND CTL pushbutton switch: • The blue ON legend goes off.

5. Close-up

Subtask 23-71-00-860-050-A

On A/C 003-226

A. Put the aircraft back to its initial configuration.

- (1) In the cockpit, on the overhead maintenance panel 50VU, disconnect the HEADSET 600 OHMS from the CVR/HEADSET jack (22RK).
- (2) Make sure that the MICROPHONE 150 OHMS - HAND is disconnected from all the crew stations.
- (3) On the ACPs of the CAPT, F/O and third occupant stations, make sure that:
 - The INT/RAD selector switch is in the neutral position
 - The green lines of all the transmission keys are off
 - All the reception pushbutton switches are pushed (no reception).
- (4) De-energize the aircraft electrical circuits
[Ref. AMM 24-41-00-862-002-A](#) [Ref. AMM 24-41-00-862-002-A-01](#)
[Ref. AMM 24-41-00-862-002-A-02.](#)

 MET.25495632		Task Card Title: NO.1 ENG (QAD)ADAPTER TENSION BOLT					
Task Card NO: 242100-01-1-001		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP			
A/C Type: 321		Block Taskcard		Main Skill: PP,II,INSP,ACFT_TECH		PROD Final Acceptance:	
Project No: 01BASPC72608		Related Task Card:		A/C Area: 01			
Internal Check:		Check Type:		Work Order NO: 31909295		Tally Sequence NO: 11	
ENCLOSURE: ☐ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐							
Type Code: FNC		Special Code: PP		MSG Code: LUR, 9, MRB		Request: JOB SETUP, TC/P, TC/P, J OB-CLOSEUP	
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRIED OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.							
Zones: 438							
DOCUMENT REFERENCES:							
Doc Typ		Reference					
AMM		24-21-00-210-043-A					
AMM		242100-210-043					
ACCESS PANELS:							
438AR							
MATERIAL REQUIREMENTS:							
Part Number		Description				Qty.	
MIL-PRF-16173		COMPOUND - CORROSION PREVENTIVE,				1.0	
MIL-PRF-23699F		TURBINE ENG OIL MRP#P001-A (USE FOR				1.0	
TOOLING REQUIREMENTS:							
Part Number		Description				Qty.	
No tools assigned to this Task Item							
Job Description							
A. PREPARATION: (1) PREPARE FOR TENSION-BOLT TORQUE CHECK PER AMM TASK 24-21-00-210-043-A .							
STEP COMPLETION SIGN OFF:							
	Qualification / Skill	Number of employees	Duration		Number./Signature	Date	
				HRS		Day	Month
Mechanic/s	PP/ACFT_TECH			HRS			
Insp.				HRS			



Task Card Title:

NO.1 ENG (QAD)ADAPTER TENSION BOLT

Task Card NO:

242100-01-1-001

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP**Job Description**

B. INSTRUCTION:

(1) LUBRICATE THE QAD ADAPTER AND THE TENSION BOLT PER AMM TASK 24-21-00-210-043-A .

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
						Day	Month	Year
Mechanic/s	PP/ACFT_TECH			HRS				
Insp.				HRS				

Job Description

B. CONTINUED:

(2) PERFORM TENSION BOLT TORQUE VALUE CHECK PER AMM TASK 24-21-00-210-043-A .

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
						Day	Month	Year
Mechanic/s	PP/ACFT_TECH			HRS				
Insp.	II/INSP			HRS				

Job Description

B. CONTINUED:

(3) RETURN AIRCRAFT BACK TO NORMAL CONFIGURATION PER AMM TASK 24-21-00-210-043-A .

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
						Day	Month	Year
Mechanic/s	PP/ACFT_TECH			HRS				
Insp.				HRS				

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below
Work order No: _____

Note: Make sure to creat the finding Work Order on task card level.

Remark's: _____



MET.25495632

Task Card Title:

NO.1 ENG (QAD)ADAPTER TENSION BOLT

Task Card NO:

242100-01-1-001

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

(1) If Task card interrupted or performed by different personals updates the system and update the following according (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 2 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 3 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 4 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 5 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____

**Task Completion Close Up And
System Update**

Station**Date****Signature**

TF-N
AIRCRAFT MAINTENANCE MANUAL
INTEGRATED DRIVE GENERATOR SYSTEM (IDG, GCU) - INSPECTION/CHECK

TASK 24-21-00-210-043 -A

On A/C 003-250

Check Torque of Quick-Attach-Detach (QAD) Tension Bolt

WARNING: BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER ENGINE SHUTDOWN. USE APPLICABLE THERMAL GLOVES. THE ENGINE PARTS AND THE ENGINE OIL CAN STAY HOT FOR ONE HOUR AFTER SHUTDOWN AND CAN BURN YOU.

WARNING: BE CAREFUL WHEN YOU USE CONSUMABLE MATERIALS. OBEY THE MATERIAL MANUFACTURER'S INSTRUCTIONS AND YOUR LOCAL REGULATIONS.

1. Reason for the Job

Refer to the MPD TASK: 242100-01

CHECK TORQUE OF QUICK-ATTACH-DETACH (QAD) TENSION BOLT

To do a torque check of the IDG QAD adapter tension-bolt without component removal.

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	1	MALLET - SOFT FACE
No specific	1	RULER - STEEL
No specific	AR	WARNING NOTICE(S)
No specific		Torque wrench: range to between 265 and 305 lbf.in (2.99 and 3.45 m.daN)

B. Consumable Materials

REFERENCE	DESIGNATION
CP8002	lockwire 0.032 in. (0.8 mm) dia.
03ABA1	Synthetic base Lubricating Oil-Turbine Engine - -
12ABE2	Corrosion Preventive Compound-Water Displacing Oily Film System

C. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
438	FAN COWL
448	FAN COWL
FOR 1000EM1(ENGINE-1)	
438AR	
FOR 1000EM2(ENGINE-2)	
448AR	

D. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 24-21-41-000-040-B	Removal of the Quick Attach/Detach (QAD) Adapter of the IDG
Ref. AMM 24-21-41-400-043-A	Installation of the Quick Attach/Detach (QAD) Adapter of the IDG (NEW EPGS)
Ref. AMM 24-21-51-000-040-B	Removal of the Integrated Drive Generator - IDG 1(2)
Ref. AMM 24-21-51-400-040-E	Installation of the Integrated Drive Generator (IDG) 1(2) (New EPGS)
Ref. AMM 70-22-00-911-002-A	Locking Procedures
Ref. AMM 71-13-00-010-040-B	Opening of the Fan Cowl Doors
Ref. AMM 71-13-00-410-040-B	Closing of the Fan Cowl Doors
Ref. AMM 78-30-00-081-041-B	Make the Thrust Reverser System Serviceable after Maintenance
Ref. AMM 78-30-00-481-041-B	Make the Thrust Reverser System Unserviceable for Maintenance
Ref. Fig. (1 / 24-21-00-991-90700)	
Ref. Fig. (1 / 24-21-00-991-91300)	

3. Job Set-up

Subtask 24-21-00-941-056-A

On A/C 003-250

TF-N
AIRCRAFT MAINTENANCE MANUAL
INTEGRATED DRIVE GENERATOR SYSTEM (IDG, GCU) - INSPECTION/CHECK

A. Safety Precautions

- (1) On the center pedestal, on ENG panel 115VU:
 - (a) Put a WARNING NOTICE(S) in position to tell persons not to start the engine.
- (2) Make sure that engine 1(2) shutdown occurs at a minimum of five minutes before you do this procedure.
- (3) On overhead maintenance panel 50VU:
 - (a) Make sure that the ON legend of the ENG/FADEC GND PWR/1(2) pushbutton switch is off.
 - (b) Put a WARNING NOTICE(S) in position to tell persons not to energize FADEC 1(2).

Subtask 24-21-00-010-053-A

On A/C 003-250

B. Open the right fan cowl door [Ref. AMM 71-13-00-010-040-B](#):

- (1) FOR 1000EM1 (ENGINE-1)
 - 438AR .
- (2) FOR 1000EM2 (ENGINE-2)
 - 448AR .

Subtask 24-21-00-040-075-A

On A/C 003-250

C. Deactivation of the Thrust Reverser

WARNING: YOU MUST DEACTIVATE THE THRUST REVERSER BEFORE YOU DO WORK ON OR AROUND IT. IF YOU DO NOT, THE THRUST REVERSER CAN OPERATE ACCIDENTALLY AND CAUSE AN INJURY AND/OR DAMAGE.

- (1) Do the deactivation of the thrust reverser system for maintenance
[Ref. AMM 78-30-00-481-041-B](#).

4. Procedure

Ref. Fig. 1 / 24-21-00-991-90700

Subtask 24-21-00-020-057-A

On A/C 003-250

A. Removal of the Lockwire

WARNING: USE PROTECTIVE GOGGLES AND GLOVES EACH TIME YOU CUT LOCKWIRE. DISCARD LOOSE OR USED LOCKWIRE IMMEDIATELY. LOOSE OR USED LOCKWIRE CAN CUT YOU OR MAKE YOU BLIND.

- (1) Cut and remove the lockwire from the tension bolt (2).

Subtask 24-21-00-640-051-A

On A/C 003-250

B. Lubrication of the QAD Adapter Tension-Bolt

- (1) Apply Corrosion Preventive Compound-Water Displacing Oily Film System (Material No. **12ABE2**) on the mating surface between the QAD ring (3) and the QAD adapter plate (1) with a spray (in the zone where you can get access).
- (2) Lubricate the tension bolt (2) with the Synthetic base Lubricating Oil-Turbine Engine - - (Material No. **03ABA1**) .

Subtask 24-21-00-220-050-B

On A/C 003-250

C. Torque Check of the QAD Adapter Tension-Bolt

Ref. Fig. (1 / 24-21-00-991-91300)

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- (1) TORQUE the QAD adapter tension-bolt (2) **to between 265 and 305 lbf.in (2.99 and 3.45 m.daN).**
- (2) Use the MALLET - SOFT FACE to lightly hit the QAD ring (3).
- (3) Measure the torque value of the QAD adapter tension-bolt (2).
 - (a) If the torque value decreases by more than 2 lbf.ft (0.27 m.daN) compared to the initial torque value, do steps (1) thru (3) again.
- (4) Lightly shake the anti-drive end of the IDG to make sure that the QAD adapter plate (1) is correctly connected:
 - (a) If the IDG is loose, do as follows:
 - 1 Remove the IDG [Ref. AMM 24-21-51-000-040-B](#).
 - 2 Send the IDG to the workshop to replace the mounting flange.
 - 3 Replace the QAD adapter plate (1) [Ref. AMM 24-21-41-000-040-B](#) and [Ref. AMM 24-21-41-400-043-A](#).
 - 4 Install the IDG [Ref. AMM 24-21-51-400-040-E](#).
 - (b) If the IDG is not loose, do as follows:
 - 1 Tap the edge of the QAD ring (3) with the MALLET - SOFT FACE.
 - 2 Measure the torque value of the QAD adapter tension-bolt (2).
 - 3 Make sure that the torque value of the QAD adapter tension-bolt (2) **to between 265 and 305 lbf.in (2.99 and 3.45 m.daN).**
- (5) Measure the distance between the QAD ring lug and the bracket of the QAD ring (3) with the RULER - STEEL or the CALIPER:
 - (a) If the distance between the QAD ring lug and the bracket of the QAD ring (3) is less than 0.312 in. (7.925 mm), do the steps that follow:
 - 1 Replace the IDG (but keep the IDG accessory hoist installed) and do not attach the tubes, the harnesses and the other components [Ref. AMM 24-21-51-000-040-B](#) and [Ref. AMM 24-21-51-400-040-E](#).
 - 2 Measure the distance between the QAD ring lug and the bracket of the QAD ring (3) with the RULER - STEEL or the CALIPER.
 - 3 If the distance between the QAD ring lug and the bracket of the QAD ring (3) is not less than 0.312 in. (7.925 mm), do the steps that follow:

WARNING: USE PROTECTIVE GOGGLES AND GLOVES EACH TIME YOU CUT LOCKWIRE. DISCARD LOOSE OR USED LOCKWIRE IMMEDIATELY. LOOSE OR USED LOCKWIRE CAN CUT YOU OR MAKE YOU BLIND.
 - a Safety the QAD adapter tension-bolt (2) with the lockwire 0.032 in. (0.8 mm) dia. (Material No. **CP8002**) [Ref. AMM 70-22-00-911-002-A](#).
 - b Remove the IDG accessory hoist and attach the components, the tubes and the harnesses.
 - c Send the worn IDG to the workshop to replace the input flange.
- 4 If the distance between the QAD ring lug and the bracket of the QAD ring (3) is less than 0.312 in. (7.925 mm), do the steps that follow:
 - a Remove the new IDG [Ref. AMM 24-21-51-000-040-B](#).
 - b Replace the QAD adapter plate (1) [Ref. AMM 24-21-41-000-040-B](#) and [Ref. AMM 24-21-41-400-043-A](#).
 - c Install the IDG that you removed [Ref. AMM 24-21-51-400-040-E](#).
- (b) If the distance between the QAD ring lug and the bracket of the QAD ring (3) is not less than 0.312 in. (7.925 mm), do the step that follows:

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WARNING: **USE PROTECTIVE GOGGLES AND GLOVES EACH TIME YOU CUT LOCKWIRE. DISCARD LOOSE OR USED LOCKWIRE IMMEDIATELY. LOOSE OR USED LOCKWIRE CAN CUT YOU OR MAKE YOU BLIND.**

- 1 Safety the QAD adapter tension-bolt (2) with the lockwire 0.032 in. (0.8 mm) dia.
(Material No. **CP8002**) [Ref. AMM 70-22-00-911-002-A](#).

5. Close-up

Subtask 24-21-00-440-072-A

[On A/C 003-250](#)

A. Reactivation of the Thrust Reverser

- (1) Do the reactivation of the thrust reverser system after maintenance
[Ref. AMM 78-30-00-081-041-B](#).

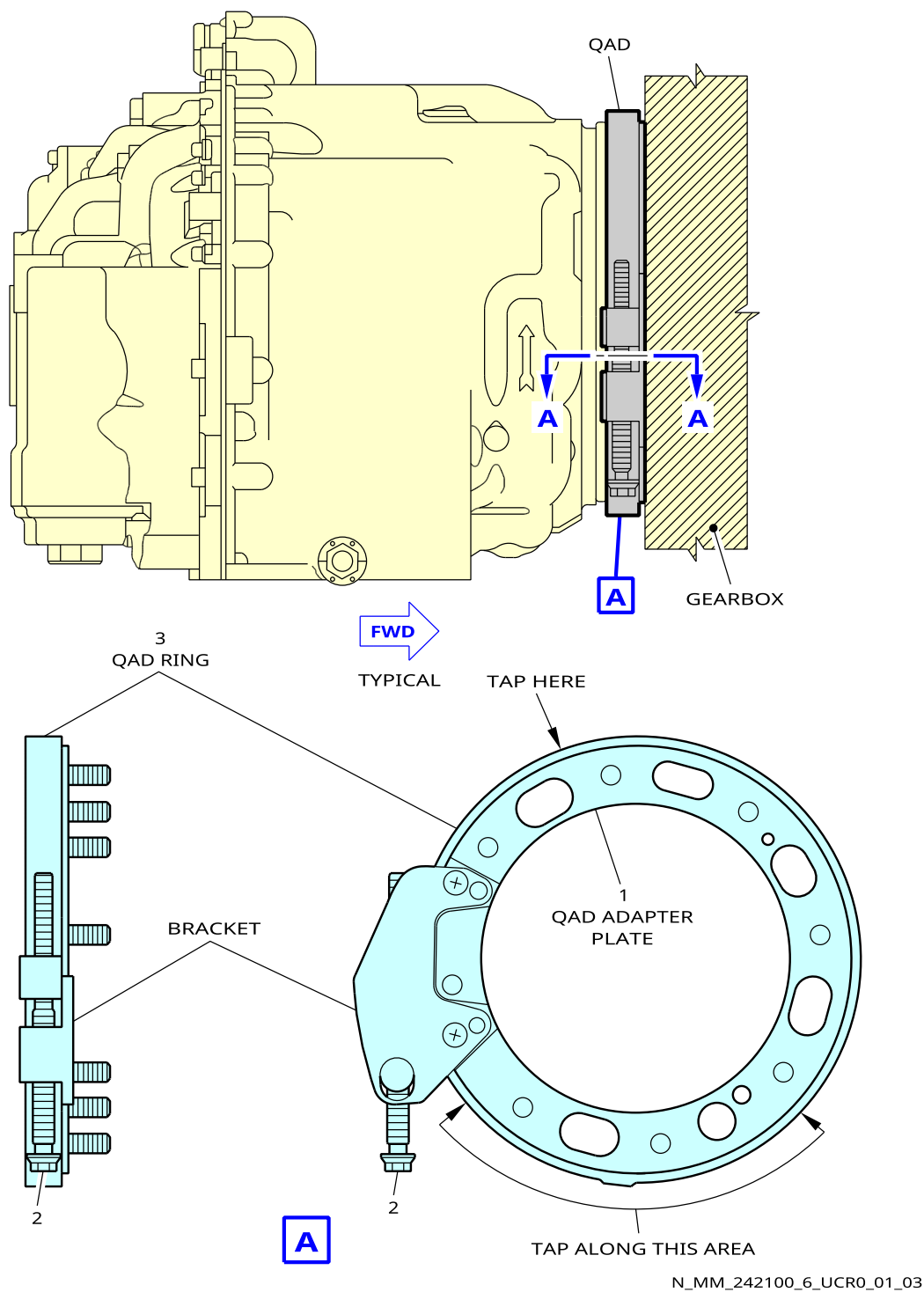
Subtask 24-21-00-410-053-A

[On A/C 003-250](#)

B. Close Access

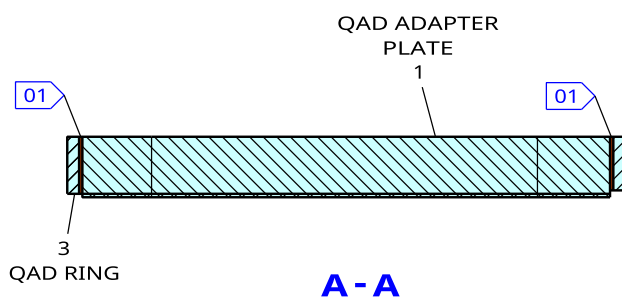
- (1) Make sure that the work area is clean and clear of tools and other items.
- (2) Close the right fan cowl door [Ref. AMM 71-13-00-410-040-B](#):
 - (a) FOR 1000EM1 (ENGINE-1)
 - 438AR .
 - (b) FOR 1000EM2 (ENGINE-2)
 - 448AR .
- (3) Remove the WARNING NOTICE(S).

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IDG System - QAD Adapter
Figure 1 / Graphic 24-21-00 -991-90700

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AIRCRAFT MAINTENANCE MANUAL
INTEGRATED DRIVE GENERATOR SYSTEM (IDG, GCU) - INSPECTION/CHECK



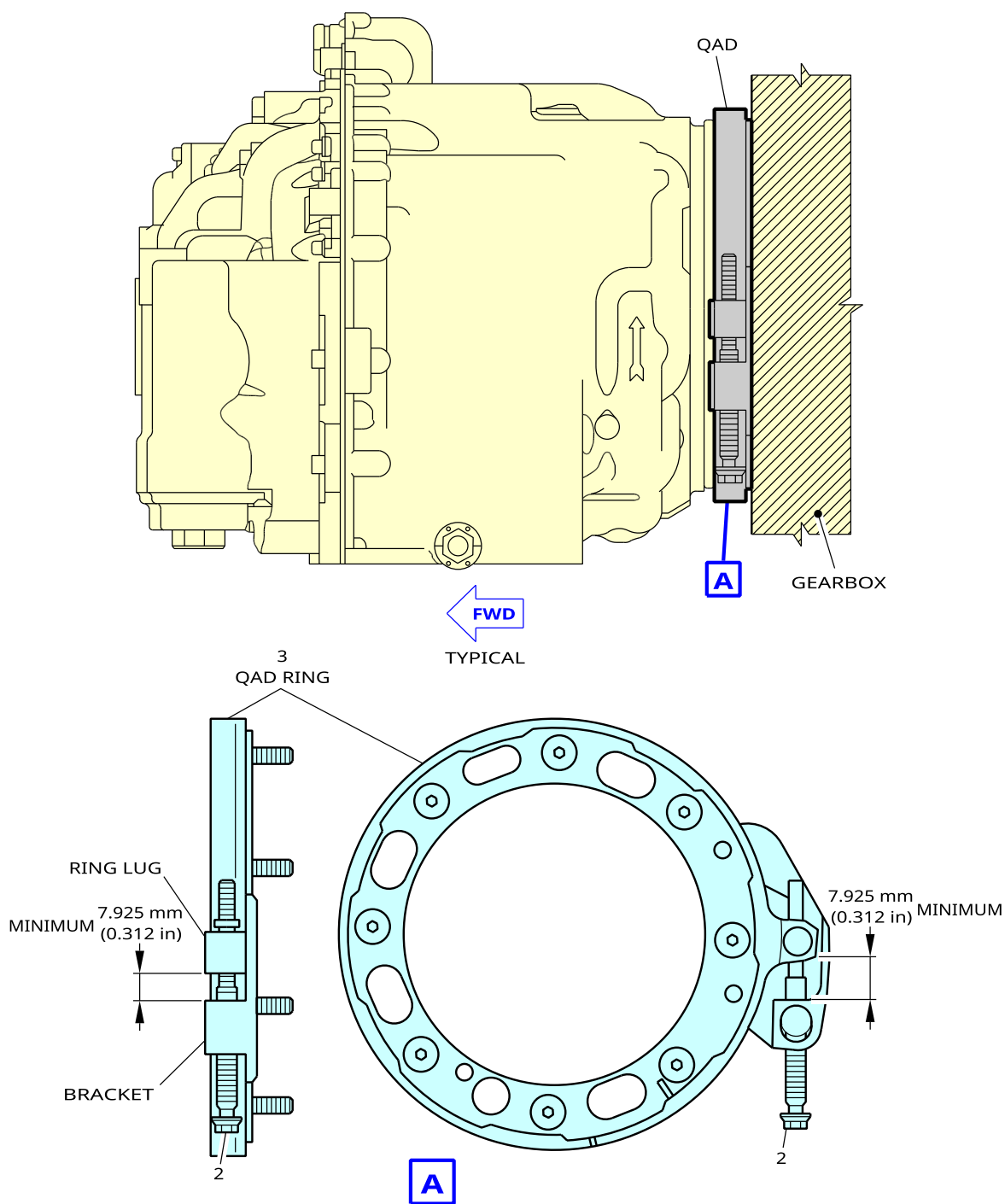
NOTE:

- 01 CORROSION PREVENTIVE COMPOUND-WATER DISPLACING OILY FILM
SYSTEM MATERIAL No 12BE2 IS APPLIED.

N_MM_242100_6_UCR0_02_00

IDG System - QAD Adapter
Figure 2 / Graphic 24-21-00 -991-90700

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AIRCRAFT MAINTENANCE MANUAL
INTEGRATED DRIVE GENERATOR SYSTEM (IDG, GCU) - INSPECTION/CHECK



N_MM_242100_6_UER0_01_03

IDG System - QAD Adapter Wear Allowance (P/N 772896)
Figure 1 / Graphic 24-21-00 -991-91300

 MET.25470465		Task Card Title: NO.2 ENG (QAD)ADAPTER TENSION BOLT							
Task Card NO: 242100-01-1-002		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP					
A/C Type: 321		Block Taskcard		Main Skill: PP,INSP,II,ACFT_TECH		PROD Final Acceptance:			
Project No: 01BASPC72608		Related Task Card:		A/C Area: 02					
Internal Check:		Check Type:		Work Order NO: 31909295		Tally Sequence NO: 12			
ENCLOSURE: ☐ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐									
Type Code: FNC		Special Code: PP		MSG Code: LUR, 9, MRB		Request: Phase: JOB SETUP,TC/P,TC/P,J OB-CLOSEUP			
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRID OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.									
Zones: 448									
DOCUMENT REFERENCES:									
Doc Typ		Reference							
AMM		24-21-00-210-043-A							
AMM		242100-210-043							
ACCESS PANELS:									
448AR									
MATERIAL REQUIREMENTS:									
Part Number		Description				Qty.			
MIL-PRF-23699F		TURBINE ENG OIL MRP#P001-A (USE FOR				1.0			
MIL-PRF-16173		COMPOUND - CORROSION PREVENTIVE,				1.0			
TOOLING REQUIREMENTS:									
Part Number		Description				Qty.			
No tools assigned to this Task Item									
Job Description									
A. PREPARATION: (1) PREPARE FOR TENSION-BOLT TORQUE CHECK PER AMM TASK 24-21-00-210-043-A .									
STEP COMPLETION SIGN OFF:									
Qualification / Skill		Number of employees		Duration		Number./Signature		Date	
Mechanic/s		PP/ACFT_TECH		HRS				Day Month Year	



Task Card Title:

NO.2 ENG (QAD)ADAPTER TENSION BOLT

Task Card NO:

242100-01-1-002

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

Insp.

HRS

Job Description

B. INSTRUCTION:

(1) LUBRICATE THE QAD ADAPTER AND THE TENSION BOLT PER AMM TASK 24-21-00-210-043-A .

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PP/ACFT_TECH		HRS				
Insp.			HRS				

Job Description

B. CONTINUED:

(2) PERFORM TENSION BOLT TORQUE VALUE CHECK PER AMM TASK 24-21-00-210-043-A .

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PP/ACFT_TECH		HRS				
Insp.	INSP/II		HRS				

Job Description

B. CONTINUED:

(3) RETURN AIRCRAFT BACK TO NORMAL CONFIGURATION PER AMM TASK 24-21-00-210-043-A .

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PP/ACFT_TECH		HRS				
Insp.			HRS				

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below

Work order No: _____

Note: Make sure to creat the finding Work Order on task card level.

Remark's: _____



Task Card Title:

NO.2 ENG (QAD)ADAPTER TENSION BOLT

Task Card NO:

242100-01-1-002

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

(1) If Task card interrupted or performed by different personals updates the system and update the following according (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 2 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 3 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 4 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 5 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____

**Task Completion Close Up And
System Update**

Station**Date****Signature**

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AIRCRAFT MAINTENANCE MANUAL
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TASK 24-21-00-210-043 -A

On A/C 003-250

Check Torque of Quick-Attach-Detach (QAD) Tension Bolt

WARNING: BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER ENGINE SHUTDOWN. USE APPLICABLE THERMAL GLOVES. THE ENGINE PARTS AND THE ENGINE OIL CAN STAY HOT FOR ONE HOUR AFTER SHUTDOWN AND CAN BURN YOU.

WARNING: BE CAREFUL WHEN YOU USE CONSUMABLE MATERIALS. OBEY THE MATERIAL MANUFACTURER'S INSTRUCTIONS AND YOUR LOCAL REGULATIONS.

1. Reason for the Job

Refer to the MPD TASK: 242100-01

CHECK TORQUE OF QUICK-ATTACH-DETACH (QAD) TENSION BOLT

To do a torque check of the IDG QAD adapter tension-bolt without component removal.

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	1	MALLET - SOFT FACE
No specific	1	RULER - STEEL
No specific	AR	WARNING NOTICE(S)
No specific		Torque wrench: range to between 265 and 305 lbf.in (2.99 and 3.45 m.daN)

B. Consumable Materials

REFERENCE	DESIGNATION
CP8002	lockwire 0.032 in. (0.8 mm) dia.
03ABA1	Synthetic base Lubricating Oil-Turbine Engine - -
12ABE2	Corrosion Preventive Compound-Water Displacing Oily Film System

C. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
438	FAN COWL
448	FAN COWL
FOR 1000EM1(ENGINE-1)	
438AR	
FOR 1000EM2(ENGINE-2)	
448AR	

D. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 24-21-41-000-040-B	Removal of the Quick Attach/Detach (QAD) Adapter of the IDG
Ref. AMM 24-21-41-400-043-A	Installation of the Quick Attach/Detach (QAD) Adapter of the IDG (NEW EPGS)
Ref. AMM 24-21-51-000-040-B	Removal of the Integrated Drive Generator - IDG 1(2)
Ref. AMM 24-21-51-400-040-E	Installation of the Integrated Drive Generator (IDG) 1(2) (New EPGS)
Ref. AMM 70-22-00-911-002-A	Locking Procedures
Ref. AMM 71-13-00-010-040-B	Opening of the Fan Cowl Doors
Ref. AMM 71-13-00-410-040-B	Closing of the Fan Cowl Doors
Ref. AMM 78-30-00-081-041-B	Make the Thrust Reverser System Serviceable after Maintenance
Ref. AMM 78-30-00-481-041-B	Make the Thrust Reverser System Unserviceable for Maintenance
Ref. Fig. (1 / 24-21-00-991-90700)	
Ref. Fig. (1 / 24-21-00-991-91300)	

3. Job Set-up

Subtask 24-21-00-941-056-A

On A/C 003-250

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AIRCRAFT MAINTENANCE MANUAL
INTEGRATED DRIVE GENERATOR SYSTEM (IDG, GCU) - INSPECTION/CHECK

A. Safety Precautions

- (1) On the center pedestal, on ENG panel 115VU:
 - (a) Put a WARNING NOTICE(S) in position to tell persons not to start the engine.
- (2) Make sure that engine 1(2) shutdown occurs at a minimum of five minutes before you do this procedure.
- (3) On overhead maintenance panel 50VU:
 - (a) Make sure that the ON legend of the ENG/FADEC GND PWR/1(2) pushbutton switch is off.
 - (b) Put a WARNING NOTICE(S) in position to tell persons not to energize FADEC 1(2).

Subtask 24-21-00-010-053-A

On A/C 003-250

B. Open the right fan cowl door [Ref. AMM 71-13-00-010-040-B](#):

- (1) FOR 1000EM1 (ENGINE-1)
 - 438AR .
- (2) FOR 1000EM2 (ENGINE-2)
 - 448AR .

Subtask 24-21-00-040-075-A

On A/C 003-250

C. Deactivation of the Thrust Reverser

WARNING: YOU MUST DEACTIVATE THE THRUST REVERSER BEFORE YOU DO WORK ON OR AROUND IT. IF YOU DO NOT, THE THRUST REVERSER CAN OPERATE ACCIDENTALLY AND CAUSE AN INJURY AND/OR DAMAGE.

- (1) Do the deactivation of the thrust reverser system for maintenance
[Ref. AMM 78-30-00-481-041-B](#).

4. Procedure

Ref. Fig. 1 / 24-21-00-991-90700

Subtask 24-21-00-020-057-A

On A/C 003-250

A. Removal of the Lockwire

WARNING: USE PROTECTIVE GOGGLES AND GLOVES EACH TIME YOU CUT LOCKWIRE. DISCARD LOOSE OR USED LOCKWIRE IMMEDIATELY. LOOSE OR USED LOCKWIRE CAN CUT YOU OR MAKE YOU BLIND.

- (1) Cut and remove the lockwire from the tension bolt (2).

Subtask 24-21-00-640-051-A

On A/C 003-250

B. Lubrication of the QAD Adapter Tension-Bolt

- (1) Apply Corrosion Preventive Compound-Water Displacing Oily Film System (Material No. **12ABE2**) on the mating surface between the QAD ring (3) and the QAD adapter plate (1) with a spray (in the zone where you can get access).
- (2) Lubricate the tension bolt (2) with the Synthetic base Lubricating Oil-Turbine Engine - - (Material No. **03ABA1**) .

Subtask 24-21-00-220-050-B

On A/C 003-250

C. Torque Check of the QAD Adapter Tension-Bolt

Ref. Fig. (1 / 24-21-00-991-91300)

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AIRCRAFT MAINTENANCE MANUAL
INTEGRATED DRIVE GENERATOR SYSTEM (IDG, GCU) - INSPECTION/CHECK

- (1) TORQUE the QAD adapter tension-bolt (2) **to between 265 and 305 lbf.in (2.99 and 3.45 m.daN).**
- (2) Use the MALLET - SOFT FACE to lightly hit the QAD ring (3).
- (3) Measure the torque value of the QAD adapter tension-bolt (2).
 - (a) If the torque value decreases by more than 2 lbf.ft (0.27 m.daN) compared to the initial torque value, do steps (1) thru (3) again.
- (4) Lightly shake the anti-drive end of the IDG to make sure that the QAD adapter plate (1) is correctly connected:
 - (a) If the IDG is loose, do as follows:
 - 1 Remove the IDG [Ref. AMM 24-21-51-000-040-B](#).
 - 2 Send the IDG to the workshop to replace the mounting flange.
 - 3 Replace the QAD adapter plate (1) [Ref. AMM 24-21-41-000-040-B](#) and [Ref. AMM 24-21-41-400-043-A](#).
 - 4 Install the IDG [Ref. AMM 24-21-51-400-040-E](#).
 - (b) If the IDG is not loose, do as follows:
 - 1 Tap the edge of the QAD ring (3) with the MALLET - SOFT FACE.
 - 2 Measure the torque value of the QAD adapter tension-bolt (2).
 - 3 Make sure that the torque value of the QAD adapter tension-bolt (2) **to between 265 and 305 lbf.in (2.99 and 3.45 m.daN).**
- (5) Measure the distance between the QAD ring lug and the bracket of the QAD ring (3) with the RULER - STEEL or the CALIPER:
 - (a) If the distance between the QAD ring lug and the bracket of the QAD ring (3) is less than 0.312 in. (7.925 mm), do the steps that follow:
 - 1 Replace the IDG (but keep the IDG accessory hoist installed) and do not attach the tubes, the harnesses and the other components [Ref. AMM 24-21-51-000-040-B](#) and [Ref. AMM 24-21-51-400-040-E](#).
 - 2 Measure the distance between the QAD ring lug and the bracket of the QAD ring (3) with the RULER - STEEL or the CALIPER.
 - 3 If the distance between the QAD ring lug and the bracket of the QAD ring (3) is not less than 0.312 in. (7.925 mm), do the steps that follow:

WARNING: USE PROTECTIVE GOGGLES AND GLOVES EACH TIME YOU CUT LOCKWIRE. DISCARD LOOSE OR USED LOCKWIRE IMMEDIATELY. LOOSE OR USED LOCKWIRE CAN CUT YOU OR MAKE YOU BLIND.
 - a Safety the QAD adapter tension-bolt (2) with the lockwire 0.032 in. (0.8 mm) dia. (Material No. **CP8002**) [Ref. AMM 70-22-00-911-002-A](#).
 - b Remove the IDG accessory hoist and attach the components, the tubes and the harnesses.
 - c Send the worn IDG to the workshop to replace the input flange.
- 4 If the distance between the QAD ring lug and the bracket of the QAD ring (3) is less than 0.312 in. (7.925 mm), do the steps that follow:
 - a Remove the new IDG [Ref. AMM 24-21-51-000-040-B](#).
 - b Replace the QAD adapter plate (1) [Ref. AMM 24-21-41-000-040-B](#) and [Ref. AMM 24-21-41-400-043-A](#).
 - c Install the IDG that you removed [Ref. AMM 24-21-51-400-040-E](#).
- (b) If the distance between the QAD ring lug and the bracket of the QAD ring (3) is not less than 0.312 in. (7.925 mm), do the step that follows:

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INTEGRATED DRIVE GENERATOR SYSTEM (IDG, GCU) - INSPECTION/CHECK

WARNING: **USE PROTECTIVE GOGGLES AND GLOVES EACH TIME YOU CUT LOCKWIRE. DISCARD LOOSE OR USED LOCKWIRE IMMEDIATELY. LOOSE OR USED LOCKWIRE CAN CUT YOU OR MAKE YOU BLIND.**

- 1 Safety the QAD adapter tension-bolt (2) with the lockwire 0.032 in. (0.8 mm) dia.
(Material No. **CP8002**) [Ref. AMM 70-22-00-911-002-A](#).

5. Close-up

Subtask 24-21-00-440-072-A

[On A/C 003-250](#)

A. Reactivation of the Thrust Reverser

- (1) Do the reactivation of the thrust reverser system after maintenance
[Ref. AMM 78-30-00-081-041-B](#).

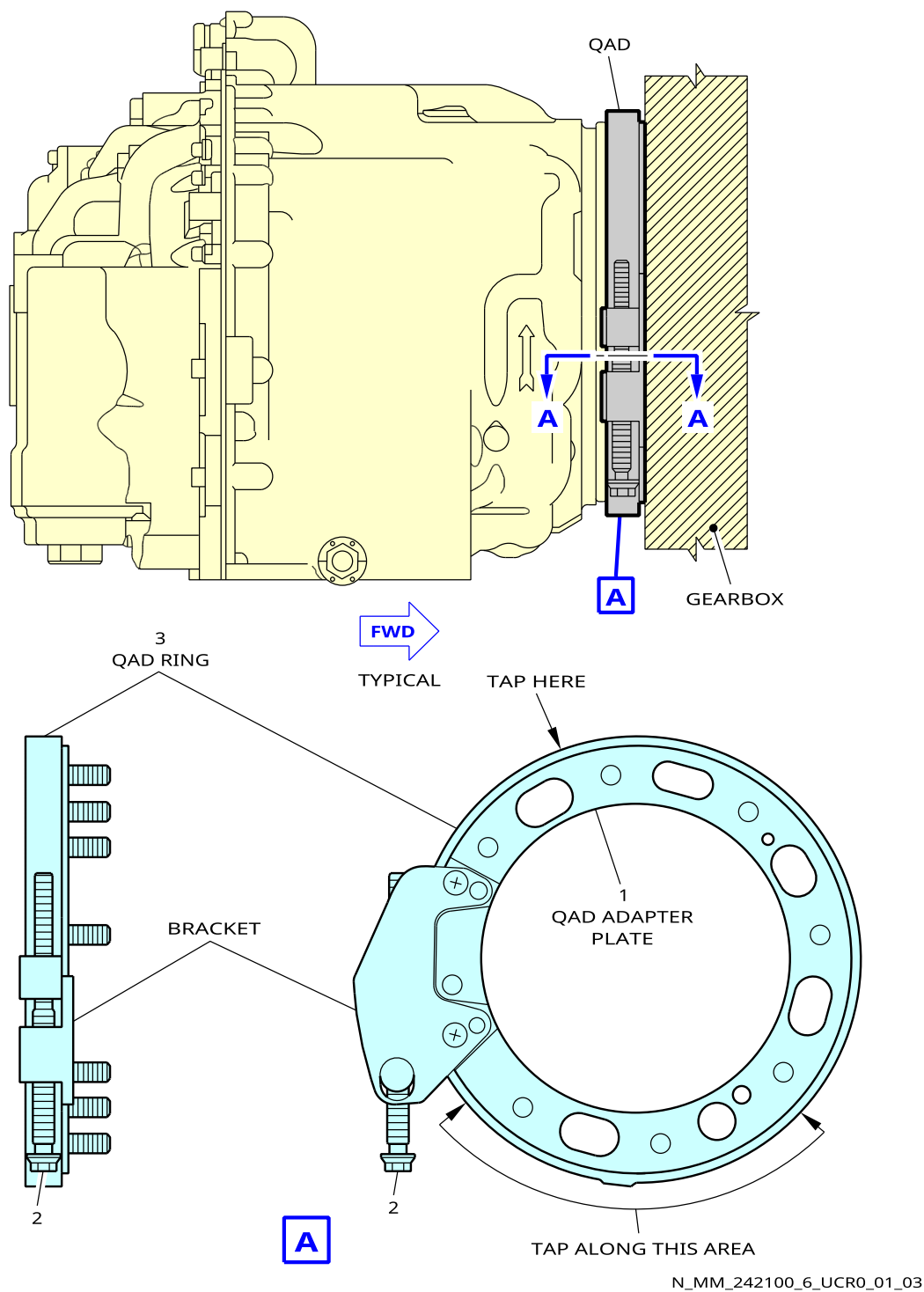
Subtask 24-21-00-410-053-A

[On A/C 003-250](#)

B. Close Access

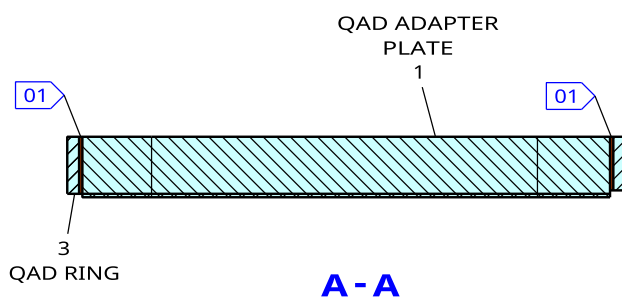
- (1) Make sure that the work area is clean and clear of tools and other items.
- (2) Close the right fan cowl door [Ref. AMM 71-13-00-410-040-B](#):
 - (a) FOR 1000EM1 (ENGINE-1)
 - 438AR .
 - (b) FOR 1000EM2 (ENGINE-2)
 - 448AR .
- (3) Remove the WARNING NOTICE(S).

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AIRCRAFT MAINTENANCE MANUAL
INTEGRATED DRIVE GENERATOR SYSTEM (IDG, GCU) - INSPECTION/CHECK



IDG System - QAD Adapter
Figure 1 / Graphic 24-21-00 -991-90700

TF-N
AIRCRAFT MAINTENANCE MANUAL
INTEGRATED DRIVE GENERATOR SYSTEM (IDG, GCU) - INSPECTION/CHECK



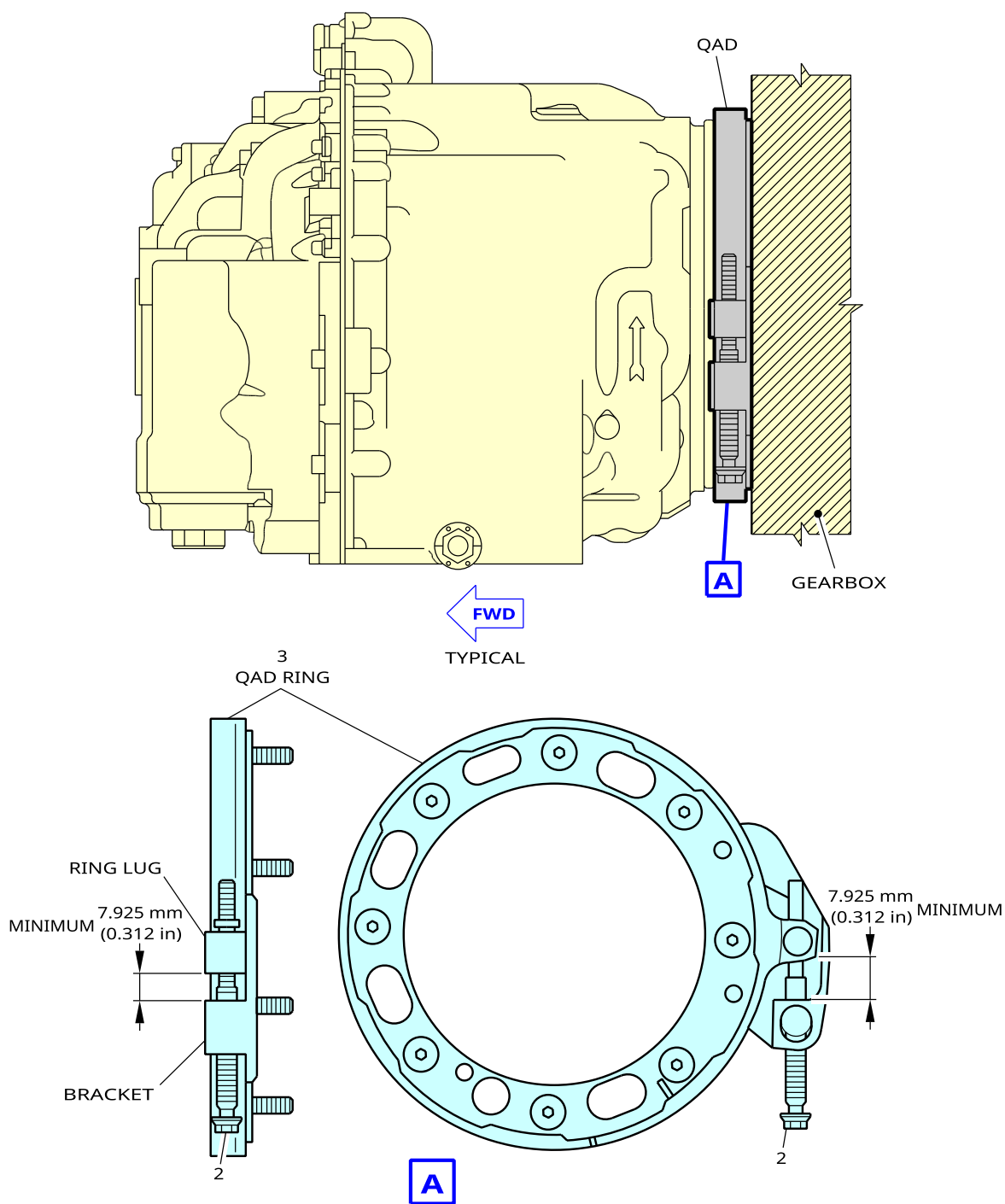
NOTE:

- 01 CORROSION PREVENTIVE COMPOUND-WATER DISPLACING OILY FILM
SYSTEM MATERIAL No 12BE2 IS APPLIED.

N_MM_242100_6_UCR0_02_00

IDG System - QAD Adapter
Figure 2 / Graphic 24-21-00 -991-90700

TF-N
AIRCRAFT MAINTENANCE MANUAL
INTEGRATED DRIVE GENERATOR SYSTEM (IDG, GCU) - INSPECTION/CHECK



N_MM_242100_6_UER0_01_03

IDG System - QAD Adapter Wear Allowance (P/N 772896)
Figure 1 / Graphic 24-21-00 -991-91300

 MET.31510236		Task Card Title: COCKPIT CHECK				
Task Card NO: 25-10-00-00-SVA		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP		
A/C Type: 321		Block Taskcard		Main Skill: CAB,ACFT_TECH		
Project No: 01BASPC72608		Related Task Card:		A/C Area:		
Internal Check:		Check Type:		Work Order NO: 31909295		
Tally Sequence NO: 15						
ENCLOSURE: ☐ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐						
Type Code: CHK		Special Code:		MSG Code: 4		
Request:		Phase:				
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRIED OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.						
Zones:						
DOCUMENT REFERENCES:						
Doc Typ		Reference				
ACCESS PANELS:						
MATERIAL REQUIREMENTS:						
Part Number		Description			Qty.	
No parts assigned to this Task Item						
TOOLING REQUIREMENTS:						
Part Number		Description			Qty.	
No tools assigned to this Task Item						
COCKPIT DEEP INSPECTION (A-CHECK):						
Job Description						
CAPTAIN SEAT: -COVERS, CUSHIONS, AND PARTS 1- PERFORM DET VISUAL CHECK (USING HAND-HELD FLASHLIGHT) THE DRESS COVER FOR ITS CONDITION, CLEANLINESS, AND PROPER FIT (NO SAGGING): A.STAINS OR DISCOLORATION ARE UNACCEPTABLE. B.PEN / INK MARKS MUST BE CLEANED. C.VERIFY PROPER ATTACHMENT OF VELCROS. D.COLOR FADING IS NOT ACCEPTABLE. E.WRINKLES PERMITTED; CRACKS AND CUTS ARE UNACCEPTABLE. F.STITCHING MUST BE INTACT AND CONTINUOUS. G.BAD SMELL NOT ACCEPTABLE.						
STEP COMPLETION SIGN OFF:						



MET.31510236

Task Card Title:

COCKPIT CHECK

Task Card NO:

25-10-00-00-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
						Day	Month	Year
Mechanic/s	CAB/ACFT_TECH			HRS				
Insp.				HRS				

Job Description

FIRST OFFICER SEAT:

-COVERS, CUSHIONS, AND PARTS

1- PERFORM DET VISUAL CHECK (USING HAND-HELD FLASHLIGHT) THE DRESS COVER FOR ITS CONDITION, CLEANLINESS, AND PROPER FIT (NO SAGGING):

H.STAINS OR DISCOLORATION ARE UNACCEPTABLE.

I.PEN / INK MARKS MUST BE CLEANED.

J.VERIFY PROPER ATTACHMENT OF VELCROS.

K.COLOR FADING IS NOT ACCEPTABLE.

L.WRINKLES PERMITTED; CRACKS AND CUTS ARE UNACCEPTABLE.

M.STITCHING MUST BE INTACT AND CONTINUOUS.

N.BAD SMELL NOT ACCEPTABLE.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
						Day	Month	Year
Mechanic/s	CAB/ACFT_TECH			HRS				
Insp.				HRS				

Job Description

SECOND AND THERED OBSERVER SEAT:

-COVERS, CUSHIONS, AND PARTS

1- PERFORM DET VISUAL CHECK OF (USING HAND-HELD FLASHLIGHT) THE DRESS COVER FOR ITS CONDITION, CLEANLINESS, AND PROPER FIT (NO SAGGING):

O.STAINS OR DISCOLORATION ARE UNACCEPTABLE.

P.PEN / INK MARKS MUST BE CLEANED.

Q.VERIFY PROPER ATTACHMENT OF VELCROS.

R.COLOR FADING IS NOT ACCEPTABLE.

S.WRINKLES PERMITTED; CRACKS AND CUTS ARE UNACCEPTABLE.

T.STITCHING MUST BE INTACT AND CONTINUOUS.

U.BAD SMELL NOT ACCEPTABLE.



MET.31510236

Task Card Title:

COCKPIT CHECK

Task Card NO:

25-10-00-00-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP**STEP COMPLETION SIGN OFF:**

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	CAB/ACFT_TECH		HRS				
Insp.			HRS				

Job Description

LIFE VEST COMPARTMENT:

1- PERFORM GENERAL VISUAL CHECK OF LIFE VEST COMPARTMENT FOR CLEANLINESS, CONDITION AND SECURITY.

- A.PULL STRAP PROPERLY INSTALLED.
- B.PARTS MUST BE ALIGNED WITH NO GAPS.
- C.VELCROS PROPERLY INSTALLED.
- D.LOOSE OR DAMAGED DOOR HINGES ARE NOT ACCEPTABLE.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	CAB/ACFT_TECH		HRS				
Insp.			HRS				

Job Description

AFT SHROUD

1- PERFORM DET VISUAL CHECK OF AFT SHROUD FOR CLEANLINESS AND CONDITION (USING HAND-HELD FLASHLIGHT).

- A.ANY CRACKS, DENTS, OR OTHER DAMAGE ARE NOT ACCEPTABLE.
- B.SURFACE SCRATCHES ARE ACCEPTABLE, BUT DEEP SCRATCHES THAT EXPOSE THE BASE MATERIAL ARE NOT ACCEPTABLE.
- C.SURFACES SHOULD BE CLEAN AND FREE OF ANY STAINS OR MARKS.
- D.MISSING, BROKEN, DAMAGED, DISBONDED, OR LOOSE PARTS ARE NOT ACCEPTABLE.
- E.SHARP EDGES OR BURRS ARE NOT ACCEPTABLE.
- F.WORN OR CHIPPED PAINT AND FINISH ARE NOT ACCEPTABLE.
- G.PARTS SHOULD BE ALIGNED PROPERLY AND THERE SHOULD BE NO GAPS.
- H.MISSING OR LOOSE FASTENERS ARE NOT ACCEPTABLE.
- I.ALL VELCROS SHOULD BE PROPERLY INSTALLED.
- J.SEALS SHOULD BE CLEAN, INSTALLED PROPERLY, AND FREE FROM MISALIGNMENT, CUTS, OR DAMAGES.
- K.PLACARDS SHOULD BE CLEAN AND IN GOOD CONDITION, WITH NO MISALIGNMENT, CUTS, OR DAMAGES.
- L. ORIENTATION OF PARTS, LAMINATES, DÉCOR, DECALS, AND PLACARDS SHOULD BE CORRECT.



Task Card Title:

COCKPIT CHECK

Task Card NO:

25-10-00-00-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP**STEP COMPLETION SIGN OFF:**

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	CAB/ACFT_TECH		HRS				
Insp.			HRS				

Job Description

SEAT ARM ASSEMBLY

1- DEPLOY AND STOW ARM-RESTS:

A.SMOOTH MOTION ACCEPTABLE. NOISY OR STIFF MOVEMENT NOT ACCEPTABLE.

NOTE : YOU MUST PUSH THE OUTER HANDLES MANY TIME.

B.CHECK THE CONDITION OF DEPLOY & STOW SWITCH/MECHANISM FOR DAMAGE OR LOOSE INSTALLATION.

C.SHOULD BE EASILY DEPLOYABLE AND STOWED WITHOUT THE NEED OF EXCESSIVE FORCE.

2- PERFORM DET VISUAL CHECK OF SEAT ARM ASSEMBLY FOR CLEANLINESS AND CONDITION (USING HAND-HELD FLASH LIGHT):

A.CRACK, DENT, OR DAMAGE ARE NOT ACCEPTABLE.

B.SURFACE SCRATCHES ARE ACCEPTABLE. DEEP SCRATCHES THAT EXPOSE THE BASE MATERIAL, ARE NOT ACCEPTABLE.

C.SURFACES ARE CLEAN AND FREE OF ANY STAINS OR MARKS.

(D) MAKE SURE THAT THE SPRINGS ARE IN GOOD CONDITION (NO CORROSION, REGULAR EXTENSION OF THE SPIRAL).

E. MAKE SURE THAT THE FITTINGS AND THE LINKS ARE IN GOOD CONDITION (NO CRACKS).

F. MAKE SURE THAT THE ROLLERS ARE NOT WORN.

G. MAKE SURE THAT THE LINKS ARE IN CORRECT CONDITION.

H.MISSING, BROKEN, DAMAGED, DISBONDED, OR LOOSE PARTS ARE NOT ACCEPTABLE.

I.SHARP EDGES OR BURRS ARE NOT ACCEPTABLE.

J.WORN OR CHIPPED PAINT AND FINISH NOT ACCEPTABLE.

K.MISALIGNED PARTS AND GAPS ARE NOT ACCEPTABLE.

L.SEALS ARE CLEAN AND INSTALLED PROPERLY. MISALIGNMENT, CUTS, OR DAMAGES ARE NOT ACCEPTABLE.

M.ARM CAPS STAINS, DAMAGE OR DISCOLORATION ARE UNACCEPTABLE.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	CAB/ACFT_TECH		HRS				
Insp.			HRS				

Job Description



Task Card Title:

COCKPIT CHECK

Task Card NO:

25-10-00-00-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP**SEAT ACTUATION****1.FUNCTION TEST THE SEAT ACTUATION SYSTEM (PERSON TO BE SEATED DURING THE TEST)**

- A.NOISY OR JERKY MOVEMENTS NOT ACCEPTABLE.
- B.INTERFERENCE OF PARTS, SHROUDS, OR OTHER ITEMS IS NOT ACCEPTABLE.
- C.FULL SEAT ACTUATION.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	CAB/ACFT_TECH		HRS				
Insp.			HRS				

Job Description**MEAL TABLE****1.DEPLOY & STOW MEAL TABLE****A. TEST THE TABLE DEPLOYMENT:**

- (1) MAKE SURE THAT THE EQUIPMENT IS LOCKED IN STOWED POSITION.
- (2) CLOSE THE HANDLE TO UNLOCK THE ARM.
- (3) PULL ON THE HANDLE UNTIL THE ROLLERS OF THE ARM ASSEMBLY ARE IN CONTACT WITH THE BODY ASSEMBLY.
- (4) PULL ON THE HANDLE UNTIL THE TABLE DEPLOYMENT.
- (5) CLOSE ON THE HANDLE TO UNLOCK THE ARM.
- (6) MAKE SURE THAT THE EQUIPMENT GOES FREELY IN TO THE STOWED POSITION.

B. TEST THE TABLE ORIENTATION:

- (1) IF NECESSARY, SET THE MODULE TO THE DEPLOYED POSITION.
- (2) DO THE UNLOCK FUNCTION:
 - (a) PUSH ON THE HANDLE TO UNLOCK THE TABLE FROM THE ARM.
 - (b) MAKE SURE THAT THE TABLE IS UNLOCKED.
- (3) TURN THE TABLE AND RELEASE THE HANDLE.
- (4) MAKE SURE THAT THE TABLE IS LOCKED.
- (5) MAKE SURE THAT THE ANGLE BETWEEN THE TABLE AND THE ARM IS 30 DEG.
- (6) DO THE STOWED FUNCTION.

NOTE : DO THIS TEST FOR THE 20 DEG. POSITION AND THE 30 DEG. POSITION OF THE TABLE.

- (a) CLOSE THE HANDLE TO UNLOCK THE ARM.
- (b) MAKE SURE THAT THE EQUIPMENT GO WITHOUT FORCE IN STOWED POSITION.

C.NOISY OR JERKY MOVEMENTS NOT ACCEPTABLE.**D.INTERFERENCE OF PARTS, SHROUDS, OR OTHER ITEMS IS NOT ACCEPTABLE.****E.CHECK FOR RIGIDITY OR ARM AND TILT MECHANISM. LOOSE INSTALLATION NOT ACCEPTABLE.****F.MISALIGNMENT OF MEAL TABLE IN STOW POSITION IS NOT ACCEPTABLE.****G.NOISY OR JERKY TABLE SLIDING MOVEMENT NOT ACCEPTABLE****2.PERFORM GENERAL VISUAL CHECK OF MEAL TABLE FOR CONDITION AND CLEANLINESS.**



Task Card Title:

COCKPIT CHECK

Task Card NO:

25-10-00-00-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

- A. CRACK, DENT, OR DAMAGE ARE NOT ACCEPTABLE.
 B. SURFACE SCRATCHES ARE ACCEPTABLE. DEEP SCRATCHES THAT EXPOSE THE BASE MATERIAL, ARE NOT ACCEPTABLE.
 C. SURFACES ARE CLEAN AND FREE OF ANY STAINS OR MARKS.
 D. MISSING, BROKEN, DAMAGED, DIS-BONDED, OR LOOSE PARTS ARE NOT ACCEPTABLE.
 E. SHARP EDGES OR BURRS ARE NOT ACCEPTABLE.
 F. WORN OR CHIPPED OR PEELED PAINT AND FINISH NOT ACCEPTABLE.
 G. PEELING OF LAMINATES OR BUBBLES IN LAMINATES ARE NOT ACCEPTABLE.
 H. TABLE SLIDING RAIL DAMAGE, SHARP EDGES, BURRS OR MISSING END CAP ARE NOT ACCEPTABLE.
 I. MAKE SURE THAT THE JUNCTION PARTS BETWEEN THE TWO TABLE ARE NOT WORN.
 J. MAKE SURE THAT THE PARTS OF THE TWO CLIP ASSEMBLIES ARE IN THE CORRECT CONDITION.
 K. MAKE SURE THAT THE LOCATING DISKS ARE IN THE CORRECT CONDITION.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	CAB/ACFT_TECH		HRS				
Insp.			HRS				

Job Description**SEAT BELT**

1. PERFORM GENERAL VISUAL CHECK OF SEAT BELT FOR CONDITION AND CLEANLINESS.

- A. STAINS OR DISCOLORATION ARE UNACCEPTABLE.
 B. WEBBING IS NOT FRAYED OR CUT.
 C. NO DENTS OR DAMAGE TO LATCH AND BUCKLE.
 D. STITCHING MUST BE INTACT AND CONTINUOUS.
 E. INSTALLATION ORIENTATION MUST BE CORRECT.

2. CHECK SEATBELT FASTEN AND UNFASTEN MECHANISM WORKS PROPERLY AND SEATBELTS INSTALLATION AT SEAT STRUCTURE IS SECURED AND NOT DAMAGED.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	CAB/ACFT_TECH		HRS				
Insp.			HRS				



MET.31510236

Task Card Title:

COCKPIT CHECK

Task Card NO:

25-10-00-00-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP**Job Description**

SEAT STRUCTURE:

1.REMOVE SEAT CUSHIONS AND PERFORM GENERAL VISUAL CHECK OF THE CONDITION FOR SERVICEABILITY OF SEATPAN, DIAPHRAGM, AND SURROUNDING STRUCTURE FOR DAMAGE, CRACKS, ETC.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
						Day	Month	Year
Mechanic/s	CAB/ACFT_TECH			HRS				
Insp.				HRS				

Job Description

COCKPIT CARPET:

1.CARPET MUST BE PROPERLY INSTALLED WITH EDGES AND CORNERS NOT LIFTED. WORN OR DAMAGED CARPET AND STAINS ARE NOT ACCEPTABLE.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
						Day	Month	Year
Mechanic/s	CAB/ACFT_TECH			HRS				
Insp.				HRS				

Job Description

SUNSHAD ROLLERS:

1. PERFORM AN OPERATIONAL AND CONDITION CHECK OF ALL FLIGHT COMPARTMENT SUNSHADE ROLLERS.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
						Day	Month	Year
Mechanic/s	CAB/ACFT_TECH			HRS				
Insp.				HRS				

Job Description

COCKPIT WALL, CEILING PANELS:

1.PERFORM DETAILED VISUAL CHECK OF THE FLIGHT COMPT INCLUDING CONTROL COLUMNS, CENTER CONSOLE, INSTRUMENT PANELS, CEILING AND SIDEWALL PANELS FOR GENERAL CONDITION AND STRUCTURAL DAMAGE. POLISH ALL EXPOSED BARE METAL AND PAINT OR TOUCH-UP AS REQUIRED.



MET.31510236

Task Card Title:

COCKPIT CHECK



Task Card NO:

25-10-00-00-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
						Day	Month	Year
Mechanic/s	CAB/ACFT_TECH			HRS				
Insp.				HRS				

BOX ASSEMBLY INSPECTION

Job Description

1. PERFORM GENERAL VISUAL INSPECTION OF BOX ASSEMBLY AS FOLLOWS:

- (1) MAKE SURE THAT THE ROLLERS ARE NOT WORN.
- (2) MAKE SURE THAT THE ECCENTRICS ARE NOT WORN.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
						Day	Month	Year
Mechanic/s	CAB/ACFT_TECH			HRS				
Insp.				HRS				

FOOTREST INSPECTION

Job Description

1. PERFORM GENERAL VISUAL INSPECTION OF FLIGHT CREW FOOTREST AS FOLLOWS:

- (1) INSPECT FOOTREST AS PER AMM TASK 25-13-14-200-002-A, SUBTASK 25-13-14-210-051-A.
- (2) MAKE SURE THAT THE BUFFERS ARE IN CORRECT CONDITION.
- (3) MAKE SURE THAT THE ROTATION AXLES ARE NOT WORN.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
						Day	Month	Year
Mechanic/s	CAB/ACFT_TECH			HRS				
Insp.				HRS				

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below

Work order No: _____

Note: Make sure to create the finding Work Order on task card level.

Remark's: _____



Task Card Title:

COCKPIT CHECK

Task Card NO:

25-10-00-00-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

(1) If Task card interrupted or performed by different personals updates the system and update the following accord (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 2 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 3 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 4 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 5 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____

**Task Completion Close Up And
System Update**

Station**Date****Signature**

 MET.31764964		Task Card Title: HZ-ASH TO ASW AISLE CARPETS R/R				
Task Card NO: 25-28-00-03-0-SVA-003		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP		
A/C Type: 321		Block Taskcard		Main Skill: CAB,ACFT_TECH		
Project No: 01BASPC72608		Related Task Card:		A/C Area: 05		
Internal Check:		Check Type:		Work Order NO: 31909295		
				Tally Sequence NO: 24		
ENCLOSURE: ☐ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐						
Type Code: INSP		Special Code: CAB		MSG Code: 4		
				Request:		
				Phase: PRE INSP		
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRID OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.						
Zones: 200						
DOCUMENT REFERENCES:						
Doc Typ		Reference				
AMM		25-28-00/41/42/43				
IPC		25-28-02/09/41/42				
ACCESS PANELS:						
MATERIAL REQUIREMENTS:						
Part Number		Description			Qty.	
No parts assigned to this Task Item						
TOOLING REQUIREMENTS:						
Part Number		Description			Qty.	
No tools assigned to this Task Item						
TASK INSTRUCTIONS						
Job Description						



Task Card Title:

HZ-ASH TO ASW AISLE CARPETS R/R

Task Card NO:

25-28-00-03-0-SVA-003

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

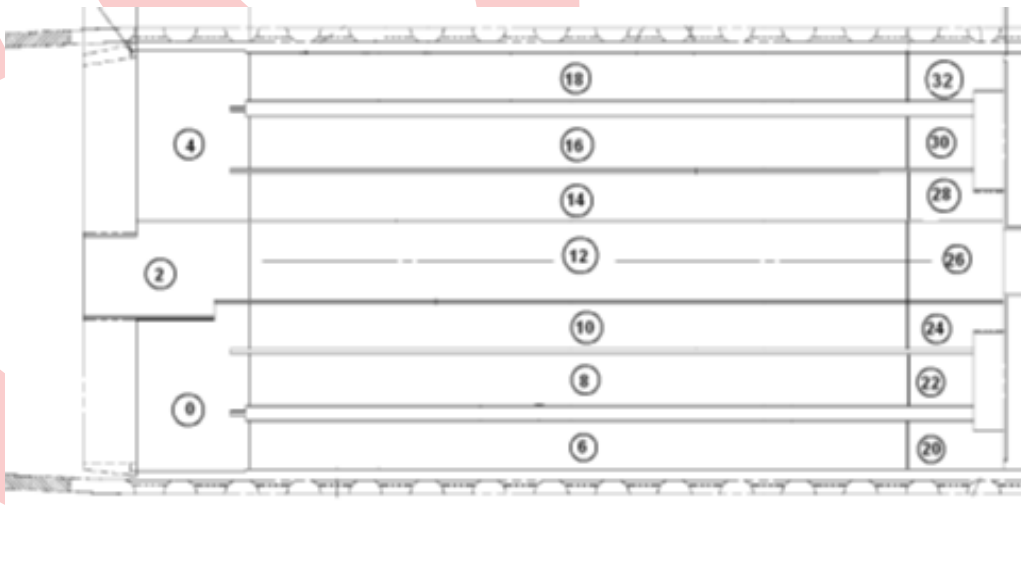
HZ-ASP

A. REPLACE PASSENGER COMPARTMENT AISLE CARPETS (REF. FIG. 1 & 2).

NOTE: ALL EXPOSED SEAT TRACKS SHOULD BE CLEANED BEFORE SEAT TRACK COVER INSTALLATION.

AISLE CARPET KIT P/N: S25-R1042-1 CONSIST OF THE FOLLOWING:

CODE NO.	CARPET P/N:	DESCRIPTION	QTY
2	D25292893202-NT1	CARPET	1
12	D25292893212-NT1	CARPET	1
26	D25292893226-NT1	CARPET	1
36	D25292893236-NT1	CARPET	1
50	D25292893250-NT1	CARPET	1
60	D25292893260-NT1	CARPET	1
70	D25292893270-NT1	CARPET	1
78	D25292893278-NT1	CARPET	1

FIGURE 1



Task Card Title:

HZ-ASH TO ASW AISLE CARPETS R/R

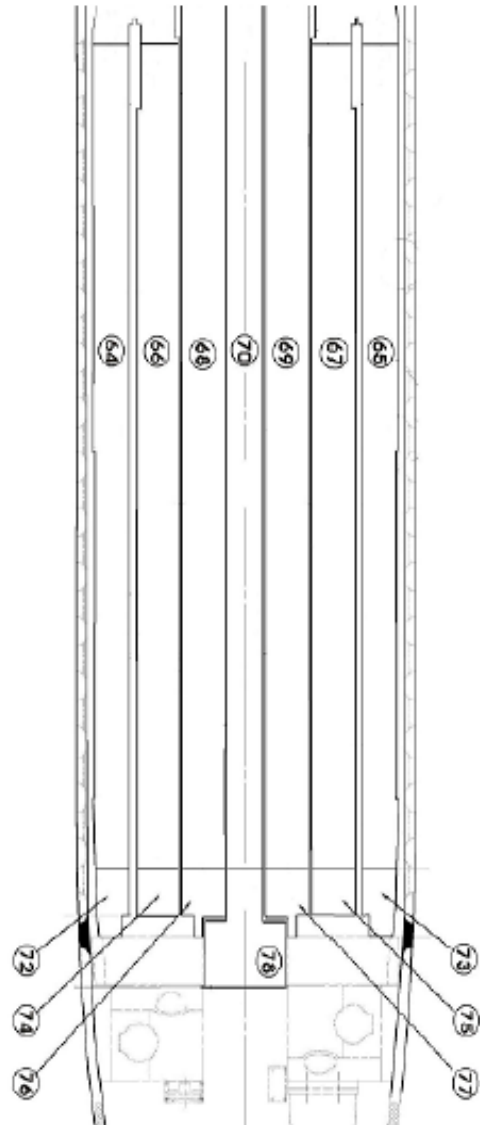
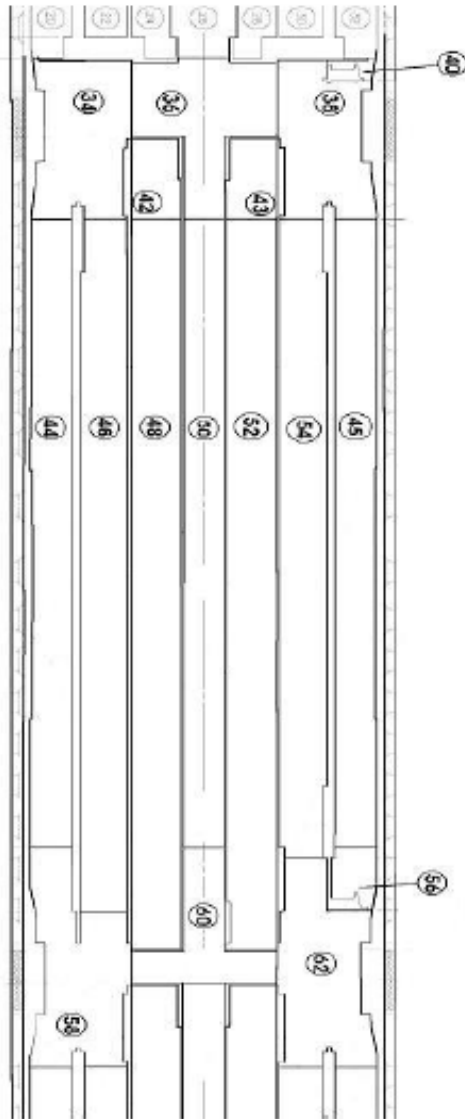
Task Card NO:

25-28-00-03-0-SVA-003

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP**FIGURE 2****STEP COMPLETION SIGN OFF:**

	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
						Day	Month	Year
Mechanic/s	CAB/ACFT_TECH			HRS				



MET.31764964

Task Card Title:

HZ-ASH TO ASW AISLE CARPETS R/R

Task Card NO:

25-28-00-03-0-SVA-003

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

Insp.

HRS

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below

Work order No: _____

Note: Make sure to creat the finding Work Order on task card level.

Remark's: _____

(1) If Task card interrupted or performed by different personals updates the system and update the following accord (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 2 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 3 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 4 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 5 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____

**Task Completion Close Up And
System Update**

Station

Date

Signature

 MET.29291052		Task Card Title: GALLY & WATER AUTO DRAIN VALVES OPC					
Task Card NO: 25-30-00-13-0-SVA		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP			
A/C Type: 321		Block Taskcard		Main Skill: CAB,ACFT_TECH		PROD Final Acceptance:	
Project No: 01BASPC72608		Related Task Card:		A/C Area: 05			
Internal Check:		Check Type:		Work Order NO: 31909295		Tally Sequence NO: 25	
ENCLOSURE: ☐ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐							
Type Code: OPC		Special Code: CAB		MSG Code: 7, 4		Request: 	
						Phase: PRE INSP	
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRIED OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.							
Zones: 200							
DOCUMENT REFERENCES:							
Doc Typ		Reference					
AMM		25-30-00					
DO		CAP# S25-11008					
ACCESS PANELS:							
MATERIAL REQUIREMENTS:							
Part Number		Description				Qty.	
No parts assigned to this Task Item							
TOOLING REQUIREMENTS:							
Part Number		Description				Qty.	
No tools assigned to this Task Item							
GALLEY SPIGOTS AND WATER AUTO DRAIN VALVES OPERATION CHECK							
Job Description							
A. PERFORM OPERATIONAL CHECK OF ALL GALLEY SPIGOTS AND WATER AUTO DRAIN VALVES, CHECK DRAIN LINE FOR CLOGGING AND WATER LEAK.							
STEP COMPLETION SIGN OFF:							
	Qualification / Skill	Number of employees	Duration		Number./Signature	Date	
						Day	Month
Mechanic/s	CAB/ACFT_TECH		HRS				Year
Insp.			HRS				
-Aircraft Maintenance Manual used: Revision No. _____ Dated._____.							
Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below							
Work order No: _____							
Note: Make sure to create the finding Work Order on task card level.							
Remark's: _____							



Task Card Title:

GALLY & WATER AUTO DRAIN VALVES OPC

Task Card NO:

25-30-00-13-0-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

(1) If Task card interrupted or performed by different personals updates the system and update the following according (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 2 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 3 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 4 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 5 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____

**Task Completion Close Up And
System Update**

Station**Date****Signature**

 MET.29291075		Task Card Title: EMERGENCY EQUIPMENT IN C/A SEAT STOWAGE INSPECTI					
Task Card NO: 25-60-00-03-0-SVA		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP			
A/C Type: 321		Block Taskcard		Main Skill: INSP,II		PROD Final Acceptance:	
Project No: 01BASPC72608		Related Task Card:		A/C Area: 05			
Internal Check:		Check Type:		Work Order NO: 31909295		Tally Sequence NO: 28	
ENCLOSURE: ☐ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐							
Type Code: CHK		Special Code: INSP		MSG Code: 7, 4		Request:	
Phase: PRE INSP							
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRID OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.							
Zones: 200							
DOCUMENT REFERENCES:							
Doc Typ		Reference					
DO		CAP# S25-11008					
ACCESS PANELS:							
MATERIAL REQUIREMENTS:							
Part Number		Description				Qty.	
No parts assigned to this Task Item							
TOOLING REQUIREMENTS:							
Part Number		Description				Qty.	
No tools assigned to this Task Item							
TASK INSTRUCTIONS							
Job Description							
A. CHECK EMERGENCY EQUIPMENT IN C/A SEAT STOWAGE FOR SECURITY AND CONDITION, MAKE SURE THAT: 1- THE FLASH LIGHT IS EXIST, SEAL SECURED AND RED LIGHT IS BLINKING. 2- LIFE VEST IS EXIST AND SECURED WITH THE RED STRAP.							
STEP COMPLETION SIGN OFF:							
	Qualification / Skill	Number of employees	Duration		Number./Signature	Date	
				HRS		Day	Month
Mechanic/s				HRS			
Insp.	INSP/II			HRS			



MET.29291075

Task Card Title:

EMERGENCY EQUIPMENT IN C/A SEAT STOWAGE INSPECTI**ON**

Task Card NO:

25-60-00-03-0-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below

Work order No: _____

Note: Make sure to create the finding Work Order on task card level.

Remark's: _____

(1) If Task card interrupted or performed by different persons updates the system and update the following accordingly (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 2 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 3 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 4 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 5 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

**Task Completion Close Up And
System Update****Station****Date****Signature**

 MET.29291049	Task Card Title: EMERGENCY EQUIP VALIDITY INSP			
Task Card NO: 25-65-00-01-0-SVA		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP
A/C Type: 321	Block Taskcard		Main Skill: II,INSP	PROD Final Acceptance:
Project No: 01BASPC72608	Related Task Card:	A/C Area: 05		
Internal Check:	Check Type:	Work Order NO: 31909295	Tally Sequence NO: 29	
ENCLOSURE: ☐ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐				
Type Code: CHK	Special Code: INSP	MSG Code: 4	Request:	Phase: PRE INSP
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRIED OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.				
Zones: 200				
DOCUMENT REFERENCES:				
Doc Typ		Reference		
CAP		S05-C10003 APPROVED.pdf		
ACCESS PANELS:				
MATERIAL REQUIREMENTS:				
Part Number		Description		Qty.
No parts assigned to this Task Item				
TOOLING REQUIREMENTS:				
Part Number		Description		Qty.
No tools assigned to this Task Item				
TASK INSTRUCTIONS				
Job Description				
A. CHECK VALIDITY OF ALL LIFE LIMITED EMERGENCY EQUIPMENT ACCORDING TO THE AIRCRAFT EMERGENCY EQUIPMENT CHART: [TICK ☒ THE APPROPRIATE CHECK BOX, IF (Yes) CHECK BOX IS TICKED, RECORD P/N, S/N & EXPIRY DATE]				
* HALON FIRE EXTINGUISHERS. ☒ Yes / ☒ No P/N: _____ S/N: _____ EXPIRY DATE _____ P/N: _____ S/N: _____ EXPIRY DATE _____ P/N: _____ S/N: _____ EXPIRY DATE _____ P/N: _____ S/N: _____ EXPIRY DATE _____ P/N: _____ S/N: _____ EXPIRY DATE _____				
* WATER FIRE EXTINGUISHERS. ☒ Yes / ☒ No P/N: _____ S/N: _____ EXPIRY DATE _____				
* PORTABLE OXYGEN BOTTLES. ☒ Yes / ☒ No P/N: _____ S/N: _____ EXPIRY DATE _____				



MET.29291049

Task Card Title:

EMERGENCY EQUIP VALIDITY INSP

Task Card NO:

25-65-00-01-0-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

P/N: _____ S/N: _____ EXPIRY DATE _____

P/N: _____ S/N: _____ EXPIRY DATE _____

P/N: _____ S/N: _____ EXPIRY DATE _____

P/N: _____ S/N: _____ EXPIRY DATE _____

P/N: _____ S/N: _____ EXPIRY DATE _____

P/N: _____ S/N: _____ EXPIRY DATE _____

P/N: _____ S/N: _____ EXPIRY DATE _____

* FLASH LIGHTS. ☐ Yes / ☐ No

P/N: _____ S/N: _____ EXPIRY DATE _____

P/N: _____ S/N: _____ EXPIRY DATE _____

P/N: _____ S/N: _____ EXPIRY DATE _____

P/N: _____ S/N: _____ EXPIRY DATE _____

P/N: _____ S/N: _____ EXPIRY DATE _____

P/N: _____ S/N: _____ EXPIRY DATE _____

P/N: _____ S/N: _____ EXPIRY DATE _____

P/N: _____ S/N: _____ EXPIRY DATE _____

P/N: _____ S/N: _____ EXPIRY DATE _____

P/N: _____ S/N: _____ EXPIRY DATE _____

P/N: _____ S/N: _____ EXPIRY DATE _____

NOTE: CHECK OPERATION, IF FOUND WEAK, BATTERIES MUST BE REPLACED.* EMERGENCY LOCATOR TRANSMITTERS. ☐ Yes / ☐ No

P/N: _____ S/N: _____ EXPIRY DATE _____

P/N: _____ S/N: _____ EXPIRY DATE _____

* LIFE RAFTS (IF INSTALLED). ☐ Yes / ☐ No

P/N: _____ S/N: _____ EXPIRY DATE _____

P/N: _____ S/N: _____ EXPIRY DATE _____

P/N: _____ S/N: _____ EXPIRY DATE _____

P/N: _____ S/N: _____ EXPIRY DATE _____

P/N: _____ S/N: _____ EXPIRY DATE _____

P/N: _____ S/N: _____ EXPIRY DATE _____

P/N: _____ S/N: _____ EXPIRY DATE _____

P/N: _____ S/N: _____ EXPIRY DATE _____

* DESCENT DEVICE. ☐ Yes / ☐ No

P/N: _____ S/N: _____ EXPIRY DATE _____

(1) IF THE EXPIRY DATE OF THE AIRCRAFT EMERGENCY EQUIPMENT IS WITHIN 80 DAYS, AND PART IS AVAILABLE, REPLACE AFFECTED PART.

(2) IF THE EXPIRY DATE OF THE THE AIRCRAFT EMERGENCY EQUIPMENT IS WITHIN 80 DAYS AND THE PART IS S/D, THE PRODUCTION

CONTROL PLANNING MUST GENERATE E-MAIL WITH ABOVE INFORMATION AND SEND TO THE FOLLOWING DEPARTMENTS:

- SV AMP SVHMP@saudia.com
- Line maintenance planning SVLMP@saudia.com



Task Card Title:

EMERGENCY EQUIP VALIDITY INSP

Task Card NO:

25-65-00-01-0-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

(PROD PLNR PRN: _____ SIGNATURE: _____.)

NOTE: OVER RIDE PROCESS MUST BE ACCOMPLISHED BY INSPECTION SUPERVISOR WHEN E-MAIL IS ISSUED.

B. CHECK VALIDITY OF MEGAPHONES ACCORDING TO THE AIRCRAFT EMERGENCY EQUIPMENT CHART:

[TICK ☒ THE APPROPRIATE CHECK BOX, IF (Yes) CHECK BOX IS TICKED, RECORD P/N, S/N & EXPIRY DATE]* MEGAPHONES. ☒ Yes / ☒ No

P/N: _____ S/N: _____ EXPIRY DATE _____.

P/N: _____ S/N: _____ EXPIRY DATE _____.

NOTE: CHECK OPERATION, IF FOUND WEAK, BATTERIES MUST BE REPLACED.**STEP COMPLETION SIGN OFF:**

	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
						Day	Month	Year
Mechanic/s				HRS				
Insp.	II/INSP			HRS				

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [☐] Yes [☐] No. If yes update the system and list Task card finding WO below

Work order No: _____

_____, _____, _____, _____, _____, _____, _____, _____

Note: Make sure to create the finding Work Order on task card level.

Remark's: _____

_____, _____



MET.29291049

Task Card Title:

EMERGENCY EQUIP VALIDITY INSP

Task Card NO:

25-65-00-01-0-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

(1) If Task card interrupted or performed by different personals updates the system and update the following according (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 2 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 3 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 4 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 5 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____

**Task Completion Close Up And
System Update**

Station**Date****Signature**

 MET.29290983		Task Card Title: SUPPLEMENTAL LIFE/RAFT PRES CHK					
Task Card NO: 256600-02-1		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP			
A/C Type: 321		Block Taskcard		Main Skill: IL,INSP,CAB,ACFT_TECH		PROD Final Acceptance:	
Project No: 01BASPC72608		Related Task Card:		A/C Area: 05			
Internal Check:		Check Type:		Work Order NO: 31909295		Tally Sequence NO: 39	
ENCLOSURE: ☐ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐							
Type Code: VCK		Special Code: CAB		MSG Code: MRB, 8		Request: Phase: JOB-SETUP,INSPECTIO N,JOB-CLOSEUP	
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRID OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.							
Zones: 200							
DOCUMENT REFERENCES:							
Doc Typ		Reference					
AMM		25-66-41-210-001-A					
AMM		256641-210-001					
ACCESS PANELS:							
MATERIAL REQUIREMENTS:							
Part Number		Description				Qty.	
No parts assigned to this Task Item							
TOOLING REQUIREMENTS:							
Part Number		Description				Qty.	
No tools assigned to this Task Item							
SUPPLEMENTAL LIFE RAFT PRESSURE CHECK							
Job Description							
A. PREPARE FOR SUPPLEMENTAL LIFE RAFT PRESSURE CHECK AS PER AMM TASK 25-66-41-210-001.							
STEP COMPLETION SIGN OFF:							
	Qualification / Skill	Number of employees	Duration		Number./Signature	Date	
				HRS		Day	Month Year
Mechanic/s	CAB/ACFT_TECH			HRS			
Insp.				HRS			
Job Description							
B. PERFORM A SUPPLEMENTAL LIFE RAFT PRESSURE GAGE READING CHECK (IF CONFIGURATION INCLUDES VISIBLE GAUGE) AS PER AMM TASK 25-66-41-210-001.							
STEP COMPLETION SIGN OFF:							
	Qualification / Skill	Number of employees	Duration		Number./Signature	Date	
				HRS		Day	Month Year

 MET.29290983	Task Card Title: SUPPLEMENTAL LIFE/RAFT PRES CHK						
Task Card NO: 256600-02-1		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP			
Mechanic/s				HRS			
Insp.	II/INSP			HRS			
Job Description							
C. RETURN THE AIRCRAFT TO ITS NORMAL CONFIGURATION AS PER AMM TASK 25-66-41-210-001.							
STEP COMPLETION SIGN OFF:							
	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	CAB/ACFT_TECH			HRS			
Insp.	INSP/II			HRS			
-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____. Task card finding (Tick Applicable Box) [☐] Yes [☐] No. If yes update the system and list Task card finding WO below Work order No: _____ _____ Note: Make sure to creat the finding Work Order on task card level. Remark's: _____ _____							



Task Card Title:

SUPPLEMENTAL LIFE/RAFT PRES CHK

Task Card NO:

256600-02-1

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

(1) If Task card interrupted or performed by different personals updates the system and update the following according (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 2 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 3 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 4 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 5 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____

**Task Completion Close Up And
System Update**

Station**Date****Signature**

TASK 25-66-41-210-001 -A

On A/C 101-200; 301-350

Check Pressure at Life Raft Pressure Gauge (If Configuration Includes Visible Gauge)

FIN: 7517MM

1. Reason for the Job

Refer to the MPD TASK: 256600-02

CHECK PRESSURE AT LIFE RAFT PRESSURE GAUGE (IF CONFIGURATION INCLUDES VISIBLE GAUGE)

2. Job Set-up Information

A. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
200	UPPER HALF FUSELAGE

B. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 24-42-00-861-001-A	Energize the Ground Service Network from the External Power
Ref. AMM 24-42-00-862-001-A	De-energize the Ground Service Network Supplied from the External Power
Ref. AMM 25-66-00-00	FLOATATION AND SURVIVAL EQUIPMENT - DESCRIPTION AND OPERATION
Ref. Fig. (1 / 25-66-41-991-00400)	

3. Job Set-up

Subtask 25-66-41-861-050-A

On A/C 101-200; 301-350

A. Energization of the Electrical Network

- (1) Energize the ground service network

[Ref. AMM 24-42-00-861-001-A.](#)

Subtask 25-66-41-010-051-A

On A/C 101-200; 301-350

B. Get Access

- (1) For the location of the supplemental life raft(s), refer to Para. Component Location

[Ref. AMM 25-66-00-00.](#)

- (2) Remove the supplemental life raft(s) from the related compartment(s).

NOTE : The location of the supplemental life raft(s) is shown with placard(s), which is (are) installed on the outside of the related compartment(s).

4. Procedure

Subtask 25-66-41-210-052-A

On A/C 101-200; 301-350

A. Visual Check of the Pressure at Life-Raft Pressure Gauge

- (1) For the correct limits of the pressure values, .

Ref. Fig. (1 / 25-66-41-991-00400)

- (2) Make sure that the life-raft pressure gauge shows the pressure value in the correct limits.

- (3) If the pressure value shown on the life-raft pressure gauge is not in the correct limits, replace the supplemental life raft.

5. Close-up

Subtask 25-66-41-410-051-A

On A/C 101-200; 301-350

A. Close Access

- (1) Install the supplemental life raft(s) in the related compartment(s).

- (2) Make sure that the work area is clean and clear of tools and other items.

TF-N
AIRCRAFT MAINTENANCE MANUAL
RAFT - LIFE - INSPECTION/CHECK

Subtask 25-66-41-862-050-A

On [A/C 101-200; 301-350](#)

B. De-Energization of the Electrical Network

(1) De-energize the ground service network

[Ref. AMM 24-42-00-862-001-A.](#)

TF-N
AIRCRAFT MAINTENANCE MANUAL
RAFT - LIFE - INSPECTION/CHECK

AMBIENT TEMPERATURE		GAUGE READING psi	
°F	°C	MINIMUM	MAXIMUM
0	-17.8	2010	2210
1	-17.2	2022	2222
2	-16.7	2034	2234
3	-16.1	2046	2246
4	-15.6	2058	2258
5	-15.0	2070	2270
6	-14.4	2082	2282
7	-13.9	2094	2294
8	-13.3	2106	2306
9	-12.8	2118	2317
10	-12.2	2130	2330
11	-11.7	2142	2342
12	-11.1	2154	2354
13	-10.6	2166	2366
14	-10.0	2178	2378
15	-9.4	2190	2390
16	-8.9	2202	2402
17	-8.3	2214	2414
18	-7.8	2226	2426
19	-7.2	2238	2438
20	-6.7	2250	2450
21	-6.1	2262	2462
22	-5.6	2274	2474
23	-5.0	2286	2486
24	-4.4	2298	2498
25	-3.9	2310	2510
26	-3.3	2322	2522
27	-2.8	2334	2534
28	-2.2	2346	2546
29	-1.7	2358	2558
30	-1.1	2370	2570
31	-0.6	2382	2582
32	0.0	2394	2594
33	0.6	2406	2606
34	1.1	2418	2618
35	1.7	2430	2630

NOTE:

THIS TABLE IS APPLICABLE FOR PN: 66536-1XX

N_MM_256641_6_SLP7_01_03

Supplemental Life Raft - Pressure Chart
Figure 1 / Graphic 25-66-41 -991-00400

TF-N
AIRCRAFT MAINTENANCE MANUAL
RAFT - LIFE - INSPECTION/CHECK

AMBIENT TEMPERATURE		GAUGE READING psi	
°F	°C	MINIMUM	MAXIMUM
36	2.2	2442	2642
37	2.8	2454	2654
38	3.3	2466	2666
39	3.9	2478	2678
40	4.4	2490	2690
41	5.0	2502	2702
42	5.6	2514	2714
43	6.1	2526	2726
44	6.7	2538	2738
45	7.2	2550	2750
46	7.8	2562	2762
47	8.3	2574	2774
48	8.9	2586	2786
49	9.4	2598	2798
50	10.0	2610	2810
51	10.6	2622	2822
52	11.1	2634	2834
53	11.7	2646	2846
54	12.2	2658	2858
55	12.8	2670	2870
56	13.3	2682	2882
57	13.9	2694	2894
58	14.4	2706	2906
59	15.0	2718	2918
60	15.6	2730	2930
61	16.1	2742	2942
62	16.7	2754	2954
63	17.2	2766	2966
64	17.8	2778	2978
65	18.3	2790	2990
66	18.9	2802	3002
67	19.4	2814	3014
68	20.0	2826	3026
69	20.6	2838	3038
70	21.1	2850	3050
71	21.7	2862	3062

NOTE:

THIS TABLE IS APPLICABLE FOR PN: 66536-1XX

N_MM_256641_6_SLP7_02_03

Supplemental Life Raft - Pressure Chart
Figure 2 / Graphic 25-66-41 -991-00400

TF-N
AIRCRAFT MAINTENANCE MANUAL
RAFT - LIFE - INSPECTION/CHECK

AMBIENT TEMPERATURE		GAUGE READING psi	
°F	°C	MINIMUM	MAXIMUM
72	22.2	2874	3074
73	22.8	2886	3086
74	23.3	2898	3098
75	23.9	2910	3110
76	24.4	2922	3122
77	25.0	2934	3134
78	25.6	2946	3146
79	26.1	2958	3158
80	26.7	2970	3170
81	27.2	2982	3182
82	27.8	2994	3194
83	28.3	3006	3206
84	28.9	3018	3218
85	29.4	3030	3230
86	30.0	3042	3242
87	30.6	3054	3254
88	31.1	3066	3266
89	31.7	3078	3278
90	32.2	3090	3290
91	32.8	3102	3302
92	33.3	3114	3314
93	33.9	3126	3326
94	34.4	3138	3338
95	35.0	3150	3350
96	35.6	3162	3362
97	36.1	3174	3374
98	36.7	3186	3386
99	37.2	3198	3398
100	37.8	3210	3410
101	38.3	3222	3422
102	38.9	3234	3434
103	39.4	3246	3446
104	40.0	3258	3458
105	40.6	3270	3470
106	41.1	3282	3482
107	41.7	3294	3494

NOTE:

THIS TABLE IS APPLICABLE FOR PN: 66536-1XX

N_MM_256641_6_SLP7_03_02

Supplemental Life Raft - Pressure Chart
Figure 3 / Graphic 25-66-41 -991-00400

TF-N
AIRCRAFT MAINTENANCE MANUAL
RAFT - LIFE - INSPECTION/CHECK

AMBIENT TEMPERATURE		GAUGE READING psi	
°F	°C	MINIMUM	MAXIMUM
108	42.2	3306	3506
109	42.8	3318	3518
110	43.3	3330	3530
111	43.9	3342	3542
112	44.4	3354	3554
113	45.0	3366	3566
114	45.6	3378	3578
115	46.1	3390	3590
116	46.7	3402	3602
117	47.2	3414	3614
118	47.8	3426	3626
119	48.3	3438	3638
120	48.9	3450	3650

NOTE:

THIS TABLE IS APPLICABLE FOR PN: 66536-1XX

N_MM_256641_6_SLP7_04_03

Supplemental Life Raft - Pressure Chart
Figure 4 / Graphic 25-66-41 -991-00400

TF-N
AIRCRAFT MAINTENANCE MANUAL
RAFT - LIFE - INSPECTION/CHECK

AMBIENT TEMPERATURE		GAUGE READING psig (bar)
°F	°C	
110	43	4700 (327)
100	38	4600 (318)
90	32	4500 (311)
80	27	4400 (305)
70	21	4350 (299)
60	16	4250 (293)
50	10	4200 (290)
40	4	4100 (284)
30	-1	4000 (278)

MINIMUM PRESSURE GAUGE READING VS. AMBIENT TEMPERATURE

NOTE:

THIS TABLE IS APPLICABLE FOR PN: R1700-1XX

N_MM_256641_6_SLP7_05_01

Supplemental Life Raft - Pressure Chart
Figure 5 / Graphic 25-66-41 -991-00400

TF-N
AIRCRAFT MAINTENANCE MANUAL
RAFT - LIFE - INSPECTION/CHECK

AMBIENT TEMPERATURE		GAUGE READING psig (bar)
°F	°C	
110	43	3190 (219)
100	38	3100 (213)
90	32	3010 (207)
80	27	2940 (202)
70	21	2850 (196)
60	16	2770 (190)
50	10	2690 (185)
40	4	2610 (179)
30	-1	2520 (173)

MINIMUM PRESSURE GAUGE READING VS. AMBIENT TEMPERATURE

NOTE:

THIS TABLE IS APPLICABLE FOR PN: R1900-XXX

N_MM_256641_6_SLP7_06_01

Supplemental Life Raft - Pressure Chart
Figure 6 / Graphic 25-66-41 -991-00400

TF-N
AIRCRAFT MAINTENANCE MANUAL
RAFT - LIFE - INSPECTION/CHECK

AMBIENT TEMPERATURE		GAUGE READING psig (bar)
°F	°C	
110	43	4850 (335)
100	38	4750 (328)
90	32	4650 (321)
80	27	4550 (314)
70	21	4500 (311)
60	16	4400 (304)
50	10	4350 (300)
40	4	4250 (293)
30	-1	4150 (287)

MINIMUM PRESSURE GAUGE READING VS. AMBIENT TEMPERATURE

NOTE:

THIS TABLE IS APPLICABLE FOR PN: R0372-XXX AND R0372AXXX

N_MM_256641_6_SLP7_07_02

Supplemental Life Raft - Pressure Chart
Figure 7 / Graphic 25-66-41 -991-00400

TF-N
AIRCRAFT MAINTENANCE MANUAL
RAFT - LIFE - INSPECTION/CHECK

AMBIENT TEMPERATURE		GAUGE READING psig (bar)
°F	°C	
110	43	4700 (327)
100	38	4600 (318)
90	32	4500 (311)
80	27	4400 (305)
70	21	4350 (299)
60	16	4250 (293)
50	10	4200 (290)
40	4	4100 (284)
30	-1	4000 (278)

MINIMUM PRESSURE GAUGE READING VS. AMBIENT TEMPERATURE

NOTE:

THIS TABLE IS FOR PN: R0425-XXX AND R0425AXXX

N_MM_256641_6_SLP7_08_01

Supplemental Life Raft - Pressure Chart
Figure 8 / Graphic 25-66-41 -991-00400

TF-N
AIRCRAFT MAINTENANCE MANUAL
RAFT - LIFE - INSPECTION/CHECK

AMBIENT TEMPERATURE		GAUGE READING psi	
°F	°C	MINIMUM	MAXIMUM
0	-17.9	1983	2183
1	-17.4	1995	2195
2	-16.8	2007	2207
3	-16.2	2019	2219
4	-15.7	2031	2231
5	-15.1	2043	2243
6	-14.6	2055	2255
7	-14.0	2067	2267
8	-13.4	2079	2279
9	-12.9	2091	2291
10	-12.3	2103	2303
11	-11.8	2115	2315
12	-11.2	2127	2327
13	-10.6	2139	2339
14	-10.1	2151	2351
15	-9.5	2163	2363
16	-9.0	2175	2375
17	-8.4	2187	2387
18	-7.8	2199	2399
19	-7.3	2211	2411
20	-6.7	2223	2423
21	-6.2	2235	2435
22	-5.6	2247	2447
23	-5.0	2259	2459
24	-4.5	2271	2471
25	-3.9	2283	2483
26	-3.4	2295	2495
27	-2.8	2307	2507
28	-2.2	2319	2519
29	-1.7	2331	2531
30	-1.1	2343	2543
31	-0.6	2355	2555
32	0.0	2367	2567
33	0.6	2379	2579
34	1.1	2391	2591

NOTE:

THIS TABLE IS APPLICABLE FOR PN: 65025-1XX

N_MM_256641_6_SLP7_09_01

Supplemental Life Raft - Pressure Chart
Figure 9 / Graphic 25-66-41 -991-00400

TF-N
AIRCRAFT MAINTENANCE MANUAL
RAFT - LIFE - INSPECTION/CHECK

AMBIENT TEMPERATURE		GAUGE READING psi	
°F	°C	MINIMUM	MAXIMUM
35	1.7	2403	2603
36	2.2	2415	2615
37	2.8	2427	2627
38	3.4	2439	2639
39	3.9	2451	2651
40	4.5	2463	2663
41	5.0	2475	2675
42	5.6	2487	2687
43	6.2	2499	2699
44	6.7	2511	2711
45	7.3	2523	2723
46	7.8	2535	2735
47	8.4	2547	2747
48	9.0	2559	2759
49	9.5	2571	2771
50	10.1	2583	2783
51	10.6	2595	2795
52	11.2	2607	2807
53	11.8	2619	2819
54	12.3	2631	2831
55	12.9	2643	2843
56	13.4	2655	2855
57	14.0	2667	2867
58	14.6	2679	2879
59	15.1	2691	2891
60	15.7	2703	2903
61	16.2	2715	2915
62	16.8	2727	2927
63	17.4	2739	2939
64	17.9	2751	2951
65	18.5	2763	2963
66	19.0	2775	2975
67	19.6	2787	2987
68	20.2	2799	2999
69	20.7	2811	3011

NOTE:

THIS TABLE IS APPLICABLE FOR PN: 65025-1XX

N_MM_256641_6_SLP7_10_01

Supplemental Life Raft - Pressure Chart
Figure 10 / Graphic 25-66-41 -991-00400

TF-N
AIRCRAFT MAINTENANCE MANUAL
RAFT - LIFE - INSPECTION/CHECK

AMBIENT TEMPERATURE		GAUGE READING psi	
°F	°C	MINIMUM	MAXIMUM
70	21.3	2823	3023
71	21.8	2835	3035
72	22.4	2847	3047
73	23.0	2859	3059
74	23.5	2871	3071
75	24.1	2883	3083
76	24.6	2895	3095
77	25.2	2907	3107
78	25.8	2919	3119
79	26.3	2931	3131
80	26.9	2943	3143
81	27.4	2955	3155
82	28.0	2967	3167
83	28.6	2979	3179
84	29.1	2991	3191
85	29.7	3003	3203
86	30.2	3015	3215
87	30.8	3027	3227
88	31.4	3039	3239
89	31.9	3051	3251
90	32.5	3063	3263
91	33.0	3075	3275
92	33.6	3087	3287
93	34.2	3099	3299
94	34.7	3111	3311
95	35.3	3123	3323
96	35.8	3135	3335
97	36.4	3147	3347
98	37.0	3159	3359
99	37.5	3171	3371
100	38.1	3183	3383
101	38.6	3195	3395
102	39.2	3207	3407
103	39.8	3219	3419
104	40.3	3231	3431

NOTE:

THIS TABLE IS APPLICABLE FOR PN: 65025-1XX

N_MM_256641_6_SLP7_11_00

Supplemental Life Raft - Pressure Chart
Figure 11 / Graphic 25-66-41 -991-00400

TF-N
AIRCRAFT MAINTENANCE MANUAL
RAFT - LIFE - INSPECTION/CHECK

AMBIENT TEMPERATURE		GAUGE READING psi	
°F	°C	MINIMUM	MAXIMUM
105	40.9	3243	3443
106	41.4	3255	3455
107	42.0	3267	3467
108	42.6	3279	3479
109	43.1	3291	3491
110	43.7	3303	3503
111	44.2	3315	3515
112	44.8	3327	3527
113	45.4	3339	3539
114	45.9	3351	3551
115	46.5	3363	3563
116	47.0	3375	3575
117	47.6	3387	3587
118	48.2	3399	3599
119	48.7	3411	3611
120	49.3	3423	3623

NOTE:

THIS TABLE IS APPLICABLE FOR PN: 65025-1XX

N_MM_256641_6_SLP7_12_00

Supplemental Life Raft - Pressure Chart
Figure 12 / Graphic 25-66-41 -991-00400

 MET.31509606		Task Card Title: CGO SMOKE DETECTION OPS CHK					
Task Card NO: 261600-04-1		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP			
A/C Type: 321		Block Taskcard		Main Skill: INSP,II,AV,ACFT_TECH		PROD Final Acceptance:	
Project No: 01BASPC72608		Related Task Card:		A/C Area: 06			
Internal Check:		Check Type:		Work Order NO: 31909295		Tally Sequence NO: 43	
ENCLOSURE: ☐ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐							
Type Code: OPC		Special Code: AV		MSG Code: MRB, 8		Request: 	
Phase: OPS/LK							
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRIED OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.							
Zones: 210							
DOCUMENT REFERENCES:							
Doc Typ		Reference					
AMM		26-16-00-710-003-A					
AMM		261600-710-003					
MRB		21.28.00/06					
MRB		26.16.00/02					
ACCESS PANELS:							
MATERIAL REQUIREMENTS:							
Part Number		Description				Qty.	
No parts assigned to this Task Item							
TOOLING REQUIREMENTS:							
Part Number		Description				Qty.	
No tools assigned to this Task Item							
TASK INSTRUCTIONS							
Job Description							
A. PERFORM CARGO COMPARTMENT SMOKE DETECTION OPERATIONAL CHECK AS PER AMM TASK 26-16-00-710-003-A.							
STEP COMPLETION SIGN OFF:							
	Qualification / Skill	Number of employees	Duration		Number./Signature	Date	
				HRS		Day	Month
Mechanic/s	AV/ACFT_TECH			HRS			
Insp.	INSP/II			HRS			



MET.31509606

Task Card Title:

CGO SMOKE DETECTION OPS CHK

Task Card NO:

261600-04-1

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

- Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below

Work order No: _____

Note: Make sure to create the finding Work Order on task card level.

Remark's: _____

(1) If Task card interrupted or performed by different persons updates the system and update the following accordingly (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 2 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 3 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 4 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 5 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

**Task Completion Close Up And
System Update****Station****Date****Signature**

TF-N
AIRCRAFT MAINTENANCE MANUAL
CARGO COMPARTMENT SMOKE DETECTION - ADJUSTMENT/TEST

TASK 26-16-00-710-003 -A

On A/C 003-200; 301-350

Operational Check of Cargo Compartment Smoke Detection by CFDS to Confirm Isolation Valves Latching Circuit Integrity.

1. Reason for the Job

Refer to the MPD TASK: 261600-04

OPERATIONAL CHECK OF CARGO COMPARTMENT SMOKE DETECTION BY CFDS TO CONFIRM ISOLATION VALVES LATCHING CIRCUIT INTEGRITY.

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	AR	SAFETY CLIP - CIRCUIT BREAKER

B. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
210	CKPT,FWD COMPT BHD TO FLT COMPT BULKHEAD

C. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 24-41-00-861-002-A	Energize the Aircraft Electrical Circuits from the External Power
Ref. AMM 24-41-00-861-002-A-01	Energize the Aircraft Electrical Circuits from the APU
Ref. AMM 24-41-00-861-002-A-02	Energize the Aircraft Electrical Circuits from Engine 1(2)
Ref. AMM 24-41-00-862-002-A	De-energize the Aircraft Electrical Circuits Supplied from the External Power
Ref. AMM 24-41-00-862-002-A-01	De-energize the Aircraft Electrical Circuits Supplied from the APU
Ref. AMM 24-41-00-862-002-A-02	De-energize the Aircraft Electrical Circuits Supplied from the Engine 1(2)
Ref. AMM 31-32-00-860-005-A	Procedure to Get Access to the SYSTEM REPORT/TEST FIRE PROT Page
Ref. AMM 31-60-00-860-001-A	EIS Start Procedure
Ref. AMM 31-60-00-860-002-A	EIS Stop Procedure

3. Job Set-up

Subtask 26-16-00-860-052-D

On A/C 003-200; 301-350

A. Aircraft Maintenance Configuration

(1) Energize the aircraft electrical circuits

[Ref. AMM 24-41-00-861-002-A](#) [Ref. AMM 24-41-00-861-002-A-01](#)

[Ref. AMM 24-41-00-861-002-A-02](#).

(2) On the overhead panel 22VU, make sure that the AFT ISOL VALVE pushbutton switch is on (the OFF light is off).

(3) Do the EIS start procedure (upper ECAM DU and lower ECAM DU only)

[Ref. AMM 31-60-00-860-001-A](#).

(4) Make sure that the ENGINE/WARNING page comes into view on the ECAM upper display unit.

(5) On the ECAM control panel, push the COND pushbutton switch and make sure that the COND page comes into view on the ECAM lower display unit.

(6) On one MCDU get the SYSTEM REPORT/TEST FIRE PROT menu

[Ref. AMM 31-32-00-860-005-A](#).

Subtask 26-16-00-865-066-A

On A/C 003-200; 301-350

B. Make sure that this (these) circuit breaker(s) is (are) closed:

PANEL	DESIGNATION	FIN	LOCATION
On A/C 003-200			
49VU	COM/CIDS/DIR ESS/1	150RH	G01

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AIRCRAFT MAINTENANCE MANUAL
CARGO COMPARTMENT SMOKE DETECTION - ADJUSTMENT/TEST

PANEL	DESIGNATION	FIN	LOCATION
On A/C 003-200; 301-350			
49VU	AIR COND/AFT CARGO/ VENT & HTG/CTL & MONG	31HN	C07
49VU	AIR COND/CIDS/SDF/DIR2 ESS	18WH	C06
49VU	AIR COND/CIDS/SDF/DIR1 ESS	15WH	C05
On A/C 101-200			
49VU	COM DIR ESS/2	157RH	G02
121VU	COM NAV/CIDS/DIR NORM/2	156RH	M07
121VU	COM NAV/CIDS/DIR NORM/1	151RH	M06
On A/C 003-200			
122VU	AIR COND/CARGO/FIRE/ EXTIG	1WX	T19
On A/C 003-200; 301-350			
122VU	AFT CARGO VENT AND HTG/FAN	30HN	S22
122VU	AIR COND/CIDS/SDF/DIR NORM/2	14WH	T18
122VU	AIR COND/CIDS/SDF/DIR NORM/1	17WH	T17

4. Procedure

Subtask 26-16-00-710-052-AB

On A/C 003-200

A. Test of the Cargo-Compartment Smoke-Detection System

- (1) Both CIDS-SDF channels and all cargo-compartment smoke-detectors must be supplied with power for a minimum of 30 seconds before you start the test.
- (2) Do this test procedure:

NOTE : Procedures related to the red message will be shown during the test on the ECAM upper display unit. These procedures are automatic and can be ignored.

ACTION	RESULT
1. On the MCDU: • Push the line key adjacent to the SDCU or SMOKE indication.	On the MCDU: If SDCU indication is displayed: • The CIDS-SDF page come into view and shows: SDCU FUNCTION IS INTEGRATED IN CIDS DIRECTORS 101RH & 102RH CALLED CIDS-SDFx
	If SMOKE indication is displayed on the MCDU: • The CIDS-SDF main menu comes into view.
If SDCU indication is displayed on the MCDU: • Push the NEXT PAGE button.	On the MCDU: • The CIDS-SDF main menu comes into view.

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AIRCRAFT MAINTENANCE MANUAL
CARGO COMPARTMENT SMOKE DETECTION - ADJUSTMENT/TEST

ACTION	RESULT
2. On the MCDU: • Push the line key adjacent to the SMOKE WARN TESTS indication.	On the MCDU: • The SMOKE WARNING TESTS menu comes into view.
• Push the line key adjacent to the FWD LDCC TEST indication.	• The SMK DET FWD LDCC TEST page comes into view and the message: ON OVHD PANEL AND ECAM WARNING DISPLAY MAKE SURE THAT "SMOKE FWD CARGO SMOKE" IS INDICATED shows.
	On the glareshield panels 130VU and 131VU: • The MASTER WARN lights flash. • You hear the Continuous Repetitive Chime (CRC).
	On the ECAM upper display unit: • The message SMOKE FWD CARGO SMOKE (red) with associated procedure comes into view.
	On the CARGO SMOKE panel 22VU: • The SMOKE FWD CARGO warning light comes on.
• Push the line key adjacent to the YES indication.	On the MCDU: The SMK DET FWD LDCC TEST page comes into view and the message: TEST OK shows.
	In the cockpit: • All smoke warnings and indications are canceled. <u>NOTE</u> : There is a delay of some seconds between the time when you push the YES indication and until the warnings go out of view and the chime stops.
3. On the MCDU: • Push the line key adjacent to the RETURN indication.	On the MCDU: • The SMOKE WARNING TESTS menu comes into view.
• Push the line key adjacent to the AFT LDCC TEST indication.	On the MCDU: • The SMK DET AFT LDCC TEST page comes into view and the message: INITIAL CONDITIONS: ON SYSTEM PAGE AIR COND MAKE SURE THAT THE AFT CARGO ISOL VALVES ARE IN OPEN POSITION shows.
	On the ECAM lower display: • The aft isolation valve symbols show open.
• Push the line key adjacent to the YES indication.	
• Push the line key adjacent to the START TEST indication.	On the glareshield panels 130VU and 131VU: • The MASTER WARN lights flash. • You hear the CRC.
	On the ECAM upper display unit: • The message SMOKE AFT CARGO SMOKE (red) with associated procedure comes into view.
	On the CARGO SMOKE panel 22VU: • The SMOKE AFT CARGO warning light comes on.
	On the ECAM lower display unit: • After approximately 15 seconds, the two aft isolation valve symbols close.

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AIRCRAFT MAINTENANCE MANUAL
CARGO COMPARTMENT SMOKE DETECTION - ADJUSTMENT/TEST

ACTION	RESULT
	On the MCDU: • The SMK DET AFT LDCC TEST page with the message: ON OVHD PANEL AND ECAM WARNING DISPLAY MAKE SURE THAT "SMOKE AFT CARGO SMOKE" IS INDICATED shows.
• Push the line key adjacent to the YES indication.	• The SMK DET AFT LDCC TEST page with the message: ON SYSTEM PAGE AIR COND MAKE SURE THAT THE AFT CARGO ISOL VALVES ARE IN CLOSED POSITION shows.
• Push the line key adjacent to the YES indication.	In the cockpit: • All smoke warnings and indications are canceled.
	On the ECAM lower display unit: • The two aft isolation valve symbols are closed.
	On the MCDU: • The SMK DET AFT LDCC TEST page with the message: RESET AFT CARGO ISOL VALVES MAKE SURE THAT THE AFT CARGO ISOL VALVES ARE IN OPEN POSITION shows.
4. On the circuit breaker panel 49VU: • Open then close circuit breaker 31HN.	On the ECAM lower display unit: • The two aft isolation valves are open.
• Push the line key adjacent to the YES indication.	On the MCDU: • The SMK DET AFT LDCC page with the message: TEST OK shows.
5. On the MCDU: • Push the line key adjacent to the RETURN indication.	On the MCDU: • The SMOKE WARNING TESTS menu comes into view.
• Push the line key adjacent to the RETURN indication.	• The CIDS-SDF main menu comes on.
Open, safety and tag the circuit breaker(s) that follow(s). Use the SAFETY CLIP - CIRCUIT BREAKER as necessary. • 15WH on the panel 49VU. • 17WH on the panel 122VU.	• The CIDS-SDF1 is de-energized.
6. On the MCDU: • Push the line key adjacent to the SDCU or SMOKE indication.	On the MCDU: If SDCU indication is displayed: • The CIDS-SDF page come into view and shows: SDCU FUNCTION IS INTEGRATED IN CIDS DIRECTORS 101RH & 102RH CALLED CIDS-SDFx
	If SMOKE indication is displayed on the MCDU: • The CIDS-SDF2 main menu comes into view.
If SDCU indication is displayed on the MCDU: • Push the NEXT PAGE button.	On the MCDU: • The CIDS-SDF2 main menu comes into view.

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AIRCRAFT MAINTENANCE MANUAL
CARGO COMPARTMENT SMOKE DETECTION - ADJUSTMENT/TEST

ACTION	RESULT
7. On the MCDU: • Push the line key adjacent to the SMOKE WARN TESTS indication.	• The SMOKE WARNING TESTS menu comes into view.
• Push the line key adjacent to the AFT LDCC TEST indication.	On the MCDU: • The SMK DET AFT LDCC TEST page with the message: INITIAL CONDITIONS: ON SYSTEM PAGE AIR COND MAKE SURE THAT THE AFT CARGO ISOL VALVES ARE IN OPEN POSITION shows.
	On the ECAM lower display: • The aft isolation valve symbols show open.
• Push the line key adjacent to the YES indication.	
• Push the line key adjacent to the START TEST indication.	On the glareshield panels 130VU and 131VU: • The MASTER WARN lights flash. • You hear the CRC.
	On the ECAM upper display unit: • The message SMOKE AFT CARGO SMOKE (red) with associated procedure comes into view.
	On the CARGO SMOKE panel 22VU: • The SMOKE AFT CARGO warning light comes on.
	On the ECAM lower display unit: • After approximately 15 seconds, the two aft isolation valve symbols close.
	On the MCDU: • The SMK DET AFT LDCC TEST page with the message: ON OVHD PANEL AND ECAM WARNING DISPLAY MAKE SURE THAT "SMOKE AFT CARGO SMOKE" IS INDICATED shows.
• Push the line key adjacent to the YES indication.	• The SMK DET AFT LDCC TEST page with the message: ON SYSTEM PAGE AIR COND MAKE SURE THAT THE AFT CARGO ISOL VALVES ARE IN CLOSED POSITION shows.
• Push the line key adjacent to the YES indication.	In the cockpit: • All smoke warnings and indications are canceled.
	On the ECAM lower display unit: • The two aft isolation valve symbols are closed.
	On the MCDU: • The SMK DET AFT LDCC TEST page with the message: RESET AFT CARGO ISOL VALVES MAKE SURE THAT THE AFT CARGO ISOL VALVES ARE IN OPEN POSITION shows.
8. On the circuit breaker panel 49VU: • Open then close circuit breaker 31HN.	On the ECAM lower display unit: • The two isolation valves are open.

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CARGO COMPARTMENT SMOKE DETECTION - ADJUSTMENT/TEST

ACTION	RESULT
Push the line key adjacent to the YES indication.	On the MCDU: The SMK DET AFT LDCC page with the message: 261000 CIRCUIT BREAKER 15WH/ CAN BUS A shows.
9. On the MCDU: Push the line key adjacent to the RETURN indication.	On the MCDU: The SMOKE WARNING TESTS menu comes into view.
Push the line key adjacent to the RETURN indication.	The CIDS-SDF2 main menu comes into view.
Remove the SAFETY CLIP - CIRCUIT BREAKER and the tag(s) and close this (these) circuit breaker(s): 15WH on the panel 49VU. 17WH on the panel 122VU.	The CIDS-SDF1 is re-energized.

5. Close-up

Subtask 26-16-00-860-053-A

On A/C 003-200; 301-350

A. Put the aircraft back to its initial configuration.

- (1) On the MCDU, push the line select key adjacent to the RETURN indication until the CFDS menu page comes into view.
- (2) Do the EIS stop procedure
[Ref. AMM 31-60-00-860-002-A.](#)
- (3) De-energize the aircraft electrical circuits
[Ref. AMM 24-41-00-862-002-A](#) [Ref. AMM 24-41-00-862-002-A-01](#)
[Ref. AMM 24-41-00-862-002-A-02.](#)

 MET.28379426		Task Card Title: CAPTAIN AND FIRST OFFICER RADIO PTT SWITCH REPLACEMENT					
Task Card NO: 27-92-00-01-SVA		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP			
A/C Type: 321		Block Taskcard		Main Skill: AV,ACFT_TECH		PROD Final Acceptance:	
Project No: 01BASPC72608		Related Task Card:		A/C Area:			
Internal Check:		Check Type:		Work Order NO: 31909295		Tally Sequence NO: 45	
ENCLOSURE: ☐ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐							
Type Code: R/R		Special Code: CAB		MSG Code: 4		Request: JOB SETUP, CLEAN-UP, OPS/LK	
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRIED OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.							
Zones: 800							
DOCUMENT REFERENCES:							
Doc Typ		Reference					
AMM		27-92-41-000-010-A					
AMM		27-92-41-400-010-A					
ACCESS PANELS:							
MATERIAL REQUIREMENTS:							
Part Number		Description				Qty.	
D2731009900000		SWITCH-TOGGLE				2.0	
D2731009900000		SWITCH-TOGGLE				1.0	
TOOLING REQUIREMENTS:							
Part Number		Description				Qty.	
98D27903500000		PIN-SIDE STICK LOCKING				1.0	
DMC519		ELECTRICIAN SERVICE TOOL KIT (TOOL/H				1.0	
98D27303001001		HOLDER SIDE STICK				1.0	
TASK INSTRUCTIONS							
Job Description							
A. GAIN ACCESS TO THE RADIO PTT SWITCH (REF. AMM 27-92-41) AND REMOVE & DISCARD THE CAPTAIN & FIRST OFFICER RADIO PTT SWITCH AS PER AMM TASK 27-92-41-000-010-A.							
STEP COMPLETION SIGN OFF:							
	Qualification / Skill	Number of employees	Duration		Number./Signature	Date	
						Day	Month
Mechanic/s	AV/ACFT_TECH		HRS				
Insp.			HRS				
TASK INSTRUCTIONS							
Job Description							
B. INSTALL A BRAND NEW CAPTAIN & FIRST OFFICER RADIO PTT SWITCH AS PER AMM TASK 27-92-41-400-010-A.							



Task Card Title:
CAPTAIN AND FIRST OFFICER RADIO PTT SWITCH REPLACEMENT



Task Card NO:

27-92-00-01-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP**STEP COMPLETION SIGN OFF:**

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	AV/ACFT_TECH		HRS				
Insp.			HRS				

TASK INSTRUCTIONS**Job Description**

C. RETURN ALL ITEMS DISTURBED PER THIS TASK TO SERVICEABLE CONDITION.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	AV/ACFT_TECH		HRS				
Insp.			HRS				

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below

Work order No: _____

Note: Make sure to create the finding Work Order on task card level.

Remark's: _____



MET.28379426

Task Card Title:

CAPTAIN AND FIRST OFFICER RADIO PTT SWITCH REPLAC**EMENT**

Task Card NO:

27-92-00-01-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

(1) If Task card interrupted or performed by different personals updates the system and update the following according (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 2 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 3 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 4 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 5 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____

**Task Completion Close Up And
System Update**

Station**Date****Signature**

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AIRCRAFT MAINTENANCE MANUAL
SIDE STICK ASSEMBLY - REMOVAL/INSTALLATION

TASK 27-92-41-000-010 -A

On A/C 001-999

Removal of the Radio PTT Switch

NOTE : This task is not applicable for the new-generation side-stick unit. For the new-generation side-stick unit, refer to the procedure (Ref. AMM TASK 27-92-41-000-803). To know if the side-stick installed on the aircraft is a new generation side-stick unit, refer to the procedure (Ref. AMM TASK 27-92-41-860-801).

FIN: 11RN, 12RN

1. Reason for the Job

Self Explanatory

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
DMC519	1	ELECTRICIAN SERVICE TOOL KIT

B. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
211	COCKPIT
212	COCKPIT

C. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 27-92-41-000-006-A	Removal of the Control Handle
Ref. ESPM 20-56-42	
Ref. Fig. (1 / 27-92-41-991-00800)	
Ref. Fig. (1 / 27-92-41-991-00900)	
Ref. Fig. (1 / 27-92-41-991-01000)	

3. Job Set-up

Subtask 27-92-41-010-061-A

On A/C 001-999

A. Remove the control handle [Ref. AMM 27-92-41-000-006-A](#).

4. Procedure

Subtask 27-92-41-020-059-A

On A/C 001-999

A. Removal of the Radio Push-to-Talk (PTT) Switch

Ref. Fig. (1 / 27-92-41-991-00800)

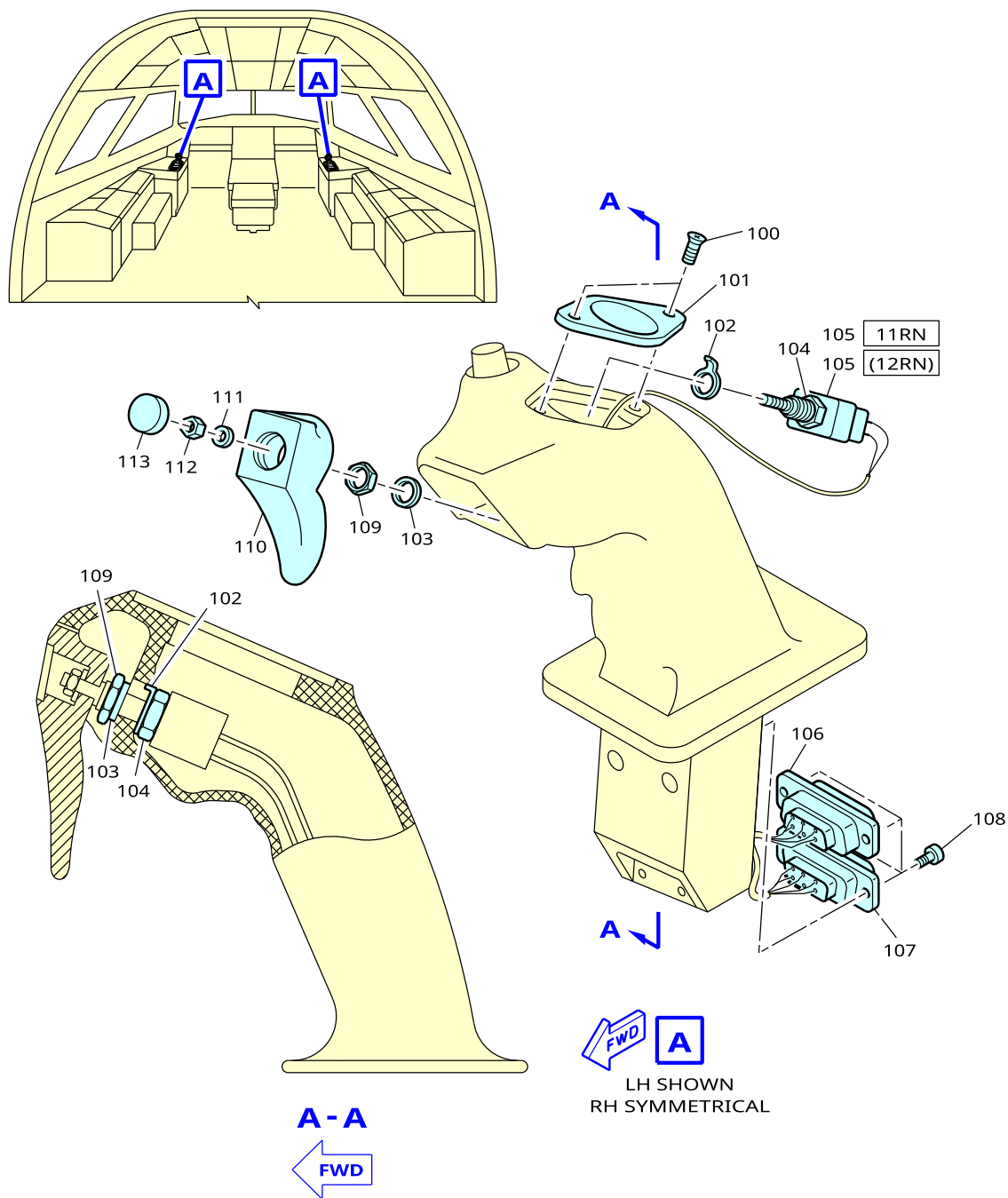
CAUTION: BEFORE YOU DISCONNECT OR CONNECT THE ELECTRICAL CONNECTORS AND RECEPTACLES, MAKE SURE THAT THEY ARE CLEAN. CONTAMINATION ON THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO THE EQUIPMENT.

- (1) Remove the screws (108) and disengage the electrical connectors (106) and (107).
- (2) On the electrical connector (107), disconnect the cables of the PTT switch (105) with the extraction/insertion tool with Part Number (P/N) M81969/14.01 **ELECTRICIAN SERVICE TOOL KIT** (DMC519)
Ref. ESPM 20-56-42.
- (3) Make a mark on the disconnected cables of the PTT switch (105).
Ref. Fig. (1 / 27-92-41-991-00900)
Ref. Fig. (1 / 27-92-41-991-01000)
- (4) Remove the screws (100) and the cover plate (101).

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SIDE STICK ASSEMBLY - REMOVAL/INSTALLATION

- (5) Remove the blanking plug (113), nut (112), washer (111) and trigger (110).
- (6) Remove the nut (109), washer (103) and PTT switch (105) together with the washer (102).

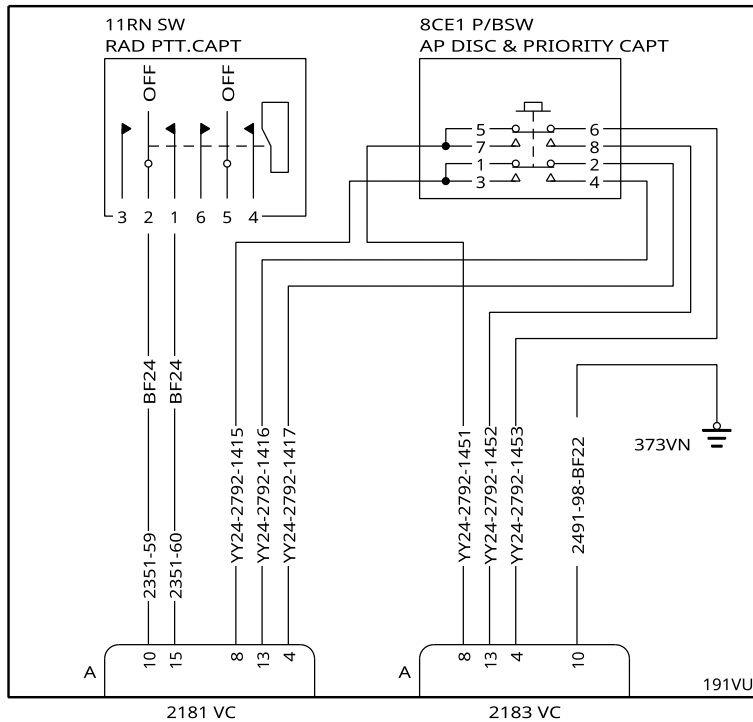
TF-N
AIRCRAFT MAINTENANCE MANUAL
SIDE STICK ASSEMBLY - REMOVAL/INSTALLATION



N_MM_279241_4_APM0_01_03

Radio Push-to-Talk (PTT) Switch
Figure 1 / Graphic 27-92-41 -991-00800

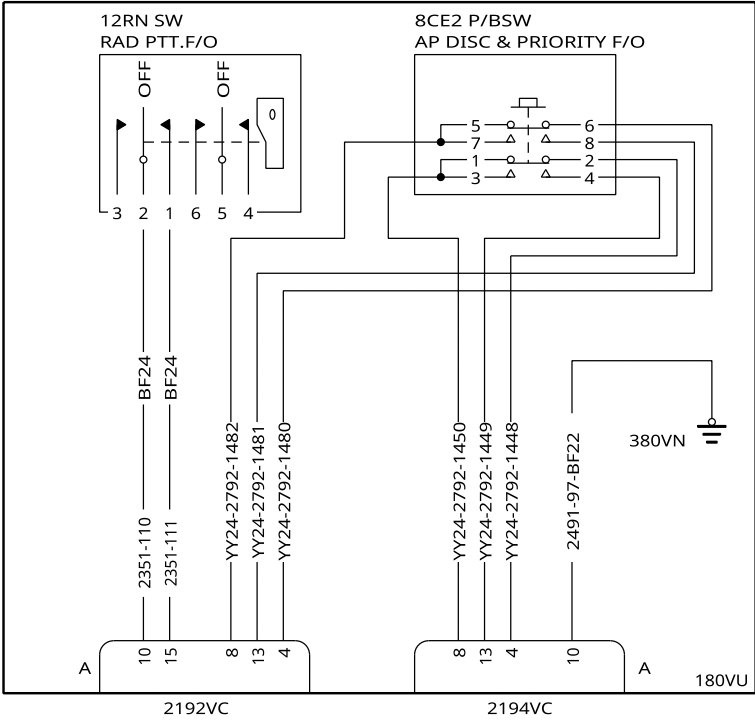
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AIRCRAFT MAINTENANCE MANUAL
SIDE STICK ASSEMBLY - REMOVAL/INSTALLATION



N_MM_279241_4_ARM0_01_00

Electrical Wiring of the Captain Side-Stick Handle
Figure 1 / Graphic 27-92-41 -991-00900

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SIDE STICK ASSEMBLY - REMOVAL/INSTALLATION



N_MM_279241_4_ATM0_01_00

Electrical Wiring of the First Officer Side-Stick-Handle
Figure 1 / Graphic 27-92-41 -991-01000

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AIRCRAFT MAINTENANCE MANUAL
SIDE STICK ASSEMBLY - REMOVAL/INSTALLATION

TASK 27-92-41-400-010 -A

On A/C 001-999

Installation of the Radio PTT Switch

WARNING: BE CAREFUL WHEN YOU USE CONSUMABLE MATERIALS. OBEY THE MATERIAL MANUFACTURER'S INSTRUCTIONS AND YOUR LOCAL REGULATIONS.

NOTE : This task is not applicable for the new-generation side-stick unit. For the new-generation side-stick unit, [Ref. AMM 27-92-41-400-803-A](#). To know if the side-stick installed on the aircraft is a new generation side-stick unit, [Ref. AMM 27-92-41-860-801-A](#).

FIN: 11RN, 12RN

1. Reason for the Job

Self Explanatory

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	1	ACCESS PLATFORM 2M (6 FT) - ADJUSTABLE
No specific		Torque wrench: range to between 0.0315 and 0.0385 m.daN (2.79 and 3.41 lbf.in)
DMC519	1	ELECTRICIAN SERVICE TOOL KIT

B. Consumable Materials

REFERENCE	DESIGNATION
05BAA1	Threadlocking Adhesive-- Low strength -
06BHG1	Silicone Sealant-Firewall High Temp. -

C. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
211	COCKPIT
212	COCKPIT
822	

D. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 24-42-00-861-001-A	Energize the Ground Service Network from the External Power
Ref. AMM 27-92-41-400-006-A	Installation of the Control Handle
Ref. AMM 27-92-41-400-803-A	Installation of the Control-Handle of the New-generation Side Stick Unit (SSU)
Ref. AMM 27-92-41-860-801-A	Identification of the Side Stick Installed on the Aircraft
Ref. ESPM 20-56-42	
Ref. ESPM 20-48-24	(for corrective action)
Ref. Fig. (1 / 27-92-41-991-00800)	
Ref. Fig. (1 / 27-92-41-991-00900)	
Ref. Fig. (1 / 27-92-41-991-01000)	

3. Job Set-up

Subtask 27-92-41-860-058-A

On A/C 001-999

A. Aircraft Maintenance Configuration

- (1) Make sure that the ground service network is energized [Ref. AMM 24-42-00-861-001-A](#).
- (2) Make sure that the ACCESS PLATFORM 2M (6 FT) - ADJUSTABLE is in position at access door 822.
- (3) Make sure that access door 822 is open.
- (4) On the battery power center 105VU:
 - Make sure that the protective cover is removed.

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SIDE STICK ASSEMBLY - REMOVAL/INSTALLATION

4. Procedure

Ref. Fig. 1 / 27-92-41-991-00800

Subtask 27-92-41-420-059-A

On A/C 001-999

A. Installation of the Radio Push-to-Talk (PTT) Switch

- (1) Clean the component interfaces and the adjacent area.
- (2) Do an inspection of the component interfaces and the adjacent area.
- (3) If you install a new PTT switch (105), you must install a pin contact at the end of each wire
Ref. ESPM 20-48-24.

NOTE : The Part Number (P/N) of the pin contact is available in the Aircraft Wiring List (AWL).

- (4) Apply the Threadlocking Adhesive-- Low strength - (Material No. **05BAA1**) between the nut (104) and the thread of the PTT switch (105).
- (5) Install the nut (104) and the lock plate (102) on the PTT switch (105).
- (6) Install the PTT switch (105) in the handle.
- (7) Make sure that the stop of the lock plate (102) is in its housing.
- (8) Install the washer (103) and the nut (109), and tighten the nut (109).
- (9) Apply the Threadlocking Adhesive-- Low strength - (Material No. **05BAA1**) on the thread of the switch.
- (10) Install the trigger (110), washer (111) and nut (112).
- (11) TORQUE the nut (112) **to between 0.0315 and 0.0385 m.daN (2.79 and 3.41 lbf.in)**.
- (12) Make sure that the top of the trigger is flush with the handle:
 - (a) If the top of the trigger is not flush with the handle, do the steps that follow:
 - 1 Remove the nut (112), washer (111) and trigger (110).
 - 2 Loosen the nut (109).
 - 3 Temporarily install the trigger and turn the nut (104) until the trigger is in the correct position.
 - 4 Tighten the nut (109).
 - 5 Install the trigger (110), washer (111) and nut (112).
 - 6 TORQUE the nut (112) **to between 0.0315 and 0.0385 m.daN (2.79 and 3.41 lbf.in)**.
- (13) Install the blanking plug (113).
- (14) Apply a thin layer of Silicone Sealant-Firewall High Temp. - (Material No. **06BHG1**) in the recess for the cover plate (101).
- (15) Install the cover plate (101).
- (16) Apply the Threadlocking Adhesive-- Low strength - (Material No. **05BAA1**) on the threads of the screws (100).
- (17) Install the screws (100) and tighten them.
- (18) Cut and remove the unwanted sealant from around the cover plate (101).
- (19) Apply a thin layer of black paint, if necessary.

CAUTION: MAKE SURE THAT THE ELECTRICAL CONNECTORS ARE CLEAN BEFORE YOU DISCONNECT OR CONNECT THEM. CONTAMINATION ON THE ELECTRICAL CONNECTORS CAN CAUSE DAMAGE TO EQUIPMENT.

- (20) Connect the cables of the PTT switch (105) to the electrical connector (107) with the extraction/insertion tool with P/N M81969/14.01 from the **ELECTRICIAN SERVICE TOOL KIT** (DMC519)

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AIRCRAFT MAINTENANCE MANUAL
SIDE STICK ASSEMBLY - REMOVAL/INSTALLATION

Ref. ESPM 20-56-42.

Ref. Fig. (1 / 27-92-41-991-00900)

Ref. Fig. (1 / 27-92-41-991-01000)

(21) Put the electrical connectors (106) and (107) in position.

(22) Apply the Threadlocking Adhesive-- Low strength - (Material No. **05BAA1**) on the threads of the screws (108).

(23) Install the screws (108) and tighten them.

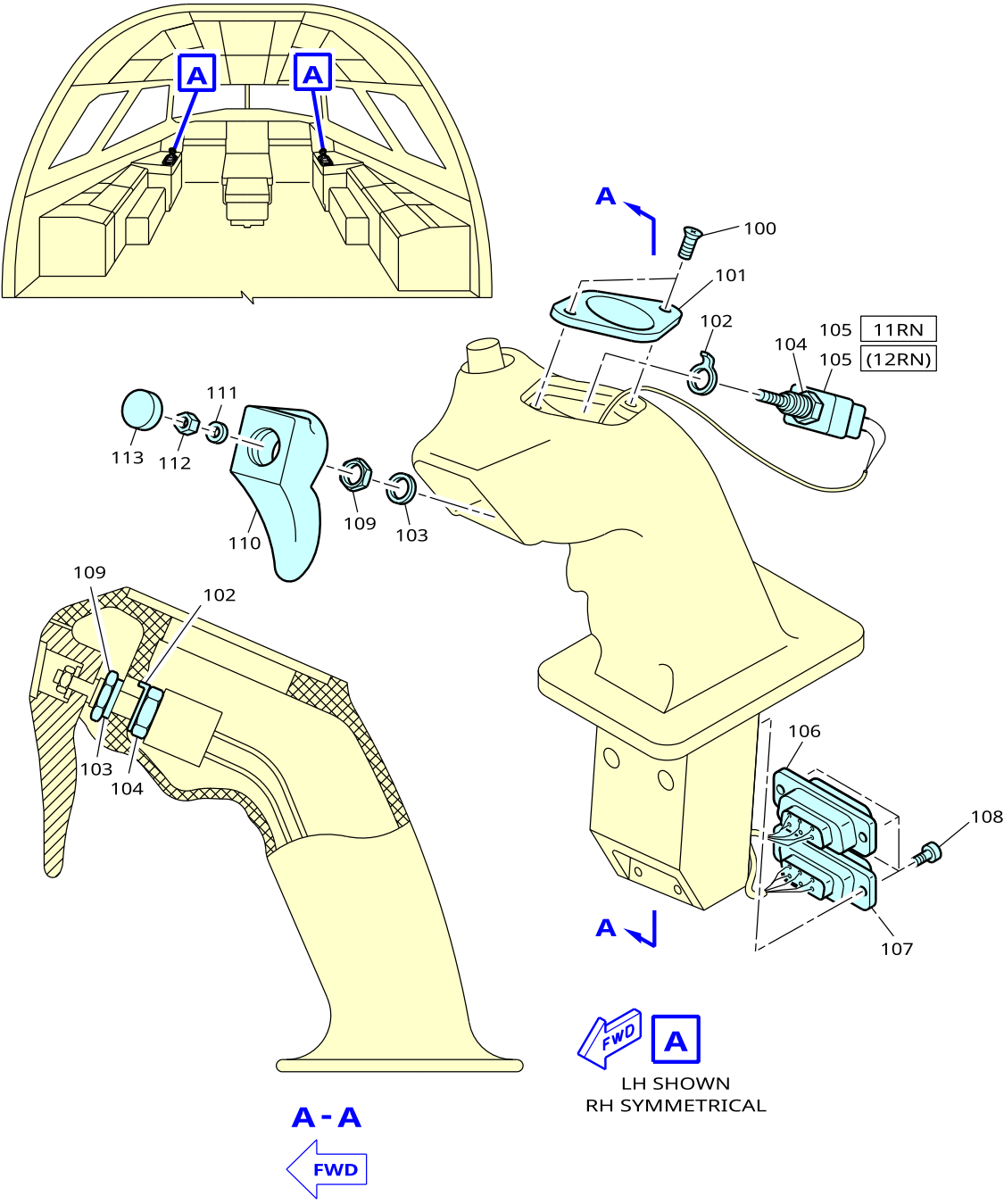
5. Close-up

Subtask 27-92-41-410-061-A

On A/C 001-999

A. Install the control handle [Ref. AMM 27-92-41-400-006-A](#).

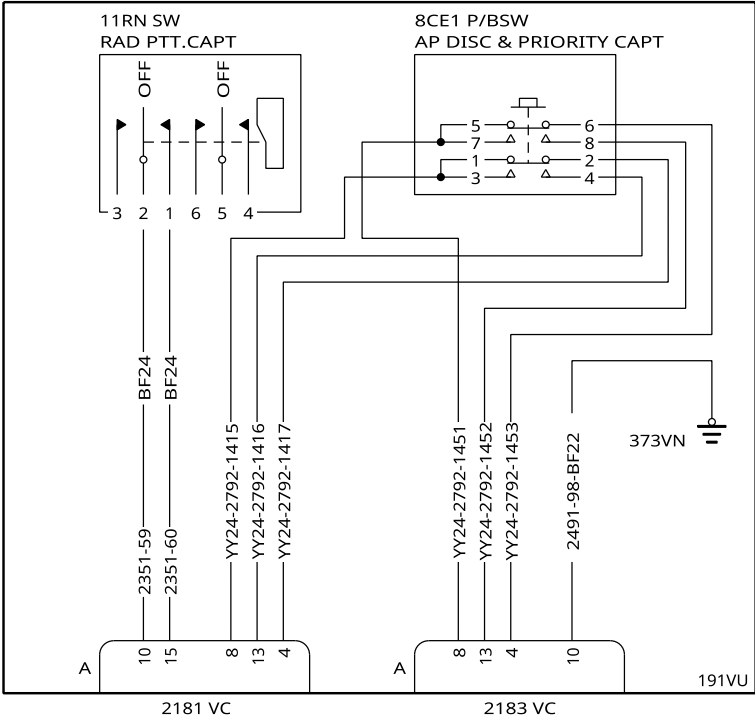
TF-N
AIRCRAFT MAINTENANCE MANUAL
SIDE STICK ASSEMBLY - REMOVAL/INSTALLATION



N_MM_279241_4_APM0_01_03

Radio Push-to-Talk (PTT) Switch
Figure 1 / Graphic 27-92-41 -991-00800

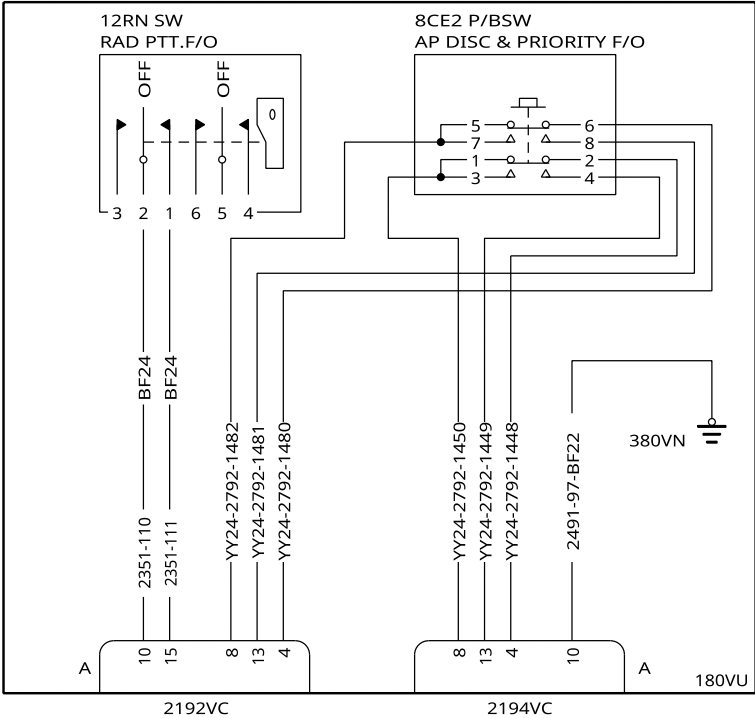
TF-N
AIRCRAFT MAINTENANCE MANUAL
SIDE STICK ASSEMBLY - REMOVAL/INSTALLATION



N_MM_279241_4_ARM0_01_00

Electrical Wiring of the Captain Side-Stick Handle
Figure 1 / Graphic 27-92-41 -991-00900

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AIRCRAFT MAINTENANCE MANUAL
SIDE STICK ASSEMBLY - REMOVAL/INSTALLATION



N_MM_279241_4_ATM0_01_00

Electrical Wiring of the First Officer Side-Stick-Handle
Figure 1 / Graphic 27-92-41 -991-01000

 MET.29291011		Task Card Title: COCKPIT AND AVIONICS COMPARTMENT AREA AND EQUIPMENT'S CLEANING.				
Task Card NO: 31-63-00-01-SVA-001		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP		
A/C Type: 321		Block Taskcard		Main Skill: AV,ACFT_TECH		
Project No: 01BASPC72608		Related Task Card:		A/C Area: 06		
Internal Check:		Check Type:		Work Order NO: 31909295		
				Tally Sequence NO: 63		
ENCLOSURE: ☐ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐						
Type Code: CLN		Special Code: AVN		MSG Code: 4		
				Request:		
				Phase: CLEAN-UP		
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRIED OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.						
Zones: 127,128,200,210,211						
DOCUMENT REFERENCES:						
Doc Typ		Reference				
AMM		31-63-22-100-001-A				
AMM		12-21-13-100-001-A				
AMM		31-63-22-000-001-A				
AMM		31-63-22-400-001-A				
ACCESS PANELS:						
MATERIAL REQUIREMENTS:						
Part Number		Description			Qty.	
TT-I-735 GRADE A		ISPOROPYL ALCOHOL S/L1YR PN:TT-I-735			1.0	
TOOLING REQUIREMENTS:						
Part Number		Description			Qty.	
No tools assigned to this Task Item						
TASK INSTRUCTION						
Job Description						
A. PERFORM THE FOLLOWING AS REQUIRED: (1) CLEAN ALL COCKPIT LCD DISPLAY UNITS AS PER AMM TASK 31-63-22-100-001-A USING "ANTISTATIC WIPE". (2) USE WITH VACUUM CLEANER AND BRUSH CLEAN PEDESTAL CONTROL PANELS AREA SURROUNDING. (3) USE COTTON TEXTILE AND BRUSH, LIFT PANELS UP ONE BY ONE AND CLEAN EACH PANEL AND AREA BELOW. (4) USE VACUUM CLEANER AND BRUSH CLEAN AVIONICS COMPARTMENT AREA ELECTRONICS AND COMPUTERS SURFACES SURROUNDING AREA, REMOVE AND REINSTALL - ON WING - ONLY FOR THE FOLLOWING SPECIFIC COMPONENTS (REF AMM TASK 12-21-13-100-001-A): (a) AVIONICS EQUIPMENT VENTILATION COMPUTER (AEVC). (b) SEC-SPOILR ELEVTR COMPUTER (SEC). (c) ELAC HARDWARE (ELAC). (d) COMPUTER - FUEL QUANTITY IND (FQIC). (e) CNTRL-INTRFACE UNT, LG (LGCIU). (f) BRAKING/STEERING CONTROL UNIT (BSCU).						



Task Card Title:

COCKPIT AND AVIONICS COMPARTMENT AREA AND EQUIPMENT'S CLEANING.

Task Card NO:

31-63-00-01-SVA-001

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

(g) BLEED MONITORING COMPUTER (BMC).

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	AV/ACFT_TECH		HRS				
Insp.			HRS				

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below

Work order No: _____

Note: Make sure to creat the finding Work Order on task card level.

Remark's: _____

(1) If Task card interrupted or performed by different personals updates the system and update the following according (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 2 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 3 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 4 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 5 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

 MET.29291011	Task Card Title: COCKPIT AND AVIONICS COMPARTMENT AREA AND EQUIPMENT'S CLEANING.			
Task Card NO: 31-63-00-01-SVA-001	Work Package: ASP/H-26 C7-CHK	A/C Registration: HZ-ASP		
Task Completion Close Up And System Update	Station	Date	Signature	

TF-N
AIRCRAFT MAINTENANCE MANUAL
DUST REMOVAL - CLEANING/PAINTING

TASK 12-21-13-100-001 -A

On A/C 001-999

Cleaning of the Avionics Compartment and the Cockpit

1. Reason for the Job

To remove the dust contamination from the avionics compartment and from the cockpit (panels 10VU, 11VU, 13VU and 101VU).

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	AR	ACCESS PLATFORM 2M (6 FT) - ADJUSTABLE
No specific	1	ACCESS PLATFORM 3M (10 FT)- ADJUSTABLE
No specific	AR	BRUSH - BRISTLED, SOFT
No specific	AR	VACUUM CLEANER

B. Consumable Materials

REFERENCE	DESIGNATION
14SBA1	Textile-Lint free Cotton -

C. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
811 , 824	

D. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 21-26-34-000-001-A	Removal of the Avionics Equipment Ventilation Computer (AEVC)
Ref. AMM 21-26-34-400-001-A	Installation of the Avionics Equipment Ventilation Computer (AEVC)
Ref. AMM 21-31-34-000-001-A	Removal of the Cabin Pressure Controller (CPC)
Ref. AMM 21-31-34-400-001-A	Installation of the Cabin Pressure Controller (CPC)
On A/C 003-200; 301-350	
Ref. AMM 21-43-34-000-002-A	Removal of the AFT Cargo Heat Controller
Ref. AMM 21-43-34-400-002-A	Installation of the AFT Cargo Heat Controller
On A/C 001-999	
Ref. AMM 21-58-34-000-001-A	Removal of the CSAS Control Unit
Ref. AMM 21-58-34-400-001-A	Installation of the Conditioned Service Air System (CSAS) Control Unit
Ref. AMM 21-61-34-000-002-A	Removal of the Air Conditioning System Controller (ACSC)
Ref. AMM 21-61-34-400-002-A	Installation of the Air Conditioning System Controller (ACSC)
On A/C 003-250; 301-350	
Ref. AMM 22-66-34-000-002-A	Removal of the Flight Augmentation Computer (FAC)
Ref. AMM 22-66-34-400-002-A	Installation of the Flight Augmentation Computer (FAC)
On A/C 001-999	
Ref. AMM 22-81-12-000-001-A	Removal of the FCU
Ref. AMM 22-81-12-400-001-A	Installation of the Flight Control Unit (FCU)
Ref. AMM 22-82-12-000-001-A	Removal of the Multipurpose Control & Display Unit (MCDU)
Ref. AMM 22-82-12-100-001-A	Cleaning of the Multipurpose Control Display Unit (MCDU)
Ref. AMM 22-82-12-400-002-A	Installation of the Multipurpose Control & Display Unit (MCDU)
Ref. AMM 22-83-34-000-001-A	Removal of the Flight Management and Guidance Computer (FMGC)
Ref. AMM 22-83-34-400-001-C	Installation of the Flight Management and Guidance Computer (FMGC)
Ref. AMM 23-11-33-000-001-D	Removal of the High Frequency (HF) Transceiver
Ref. AMM 23-11-33-400-001-D	Installation of the High Frequency (HF) Transceiver
On A/C 003-200	
Ref. AMM 23-12-33-000-001-A	Removal of the Very High Frequency (VHF) Transceiver
Ref. AMM 23-12-33-400-001-A	Installation of the Very High Frequency (VHF) Transceiver
On A/C 001-999	
Ref. AMM 23-28-34-000-003-A	Removal of the Satellite Data Unit (SDU)
Ref. AMM 23-28-34-400-003-A	Installation of the Satellite Data Unit (SDU)

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AIRCRAFT MAINTENANCE MANUAL
DUST REMOVAL - CLEANING/PAINTING

REFERENCE	DESIGNATION
On A/C 001-999	
Ref. AMM 23-51-34-000-001-A	Removal of the Audio Management Unit (AMU)
Ref. AMM 23-51-34-400-001-A	Installation of the Audio Management Unit (AMU)
Ref. AMM 23-73-34-000-001-B	Removal of the Cabin Intercommunication Data System (CIDS) Director
Ref. AMM 23-73-34-400-001-B	Installation of the Cabin Intercommunication Data System (CIDS) Director
Ref. AMM 24-22-34-000-001-A	Removal of the GCU-1(2)
Ref. AMM 24-22-34-400-001-A	Installation of the Generator Control Unit (GCU)
Ref. AMM 24-41-34-000-001-B	Removal of the Ground Auxiliary Power Control Unit (GAPCU)
Ref. AMM 24-41-34-400-001-B	Installation of the Ground Auxiliary Power Control Unit (GAPCU)
Ref. AMM 24-42-00-861-001-A	Energize the Ground Service Network from the External Power
Ref. AMM 24-42-00-862-001-A	De-energize the Ground Service Network Supplied from the External Power
Ref. AMM 27-51-34-000-001-A	Removal of the SFCC
Ref. AMM 27-51-34-400-001-A	Installation of the SFCC
Ref. AMM 27-93-34-000-001-A	Removal of the ELAC
Ref. AMM 27-93-34-400-001-A	Installation of the ELAC
Ref. AMM 27-94-34-000-001-A	Removal of the SEC
Ref. AMM 27-94-34-400-001-A	Installation of the SEC
Ref. AMM 27-95-34-000-001-A	Removal of the Flight Control Data Concentrator (FCDC)
Ref. AMM 27-95-34-400-001-A	Installation of the Flight Control Data Concentrator (FCDC)
Ref. AMM 28-42-34-000-001-A	Removal of the Fuel Quantity-Indicating Computer (FQIC)
Ref. AMM 28-42-34-400-001-A	Installation of the Fuel Quantity-Indicating Computer (FQIC)
On A/C 101-350	
Ref. AMM 28-46-34-000-001-A	Removal of the Fuel Level Sensing Control Unit (FLSCU)
Ref. AMM 28-46-34-400-001-A	Installation of the Fuel Level Sensing Control Unit (FLSCU)
On A/C 001-999	
Ref. AMM 31-32-34-000-001-A	Removal of the Centralized Fault Display Interface Unit (CFDIU)
Ref. AMM 31-32-34-400-001-A	Installation of the Centralized Fault Display Interface Unit (CFDIU)
On A/C 003-250	
Ref. AMM 31-33-52-000-002-A	Removal of the Wireless Ground Link - Quick Access Recorder (WGL-QAR)
Ref. AMM 31-33-52-400-002-A	Installation of the Wireless Ground Link - Quick Access Recorder (WGL-QAR)
On A/C 003-250; 301-350	
Ref. AMM 31-36-34-000-002-A	Removal of the FDIU/DMU (FDIMU)
Ref. AMM 31-36-34-400-002-A	Installation of the FDIU/DMU (FDIMU)
On A/C 001-999	
Ref. AMM 31-53-34-000-001-A	Removal of the Flight Warning Computer (FWC)
Ref. AMM 31-53-34-400-001-A	Installation of the Flight Warning Computer (FWC)
Ref. AMM 31-55-34-000-001-A	Removal of the SDAC
Ref. AMM 31-55-34-400-001-A	Installation of the SDAC
Ref. AMM 31-63-22-000-001-A	Removal of the Display Unit (DU)
Ref. AMM 31-63-22-400-001-A	Installation of the Display Unit (DU)
Ref. AMM 31-63-34-000-001-A	Removal of the Display Management Computer (DMC)
Ref. AMM 31-63-34-400-001-A	Installation of the Display Management Computer (DMC)
Ref. AMM 32-31-71-000-001-A	Removal of the LGCIU
Ref. AMM 32-31-71-400-001-A	Installation of the LGCIU
Ref. AMM 32-42-34-000-001-A	Removal of the Braking/Steering Control Unit (BSCU)
Ref. AMM 32-42-34-400-001-A	Installation of the Braking/Steering Control Unit (BSCU)
On A/C 003-250; 301-350	
Ref. AMM 32-49-34-000-001-A	Removal of the Tire Pressure Indicating System (TPIS) Detection Unit
Ref. AMM 32-49-34-400-001-A	Installation of the Tire Pressure Indicating System (TPIS) Detection Unit
On A/C 001-999	
Ref. AMM 34-12-34-000-001-A	Removal of the Air Data/Inertial Reference Unit (ADIRU)
Ref. AMM 34-12-34-400-001-A	Installation of the Air Data/Inertial Reference Unit (ADIRU)

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REFERENCE	DESIGNATION
On A/C 003-250	
Ref. AMM 34-36-31-000-001-B	Removal of the Multi Mode Receiver
Ref. AMM 34-36-31-400-001-B	Installation of the Multi-Mode Receiver (MMR)
On A/C 001-999	
Ref. AMM 34-41-33-000-004-A	Removal of the Weather Radar (WXR) Transceiver
Ref. AMM 34-41-33-400-003-A	Installation of the Weather Radar (WXR) Transceiver
On A/C 003-200	
Ref. AMM 34-43-34-000-004-A	Removal of the Traffic and Terrain Collision Avoidance System (T2CAS) Computer
Ref. AMM 34-43-34-400-004-A	Installation of the Traffic and Terrain Collision Avoidance System (T2CAS) Computer
On A/C 001-999	
Ref. AMM 34-51-33-000-001-A	Removal of the DME Interrogator
Ref. AMM 34-51-33-400-001-A	Installation of the DME Interrogator
Ref. AMM 34-52-33-000-001-A	Removal of the ATC/Mode S Transponder
Ref. AMM 34-52-33-400-001-A	Installation of the ATC/Mode S Transponder
On A/C 003-250; 301-350	
Ref. AMM 34-53-31-000-002-A	Removal of the ADF Receiver
Ref. AMM 34-53-31-400-002-A	Installation of the ADF Receiver
On A/C 003-250; 301-308	
Ref. AMM 34-55-31-000-001-A	Removal of the VOR/MKR Receiver
Ref. AMM 34-55-31-400-001-A	Installation of the VOR/MKR Receiver
On A/C 001-999	
Ref. AMM 36-11-34-000-001-A	Removal of the BMC
Ref. AMM 36-11-34-400-001-A	Installation of the BMC
Ref. AMM 46-21-34-000-001-A	Removal of the Air Traffic Service Unit (ATSU)
Ref. AMM 46-21-34-400-001-A	Installation of the Air Traffic Service Unit (ATSU)
Ref. AMM 46-26-32-000-801-A	Removal of the Flight Operations and MAintenance eXchanger (FOMAX)
Ref. AMM 46-26-32-400-801-A	Installation of the Flight Operations and MAintenance eXchanger (FOMAX)
On A/C 001-999	
Ref. AMM 52-41-00-010-002-A	Open the Avionics Compartment Doors for Access
Ref. AMM 52-41-00-410-002-A	Close the Avionics Compartment Doors after Access
On A/C 003-250	
Ref. AMM 73-25-34-000-040-B	Removal of the Engine Interface Unit (EIU)
Ref. AMM 73-25-34-400-040-B	Installation of the Engine Interface Unit (EIU)
Ref. AMM 77-32-34-000-042-A	Removal of the Engine Vibration Monitoring Unit (EVMU)
Ref. AMM 77-32-34-400-042-A	Installation of the Engine Vibration Monitoring Unit (EVMU)
Ref. Fig. (1 / 12-21-13-991-00100)	
Ref. Fig. (1 / 12-21-13-991-00200)	

3. Job Set-up

Subtask 12-21-13-861-050-A

On A/C 001-999

A. Energize the ground service network

Ref. AMM 24-42-00-861-001-A.

Subtask 12-21-13-941-050-A

On A/C 001-999

B. Get Access

- (1) Put an ACCESS PLATFORM 3M (10 FT)- ADJUSTABLE in position at access door 824 .
- (2) Open rear avionics-compartment access-door 824 Ref. AMM 52-41-00-010-002-A.
- (3) Put an ACCESS PLATFORM 2M (6 FT) - ADJUSTABLE in position at access door 811 .

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(4) Open forward avionics-compartment access-door 811 [Ref. AMM 52-41-00-010-002-A](#).
Subtask 12-21-13-020-051-A

On A/C 001-999

C. Removal of the Units in the Cockpit

Ref. Fig. (1 / 12-21-13-991-00100)

(1) On main instrument panel 10VU:

- Remove the display units (2WT1, 2WT2, 3WT1, 3WT2, 4WT1, 4WT2)
[Ref. AMM 31-63-22-000-001-A](#).

(2) On glareshield 13VU:

- Remove the Flight Control Unit (FCU) (2CA) [Ref. AMM 22-81-12-000-001-A](#).

(3) On center pedestal 11VU:

- Remove the Multipurpose Control and Display Units (MCDUs) (3CA1 and 3CA2)
[Ref. AMM 22-82-12-000-001-A](#).

(4) On left console 101VU:

- If the MCDU (3CA3) is installed, remove it [Ref. AMM 22-82-12-000-001-A](#).

Subtask 12-21-13-020-052-A

On A/C 001-999

D. Removal of the Computers in the Avionics Compartment From Shelf 81VU

Ref. Fig. (1 / 12-21-13-991-00200)

(1) If HF transceiver 1 (3RE1) is installed, remove it [Ref. AMM 23-11-33-000-001-D](#).

(2) If Air Traffic Control (ATC)/Mode-S transponder 1 (1SH1) is installed, remove it
[Ref. AMM 34-52-33-000-001-A](#).

(3) If Distance Measuring Equipment (DME) interrogator 1 (2SD1) is installed, remove it
[Ref. AMM 34-51-33-000-001-A](#).

(4) If VOR/MKR receiver 1 (3RS1) is installed, remove it

On A/C 003-250; 301-308

[Ref. AMM 34-55-31-000-001-A](#).

(5) If VHF transceiver 3 (1RC3) and VHF transceiver 1 (1RC1) are installed, remove them

On A/C 003-200

[Ref. AMM 23-12-33-000-001-A](#).

(6) If Multiple VHF Data Radio (MVDR) transceiver 1 (10RC1) is installed, remove it

On A/C 201-350

[Ref. AMM 23-12-33-000-802-A](#).

(7) If the Audio Management Unit (AMU) (1RN) is installed, remove it

On A/C 001-999

[Ref. AMM 23-51-34-000-001-A](#).

(8) If Automatic Direction Finder (ADF) receiver 1 (2RP1) is installed, remove it

On A/C 003-250; 301-350

[Ref. AMM 34-53-31-000-002-A](#).

(9) If Air Traffic Service Unit (ATSU) (1TX1) is installed, remove it

On A/C 001-999

[Ref. AMM 46-21-34-000-001-A](#).

(10) If the Flight Operations and MAintenance eXchanger (FOMAX) (64TA) is installed, remove it
[Ref. AMM 46-26-32-000-801-A](#).

Subtask 12-21-13-020-053-B

On A/C 003-200

E. Removal of the Computers in the Avionics Compartment From Shelf 82VU

Ref. Fig. (1 / 12-21-13-991-00200)

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- (1) If ADF receiver 2 (2RP2) is installed, remove it [Ref. AMM 34-53-31-000-002-A](#).
- (2) Remove VHF transceiver 2 (1RC2) [Ref. AMM 23-12-33-000-001-A](#).
- (3) Remove VOR/MKR receiver 2 (3RS2) [Ref. AMM 34-55-31-000-001-A](#).
- (4) Remove DME interrogator 2 (2SD2) [Ref. AMM 34-51-33-000-001-A](#).
- (5) Remove ATC/Mode S transponder-2 (1SH2) [Ref. AMM 34-52-33-000-001-A](#).
- (6) If the Traffic and Terrain Collision Avoidance System (T2CAS) computer (100SG) is installed, remove it [Ref. AMM 34-43-34-000-004-A](#).
- (7) If HF transceiver 2 (3RE2) is installed, remove it [Ref. AMM 23-11-33-000-001-D](#).
- (8) If the Aircraft Communication Addressing and Reporting System (ACARS) Management Unit (MU) (1RB) is installed, remove it [23-24-34-000-001](#).

Subtask 12-21-13-020-054-A

On A/C 001-999

- F. Removal of the Computers in the Avionics Compartment on the Shelf 83VU
Ref. Fig. (1 / 12-21-13-991-00200)

- (1) Remove the ELAC 1 (2CE1) [Ref. AMM 27-93-34-000-001-A](#).
- (2) Remove the SEC 1 (1CE1) [Ref. AMM 27-94-34-000-001-A](#).
- (3) Remove the FCDC 1 (3CE1) [Ref. AMM 27-95-34-000-001-A](#).
- (4) Remove the FMGC 1 (1CA1) [Ref. AMM 22-83-34-000-001-A](#).
- (5) Remove the FAC 1 (1CC1)

On A/C 003-250; 301-350

[Ref. AMM 22-66-34-000-002-A](#).

Subtask 12-21-13-020-055-A

On A/C 001-999

- G. Removal of the Computers in the Avionics Compartment on the Shelf 84VU
Ref. Fig. (1 / 12-21-13-991-00200)

- (1) Remove the FAC 2 (1CC2)

On A/C 003-250; 301-350

[Ref. AMM 22-66-34-000-002-A](#).

- (2) Remove the FMGC 2 (1CA2)

On A/C 001-999

[Ref. AMM 22-83-34-000-001-A](#).

- (3) Remove the FCDC 2 (3CE2) [Ref. AMM 27-95-34-000-001-A](#).
- (4) Remove the SEC 2 (1CE2) [Ref. AMM 27-94-34-000-001-A](#).
- (5) Remove the ELAC 2 (2CE2) [Ref. AMM 27-93-34-000-001-A](#).

Subtask 12-21-13-020-056-C

On A/C 003-250

- H. Removal of the Computers in the Avionics Compartment on the Shelf 85VU
Ref. Fig. (1 / 12-21-13-991-00200)

- (1) Remove the SFCC 1 (21CV) [Ref. AMM 27-51-34-000-001-A](#).
- (2) Remove the EIU1 (1KS1) [Ref. AMM 73-25-34-000-040-B](#).
- (3) Remove the DMC 1 and the DMC 3 (1WT1, 1WT3) [Ref. AMM 31-63-34-000-001-A](#).
- (4) Remove the FWC 1 (1WW1) [Ref. AMM 31-53-34-000-001-A](#).
- (5) Remove the SDAC 1 (1WV1) [Ref. AMM 31-55-34-000-001-A](#).

Subtask 12-21-13-020-057-C

On A/C 003-250

- J. Removal of the Computers in the Avionics Compartment on the Shelf 86VU
Ref. Fig. (1 / 12-21-13-991-00200)

- (1) Remove the SDAC 2 (1WV2) [Ref. AMM 31-55-34-000-001-A](#).

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- (2) Remove the FWC 2 (1WW2) [Ref. AMM 31-53-34-000-001-A](#).
- (3) Remove the DMC 2 (1WT2) [Ref. AMM 31-63-34-000-001-A](#).
- (4) Remove the EIU 2 (1KS2) [Ref. AMM 73-25-34-000-040-B](#).
- (5) Remove the EVMU (2EV) [Ref. AMM 77-32-34-000-042-A](#).
- (6) Remove the SFCC 2 (22CV) [Ref. AMM 27-51-34-000-001-A](#).

Subtask 12-21-13-020-058-A

On A/C 003-200; 251-350

K. Removal of the Computers in the Avionics Compartment on the Shelf 87VU

Ref. Fig. (1 / 12-21-13-991-00200)

- (1) Remove the Flight Data Interface Unit (FDIU) (2TU) 31-33-34-000-001, if installed.
- (2) Remove the Quick Access Recorder (QAR) (3TU) 31-33-52-000-001 or

On A/C 003-200

[Ref. AMM 31-33-52-000-002-A](#), if installed.

- (3) Remove the Data Management Unit (DMU) (1TV) 31-36-34-000-001, if installed.
- (4) Remove the Centralized Fault Display Interface Unit (CFDIU) (1TW)

On A/C 003-200; 251-350

[Ref. AMM 31-32-34-000-001-A](#).

- (5) Remove the Cabin Intercommunication Data System (CIDS) Director 1 (101RH)
[Ref. AMM 23-73-34-000-001-B](#).
- (6) Remove the Digital AIDS Recorder (DAR) (2TV) 31-36-52-000-001 or 31-36-52-000-002, if installed.
- (7) Remove the Flight Data Interface and Management Unit (FDIMU) (10TV)

On A/C 003-200; 301-350

[Ref. AMM 31-36-34-000-002-A](#)

On A/C 251-300

[Ref. AMM 31-36-34-000-002-B](#), if installed.

- (8) Remove the Avionics Equipment Ventilation Computer (AEVC) (10HQ)

On A/C 003-200; 251-350

[Ref. AMM 21-26-34-000-001-A](#), if installed.

Subtask 12-21-13-020-059-B

On A/C 001-999

L. Removal of the Computers in the Avionics Compartment From Shelf 88VU

Ref. Fig. (1 / 12-21-13-991-00200)

- (1) Remove Cabin Intercommunication Data System (CIDS) director 2 (102RH)
[Ref. AMM 23-73-34-000-001-B](#).
- (2) If the Avionics Equipment Ventilation Computer (AEVC) (10HQ) is installed, remove it
[Ref. AMM 21-26-34-000-001-A](#).
- (3) If the Ground Proximity Warning Computer (GPWC) (1WZ) is installed, remove it
34-48-34-000-001 or 34-48-34-000-003.
- (4) If the announce/music tape reproducer (10RX) is installed, remove it 23-32-34-000-001.
- (5) If the Digital Interface Unit (DIU) (190MH) is installed, remove it 23-34-34-000-001.
- (6) If the Multiplexer (MUX) (8MK) is installed, remove it 23-33-31-000-001.
- (7) If the audio reproducer (10MK) is installed, remove it 23-33-52-000-001.
- (8) If the media server (170MH) is installed, remove it 23-36-62-000-001.
- (9) If the In-Flight Entertainment (IFE) controller (311MK) is installed, remove it

On A/C 003-050; 201-350

[Ref. AMM 23-33-37-000-001-A](#).

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(10) If the IFE digital server-unit(s) (312MK and/or 317MK) is (are) installed, remove it (them)
23-33-39-000-001.

(11) If the IFE server(s) (313MK and/or 314MK and/or 315MK) is (are) installed, remove it (them)

On A/C 003-050

Ref. AMM 23-33-39-000-002-A.

(12) If the Low Earth Orbit (LEO) Satellite Data Unit (SDU) (1005RV1) is installed, remove it
23-28-34-000-804.

(13) If the LEO passive antenna (1016RV1) is installed, remove it 23-28-13-000-802 .

(14) If the SDU (5RV1) is installed, remove it

On A/C 001-999

Ref. AMM 23-28-34-000-003-A or 23-28-34-000-002 or 23-28-34-000-001.

Subtask 12-21-13-020-060-A

On A/C 001-999

M. Removal of the ADIRU (1FP1, 1FP2, 1FP3)

Ref. Fig. (1 / 12-21-13-991-00200)

(1) Remove the ADIRU 1, the ADIRU 2 and the ADIRU 3 (1FP1, 1FP2, 1FP3)

Ref. AMM 34-12-34-000-001-A.

Subtask 12-21-13-020-061-A

On A/C 001-999

N. Removal of the Computers in the Avionics Compartment on the Shelf 91VU

Ref. Fig. (1 / 12-21-13-991-00200)

(1) Remove the GCU1 and the GCU2 (1XU1, 1XU2) Ref. AMM 24-22-34-000-001-A.

(2) Remove the APU GCU (1XS) 24-23-34-000-001, if installed.

Subtask 12-21-13-020-062-C

On A/C 001-999

EMB SB 47-1029 for A/C 003-004; 051-053

P. Removal of the Computers in the Avionics Compartment From Shelf 92VU

Ref. Fig. (1 / 12-21-13-991-00200)

(1) If Fuel Level Sensing Control Unit (FLSCU) 1 (7QJ), FLSCU 2 (9QJ) and FLSCU 3 (47QJ) are installed, remove them

On A/C 101-350

Ref. AMM 28-46-34-000-001-A

On A/C 003-100

Ref. AMM 28-46-34-000-001-B.

(2) If the Fuel Quantity Indication Computer (FQIC) (3QT) is installed, remove it

On A/C 001-999

Ref. AMM 28-42-34-000-001-A.

(3) If the Ground Power Control Unit (GPCU) (1XG) is installed, remove it

Ref. AMM 24-41-34-000-001-B.

(4) If the Ground and Auxiliary Power Control Unit (GAPCU) (24XG) is installed, remove it

Ref. AMM 24-41-34-000-001-B.

(5) If the Conditioned Service Air System (CSAS) Control Unit (CU) (100HW) is installed, remove it Ref. AMM 21-58-34-000-001-A.

(6) If the Auxiliary Fuel Management Computer (AFMC) (2QN) is installed, remove it
28-51-35-000-001.

(7) If the Auxiliary Level Sensing Control Unit (ALSCU) (60QJ) is installed, remove it
28-46-35-000-001.

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Subtask 12-21-13-020-063-B

On A/C 001-999

Q. Removal of the Computers in the Avionics Compartment on the Shelf 93VU

Ref. Fig. (1 / 12-21-13-991-00200)

- (1) Remove the SEC 3 (1CE3) [Ref. AMM 27-94-34-000-001-A.](#)
- (2) Remove the LGCIU 1 (5GA1) [Ref. AMM 32-31-71-000-001-A.](#)

Subtask 12-21-13-020-064-B

On A/C 003-200; 301-350

R. Removal of the Computers in the Avionics Compartment From Shelf 94VU

Ref. Fig. (1 / 12-21-13-991-00200)

- (1) Remove LGCIU 2 (5GA2) [Ref. AMM 32-31-71-000-001-A.](#)
- (2) If the SDCU (10WQ) is installed, remove it [26-17-34-000-001.](#)
- (3) Remove the BSCU (10GG) [Ref. AMM 32-42-34-000-001-A.](#)
- (4) If the TPIS detection unit (2GV) is installed, remove it [Ref. AMM 32-49-34-000-001-A.](#)
- (5) If the aft cargo heat-controller (10HC) is installed, remove it [Ref. AMM 21-43-34-000-002-A.](#)
- (6) If the GAPCU (24XG) is installed, remove it [Ref. AMM 24-41-34-000-001-B.](#)

Subtask 12-21-13-020-065-G

On A/C 001-999

S. Removal of the Computers in the Avionics Compartment From Shelf 95VU

Ref. Fig. (1 / 12-21-13-991-00200)

- (1) If Instrument Landing System (ILS) receiver 1 (2RT1) or Multi Mode Receiver (MMR) 1 (40RT1) is installed, remove it

On A/C 003-250

[Ref. AMM 34-36-31-000-001-B](#)

On A/C 251-350

[Ref. AMM 34-36-31-000-001-E.](#)

- (2) If Bleed Monitoring Computer (BMC) 1 (1HA1) is installed, remove it

On A/C 001-999

[Ref. AMM 36-11-34-000-001-A.](#)

- (3) If Cabin Pressure Controller (CPC) 1 (11HL) is installed, remove it [Ref. AMM 21-31-34-000-001-A.](#)
- (4) If Air Conditioning System Controller (ACSC) 1 (47HH) is installed, remove it [Ref. AMM 21-61-34-000-002-A.](#)
- (5) If weather radar-transceiver 1 (1SQ1) is installed, remove it [Ref. AMM 34-41-33-000-004-A.](#)

Subtask 12-21-13-020-066-H

On A/C 003-200; 301-350

T. Removal of the Computers in the Avionics Compartment on the Shelf 96VU

Ref. Fig. (1 / 12-21-13-991-00200)

- (1) Remove the pack 2 temperature controller (57HH) [Ref. AMM 21-61-34-000-002-A](#), if installed.
- (2) Remove the cabin pressure controller 2 (12HL) [Ref. AMM 21-31-34-000-001-A.](#)
- (3) Remove the BMC 2 (1HA2) [Ref. AMM 36-11-34-000-001-A.](#)
- (4) Remove the AFT cargo heat controller (10HC) [Ref. AMM 21-43-34-000-002-A](#), if installed.
- (5) Remove the ILS Receiver 2 (2RT2)

On A/C 003-200

[Ref. AMM 34-36-31-000-001-B](#)

On A/C 301-350

[Ref. AMM 34-36-31-000-001-E](#), or the Multi Mode Receiver 2 (40RT2) , .

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Subtask 12-21-13-020-067-A

On A/C 001-999

U. Removal of the Units in the Avionics Compartment From Shelf 109VU

Ref. Fig. (1 / 12-21-13-991-00200)

- (1) If Weather Radar (WXR) transceivers 1 and 2 (1SQ1, 1SQ2) are installed, remove them

[Ref. AMM 34-41-33-000-004-A.](#)

Subtask 12-21-13-020-068-A

On A/C 001-999

V. Removal of the Rack Assembly

Ref. Fig. (1 / 12-21-13-991-00200)

- (1) On shelves 81VU, 82VU, 83VU, 84VU, 85VU, 86VU, 87VU, 88VU, 109VU, 91VU, 92VU, 93VU, 94VU, 95VU and 96VU:

- (a) Remove the nuts, the washers (if installed) and the screws from the rack assembly.
(b) Move the rack assembly to clean the area.

NOTE : It is not necessary to disconnect the electrical connectors or the cables from the rack assembly.

- (c) Make sure that the rack assembly is not removed from the shelf.

4. Procedure

Subtask 12-21-13-160-050-A

On A/C 001-999

A. Cleaning of the Avionics Compartment and the Cockpit (panels 10VU, 11VU, 13VU and 101VU)

- (1) Use a VACUUM CLEANER and a BRUSH - BRISTLED, SOFT or a Textile-Lint free Cotton - (Material No. **14SBA1**) to clean:

- (a) The area where the units and computers are installed
(b) The area of the rack assy
(c) All the faces of the removed units and computers
(d) The avionics compartment
(e) The ducts
(f) The shelves.

NOTE : Keep the vacuum cleaner out of the avionics compartment to prevent new dust contamination in the avionics compartment.

- (2) Clean the screen of the MCDUs (3CA1, 3CA2, 3CA3) [Ref. AMM 22-82-12-100-001-A.](#)

5. Close-up

Subtask 12-21-13-420-051-A

On A/C 001-999

A. Installation of the Units in the Cockpit

Ref. Fig. (1 / 12-21-13-991-00100)

NOTE : For computers with pre-loaded software, software loading is not necessary if you install the computer removed for access..

- (1) On main instrument panel 10VU:

- (a) Install the display units (2WT1, 2WT2, 3WT1, 3WT2, 4WT1, 4WT2)

[Ref. AMM 31-63-22-400-001-A.](#)

- (b) On glareshield 13VU:

- Install the FCU (2CA) [Ref. AMM 22-81-12-400-001-A.](#)

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- (c) On center pedestal 11VU:
 - Install the MCDUs (3CA1 and 3CA2) [Ref. AMM 22-82-12-400-002-A](#).
- (d) On left side console 101VU:
 - If removed, install the MCDU (3CA3) [Ref. AMM 22-82-12-400-002-A](#).

Subtask 12-21-13-420-069-A

On A/C 001-999

B. Installation of the Rack Assembly

Ref. Fig. (1 / 12-21-13-991-00200)

- (1) On shelves 81VU, 82VU, 83VU, 84VU, 85VU, 86VU, 87VU, 88VU, 109VU, 91VU, 92VU, 93VU, 94VU, 95VU and 96VU:
 - (a) Put the rack assembly in its correct installation position.
 - (b) Make sure that the electrical connectors and the cables are connected correctly.
 - (c) Install the screws, the washers (if removed) and the nuts on the rack assembly.
 - (d) Tighten the nuts.

Subtask 12-21-13-420-052-A

On A/C 001-999

C. Installation of the Computers in the Avionics Compartment on Shelf 81VU

Ref. Fig. (1 / 12-21-13-991-00200)

NOTE : If you install the computer which is removed for the access of the components, the software loading is not necessary (for the computers with the loadable software).

- (1) If HF transceiver 1 (3RE1) is removed, install it [Ref. AMM 23-11-33-400-001-D](#).
- (2) If ATC/Mode-S transponder 1 (1SH1) is removed, install it [Ref. AMM 34-52-33-400-001-A](#).
- (3) If DME interrogator 1 (2SD1) is removed, install it [Ref. AMM 34-51-33-400-001-A](#).
- (4) If VOR/MKR receiver 1 (3RS1) is removed, install it

On A/C 003-250; 301-308

[Ref. AMM 34-55-31-400-001-A](#).

- (5) If VHF transceiver 3 (1RC3) and VHF transceiver 1 (1RC1) are removed, install them

On A/C 003-200

[Ref. AMM 23-12-33-400-001-A](#).

- (6) If MVDR transceiver 1 (10RC1) is removed, install it

On A/C 201-350

[Ref. AMM 23-12-33-400-802-A](#).

- (7) If the AMU (1RN) is removed, install it

On A/C 001-999

[Ref. AMM 23-51-34-400-001-A](#).

- (8) If ADF receiver 1 (2RP1) is removed, install it

On A/C 003-250; 301-350

[Ref. AMM 34-53-31-400-002-A](#).

- (9) If ATSU (1TX1) is removed, install it

On A/C 001-999

[Ref. AMM 46-21-34-400-001-A](#).

- (10) If the FOMAX (64TA) is removed, install it [Ref. AMM 46-26-32-400-801-A](#).

Subtask 12-21-13-420-053-B

On A/C 003-200

D. Installation of the Computers in the Avionics Compartment on Shelf 82VU

Ref. Fig. (1 / 12-21-13-991-00200)

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NOTE : If you install the computer which is removed for the access of the components, the software loading is not necessary (for the computers with the loadable software).

- (1) If ADF receiver 2 (2RP2) is removed, install it [Ref. AMM 34-53-31-400-002-A](#).
- (2) Install VHF transceiver 2 (1RC2) [Ref. AMM 23-12-33-400-001-A](#).
- (3) Install VOR/MKR receiver 2 (3RS2) [Ref. AMM 34-55-31-400-001-A](#).
- (4) Install DME interrogator 2 (2SD2) [Ref. AMM 34-51-33-400-001-A](#).
- (5) Install ATC/Mode S transponder-2 (1SH2) [Ref. AMM 34-52-33-400-001-A](#).
- (6) If the T2CAS computer (100SG) is removed, install it [Ref. AMM 34-43-34-400-004-A](#).
- (7) If HF transceiver 2 (3RE2) is removed, install it [Ref. AMM 23-11-33-400-001-D](#).
- (8) If the ACARS MU (1RB) is removed, install it [23-24-34-400-001](#).

Subtask 12-21-13-420-054-A

On A/C 001-999

- E. Installation of the Computers in the Avionics Compartment on the Shelf 83VU
Ref. Fig. (1 / 12-21-13-991-00200)

NOTE : For computers with loadable software, software loading is not necessary if you install the computer removed for access.

- (1) Install the ELAC 1 (2CE1) [Ref. AMM 27-93-34-400-001-A](#).
- (2) Install the SEC 1 (1CE1) [Ref. AMM 27-94-34-400-001-A](#).
- (3) Install the FCDC 1 (3CE1) [Ref. AMM 27-95-34-400-001-A](#).
- (4) Install the FMGC 1 (1CA1) [Ref. AMM 22-83-34-400-001-C](#).
- (5) Install the FAC 1 (1CC1)

On A/C 003-250; 301-350

[Ref. AMM 22-66-34-400-002-A](#).

Subtask 12-21-13-420-055-A

On A/C 001-999

- F. Installation of the Computers in the Avionics Compartment on the Shelf 84VU
Ref. Fig. (1 / 12-21-13-991-00200)

NOTE : For computers with loadable software, software loading is not necessary if you install the computer removed for access.

- (1) Install the FAC 2 (1CC2)

On A/C 003-250; 301-350

[Ref. AMM 22-66-34-400-002-A](#).

- (2) Install the FMGC 2 (1CA2)

On A/C 001-999

[Ref. AMM 22-83-34-400-001-C](#).

- (3) Install the FCDC 2 (3CE2) [Ref. AMM 27-95-34-400-001-A](#).
- (4) Install the SEC 2 (1CE2) [Ref. AMM 27-94-34-400-001-A](#).
- (5) Install the ELAC 2 (2CE2) [Ref. AMM 27-93-34-400-001-A](#).

Subtask 12-21-13-420-056-C

On A/C 003-250

- G. Installation of the Computers in the Avionics Compartment on the Shelf 85VU
Ref. Fig. (1 / 12-21-13-991-00200)

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NOTE : For computers with loadable software, software loading is not necessary if you Install the computer removed for access.

- (1) Install the SFCC 1 (21CV) [Ref. AMM 27-51-34-400-001-A](#).
- (2) Install the EIU1 (1KS1) [Ref. AMM 73-25-34-400-040-B](#).
- (3) Install the DMC 1 and the DMC 3 (1WT1, 1WT3) [Ref. AMM 31-63-34-400-001-A](#).
- (4) Install the FWC 1 (1WW1) [Ref. AMM 31-53-34-400-001-A](#).
- (5) Install the SDAC 1 (1WV1) [Ref. AMM 31-55-34-400-001-A](#).

Subtask 12-21-13-420-057-C

On A/C 003-250

- H. Installation of the Computers in the Avionics Compartment on the shelf 86VU
Ref. Fig. (1 / 12-21-13-991-00200)

NOTE : For computers with loadable software, software loading is not necessary if you install the computer removed for access.

- (1) Install the SDAC 2 (1WV2) [Ref. AMM 31-55-34-400-001-A](#).
- (2) Install the FWC 2 (1WW2) [Ref. AMM 31-53-34-400-001-A](#).
- (3) Install the DMC 2 (1WT2) [Ref. AMM 31-63-34-400-001-A](#).
- (4) Install the EIU 2 (1KS2) [Ref. AMM 73-25-34-400-040-B](#).
- (5) Install the EVMU (2EV) [Ref. AMM 77-32-34-400-042-A](#).
- (6) Install the SFCC 2 (22CV) [Ref. AMM 27-51-34-400-001-A](#).

Subtask 12-21-13-420-058-A

On A/C 003-200; 251-350

- J. Installation of the Computers in the Avionics Compartment on the shelf 87VU
Ref. Fig. (1 / 12-21-13-991-00200)

NOTE : For computers with loadable software, software loading is not necessary if you install the computer removed for access.

- (1) Install the FDIU (2TU) 31-33-34-400-001, if removed.
- (2) Install the QAR (3TU) 31-33-52-400-001 or

On A/C 003-200

[Ref. AMM 31-33-52-400-002-A](#) , if removed.

- (3) Install the DMU (1TV) 31-36-34-400-001, if removed.
- (4) Install the CFDIU (1TW)

On A/C 003-200; 251-350

[Ref. AMM 31-32-34-400-001-A](#).

- (5) Install the CIDS Director 1 (101RH) [Ref. AMM 23-73-34-400-001-B](#).
- (6) Install the DAR (2TV) 31-36-52-400-001 or 31-36-52-400-002 , if removed.
- (7) Install the FDIMU (10TV)

On A/C 003-200; 301-350

[Ref. AMM 31-36-34-400-002-A](#)

On A/C 251-300

[Ref. AMM 31-36-34-400-002-B](#), if removed.

- (8) Install the AEVC (10HQ)

On A/C 003-200; 251-350

[Ref. AMM 21-26-34-400-001-A](#), if removed.

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Subtask 12-21-13-420-059-B

On A/C 001-999

- K. Installation of the Computers in the Avionics Compartment on Shelf 88VU
Ref. Fig. (1 / 12-21-13-991-00200)

NOTE : Software loading is not necessary for computers with loadable software, if the removed computer is installed to get access.

- (1) Install CIDS director 2 (102RH) [Ref. AMM 23-73-34-400-001-B](#).
- (2) If the AEVC (10HQ) is removed, install it [Ref. AMM 21-26-34-400-001-A](#).
- (3) If the GPWC (1WZ) is removed, install it 34-48-34-400-001 or 34-48-34-400-002.
- (4) If the announce/music tape reproducer (10RX) is removed, install it 23-32-34-400-001.
- (5) If the DIU (190MH) is removed, install it 23-34-34-400-001.
- (6) If the MUX (8MK) is removed, install it 23-33-31-400-001.
- (7) If the audio reproducer (10MK) is removed, install it 23-33-52-400-001.
- (8) If the media server (170MH) is removed, install it 23-36-62-400-001.
- (9) If the IFE controller (311MK) is removed, install it

On A/C 003-050; 201-350

[Ref. AMM 23-33-37-400-001-A](#).

- (10) If the IFE digital server-unit(s) (312MK and/or 317MK) is (are) removed, install it (them) 23-33-39-400-001.
- (11) If the IFE server(s) (313MK and/or 314MK and/or 315MK) is (are) removed, install it (them)

On A/C 003-050

[Ref. AMM 23-33-39-400-002-A](#).

- (12) If the LEO SDU (1005RV1) is removed, install it 23-28-34-400-802.
- (13) If the LEO passive antenna (1016RV1) is removed, install it 23-28-13-400-802.
- (14) If the SDU (5RV1) is removed, install it

On A/C 001-999

[Ref. AMM 23-28-34-400-003-A](#) or 23-28-34-400-002 or 23-28-34-400-001.

Subtask 12-21-13-420-060-A

On A/C 001-999

- L. Installation of the ADIRU (1FP1, 1FP2, 1FP3)
Ref. Fig. (1 / 12-21-13-991-00200)

NOTE : For computers with loadable software, software loading is not necessary if you install the computer removed for access.

- (1) Install the ADIRU 1, the ADIRU 2 and the ADIRU 3 (1FP1, 1FP2, 1FP3)
[Ref. AMM 34-12-34-400-001-A](#).

Subtask 12-21-13-420-061-A

On A/C 001-999

- M. Installation of the Computers in the Avionics Compartment on the Shelf 91VU
Ref. Fig. (1 / 12-21-13-991-00200)

NOTE : For computers with loadable software, software loading is not necessary if you install the computer removed for access.

- (1) Install the GCU1 and the GCU2 (1XU1, 1XU2) [Ref. AMM 24-22-34-400-001-A](#).
- (2) Install the APU GCU (1XS) 24-23-34-400-001, if removed.

Subtask 12-21-13-420-062-B

On A/C 001-999

EMB SB 47-1029 for A/C 003-004; 051-053

N. Installation of the Computers in the Avionics Compartment on Shelf 92VU

Ref. Fig. (1 / 12-21-13-991-00200)

NOTE : If you install the computer, which is removed to get access to the components, the software loading is not necessary (for the computers with the loadable software).

(1) If FLSCU 1 (7QJ), FLSCU 2 (9QJ) and FLSCU 3 (47QJ) are removed, install them

On A/C 101-350

Ref. AMM 28-46-34-400-001-A

On A/C 003-100

Ref. AMM 28-46-34-400-001-B.

(2) If the FQIC (3QT) is removed, install it

On A/C 001-999

Ref. AMM 28-42-34-400-001-A.

(3) If the GPCU (1XG) is removed, install it Ref. AMM 24-41-34-400-001-B.

(4) If the GAPCU (24XG) is removed, install it Ref. AMM 24-41-34-400-001-B.

(5) If the CSAS CU (100HW) is removed, install it Ref. AMM 21-58-34-400-001-A.

(6) If the AFMC (2QN) is removed, install it 28-51-35-400-001.

(7) If the ALSCU (60QJ) is removed, install it 28-46-35-400-001.

Subtask 12-21-13-420-063-B

On A/C 001-999

P. Installation of the Computers in the Avionics Compartment on the Shelf 93VU

Ref. Fig. (1 / 12-21-13-991-00200)

NOTE : For computers with loadable software, software loading is not necessary if you install the computer removed for access.

(1) Install the SEC 3 (1CE3) Ref. AMM 27-94-34-400-001-A.

(2) Install the LGCIU 1 (5GA1) Ref. AMM 32-31-71-400-001-A.

Subtask 12-21-13-420-064-B

On A/C 003-200; 301-350

Q. Installation of the Computers in the Avionics Compartment on Shelf 94VU

Ref. Fig. (1 / 12-21-13-991-00200)

NOTE : If you install the computer, which is removed to get access to the components, the software loading is not necessary (for the computers with the loadable software).

(1) Install LGCIU 2 (5GA2) Ref. AMM 32-31-71-400-001-A.

(2) If the SDCU (10WQ) is removed, install it 26-17-34-400-001.

(3) Install the BSCU (10GG) Ref. AMM 32-42-34-400-001-A.

(4) If the TPIS detection unit (2GV) is removed, install it Ref. AMM 32-49-34-400-001-A.

(5) If the aft cargo heat-controller (10HC) is removed, install it Ref. AMM 21-43-34-400-002-A.

(6) If the GAPCU (24XG) is removed, install it Ref. AMM 24-41-34-400-001-B.

Subtask 12-21-13-420-065-G

On A/C 001-999

R. Installation of the Computers in the Avionics Compartment on Shelf 95VU

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Ref. Fig. (1 / 12-21-13-991-00200)

NOTE : If you install the computer which is removed for the access of the components, the software loading is not necessary (for the computers with the loadable software).

- (1) If ILS receiver 1 (2RT1) or MMR 1 (40RT1) is removed, install it

On A/C 003-250

Ref. AMM 34-36-31-400-001-B

On A/C 251-350

Ref. AMM 34-36-31-400-001-G.

- (2) If BMC 1 (1HA1) is removed, install it

On A/C 001-999

Ref. AMM 36-11-34-400-001-A.

- (3) If CPC 1 (11HL) is removed, install it Ref. AMM 21-31-34-400-001-A.

- (4) If ACSC 1 (47HH) is removed, install it Ref. AMM 21-61-34-400-002-A.

- (5) If weather radar-transceiver 1 (1SQ1) is removed, install it Ref. AMM 34-41-33-400-003-A.

Subtask 12-21-13-420-066-H

On A/C 003-200; 301-350

S. Installation of the Computers in the Avionics Compartment on the Shelf 96VU

Ref. Fig. (1 / 12-21-13-991-00200)

NOTE : For computers with loadable software, software loading is not necessary if you install the computer removed for access.

- (1) Install the pack 2 temperature controller (57HH) Ref. AMM 21-61-34-400-002-A, if removed.

- (2) Install the cabin pressure controller 2 (12HL) Ref. AMM 21-31-34-400-001-A.

- (3) Install the BMC 2 (1HA2) Ref. AMM 36-11-34-400-001-A.

- (4) Install the AFT cargo heat controller (10HC) Ref. AMM 21-43-34-400-002-A, if removed.

- (5) Install the ILS Receiver 2 (2RT2)

On A/C 003-200

Ref. AMM 34-36-31-400-001-B

On A/C 301-350

Ref. AMM 34-36-31-400-001-G or the Multi Mode Receiver 2 (40RT2) .

Subtask 12-21-13-420-067-A

On A/C 001-999

T. Installation of the Computers in the Avionics Compartment on Shelf 109VU

Ref. Fig. (1 / 12-21-13-991-00200)

NOTE : For computers with loadable software, software loading is not necessary if you install the computer removed for access.

- (1) If WXR transceivers 1 and 2 (1SQ1, 1SQ2) are removed, install them

Ref. AMM 34-41-33-400-003-A.

Subtask 12-21-13-942-050-A

On A/C 001-999

U. Close Access

- (1) Close rear avionics-compartment access-door 824 Ref. AMM 52-41-00-410-002-A.

- (2) Close forward avionics-compartment access-door 811 Ref. AMM 52-41-00-410-002-A.

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(3) Remove the ground support and maintenance equipment, the special and standard tools and all other items.

(4) Remove the access platform(s).

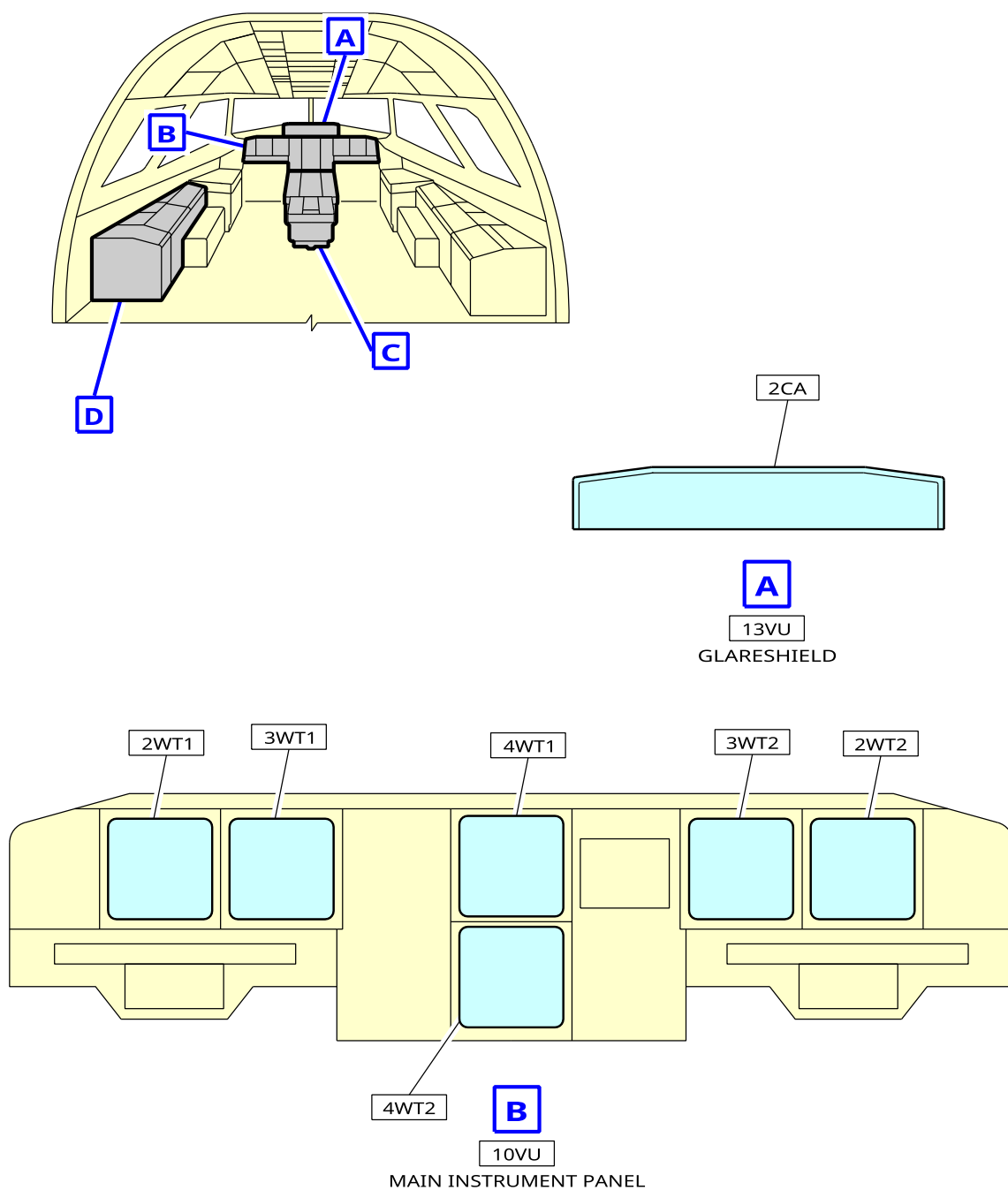
Subtask 12-21-13-862-050-A

On A/C 001-999

V. De-energize the ground service network

Ref. AMM 24-42-00-862-001-A.

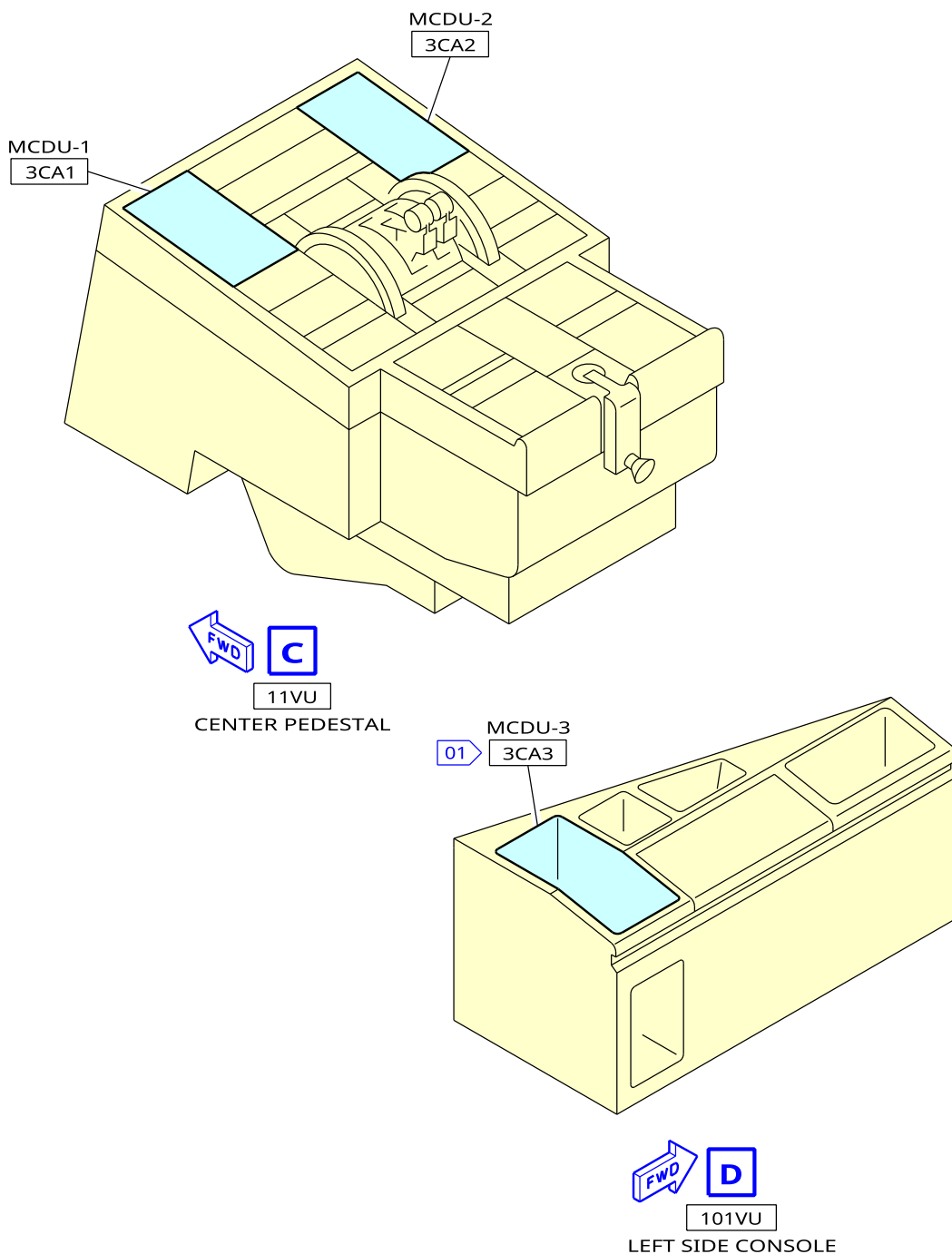
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N_MM_122113_7_AAM0_01_01

Location of Units in the Cockpit
Figure 1 / Graphic 12-21-13 -991-00100

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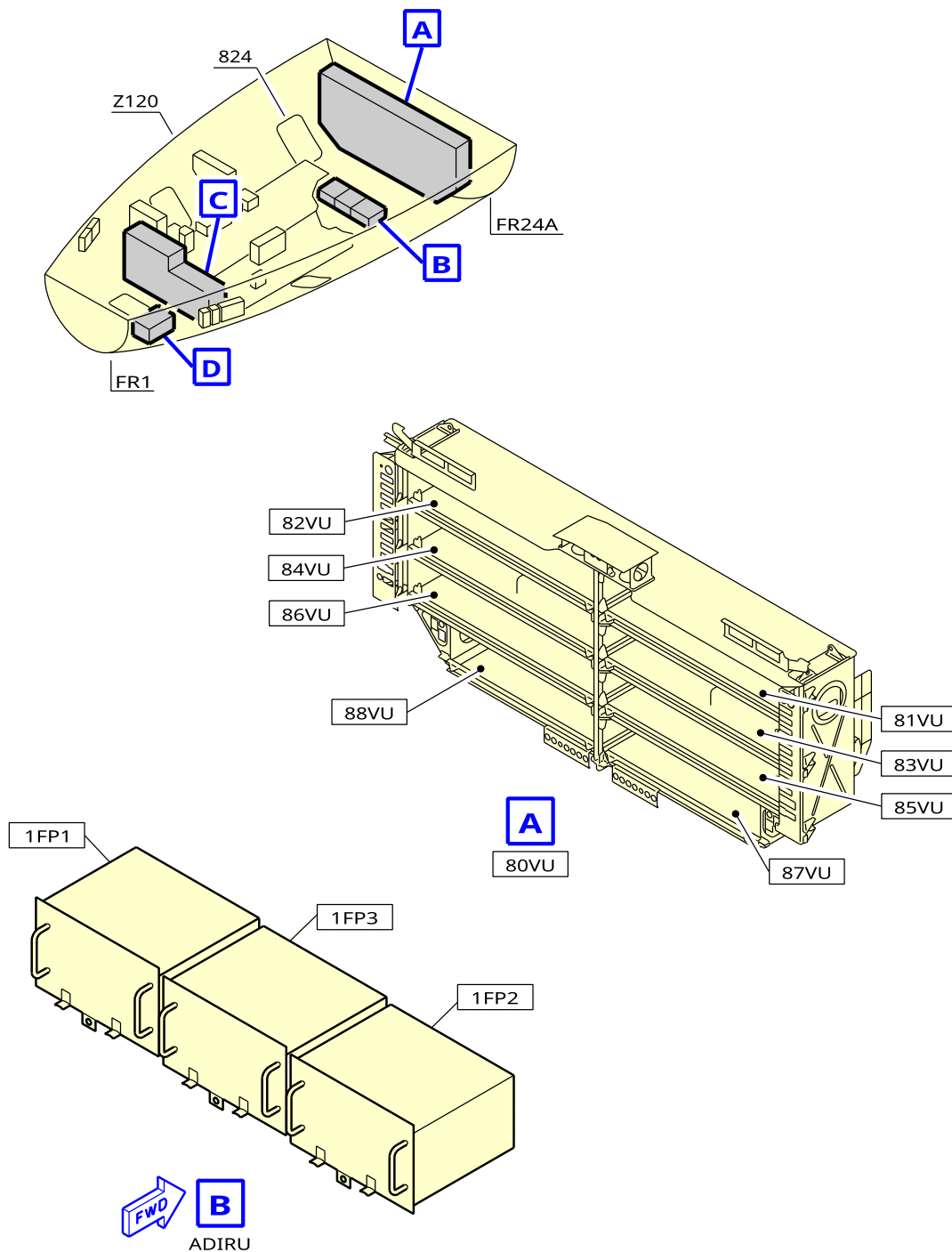


NOTE:
01 IF INSTALLED

N_MM_122113_7_AAM0_02_00

Location of Units in the Cockpit
Figure 2 / Graphic 12-21-13 -991-00100

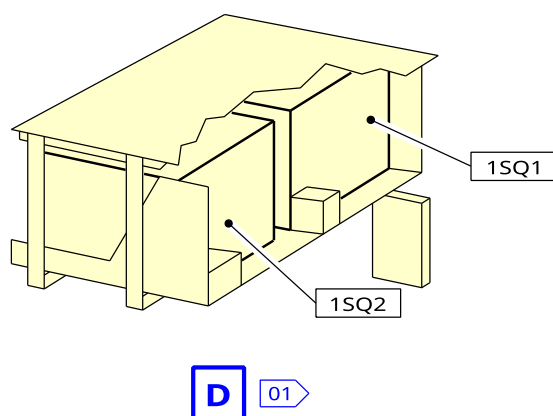
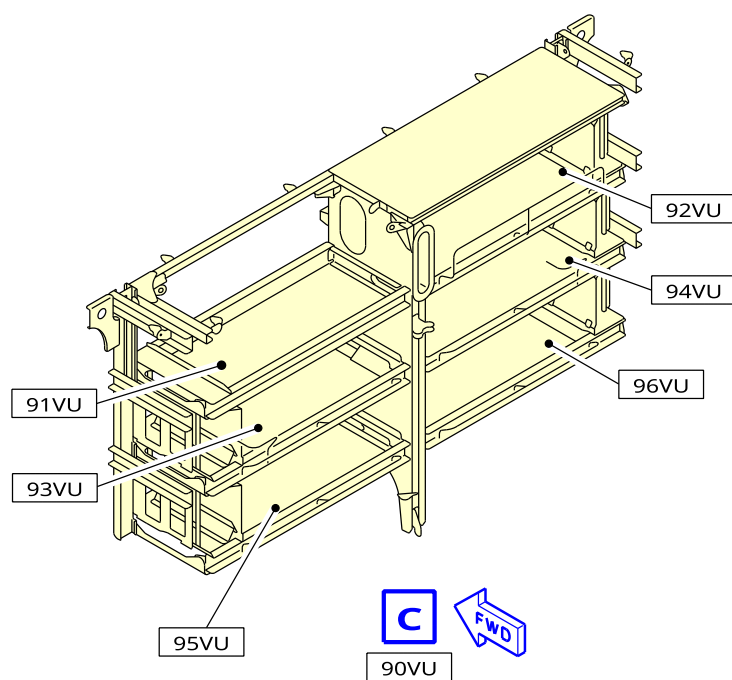
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N_MM_122113_7_ABM0_01_02

Location of Shelves and Computers in the Avionics Compartment
Figure 1 / Graphic 12-21-13 -991-00200

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NOTE:

- 01 THE LOCATION OF 1S1 AND 1S2 CAN BE 109VU OR 95VU BASED ON THE AIRCRAFT CONFIGURATION.

N_MM_122113_7_ABM0_02_01

Location of Shelves and Computers in the Avionics Compartment
Figure 2 / Graphic 12-21-13 -991-00200

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DISPLAY UNIT - CLEANING/PAINTING

TASK 31-63-22-100-001 -A

On A/C 001-999

Cleaning of the Display Unit

FIN: 2WT1, 2WT2, 3WT1, 3WT2, 4WT1, 4WT2

1. Reason for the Job

Self explanatory

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	AR	OPTICAL PRECISION PAD

B. Consumable Materials

REFERENCE	DESIGNATION
08AGA1	Aqueous Cleaner-Visual Display Unit - -
08BBD1	Non Aqueous Cleaner-- Isopropyl Alcohol -
14SBA1	Textile-Lint free Cotton -
No specific	Pad - sterile gauze
No specific	impregnated screen pad 2000

C. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
210	CKPT,FWD COMPT BHD TO FLT COMPT BULKHEAD

3. Job Set-up

Subtask 31-63-22-869-050-A

On A/C 001-999

A. No special job set-up is necessary for this task.

4. Procedure

Subtask 31-63-22-100-051-A

On A/C 001-999

A. Cleaning of the Screen of the Display Unit

CAUTION: BE VERY CAREFUL WHEN YOU CLEAN THE FRONT GLASS OF LCDU'S. IT IS NOT POSSIBLE TO REPLACE THE GLASS. IF YOU CAUSE DAMAGE TO LCDU'S, THE DAMAGE WILL BE PERMANENT.

CAUTION: MAKE SURE THAT YOU OBEY THE INSTRUCTIONS IN THE CLEANING PROCEDURE THAT FOLLOWS. INCORRECT CLEANING CAN CAUSE DAMAGE TO THE ANTI-REFLECTIVE COATING. IT IS NOT PERMITTED TO USE TISSUES SUCH AS KLEENEX TISSUES. SUCH TISSUES CAN CAUSE LOCAL REMOVAL OF THE SURFACE TREATMENT, AND THUS HALOS OR SCRATCHES. ALWAYS APPLY THE CLEANING AGENT TO THE PAD AND NOT TO THE SCREEN.

CAUTION: ONLY USE THE MATERIALS SPECIFIED IN THE CLEANING PROCEDURE. YOU CAN CAUSE DAMAGE TO THE DISPLAY UNIT SCREENS IF YOU USE OTHER MATERIALS (FOR EXAMPLE ACETONE- AND/OR AMMONIA-BASE PRODUCTS).

- (1) Make an absorbent surgical cotton wadding Textile-Lint free Cotton - (Material No. **14SBA1**) or a Pad - sterile gauze or a OPTICAL PRECISION PAD moist with Aqueous Cleaner- Visual Display Unit - - (Material No. **08AGA1**) or Non Aqueous Cleaner-- Isopropyl Alcohol - (Material No. **08BBD1**) or use a impregnated screen pad 2000.

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- (2) Move the moist wad (or pad) lightly across the surface of the screen (one time only).
- (3) If you find again marks or dust on the screen, do steps (1) and (2) again with a new piece of moist wad (or pad). Do these steps again until the screen is fully clean.

NOTE : Use a new pad (or wad) each time to prevent scratches that can be caused by a dusty pad.

- (4) Dry the screen with a clean wad (or pad). To do this, move the clean wad (or pad) lightly across the surface of the screen (one time only).

NOTE : We recommend to frequently clean the screen to remove finger marks or dust.

5. Close-up

Subtask 31-63-22-860-051-A

On A/C 001-999

- A. Make sure that the work area is clean and clear of tools and other items.

 MET.25495763		Task Card Title: BRAKE TEMPRETURE SENSOR CLEANING					
Task Card NO: 32-47-00-SVA		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP			
A/C Type: 321		Block Taskcard		Main Skill: AF,ACFT_TECH		PROD Final Acceptance:	
Project No: 01BASPC72608		Related Task Card:		A/C Area:			
Internal Check:		Check Type:		Work Order NO: 31909295		Tally Sequence NO: 65	
ENCLOSURE: ☐ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐							
Type Code: CLN		Special Code: AF		MSG Code: 4		Request: JOB SETUP, TC/P, CLOSE-UP, CLOSE-UP	
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRID OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.							
Zones:							
DOCUMENT REFERENCES:							
Doc Typ		Reference					
AMM		32-47-15					
CAP		CAP S32-C10005.pdf					
ACCESS PANELS:							
MATERIAL REQUIREMENTS:							
Part Number		Description				Qty.	
LINTFREECOTTONCLO*		LINT-FREE COTTON CLOTH				1.0	
ISOPROPYL-ALCOHOL-99		ISOPROPANOL 99.9%				1.0	
TOOLING REQUIREMENTS:							
Part Number		Description				Qty.	
No tools assigned to this Task Item							
JOB SET-UP							
Job Description							
A. GET ACCESS TO THE BRAKE TEMPERATURE SENSOR AS PER AMM 32-47-15.							
STEP COMPLETION SIGN OFF:							
Qualification / Skill		Number of employees		Duration		Number./Signature	
Date							
Day Month Year							
Mechanic/s		AF/ACFT_TECH		HRS			
Insp.				HRS			
TASK INSTRUCTION							
Job Description							
A. REMOVE THE BRAKE TEMPERATURE SENSOR AS PER AMM 32-47-15.							
STEP COMPLETION SIGN OFF:							



Task Card Title:

BRAKE TEMPRETURE SENSOR CLEANING

Task Card NO:

32-47-00-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	AF/ACFT_TECH		HRS				
Insp.			HRS				

WORK INSTRUCTION**Job Description**

B. CLEAN THE BRAKE TEMPERATURE SENSOR AS FOLLOWS:

- CLEAN ALL THE EXTERNAL PARTS INCLUDING THE CONNECTOR USING CLEAN AND SOFT LINT-FREE CLOTH DAMPED (PN LINTFREECOTTONCLO*) WITH CLEANING PRODUCT (PN: ISOPROPYL-ALCOHOL-99).
- REMOVE DUST AND UNWANTED MATERIALS FROM SURFACES USING CLEANING PRODUCTS AND A SOFT NYLON BRUSH.
- DRY ALL THE PARTS WITH COMPRESSED AIR OR WITH A CLEAN AND SOFT LINT-FREE CLOTH (PN: LINTFREECOTTONCLO*)

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	AF/ACFT_TECH		HRS				
Insp.			HRS				

CLOSE-UP**Job Description**

A. INSTALL THE BRAKE TEMPERATURE SENSOR AS PER AMM 32-47-15

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	AF/ACFT_TECH		HRS				
Insp.			HRS				

CLOSE-UP**Job Description**

B. RETURN ALL DISTURBED ITEMS TO ITS SERVICEABLE CONDITION.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	AF/ACFT_TECH		HRS				
Insp.			HRS				



MET.25495763

Task Card Title:

BRAKE TEMPRETURE SENSOR CLEANING

Task Card NO:

32-47-00-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below

Work order No: _____

Note: Make sure to creat the finding Work Order on task card level.

Remark's: _____

(1) If Task card interrupted or performed by different personals updates the system and update the following accord (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 2 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 3 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 4 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 5 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

**Task Completion Close Up And
System Update****Station****Date****Signature**

TF-N
AIRCRAFT MAINTENANCE MANUAL
SENSOR - BRAKE TEMPERATURE - REMOVAL/INSTALLATION

On A/C 001-999

SENSOR - BRAKE TEMPERATURE - REMOVAL/INSTALLATION

TF-N
AIRCRAFT MAINTENANCE MANUAL
SENSOR - BRAKE TEMPERATURE - REMOVAL/INSTALLATION

TASK 32-47-15-000-001 -A

On A/C 001-999

Removal of the Brake Temperature Sensor

WARNING: MAKE SURE THAT THE SAFETY DEVICES AND THE WARNING NOTICES ARE IN POSITION BEFORE YOU START A TASK ON OR NEAR:

- THE FLIGHT CONTROLS
- THE FLIGHT CONTROL SURFACES
- THE LANDING GEAR AND RELATED DOORS
- THE CARGO DOORS
- OTHER COMPONENTS THAT MOVE.

MOVEMENT OF COMPONENTS CAN KILL OR CAUSE INJURY TO PERSONS AND/OR DAMAGE TO EQUIPMENT.

WARNING: MAKE SURE THAT THE GROUND SAFETY LOCKS ARE CORRECTLY INSTALLED ON THE MLG AND NLG. MOVEMENTS OF THE LANDING GEARS CAN CAUSE INJURY TO PERSONS AND CAUSE DAMAGE TO THE AIRCRAFT AND/OR EQUIPMENT.

WARNING: MAKE SURE THAT THE WHEEL CHOCKS ARE IN POSITION. MOVEMENT OF THE AIRCRAFT CAN BE DANGEROUS.

WARNING: LET THE BRAKES AND THE WHEELS BECOME COOL BEFORE YOU GO NEAR THE LANDING GEAR. DO NOT APPLY A LIQUID OR GAS DIRECTLY ON A HOT WHEEL OR BRAKE UNIT. THIS CAN CAUSE A TIRE EXPLOSION AND/OR A WHEEL RIM BURST.

FIN: 4GW, 5GW, 6GW, 7GW

1. Reason for the Job

Self explanatory

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	AR	CAP - BLANKING
No specific	AR	CHOCK - WHEEL(S)
No specific	AR	SAFETY CLIP - CIRCUIT BREAKER
No specific	AR	WARNING NOTICE(S)

B. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
731	L.H MAIN GEAR
741	R.H MAIN GEAR

C. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 32-00-00-481-001-A	Installation of the Safety Devices on the Landing Gears
On A/C 003-250	
Ref. Fig. (1 / 32-47-15-991-00100)	

3. Job Set-up

Subtask 32-47-15-941-052-A

On A/C 001-999

A. Safety Precautions

TF-N
AIRCRAFT MAINTENANCE MANUAL
SENSOR - BRAKE TEMPERATURE - REMOVAL/INSTALLATION

- (1) Make sure that the safety devices are installed on the landing gears
[Ref. AMM 32-00-00-481-001-A.](#)
- (2) Make sure that the CHOCK - WHEEL(S) are in position in front of each forward outboard MLG wheel.
- (3) In the cockpit:
 - (a) On right center instrument-panel 400VU:
 - 1 Make sure that the landing-gear normal control-lever is in the DOWN position.
 - 2 Put the WARNING NOTICE(S) in position adjacent to the landing-gear normal control-lever to tell persons not to operate the landing gear.
 - (b) At the rear of center pedestal 11VU:
 - 1 Make sure that the GRAVITY GEAR EXTN handle is correctly folded.
- (4) In the cockpit, on the center pedestal, on panel 110VU, set the PARK BRK selector switch to the OFF position.
- (5) In the cockpit, put the WARNING NOTICE(S) in position to tell persons not to operate:
 - The landing gear control-lever
 - The brake pedals
 - The PARK BRK selector switch.

Subtask 32-47-15-865-050-A

[On A/C 001-999](#)

- B. Open, safety and tag the circuit breaker(s) that follow(s). Use the SAFETY CLIP - CIRCUIT BREAKER as necessary.

PANEL	DESIGNATION	FIN	LOCATION
121VU	HYDRAULIC/BRK/TEMP/ DET/UNIT	1GW	M37

4. Procedure

[On A/C 003-250](#)

Ref. Fig. 1 / 32-47-15-991-00100

Subtask 32-47-15-020-050-J

[On A/C 101-200; 251-350](#)

A. Removal of the Brake Temperature Sensor

- (1) Move the rubber boot (4) sufficiently to get access to the electrical connector (3).
- (2) If the heat shrink sleeve is installed on the electrical connector (3), cut and remove it.

CAUTION: BEFORE YOU DISCONNECT OR CONNECT THE ELECTRICAL CONNECTORS AND RECEPTACLES, MAKE SURE THAT THEY ARE CLEAN. CONTAMINATION ON THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO THE EQUIPMENT.

- (3) Disconnect the electrical connector (3) from the temperature sensor (1).
- (4) Put the CAP - BLANKING on the electrical connector (3).

WARNING: USE PROTECTIVE GOGGLES AND GLOVES EACH TIME YOU CUT LOCKWIRE. DISCARD LOOSE OR USED LOCKWIRE IMMEDIATELY. LOOSE OR USED LOCKWIRE CAN CUT YOU OR MAKE YOU BLIND.

- (5) Cut and remove the lockwire from the screw (2) and discard it.
- (6) Remove the screw (2).
- (7) Release the temperature sensor (1) from its housing along its axis.
- (8) Put the protective cap (5) on the temperature sensor (1).

TF-N
AIRCRAFT MAINTENANCE MANUAL
SENSOR - BRAKE TEMPERATURE - REMOVAL/INSTALLATION

TASK 32-47-15-400-001 -A

On A/C 001-999

Installation of the Brake Temperature Sensor

WARNING: MAKE SURE THAT THE SAFETY DEVICES AND THE WARNING NOTICES ARE IN POSITION BEFORE YOU START A TASK ON OR NEAR:

- THE FLIGHT CONTROLS
- THE FLIGHT CONTROL SURFACES
- THE LANDING GEAR AND RELATED DOORS
- THE CARGO DOORS
- OTHER COMPONENTS THAT MOVE.

MOVEMENT OF COMPONENTS CAN KILL OR CAUSE INJURY TO PERSONS AND/OR DAMAGE TO EQUIPMENT.

WARNING: MAKE SURE THAT THE GROUND SAFETY LOCKS ARE CORRECTLY INSTALLED ON THE MLG AND NLG. MOVEMENTS OF THE LANDING GEARS CAN CAUSE INJURY TO PERSONS AND CAUSE DAMAGE TO THE AIRCRAFT AND/OR EQUIPMENT.

WARNING: MAKE SURE THAT THE WHEEL CHOCKS ARE IN POSITION. MOVEMENT OF THE AIRCRAFT CAN BE DANGEROUS.

WARNING: BE CAREFUL WHEN YOU USE CONSUMABLE MATERIALS. OBEY THE MATERIAL MANUFACTURER'S INSTRUCTIONS AND YOUR LOCAL REGULATIONS.

FIN: 4GW, 5GW, 6GW, 7GW

1. Reason for the Job

Self explanatory

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	AR	CAP - BLANKING
No specific	AR	CHOCK - WHEEL(S)
On A/C 003-200; 251-350		
No specific	1	HOT AIR BLOWER MAX OPERATING 370 DEG C (700 DEG F)
On A/C 001-999		
No specific	AR	SAFETY CLIP - CIRCUIT BREAKER
No specific	AR	WARNING NOTICE(S)
On A/C 003-200; 251-350		
No specific		Torque wrench: range to between 0.65 and 0.8 m.daN (57.52 and 70.80 lbf.in)

B. Consumable Materials

REFERENCE	DESIGNATION
03LBB1	Grease-Thread Anti-seize Graphited and Petrolatum -
14QFC1	Wire-Locking Dia: 0.8 mm CRES -

C. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
731	L.H MAIN GEAR
741	R.H MAIN GEAR

D. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 32-00-00-081-001-A	Removal of the Safety Devices from the Landing Gears

TF-N
AIRCRAFT MAINTENANCE MANUAL
SENSOR - BRAKE TEMPERATURE - REMOVAL/INSTALLATION

REFERENCE	DESIGNATION
Ref. AMM 32-00-00-481-001-A	Installation of the Safety Devices on the Landing Gears
Ref. AMM 32-47-00-710-001-A	Operational Test of the Brake Temperature System
On A/C 101-200; 251-350	
Ref. ESPM 20-33-34	
Ref. ESPM 20-55-50	
On A/C 003-250	
Ref. Fig. (1 / 32-47-15-991-00100)	

3. Job Set-up

Subtask 32-47-15-941-053-A

On A/C 001-999

A. Safety Precautions

- (1) Make sure that the safety devices are installed on the landing gears
Ref. AMM 32-00-00-481-001-A.
- (2) Make sure that the CHOCK - WHEEL(S) are in position in front of each forward outboard MLG wheel.
- (3) In the cockpit:
 - (a) On right center instrument-panel 400VU:
 - 1 Make sure that the landing-gear normal control-lever is in the DOWN position.
 - 2 Make sure that the WARNING NOTICE(S) is (are) in position adjacent to the landing-gear normal control-lever to tell persons not to operate the landing gear.
 - (b) At the rear of center pedestal 11VU:
 - 1 Make sure that the GRAVITY GEAR EXTN handle is correctly folded.
- (4) In the cockpit, on the center pedestal, on panel 110VU, make sure that the PARK BRK selector switch is in the OFF position.
- (5) In the cockpit, make sure that the WARNING NOTICE(S) is (are) in position to tell persons not to operate:
 - The landing gear control-lever
 - The brake pedals
 - The PARK BRK selector switch.

Subtask 32-47-15-865-051-A

On A/C 001-999

- B. Make sure that the circuit breaker(s) that follow(s) is (are) open, safetied and tagged. Use the SAFETY CLIP - CIRCUIT BREAKER as necessary.

PANEL	DESIGNATION	FIN	LOCATION
121VU	HYDRAULIC/BRK/TEMP/ DET/UNIT	1GW	M37

4. Procedure

On A/C 003-250

Ref. Fig. 1 / 32-47-15-991-00100

Subtask 32-47-15-420-052-L

On A/C 101-200; 251-350

A. Installation of the Brake Temperature Sensor

CAUTION: BEFORE YOU DISCONNECT OR CONNECT THE ELECTRICAL CONNECTORS AND RECEPTACLES, MAKE SURE THAT THEY ARE CLEAN. CONTAMINATION ON THE ELECTRICAL CONNECTORS AND RECEPTACLES CAN CAUSE DAMAGE TO THE EQUIPMENT.

TF-N
AIRCRAFT MAINTENANCE MANUAL
SENSOR - BRAKE TEMPERATURE - REMOVAL/INSTALLATION

- (1) Remove the CAP - BLANKING from the electrical connector (3).
- (2) Remove the protective cap (5) from the temperature sensor (1).
- (3) Clean the electrical connector (3) and the receptacle of the temperature sensor (1)
Ref. ESPM 20-55-50.
- (4) Install the temperature sensor (1).
- (5) Apply a thin layer of Grease-Thread Anti-seize Graphited and Petrolatum - (Material No. **03LBB1**) to the threads of the screw (2).
- (6) Install the screw (2) and TORQUE it **to between 0.65 and 0.8 m.daN (57.52 and 70.80 lbf.in).**

WARNING: **USE PROTECTIVE GOGGLES AND GLOVES EACH TIME YOU CUT LOCKWIRE. DISCARD LOOSE OR USED LOCKWIRE IMMEDIATELY. LOOSE OR USED LOCKWIRE CAN CUT YOU OR MAKE YOU BLIND.**

- (7) Safety the screw (2) with Wire-Locking Dia: 0.8 mm CRES - (Material No. **14QFC1**) .
- (8) Do the visual inspection of the rubber boot (4).
 - (a) If the rubber boot (4) is in correct condition, do one of the steps that follow:
 - 1 If you want to install the heat shrink sleeve on the electrical connector (3), do as follows:

NOTE : The installation of the heat shrink sleeve is optional.

- a Move the rubber boot (4) sufficiently to get access to the electrical connector (3).
 - b Put an approximately 50 MM (1.97 IN.) length of heat shrink sleeve (ABS0916B09 or NSA937502-08) in position on the harness of the electrical connector (3)
Ref. ESPM 20-33-34.
 - c Connect the electrical connector (3) to the temperature sensor (1).
 - d Move the heat shrink sleeve in position on the electrical connector (3).
 - e Use the HOT AIR BLOWER MAX OPERATING 370 DEG C (700 DEG F) that can supply air at a minimum temperature of 135 DEG.C (275 DEG.F) to:
 - Decrease the size of the heat shrink sleeve fully along its full length.
 - f Move the rubber boot (4) back in position on the electrical connector (3).
 - 2 If you do not want to install the heat shrink sleeve electrical connector (3), do as follows:
 - a Connect the electrical connector (3) to the temperature sensor (1).
 - b Move the rubber boot (4) back in position on the electrical connector (3).
- (b) If the rubber boot (4) is not in correct condition, remove it and install the heat shrink sleeve (ABS0836B06) as follows:

NOTE : This procedure gives the instructions for the temporary repair of the brake-temperature-sensor connector protection.

- 1 Cut a piece of heat shrink sleeve (ABS0836B06) to a length between 52 MM (2.0472 IN.) to 58 MM (2.2835 IN.).
- 2 Install the heat shrink sleeve on the electrical connector (3).
- 3 Connect the electrical connector (3) to the temperature sensor (1).

TF-N
AIRCRAFT MAINTENANCE MANUAL
SENSOR - BRAKE TEMPERATURE - REMOVAL/INSTALLATION

- 4 Move the heat shrink sleeve onto the electrical connector (3). But do not put the heat shrink sleeve on the milled external part of the electrical connector (3).
- 5 Use the HOT AIR BLOWER MAX OPERATING 370 DEG C (700 DEG F) that can supply air at a minimum temperature of 135 DEG.C (275 DEG.F):
 - a To decrease the size of the heat shrink sleeve fully around the backshell of the electrical connector (3) only.
 - b Make sure that you can disconnect the electrical connector (3) and do not cause damage to the heat shrink sleeve.
- 6 Make sure that you replace the rubber boot (4) in less than 400 flight hours.

Subtask 32-47-15-865-052-A

On A/C 001-999

- B. Remove the SAFETY CLIP - CIRCUIT BREAKER and the tag(s) and close this (these) circuit breaker(s):

PANEL	DESIGNATION	FIN	LOCATION
121VU	HYDRAULIC/BRK/TEMP/ DET/UNIT	1GW	M37

Subtask 32-47-15-710-050-A

On A/C 001-999

- C. Test

- (1) Do the operational test of the brake temperature system. [Ref. AMM 32-47-00-710-001-A](#)

5. Close-up

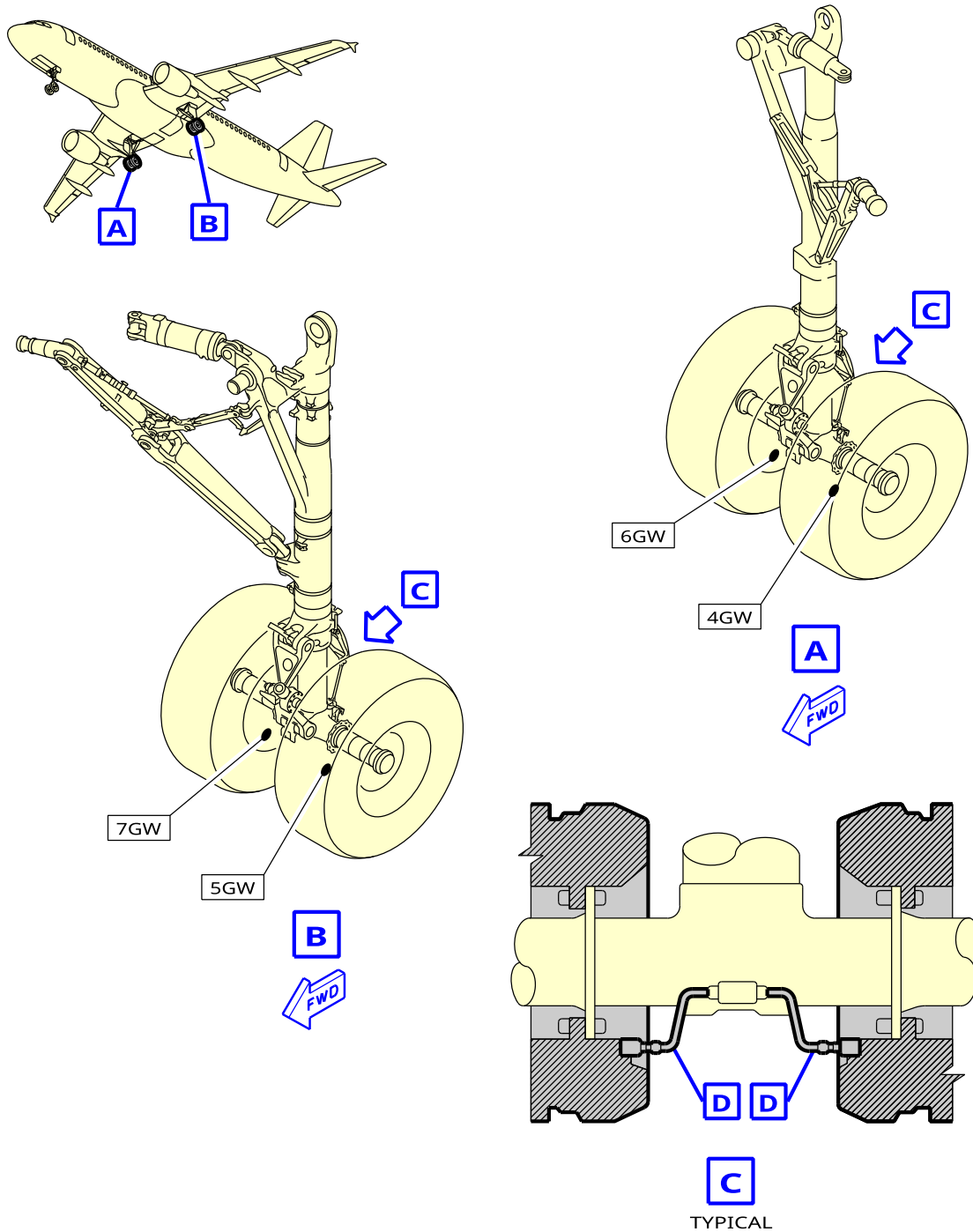
Subtask 32-47-15-942-050-A

On A/C 001-999

- A. Removal of Equipment

- (1) Make sure that the work area is clean and clear of tools and other items.
- (2) Remove the safety devices from the landing gears [Ref. AMM 32-00-00-081-001-A](#).
- (3) Remove the WARNING NOTICE(S).
- (4) Remove all the fixtures, tools, test and support equipment used during this procedure.

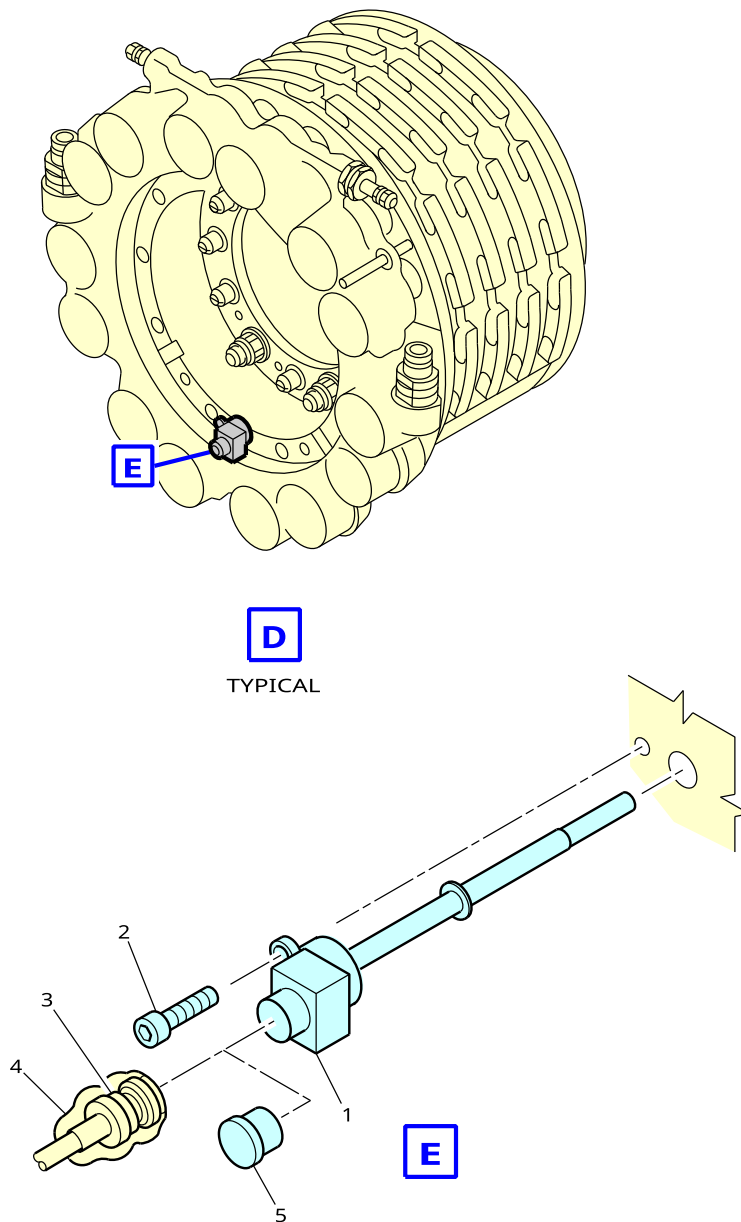
TF-N
AIRCRAFT MAINTENANCE MANUAL
SENSOR - BRAKE TEMPERATURE - REMOVAL/INSTALLATION



N_MM_324715_4_AAM2_01_02

Brake Temperature Sensor
Figure 1 / Graphic 32-47-15 -991-00100
On A/C 003-250

TF-N
AIRCRAFT MAINTENANCE MANUAL
SENSOR - BRAKE TEMPERATURE - REMOVAL/INSTALLATION



N_MM_324715_4_AAM2_02_00

Brake Temperature Sensor
Figure 2 / Graphic 32-47-15 -991-00100

 MET.29291009		Task Card Title: NO. 1 ENGINE HIGH PRESSURE DUCT INSPECTION					
Task Card NO: 361100-08-2-001-SVA		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP			
A/C Type: 321		Block Taskcard		Main Skill: INSP,IL,NDT_INSP,ACFT_TECH, AP		PROD Final Acceptance:	
Project No: 01BASPC72608		Related Task Card:		A/C Area: 01			
Internal Check:		Check Type:		Work Order NO: 31909295		Tally Sequence NO: 75	
ENCLOSURE: ☐ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐							
Type Code: DVI		Special Code: AP, INSP		MSG Code: 4		Request: JOB-SETUP,INSPECTION,INSPECTION,JOEUP	
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRID OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.							
Zones:							
DOCUMENT REFERENCES:							
Doc Typ		Reference					
AMM		36-11-49-200-041-A					
AMM		36-11-49-200-041					
CAP		S36-C10003					
DO		VSB RA32036-13					
IPC		36-11-49-01A					
IPC		36-11-49-01A					
ACCESS PANELS:							
MATERIAL REQUIREMENTS:							
Part Number		Description				Qty.	
No parts assigned to this Task Item							
TOOLING REQUIREMENTS:							
Part Number		Description				Qty.	
No tools assigned to this Task Item							
TASK SET UP							
Job Description							
A. PERFORM JOB SET UP FOR VISUAL INSPECTION OF THE PNEUMATIC SYSTEM HOSES, TUBES AND DUCTS AS PER AMM 36-11-49-200-041-A. (REF. AMM TASK 78-30-00-081-041-B)							
STEP COMPLETION SIGN OFF:							
Qualification / Skill		Number of employees		Duration		Number./Signature	
Date							
Day Month Year							
Mechanic/s		ACFT_TECH/AP		HRS			
Insp.				HRS			
UPPER 9TH STAGE HP DUCT AND DUCT SUPPORT LINK DETAIL INSPECTION							
Job Description							



MET.29291009

Task Card Title:

NO. 1 ENGINE HIGH PRESSURE DUCT INSPECTION

Task Card NO:

361100-08-2-001-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP**! NOTE**

The required parts may need replacement in case of findings.

All PNs are mentioned in the work template instructions and should be requested upon findings.

- PN 642-5508-505 >>> 9 STG HP PNEUMATIC DUCT
- PN P40560 >>> SUPPORT LINK OF HP DUCT
- PN P40570 >>> SUPPORT LINK OF HP DUCT

A. USE THE FOLLOWING INSTRUCTIONS FOR UPPER 9TH STAGE HP DUCTS INSTALLED ON AN AIRCRAFT.

1. OPEN FAN COWL DOORS. REFER TO AIRBUS A318/A319/A320/A321 AIRCRAFT MAINTENANCE MANUAL CHAPTER 71-13-00.
2. MAKE THE THRUST REVERSER UNSERVICEABLE FOR MAINTENANCE. REFER TO AIRBUS A318/A319 /A320/A321 AIRCRAFT MAINTENANCE MANUAL CHAPTER 78-30-00.
3. OPEN THE THRUST REVERSER HALVES. REFER TO AIRBUS A318/A319/A320/A321 AIRCRAFT MAINTENANCE MANUAL CHAPTER 78-36-00.
4. VERIFY THE **UPPER 9TH STAGE HIGH PRESSURE DUCT (HP) PN & SN** THAT IS INSTALLED IN THIS AIRCRAFT FIRSTLY. USE ATTACHED FIGURE 1 & 2 & IPC FIGURE 36-11-49-01A AS REFERENCE. PLEASE RECORD THE PN & SN BELOW AND ATTACH CLEAR PHOTO IN THE WO

PN:.....

SN:.....

5. PERFORM A DETAILED VISUAL INSPECTION OF THE UPPER 9TH STAGE HP DUCT BEND AREA **PN 642 5508-503 OR PN 642-5508-505**(ADJACENT TO ENGINE INTERFACE) FOR CRACKS. REFER TO (ATTACHED FIGURE 1) FOR INSPECTION AREA.

PN: 9th stage HP duct	Quantity	Remark
642-5508-505	1	This PN might be needed for re findings. please request upon 1

- a. USE SUFFICIENT LIGHTING AND A 10-POWER MAGNIFICATION FOR THE INSPECTION.
- b. INSPECT THE ELBOW AND CIRCUMFERENTIAL WELDS AT EACH END FOR CRACKS.



Task Card Title:

NO. 1 ENGINE HIGH PRESSURE DUCT INSPECTION

Task Card NO:

361100-08-2-001-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

INSPECT THE EXTERNAL SURFACES ALL THE WAY AROUND THE UPPER 9TH STAGE HP DUCT'S ELBOW.

- c. IF A CRACK IS FOUND, REPLACE THE UPPER 9TH STAGE HP DUCT PER SB RA32036-12.
IF THERE ARE NO CRACKS, PROCEED TO NEXT PARAGRAPH A.6.

6. PERFORM A DETAILED VISUAL INSPECTION OF THE UPPER 9TH STAGE HP DUCT WHERE IT ATTACHES TO THE LINK(S) FOR CRACKS. REFER TO (ATTACHED FIGURE 2) FOR INSPECTION AREA.

- a. USE SUFFICIENT LIGHTING AND A 10-POWER MAGNIFICATION FOR THE INSPECTION
b. INSPECT THE CIRCUMFERENTIAL WELDS AT EACH END FOR CRACKS. INSPECT THE EXTERNAL SURFACES ALL THE WAY AROUND THE UPPER 9TH STAGE HP DUCT AT THE ATTACHMENT POINT.
c. IF A CRACK IS FOUND, REPLACE THE UPPER 9TH STAGE HP DUCT PER SB RA32036-12.
IF THERE ARE NO CRACKS, PROCEED TO NEXT PARAGRAPH A.7.

7. VERIFY IF THE INSTALLED SUPPORT LINK ATTACHED TO THE UPPER 9TH STAGE HP DUCT IS 1-LINK OR 2-LINK CONFIGURATION. USE ATTACHED FIGURE 2 & FIGURE 4 AS REFERENCES. RECORD IT BELOW AND ATTACH CLEAR PHOTO IN THE WO

TYPE OF SUPPORT LINK CONFIGURATION (1-LINK OR 2-LINK CONFIGURATION):

ALSO, WHETHER IT IS 1-LINK OR 2-LINK CONFIGURATION, PLEASE VERIFY THE SUPPORT LINK PN (USE FIGURE 2 AS REFERENCE TO LOCATE THE SAME PN AS SHOWN IN THE FIGURES) AND RECORD IT BELOW IF IT IS MARKED ON THE SUPPORT LINK ITSELF. IF NOT MARKED, PUT IT AS N/A AND GO TO THE NEXT STEP A.8

IF IT IS 1-LINK CONFIGURATION

SUPPORT LINK PN:

IF IT IS 2-LINK CONFIGURATION

SUPPORT LINK PN:

SUPPORT LINK PN:

8. INSPECT THE SUPPORT LINK **PN P40570** (AND **PN P40560** IF 2-LINK CONFIGURATION IS INSTALLED) FOR WEAR OR EXCESSIVE PLAY IN THE AXIAL OR RADIAL DIRECTION (SEE FIGURE 2). IF THERE IS FREE PLAY BETWEEN THE BALL AND RACE, COMPLETE RA32036-12, PART B. AS FOLLOW;

AS PER SB RA32036-12 PART B:

- a. INSPECT THE SUPPORT LINK PN P40570 FOR WEAR OR EXCESSIVE PLAY IN THE AXIAL OR RADIAL



Task Card Title:

NO. 1 ENGINE HIGH PRESSURE DUCT INSPECTION

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Work Package:

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A/C Registration:

HZ-ASP

DIRECTION (SEE FIGURE 1). IF THERE IS FREE PLAY BETWEEN THE BALL AND RACE, CHANGE THE BALL AND/OR FULL LINK ASSEMBLY AS REQUIRED. FOLLOW AMM INSTRUCTIONS IN CHAPTER 36-11-49, TASK 36-11-49-400-046A. FOR INSTALLATION OF THE HP DUCT & AFT LINK AND TASK 36-11-49-000-046A FOR THE REMOVAL.

- b. WRITE IN THE ENGINE LOG AND/OR OTHER APPLICABLE RECORDS THAT SERVICE BULLETIN RA32036-12, REV. 3 PART B HAS BEEN INCORPORATED.

PN: Support Link for HP Duct	Quantity	Remark
P40560	1	This PN might require replacement. Please request.
P40570	1	This PN might require replacement. Please request.

9. CLOSE THE THRUST REVERSER HALVES. REFER TO AIRBUS A318/A319/A320/A321 AIRCRAFT MAINTENANCE MANUAL CHAPTER 78-36-00, TASK 78-36-00-410-040-B.
10. Make the thrust reverser serviceable for maintenance. Refer to Airbus A318/A319/A320/A321 Aircraft Maintenance Manual Chapter 78-30-00.
11. Close Fan Cowl doors. Refer to Airbus A318/A319/A320/A321 Aircraft Maintenance Manual Chapter 71-13-00.
12. Write in the engine log and/or other records that Goodrich Service bulletin RA32036-13 has been incorporated.
13. Re-inspect the upper 9th stage HP duct per the inspection interval listed in the Compliance section on the Summary page.

B. Report Inspection Results

1. Report to Goodrich the inspection results including no findings, cracks, or worn links and actions taken. Include aircraft/engine serial numbers, hours, and cycles, and upper 9th stage HP duct serial number information.
2. Use the reporting sheet (see Attached Figure 3) at the end of this bulletin to transmit inspection results.





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Please when you fill the reporting sheet, attach it in the work order and The Engineering will collect all these reporting sheets and provide it to Goodrich

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	ACFT_TECH/AP			HRS			
Insp.	INSP/II			HRS			

DYE PENETRANT TEST (NDT) ON THE 9TH STAGE UPPER HP DUCT:**Job Description**

(1) PERFORM THE DYE PENETRANT TEST AS PER REFERENCE NTM 51-80-00 ON THE 9TH STAGE UPPER HP DUCT ITEM 190 (AT THE DUCT BEND / ELBOW AREA) FOR THE SUBJECT ENGINE.

(2) IF YOU FIND DURING THE TEST ANY FINDINGS, OBSERVATIONS, CRACKS, LEAKS, ETC., PLEASE RECORD IT IN THIS WO OR SEND A QUERY TO ENG-CON-SYS GROUP (NOTE: TQ IS ONLY NECESSARY IF THERE ARE FINDINGS DURING THE INSPECTION PROCESS).

(3) IN CASE OF FINDING THAT REQUIRE REPLACEMENT OF THE AFFECTED PARTS, PLEASE DO THE NECESSARY ACTION FOR THE REPLACEMENT OF THE AFFECTED PARTS AS PER REFERENCE AMM 36-11-49.

(4) ENSURE YOU CLEAN THE TESTED / INSPECTED AREA AFTER COMPLETING THE TEST.

NOTES:

- ◆ THE FOLLOWING ARE THE SUSPECTED/AFFECTED PARTS THAT MIGHT NEED REPLACEMENT ONLY IN CASE OF FINDINGS.
- ◆ PN: 642-5508-505 OR 642-5508-503 >>> 9 STAGE UPPER HP DUCT ITEM 190.
- ◆ THE ABOVE PARTS SHOULD BE REQUESTED ONLY UPON FINDINGS.
- ◆ PLEASE REFER TO ATTACHMENTS FOR 9TH STAGE UPPER HP DUCT ITEM 190 (DUCT BEND / ELBOW AREA) FOR YOUR REFERENCE.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	ACFT_TECH/AP			HRS			



Task Card Title:
NO. 1 ENGINE HIGH PRESSURE DUCT INSPECTION



Task Card NO:

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Work Package:

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A/C Registration:

HZ-ASP

Insp.

II/NDT_INSP

HRS

LOWER HP DUCT DETAILED INSPECTION**Job Description**

- (1) Get access to the LOWER HIGH-PRESSURE DUCT (HP). Use attached IPC Figure 36-11-49-01A item 180 as reference.
- (2) Perform a detailed visual inspection of the lower HP duct (adjacent to engine interface) for cracks. As per the following steps:
- a) Use sufficient lighting and a 10-power magnification for the inspection.
 - b) Inspect the elbow/bend, circumferential welds and all the external surfaces all the way around the lower HP duct's areas.
 - c) If findings are found during the inspection please do the necessary action as per AMM 36-11-49.

NOTE: refer to AMM 36-11-49-200-041-A for possible findings that you might find on the lower HP duct (table item# 6 in section# 4 procedure).

PN	QTY	Remarks
642-5510-501 Lower HP duct	1	This PN might be needed for replacement in case of findings. please request upon finding only

- (3) Please raise a technical query to ENG-CON-SYS group for the inspection results (Note: TQ is only necessary if there are findings during the inspection process).

The time required to perform this job is 30 mins for ACFT_TECH and 15 mins INSP.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	II/ACFT_TECH/AP			HRS			
Insp.	INSP			HRS			

DYE PENETRANT TEST (NDT) ON THE LOWER HP DUCT:**Job Description**

- (1) PERFORM THE DYE PENETRANT TEST AS PER REFERENCE NTM 51-80-00 ON THE LOWER HP DUCT ITEM 180 (AT THE BEND AREA) FOR THE SUBJECT ENGINE.
- (2) IF YOU FIND DURING THE TEST ANY FINDINGS, OBSERVATIONS, CRACKS, LEAKS, ETC., PLEASE RECORD IT IN THIS WO OR SEND A QUERY TO ENG-CON-SYS GROUP (NOTE: TQ IS ONLY NECESSARY IF THERE ARE FINDINGS DURING THE INSPECTION PROCESS).



Task Card Title:

NO. 1 ENGINE HIGH PRESSURE DUCT INSPECTION

Task Card NO:

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Work Package:

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A/C Registration:

HZ-ASP

(3) IN CASE OF FINDINGS THAT REQUIRE REPLACEMENT OF THE AFFECTED PARTS, PLEASE DO THE NECESSARY ACTION FOR THE REPLACEMENT OF THE AFFECTED PARTS AS PER REFERENCE AMM 36-11-49 .

(4) ENSURE YOU CLEAN THE TESTED / INSPECTED AREA AFTER COMPLETING THE TEST.

NOTES :

- ❖ THE FOLLOWING IS THE SUSPECTED/AFFECTED PART THAT MIGHT NEED REPLACEMENT ONLY IN CASE OF FINDINGS.
- ❖ PN 642-5510-501 >>> LOWER HP DUCT ITEM 180.
- ❖ THE ABOVE PART SHOULD BE REQUESTED ONLY UPON FINDINGS.
- ❖ PLEASE REFER TO ATTACHMENTS FOR LOWER HP DUCT ITEM 180 (BEND AREA) FOR YOUR REFERENCE.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	ACFT_TECH/AP		HRS				
Insp.	II/NDT_INSP		HRS				

VISUAL INSPECTION OF THE PNEUMATIC SYSTEM**Job Description**

1. Perform visual Inspection of the Pneumatic System Hoses, Tubes and Ducts AS PER AMM 36-11-49-200-041-A.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s			HRS				
Insp.	II/INSP		HRS				

TASK CLOSE UP**Job Description**

A. PERFORM JOB-CLOSEUP AND RETURN AIRCRAFT TO NORMAL CONFIGURATION AS PER AMM TASK 36-11-49-200-041-A. (REF. AMM TASK 78-30-00-081-041-B)

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	ACFT_TECH/AP		HRS				
Insp.			HRS				



Task Card Title:

NO. 1 ENGINE HIGH PRESSURE DUCT INSPECTION

Task Card NO:

361100-08-2-001-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below

Work order No: _____

Note: Make sure to create the finding Work Order on task card level.

Remark's: _____

(1) If Task card interrupted or performed by different persons updates the system and update the following accordingly (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 2 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 3 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 4 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 5 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

**Task Completion Close Up And
System Update****Station****Date****Signature**

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DUCTS - BLEED AIR - INSPECTION/CHECK

TASK 36-11-49-200-041 -A

On A/C 003-250

Visual Inspection of the Pneumatic System Hoses, Tubes and Ducts

WARNING: MAKE SURE THAT THE PNEUMATIC SYSTEM IS DEPRESSURIZED BEFORE YOU START WORK. PRESSURIZED AIR CAN CAUSE INJURY TO PERSONNEL.

WARNING: BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER ENGINE SHUTDOWN. USE APPLICABLE THERMAL GLOVES. THE ENGINE PARTS AND THE ENGINE OIL CAN STAY HOT FOR ONE HOUR AFTER SHUTDOWN AND CAN BURN YOU.

1. Reason for the Job

Self explanatory

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	AR	ACCESS PLATFORM 1M(3 FT)
No specific	AR	SAFETY CLIP - CIRCUIT BREAKER
No specific	AR	WARNING NOTICE(S)

B. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
FOR 1000EM1(ENGINE-1)	
437AL , 438AR , 451AL , 452AR	
FOR 1000EM2(ENGINE-2)	
447AL , 448AR , 461AL , 462AR	

C. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 36-11-49-000-042-A	Removal of the Pylon Interface Duct
Ref. AMM 36-11-49-000-043-A	Removal of the Mid Stage Tee Duct
Ref. AMM 36-11-49-000-044-A	Removal of the High Pressure Tee Duct
Ref. AMM 36-11-49-000-045-A	Removal of the Lower High Pressure Duct
Ref. AMM 36-11-49-000-046-A	Removal of the Upper HP Duct
Ref. AMM 36-11-49-000-047-A	Removal of the Pneumatic System Tubes and Hoses
Ref. AMM 36-11-49-000-048-A	Removal of the Fifth Stage Elbow Duct
Ref. AMM 36-11-49-000-842-A	Removal of the Upper HP Duct and Mid Stage T-Duct Assembly Links
Ref. AMM 36-11-49-200-041-A	Visual Inspection of the Pneumatic System Hoses, Tubes and Ducts
Ref. AMM 36-11-49-400-042-A	Installation of the Pylon Interface Duct
Ref. AMM 36-11-49-400-043-A	Installation of the Mid Stage Tee Duct
Ref. AMM 36-11-49-400-044-A	Installation of the High-Pressure Tee Duct
Ref. AMM 36-11-49-400-045-A	Installation of the Lower High-Pressure Duct
Ref. AMM 36-11-49-400-046-A	Installation of the Upper HP Duct
Ref. AMM 36-11-49-400-047-A	Installation of the Pneumatic System Tubes and Hoses
Ref. AMM 36-11-49-400-048-A	Installation of the Fifth-Stage Elbow Duct
Ref. AMM 36-11-49-400-842-A	Installation of the Upper HP Duct and Mid Stage T-Duct Assembly Links
Ref. AMM 71-13-00-010-040-B	Opening of the Fan Cowl Doors
Ref. AMM 71-13-00-410-040-B	Closing of the Fan Cowl Doors
Ref. AMM 78-30-00-081-041-B	Make the Thrust Reverser System Serviceable after Maintenance
Ref. AMM 78-30-00-481-041-B	Make the Thrust Reverser System Unserviceable for Maintenance
Ref. AMM 78-32-11-220-040-B	General Visual Inspection of the Inner Barrel Heatshield, Fire Protective Coating and Fire Seals (C-Ducts Open)
Ref. AMM 78-36-00-010-040-B	Opening of the Thrust Reverser Doors
Ref. AMM 78-36-00-410-040-B	Closing of the Thrust Reverser Doors

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REFERENCE	DESIGNATION
Ref. AMM 80-11-10-210-003-A	Starter Inspection for a Probably FOD (Foreign Object Damage)
Ref. Fig. (1 / 36-11-49-991-91400)	
Ref. Fig. (1 / 36-11-49-991-91500)	
Ref. Fig. (1 / 36-11-49-991-91600)	
Ref. Fig. (1 / 36-11-49-991-91700)	
Ref. Fig. (1 / 36-11-49-991-91800)	
Ref. Fig. (1 / 36-11-49-991-91900)	

3. Job Set-up

Subtask 36-11-49-941-091-A

On A/C 003-250

A. Safety Precautions

- (1) On the ENG panel 115VU:
 - put a WARNING NOTICE(S) to tell persons not to start the engine 1(2).
- (2) Make sure that the engine shutdown occurred five minutes or more before you do this procedure.
- (3) On the ENG section of the maintenance panel 50VU:
 - make sure that the ON legend of the ENG/FADEC GND PWR 1(2) pushbutton switch is off.
 - put a WARNING NOTICE(S) to tell persons not to energize the FADEC 1(2).

Subtask 36-11-49-010-122-A

On A/C 003-250

B. Get Access

- (1) Open the fan cowl doors [Ref. AMM 71-13-00-010-040-B](#):
 - (a) FOR 1000EM1 (ENGINE-1)
437AL 438AR
 - (b) FOR 1000EM2 (ENGINE-2)
447AL 448AR
- (2) Put the ACCESS PLATFORM 1M(3 FT) in position below the engine.

Subtask 36-11-49-040-094-A

On A/C 003-250

C. Make the thrust reverser unserviceable [Ref. AMM 78-30-00-481-041-B](#).

Subtask 36-11-49-010-123-A

On A/C 003-250

D. Open the thrust reverser doors [Ref. AMM 78-36-00-010-040-B](#):

- (1) FOR 1000EM1 (ENGINE-1)
451AL 452AR
- (2) FOR 1000EM2 (ENGINE-2)
461AL 462AR

Subtask 36-11-49-865-129-A

On A/C 003-250

E. Open, safety and tag the circuit breaker(s) that follow(s). Use the SAFETY CLIP - CIRCUIT BREAKER as necessary.

PANEL	DESIGNATION	FIN	LOCATION
49VU	AIR BLEED/ENG 1/CTL	3HA1	D12
49VU	AIR BLEED/ENG 1/MONG	2HA1	D11
122VU	AIR BLEED/ENG 2/CTL	3HA2	Z23
122VU	AIR BLEED/ENG 2/MONG	2HA2	Z22

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AIRCRAFT MAINTENANCE MANUAL
DUCTS - BLEED AIR - INSPECTION/CHECK

4. Procedure

Subtask 36-11-49-210-062-B

On A/C 003-250

A. Do a Visual Inspection of the Pneumatic System Ducts, Hoses and Tubes

Ref. Fig. (1 / 36-11-49-991-91400)

Ref. Fig. (1 / 36-11-49-991-91500)

Ref. Fig. (1 / 36-11-49-991-91600)

Ref. Fig. (1 / 36-11-49-991-91700)

Ref. Fig. (1 / 36-11-49-991-91800)

Ref. Fig. (1 / 36-11-49-991-91900)

INSPECT/CHECK	MAXIMUM SERVICEABLE LIMITS	REMARKS
1.Examine the pneumatic-system flexible-duct-joint assembly (pylon Interface duct) for: Ref. Fig. (1 / 36-11-49-991-91400)		
A.Damage	None permitted in less than 0.50 in. (12.69 mm) of links, flex joints, couplings, and flanges.	Replace the duct Ref. AMM 36-11-49-000-042-A and Ref. AMM 36-11-49-400-042-A .
B.Wrinkles in bends	Maximum depth must be less than 5 percent of the duct diameter.	If the damage is more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-042-A and Ref. AMM 36-11-49-400-042-A .
C.Dents	Maximum depth must be less than 20 percent of the tube outside diameter. Maximum radius of the dent is not less than two times of the duct wall thickness. Maximum of four dents in each one foot (305 mm) of the duct. Minimum distance of 1.0 inch (25.4 mm) between the dents. No limit to dents permitted unless in bends, no dents are permitted in bends.	If the dents are less than the permitted limits, send the duct to the shop for analysis. If the dents are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-042-A and Ref. AMM 36-11-49-400-042-A .
D.Flat areas	Maximum length must be less than three times of the outside diameter of the duct. Contour change must be less than 20 percent of the outside diameter of the duct.	If the flat areas are less than the permitted limits, send the duct to the shop for analysis. If the flat areas are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-042-A and Ref. AMM 36-11-49-400-042-A .
E.Nicks, scratches, chafing in straight section	Maximum depth must be less than 10 percent of the wall thickness of the duct. There is no limit to the number of nicks, scratches or chafing permitted in the straight section of the duct.	- If the damage is less than the permitted limits: . Remove the pylon interface duct Ref. AMM 36-11-49-000-042-A. . Send the duct to the shop for analysis. . Install the pylon interface duct Ref. AMM 36-11-49-400-042-A. - If the damage is more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-042-A and Ref. AMM 36-11-49-400-042-A.
F.Nicks, scratches, chafing in bends	Maximum depth must be less than 10 percent of the wall thickness of the duct.	- If the damage is less than the permitted limits: . Remove the pylon interface duct Ref. AMM 36-11-49-000-042-A.

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INSPECT/CHECK	MAXIMUM SERVICEABLE LIMITS	REMARKS
		. Send the duct to the shop for analysis. . Install the pylon interface duct Ref. AMM 36-11-49-400-042-A. • If the damage is more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-042-A and Ref. AMM 36-11-49-400-042-A.
G.Splits, cracks, kinks	Maximum length must be less than the duct outside diameter.	If the damage is less than the duct outer diameter, send the duct to the shop for analysis.
	Maximum length same as or more than the duct outside diameter.	If the damage is the same as or more than the duct outer diameter, send the duct to the shop for analysis.
H.Internal splits, cracks of flexible joints (bellows)	None permitted	Replace the duct Ref. AMM 36-11-49-000-042-A and Ref. AMM 36-11-49-400-042-A . Inspect the pneumatic system for FOD (foreign object damage) Ref. AMM 36-11-49-200-041-A . Inspect the starter motor for FOD (foreign object damage) Ref. AMM 80-11-10-210-003-A .
2.Examine the pneumatic-system mid-stage-tee assembly for: Ref. Fig. (1 / 36-11-49-991-91500)		
A.Damage	None permitted in less than 0.50 in. (12.69 mm) of links, flex joints, couplings, and flanges.	Replace the duct Ref. AMM 36-11-49-000-043-A and Ref. AMM 36-11-49-400-043-A .
B.Cracks, splits or kinks	None permitted	Replace the duct Ref. AMM 36-11-49-000-043-A and Ref. AMM 36-11-49-400-043-A .
C.Wrinkles in bends	Maximum depth cannot be more than 3 percent of the duct diameter or 0.060 inch (1.52 mm). The smaller value is the limit.	If wrinkles are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-043-A and Ref. AMM 36-11-49-400-043-A .
D.Dents	Maximum depth of 0.3 inch (7.62 mm) or Maximum depth cannot be more than 10 percent of the duct outside diameter. Minimum width to depth ratio to be 5:1. Dent has to be free from cracks, abrasion and sudden contour changes. Radius is not less than two times of the wall thickness. Maximum of two dents for each one foot (305mm) of the duct length.	If the dents are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-043-A and Ref. AMM 36-11-49-400-043-A .
E.Flat areas	Maximum flat area cannot be more than two times of the duct outside diameter. The contour change cannot be more than 10 percent of the duct outside diameter.	If the flat areas are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-043-A and Ref. AMM 36-11-49-400-043-A .
F.Nicks, scratches, chafing	Maximum depth cannot be more than 10 percent of the wall thickness.	If the damage is more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-043-A and Ref. AMM 36-11-49-400-043-A .

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AIRCRAFT MAINTENANCE MANUAL
DUCTS - BLEED AIR - INSPECTION/CHECK

INSPECT/CHECK	MAXIMUM SERVICEABLE LIMITS	REMARKS
3.Examine the pneumatic-system mid-stage-elbow assembly for: Ref. Fig. (1 / 36-11-49-991-91500)		
A.Damage	None permitted in less than 0.50 in. (12.69 mm) of links, flex joints, couplings, and flanges.	Replace the duct Ref. AMM 36-11-49-000-048-A and Ref. AMM 36-11-49-400-048-A .
B.Cracks, splits or kinks	None permitted	Replace the duct Ref. AMM 36-11-49-000-048-A and Ref. AMM 36-11-49-400-048-A .
C.Wrinkles in bends	Maximum depth cannot be more than 3 percent of the duct diameter or 0.060 inch (1.52 mm). The smaller value is the limit.	If wrinkles are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-048-A and Ref. AMM 36-11-49-400-048-A .
D.Dents	Maximum depth cannot be more than 10 percent of the duct outside diameter. Maximum radius is not less than two times of the wall thickness. Maximum of two dents for each one foot (305 mm) of the duct length.	If the dents are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-048-A and Ref. AMM 36-11-49-400-048-A .
E.Flat areas	Maximum flat area cannot be more than two times of the duct outside diameter. The contour change cannot be more than 10 percent of the duct outside diameter.	If the flat areas are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-048-A and Ref. AMM 36-11-49-400-048-A .
F.Nicks, scratches, chafing	Maximum depth cannot be more than 10 percent of the wall thickness.	If the damage is more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-048-A and Ref. AMM 36-11-49-400-048-A .
4.Examine the pneumatic-system upper high-pressure manifold-duct assembly for: Ref. Fig. (1 / 36-11-49-991-91600)		
A.Damage	None permitted in less than 0.50 in. (12.69 mm) of links, flex joints, couplings, and flanges.	Replace the duct Ref. AMM 36-11-49-000-046-A and Ref. AMM 36-11-49-400-046-A .
B.Cracks, splits or kinks	None permitted	Replace the duct Ref. AMM 36-11-49-000-046-A and Ref. AMM 36-11-49-400-046-A .
C.Wrinkles in bends	Maximum depth cannot be more than 3 percent of the duct diameter or 0.060 inch (1.52 mm). The smaller value is the limit.	If wrinkles are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-046-A and Ref. AMM 36-11-49-400-046-A .
D.Dents	Maximum depth cannot be more than 10 percent of the duct outside diameter. Maximum radius is not less than two times of the wall thickness. Maximum of two dents for each one foot (305 mm) of the duct length.	If the dents are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-046-A and Ref. AMM 36-11-49-400-046-A .
E.Flat areas	Maximum flat area cannot be more than two times the duct outside diameter. The contour change cannot be more than 10 percent of the duct outside diameter.	If the flat areas are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-046-A and Ref. AMM 36-11-49-400-046-A .
F.Nicks, scratches, chafing	Maximum depth cannot be more than 10 percent of the wall thickness.	If the damage is more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-046-A and Ref. AMM 36-11-49-400-046-A .

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AIRCRAFT MAINTENANCE MANUAL
DUCTS - BLEED AIR - INSPECTION/CHECK

INSPECT/CHECK	MAXIMUM SERVICEABLE LIMITS	REMARKS
G.Worn bushing holes	Maximum diameter of the bushing hole is 0.26 in. (6.6040 mm)	If the bushing hole diameter more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-046-A and Ref. AMM 36-11-49-400-046-A .
H.Loose support links	None permitted	Replace the support link(s) Ref. AMM 36-11-49-000-842-A and Ref. AMM 36-11-49-400-842-A .
5.Examine the pneumatic-system center high-pressure manifold-duct assembly for: Ref. Fig. (1 / 36-11-49-991-91700)		
A.Damage	None permitted in less than 0.50 in. (12.69 mm) of links, flex joints, couplings, and flanges.	Replace the duct Ref. AMM 36-11-49-000-044-A and Ref. AMM 36-11-49-400-044-A .
B.Cracks, splits or kinks	None permitted	Replace the duct Ref. AMM 36-11-49-000-044-A and Ref. AMM 36-11-49-400-044-A .
C.Wrinkles in bends	Maximum depth cannot be more than 3 percent of the duct diameter or 0.060 inch (1.52 mm).The smaller value is the limit.	If wrinkles are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-044-A and Ref. AMM 36-11-49-400-044-A .
D.Dents	Maximum depth cannot be more than 10 percent of the duct outside diameter. Maximum radius is not less than two times of the wall thickness. Maximum of two dents for each one foot (305 mm) of the duct length.	If the dents are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-044-A and Ref. AMM 36-11-49-400-044-A .
E.Flat areas	Maximum flat area cannot be more than two times the duct outside diameter. The contour change cannot be more than 10 percent of the duct outside diameter.	If the flat areas are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-044-A and Ref. AMM 36-11-49-400-044-A .
F.Nicks, scratches, chafing	Maximum depth cannot be more than 10 percent of the wall thickness.	If the damage is more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-044-A and Ref. AMM 36-11-49-400-044-A .
6.Examine the pneumatic-system lower high-pressure-duct assembly for: Ref. Fig. (1 / 36-11-49-991-91800)		
A.Damage	None permitted in less than 0.50 in. (12.69 mm) of links, flex joints, couplings, and flanges.	Replace the duct Ref. AMM 36-11-49-000-045-A and Ref. AMM 36-11-49-400-045-A .
B.Cracks, splits or kinks	None permitted	Replace the duct Ref. AMM 36-11-49-000-045-A and Ref. AMM 36-11-49-400-045-A .
C.Wrinkles in bends	Maximum depth cannot be more than 3 percent of the duct diameter or 0.060 inch (1.52 mm). The smaller value is the limit.	If wrinkles are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-045-A and Ref. AMM 36-11-49-400-045-A .
D.Dents	Maximum depth cannot be more than 10 percent of the duct outside diameter. Maximum radius is not less than two times of the wall thickness. Maximum of two dents for each one foot (305 mm) of the duct length.	If the dents are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-045-A and Ref. AMM 36-11-49-400-045-A .

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DUCTS - BLEED AIR - INSPECTION/CHECK

INSPECT/CHECK	MAXIMUM SERVICEABLE LIMITS	REMARKS
E.Flat areas	Maximum flat area cannot be more than two times the duct outside diameter. The contour change cannot be more than 10 percent of the duct outside diameter.	If the flat areas are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-045-A and Ref. AMM 36-11-49-400-045-A .
F.Nicks, scratches, chafing	Maximum depth cannot be more than 10 percent of the wall thickness.	If the damage is more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-045-A and Ref. AMM 36-11-49-400-045-A .
7.Examine the pressure regulator valve to interface hose assembly for the flexible part of the sense line for: Ref. Fig. (1 / 36-11-49-991-91900)		
A.Broken wire braids	Maximum of three broken wires in each one plait. No more than six broken wires in an assembly. No more than six broken wires in each one foot (305 mm) of hose.	If the damage is more than the permitted limits, replace the hose 36-11-49-000-026 and 36-11-49-400-026.
B.Cracks	None permitted	If cracks are found, replace the hose 36-11-49-000-026 and 36-11-49-400-026.
C.Crimps	None permitted	Replace the hose assembly 36-11-49-000-026 and 36-11-49-400-026.
D.Internal blockage	None permitted	If blockage is found, remove the blockage or replace the hose 36-11-49-000-026 and 36-11-49-400-026.
E.Leakage	None permitted	Replace the hose 36-11-49-000-026 and 36-11-49-400-026.
8.Examine the flexible hose assemblies for: Ref. Fig. (1 / 36-11-49-991-91900)		
A.Broken wire braids	Maximum of three broken wires in each one plait. No more than six broken wires in an assembly. No more than six broken wires in each one foot (305 mm) of hose.	If the damage is more than the permitted limits, replace the hose Ref. AMM 36-11-49-000-047-A and Ref. AMM 36-11-49-400-047-A .
B.Cracks	None permitted	If cracks are found, replace the hose Ref. AMM 36-11-49-000-047-A and Ref. AMM 36-11-49-400-047-A .
C.Crimps	None permitted	Replace the hose assembly Ref. AMM 36-11-49-000-047-A and Ref. AMM 36-11-49-400-047-A .
D.Internal blockage	None permitted	If blockage is found, remove the blockage or replace the hose Ref. AMM 36-11-49-000-047-A and Ref. AMM 36-11-49-400-047-A .
E.Leakage	None permitted	Replace the hose Ref. AMM 36-11-49-000-047-A and Ref. AMM 36-11-49-400-047-A .
9.Examine the fittings on the two ends of the hose for:		
A.Thread damage	Damage is permitted if the threads that are good are not decreased by more than one thread.	If the damage is more than the permitted limits, replace the hose Ref. AMM 36-11-49-000-042-A and Ref. AMM 36-11-49-400-042-A .

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AIRCRAFT MAINTENANCE MANUAL
DUCTS - BLEED AIR - INSPECTION/CHECK

INSPECT/CHECK	MAXIMUM SERVICEABLE LIMITS	REMARKS
B.Damaged flats	Damage is permitted if the end fittings can keep the correct torque value.	If the end fittings cannot keep the correct torque, replace the hose Ref. AMM 36-11-49-000-043-A and Ref. AMM 36-11-49-400-043-A .
C.Safety wire hole damaged or worn	Damage is permitted if the metal around the hole is not cracked or torn. Make sure that one corner stays undrilled.	Blend out damage. If necessary, drill a hole in the undrilled corner. The new hole must be the same diameter and distance from the end of the nut as the old hole.
10.Examine the high pressure bleed-valve to pressure regulator tube-assembly for the rigid part of the sense line for: Ref. Fig. (1 / 36-11-49-991-91900)		
A.Wrinkles in bends	Maximum depth must be less than 5 percent of the tube diameter.	If the damage is more than the permitted limits, replace the tube 36-11-49-000-026 and 36-11-49-400-026.
B.Dents	Maximum depth must be less than 20 percent of the tube outside diameter. Maximum radius of the dent is not less than two times of the tube wall thickness. Maximum of four dents in each one foot (305 mm) of the tube. Minimum distance of 1.0 inch (25.4 mm) between the dents. No limit to dents permitted unless in bends, no dents are permitted in bends.	If the dents are less than the permitted limits, send the tube to the shop for analysis. If the dents are more than the permitted limits, replace the tube 36-11-49-000-026 and 36-11-49-400-026.
C.Flat areas	Maximum length must be less than three times the outside diameter of the tube. Contour change must be less than 20 percent of the outside diameter of the tube.	If the flat areas are less than the permitted limits, send the tube to the shop for analysis. If the flat areas are more than the permitted limits, replace the tube 36-11-49-000-026 and 36-11-49-400-026.
D.Nicks, scratches, chafing in straight section	Maximum depth must be less than 10 percent of the wall thickness of the tube. There is no limit to the number of nicks, scratches or chafing allowed in the straight section of the tube.	If the damage is less than the permitted limits, send the tube to the shop for analysis. If the damage is more than the permitted limits, replace the tube 36-11-49-000-026 and 36-11-49-400-026.
E.Nicks, scratches, chafing in bends	Maximum depth must be less than 10 percent of the wall thickness of the tube.	If the damage is less than the permitted limits, send the tube to the shop for analysis. If the damage is more than the permitted limits, replace the tube 36-11-49-000-026 and 36-11-49-400-026.
F.Splits, cracks, kinks	Maximum length must be less than the tube outside diameter.	If the damage is less than the tube outer diameter, send the tube to the shop for analysis.
	Maximum length same as or more than the tube outside diameter.	If the damage is the same as or more than the tube outer diameter, send the tube to the shop for analysis.
11.Examine the tee assembly to hose tube assembly for: Ref. Fig. (1 / 36-11-49-991-91900)		
A.Wrinkles in bends	Maximum depth must be less than 5 percent of the tube diameter.	If the damage is more than the permitted limits, replace the tube Ref. AMM 36-11-49-000-047-A and Ref. AMM 36-11-49-400-047-A .

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AIRCRAFT MAINTENANCE MANUAL
DUCTS - BLEED AIR - INSPECTION/CHECK

INSPECT/CHECK	MAXIMUM SERVICEABLE LIMITS	REMARKS
B.Dents	Maximum depth must be less than 20 percent of the tube outside diameter. Maximum radius of the dent is not less than two times of the tube wall thickness. Maximum of four dents in each one foot (305 mm) of the tube. Minimum distance of 1.0 inch (25.4 mm) between the dents. No limit to dents permitted unless in bends, no dents are permitted in bends.	If the dents are less than the permitted limits, send tube to the shop for analysis. If the dents are more than the permitted limits, replace the tube Ref. AMM 36-11-49-000-047-A and Ref. AMM 36-11-49-400-047-A .
C.Flat areas	Maximum length must be less than three times the outside diameter of the tube. Contour change must be less than 20 percent of the outside diameter of the tube.	If the flat areas are less than the permitted limits, send the tube to the shop for analysis. If the flat areas are more than the permitted limits, replace the tube Ref. AMM 36-11-49-000-047-A and Ref. AMM 36-11-49-400-047-A .
D.Nicks, scratches, chafing in straight section	Maximum depth must be less than 10 percent of the wall thickness of the tube. There is no limit to the number of nicks, scratches or chafing allowed in the straight section of the tube.	If the damage is less than the permitted limits, send the tube to the shop for analysis. If the damage is more than the permitted limits, replace the tube Ref. AMM 36-11-49-000-047-A and Ref. AMM 36-11-49-400-047-A .
E.Nicks, scratches, chafing in bends	Maximum depth must be less than 10 percent of the wall thickness of the tube.	If the damage is less than the permitted limits, send the tube to the shop for analysis. If the damage is more than the permitted limits, replace the tube Ref. AMM 36-11-49-000-047-A and Ref. AMM 36-11-49-400-047-A .
F.Splits, cracks, kinks	Maximum length must be less than the tube outside diameter.	If the damage is less than the tube outer diameter, send the tube to the shop for analysis.
	Maximum length same as or more than the tube outside diameter.	If the damage is the same as or more than the tube outer diameter, send the tube to the shop for analysis.

Subtask 36-11-49-210-072-A

On A/C 003-250

B. Visual Inspection of the Inner Barrel Heatshield of the Thrust Reverser

- (1) If there is a leak because of the damaged part(s), do the visual inspection of the inner barrel heatshield of the thrust reverser [Ref. AMM 78-32-11-220-040-B](#).

5. Close-up

Subtask 36-11-49-410-104-A

On A/C 003-250

A. Close Access

- (1) Make sure that the work area is clean and clear of tools and other items.
- (2) Close the thrust reverser doors [Ref. AMM 78-36-00-410-040-B](#):
 - (a) FOR 1000EM1 (ENGINE-1)
451AL 452AR
 - (b) FOR 1000EM2 (ENGINE-2)
461AL 462AR

Subtask 36-11-49-440-073-A

On A/C 003-250

B. Make the thrust reverser serviceable [Ref. AMM 78-30-00-081-041-B](#).

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Subtask 36-11-49-410-105-A

On A/C 003-250

C. Close Access

(1) Close the fan cowl doors [Ref. AMM 71-13-00-410-040-B](#):

(a) FOR 1000EM1 (ENGINE-1)

437AL 438AR

(b) FOR 1000EM2 (ENGINE-2)

447AL 448AR

(2) Remove the access platform(s).

(3) Remove the WARNING NOTICE(S).

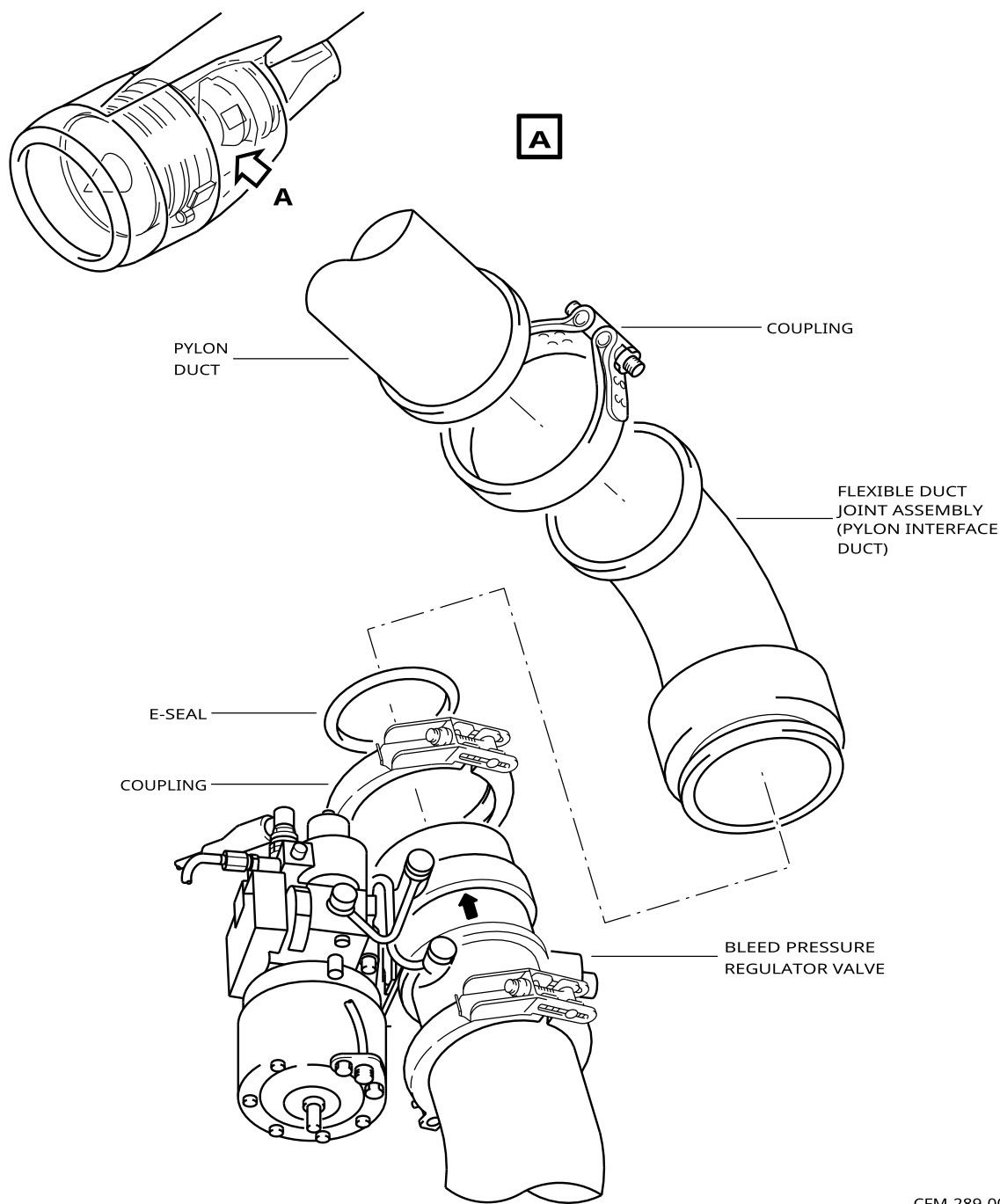
Subtask 36-11-49-865-130-A

On A/C 003-250

D. Remove the SAFETY CLIP - CIRCUIT BREAKER and the tag(s) and close this (these) circuit breaker(s):

PANEL	DESIGNATION	FIN	LOCATION
49VU	AIR BLEED/ENG 1/CTL	3HA1	D12
49VU	AIR BLEED/ENG 1/MONG	2HA1	D11
122VU	AIR BLEED/ENG 2/CTL	3HA2	Z23
122VU	AIR BLEED/ENG 2/MONG	2HA2	Z22

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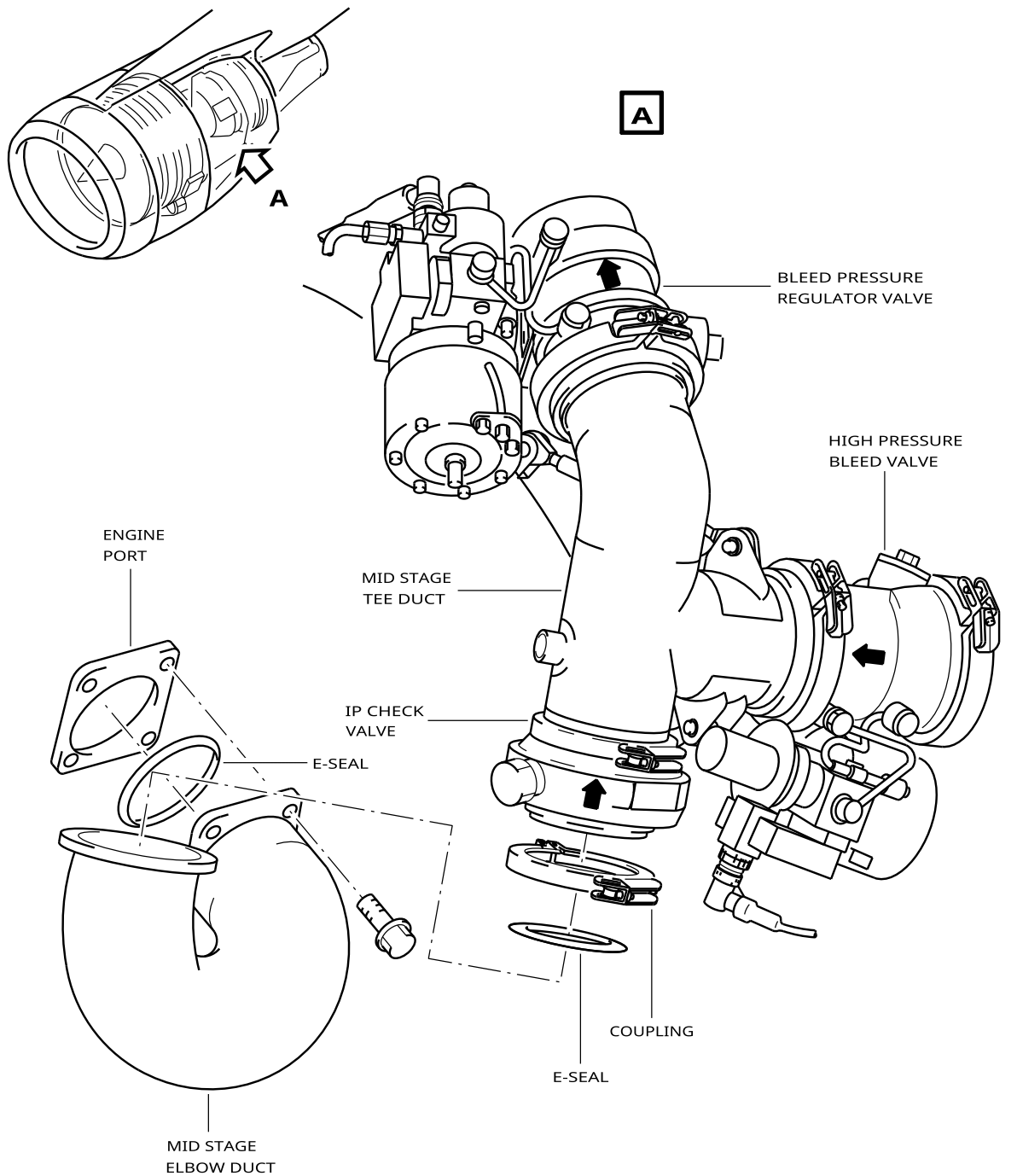


CFM-289-00

N_MM_361149_6_BAM0_01_00

Pneumatic System Flexible Duct Joint Assembly
Figure 1 / Graphic 36-11-49 -991-91400

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AIRCRAFT MAINTENANCE MANUAL
DUCTS - BLEED AIR - INSPECTION/CHECK

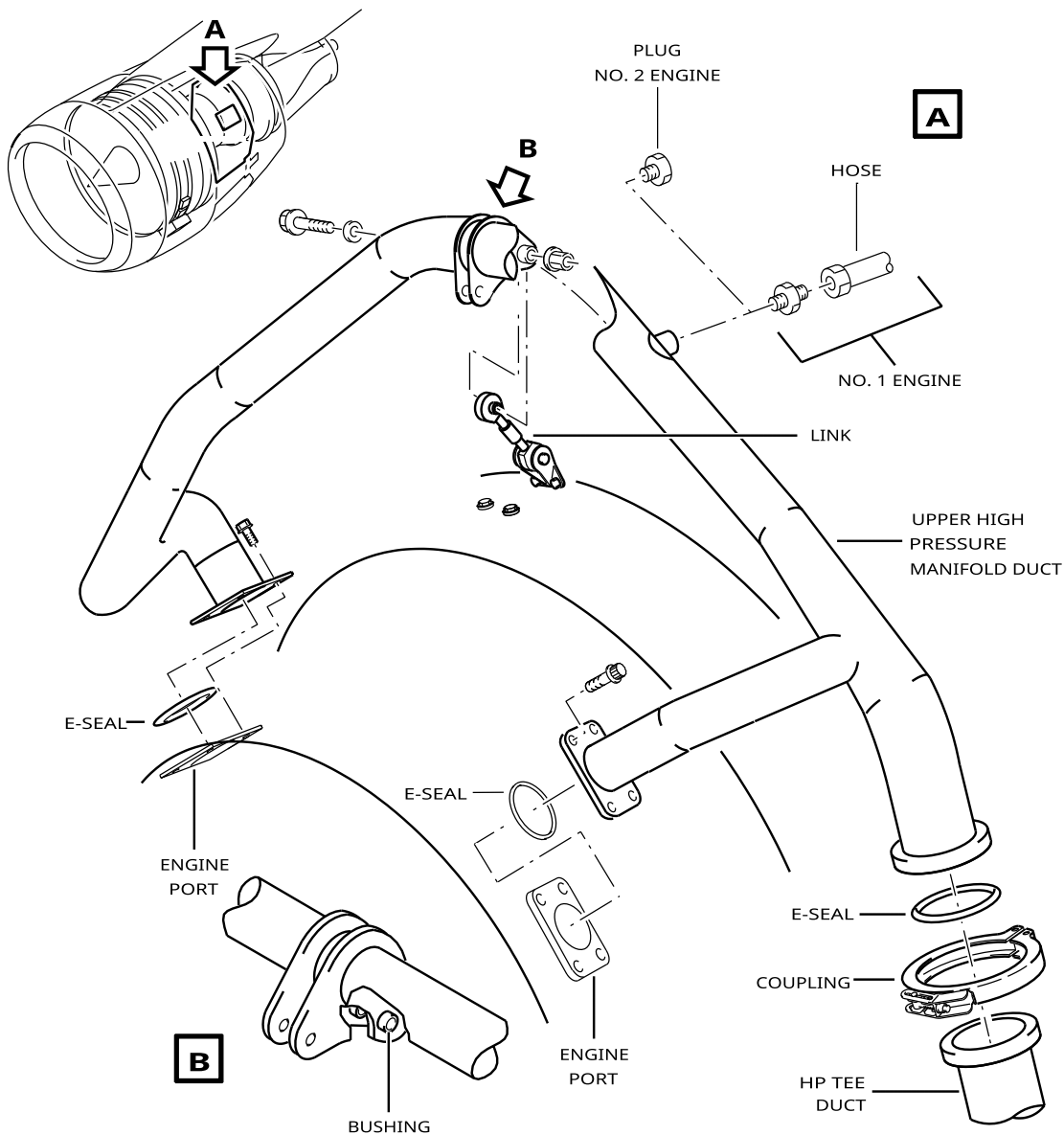


CFM-290-00

N_MM_361149_6_BCM0_01_00

Mid Stage Tee and Elbow Duct Assemblies
Figure 1 / Graphic 36-11-49 -991-91500

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AIRCRAFT MAINTENANCE MANUAL
DUCTS - BLEED AIR - INSPECTION/CHECK

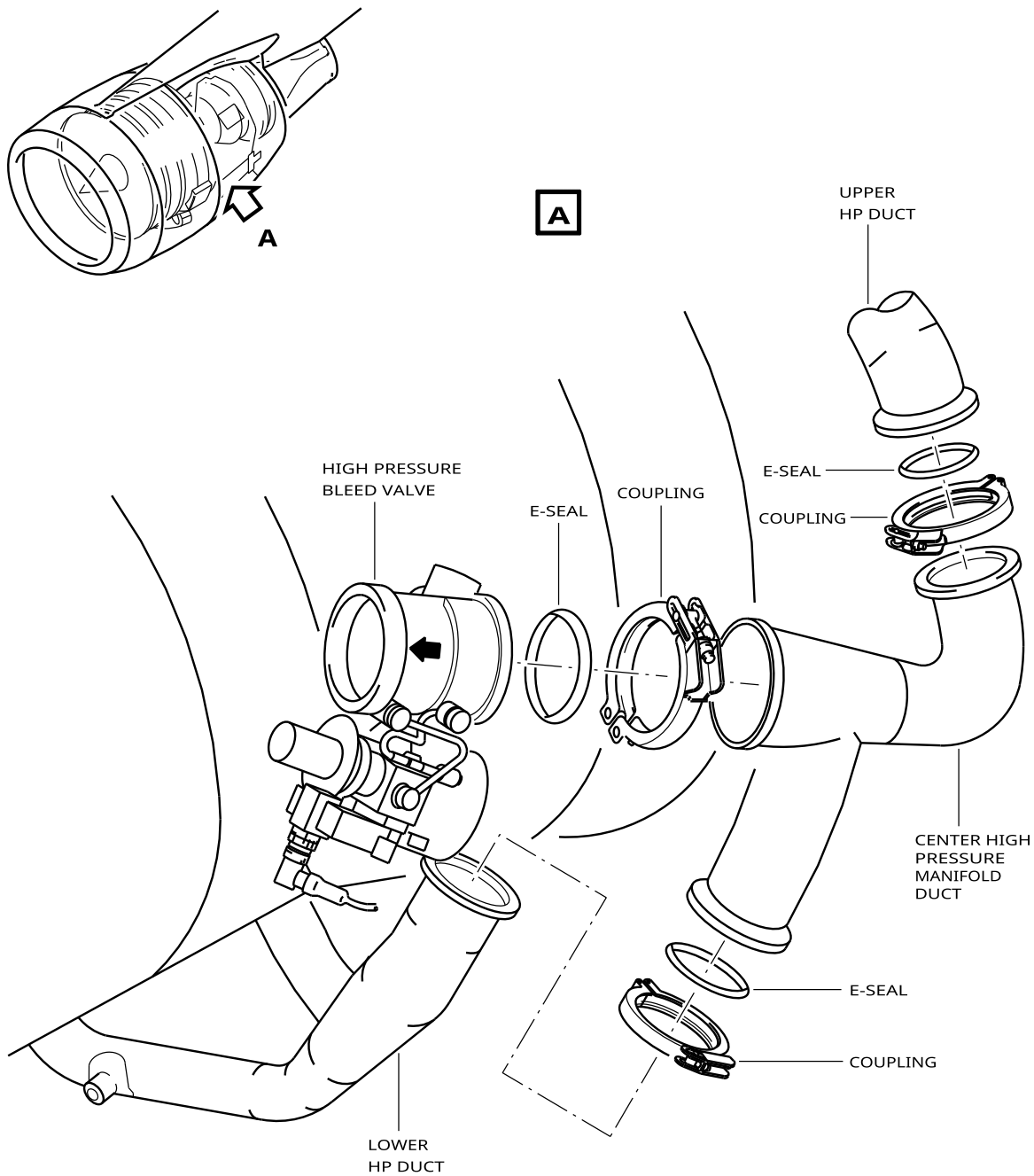


CFM-389-00

N_MM_361149_6_BFM0_01_00

Upper High Pressure Manifold Duct Assembly
Figure 1 / Graphic 36-11-49 -991-91600

TF-N
AIRCRAFT MAINTENANCE MANUAL
DUCTS - BLEED AIR - INSPECTION/CHECK

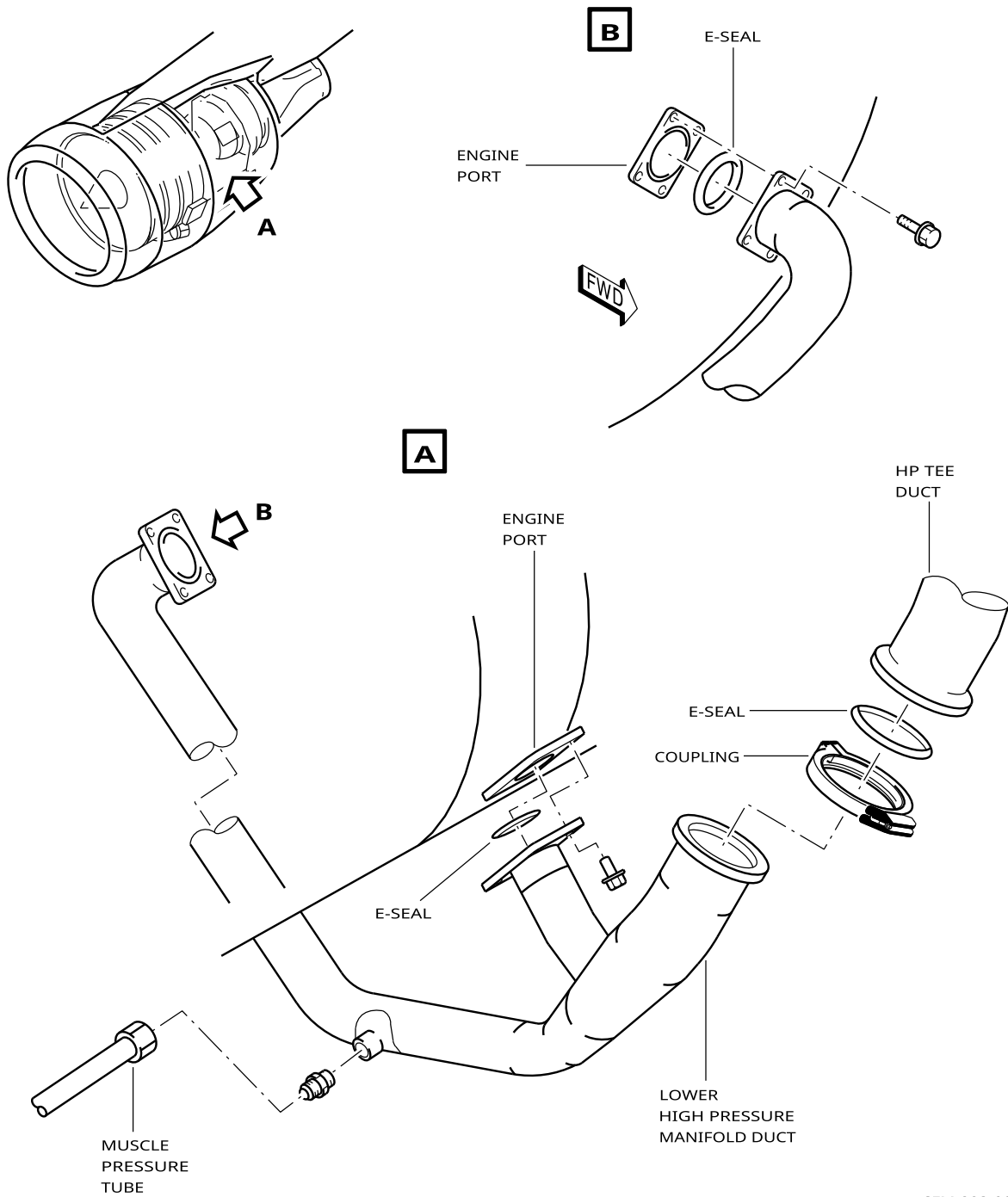


CFM-292-00

N_MM_361149_6_BGM0_01_00

Center High Pressure Manifold Duct Assembly
Figure 1 / Graphic 36-11-49 -991-91700

TF-N
AIRCRAFT MAINTENANCE MANUAL
DUCTS - BLEED AIR - INSPECTION/CHECK

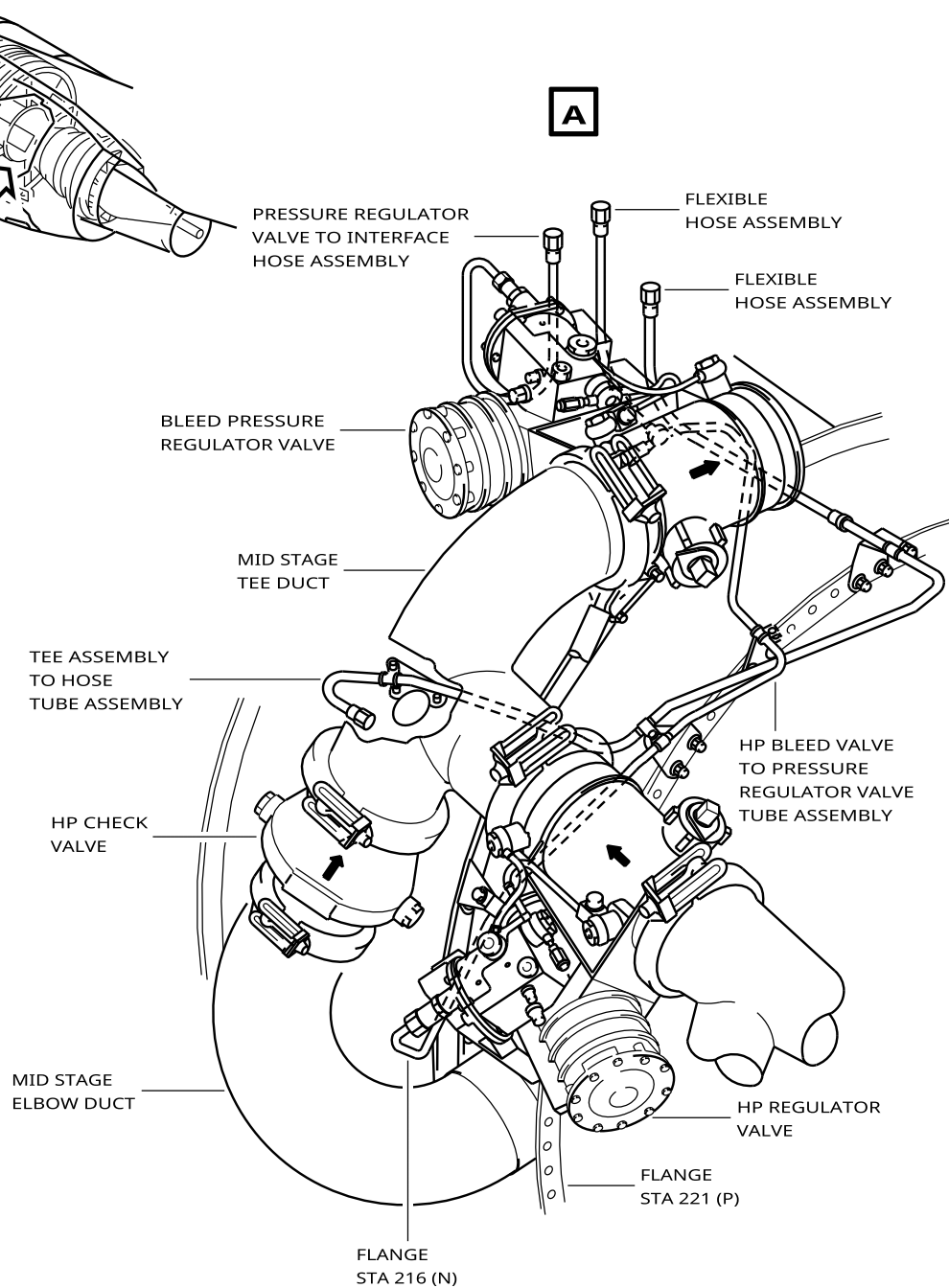


CFM-293-00

N_MM_361149_6_BJM0_01_00

Lower High Pressure Manifold Duct Assembly
Figure 1 / Graphic 36-11-49 -991-91800

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CFM-294-00

N_MM_361149_6_BLM0_01_00

Pneumatic System Hose and Tube Assemblies
Figure 1 / Graphic 36-11-49 -991-91900

 MET.29291060		Task Card Title: NO. 2 ENGINE HIGH PRESSURE DUCT INSPECTION								
Task Card NO: 361100-08-2-002-SVA		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP						
A/C Type: 321		Block Taskcard		Main Skill: INSP,IL,NDT_INSP,ACFT_TECH, AP		PROD Final Acceptance:				
Project No: 01BASPC72608		Related Task Card:		A/C Area: 02						
Internal Check:		Check Type:		Work Order NO: 31909295		Tally Sequence NO: 76				
ENCLOSURE: ☐ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐										
Type Code: DVI		Special Code: AP, INSP		MSG Code: 4		Request: JOB-SETUP,INSPECTION,INSPECTION,JOEUP				
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRID OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.										
Zones:										
DOCUMENT REFERENCES:										
Doc Typ		Reference								
AMM		36-11-49-200-041-A								
AMM		36-11-49-200-041								
CAP		S36-C10003								
DO		VSB RA32036-13								
IPC		36-11-49-01A								
IPC		36-11-49-01A								
ACCESS PANELS:										
MATERIAL REQUIREMENTS:										
Part Number		Description				Qty.				
No parts assigned to this Task Item										
TOOLING REQUIREMENTS:										
Part Number		Description				Qty.				
No tools assigned to this Task Item										
TASK SET UP										
Job Description										
A. PERFORM JOB SET UP FOR VISUAL INSPECTION OF THE PNEUMATIC SYSTEM HOSES, TUBES AND DUCTS AS PER AMM 36-11-49-200-041-A. (REF. AMM TASK 78-30-00-081-041-B)										
STEP COMPLETION SIGN OFF:										
		Qualification / Skill	Number of employees	Duration		Number./Signature		Date		
					HRS			Day	Month	Year
Mechanic/s		ACFT_TECH/AP			HRS					
Insp.					HRS					
UPPER 9TH STAGE HP DUCT AND DUCT SUPPORT LINK DETAIL INSPECTION										
Job Description										



Task Card Title:

NO. 2 ENGINE HIGH PRESSURE DUCT INSPECTION

Task Card NO:

361100-08-2-002-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP**UPPER 9TH STAGE HP DUCT AND DUCT SUPPORT LINK DETAIL INSPECTION****! NOTE**

The required parts may need replacement in case of findings.

All PNs are mentioned in the work template instructions and should be requested upon findings.

- PN 642-5508-505 >>> 9 STG HP PNEUMATIC DUCT
- PN P40560 >>> SUPPORT LINK OF HP DUCT
- PN P40570 >>> SUPPORT LINK OF HP DUCT

A. USE THE FOLLOWING INSTRUCTIONS FOR UPPER 9TH STAGE HP DUCTS INSTALLED ON AN AIRCRAFT.

1. OPEN FAN COWL DOORS. REFER TO AIRBUS A318/A319/A320/A321 AIRCRAFT MAINTENANCE MANUAL CHAPTER 71-13-00.
2. MAKE THE THRUST REVERSER UNSERVICEABLE FOR MAINTENANCE. REFER TO AIRBUS A318/A319 /A320/A321 AIRCRAFT MAINTENANCE MANUAL CHAPTER 78-30-00.
3. OPEN THE THRUST REVERSER HALVES. REFER TO AIRBUS A318/A319/A320/A321 AIRCRAFT MAINTENANCE MANUAL CHAPTER 78-36-00.
4. VERIFY THE **UPPER 9TH STAGE HIGH PRESSURE DUCT (HP) PN & SN** THAT IS INSTALLED IN THIS AIRCRAFT FIRSTLY. USE ATTACHED FIGURE 1 & 2 & IPC FIGURE 36-11-49-01A AS REFERENCE. PLEASE RECORD THE PN & SN BELOW AND ATTACH CLEAR PHOTO IN THE WO

PN:..... SN:.....

5. PERFORM A DETAILED VISUAL INSPECTION OF THE UPPER 9TH STAGE HP DUCT BEND AREA **PN 642 5508-503 OR PN 642-5508-505**(ADJACENT TO ENGINE INTERFACE) FOR CRACKS. REFER TO (ATTACHED FIGURE 1) FOR INSPECTION AREA.

PN: 9th stage HP duct	Quantity	Remark
642-5508-505	1	This PN might be needed for re findings. please request upon 1

a. USE SUFFICIENT LIGHTING AND A 10-POWER MAGNIFICATION FOR THE INSPECTION.



Task Card Title:

NO. 2 ENGINE HIGH PRESSURE DUCT INSPECTION

Task Card NO:

361100-08-2-002-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

- b. INSPECT THE ELBOW AND CIRCUMFERENTIAL WELDS AT EACH END FOR CRACKS. INSPECT THE EXTERNAL SURFACES ALL THE WAY AROUND THE UPPER 9TH STAGE HP DUCT'S ELBOW.
- c. IF A CRACK IS FOUND, REPLACE THE UPPER 9TH STAGE HP DUCT PER SB RA32036-12. IF THERE ARE NO CRACKS, PROCEED TO NEXT PARAGRAPH A.6.

6. PERFORM A DETAILED VISUAL INSPECTION OF THE UPPER 9TH STAGE HP DUCT WHERE IT ATTACHES TO THE LINK(S) FOR CRACKS. REFER TO (ATTACHED FIGURE 2) FOR INSPECTION AREA.

- a. USE SUFFICIENT LIGHTING AND A 10-POWER MAGNIFICATION FOR THE INSPECTION
- b. INSPECT THE CIRCUMFERENTIAL WELDS AT EACH END FOR CRACKS. INSPECT THE EXTERNAL SURFACES ALL THE WAY AROUND THE UPPER 9TH STAGE HP DUCT AT THE ATTACHMENT POINT.
- c. IF A CRACK IS FOUND, REPLACE THE UPPER 9TH STAGE HP DUCT PER SB RA32036-12. IF THERE ARE NO CRACKS, PROCEED TO NEXT PARAGRAPH A.7.

7. VERIFY IF THE INSTALLED SUPPORT LINK ATTACHED TO THE UPPER 9TH STAGE HP DUCT IS 1-LINK OR 2-LINK CONFIGURATION. USE ATTACHED FIGURE 2 & FIGURE 4 AS REFERENCES. RECORD IT BELOW AND ATTACH CLEAR PHOTO IN THE WO

TYPE OF SUPPORT LINK CONFIGURATION (1-LINK OR 2-LINK CONFIGURATION):

ALSO, WHETHER IT IS 1-LINK OR 2-LINK CONFIGURATION, PLEASE VERIFY THE SUPPORT LINK PN (USE FIGURE 2 AS REFERENCE TO LOCATE THE SAME PN AS SHOWN IN THE FIGURES) AND RECORD IT BELOW IF IT IS MARKED ON THE SUPPORT LINK ITSELF. IF NOT MARKED, PUT IT AS N/A AND GO TO THE NEXT STEP A.8

IF IT IS 1-LINK CONFIGURATION

SUPPORT LINK PN:

IF IT IS 2-LINK CONFIGURATION

SUPPORT LINK PN:

SUPPORT LINK PN:

8. INSPECT THE SUPPORT LINK **PN P40570** (AND **PN P40560** IF 2-LINK CONFIGURATION IS INSTALLED) FOR WEAR OR EXCESSIVE PLAY IN THE AXIAL OR RADIAL DIRECTION (SEE FIGURE 2). IF THERE IS FREE PLAY BETWEEN THE BALL AND RACE, COMPLETE RA32036-12, PART B. AS FOLLOW;

AS PER SB RA32036-12 PART B:



Task Card Title:

NO. 2 ENGINE HIGH PRESSURE DUCT INSPECTION

Task Card NO:

361100-08-2-002-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

- a. INSPECT THE SUPPORT LINK PN P40570 FOR WEAR OR EXCESSIVE PLAY IN THE AXIAL OR RADIAL DIRECTION (SEE FIGURE 1). IF THERE IS FREE PLAY BETWEEN THE BALL AND RACE, CHANGE THE BALL AND/OR FULL LINK ASSEMBLY AS REQUIRED. FOLLOW AMM INSTRUCTIONS IN CHAPTER 36-11-49, TASK 36-11-49-400-046A. FOR INSTALLATION OF THE HP DUCT & AFT LINK AND TASK 36-11-49-000-046A FOR THE REMOVAL.
- b. WRITE IN THE ENGINE LOG AND/OR OTHER APPLICABLE RECORDS THAT SERVICE BULLETIN RA32036-12, REV. 3 PART B HAS BEEN INCORPORATED.

PN: Support Link for HP Duct	Quantity	Remark
P40560	1	This PN might be replaced. Please request.
P40570	1	This PN might be replaced. Please request.

9. CLOSE THE THRUST REVERSER HALVES. REFER TO AIRBUS A318/A319/A320/A321 AIRCRAFT MAINTENANCE MANUAL CHAPTER 78-36-00, TASK 78-36-00-410-040-B.
10. Make the thrust reverser serviceable for maintenance. Refer to Airbus A318/A319/A320/A321 Aircraft Maintenance Manual Chapter 78-30-00.
11. Close Fan Cowl doors. Refer to Airbus A318/A319/A320/A321 Aircraft Maintenance Manual Chapter 71-13-00.
12. Write in the engine log and/or other records that Goodrich Service bulletin RA32036-13 has been incorporated.
13. Re-inspect the upper 9th stage HP duct per the inspection interval listed in the Compliance section on the Summary page.

B. Report Inspection Results

- Report to Goodrich the inspection results including no findings, cracks, or worn links and actions taken. Include aircraft/engine serial numbers, hours, and cycles, and upper 9th stage HP duct serial number information.
- Use the reporting sheet (see Attached Figure 3) at the end of this bulletin to transmit inspection results.



Task Card Title:

NO. 2 ENGINE HIGH PRESSURE DUCT INSPECTION

Task Card NO:

361100-08-2-002-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP**NOTE**

Please when you fill the reporting sheet, attach it in the work order and The Engineering will collect all these reporting sheets and provide it to Goodrich

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	ACFT_TECH/AP		HRS				
Insp.	INSP/II		HRS				

DYE PENETRANT TEST (NDT) ON THE 9TH STAGE UPPER HP DUCT:**Job Description**

(1) PERFORM THE DYE PENETRANT TEST AS PER REFERENCE NTM 51-80-00 ON THE 9TH STAGE UPPER HP DUCT ITEM 190 (AT THE DUCT BEND / ELBOW AREA) FOR THE SUBJECT ENGINE.

(2) IF YOU FIND DURING THE TEST ANY FINDINGS, OBSERVATIONS, CRACKS, LEAKS, ETC., PLEASE RECORD IT IN THIS WO OR SEND A QUERY TO ENG-CON-SYS GROUP.

NOTE: TQ IS ONLY NECESSARY IF THERE ARE FINDINGS DURING THE INSPECTION PROCESS.

(3) IN CASE OF FINDING THAT REQUIRE REPLACEMENT OF THE AFFECTED PARTS, PLEASE DO THE NECESSARY ACTION FOR THE REPLACEMENT OF THE AFFECTED PARTS AS PER REFERENCE AMM 36-11-49.

(4) ENSURE YOU CLEAN THE TESTED / INSPECTED AREA AFTER COMPLETING THE TEST.

NOTES:

❖ THE FOLLOWING ARE THE SUSPECTED/AFFECTED PARTS THAT MIGHT NEED REPLACEMENT ONLY IN CASE OF FINDINGS.

❖ PN: 642-5508-505 OR 642-5508-503 >>> 9 STAGE UPPER HP DUCT ITEM 190.

❖ THE ABOVE PARTS SHOULD BE REQUESTED ONLY UPON FINDINGS.

❖ PLEASE REFER TO ATTACHMENTS FOR 9TH STAGE UPPER HP DUCT ITEM 190 (DUCT BEND / ELBOW AREA) FOR YOUR REFERENCE.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	ACFT_TECH/AP		HRS				



Task Card Title:
NO. 2 ENGINE HIGH PRESSURE DUCT INSPECTION



Task Card NO:

361100-08-2-002-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

Insp.

II/NDT_INSP

HRS

LOWER HP DUCT DETAILED INSPECTION**Job Description**

- (1) Get access to the LOWER HIGH-PRESSURE DUCT (HP). Use attached IPC Figure 36-11-49-01A item 180 as reference.
- (2) Perform a detailed visual inspection of the lower HP duct (adjacent to engine interface) for cracks. As per the following steps:
- a) Use sufficient lighting and a 10-power magnification for the inspection.
 - b) Inspect the elbow/bend, circumferential welds and all the external surfaces all the way around the lower HP duct's areas.
 - c) If findings are found during the inspection please do the necessary action as per AMM 36-11-49.

NOTE: refer to AMM 36-11-49-200-041-A for possible findings that you might find on the lower HP duct (table item# 6 in section# 4 procedure).

PN	QTY	Remarks
642-5510-501 Lower HP duct	1	This PN might be needed for replacement in case of findings. please request upon finding only

- (3) Please raise a technical query to ENG-CON-SYS group for the inspection results.
Note: TQ is only necessary if there are findings during the inspection process.

The time required to perform this job is 30 mins for ACFT_TECH and 15 mins INSP.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	II/ACFT_TECH/AP			HRS			
Insp.	INSP			HRS			

DYE PENETRANT TEST (NDT) ON THE LOWER HP DUCT:**Job Description**

- (1) PERFORM THE DYE PENETRANT TEST AS PER REFERENCE NTM 51-80-00 ON THE LOWER HP DUCT ITEM 180 (AT THE BEND AREA) FOR THE SUBJECT ENGINE.
- (2) IF YOU FIND DURING THE TEST ANY FINDINGS, OBSERVATIONS, CRACKS, LEAKS, ETC., PLEASE RECORD IT IN THIS WO OR SEND A QUERY TO ENG-CON-SYS GROUP.



Task Card Title:

NO. 2 ENGINE HIGH PRESSURE DUCT INSPECTION

Task Card NO:

361100-08-2-002-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASPNOTE:

TQ IS ONLY NECESSARY IF THERE ARE FINDINGS DURING THE INSPECTION PROCESS.

(3) IN CASE OF FINDINGS THAT REQUIRE REPLACEMENT OF THE AFFECTED PARTS, PLEASE DO THE NECESSARY ACTION FOR THE REPLACEMENT OF THE AFFECTED PARTS AS PER REFERENCE AMM 36-11-49 .

(4) ENSURE YOU CLEAN THE TESTED / INSPECTED AREA AFTER COMPLETING THE TEST.

NOTES :

- ❖ THE FOLLOWING IS THE SUSPECTED/AFFECTED PART THAT MIGHT NEED REPLACEMENT ONLY IN CASE OF FINDINGS.
- ❖ PN 642-5510-501 >>> LOWER HP DUCT ITEM 180.
- ❖ THE ABOVE PART SHOULD BE REQUESTED ONLY UPON FINDINGS.
- ❖ PLEASE REFER TO ATTACHMENTS FOR LOWER HP DUCT ITEM 180 (BEND AREA) FOR YOUR REFERENCE.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	ACFT_TECH/AP		HRS				
Insp.	II/NDT_INSP		HRS				

VISUAL INSPECTION OF THE PNEUMATIC SYSTEM**Job Description**

1. Perform visual Inspection of the Pneumatic System Hoses, Tubes and Ducts AS PER AMM 36-11-49-200-041-A.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s			HRS				
Insp.	II/INSP		HRS				

TASK CLOSE UP**Job Description**

A. PERFORM JOB-CLOSEUP AND RETURN AIRCRAFT TO NORMAL CONFIGURATION AS PER AMM TASK 36-11-49-200-041-A. (REF. AMM TASK 78-30-00-081-041-B)

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	ACFT_TECH/AP		HRS				
Insp.			HRS				



Task Card Title:

NO. 2 ENGINE HIGH PRESSURE DUCT INSPECTION

Task Card NO:

361100-08-2-002-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below

Work order No: _____

Note: Make sure to create the finding Work Order on task card level.

Remark's: _____

(1) If Task card interrupted or performed by different persons updates the system and update the following accordingly (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 2 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 3 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 4 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 5 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

**Task Completion Close Up And
System Update****Station****Date****Signature**

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DUCTS - BLEED AIR - INSPECTION/CHECK

TASK 36-11-49-200-041 -A

On A/C 003-250

Visual Inspection of the Pneumatic System Hoses, Tubes and Ducts

WARNING: MAKE SURE THAT THE PNEUMATIC SYSTEM IS DEPRESSURIZED BEFORE YOU START WORK. PRESSURIZED AIR CAN CAUSE INJURY TO PERSONNEL.

WARNING: BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER ENGINE SHUTDOWN. USE APPLICABLE THERMAL GLOVES. THE ENGINE PARTS AND THE ENGINE OIL CAN STAY HOT FOR ONE HOUR AFTER SHUTDOWN AND CAN BURN YOU.

1. Reason for the Job

Self explanatory

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	AR	ACCESS PLATFORM 1M(3 FT)
No specific	AR	SAFETY CLIP - CIRCUIT BREAKER
No specific	AR	WARNING NOTICE(S)

B. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
FOR 1000EM1(ENGINE-1)	
437AL , 438AR , 451AL , 452AR	
FOR 1000EM2(ENGINE-2)	
447AL , 448AR , 461AL , 462AR	

C. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 36-11-49-000-042-A	Removal of the Pylon Interface Duct
Ref. AMM 36-11-49-000-043-A	Removal of the Mid Stage Tee Duct
Ref. AMM 36-11-49-000-044-A	Removal of the High Pressure Tee Duct
Ref. AMM 36-11-49-000-045-A	Removal of the Lower High Pressure Duct
Ref. AMM 36-11-49-000-046-A	Removal of the Upper HP Duct
Ref. AMM 36-11-49-000-047-A	Removal of the Pneumatic System Tubes and Hoses
Ref. AMM 36-11-49-000-048-A	Removal of the Fifth Stage Elbow Duct
Ref. AMM 36-11-49-000-842-A	Removal of the Upper HP Duct and Mid Stage T-Duct Assembly Links
Ref. AMM 36-11-49-200-041-A	Visual Inspection of the Pneumatic System Hoses, Tubes and Ducts
Ref. AMM 36-11-49-400-042-A	Installation of the Pylon Interface Duct
Ref. AMM 36-11-49-400-043-A	Installation of the Mid Stage Tee Duct
Ref. AMM 36-11-49-400-044-A	Installation of the High-Pressure Tee Duct
Ref. AMM 36-11-49-400-045-A	Installation of the Lower High-Pressure Duct
Ref. AMM 36-11-49-400-046-A	Installation of the Upper HP Duct
Ref. AMM 36-11-49-400-047-A	Installation of the Pneumatic System Tubes and Hoses
Ref. AMM 36-11-49-400-048-A	Installation of the Fifth-Stage Elbow Duct
Ref. AMM 36-11-49-400-842-A	Installation of the Upper HP Duct and Mid Stage T-Duct Assembly Links
Ref. AMM 71-13-00-010-040-B	Opening of the Fan Cowl Doors
Ref. AMM 71-13-00-410-040-B	Closing of the Fan Cowl Doors
Ref. AMM 78-30-00-081-041-B	Make the Thrust Reverser System Serviceable after Maintenance
Ref. AMM 78-30-00-481-041-B	Make the Thrust Reverser System Unserviceable for Maintenance
Ref. AMM 78-32-11-220-040-B	General Visual Inspection of the Inner Barrel Heatshield, Fire Protective Coating and Fire Seals (C-Ducts Open)
Ref. AMM 78-36-00-010-040-B	Opening of the Thrust Reverser Doors
Ref. AMM 78-36-00-410-040-B	Closing of the Thrust Reverser Doors

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DUCTS - BLEED AIR - INSPECTION/CHECK

REFERENCE	DESIGNATION
Ref. AMM 80-11-10-210-003-A	Starter Inspection for a Probably FOD (Foreign Object Damage)
Ref. Fig. (1 / 36-11-49-991-91400)	
Ref. Fig. (1 / 36-11-49-991-91500)	
Ref. Fig. (1 / 36-11-49-991-91600)	
Ref. Fig. (1 / 36-11-49-991-91700)	
Ref. Fig. (1 / 36-11-49-991-91800)	
Ref. Fig. (1 / 36-11-49-991-91900)	

3. Job Set-up

Subtask 36-11-49-941-091-A

On A/C 003-250

A. Safety Precautions

- (1) On the ENG panel 115VU:
 - put a WARNING NOTICE(S) to tell persons not to start the engine 1(2).
- (2) Make sure that the engine shutdown occurred five minutes or more before you do this procedure.
- (3) On the ENG section of the maintenance panel 50VU:
 - make sure that the ON legend of the ENG/FADEC GND PWR 1(2) pushbutton switch is off.
 - put a WARNING NOTICE(S) to tell persons not to energize the FADEC 1(2).

Subtask 36-11-49-010-122-A

On A/C 003-250

B. Get Access

- (1) Open the fan cowl doors [Ref. AMM 71-13-00-010-040-B](#):
 - (a) FOR 1000EM1 (ENGINE-1)
437AL 438AR
 - (b) FOR 1000EM2 (ENGINE-2)
447AL 448AR
- (2) Put the ACCESS PLATFORM 1M(3 FT) in position below the engine.

Subtask 36-11-49-040-094-A

On A/C 003-250

C. Make the thrust reverser unserviceable [Ref. AMM 78-30-00-481-041-B](#).

Subtask 36-11-49-010-123-A

On A/C 003-250

D. Open the thrust reverser doors [Ref. AMM 78-36-00-010-040-B](#):

- (1) FOR 1000EM1 (ENGINE-1)
451AL 452AR
- (2) FOR 1000EM2 (ENGINE-2)
461AL 462AR

Subtask 36-11-49-865-129-A

On A/C 003-250

E. Open, safety and tag the circuit breaker(s) that follow(s). Use the SAFETY CLIP - CIRCUIT BREAKER as necessary.

PANEL	DESIGNATION	FIN	LOCATION
49VU	AIR BLEED/ENG 1/CTL	3HA1	D12
49VU	AIR BLEED/ENG 1/MONG	2HA1	D11
122VU	AIR BLEED/ENG 2/CTL	3HA2	Z23
122VU	AIR BLEED/ENG 2/MONG	2HA2	Z22

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DUCTS - BLEED AIR - INSPECTION/CHECK

4. Procedure

Subtask 36-11-49-210-062-B

On A/C 003-250

A. Do a Visual Inspection of the Pneumatic System Ducts, Hoses and Tubes

Ref. Fig. (1 / 36-11-49-991-91400)

Ref. Fig. (1 / 36-11-49-991-91500)

Ref. Fig. (1 / 36-11-49-991-91600)

Ref. Fig. (1 / 36-11-49-991-91700)

Ref. Fig. (1 / 36-11-49-991-91800)

Ref. Fig. (1 / 36-11-49-991-91900)

INSPECT/CHECK	MAXIMUM SERVICEABLE LIMITS	REMARKS
1.Examine the pneumatic-system flexible-duct-joint assembly (pylon Interface duct) for: Ref. Fig. (1 / 36-11-49-991-91400)		
A.Damage	None permitted in less than 0.50 in. (12.69 mm) of links, flex joints, couplings, and flanges.	Replace the duct Ref. AMM 36-11-49-000-042-A and Ref. AMM 36-11-49-400-042-A .
B.Wrinkles in bends	Maximum depth must be less than 5 percent of the duct diameter.	If the damage is more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-042-A and Ref. AMM 36-11-49-400-042-A .
C.Dents	Maximum depth must be less than 20 percent of the tube outside diameter. Maximum radius of the dent is not less than two times of the duct wall thickness. Maximum of four dents in each one foot (305 mm) of the duct. Minimum distance of 1.0 inch (25.4 mm) between the dents. No limit to dents permitted unless in bends, no dents are permitted in bends.	If the dents are less than the permitted limits, send the duct to the shop for analysis. If the dents are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-042-A and Ref. AMM 36-11-49-400-042-A .
D.Flat areas	Maximum length must be less than three times of the outside diameter of the duct. Contour change must be less than 20 percent of the outside diameter of the duct.	If the flat areas are less than the permitted limits, send the duct to the shop for analysis. If the flat areas are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-042-A and Ref. AMM 36-11-49-400-042-A .
E.Nicks, scratches, chafing in straight section	Maximum depth must be less than 10 percent of the wall thickness of the duct. There is no limit to the number of nicks, scratches or chafing permitted in the straight section of the duct.	- If the damage is less than the permitted limits: . Remove the pylon interface duct Ref. AMM 36-11-49-000-042-A. . Send the duct to the shop for analysis. . Install the pylon interface duct Ref. AMM 36-11-49-400-042-A. - If the damage is more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-042-A and Ref. AMM 36-11-49-400-042-A.
F.Nicks, scratches, chafing in bends	Maximum depth must be less than 10 percent of the wall thickness of the duct.	- If the damage is less than the permitted limits: . Remove the pylon interface duct Ref. AMM 36-11-49-000-042-A.

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INSPECT/CHECK	MAXIMUM SERVICEABLE LIMITS	REMARKS
		. Send the duct to the shop for analysis. . Install the pylon interface duct Ref. AMM 36-11-49-400-042-A. • If the damage is more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-042-A and Ref. AMM 36-11-49-400-042-A.
G.Splits, cracks, kinks	Maximum length must be less than the duct outside diameter.	If the damage is less than the duct outer diameter, send the duct to the shop for analysis.
	Maximum length same as or more than the duct outside diameter.	If the damage is the same as or more than the duct outer diameter, send the duct to the shop for analysis.
H.Internal splits, cracks of flexible joints (bellows)	None permitted	Replace the duct Ref. AMM 36-11-49-000-042-A and Ref. AMM 36-11-49-400-042-A . Inspect the pneumatic system for FOD (foreign object damage) Ref. AMM 36-11-49-200-041-A . Inspect the starter motor for FOD (foreign object damage) Ref. AMM 80-11-10-210-003-A .
2.Examine the pneumatic-system mid-stage-tee assembly for: Ref. Fig. (1 / 36-11-49-991-91500)		
A.Damage	None permitted in less than 0.50 in. (12.69 mm) of links, flex joints, couplings, and flanges.	Replace the duct Ref. AMM 36-11-49-000-043-A and Ref. AMM 36-11-49-400-043-A .
B.Cracks, splits or kinks	None permitted	Replace the duct Ref. AMM 36-11-49-000-043-A and Ref. AMM 36-11-49-400-043-A .
C.Wrinkles in bends	Maximum depth cannot be more than 3 percent of the duct diameter or 0.060 inch (1.52 mm). The smaller value is the limit.	If wrinkles are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-043-A and Ref. AMM 36-11-49-400-043-A .
D.Dents	Maximum depth of 0.3 inch (7.62 mm) or Maximum depth cannot be more than 10 percent of the duct outside diameter. Minimum width to depth ratio to be 5:1. Dent has to be free from cracks, abrasion and sudden contour changes. Radius is not less than two times of the wall thickness. Maximum of two dents for each one foot (305mm) of the duct length.	If the dents are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-043-A and Ref. AMM 36-11-49-400-043-A .
E.Flat areas	Maximum flat area cannot be more than two times of the duct outside diameter. The contour change cannot be more than 10 percent of the duct outside diameter.	If the flat areas are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-043-A and Ref. AMM 36-11-49-400-043-A .
F.Nicks, scratches, chafing	Maximum depth cannot be more than 10 percent of the wall thickness.	If the damage is more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-043-A and Ref. AMM 36-11-49-400-043-A .

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DUCTS - BLEED AIR - INSPECTION/CHECK

INSPECT/CHECK	MAXIMUM SERVICEABLE LIMITS	REMARKS
3.Examine the pneumatic-system mid-stage-elbow assembly for: Ref. Fig. (1 / 36-11-49-991-91500)		
A.Damage	None permitted in less than 0.50 in. (12.69 mm) of links, flex joints, couplings, and flanges.	Replace the duct Ref. AMM 36-11-49-000-048-A and Ref. AMM 36-11-49-400-048-A .
B.Cracks, splits or kinks	None permitted	Replace the duct Ref. AMM 36-11-49-000-048-A and Ref. AMM 36-11-49-400-048-A .
C.Wrinkles in bends	Maximum depth cannot be more than 3 percent of the duct diameter or 0.060 inch (1.52 mm). The smaller value is the limit.	If wrinkles are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-048-A and Ref. AMM 36-11-49-400-048-A .
D.Dents	Maximum depth cannot be more than 10 percent of the duct outside diameter. Maximum radius is not less than two times of the wall thickness. Maximum of two dents for each one foot (305 mm) of the duct length.	If the dents are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-048-A and Ref. AMM 36-11-49-400-048-A .
E.Flat areas	Maximum flat area cannot be more than two times of the duct outside diameter. The contour change cannot be more than 10 percent of the duct outside diameter.	If the flat areas are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-048-A and Ref. AMM 36-11-49-400-048-A .
F.Nicks, scratches, chafing	Maximum depth cannot be more than 10 percent of the wall thickness.	If the damage is more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-048-A and Ref. AMM 36-11-49-400-048-A .
4.Examine the pneumatic-system upper high-pressure manifold-duct assembly for: Ref. Fig. (1 / 36-11-49-991-91600)		
A.Damage	None permitted in less than 0.50 in. (12.69 mm) of links, flex joints, couplings, and flanges.	Replace the duct Ref. AMM 36-11-49-000-046-A and Ref. AMM 36-11-49-400-046-A .
B.Cracks, splits or kinks	None permitted	Replace the duct Ref. AMM 36-11-49-000-046-A and Ref. AMM 36-11-49-400-046-A .
C.Wrinkles in bends	Maximum depth cannot be more than 3 percent of the duct diameter or 0.060 inch (1.52 mm). The smaller value is the limit.	If wrinkles are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-046-A and Ref. AMM 36-11-49-400-046-A .
D.Dents	Maximum depth cannot be more than 10 percent of the duct outside diameter. Maximum radius is not less than two times of the wall thickness. Maximum of two dents for each one foot (305 mm) of the duct length.	If the dents are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-046-A and Ref. AMM 36-11-49-400-046-A .
E.Flat areas	Maximum flat area cannot be more than two times the duct outside diameter. The contour change cannot be more than 10 percent of the duct outside diameter.	If the flat areas are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-046-A and Ref. AMM 36-11-49-400-046-A .
F.Nicks, scratches, chafing	Maximum depth cannot be more than 10 percent of the wall thickness.	If the damage is more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-046-A and Ref. AMM 36-11-49-400-046-A .

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DUCTS - BLEED AIR - INSPECTION/CHECK

INSPECT/CHECK	MAXIMUM SERVICEABLE LIMITS	REMARKS
G.Worn bushing holes	Maximum diameter of the bushing hole is 0.26 in. (6.6040 mm)	If the bushing hole diameter more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-046-A and Ref. AMM 36-11-49-400-046-A .
H.Loose support links	None permitted	Replace the support link(s) Ref. AMM 36-11-49-000-842-A and Ref. AMM 36-11-49-400-842-A .
5.Examine the pneumatic-system center high-pressure manifold-duct assembly for: Ref. Fig. (1 / 36-11-49-991-91700)		
A.Damage	None permitted in less than 0.50 in. (12.69 mm) of links, flex joints, couplings, and flanges.	Replace the duct Ref. AMM 36-11-49-000-044-A and Ref. AMM 36-11-49-400-044-A .
B.Cracks, splits or kinks	None permitted	Replace the duct Ref. AMM 36-11-49-000-044-A and Ref. AMM 36-11-49-400-044-A .
C.Wrinkles in bends	Maximum depth cannot be more than 3 percent of the duct diameter or 0.060 inch (1.52 mm).The smaller value is the limit.	If wrinkles are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-044-A and Ref. AMM 36-11-49-400-044-A .
D.Dents	Maximum depth cannot be more than 10 percent of the duct outside diameter. Maximum radius is not less than two times of the wall thickness. Maximum of two dents for each one foot (305 mm) of the duct length.	If the dents are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-044-A and Ref. AMM 36-11-49-400-044-A .
E.Flat areas	Maximum flat area cannot be more than two times the duct outside diameter. The contour change cannot be more than 10 percent of the duct outside diameter.	If the flat areas are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-044-A and Ref. AMM 36-11-49-400-044-A .
F.Nicks, scratches, chafing	Maximum depth cannot be more than 10 percent of the wall thickness.	If the damage is more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-044-A and Ref. AMM 36-11-49-400-044-A .
6.Examine the pneumatic-system lower high-pressure-duct assembly for: Ref. Fig. (1 / 36-11-49-991-91800)		
A.Damage	None permitted in less than 0.50 in. (12.69 mm) of links, flex joints, couplings, and flanges.	Replace the duct Ref. AMM 36-11-49-000-045-A and Ref. AMM 36-11-49-400-045-A .
B.Cracks, splits or kinks	None permitted	Replace the duct Ref. AMM 36-11-49-000-045-A and Ref. AMM 36-11-49-400-045-A .
C.Wrinkles in bends	Maximum depth cannot be more than 3 percent of the duct diameter or 0.060 inch (1.52 mm). The smaller value is the limit.	If wrinkles are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-045-A and Ref. AMM 36-11-49-400-045-A .
D.Dents	Maximum depth cannot be more than 10 percent of the duct outside diameter. Maximum radius is not less than two times of the wall thickness. Maximum of two dents for each one foot (305 mm) of the duct length.	If the dents are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-045-A and Ref. AMM 36-11-49-400-045-A .

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AIRCRAFT MAINTENANCE MANUAL
DUCTS - BLEED AIR - INSPECTION/CHECK

INSPECT/CHECK	MAXIMUM SERVICEABLE LIMITS	REMARKS
E.Flat areas	Maximum flat area cannot be more than two times the duct outside diameter. The contour change cannot be more than 10 percent of the duct outside diameter.	If the flat areas are more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-045-A and Ref. AMM 36-11-49-400-045-A .
F.Nicks, scratches, chafing	Maximum depth cannot be more than 10 percent of the wall thickness.	If the damage is more than the permitted limits, replace the duct Ref. AMM 36-11-49-000-045-A and Ref. AMM 36-11-49-400-045-A .
7.Examine the pressure regulator valve to interface hose assembly for the flexible part of the sense line for: Ref. Fig. (1 / 36-11-49-991-91900)		
A.Broken wire braids	Maximum of three broken wires in each one plait. No more than six broken wires in an assembly. No more than six broken wires in each one foot (305 mm) of hose.	If the damage is more than the permitted limits, replace the hose 36-11-49-000-026 and 36-11-49-400-026.
B.Cracks	None permitted	If cracks are found, replace the hose 36-11-49-000-026 and 36-11-49-400-026.
C.Crimps	None permitted	Replace the hose assembly 36-11-49-000-026 and 36-11-49-400-026.
D.Internal blockage	None permitted	If blockage is found, remove the blockage or replace the hose 36-11-49-000-026 and 36-11-49-400-026.
E.Leakage	None permitted	Replace the hose 36-11-49-000-026 and 36-11-49-400-026.
8.Examine the flexible hose assemblies for: Ref. Fig. (1 / 36-11-49-991-91900)		
A.Broken wire braids	Maximum of three broken wires in each one plait. No more than six broken wires in an assembly. No more than six broken wires in each one foot (305 mm) of hose.	If the damage is more than the permitted limits, replace the hose Ref. AMM 36-11-49-000-047-A and Ref. AMM 36-11-49-400-047-A .
B.Cracks	None permitted	If cracks are found, replace the hose Ref. AMM 36-11-49-000-047-A and Ref. AMM 36-11-49-400-047-A .
C.Crimps	None permitted	Replace the hose assembly Ref. AMM 36-11-49-000-047-A and Ref. AMM 36-11-49-400-047-A .
D.Internal blockage	None permitted	If blockage is found, remove the blockage or replace the hose Ref. AMM 36-11-49-000-047-A and Ref. AMM 36-11-49-400-047-A .
E.Leakage	None permitted	Replace the hose Ref. AMM 36-11-49-000-047-A and Ref. AMM 36-11-49-400-047-A .
9.Examine the fittings on the two ends of the hose for:		
A.Thread damage	Damage is permitted if the threads that are good are not decreased by more than one thread.	If the damage is more than the permitted limits, replace the hose Ref. AMM 36-11-49-000-042-A and Ref. AMM 36-11-49-400-042-A .

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AIRCRAFT MAINTENANCE MANUAL
DUCTS - BLEED AIR - INSPECTION/CHECK

INSPECT/CHECK	MAXIMUM SERVICEABLE LIMITS	REMARKS
B.Damaged flats	Damage is permitted if the end fittings can keep the correct torque value.	If the end fittings cannot keep the correct torque, replace the hose Ref. AMM 36-11-49-000-043-A and Ref. AMM 36-11-49-400-043-A .
C.Safety wire hole damaged or worn	Damage is permitted if the metal around the hole is not cracked or torn. Make sure that one corner stays undrilled.	Blend out damage. If necessary, drill a hole in the undrilled corner. The new hole must be the same diameter and distance from the end of the nut as the old hole.
10.Examine the high pressure bleed-valve to pressure regulator tube-assembly for the rigid part of the sense line for: Ref. Fig. (1 / 36-11-49-991-91900)		
A.Wrinkles in bends	Maximum depth must be less than 5 percent of the tube diameter.	If the damage is more than the permitted limits, replace the tube 36-11-49-000-026 and 36-11-49-400-026.
B.Dents	Maximum depth must be less than 20 percent of the tube outside diameter. Maximum radius of the dent is not less than two times of the tube wall thickness. Maximum of four dents in each one foot (305 mm) of the tube. Minimum distance of 1.0 inch (25.4 mm) between the dents. No limit to dents permitted unless in bends, no dents are permitted in bends.	If the dents are less than the permitted limits, send the tube to the shop for analysis. If the dents are more than the permitted limits, replace the tube 36-11-49-000-026 and 36-11-49-400-026.
C.Flat areas	Maximum length must be less than three times the outside diameter of the tube. Contour change must be less than 20 percent of the outside diameter of the tube.	If the flat areas are less than the permitted limits, send the tube to the shop for analysis. If the flat areas are more than the permitted limits, replace the tube 36-11-49-000-026 and 36-11-49-400-026.
D.Nicks, scratches, chafing in straight section	Maximum depth must be less than 10 percent of the wall thickness of the tube. There is no limit to the number of nicks, scratches or chafing allowed in the straight section of the tube.	If the damage is less than the permitted limits, send the tube to the shop for analysis. If the damage is more than the permitted limits, replace the tube 36-11-49-000-026 and 36-11-49-400-026.
E.Nicks, scratches, chafing in bends	Maximum depth must be less than 10 percent of the wall thickness of the tube.	If the damage is less than the permitted limits, send the tube to the shop for analysis. If the damage is more than the permitted limits, replace the tube 36-11-49-000-026 and 36-11-49-400-026.
F.Splits, cracks, kinks	Maximum length must be less than the tube outside diameter.	If the damage is less than the tube outer diameter, send the tube to the shop for analysis.
	Maximum length same as or more than the tube outside diameter.	If the damage is the same as or more than the tube outer diameter, send the tube to the shop for analysis.
11.Examine the tee assembly to hose tube assembly for: Ref. Fig. (1 / 36-11-49-991-91900)		
A.Wrinkles in bends	Maximum depth must be less than 5 percent of the tube diameter.	If the damage is more than the permitted limits, replace the tube Ref. AMM 36-11-49-000-047-A and Ref. AMM 36-11-49-400-047-A .

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INSPECT/CHECK	MAXIMUM SERVICEABLE LIMITS	REMARKS
B.Dents	Maximum depth must be less than 20 percent of the tube outside diameter. Maximum radius of the dent is not less than two times of the tube wall thickness. Maximum of four dents in each one foot (305 mm) of the tube. Minimum distance of 1.0 inch (25.4 mm) between the dents. No limit to dents permitted unless in bends, no dents are permitted in bends.	If the dents are less than the permitted limits, send tube to the shop for analysis. If the dents are more than the permitted limits, replace the tube Ref. AMM 36-11-49-000-047-A and Ref. AMM 36-11-49-400-047-A .
C.Flat areas	Maximum length must be less than three times the outside diameter of the tube. Contour change must be less than 20 percent of the outside diameter of the tube.	If the flat areas are less than the permitted limits, send the tube to the shop for analysis. If the flat areas are more than the permitted limits, replace the tube Ref. AMM 36-11-49-000-047-A and Ref. AMM 36-11-49-400-047-A .
D.Nicks, scratches, chafing in straight section	Maximum depth must be less than 10 percent of the wall thickness of the tube. There is no limit to the number of nicks, scratches or chafing allowed in the straight section of the tube.	If the damage is less than the permitted limits, send the tube to the shop for analysis. If the damage is more than the permitted limits, replace the tube Ref. AMM 36-11-49-000-047-A and Ref. AMM 36-11-49-400-047-A .
E.Nicks, scratches, chafing in bends	Maximum depth must be less than 10 percent of the wall thickness of the tube.	If the damage is less than the permitted limits, send the tube to the shop for analysis. If the damage is more than the permitted limits, replace the tube Ref. AMM 36-11-49-000-047-A and Ref. AMM 36-11-49-400-047-A .
F.Splits, cracks, kinks	Maximum length must be less than the tube outside diameter.	If the damage is less than the tube outer diameter, send the tube to the shop for analysis.
	Maximum length same as or more than the tube outside diameter.	If the damage is the same as or more than the tube outer diameter, send the tube to the shop for analysis.

Subtask 36-11-49-210-072-A

On A/C 003-250

B. Visual Inspection of the Inner Barrel Heatshield of the Thrust Reverser

- (1) If there is a leak because of the damaged part(s), do the visual inspection of the inner barrel heatshield of the thrust reverser [Ref. AMM 78-32-11-220-040-B](#).

5. Close-up

Subtask 36-11-49-410-104-A

On A/C 003-250

A. Close Access

- (1) Make sure that the work area is clean and clear of tools and other items.
- (2) Close the thrust reverser doors [Ref. AMM 78-36-00-410-040-B](#):
 - (a) FOR 1000EM1 (ENGINE-1)
451AL 452AR
 - (b) FOR 1000EM2 (ENGINE-2)
461AL 462AR

Subtask 36-11-49-440-073-A

On A/C 003-250

B. Make the thrust reverser serviceable [Ref. AMM 78-30-00-081-041-B](#).

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DUCTS - BLEED AIR - INSPECTION/CHECK

Subtask 36-11-49-410-105-A

On A/C 003-250

C. Close Access

(1) Close the fan cowl doors [Ref. AMM 71-13-00-410-040-B](#):

(a) FOR 1000EM1 (ENGINE-1)

437AL 438AR

(b) FOR 1000EM2 (ENGINE-2)

447AL 448AR

(2) Remove the access platform(s).

(3) Remove the WARNING NOTICE(S).

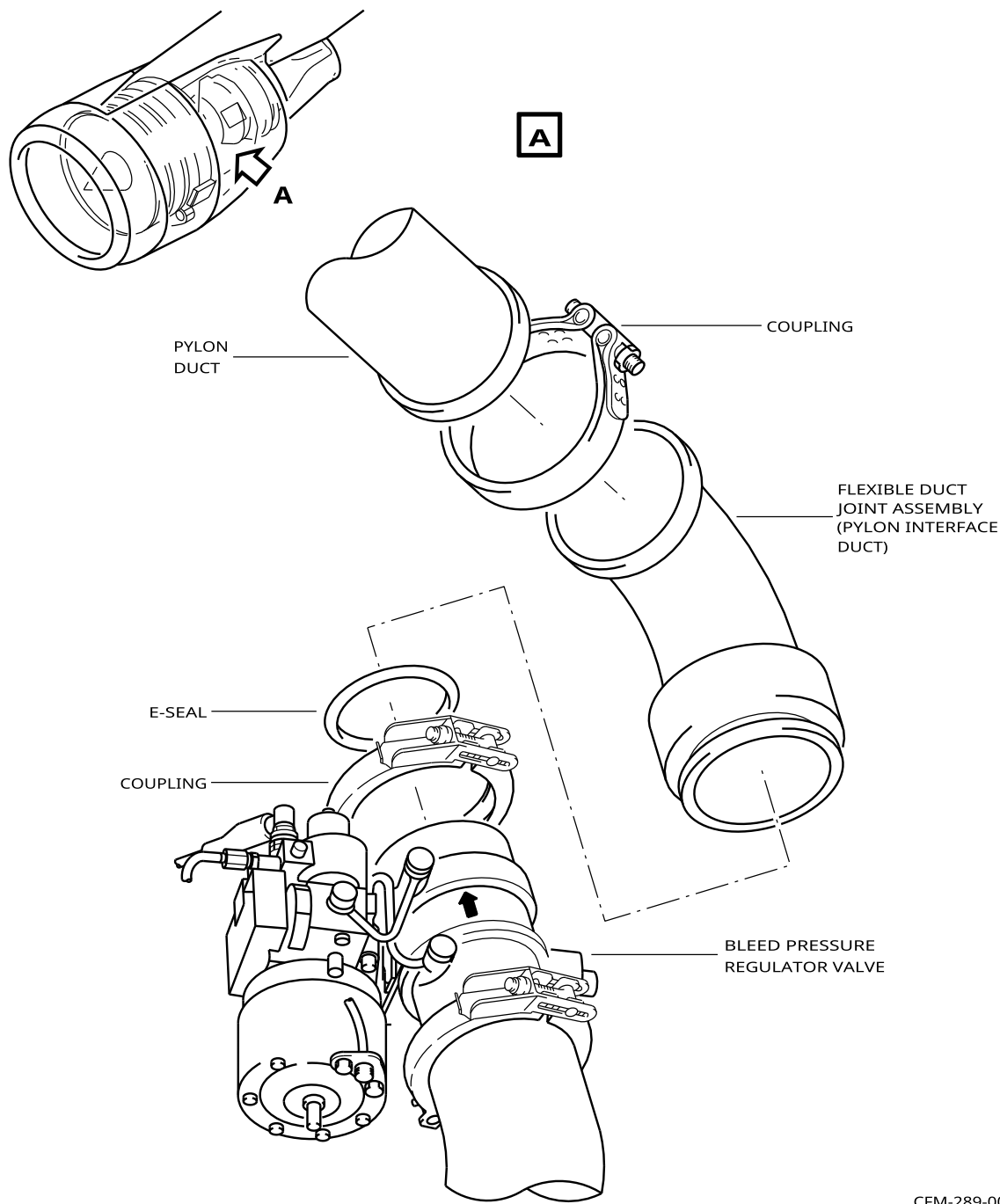
Subtask 36-11-49-865-130-A

On A/C 003-250

D. Remove the SAFETY CLIP - CIRCUIT BREAKER and the tag(s) and close this (these) circuit breaker(s):

PANEL	DESIGNATION	FIN	LOCATION
49VU	AIR BLEED/ENG 1/CTL	3HA1	D12
49VU	AIR BLEED/ENG 1/MONG	2HA1	D11
122VU	AIR BLEED/ENG 2/CTL	3HA2	Z23
122VU	AIR BLEED/ENG 2/MONG	2HA2	Z22

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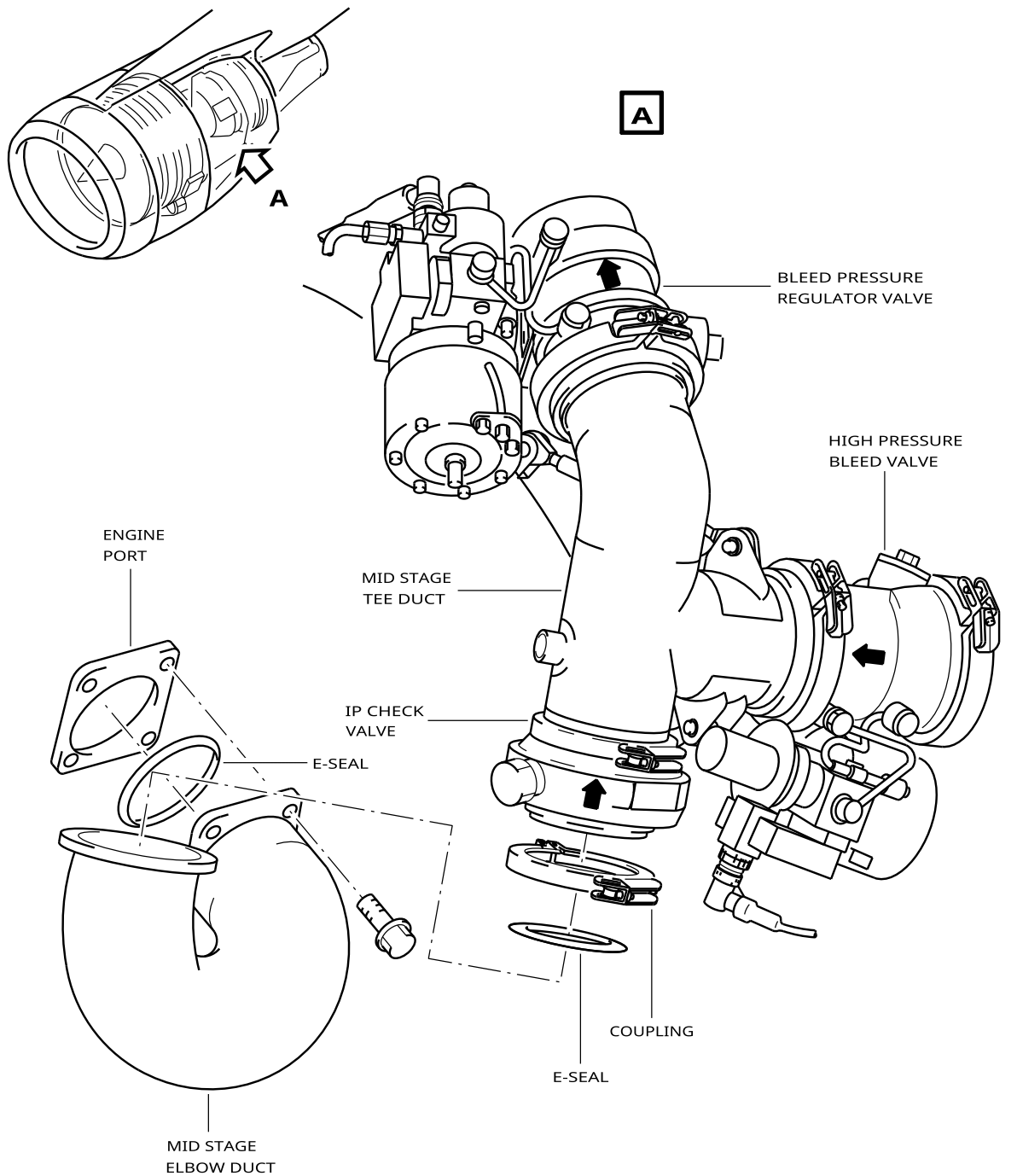


CFM-289-00

N_MM_361149_6_BAM0_01_00

Pneumatic System Flexible Duct Joint Assembly
Figure 1 / Graphic 36-11-49 -991-91400

TF-N
AIRCRAFT MAINTENANCE MANUAL
DUCTS - BLEED AIR - INSPECTION/CHECK

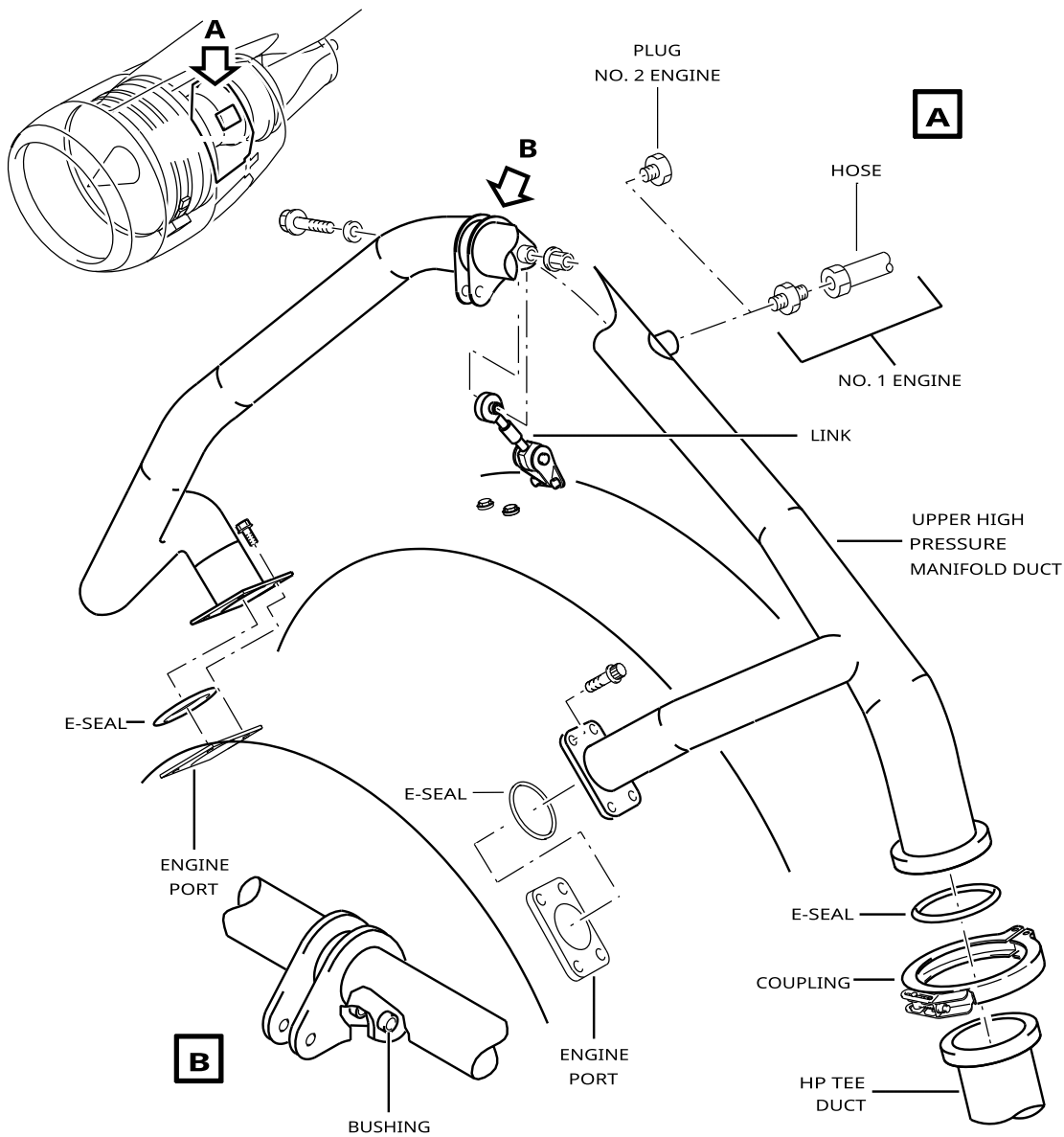


CFM-290-00

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Mid Stage Tee and Elbow Duct Assemblies
Figure 1 / Graphic 36-11-49 -991-91500

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AIRCRAFT MAINTENANCE MANUAL
DUCTS - BLEED AIR - INSPECTION/CHECK

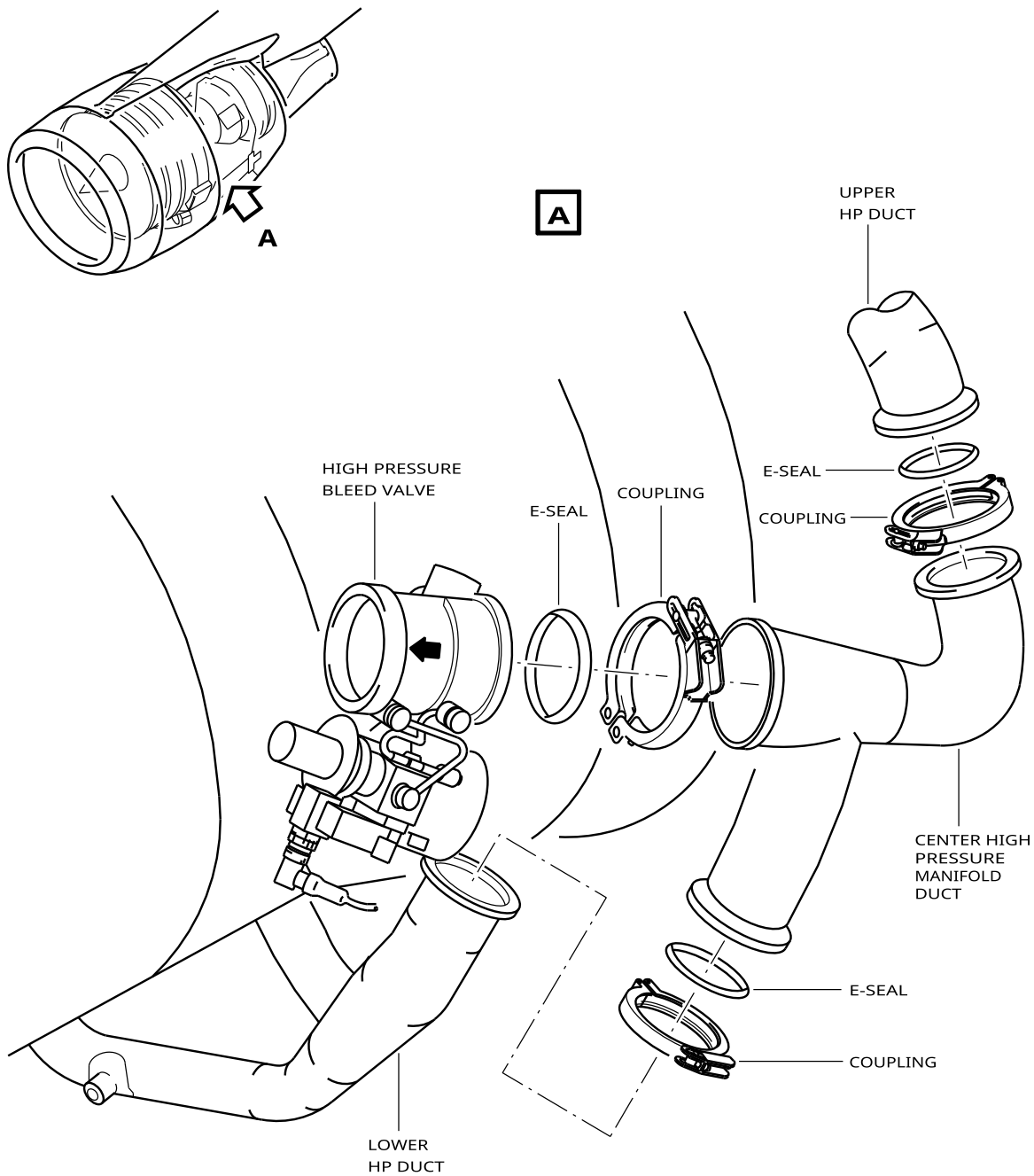


CFM-389-00

N_MM_361149_6_BFM0_01_00

Upper High Pressure Manifold Duct Assembly
Figure 1 / Graphic 36-11-49 -991-91600

TF-N
AIRCRAFT MAINTENANCE MANUAL
DUCTS - BLEED AIR - INSPECTION/CHECK

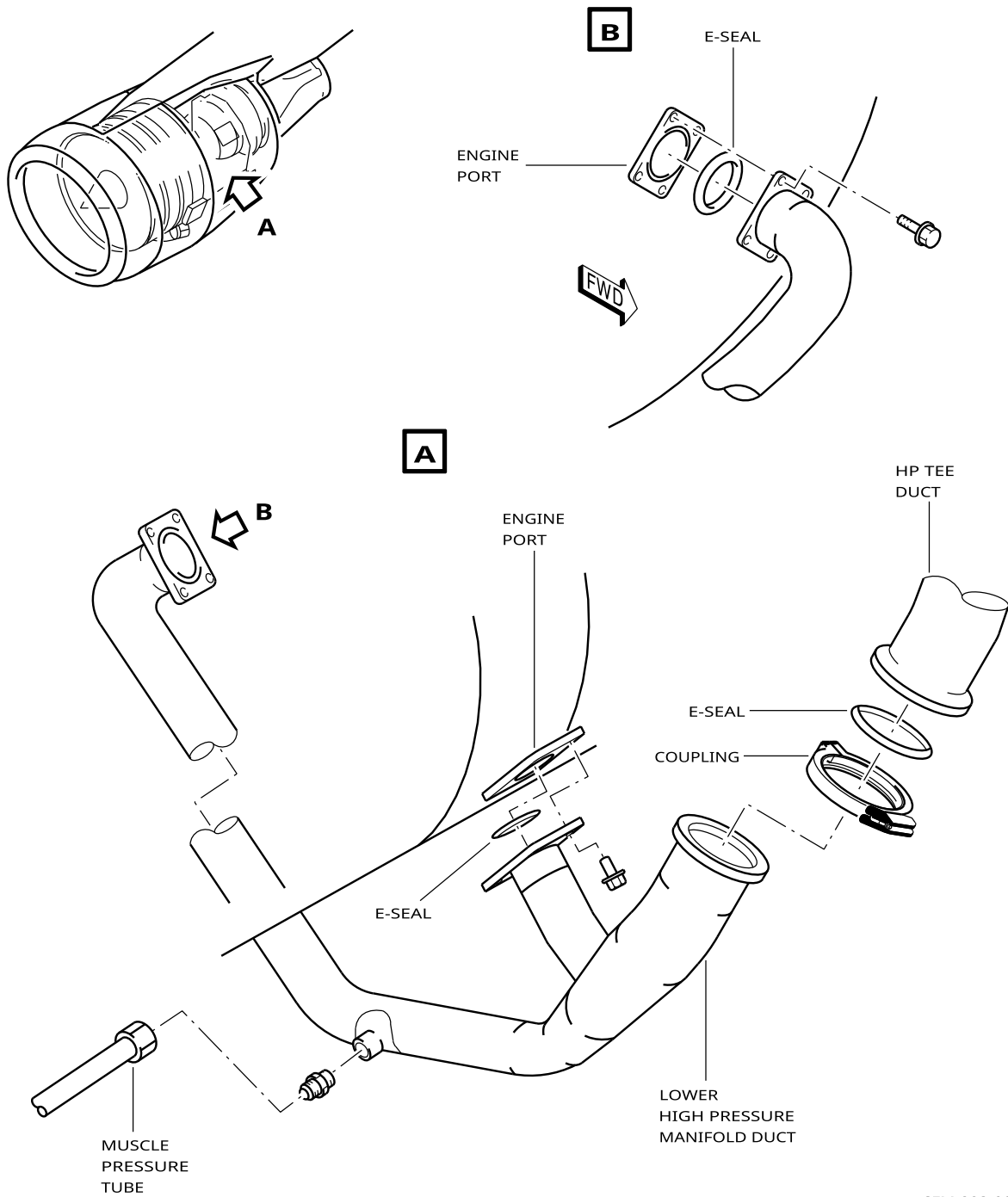


CFM-292-00

N_MM_361149_6_BGM0_01_00

Center High Pressure Manifold Duct Assembly
Figure 1 / Graphic 36-11-49 -991-91700

TF-N
AIRCRAFT MAINTENANCE MANUAL
DUCTS - BLEED AIR - INSPECTION/CHECK

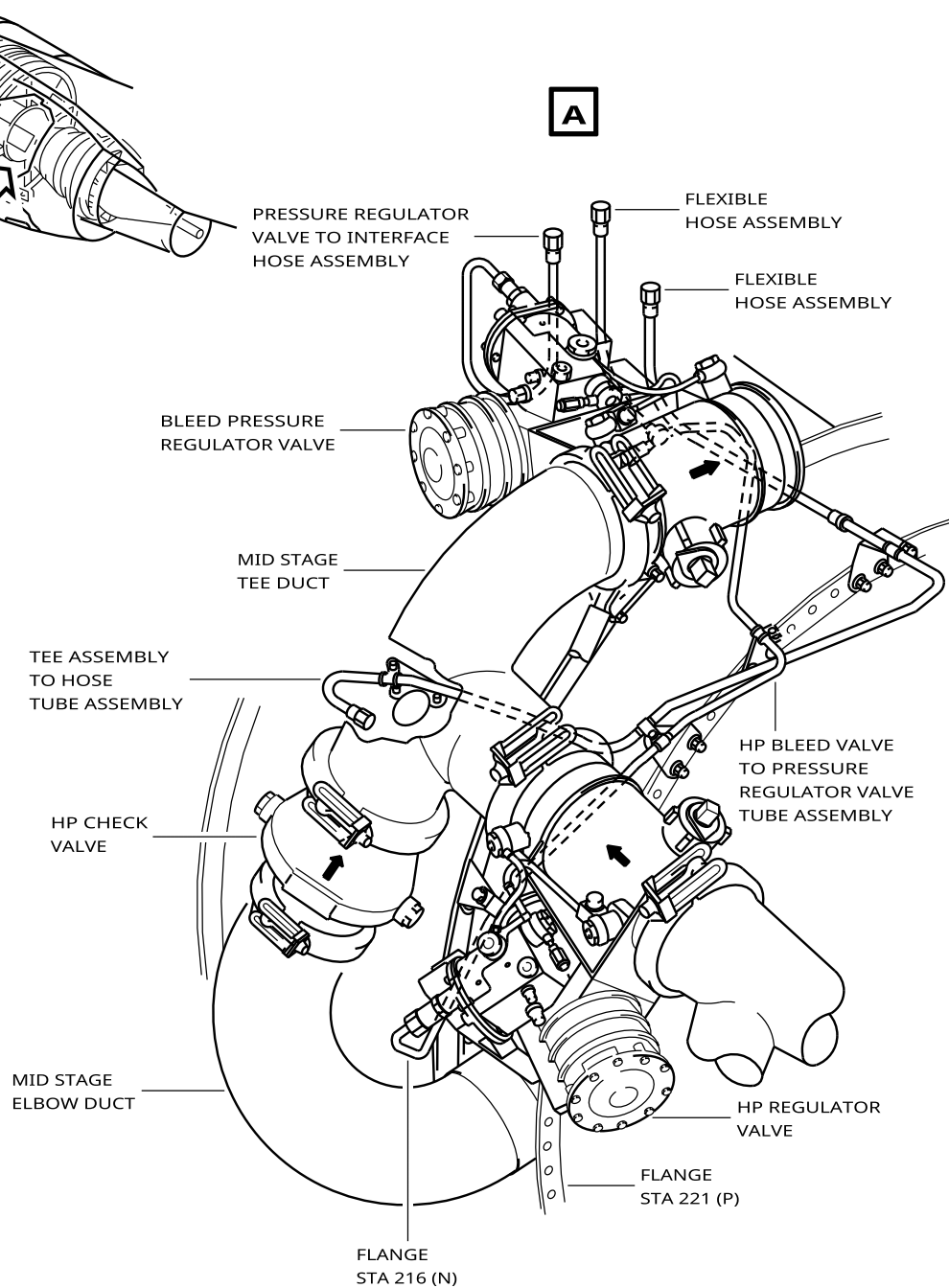


CFM-293-00

N_MM_361149_6_BJM0_01_00

Lower High Pressure Manifold Duct Assembly
Figure 1 / Graphic 36-11-49 -991-91800

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DUCTS - BLEED AIR - INSPECTION/CHECK



CFM-294-00

N_MM_361149_6_BLM0_01_00

Pneumatic System Hose and Tube Assemblies
Figure 1 / Graphic 36-11-49 -991-91900

 MET.31510223		Task Card Title: APU OIL COOLER CLEANING					
Task Card NO: 49-90-00-01-1-SVA		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP			
A/C Type: 321		Block Taskcard		Main Skill: ACFT_TECH,AP		PROD Final Acceptance:	
Project No: 01BASPC72608		Related Task Card:		A/C Area: 11			
Internal Check:		Check Type:		Work Order NO: 31909295		Tally Sequence NO: 79	
ENCLOSURE: ☐ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐							
Type Code: CLN		Special Code: AP		MSG Code: 4		Request: JOB-SETUP,CLEAN-UP,JOB-CLOSEUP	
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRID OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.							
Zones: 315							
DOCUMENT REFERENCES:							
Doc Typ		Reference					
AMM		49-91-44-100-003-A					
ACCESS PANELS:							
315AL, 316AR							
MATERIAL REQUIREMENTS:							
Part Number		Description				Qty.	
08AJA9		AQUEOUS CLEANER, USE PN: TC-100				1.0	
TOOLING REQUIREMENTS:							
Part Number		Description				Qty.	
No tools assigned to this Task Item							
PREPARATION							
Job Description							
A. PERFORM JOB-SETUP FOR REMOVE AIR/OIL COOLER FOR CLEANING (131-9(A)) PER AMM 49-91-44-100-003-A.							
STEP COMPLETION SIGN OFF:							
	Qualification / Skill	Number of employees	Duration		Number./Signature	Date	
				HRS		Day	Month
Mechanic/s	ACFT_TECH/AP			HRS			
Insp.				HRS			
WORK INSTRUCTION							
Job Description							
B. PERFORM REMOVE AIR/OIL COOLER FOR CLEANING (131-9(A)) PER AMM TASK 49-91-44-100-003-A.							
STEP COMPLETION SIGN OFF:							
	Qualification / Skill	Number of employees	Duration		Number./Signature	Date	
				HRS		Day	Month
Mechanic/s	ACFT_TECH/AP			HRS			
Insp.				HRS			



Task Card Title:

APU OIL COOLER CLEANING



Task Card NO:

49-90-00-01-1-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

TASK CLOSE UP

Job Description

C. PERFORM JOB-CLOSEUP AND RETURN THE AIRCRAFT BACK TO NORMAL CONFIGURATION PER AMM TASK 49-91-44-100-003-A.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	ACFT_TECH/AP		HRS				
Insp.			HRS				

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below
Work order No: _____

Note: Make sure to create the finding Work Order on task card level.

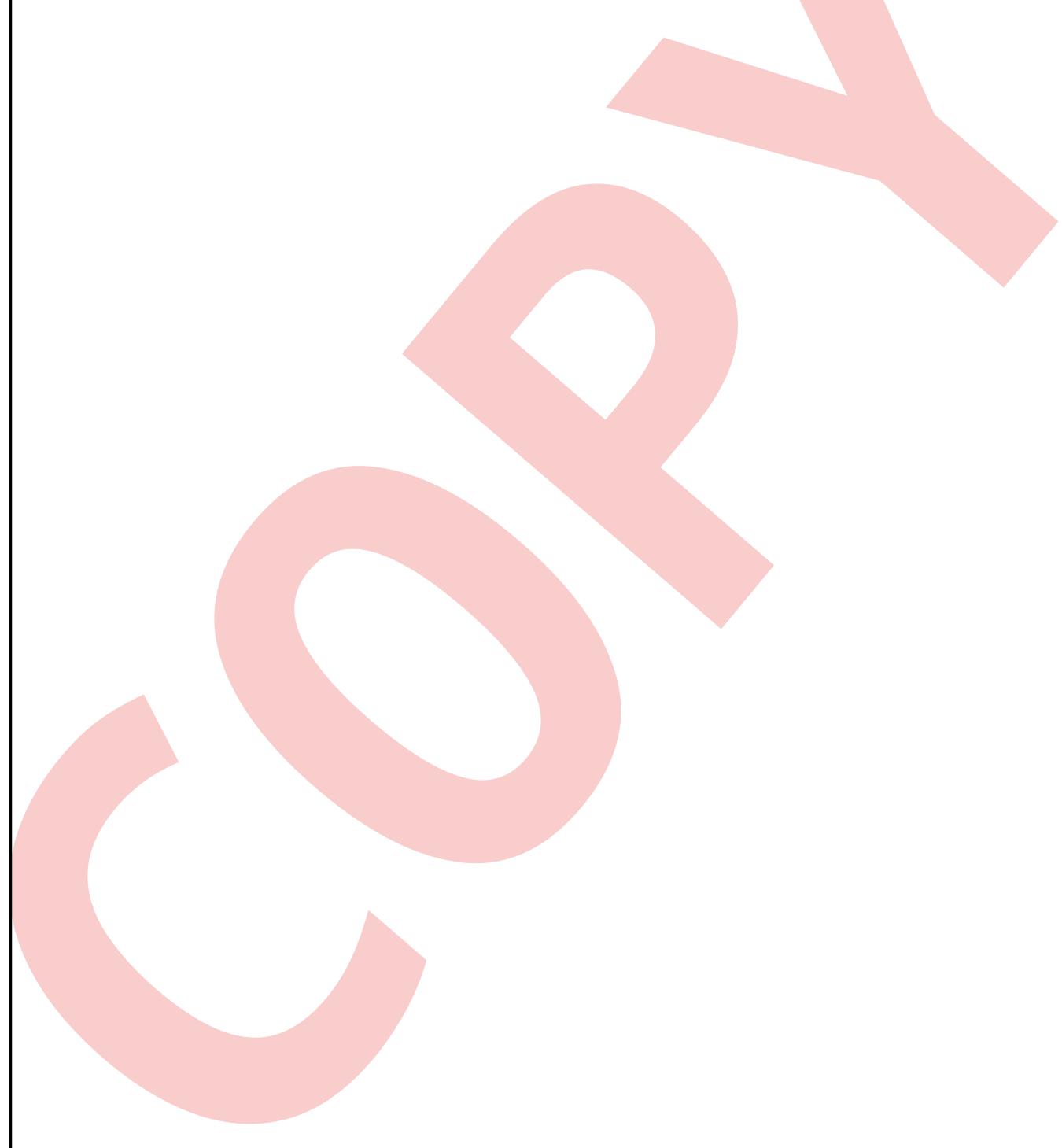
Remark's: _____

(1) If Task card interrupted or performed by different personals updates the system and update the following according (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____
Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 2 [] Task card [] AMM [] Other specify. _____
Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 3 [] Task card [] AMM [] Other specify. _____
Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 4 [] Task card [] AMM [] Other specify. _____
Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 5 [] Task card [] AMM [] Other specify. _____
Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
Remark's: _____

 MET.31510223	Task Card Title: APU OIL COOLER CLEANING			
Task Card NO: 49-90-00-01-1-SVA		Work Package: ASP/H-26 C7-CHK	A/C Registration: HZ-ASP	
Task Completion Close Up And System Update		Station	Date	Signature
				

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COOLER - OIL ((131-9(A))) - CLEANING/PAINTING

TASK 49-91-44-100-003 -A

On A/C 001-999

EMB SB 49-1121 for A/C 003-004; 051-055; 101-115; 201-230

Remove Air/Oil Cooler for Cleaning (131-9(A))

WARNING: DO NOT TOUCH THE APU UNTIL IT IS SUFFICIENTLY COOL TO PREVENT BURNS WHEN YOU DO THE MAINTENANCE TASK(S).

WARNING: WHEN YOU USE COMPRESSED AIR:
- USE THE CORRECT PROTECTION (FACE SHIELD OR GOGGLES)
- DO NOT POINT THE AIRFLOW AT YOURSELF OR OTHER PERSONS.
LOOSE PARTICLES AND/OR HIGH PRESSURE AIR CAN CAUSE INJURY.

WARNING: BE CAREFUL WHEN YOU USE CONSUMABLE MATERIALS. OBEY THE MATERIAL MANUFACTURER'S INSTRUCTIONS AND YOUR LOCAL REGULATIONS.

WARNING: MAKE SURE THAT THE SAFETY DEVICES AND THE WARNING NOTICES ARE IN POSITION BEFORE YOU START A TASK ON OR NEAR:
- THE FLIGHT CONTROLS
- THE FLIGHT CONTROL SURFACES
- THE LANDING GEAR AND RELATED DOORS
- THE CARGO DOORS
- OTHER COMPONENTS THAT MOVE.
MOVEMENT OF COMPONENTS CAN KILL OR CAUSE INJURY TO PERSONS AND/OR DAMAGE TO EQUIPMENT.

CAUTION: PUT PLASTIC CAPS OR PLUGS ON ALL FUEL AND/OR LUBRICATION SYSTEM COMPONENTS AND LINES. IF YOU USE TAPE, CLOTH, OR METAL CAPS ON THEM, YOU CAN DAMAGE THE APU.

1. Reason for the Job

Self explanatory

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	1	ACCESS PLATFORM 4M (13 FT)-ADJUSTABLE
No specific	1	CLEANING EQUIPMENT - STEAM
No specific	1	LIGHT SOURCE - INSPECTION
No specific	1	OVEN - CURING, PORTABLE
No specific	AR	PLUG - BLANKING
No specific	AR	SAFETY CLIP - CIRCUIT BREAKER
No specific	AR	WARNING NOTICE(S)

B. Consumable Materials

REFERENCE	DESIGNATION
08AJA9	Aqueous Cleaner-Turbine Engine Gas Path - Legacy

C. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
210	COCKPIT FWD COMPARTMENT BULKHEAD TO FLIGHT COMPARTMENT BULKHEAD
315	TAIL CONE APU AND ACCESSORY COMPARTMENT

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ZONE/ACCESS	ZONE DESCRIPTION
316	TAIL CONE APU AND ACCESSORY COMPARTMENT
315AL , 316AR	

D. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 24-42-00-861-001-A	Energize the Ground Service Network from the External Power
Ref. AMM 24-42-00-862-001-A	De-energize the Ground Service Network Supplied from the External Power
Ref. AMM 29-00-00-864-001-A	Put the Related Hydraulic System in the Depressurized Configuration before Maintenance Action
Ref. AMM 49-00-00-860-008-A	APU Start By External Power
Ref. AMM 49-00-00-860-008-A-01	APU Start By Batteries
Ref. AMM 49-20-00-600-803-A	APU Preservation / Storage / Return to Service (131-9(A))
Ref. AMM 49-91-44-000-004-A	Removal of the Oil Cooler (131-9(A))
Ref. AMM 49-91-44-400-004-A	Installation of the Oil Cooler (131-9(A))
Ref. AMM 52-41-00-010-001-A	Opening of the APU Access Doors
Ref. AMM 52-41-00-410-001-A	Closing of the APU Access Doors

3. Job Set-up

Subtask 49-91-44-860-065-A

On A/C 001-999

EMB SB 49-1121 for A/C 003-004; 051-055; 101-115; 201-230

A. Aircraft Maintenance Configuration

(1) Energize the ground service network

[Ref. AMM 24-42-00-861-001-A.](#)

(2) On the panel 25VU, make sure that the APU MASTER SW p/bsw is OFF.

(3) Put a WARNING NOTICE(S) in the cockpit to tell persons not to start the APU.

(4) Make sure that the hydraulic systems are depressurized

[Ref. AMM 29-00-00-864-001-A.](#)

Subtask 49-91-44-865-073-A

On A/C 001-999

EMB SB 49-1121 for A/C 003-004; 051-055; 101-115; 201-230

B. Open, safety and tag the circuit breaker(s) that follow(s). Use the SAFETY CLIP - CIRCUIT BREAKER as necessary.

PANEL	DESIGNATION	FIN	LOCATION
121VU	APU/ECB/SPLY	1KD	L41
On A/C 003-250; 301-307			
121VU	APU/APU/CTL	2KD	L42

Subtask 49-91-44-010-060-A

On A/C 001-999

EMB SB 49-1121 for A/C 003-004; 051-055; 101-115; 201-230

C. Get Access

(1) Put the ACCESS PLATFORM 4M (13 FT)-ADJUSTABLE below the APU compartment.

(2) Open the APU access doors 315AL 316AR and [Ref. AMM 52-41-00-010-001-A.](#)

(3) Set the LIGHT SW 11LJ to the ON position.

4. Procedure

Subtask 49-91-44-100-053-A

On A/C 001-999

EMB SB 49-1121 for A/C 003-004; 051-055; 101-115; 201-230

A. Procedure to Clean the Oil Cooler

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COOLER - OIL ((131-9(A))) - CLEANING/PAINTING

NOTE : Two alternative procedures are used to clean the oil cooler, by immersion or with steam.
The two procedures are given as follows.

(1) Clean the oil cooler by immersion as follows:

- (a) Remove the oil cooler [Ref. AMM 49-91-44-000-004-A](#).
- (b) Put the PLUG - BLANKING on the oil inlet and outlet ports.
- (c) Prepare the cleaning agent as given in the manufacturers instructions.
- (d) Fill the cleaning tank with sufficient Aqueous Cleaner-Turbine Engine Gas Path - Legacy (Material No. **08AJA9**) to immerse the oil cooler.
- (e) If the oil cooler has a large quantity of contamination, heat the cleaning agent to the recommended temperature, refer to the manufacturers instructions.

CAUTION: DO NOT LET THE OIL COOLER STAY IN THE CLEANING AGENT FOR MORE THAN FOUR HOURS. AFTER THIS TIME, THE CLEANING AGENT WILL CAUSE DAMAGE TO THE OIL COOLER COVERING.

- (f) Immerse the oil cooler in the cleaning agent for a maximum working time of four hours.
- (g) Shake the oil cooler to remove any remaining air.
- (h) Flush the cleaning agent through the cooler matrix to help make the cleaning time faster.
- (i) Remove the oil cooler from the cleaning tank and drain fully.
- (j) Flush the oil cooler with clean water.
- (k) Do a visual inspection as given in paragraph 4.A.(3).

(2) Clean the oil cooler with steam as follows:

CAUTION: KEEP THE WAND TIP 30 CENTIMETERS (12 INCHES) AWAY FROM THE OIL COOLER MATRIX. THE STEAM PRESSURE FROM THE WAND CAN BEND THE MATRIX VANES AND REMOVE THE PROTECTIVE COATING.

- (a) Remove the oil cooler [Ref. AMM 49-91-44-000-004-A](#).
- (b) Put the PLUG - BLANKING on the oil inlet and outlet ports.
- (c) Prepare steam cleaning solution as follows:
 - 1 Steam cleaning must be performed mainly with water.
 - 2 Add a small quantity of Aqueous Cleaner-Turbine Engine Gas Path - Legacy (Material No. **08AJA9**) to water to help remove large quantities of contamination from the oil cooler.
- (d) Set the pressure of the CLEANING EQUIPMENT - STEAM to between 1.7 bar (24.6564 psi) and 2.0 bar (29.0075 psi).

WARNING: WHEN YOU USE COMPRESSED AIR:
- USE THE CORRECT PROTECTION (FACE SHIELD OR GOGGLES)
- DO NOT POINT THE AIRFLOW AT YOURSELF OR OTHER PERSONS.
LOOSE PARTICLES AND/OR HIGH PRESSURE AIR CAN CAUSE INJURY.

- (e) Point the cleaner spray through the matrix of the oil cooler until all signs of contamination are removed, turn the oil cooler as you clean it.
- (f) Flush the oil cooler with clean water until the cleaning agent is fully removed.
- (g) Do a visual inspection given in paragraph 4.A.(3).

(3) Do a visual inspection of the oil cooler as follows:

WARNING: WHEN YOU USE COMPRESSED AIR:

TF-N
AIRCRAFT MAINTENANCE MANUAL
COOLER - OIL ((131-9(A))) - CLEANING/PAINTING

- USE THE CORRECT PROTECTION (FACE SHIELD OR GOGGLES)
 - DO NOT POINT THE AIRFLOW AT YOURSELF OR OTHER PERSONS.
- LOOSE PARTICLES AND/OR HIGH PRESSURE AIR CAN CAUSE INJURY.

- (a) Blow any remaining water out of the cooler matrix with compressed air, use a maximum pressure of 2.0 bar (29.0075 psi).
- (b) Point the LIGHT SOURCE - INSPECTION through the matrix and do a check to make sure that it is clean and free from contamination as follows:
 - 1 If you find signs of contamination and only if the total immersion time is in the four hours limit, do the cleaning of the oil cooler again:
 - a If the oil cooler cannot be cleaned correctly, return the oil cooler to the workshop.
 - 2 If you do not find signs of contamination, do the drying of the oil cooler.
- (4) Dry the oil cooler as follows:

WARNING: WHEN YOU USE COMPRESSED AIR:

- USE THE CORRECT PROTECTION (FACE SHIELD OR GOGGLES)
 - DO NOT POINT THE AIRFLOW AT YOURSELF OR OTHER PERSONS.
- LOOSE PARTICLES AND/OR HIGH PRESSURE AIR CAN CAUSE INJURY.

WARNING: USE THERMAL GLOVES WHEN YOU MOVE OR TOUCH HOT OR COLD PARTS. HOT OR COLD PARTS CAN CAUSE INJURY.

CAUTION: DO THE DRYING PROCEDURE MAXIMUM FOUR HOURS AFTER THE WET CLEANING PROCEDURE TO PREVENT CORROSION.

- (a) Remove as much water as possible from the oil cooler with compressed air.
- (b) Remove the PLUG - BLANKING from the oil inlet and outlet ports.
- (c) If the oil cooler is to be installed on an APU to go on storage, do the steps that follow:
 - 1 Put the oil cooler into an OVEN - CURING, PORTABLE at the temperature of 121 deg.C (249.80 deg.F) plus or minus 14 deg.C (57.20 deg.F).
 - 2 Keep the oil cooler in the OVEN - CURING, PORTABLE for a minimum of two hours or until the oil cooler is dry.
 - 3 Remove the oil cooler from the OVEN - CURING, PORTABLE and while hot, blow compressed air through it to remove any moisture.
 - 4 Do the drying process again until the cooler is fully dry.
 - 5 Install the oil cooler [Ref. AMM 49-91-44-400-004-A](#).
 - 6 Do the preservation or storage procedure of the APU [Ref. AMM 49-20-00-600-803-A](#).
- (d) If the APU is on-wing, do the steps that follow:
 - 1 Blow compressed air through it to remove any moisture.
 - 2 Install the oil cooler [Ref. AMM 49-91-44-400-004-A](#).
 - 3 Start and operate the APU for 15 minutes to dry the oil cooler
[Ref. AMM 49-00-00-860-008-A](#) [Ref. AMM 49-00-00-860-008-A-01](#).

TF-N
AIRCRAFT MAINTENANCE MANUAL
COOLER - OIL ((131-9(A))) - CLEANING/PAINTING

5. Close-up

Subtask 49-91-44-410-059-A

On A/C 001-999

EMB SB 49-1121 for A/C 003-004; 051-055; 101-115; 201-230

A. Close Access

- (1) Make sure that the work area is clean and clear of tools and other items.
- (2) Remove the ground support and maintenance equipment, the special and standard tools and all other items.
- (3) Set the LIGHT SW 11LJ to the OFF position.
- (4) Close the APU access doors 315AL 316AR Ref. AMM 52-41-00-410-001-A.
- (5) Remove the access platform(s).

Subtask 49-91-44-865-076-A

On A/C 001-999

EMB SB 49-1121 for A/C 003-004; 051-055; 101-115; 201-230

B. Remove the SAFETY CLIP - CIRCUIT BREAKER and the tag(s) and close this (these) circuit breaker(s):

PANEL	DESIGNATION	FIN	LOCATION
121VU	APU/ECB/SPLY	1KD	L41
On A/C 003-250; 301-307			
121VU	APU/APU/CTL	2KD	L42

Subtask 49-91-44-860-066-A

On A/C 001-999

EMB SB 49-1121 for A/C 003-004; 051-055; 101-115; 201-230

C. Aircraft Maintenance Configuration

- (1) De-energize the ground service network
Ref. AMM 24-42-00-862-001-A.
- (2) Remove the WARNING NOTICE(S).

 MET.29290988		Task Card Title: FWD CARGO COMPARTMENT DOOR PROXIMITY (28WV & 30WV) SWITCH AND TARGET CLEANING					
Task Card NO: 52-31-00-01-0-SVA		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP			
A/C Type: 321		Block Taskcard		Main Skill: ACFT_TECH,AP		PROD Final Acceptance:	
Project No: 01BASPC72608		Related Task Card:		A/C Area:			
Internal Check:		Check Type:		Work Order NO: 31909295		Tally Sequence NO: 81	
ENCLOSURE: ☐ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐							
Type Code: CLN		Special Code: AP, AV		MSG Code: 4		Request: JOB-SETUP,	
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRIED OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.							
Zones: 821							
DOCUMENT REFERENCES:							
Doc Typ		Reference					
DO		20-55-10.pdf					
ACCESS PANELS:							
MATERIAL REQUIREMENTS:							
Part Number		Description				Qty.	
TT-I-735		ISOPROPYL ALCOHOL				1.0	
MIL-PRF-7870		OIL LOW FREEZING*HAZMAT*IF SD U SV729				1.0	
LOTOXANE		NON-AQUEOUS CLEANER				1.0	
MIL-PRF-680		NON AQUEOUS CLEANER- PETROLEUM BASED				1.0	
14SBA1		TEXTILE - LINT FREE COTTON AMS3819				1.0	
TOOLING REQUIREMENTS:							
Part Number		Description				Qty.	
No tools assigned to this Task Item							
TASK INSTRUCTIONS							
Job Description							
A. CLEAN FWD CARGO DOOR PROXIMITY SENSORS (28WV & 30WV) AND TARGETS AS PER ESPM 20-55-10-02.							
(1) CLEAN THE PROXIMITY-SENSOR ELECTRICAL COMPONENTS.							
(2) CLEAN THE PROXIMITY-SENSOR TARGETS AND THE BRACKETS							
STEP COMPLETION SIGN OFF:							
	Qualification / Skill	Number of employees	Duration		Number./Signature	Date	
				HRS		Day	Month
Mechanic/s	ACFT_TECH/AP			HRS			
Insp.				HRS			
B.							
Job Description							
B. MEASURE AND RECORD THE CLEARANCE J1 BETWEEN THE TARGET AND THE							



Task Card Title:

FWD CARGO COMPARTMENT DOOR PROXIMITY (28WV & 3**0WV) SWITCH AND TARGET CLEANING**

Task Card NO:

52-31-00-01-0-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

RELATED PROXIMITY SWITCH (30WV) AS PER AMM TASK 52-71-00-820-003 –A:

RECORD THE CLEARANCE J1 VALUE BEFORE ADJUSTMENT: -----

NOTE: MAKE SURE THAT CLEARANCE J1 IS BETWEEN 1.0 MM (0.0394 IN.) AND 1.8 MM (0.0709 IN.)

IT IS RECOMMENDED THAT YOU KEEP A MINIMUM DISTANCE OF 1.2 MM (0.0472 IN.) BETWEEN THE SENSOR AND THE TARGET TO PREVENT THE FALSE OPEN-DOOR INDICATION.

* IF THE CLEARANCE J1 IS NOT IN THE TOLERANCES, ADJUST AS PER
AMM TASK 52-71-00-820-003 –A.

RECORD THE CLEARANCE J1 VALUE AFTER ADJUSTMENT: -----

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	ACFT_TECH/AP		HRS				
Insp.			HRS				

C.**Job Description**C. MEASURE AND RECORD THE CLEARANCE J2 BETWEEN THE TARGET AND THE
RELATED PROXIMITY SWITCH 28WV AS PER AMM TASK 52-71-00-820-004 –A:

RECORD THE CLEARANCE J2 VALUE BEFORE ADJUSTMENT: -----

NOTE: MAKE SURE THAT CLEARANCE J2 IS BETWEEN 1.60 MM (0.0630 IN.) AND 2.00 MM (0.0787 IN.)

* IF THE CLEARANCE J2 IS NOT IN THE TOLERANCES, ADJUST AS PER AMM TASK 52-71-00-820-004 –A.

RECORD THE CLEARANCE J2 VALUE AFTER ADJUSTMENT: -----

* IF NEEDED DO THE LUBRICATION OF THE FORWARD CARGO-COMPARTMENT DOOR-HANDLE REF.
AMM 12-22-52-640-009.**STEP COMPLETION SIGN OFF:**

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	ACFT_TECH/AP		HRS				
Insp.			HRS				

D&E**Job Description**D. DO AN OPERATIONAL TEST OF THE DOOR WARNING SYSTEM RELATED TO
THE CARGO DOORS AS PER AMM TASK 52-71-00-710-001-A.

E. RAISE TQ TO ENG-CON-STR WITH RECORDED MEASUREMENT.

STEP COMPLETION SIGN OFF:



Task Card Title:

FWD CARGO COMPARTMENT DOOR PROXIMITY (28WV & 3**0WV) SWITCH AND TARGET CLEANING**

Task Card NO:

52-31-00-01-0-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	ACFT_TECH/AP		HRS				
Insp.			HRS				

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below

Work order No: _____

Note: Make sure to creat the finding Work Order on task card level.

Remark's: _____

(1) If Task card interrupted or performed by different personals updates the system and update the following accord (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 2 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 3 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 4 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 5 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

**Task Completion Close Up And
System Update****Station****Date****Signature**

 MET.29290996	Task Card Title: AFT CARGO COMPARTMENT DOOR PROXIMITY SWITCH A							
ND TARGET CLEANING								
Task Card NO: 52-32-00-01-0-SVA		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP				
A/C Type: 321	Block Taskcard		Main Skill: AV,ACFT_TECH,AP		PROD Final Acceptance:			
Project No: 01BASPC72608	Related Task Card:		A/C Area:					
Internal Check:	Check Type:		Work Order NO: 31909295		Tally Sequence NO: 82			
ENCLOSURE: ☐ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐								
Type Code: CLN	Special Code: AP	MSG Code: 4	Request:	Phase: CLEAN-UP,TC/P,TC/P,J OB-CLOSEUP				
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRID OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.								
Zones:								
DOCUMENT REFERENCES:								
Doc Typ		Reference						
DO		20-55-10.pdf						
ACCESS PANELS:								
MATERIAL REQUIREMENTS:								
Part Number		Description			Qty.			
No parts assigned to this Task Item								
TOOLING REQUIREMENTS:								
Part Number		Description			Qty.			
No tools assigned to this Task Item								
A. CLEAN								
Job Description								
A. CLEAN FWD CARGO DOOR PROXIMITY SENSORS (32WV & 34WV) AND TARGETS AS PER ESPM 20-55-10-02.								
(1) CLEAN THE PROXIMITY-SENSOR ELECTRICAL COMPONENTS.								
(2) CLEAN THE PROXIMITY-SENSOR TARGETS AND THE BRACKETS								
.								
STEP COMPLETION SIGN OFF:								
	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
				HRS		Day	Month	Year
Mechanic/s	ACFT_TECH/AP			HRS				
Insp.				HRS				
MEASURE AND RECORD								



Task Card Title:

AFT CARGO COMPARTMENT DOOR PROXIMITY SWITCH A**ND TARGET CLEANING**

Task Card NO:

52-32-00-01-0-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP**Job Description**

B. MEASURE AND RECORD THE CLEARANCE J1 BETWEEN THE TARGET AND THE RELATED PROXIMITY SWITCH (32WV) AS PER AMM TASK 52-71-00-820-003 –A:

RECORD THE CLEARANCE J1 VALUE BEFORE ADJUSTMENT: -----

NOTE: MAKE SURE THAT CLEARANCE J1 IS BETWEEN 1.0 MM (0.0394 IN.) AND 1.8 MM (0.0709 IN.)

IT IS RECOMMENDED THAT YOU KEEP A MINIMUM DISTANCE OF 1.2 MM (0.0472 IN.) BETWEEN THE SENSOR AND THE TARGET TO PREVENT THE FALSE OPEN-DOOR INDICATION.

◆ IF THE CLEARANCE J1 IS NOT IN THE TOLERANCES, ADJUST AS PER AMM TASK 52-71-00-820-003 –A.

RECORD THE CLEARANCE J1 VALUE AFTER ADJUSTMENT: -----

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	AV/ACFT_TECH			HRS			
Insp.				HRS			

MEASURE AND RECORD THE CLEARANCE J2**Job Description**

C. MEASURE AND RECORD THE CLEARANCE J2 BETWEEN THE TARGET AND THE RELATED PROXIMITY SWITCH (34WV) AS PER AMM TASK 52-71-00-820-004 –A:

RECORD THE CLEARANCE J2 VALUE BEFORE ADJUSTMENT: -----

NOTE: MAKE SURE THAT CLEARANCE J2 IS BETWEEN 1.60 MM (0.0630 IN.) AND 2.00 MM (0.0787 IN.)

◆ IF THE CLEARANCE J2 IS NOT IN THE TOLERANCES, ADJUST AS PER AMM TASK 52-71-00-820-004 –A.

RECORD THE CLEARANCE J2 VALUE AFTER ADJUSTMENT: -----

◆ IF NEEDED DO THE LUBRICATION OF THE AFT CARGO-COMPARTMENT DOOR-HANDLE REF. AMM 12-22-52-640-009.



Task Card Title:

AFT CARGO COMPARTMENT DOOR PROXIMITY SWITCH A**ND TARGET CLEANING**

Task Card NO:

52-32-00-01-0-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP**STEP COMPLETION SIGN OFF:**

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	AV/ACFT_TECH		HRS				
Insp.			HRS				

OPERATIONAL TEST AND RAISE TO**Job Description**

D. DO AN OPERATIONAL TEST OF THE DOOR WARNING SYSTEM RELATED TO THE CARGO DOORS AS PER AMM TASK 52-71-00-710-001-A.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	ACFT_TECH/AP		HRS				
Insp.			HRS				

RAISE TQ ENGG**Job Description**

E. RAISE TQ TO ENG-CON-STR WITH RECORDED MEASUREMENT.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	ACFT_TECH/AP		HRS				
Insp.			HRS				



Task Card Title:

AFT CARGO COMPARTMENT DOOR PROXIMITY SWITCH A**ND TARGET CLEANING**

Task Card NO:

52-32-00-01-0-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below

Work order No: _____

Note: Make sure to creat the finding Work Order on task card level.

Remark's: _____

(1) If Task card interrupted or performed by different personals updates the system and update the following accord (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 2 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 3 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 4 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

- Interruption No. 5 [] Task card [] AMM [] Other specify. _____

Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____

Remark's: _____

**Task Completion Close Up And
System Update****Station****Date****Signature**

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 MET.28388283		Task Card Title: MLG ACT FITTINGS KEEL BEAM INSP				
Task Card NO: 533154-05-1-A		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP		
A/C Type: 321		Block Taskcard		Main Skill: INSP,II,NDT_INSP,AF,ACFT_TEC H,NDT		
Project No: 01BASPC72608		Related Task Card:		A/C Area:		
Internal Check:		Check Type:		Work Order NO: 31909295		
				Tally Sequence NO: 88		
ENCLOSURE: ☒ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☒ ALI: ☐						
Type Code: DET		Special Code: NDT		MSG Code: A		
				Request: JOB-SETUP, JOB-SETUP, X-RAY/NDT, INSPECTI ON,		
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRID OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.						
Zones: 147,148						
DOCUMENT REFERENCES:						
Doc Typ		Reference				
DO		SB_53-1195-13_ENV.pdf				
DO		ISB 53-1195				
EASA		AD2012-0118				
EASA		AD 2024-0097R2				
EO.		EO 531680				
EO.		EO531681				
FAA		AD2009-20-05				
ACCESS PANELS:						
MATERIAL REQUIREMENTS:						
Part Number		Description			Qty.	
08BAA9		NON AQUEOUS CLEANER - GENERAL			1.0	
14SBA1		TEXTILE - LINT FREE COTTON AMS3819			1.0	
D5337261321600		FITTING			1.0	
D5337261321700		FITTING			1.0	
D5337076420400		BUSHING			1.0	
D5337076420200		BUSHING			1.0	
LOTOXANE		NON-AQUEOUS CLEANER			1.0	
14SBA1		TEXTILE - LINT FREE COTTON AMS3819			1.0	
D5337261321600		FITTING			1.0	
D5337261321700		FITTING			1.0	
D5337076420400		BUSHING			1.0	
D5337076420200		BUSHING			1.0	
R5307111620000		BUSHING			2.0	
R5307111620200		BUSHING, SHOP REQ (FAD)			2.0	
98D53308002000		KIT LUG REPAIR REAMER			1.0	
R5307111620400		BUSH			1.0	
R5307111620600		BUSH			2.0	
NAS1149C1016R		WASHER U/O B747			1.0	

 MET.28388283	Task Card Title: MLG ACT FITTINGS KEEL BEAM INSP						
Task Card NO: 533154-05-1-A	Work Package: ASP/H-26 C7-CHK	A/C Registration: HZ-ASP					
MS24665-300	* PIN-COTTER * ALSO USED ON L1011/		1.0				
TOOLING REQUIREMENTS:							
Part Number	Description	Qty.					
98D53308001000	KIT LUG REPAIR REAMER	1.0					
INSTRUCTIONS							
Job Description							
A. PERFORM TASK SET A320-A-53-XX-1195-01000-93BA-A (A320-A-53-XX-1195-01ZZZ-93BZ-A)							
STEP COMPLETION SIGN OFF:							
	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	AF/ACFT_TECH		HRS				
Insp.			HRS				
TASK PREPARATION							
Job Description							
B. PERFORM TASK A320-A-53-XX-1195-01001-540A-A (A320-A-53-XX-1195-01ZZZ-540Z-A)							
STEP COMPLETION SIGN OFF:							
	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	AF/ACFT_TECH		HRS				
Insp.			HRS				
TASK PREPARATION							
Job Description							
C. DISCONNECT THE MLG DOOR ACTUATOR FROM ACTUATOR FITTING ON LH AND RH SIDES. PERFORM TASK A320-A-53-XX-1195-02001-540A-A (A320-A-53-XX-1195-02ZZZ-540Z-A).							
STEP COMPLETION SIGN OFF:							
	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	AF/ACFT_TECH		HRS				
Insp.			HRS				
TASK INSTRUCTIONS							
Job Description							
D. PERFORM A SPECIAL DETAILED INSPECTION OF THE KEEL BEAM MLG DOOR ACTUATOR FITTINGS LH AND RH. PERFORM TASK A320-A-53-XX-1195-01001-300A-A (A320-A-53-XX-1195-01ZZZ-300Z-A),							
- FILL NDT REPORT (ATTACHED)							
- PLEASE RAISE TQ TO (ENG-CON-STR) WITH NDT REPORT (ATTACHED AFTER FILL) .							



MET.28388283

Task Card Title:

MLG ACT FITTINGS KEEL BEAM INSP



Task Card NO:

533154-05-1-A

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
						Day	Month	Year
Mechanic/s	AF/ACFT_TECH			HRS				
Insp.	NDT_INSP/NDT			HRS				

Job Description

E. PERFORM INSPECTION OF THE KEEL BEAM MLG DOOR ACTUATOR FITTINGS LH AND RH.
PERFORM TASK A320-A-53-XX-1195-01001-300A-A (A320-A-53-XX-1195-01ZZZ-300Z-A),

- FILL INSPECTION REPORT (ATTACHED)

- PLEASE RAISE TQ TO (ENG-CON-STR) WITH INSPECTION REPORT (ATTACHED AFTER FILL) .

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
						Day	Month	Year
Mechanic/s				HRS				
Insp.	INSP/II			HRS				

TASK INSTRUCTIONS

Job Description

E. DEPENDING ON FINDING, REPLACE THE MLG DOOR ACTUATOR FITTING ON LH AND RH SIDES.
PERFORM TASK A320-A-53-XX-1195-01001-920A-A (A320-A-53-XX-1195-01ZZZ-920Z-A)

IF ACTUATOR FITTING REPLACED , PLEASE RAISE TQ TO (ENG-CON-STR).

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
						Day	Month	Year
Mechanic/s	AF/ACFT_TECH			HRS				
Insp.	INSP/II			HRS				

TASK INSTRUCTIONS

Job Description

F. DEPENDING ON THE FINDING, REPAIR THE PLAIN BUSH MIGRATION ON LH AND RH SIDES.
PERFORM TASK A320-A-53-XX-1195-01001-690A-A (A320-A-53-XX-1195-01ZZZ-690Z-A).

IF PERFORMED THE REPAIR , PLEASE RAISE TQ TO (ENG-CON-STR).

STEP COMPLETION SIGN OFF:



Task Card Title:

MLG ACT FITTINGS KEEL BEAM INSP

Task Card NO:

533154-05-1-A

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
						Day	Month	Year
Mechanic/s	AF/ACFT_TECH			HRS				
Insp.	II/INSP			HRS				

TASK INSTRUCTIONS**Job Description**

G. DEPENDING ON FINDING, REPAIR THE FLANGED BUSH AXIAL LOOSE ON LH AND RH SIDES.
PERFORM TASK A320-A-53-XX-1195-02001-690A-A (A320-A-53-XX-1195-02ZZZ-690Z-A).

IF PERFORMED THE REPAIR , PLEASE RAISE TQ TO (ENG-CON-STR).

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
						Day	Month	Year
Mechanic/s	AF/ACFT_TECH			HRS				
Insp.	INSP/II			HRS				

TASK INSTRUCTIONS**Job Description**

H. DEPENDING ON FINDING , REPAIR THE WEAR MARKS UNDERNEATH BOLT HEAD ON LH AND RH SIDES
PERFORM TASK A320-A-53-XX-1195-03001-690A-A (A320-A-53-XX-1195-03ZZZ-690Z-A)

IF PERFORMED THE REPAIR , PLEASE RAISE TQ TO (ENG-CON-STR).

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
						Day	Month	Year
Mechanic/s	AF/ACFT_TECH			HRS				
Insp.	INSP/II			HRS				

TASK INSTRUCTIONS**Job Description**

I. DEPENDING ON THE FINDING, REPAIR THE OTHER DAMAGES ON LH AND RH SIDES
PERFORM TASK A320-A-53-XX-1195-04001-690A-A (A320-A-53-XX-1195-04ZZZ-690Z-A)

IF PERFORMED THE REPAIR , PLEASE RAISE TQ TO (ENG-CON-STR).

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
						Day	Month	Year
Mechanic/s	AF/ACFT_TECH			HRS				
Insp.	INSP/II			HRS				

PREPARATION FOR TEST**Job Description**



Task Card Title:

MLG ACT FITTINGS KEEL BEAM INSP

Task Card NO:

533154-05-1-A

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

J. PERFORM TASK A320-A-53-XX-1195-01001-331A-A (A320-A-53-XX-1195-01ZZZ-331Z-A).

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	AF/ACFT_TECH			HRS			
Insp.	INSP/II			HRS			

TEST**Job Description**

K. PERFORM TASK A320-A-53-XX-1195-01001-390A-A (A320-A-53-XX-1195-01ZZZ-390Z-A)

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	AF/ACFT_TECH			HRS			
Insp.	INSP/II			HRS			

CLOSE-UP**Job Description**

L. PERFORM TASK A320-A-53-XX-1195-02001-740A-A (A320-A-53-XX-1195-02ZZZ-740Z-A)

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	AF/ACFT_TECH			HRS			
Insp.	INSP/II			HRS			

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below

Work order No: _____

_____, _____, _____, _____, _____, _____, _____, _____

Note: Make sure to creat the finding Work Order on task card level.

Remark's: _____

_____, _____, _____, _____, _____, _____, _____, _____



Task Card Title:

MLG ACT FITTINGS KEEL BEAM INSP

Task Card NO:

533154-05-1-A

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

(1) If Task card interrupted or performed by different personals updates the system and update the following according (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 2 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 3 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 4 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 5 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____

**Task Completion Close Up And
System Update**

Station**Date****Signature**

 MET.29291089		Task Card Title: COCKPIT WINDOWS INSPECTION						
Task Card NO: 561000-01-1-SVA		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP				
A/C Type: 321		Block Taskcard		Main Skill: II,INSP,STR,AF,ACFT_TECH		PROD Final Acceptance:		
Project No: 01BASPC72608		Related Task Card:		A/C Area: 06				
Internal Check:		Check Type:		Work Order NO: 31909295		Tally Sequence NO: 90		
ENCLOSURE: ☒ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐								
Type Code: DVI		Special Code: INSP		MSG Code: 8		Request: Phase: PRE HGR,INSPECTION, INSPECTION,JOB-CLOS EUP		
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRID OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.								
Zones: 210								
DOCUMENT REFERENCES:								
Doc Typ		Reference						
AMM		56-11-11-200-001-A						
AMM		56-11-12-200-001-A						
AMM		56-12-11-200-001-A						
CAP		CAP S56-C10001 escalation to match new A-CHK interval.pdf						
CAP		S56-C10001						
CAP		S56-C10001.pdf						
DO		SIL 56-017						
ACCESS PANELS:								
MATERIAL REQUIREMENTS:								
Part Number		Description				Qty.		
No parts assigned to this Task Item								
TOOLING REQUIREMENTS:								
Part Number		Description				Qty.		
No tools assigned to this Task Item								
TASK INSTRUCTIONS								
Job Description								
A. PREPARE FOR THE COCKPIT WINDOWS INSPECTIONS AS PER THE FOLLOWING:								
(1) THE WINDSHIELDS (REF. AMM TASK 56-11-11-200-001). (2) THE FIXED WINDOWS (REF. AMM TASK 56-11-12-200-001). (3) THE SLIDING WINDOWS (REF. AMM TASK 56-12-11-200-001).								
STEP COMPLETION SIGN OFF:								
		Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
						Day	Month	Year
Mechanic/s		AF/ACFT_TECH			HRS			



Task Card Title:

COCKPIT WINDOWS INSPECTION

Task Card NO:

561000-01-1-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

Insp.

HRS

TASK INSTRUCTIONS**Job Description**

B. PERFORM A DETAILED VISUAL INSPECTION OF THE FOLLOWING COCKPIT WINDOWS (FROM INSIDE).

- (1) THE WINDSHIELDS (REF. AMM TASK 56-11-11-200-001).
- (2) THE FIXED WINDOWS (REF. AMM TASK 56-11-12-200-001).
- (3) THE SLIDING WINDOWS (REF. AMM TASK 56-12-11-200-001).

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s			HRS				
Insp.	II/INSP		HRS				

TASK INSTRUCTIONS**Job Description**

C. IF DELAMINATION , INTERLAYER MICROFLAKES ,BUS BAR BURN SPOT FOUND WITHIN ALLOWABLE DAMAGE LIMITATION IN THE APPLICABLE AMM ON FLIGHT COMPARTMENT NO.1, NO.2 AND NO.3 WINDOWS ,THEN REFER TO THE FOLLOWING INSTRUCTION FOR DAMAGE MONITORING AND CONTROLLING:

- a. DELAMINATION AND/OR INTERLAYER MICROFLAKES: IF DELAMINATION AND/OR INTERLAYER MICROFLAKES IS FOUND WITHIN APPLICABLE AMM LIMITATION, THEN PERFORM THE FOLLOWINGS:
 1. RECORD SIZE AND LOCATION OF DELAMINATION AND/OR INTERLAYER MICROFLAKES IN LOG BOOK WO ,
 2. CHECK THE WEATHER SEAL (MOISTURE SEAL) FROM OUTSIDE, REPAIR AS REQUIRED PER APPLICABLE AMM AT NEXT R.O.N.
 3. PERFORM REPETITIVE INSPECTION AS FOLLOWING:
 - CHECK THE AFFECTED AREA FOR ANY SIGNS OF DAMAGE PROPAGATION EVERY 7-DAYS.
 - UPDATE THE DELAMINATION DIMENSION EVERY REPEAT INSPECTION IN THE RELATED WORK ORDER.
 - IF DELAMINATED AREA AND/OR INTERLAYER MICROFLAKES HAS REACHED MAX ALLOWABLE LIMITATION IN APPLICABLE AMM, R/R THE AFFECTED WINDOW WITHIN 7 DAYS. CHECK THE AFFECTED AREA FOR ANY SIGNS OF DAMAGE PROPAGATION EVERY DAY. IN CASE OF ANY DELAMINATION PROPAGATION EXCEED MAX ALLOWABLE LIMITATION IN APPLICABLE AMM DURING THE 7 DAYS PERIOD, THE AFFECTED WINDOW MUST BE REPLACED BEFORE NEXT FLIGHT.
 4. REPLACE THE AFFECTED WINDOW DURING NEXT C-CHECK.

b- BUS BAR BURN SPOT:

IF BURN SPOT AT THE BUS BAR AREA IS FOUND BROWN AND WITHIN APPLICABLE AMM LIMITATION, THEN PERFORM THE FOLLOWING:

1. RECORD SIZE AND LOCATION OF DEFECTED AREA IN LOGBOOK WO.
2. PERFORM REPETITIVE INSPECTION AS FOLLOWING:
 - CHECK THE AFFECTED AREA FOR ANY SIGNS OF DAMAGE PROPAGATION EVERY 7-DAYS.
 - UPDATE THE BUS BAR BURN SPOT DIMENSION EVERY REPEAT INSPECTION IN THE RELATED WORK ORDER.
 - IF BUS BAR BROWN BURN SPOT PROPAGATION EXCEED APPLICABLE AMM LIMITATION OR BLACK SPOTS APPEARED ON BUS BAR OR AT THE EDGE OF THE BUS BAR, REPLACE WINDOW BEFORE FURTHER FLIGHT.
3. REPLACE THE THE AFFECTED WINDOW DURING NEXT C-CHECK.



Task Card Title:

COCKPIT WINDOWS INSPECTION



Task Card NO:

561000-01-1-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	STR/ACFT_TECH		HRS				
Insp.	INSP/II		HRS				

TASK INSTRUCTIONS

Job Description

D. RETURN AIRCRAFT TO NORMAL AS PER THE FOLLOWING:

- (1) THE WINDSHIELDS (REF. AMM TASK 56-11-11-200-001).
- (2) THE FIXED WINDOWS (REF. AMM TASK 56-11-12-200-001).
- (3) THE SLIDING WINDOWS (REF. AMM TASK 56-12-11-200-001)

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	AF/ACFT_TECH		HRS				
Insp.			HRS				

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below

Work order No: _____

Note: Make sure to creat the finding Work Order on task card level.

Remark's: _____



Task Card Title:

COCKPIT WINDOWS INSPECTION

Task Card NO:

561000-01-1-SVA

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

(1) If Task card interrupted or performed by different personals updates the system and update the following according (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 2 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 3 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 4 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 5 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____

**Task Completion Close Up And
System Update**

Station**Date****Signature**

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WINDSHIELD - INSPECTION/CHECK

TASK 56-11-11-200-001 -A

On A/C 001-999

Inspection of the Windshields

FIN: 3DG1, 3DG2

1. Reason for the Job

Self explanatory

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	AR	ACCESS PLATFORM 5M (16 FT)- ADJUSTABLE
No specific	AR	SAFETY CLIP - CIRCUIT BREAKER
SJ201	1	SURFACE TESTER
SJ210	1	SURFACE TESTER

B. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
210	CKPT,FWD COMPT BHD TO FLT COMPT BULKHEAD
211	COCKPIT
212	COCKPIT

C. Referenced Information

REFERENCE	DESIGNATION
On A/C 001-999	
Ref. AMM 56-10-00-110-001-A	Cleaning of the L/R Windshields
Ref. AMM 56-10-00-200-004-A	Measurement of the Scratch Depth With Mitutoyo Surface Tester SJ201
Ref. AMM 56-10-00-200-004-A-01	Measurement of the Scratch Depth with Mitutoyo Surface Tester SJ210
Ref. AMM 56-11-11-000-001-A	Removal of the Windshield
Ref. AMM 56-11-11-000-001-A-01	Removal of the Windshield
Ref. AMM 56-11-11-300-002-A	Repair of the Soft Liner (SGS (formerly SPS) Windshield)
Ref. AMM 56-11-11-400-001-A	Installation of the Windshield
Ref. AMM 56-11-11-400-001-A-01	Installation of the Windshield
Ref. Fig. (1 / 56-11-11-991-00500)	
Ref. Fig. (1 / 56-11-11-991-00600)	
Ref. Fig. (1 / 56-11-11-991-00700)	
Ref. Fig. (1 / 56-11-11-991-00800)	
Ref. Fig. (1 / 56-11-11-991-00900)	
Ref. Fig. (1 / 56-11-11-991-01000)	
Ref. Fig. (1 / 56-11-11-991-01700)	
Ref. Fig. (1 / 56-11-11-991-01900)	
Ref. Fig. (1 / 56-11-11-991-02400)	
Ref. Fig. (1 / 56-11-11-991-03400)	

3. Job Set-up

Subtask 56-11-11-860-052-A

On A/C 001-999

A. Aircraft Maintenance Configuration

WARNING: BEFORE YOU DO MAINTENANCE WORK WITHIN A RADIUS OF 29 METERS (96 FEET) AROUND AND ABOVE THE BROADBAND ANTENNA, MAKE SURE THAT THE ANTENNA IS DE-ENERGIZED. THE ANTENNA USES HIGH POWER WHEN IT TRANSMITS AND CAN CAUSE INJURY TO PERSONS.

(1) If the broadband antenna is installed, do its deactivation for maintenance

On A/C 251-310; 315-350

Ref. AMM 44-33-00-040-804-A.

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WINDSHIELD - INSPECTION/CHECK

Subtask 56-11-11-941-055-A

On A/C 001-999

B. Get Access

- (1) Put the ACCESS PLATFORM 5M (16 FT)- ADJUSTABLE in position in front of the windshield at zone 210 (only if you must do a detailed inspection from the outer side of the aircraft).
- (2) Clean the windshields Ref. AMM 56-10-00-110-001-A.

Subtask 56-11-11-865-059-A

On A/C 001-999

WARNING: MAKE SURE THAT THE CIRCUIT BREAKERS RELATED TO THE WEATHER RADAR ARE OPENED, SAFETIED AND TAGGED.

- C. Open, safety and tag the circuit breaker(s) that follow(s). Use the SAFETY CLIP - CIRCUIT BREAKER as necessary.

PANEL	DESIGNATION	FIN	LOCATION
121VU	COM NAV/RADAR/2	5SQ2	K14
121VU	COM NAV/RADAR/1	5SQ1	K13
122VU	ANTI ICE/WINDOWS/R	4DG2	W14
122VU	ANTI ICE/WHC/2	5DG2	W13
122VU	ANTI ICE/WINDOWS/L	4DG1	X14
122VU	ANTI ICE/WHC/1	5DG1	X13
123VU	ANTI ICE/L/WSHLD	1DG1	AF10
123VU	ANTI ICE/R/WSHLD	1DG2	AF03

4. Procedure

Subtask 56-11-11-220-051-A

On A/C 001-999

A. Inspection of the Windshields

NOTE : In the inspection tables, the word impaired refers to the visibility. The visibility is impaired when the defects have an unwanted effect on it.
If the visibility is impaired, it is necessary to replace the windshields.

- (1) Examine the windshields for:
 - (a) Cracks:
Ref. Fig. (1 / 56-11-11-991-00500)
 - (b) Scratches:
Ref. Fig. (1 / 56-11-11-991-00600)

NOTE : There are two procedures to measure the depth of the scratches. The fingernail procedure measures the approximate depth of the scratches. The MITUTOYO surface-tester procedure measures the accurate depth of the scratches. If a MITUTOYO surface tester is available, Airbus recommends that you use the MITUTOYO surface-tester procedure because it gives accurate results.

- 1 If you see scratches, do a check of their depth with one of the two procedures that follow:

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- The MITUTOYO **SURFACE TESTER** (SJ201) (SJ201) or the MITUTOYO **SURFACE TESTER** (SJ210) (SJ210) procedure [Ref. AMM 56-10-00-200-004-A](#) [Ref. AMM 56-10-00-200-004-A-01](#), or
 - The fingernail procedure.
- (c) Chips:
Ref. Fig. (1 / 56-11-11-991-00700)
- (d) Delamination:
Ref. Fig. (1 / 56-11-11-991-00800)

NOTE : There is a tape on the windshield. This tape is necessary to prevent cracks where there is no heating (in the corners of the windows).

- 1 When you measure the delamination, make sure that you do it from the outside of the aircraft.
- (e) Discoloration:
Ref. Fig. (1 / 56-11-11-991-00900)
- (f) Interlayer microflakes:
Ref. Fig. (1 / 56-11-11-991-01000)
- (g) Bubbles:
Ref. Fig. (1 / 56-11-11-991-01000)
- 1 When you measure the bubbles, make sure that you do it from the outside of the aircraft.
- (h) Burn signs:
Ref. Fig. (1 / 56-11-11-991-01700)
- (i) Transparency:
Ref. Fig. (1 / 56-11-11-991-01700)
- (j) Rain-repellent fluid residue:
Ref. Fig. (1 / 56-11-11-991-01700)
- (k) Damage on the soft liner:
Ref. Fig. (1 / 56-11-11-991-01900)

NOTE : This damage is only applicable to the windshield panels that have a soft liner (self-healing liner), SGS supplier, P/Ns STA320-1-5-1 and STA320-2-5-1.

- 1 If you find scratches or other damage, repair the soft liner
[Ref. AMM 56-11-11-300-002-A](#).
- (l) Heating film cracks:
Ref. Fig. (1 / 56-11-11-991-02400)
- (m) Distortion and optical deviations:
Ref. Fig. (1 / 56-11-11-991-03400)

NOTE : Distortion and optical deviations have no effect on the structural strength of the glass. The windshield can have distortion and optical deviations in zone 1 and/or zone 2 because of the manufacturing process of the glass.

- 1 Make sure that the visibility is not impaired because of distortion and optical deviations in zone 1. If the visibility is impaired, replace the windshield
[Ref. AMM 56-11-11-000-001-A](#) [Ref. AMM 56-11-11-000-001-A-01](#) and
[Ref. AMM 56-11-11-400-001-A](#) [Ref. AMM 56-11-11-400-001-A-01](#).

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NOTE : An impaired visibility is permitted in zone 2.

NOTE : Zone 2 is the 50 mm (1.969 in.) wide shaded border from the daylight opening.

5. Close-up

Subtask 56-11-11-865-060-A

On A/C 001-999

- A. Remove the SAFETY CLIP - CIRCUIT BREAKER and the tag(s) and close this (these) circuit breaker(s):

PANEL	DESIGNATION	FIN	LOCATION
121VU	COM NAV/RADAR/2	5SQ2	K14
121VU	COM NAV/RADAR/1	5SQ1	K13
122VU	ANTI ICE/WINDOWS/R	4DG2	W14
122VU	ANTI ICE/WHC/2	5DG2	W13
122VU	ANTI ICE/WINDOWS/L	4DG1	X14
122VU	ANTI ICE/WHC/1	5DG1	X13
123VU	ANTI ICE/L/WSHLD	1DG1	AF10
123VU	ANTI ICE/R/WSHLD	1DG2	AF03

Subtask 56-11-11-942-058-A

On A/C 001-999

- B. Close Access

- (1) Remove the ground support and maintenance equipment, the special and standard tools and all other items.
- (2) Remove the access platform(s).

Subtask 56-11-11-860-053-A

On A/C 001-999

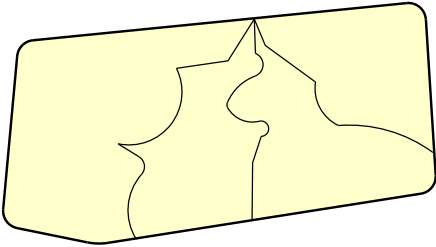
- C. Aircraft Maintenance Configuration

- (1) If the broadband antenna is installed, do its reactivation after maintenance

On A/C 251-310; 315-350

Ref. AMM 44-33-00-440-804-A.

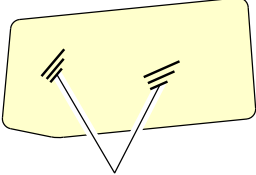
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DESCRIPTION OF DEFECT	CORRECTIVE ACTIONS
CRACKS:  LINE TYPE DEFECTS THROUGH THE DEPTH OF THE PLY OR THE PANE OF THE WINDSHIELD PANEL	OUTER PLY: TEN FLIGHT LEGS PERMITTED BUT NOT FOR SPS WINDSHIELDS P/N'S STA320-1-3 AND STA320-2-3 (ONE FLIGHT LEG PERMITTED) ON THE CONDITIONS THAT FOLLOW: - THE AIRCRAFT DOES NOT FLY IN KNOWN ICING CONDITIONS. - THE VISIBILITY IS NOT IMPAIRED. - THE OTHER WINDSHIELD PANEL IS SERVICEABLE. OPEN, SAFETY AND TAG THESE CIRCUIT BREAKERS: - FOR THE LEFT WINDSHIELD: 1DG1 - FOR THE RIGHT WINDSHIELD: 1DG2. YOU MUST NOT APPLY ELECTRICAL HEAT TO CRACKED WINDSHIELD IF THERE IS ARCING OR UNSATISFACTORY POWER TEST.
	MIDDLE/INNER PLY: MIDDLE OR INNER PLY OR THE TWO PLIES CRACKED: NOT PERMITTED

N_MM_561111_6_ACM0_01_01

Windshields - Permitted Damage
Figure 1 / Graphic 56-11-11 -991-00500
On A/C 001-999

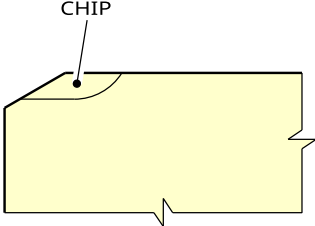
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DESCRIPTION OF DEFECT	SUPPLIER	DEFECT LOCATION	PERMITTED/NOT PERMITTED DEFECT
SCRATCHES: LINE TYPE DEFECTS ON THE EXTERNAL SURFACE OF THE WINDSHIELD PANEL DETAIL OF THE SCRATCHES: 	GKN SUPPLIER, SGS SUPPLIER AND PPG SUPPLIER BUT NOT THE SGS SUPPLIER (WINDSHIELD WITH A SOFT LINER) - P/N'S STA320-1-5-1 AND STA320-2-5-1	OUTER PLY	PERMITTED IF THE VISIBILITY IS NOT DECREASED
		INNER PLY	PERMITTED: - IF YOU CANNOT FEEL THE SCRATCHES WHEN YOU TOUCH THE WINDSHIELD WITH YOUR FINGERNAIL OR - IF THE DEPTH OF THE SCRATCHES MEASURED WITH MITUTOYO SURFACE-TESTER SJ201/SJ210 IS LESS THAN 0.05 mm (0.0019 in) (REF. TASK 561000-200-004-A) OR (REF. TASK 561000-200-004-A01). NOT PERMITTED: - IF YOU CAN FEEL THE SCRATCHES WHEN YOU TOUCH THE WINDSHIELD WITH YOUR FINGERNAIL OR - IF THE DEPTH OF THE SCRATCHES MEASURED WITH MITUTOYO SURFACE-TESTER SJ201/SJ210 IS MORE THAN 0.05 mm (0.0019 in) (REF. TASK 561000-200-004-A) OR (REF. TASK 561000-200-004-A01). NOTE: THERE ARE TWO PROCEDURES TO MEASURE THE DEPTH OF THE SCRATCHES. THE FINGERNAIL PROCEDURE HELPS YOU TO MEASURE AN APPROXIMATE DEPTH OF THE SCRATCHES. THE MITUTOYO SURFACE-TESTER PROCEDURE HELPS YOU TO MEASURE THE ACCURATE DEPTH OF THE SCRATCHES. IF THE MITUTOYO SURFACE TESTER IS AVAILABLE, THE MITUTOYO SURFACE-TESTER PROCEDURE IS RECOMMENDED BECAUSE IT GIVES ACCURATE MEASUREMENT RESULTS.
	SGS SUPPLIER (WINDSHIELD WITH A SOFT LINER) - P/N'S STA320-1-5-1 AND STA320-2-5-1	OUTER PLY	PERMITTED IF THE VISIBILITY IS NOT DECREASED
		INNER PLY	PERMITTED IF THE SCRATCHES DO NOT TOUCH THE CHEMICALLY STRENGTHENED GLASS AND IF THE VISIBILITY IS NOT DECREASED

N_MM_561111_6_AEM0_01_04

Windshields - Permitted Damage
Figure 1 / Graphic 56-11-11 -991-00600

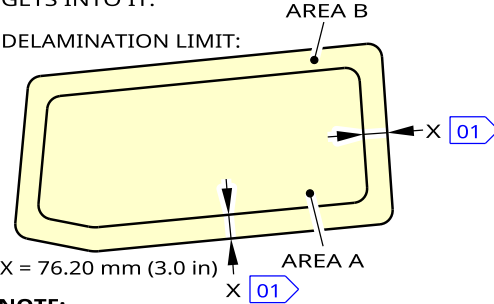
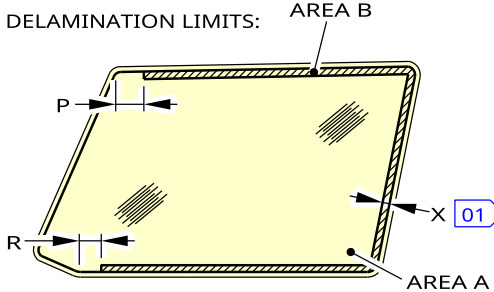
TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDSHIELD - INSPECTION/CHECK

DESCRIPTION OF DEFECT	CORRECTIVE ACTION
CHIPS: FLAKES OF GLASS BROKEN FROM THE SURFACE AND THE EDGES OF THE WINDSHIELD PANEL DETAIL OF A CHIP 	OUTER PLY: PERMITTED IF THE VISIBILITY IS NOT IMPAIRED
	INNER PLY: NOT PERMITTED
	MIDDLE PLY: NOT PERMITTED

N_MM_561111_6_AGM0_01_01

Windshields - Permitted Damage
Figure 1 / Graphic 56-11-11 -991-00700

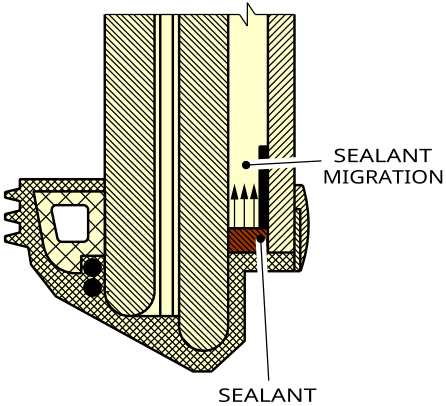
TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDSHIELD - INSPECTION/CHECK

DESCRIPTION OF DEFECT	SUPPLIER	CORRECTIVE ACTION
DELAMINATION: THE SEPARATION OF THE GLASS FROM THE ADJACENT VINYL INTERLAYER MOST FREQUENTLY OCCURS AROUND THE EDGES, THE SENSORS AND IN THE AREAS WHERE THERE IS NO HEAT. AN AREA WITH DELAMINATION CAN BE TRANSPARENT, BUT IT WILL BECOME LESS TRANSPARENT IF THE MOISTURE GETS INTO IT. DELAMINATION LIMIT:  X = 76.20 mm (3.0 in) NOTE: 01 DIMENSION X IS MEASURED FROM THE INNER EDGE OF THE SEAL OF THE TRANSPARENT WINDOW AREA.	GKN SUPPLIER, PPG SUPPLIER AND SGS SUPPLIER - P/N'S STA320-1-8-2, STA320-2-8-2 AND STA320-1-9-3, STA320-2-9-3	MIDDLE, OUTER AND INNER PLY: AREA A: NOT PERMITTED AREA B: PERMITTED IF IT DOES NOT DECREASE THE VISIBILITY.
DELAMINATION: THE SEPARATION OF THE GLASS FROM THE ADJACENT VINYL INTERLAYER MOST FREQUENTLY OCCURS AROUND THE EDGES, THE SENSORS AND IN THE AREAS WHERE THERE IS NO HEAT. AN AREA WITH DELAMINATION CAN BE TRANSPARENT, BUT IT WILL BECOME LESS TRANSPARENT IF THE MOISTURE GETS INTO IT. DELAMINATION LIMITS:  X = 20 mm (0.787 in) P = 75 mm (2.953 in) R = 60 mm (2.362 in) NOTE: 01 DIMENSION X IS MEASURED FROM THE INNER EDGE OF THE SEAL OF THE TRANSPARENT WINDOW AREA.	SGS SUPPLIER - P/N'S STA320-1-3-1, STA320-2-3-1, STA320-1-4-1, STA320-2-4-1, STA320-1-5-1, STA320-2-5-1, STA320-1-6-1, STA320-2-6-1 AND STA320-1-7-1, STA320-2-7-1	MIDDLE, OUTER AND INNER PLY: AREA A: NOT PERMITTED AREA B: PERMITTED IF IT DOES NOT DECREASE THE VISIBILITY.

N_MM_561111_6_AJMO_01_04

Windshields - Permitted Damage
Figure 1 / Graphic 56-11-11 -991-00800

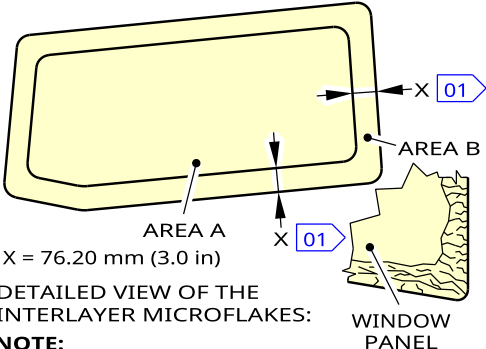
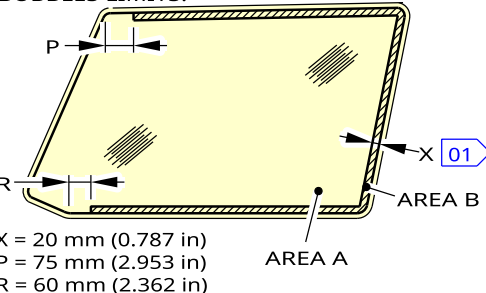
TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDSHIELD - INSPECTION/CHECK

DESCRIPTION OF DEFECTS	CORRECTIVE ACTION
DISCOLORATION DISCOLORATION OCCURS WHEN: - THERE IS A REFLECTION FROM THE BUSBAR AND THE CONDUCTIVE FILM (CARBON DISCOLORATION). THIS REFLECTION CHANGES COLOR WHEN YOU LOOK AT IT FROM AN OBLIQUE ANGLE. - THERE IS MIGRATION OF THE BROWN SEALANT TO THE EDGES OF THE OUTER INTERLAYER WHICH CAN EXTEND TO A WIDTH BETWEEN 10 mm (0.39 in) AND 20 mm (0.78 in) AT THE EDGES OF THE WINDOW PANEL. SEALANT MIGRATION 	PERMITTED IF THE VISIBILITY IS NOT IMPAIRED

N_MM_561111_6_ALM0_01_01

Windshields - Permitted Damage
Figure 1 / Graphic 56-11-11 -991-00900

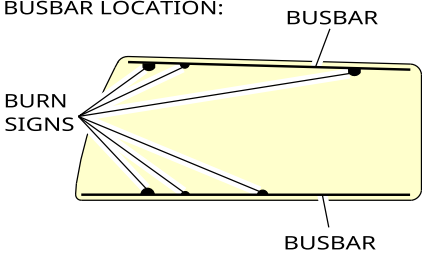
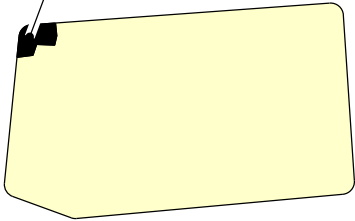
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AIRCRAFT MAINTENANCE MANUAL
WINDSHIELD - INSPECTION/CHECK

DESCRIPTION OF DEFECT	SUPPLIER	CORRECTIVE ACTION
INTERLAYER MICROFLAKES: MICROFLAKES ARE CHEMICAL REACTIONS AT THE EDGES OF THE INTERLAYER CAUSED BECAUSE OF THE MOISTURE AND WEAR OF THE WINDOW. INTERLAYER MICROFLAKES LIMIT:  X = 76.20 mm (3.0 in) DETAILED VIEW OF THE INTERLAYER MICROFLAKES: NOTE: 01 DIMENSION X IS MEASURED FROM THE INNER EDGE OF THE SEAL OF THE TRANSPARENT WINDOW AREA.		AREA A: NOT PERMITTED AREA B: PERMITTED IF THE VISIBILITY DOES NOT DECREASE. NOTE: - MICROFLAKES OCCURS ONLY IN THE INTERLAYERS. - EQUIVALENT DAMAGE IN ONE OF THE GLASS PLIES WILL CAUSE THE PLY TO BREAK IMMEDIATELY.
BUBBLES: GAS RELEASES SMALL BUBBLES WHEN THE VINYL BECOMES TOO HOT. THE BUBBLES DO NOT DECREASE THE STRUCTURAL STRENGTH. BUBBLES LIMITS:  X = 20 mm (0.787 in) P = 75 mm (2.953 in) R = 60 mm (2.362 in) NOTE: 01 DIMENSION X IS MEASURED FROM THE INNER EDGE OF THE SEAL OF THE TRANSPARENT WINDOW AREA.	SGS SUPPLIER - P/N'S STA320-1-3-1, STA320-2-3-1, STA320-1-4-1, STA320-2-4-1, STA320-1-5-1, STA320-2-5-1, STA320-1-6-1, STA320-2-6-1 AND STA320-1-7-1, STA320-2-7-1	MIDDLE, OUTER AND INNER PLY: AREA A: NOT PERMITTED AREA B: PERMITTED IF THE VISIBILITY DOES NOT DECREASE.
BUBBLES: GAS RELEASES SMALL BUBBLES WHEN THE VINYL BECOMES TOO HOT. THE BUBBLES DO NOT DECREASE THE STRUCTURAL STRENGTH.	GKN SUPPLIER, PPG SUPPLIER AND SGS SUPPLIER - P/N'S STA320-1-8-2, STA320-2-8-2 AND STA320-1-9-3, STA320-2-9-3	MIDDLE, OUTER AND INNER PLY: PERMITTED IF THE VISIBILITY DOES NOT DECREASE.

N_MM_561111_6_ANM0_01_03

Windshields - Permitted Damage
Figure 1 / Graphic 56-11-11 -991-01000

TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDSHIELD - INSPECTION/CHECK

DESCRIPTION OF DEFECTS	CORRECTIVE ACTIONS
BURN SIGN: - BURN SIGN IN THE BUSBAR AREA: THE WINDSHIELD HAS BURN SIGNS ALONG A BUSBAR IF THE ARCING NEXT TO A BUSBAR DAMAGE CAUSED DAMAGE TO THE WINDSHIELD. BUSBAR LOCATION: 	PERMITTED IF THE TOTAL LENGTH OF THE BURN SIGNS ON THE BUSBAR IS NOT MORE THAN 200 mm (7.874 in). IF THE TOTAL LENGTH OF THE BURN SIGNS ON ONE BUSBAR IS MORE THAN 200 mm (7.874 in), STOP THE WINDOW HEATING SYSTEM UNTIL YOU REPLACE THE WINDOW AS FOLLOWS: - OPEN, SAFETY AND TAG THESE CIRCUIT BREAKERS: - FOR THE LEFT SIDE: 1DG1 - FOR THE RIGHT SIDE: 1DG2. DO NOT OPERATE THE AIRCRAFT IN KNOWN ICING CONDITIONS.
BURNS: FREQUENT HIGH TEMPERATURES THAT GIVE THE INTERLAYER A BROWN COLOR ALSO CAUSE BURNS. BURNED DAMAGE 	PERMITTED IF THE VISIBILITY IS NOT IMPAIRED
TRANSPARENCY: HALOS ON THE SURFACE OF THE WINDSHIELD PANEL CAN MAKE IT LESS TRANSPARENT.	PERMITTED IF THE VISIBILITY IS NOT IMPAIRED
RAIN-REPELLENT FLUID RESIDUE: DISCOLORATION OF THE EXTERNAL SURFACE OF THE WINDOWS.	CLEAN THE EXTERNAL SURFACE OF THE WINDSHIELD PANEL.

N_MM_561111_6_AQM0_01_01

Windshields - Permitted Damage
Figure 1 / Graphic 56-11-11 -991-01700

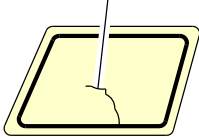
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AIRCRAFT MAINTENANCE MANUAL
WINDSHIELD - INSPECTION/CHECK

DESCRIPTION OF DEFECT	CORRECTIVE ACTION
DAMAGE ON THE SOFT LINER: - THE SOFT LINER IS INSTALLED ON THE INNER SURFACE OF THE WINDSHIELD PANEL. - THE DAMAGE TYPES ON THE SOFT LINER ARE: CHIPS, BUBBLES AND SCRATCHES. NOTE: THIS DAMAGE IS ONLY APPLICABLE TO WINDSHIELD PANELS WITH A SOFT LINER (SPS SUPPLIER)	PERMITTED IF THE VISIBILITY IS NOT IMPAIRED CAUTION: IF THE DAMAGE EXTEND TO THE CHEMICALLY-STRENGTHENED GLASS INNER-PLY, REFER TO THE RELATED PERMITTED DAMAGE.

N_MM_561111_6_ASM0_01_01

Windshields - Permitted Damage
Figure 1 / Graphic 56-11-11 -991-01900

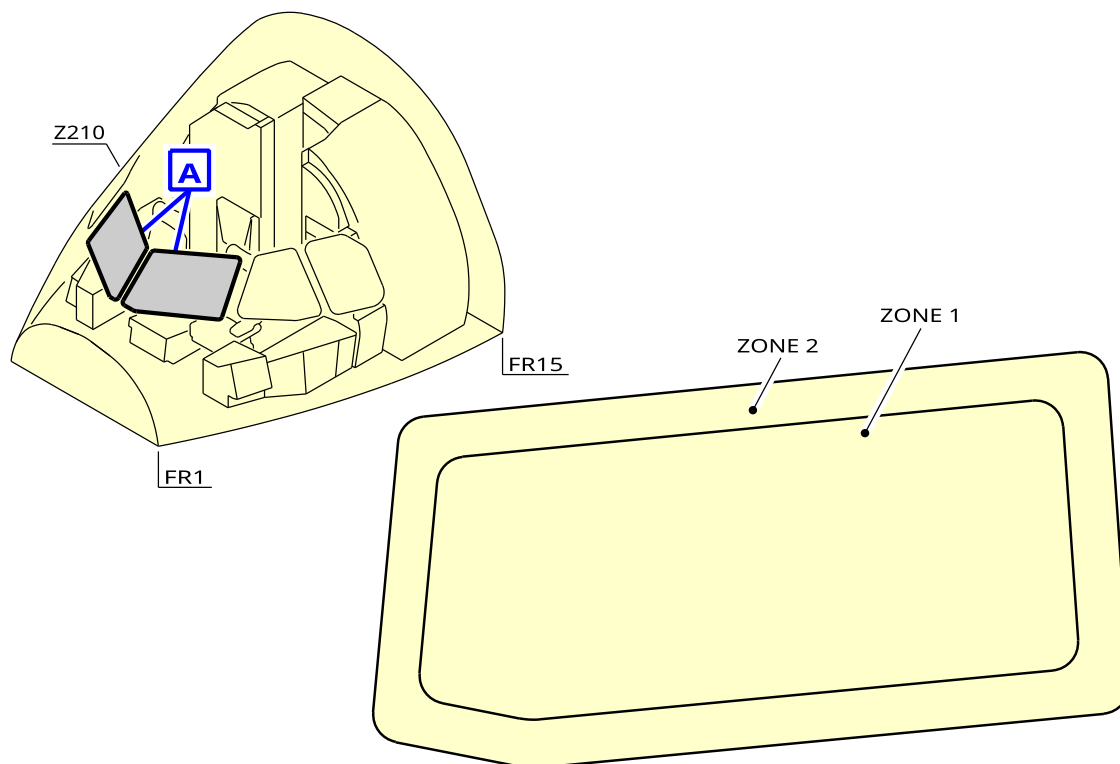
TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDSHIELD - INSPECTION/CHECK

DESCRIPTION OF DEFECT	CORRECTIVE ACTION
HEATING FILM CRACKS: THE HEATING FILM IS BETWEEN THE OUTER PLY AND THE MIDDLE PLY OF THE WINDOW. HEATING FILM CRACKS CAN CAUSE THE ELECTRICAL FAILURE OF THE HEATING SYSTEM BUT HAVE NO EFFECT ON ITS STRUCTURAL INTEGRITY. HEATING FILM CRACKS 	THE AIRCRAFT DISPATCH IS AUTHORIZED IF: - THE AIRCRAFT DOES NOT FLY IN KNOWN ICING CONDITIONS. - THE VISIBILITY IS NOT DECREASED. - THE OTHER WINDSHIELD PANEL IS SERVICEABLE. NOTE: - IT IS POSSIBLE TO THINK THAT MIDDLE PLY CRACKS ARE HEATING FILM CRACKS.

N_MM_561111_6_AFM0_01_02

Windshields - Permitted Damage
Figure 1 / Graphic 56-11-11 -991-02400

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AIRCRAFT MAINTENANCE MANUAL
WINDSHIELD - INSPECTION/CHECK



ZONE 2 = 50 mm (1.969 in) WIDE SHADED BORDER
FROM THE DAYLIGHT OPENING



LH SHOWN
RH SYMMETRICAL

DESCRIPTION OF THE DEFECT	CORRECTIVE ACTION
DISTORTION AND OPTICAL DEVIATIONS THE WINDSHIELD CAN HAVE DISTORTION AND OPTICAL DEVIATIONS IN ZONE 1 AND/OR ZONE 2 BECAUSE OF THE MANUFACTURING PROCESS OF THE GLASS.	THIS DAMAGE CAN OCCUR IN ZONE 1 AND/OR ZONE 2 IF THE VISIBILITY IS IMPAIRED BECAUSE OF DISTORTION AND OPTICAL DEVIATIONS IN ZONE 1, REPLACE THE WINDOW PANEL. NOTE: AN IMPAIRED VISIBILITY IS PERMITTED IN ZONE 2.

N_MM_561111_6_DSA2_01_01

Windshield - Permitted Damage
Figure 1 / Graphic 56-11-11 -991-03400

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AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK

TASK 56-11-12-200-001 -A

On A/C 001-999

Inspection of the Fixed Windows

FIN: 7DG1, 7DG2

1. Reason for the Job

NOTE : This procedure is applicable to:

- The acrylic fixed windows (GKN supplier, P/N NH...)
- The glass fixed windows (PPG supplier, P/N NP...)
- The glass fixed windows (SGS supplier, P/N SPS...).

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	2	ACCESS PLATFORM 5M (16 FT)- ADJUSTABLE
No specific	1	FLASHLIGHT
No specific	AR	LIGHT SOURCE - WHITE
No specific	AR	MAGNIFYING GLASS POWER 10X
No specific	AR	MICROMETER - OPTICAL
No specific	AR	RULER 150 MM (6 IN) - METALLIC
No specific	AR	SAFETY CLIP - CIRCUIT BREAKER
SJ201	1	SURFACE TESTER
SJ210	1	SURFACE TESTER

B. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
210	CKPT,FWD COMPT BHD TO FLT COMPT BULKHEAD
211	COCKPIT
212	COCKPIT

C. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 56-10-00-110-002-A	Cleaning of the L/R Fixed Windows
Ref. AMM 56-10-00-200-004-A	Measurement of the Scratch Depth With Mitutoyo Surface Tester SJ201
Ref. AMM 56-10-00-200-004-A-01	Measurement of the Scratch Depth with Mitutoyo Surface Tester SJ210
Ref. AMM 56-11-12-000-001-A	Removal of the Fixed Windows
Ref. AMM 56-11-12-000-002-A	Removal of the Fixed Windows
Ref. AMM 56-11-12-400-002-A	Installation of the Fixed Windows
Ref. AMM 56-11-12-400-003-A	Installation of the Fixed Windows
Ref. Fig. (1 / 56-11-12-991-00500)	
Ref. Fig. (1 / 56-11-12-991-00600)	
Ref. Fig. (1 / 56-11-12-991-00700)	
Ref. Fig. (1 / 56-11-12-991-00800)	
Ref. Fig. (1 / 56-11-12-991-00900)	
Ref. Fig. (1 / 56-11-12-991-01000)	
Ref. Fig. (1 / 56-11-12-991-01200)	
Ref. Fig. (1 / 56-11-12-991-07400)	
Ref. Fig. (1 / 56-11-12-991-08000)	
Ref. Fig. (1 / 56-11-12-991-01300)	
Ref. Fig. (1 / 56-11-12-991-01400)	
Ref. Fig. (1 / 56-11-12-991-01500)	
Ref. Fig. (1 / 56-11-12-991-01600)	
Ref. Fig. (1 / 56-11-12-991-01700)	
Ref. Fig. (1 / 56-11-12-991-01800)	
Ref. Fig. (1 / 56-11-12-991-01900)	

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AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK

REFERENCE	DESIGNATION
Ref. Fig. (1 / 56-11-12-991-01100)	
Ref. Fig. (1 / 56-11-12-991-02000)	
Ref. Fig. (1 / 56-11-12-991-07500)	
Ref. Fig. (1 / 56-11-12-991-08100)	
Ref. Fig. (1 / 56-11-12-991-02100)	

3. Job Set-up

Subtask 56-11-12-941-051-A

On A/C 001-999

A. Get Access

- (1) Put the ACCESS PLATFORM 5M (16 FT)- ADJUSTABLE in position in front of the windshield at zone 210 (only if you must do a detailed inspection from the outer side of the aircraft).
- (2) Clean the fixed windows [Ref. AMM 56-10-00-110-002-A](#).

Subtask 56-11-12-865-059-A

On A/C 001-999

WARNING: MAKE SURE THAT THE CIRCUIT BREAKERS RELATED TO THE WEATHER RADAR ARE OPENED, SAFETIED AND TAGGED.

- #### B. Open, safety and tag the circuit breaker(s) that follow(s). Use the SAFETY CLIP - CIRCUIT BREAKER as necessary.

PANEL	DESIGNATION	FIN	LOCATION
121VU	COM NAV/RADAR/2	5SQ2	K14
121VU	COM NAV/RADAR/1	5SQ1	K13
122VU	ANTI ICE/WINDOWS/R	4DG2	W14
122VU	ANTI ICE/WHC/2	5DG2	W13
122VU	ANTI ICE/WINDOWS/L	4DG1	X14
122VU	ANTI ICE/WHC/1	5DG1	X13
123VU	ANTI ICE/L/WSHLD	1DG1	AF10
123VU	ANTI ICE/R/WSHLD	1DG2	AF03

4. Procedure

Subtask 56-11-12-220-051-A

On A/C 001-999

A. Inspection of the Fixed Windows

- (1) Do the inspection of the acrylic fixed windows (GKN supplier, P/N NH...).

NOTE : In the inspection tables, the word impaired refers to the visibility. The visibility is impaired when defects have effects on it.
If there is impaired visibility, it is necessary to replace the window panel.

- (a) Examine the acrylic fixed windows for:

- 1 Cracks
Ref. Fig. (1 / 56-11-12-991-00500)
- 2 Scratches
Ref. Fig. (1 / 56-11-12-991-00600)
- 3 Chips
Ref. Fig. (1 / 56-11-12-991-00700)
- 4 Delamination

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AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK

- Ref. Fig. (1 / 56-11-12-991-00800)
- 5 Bubbles
Ref. Fig. (1 / 56-11-12-991-00900)
- 6 Discoloration or burning
Ref. Fig. (1 / 56-11-12-991-00900)
- 7 Transparency
Ref. Fig. (1 / 56-11-12-991-00900)
- 8 Crazing
Ref. Fig. (1 / 56-11-12-991-01000)
- 9 Burn spots
Ref. Fig. (1 / 56-11-12-991-01200)
- 10 Heating film cracks
Ref. Fig. (1 / 56-11-12-991-07400)
- 11 Distortion and optical deviations
Ref. Fig. (1 / 56-11-12-991-08000)

NOTE : Distortion and optical deviations have no effect on the structural strength of the window. The fixed window can have distortion and optical deviations in zone 1 and/or zone 2 because of the manufacturing process of the window.

- a Make sure that there is no impaired visibility because of distortion and optical deviations in zone 1. If there is impaired visibility, replace the fixed windows [Ref. AMM 56-11-12-000-001-A](#) and [Ref. AMM 56-11-12-400-002-A](#).

NOTE : The impaired visibility is permitted in zone 2.

NOTE : Zone 2 is 50mm (1.969 in) wide shaded border from the daylight opening.

- (b) Examine the acrylic fixed windows for burn spots

:

Ref. Fig. (1 / 56-11-12-991-01300)

- 1 Point the FLASHLIGHT at 45 degrees to the outer surface of the window panel.
- 2 Examine the busbar for burn spots.

- (c) Examine the acrylic fixed windows for crazing

:

Ref. Fig. (1 / 56-11-12-991-01400)

- 1 Put the RULER 150 MM (6 IN) - METALLIC perpendicular to the surface of the fixed window panel where you see the maximum depth of the crazing.

NOTE : You can use the MICROMETER - OPTICAL as an alternative to the metal ruler.

- 2 Look at the ruler reflection at an angle of 45 degrees to find the depth of the crazing.

NOTE : The correct depth is approximately two times of the depth.

NOTE : You can use the MAGNIFYING GLASS POWER 10X as an optical aid where the depth of the crazing is at a minimum.

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WINDOW - FIXED - INSPECTION/CHECK

- (2) Do the inspection of the glass fixed windows (PPG supplier, P/N NP... and SGS supplier, P/N SPS...).

NOTE : In the inspection tables, the word impaired refers to the visibility. The visibility is impaired when defects have effects on it.
If there is impaired visibility, it is necessary to replace the window panel.

- (a) Examine the glass fixed windows for:

- 1 Cracks
Ref. Fig. (1 / 56-11-12-991-01500)
- 2 Scratches
Ref. Fig. (1 / 56-11-12-991-01600)

NOTE : There are two procedures to measure the depth of the scratches. The fingernail procedure measures the approximate depth of the scratches. The MITUTOYO surface-tester procedure measures the accurate depth of the scratches. If the MITUTOYO surface tester is available, the MITUTOYO surface-tester procedure is recommended because it gives accurate measurement results.

- a If you see scratches, do a check of the depth of the scratches:
- With the MITUTOYO **SURFACE TESTER** (SJ201) (SJ201)
or the MITUTOYO **SURFACE TESTER** (SJ210) (SJ210)
[Ref. AMM 56-10-00-200-004-A](#) [Ref. AMM 56-10-00-200-004-A-01](#), or
 - With your fingernail.

- 3 Chips
Ref. Fig. (1 / 56-11-12-991-01700)
- 4 Delamination
Ref. Fig. (1 / 56-11-12-991-01800)
- 5 Bubbles
Ref. Fig. (1 / 56-11-12-991-01900)
- 6 Discoloration
Ref. Fig. (1 / 56-11-12-991-01900)
- 7 Transparency
Ref. Fig. (1 / 56-11-12-991-01900)
- 8 Interlayer microflakes
Ref. Fig. (1 / 56-11-12-991-01100)
- 9 Burn spots
Ref. Fig. (1 / 56-11-12-991-02000)
- 10 Heating film cracking
.
Ref. Fig. (1 / 56-11-12-991-07500)
- 11 Distortion and optical deviations
Ref. Fig. (1 / 56-11-12-991-08100)

NOTE : Distortion and optical deviations have no effect on the structural strength of the window. The fixed window can have distortion and optical deviations in zone 1 and/or zone 2 because of the manufacturing process of the window.

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AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK

- a Make sure that there is no impaired visibility because of distortion and optical deviations in zone 1. If there is impaired visibility, replace the fixed windows [Ref. AMM 56-11-12-000-001-A](#) or [Ref. AMM 56-11-12-000-002-A](#) and [Ref. AMM 56-11-12-400-002-A](#) or [Ref. AMM 56-11-12-400-003-A](#).

NOTE : The impaired visibility is permitted in zone 2.

NOTE : Zone 2 is 50mm (1.969 in) wide shaded border from the daylight opening.

- (b) Examine the glass fixed windows for burn spots

:

Ref. Fig. (1 / 56-11-12-991-02100)

- 1 Point the LIGHT SOURCE - WHITE at 45 degrees to the outer surface of the window panel.
- 2 Examine the busbar for burn spots.

5. Close-up

Subtask 56-11-12-865-060-A

[On A/C 001-999](#)

- A. Remove the SAFETY CLIP - CIRCUIT BREAKER and the tag(s) and close this (these) circuit breaker(s):

PANEL	DESIGNATION	FIN	LOCATION
121VU	COM NAV/RADAR/2	5SQ2	K14
121VU	COM NAV/RADAR/1	5SQ1	K13
122VU	ANTI ICE/WINDOWS/R	4DG2	W14
122VU	ANTI ICE/WHC/2	5DG2	W13
122VU	ANTI ICE/WINDOWS/L	4DG1	X14
122VU	ANTI ICE/WHC/1	5DG1	X13
123VU	ANTI ICE/L/WSHLD	1DG1	AF10
123VU	ANTI ICE/R/WSHLD	1DG2	AF03

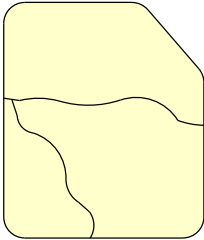
Subtask 56-11-12-942-054-A

[On A/C 001-999](#)

- B. Close Access

- (1) Remove the access platform(s).
- (2) Remove the ground support and maintenance equipment, the special and standard tools and all other items.

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AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK

DESCRIPTION OF THE DEFECT	CORRECTIVE ACTION
<u>CRACKS</u> DEFECT LINE TYPES IN THE DEPTH OF THE PLY OF THE ACRYLIC FIXED WINDOW. 	<u>OUTER/INNER PLY</u> NOT PERMITTED DO NOT APPLY ELECTRICAL HEAT TO THE ACRYLIC FIXED WINDOW WITH CRACKS IF THERE IS AN ARCING.

N_MM_561112_6_ACM0_01_01

Acrylic Fixed Window - Permitted Damage
Figure 1 / Graphic 56-11-12 -991-00500

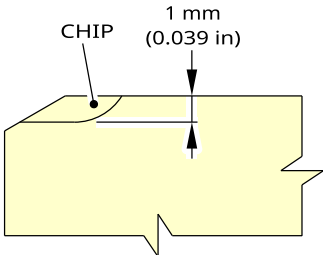
TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK

DESCRIPTION OF THE DEFECT	CORRECTIVE ACTION
<u>SCRATCHES</u> DEFECT LINE TYPE ON THE EXTERNAL SURFACE OF THE ACRYLIC FIXED WINDOW.	<u>OUTER/INNER PLY</u> MAXIMUM DEPTH: 1 mm (0.039 in) PERMITTED IF THE SCRATCHES ARE IN THE TOLERANCE. PERMITTED IF THE VISIBILITY IS NOT IMPAIRED.

N_MM_561112_6_AEM0_01_03

Acrylic Fixed Window - Permitted Damage
Figure 1 / Graphic 56-11-12 -991-00600

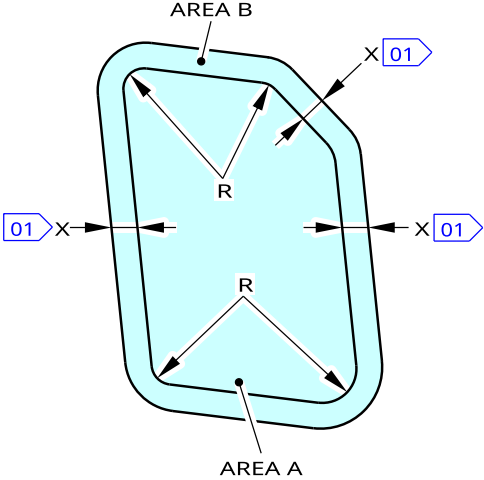
TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK

DESCRIPTION OF THE DEFECT	CORRECTIVE ACTION
<u>CHIPS</u> FLAKES OF BROKEN GLASS FROM THE SURFACE AND THE EDGES OF THE ACRYLIC FIXED WINDOW. <u>DESCRIPTION OF A CHIP</u> 	<u>OUTER/INNER PLY</u> MAXIMUM DEPTH: 1 mm (0.039 in) PERMITTED IF THE CHIPS ARE IN THE TOLERANCE. PERMITTED IF THE VISIBILITY IS NOT IMPAIRED.

N_MM_561112_6_AGM0_01_02

Acrylic Fixed Window - Permitted Damage
Figure 1 / Graphic 56-11-12 -991-00700

TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK

DESCRIPTION OF THE DEFECT	CORRECTIVE ACTION
<u>DELAMINATION:</u> THE SEPARATION OF THE MATERIAL OCCURS FREQUENTLY AROUND THE EDGES, THE SENSORS AND IN THE AREAS WITH NO HEATING. THE AREA WITH DELAMINATION CAN BE TRANSPARENT. THIS AREA WILL NOT BE TRANSPARENT IF THE MOISTURE GOES INTO IT. <u>DELAMINATION LIMITS:</u>  X = 76.20 mm (3 in) R = 76.20 mm (3 in) NOTE: 01 DIMENSION X IS MEASURED FROM THE EDGE OF THE DAYLIGHT OPENING (EDGE OF THE SEAL ON EXTERNAL SURFACE).	<u>OUTER AND INNER PLY:</u> AREA A: – SATISFACTORY IF EACH DELAMINATION AREA HAS A SURFACE OF NO MORE THAN 1290 mm² (2.0 in²). – THE TOTAL SURFACE OF THESE AREAS MUST NOT BE MORE THAN 3871 mm² (6.0 in²). – MAKE SURE THAT THE DELAMINATION AREAS DO NOT MAKE THE VISIBILITY UNSATISFACTORY. AREA B: DELAMINATION ON AREA B IS PERMITTED. IF THE DELAMINATION AREA IS ACROSS THE TWO ZONES (A AND B), YOU MUST SEE IT AS TWO DIFFERENT DELAMINATION CONDITIONS (ONE ON AREA A AND THE OTHER ON AREA B). THUS, YOU MUST DO TWO ISOLATED CORRECTIONS, ONE ON AREA A AND THE OTHER ON AREA B. PERMITTED IF THE VISIBILITY DOES NOT DECREASE.

NOTE:
THERE ARE STRESS RELEASE SEPARATORS ABOVE THE BUSBARS IN THE TRANSPARENT WINDOW AREA. IT IS POSSIBLE TO THINK THAT THEY ARE DELAMINATION AREAS.

N_MM_561112_6_AJM0_01_10

Acrylic Fixed Window - Permitted Damage
Figure 1 / Graphic 56-11-12 -991-00800

TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK

DESCRIPTION OF THE DEFECT	CORRECTIVE ACTION
<u>BUBBLES</u> WHEN THE ACRYLIC MATERIAL BECOMES TOO HOT, THE GAS CAUSES SMALL BUBBLES. THE BUBBLES DO NOT DECREASE THE STRUCTURAL INTEGRITY.	<u>OUTER/INNER PLY</u> ISOLATED BUBBLES ARE PERMITTED TO A MAXIMUM DIAMETER OF 6.4 mm (0.25 in) IF THE VISIBILITY IS NOT IMPAIRED.
<u>CAUSES OF THE DISCOLORATION:</u> - BURNING OF THE CONDUCTIVE FILM. - CARBON DISCOLORATION BECAUSE OF THE REFLECTION FROM THE BUSBAR AND THE CONDUCTIVE FILM. THIS REFLECTION CHANGES (RAINBOW COLORS) WHEN YOU LOOK AT IT FROM AN TILTED ANGLE. <u>CAUSES OF THE BURNING:</u> - FREQUENT HIGH TEMPERATURES WHICH MAKE THE INTERLAYER BECOME BROWN AT HOT POINTS. - BLACK OR BROWN DISCOLORATION OR BURNING ARE POSSIBLE DEFECTS. DISCOLORATION OR BURNING HAS NO EFFECT ON THE STRUCTURAL INTEGRITY.	PERMITTED IF THE VISIBILITY IS NOT IMPAIRED.
<u>TRANSPARENCY</u> HALOS ON THE SURFACE OF THE ACRYLIC FIXED WINDOW CAN MAKE IT LESS TRANSPARENT.	PERMITTED IF THE VISIBILITY IS NOT IMPAIRED.

N_MM_561112_6_ALM0_01_02

Acrylic Fixed Window - Permitted Damage
Figure 1 / Graphic 56-11-12 -991-00900

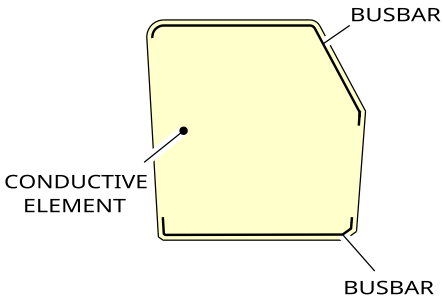
TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK

DESCRIPTION OF THE DEFECT	CORRECTIVE ACTION
<u>CRAZING</u> - SERIES OF CRACKS PERPENDICULAR TO THE SURFACE BUT WHICH DO NOT EXTEND THROUGH THE PANEL. - YOU CAN SEE SOME SURFACE CRACKS ONLY FROM A SPECIFIED ANGLE (FOR EXAMPLE WHEN THERE IS A LIGHT REFLECTION FROM THE CRACK SURFACE). - TENSION AND CONTAMINATION CAUSE CRAZING.	<u>INNER/OUTER PLY</u> CRAZING ON THE FULL SURFACE IS PERMITTED: - IF THE DEPTH IS NOT MORE THAN 0.5 mm (0.02 in) - IF THE VISIBILITY IS NOT IMPAIRED. CRAZING ON THE OUTER SURFACE OF AN AREA OR A NUMBER OF ISOLATED AREAS IS PERMITTED: - WHERE THE TOTAL AREA IS NOT MORE THAN 500 mm² (0.78 in²) - IF THE DEPTH IS NOT MORE THAN 1.0 mm (0.039 in) - IF THE VISIBILITY IS NOT IMPAIRED. CRAZING ON ONE AREA OF THE OUTER SURFACE WHICH IS NOT MORE THAN 100 mm² (0.16 in²) IS PERMITTED: - IF THE DEPTH IS NOT MORE THAN 1.5 mm (0.06 in) - IF THE VISIBILITY IS NOT IMPAIRED. - NOT PERMITTED IF THE CRAZING IS MORE THAN THE OUTER SURFACE OR NOT IN THE TOLERANCE.

N_MM_561112_6_AMM0_01_02

Acrylic Fixed Window - Permitted Damage
Figure 1 / Graphic 56-11-12 -991-01000

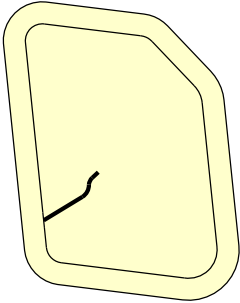
TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK

DESCRIPTION OF THE DEFECTS	CORRECTIVE ACTION
<u>BURN SPOT</u> - <u>BUSBAR AND CONDUCTIVE ELEMENT LOCATION</u> 	
- <u>BURN SPOT ON THE BUSBAR</u> AN ARCING CAUSES BURN SPOT ON THE BUSBAR AREA AFTER A BUSBAR DAMAGE.	PERMITTED IF THE DIAMETER OF THE SPOT IS LESS THAN 10 mm (0.39 in). IF THE DIAMETER OF THE SPOT IS MORE THAN 10 mm (0.39 in), STOP THE SIDE FIXED WINDOW HEATING AS FOLLOWS: - OPEN, SAFETY AND TAG THE CIRCUIT BREAKERS: - FOR THE LEFT SIDE: 4DG1 (122VU) - FOR THE RIGHT SIDE: 4DG2 (122VU).
- <u>BURN SPOT ON THE CONDUCTIVE ELEMENT</u> CAUSES OF THE BURN SPOT: - DAMAGE TO THE HEATING ELEMENT CAUSES BURN SPOT ON THE CONDUCTIVE ELEMENT. - FREQUENT HIGH TEMPERATURES WHICH MAKE THE INTERLAYER BECOME BROWN AT HOT POINTS.	PERMITTED IF THE VISIBILITY IS NOT IMPAIRED.

N_MM_561112_6_AQM0_01_01

Acrylic Fixed Window - Permitted Damage
Figure 1 / Graphic 56-11-12 -991-01200

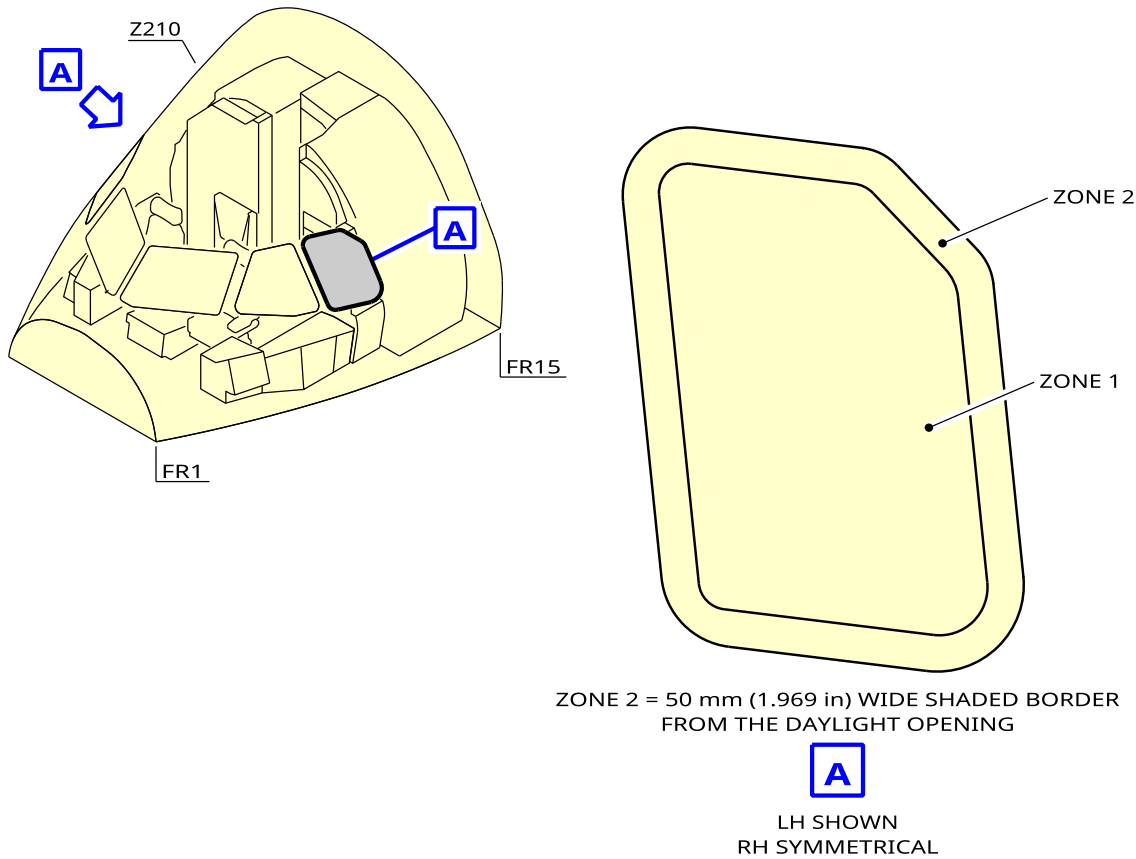
TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK

DESCRIPTION OF THE DEFECT	DAMAGE LOCATION	PERMITTED/NOT PERMITTED DAMAGE
<u>HEATING FILM CRACKS</u> THE HEATING FILM CRACKS CAN CAUSE AN ELECTRICAL FAILURE OF THE HEATING SYSTEM. THIS TYPE OF DAMAGE HAS NO EFFECT ON THE STRUCTURAL INTEGRITY. NOTE: THE CRACKS OF THE HEATING FILM CAN LOOK THE SAME AS THE CRACKS OF THE MIDDLE PLY. <u>DESCRIPTION OF THE HEATING FILM CRACKS</u> 	BETWEEN THE OUTER PLY AND THE MIDDLE PLY OF THE ACRYLIC FIXED WINDOW.	THE AIRCRAFT DISPATCH IS PERMITTED IF: - THE VISIBILITY DOES NOT DECREASE. - THE CIRCUIT BREAKERS OF THE FIXED WINDOW ARE OPEN BEFORE THE FLIGHT. - THE VISIBILITY IS NOT IMPAIRED.

N_MM_561112_6_FAQ0_01_02

Acrylic Fixed Window - Permitted Damage
Figure 1 / Graphic 56-11-12 -991-07400

TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK

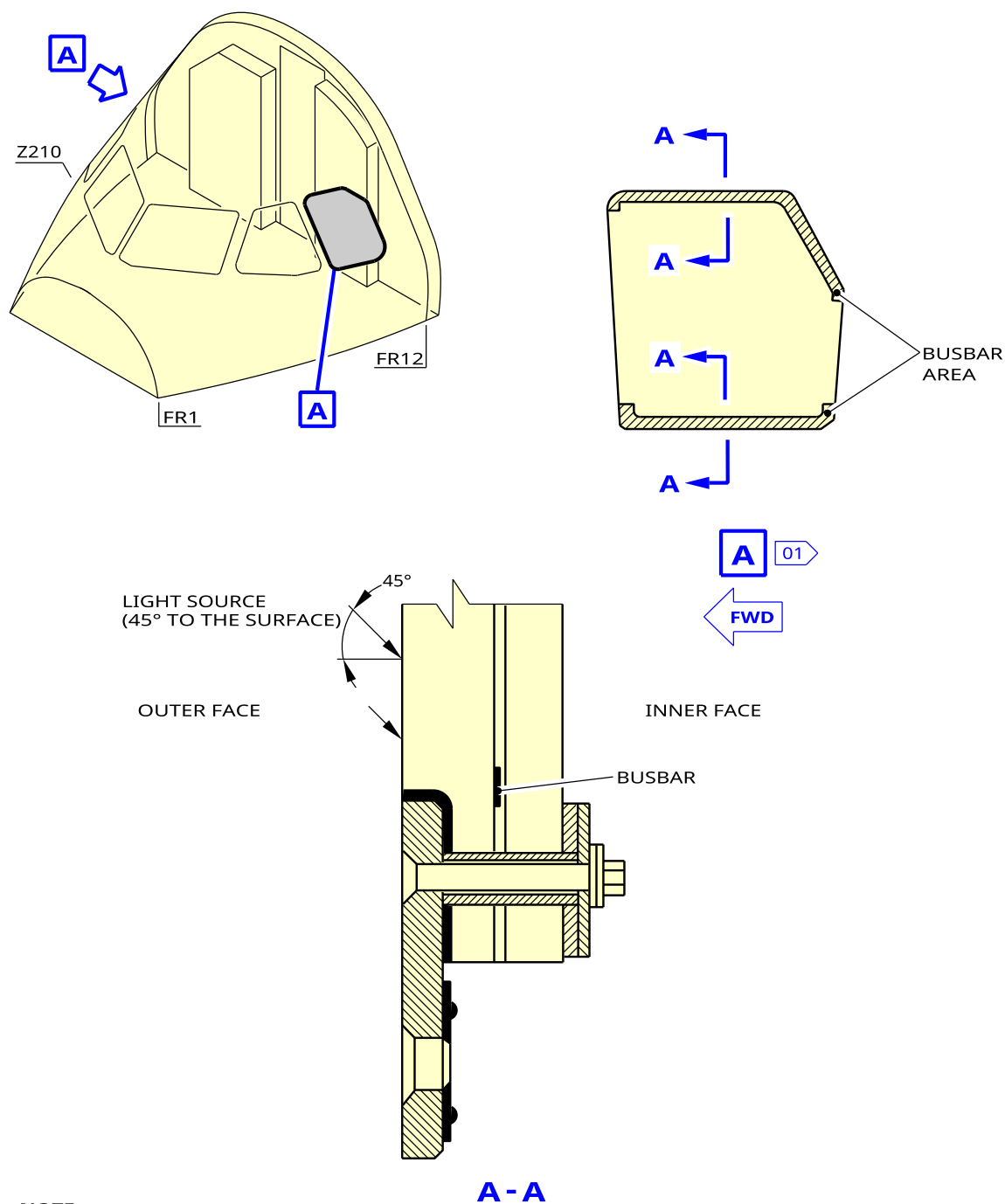


DESCRIPTION OF THE DEFECT	CORRECTIVE ACTION
DISTORTION AND OPTICAL DEVIATIONS THE FIXED WINDOW CAN HAVE DISTORTION AND OPTICAL DEVIATIONS IN ZONE 1 AND/OR ZONE 2 BECAUSE OF THE MANUFACTURING PROCESS OF THE WINDOW.	THIS DAMAGE CAN OCCUR IN ZONE 1 AND/OR ZONE 2. IF THERE IS IMPAIRED VISIBILITY BECAUSE OF DISTORTION AND OPTICAL DEVIATIONS IN ZONE 1, REPLACE THE WINDOW PANEL. NOTE: THE IMPAIRED VISIBILITY IS PERMITTED IN ZONE 2.

N_MM_561112_6_SHM1_01_00

Acrylic Fixed Window - Distortion and Optical Deviations
Figure 1 / Graphic 56-11-12 -991-08000

TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK



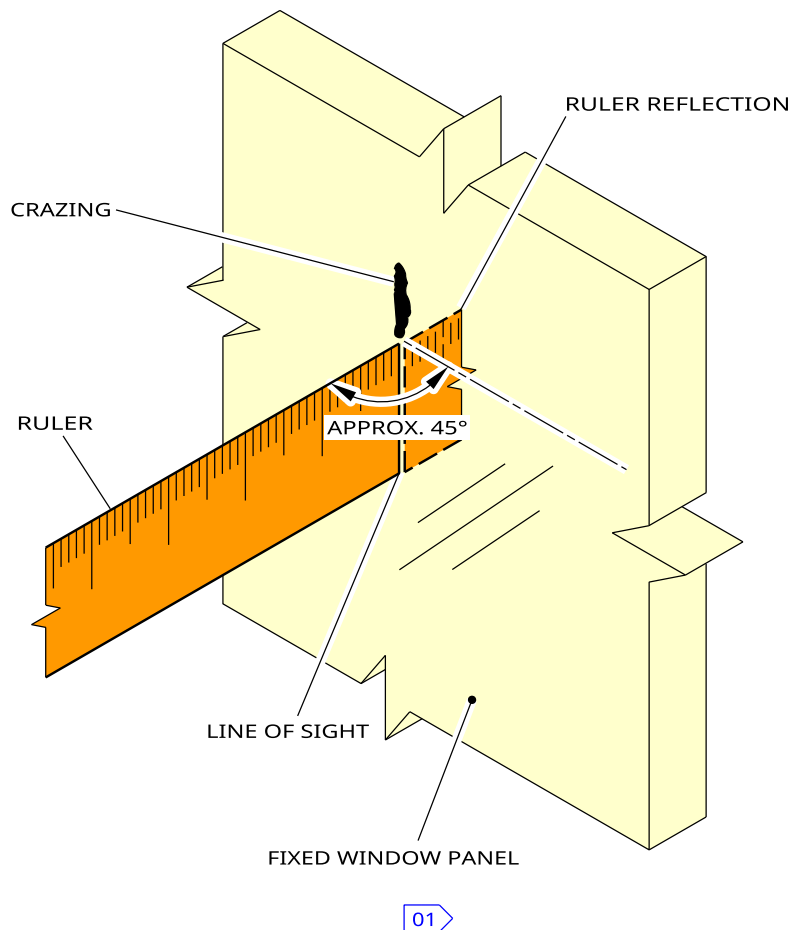
NOTE:
01 LH SHOWN, RH SYMMETRICAL

VIEW LOOKING FWD

N_MM_561112_6_FAM0_01_01

Acrylic Fixed Window - Burn Spot Inspection-Procedure
Figure 1 / Graphic 56-11-12 -991-01300

TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK



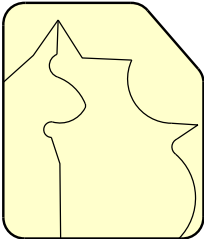
NOTE:

01 TYPICAL

N_MM_561112_6_GAM0_01_02

Acrylic Fixed Window - Crazing Inspection Procedure
Figure 1 / Graphic 56-11-12 -991-01400

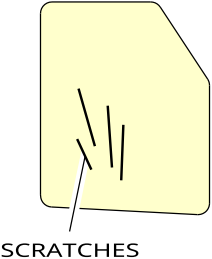
TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK

DESCRIPTION OF THE DEFECT	DAMAGE LOCATION	PERMITTED/NOT PERMITTED DAMAGE
<u>CRACKS</u> DEFECT LINE TYPES IN THE DEPTH OF THE PLY OF THE GLASS FIXED WINDOW. 	OUTER PLY	ONLY ONE FLIGHT IS PERMITTED IF: -THE VISIBILITY IS NOT IMPARED - THE RELATED SLIDING LATERAL WINDOW HAS NO CRACKS.
	INNER/MIDDLE PLY	MIDDLE AND/OR INNER PLY(IES) WITH CRACKS IS(ARE) NOT PERMITTED.

N_MM_561112_6_GCM0_01_02

Glass Fixed Window - Permitted Damage
Figure 1 / Graphic 56-11-12 -991-01500

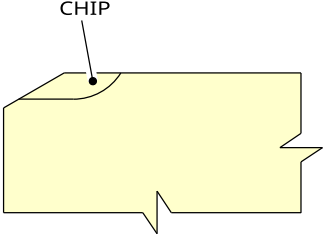
TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK

DESCRIPTION OF DEFECT	CORRECTIVE ACTION
<u>SCRATCHES</u> LINE TYPE DEFECTS IN THE EXTERNAL SURFACE OF THE FIXED WINDOW  SCRATCHES	<u>OUTER PLY</u> PERMITTED IF THE VISIBILITY IS NOT IMPAIRED.
	<u>INNER PLY</u> PERMITTED: - IF YOU CANNOT FEEL THE SCRATCHES WHEN YOU TOUCH THE FIXED WINDOW WITH YOUR FINGERNAIL, OR - IF THE DEPTH OF THE SCRATCHES MEASURED WITH THE MITUTOYO SURFACE TESTER SJ201/SJ210 IS LESS THAN 0.05 mm (0.0019 in) (REF. TASK 561000-200-004). NOT PERMITTED: - IF YOU CAN FEEL THE SCRATCHES WHEN YOU TOUCH THE FIXED WINDOW WITH YOUR FINGERNAIL, OR - IF THE DEPTH OF THE SCRATCHES MEASURED WITH THE MITUTOYO SURFACE TESTER SJ201/SJ210 IS MORE THAN 0.05 mm (0.0019 in) (REF. TASK 561000-200-004). NOTE: THERE ARE TWO PROCEDURES TO MEASURE THE DEPTH OF THE SCRATCHES. THE FINGERNAIL PROCEDURE WHICH HELPS YOU TO MEASURE AN APPROXIMATE DEPTH OF THE SCRATCHES. THE MITUTOYO SURFACE-TESTER PROCEDURE WHICH HELPS YOU TO MEASURE THE ACCURATE DEPTH OF THE SCRATCHES. IF THE MITUTOYO SURFACE TESTER IS AVAILABLE, THE MITUTOYO SURFACE-TESTER PROCEDURE IS RECOMMENDED BECAUSE IT GIVES ACCURATE MEASUREMENT RESULTS.

N_MM_561112_6_GEM0_01_04

Glass Fixed Window - Permitted Damage
Figure 1 / Graphic 56-11-12 -991-01600

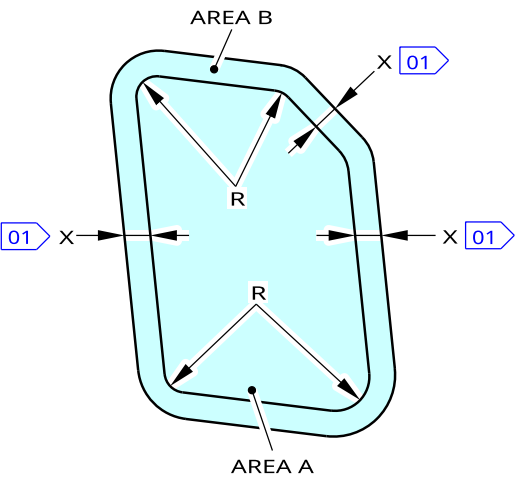
TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK

DESCRIPTION OF THE DEFECT	CORRECTIVE ACTION
<u>CHIPS</u> FLAKES OF BROKEN GLASS FROM THE SURFACE AND THE EDGES OF THE GLASS FIXED WINDOW. <u>DESCRIPTION OF A CHIP</u> 	<u>OUTER PLY</u> PERMITTED IF THE VISIBILITY IS NOT IMPAIRED.
	<u>INNER/MIDDLE PLY</u> NOT PERMITTED

N_MM_561112_6_GGM0_01_01

Glass Fixed Window - Permitted Damage
Figure 1 / Graphic 56-11-12 -991-01700

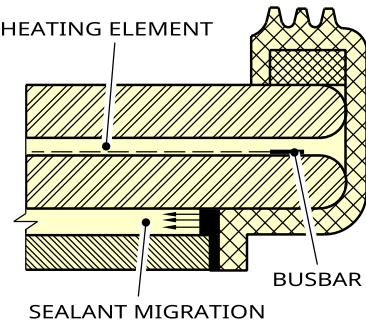
TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK

DESCRIPTION OF THE DEFECT	CORRECTIVE ACTION
<u>DELAMINATION:</u> THE SEPARATION OF THE GLASS FROM THE ADJACENT VINYL INTERLAYER OCCURS FREQUENTLY AROUND THE EDGES, THE SENSORS AND IN THE AREAS WITH NO HEATING. THE AREA WITH DELAMINATION CAN BE TRANSPARENT. THIS AREA WILL NOT BE TRANSPARENT IF THE MOISTURE GOES INTO IT. <u>DELAMINATION LIMITS:</u>  X = 76.20 mm (3.00 in) R = 76.20 mm (3.00 in) NOTE: 01 DIMENSION X IS MEASURED FROM THE EDGE OF THE DAYLIGHT OPENING (EDGE OF THE SEAL ON EXTERNAL SURFACE).	<u>MIDDLE, OUTER AND INNER PLY:</u> AREA A: NOT PERMITTED AREA B: PERMITTED IF THE VISIBILITY DOES NOT DECREASE.

N_MM_561112_6_GJM0_01_08

Glass Fixed Window - Permitted Damage
Figure 1 / Graphic 56-11-12 -991-01800

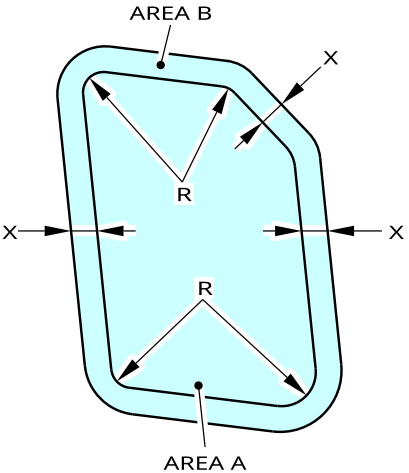
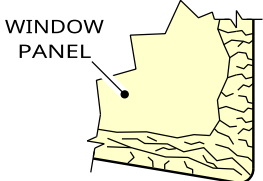
TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK

DESCRIPTION OF THE DEFECTS	CORRECTIVE ACTION
<u>BUBBLES:</u> WHEN THE VINYL MATERIAL BECOMES TOO HOT, THE GAS CAUSES SMALL BUBBLES. THESE BUBBLES DO NOT DECREASE THE STRUCTURAL INTEGRITY.	<u>INNER, MIDDLE AND OUTER PLY:</u> PERMITTED IF THE VISIBILITY DOES NOT DECREASE.
<u>DISCOLORATION:</u> CAUSES OF THE DISCOLORATION: - CARBON DISCOLORATION OCCURS BECAUSE OF THE REFLECTION FROM THE BUSBAR AND THE CONDUCTIVE FILM. THIS REFLECTION CHANGES (RAINBOW COLORS) WHEN YOU LOOK AT IT FROM AN TILTED ANGLE. - MIGRATION OF THE BROWN SEALANT TO THE EDGE OF THE OUTER INTERLAYER. THIS INTERLAYER CAN EXTEND TO A WIDTH THAT IS BETWEEN 10 mm (0.39 in) AND 20 mm (0.78 in) AT THE EDGE OF THE WINDOW PANEL. <u>SEALANT MIGRATION:</u>  HEATING ELEMENT BUSBAR SEALANT MIGRATION OUTER FACE	PERMITTED IF THE VISIBILITY DOES NOT DECREASE.
<u>TRANSPARENCY:</u> HALOS ON THE SURFACE OF THE GLASS FIXED WINDOW CAN MAKE IT LESS TRANSPARENT.	PERMITTED IF THE VISIBILITY DOES NOT DECREASE.

N_MM_561112_6_GLM0_01_02

Glass Fixed Window - Permitted Damage
Figure 1 / Graphic 56-11-12 -991-01900

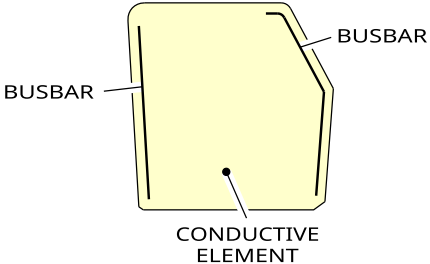
TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK

DESCRIPTION OF THE DEFECT	CORRECTIVE ACTION
<u>INTERLAYER MICROFLAKES</u> MICROFLAKES ARE CHEMICAL REACTIONS IN THE PERIPHERY OF THE INTERLAYER THAT MOISTURE AND WINDOW AGEING CAUSE. <u>INTERLAYER MICROFLAKES LIMITS</u>  X = 76.20 mm (3.00 in) R = 76.20 mm (3.00 in) <u>DESCRIPTION OF THE INTERLAYER MICROFLAKES</u> 	AREA A: NOT PERMITTED AREA B: PERMITTED IF THE VISIBILITY IS NOT IMPAIRED. NOTE: - MICROFLAKES ONLY OCCUR IN THE INTERLAYERS - AN EQUIVALENT DAMAGE IN ONE OF THE GLASS PLIES CAN IMMEDIATELY BREAK THE PLY.

N_MM_561112_6_GRM0_01_01

Glass Fixed Window - Permitted Damage
Figure 1 / Graphic 56-11-12 -991-01100

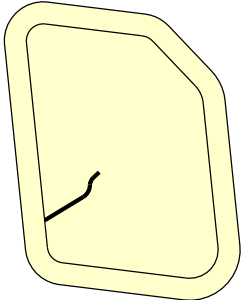
TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK

DESCRIPTION OF THE DEFECTS	CORRECTIVE ACTION
<u>BURN SPOT</u> <u>- BUSBAR AND CONDUCTIVE ELEMENT LOCATION</u>  <u>- BURN SPOT ON THE BUSBAR</u> AN ARCING CAUSES BURN SPOT ON THE BUSBAR AREA AFTER A BUSBAR DAMAGE.	PERMITTED IF THE DIAMETER OF THE SPOT IS LESS THAN 10 mm (0.39 in). IF THE DIAMETER OF THE SPOT IS MORE THAN 10 mm (0.39 in), STOP THE SIDE FIXED WINDOW HEATING AS FOLLOWS: - OPEN, SAFETY AND TAG THE CIRCUIT BREAKERS: - FOR THE LEFT SIDE: 4DG1 (122VU) - FOR THE RIGHT SIDE: 4DG2 (122VU).
<u>- BURN SPOT ON THE CONDUCTIVE ELEMENT</u> CAUSES OF THE BURN SPOT: - DAMAGE TO THE HEATING ELEMENT CAUSES BURN SPOT ON THE CONDUCTIVE ELEMENT. - FREQUENT HIGH TEMPERATURES WHICH MAKE THE INTERLAYER BECOME BROWN AT HOT POINTS.	PERMITTED IF THE DIAMETER OF THE SPOT IS LESS THAN 10 mm (0.39 in). PERMITTED IF THE VISIBILITY IS NOT IMPAIRED. IF THE DIAMETER OF THE SPOT IS MORE THAN 10 mm (0.39 in), STOP THE SIDE FIXED WINDOW HEATING AS FOLLOWS: - OPEN, SAFETY AND TAG THE CIRCUIT BREAKERS: - FOR THE LEFT SIDE: 4DG1 (122VU) - FOR THE RIGHT SIDE: 4DG2 (122VU).

N_MM_561112_6_GQM0_01_01

Glass Fixed Window - Permitted Damage
Figure 1 / Graphic 56-11-12 -991-02000

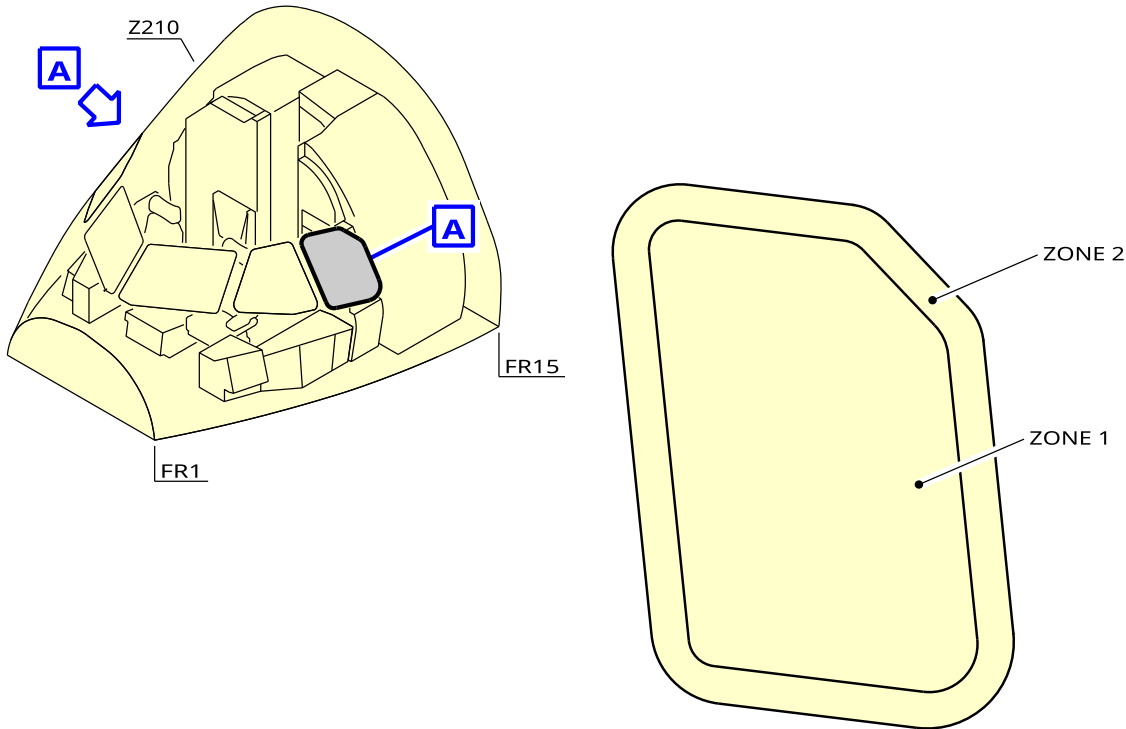
TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK

DESCRIPTION OF THE DEFECT	DAMAGE LOCATION	PERMITTED/NOT PERMITTED DAMAGE
<u>HEATING FILM CRACKS</u> THE HEATING FILM CRACKS CAN CAUSE AN ELECTRICAL FAILURE OF THE HEATING SYSTEM. THIS TYPE OF DAMAGE HAS NO EFFECT ON THE STRUCTURAL INTEGRITY. NOTE: THE CRACKS OF THE HEATING FILM CAN LOOK THE SAME AS THE CRACKS OF THE MIDDLE PLY. <u>DESCRIPTION OF THE HEATING FILM CRACKS</u> 	BETWEEN THE OUTER PLY AND THE MIDDLE PLY OF THE GLASS FIXED WINDOW.	THE AIRCRAFT DISPATCH IS PERMITTED IF: - THE VISIBILITY DOES NOT DECREASE. - THE CIRCUIT BREAKERS OF THE GLASS FIXED WINDOW ARE OPEN BEFORE THE FLIGHT. - THE VISIBILITY IS NOT IMPAIRED.

N_MM_561112_6_NAN0_01_02

Glass Fixed Window - Permitted Damage
Figure 1 / Graphic 56-11-12 -991-07500

TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK



ZONE 2 = 50 mm (1.969 in) WIDE SHADED BORDER
FROM THE DAYLIGHT OPENING



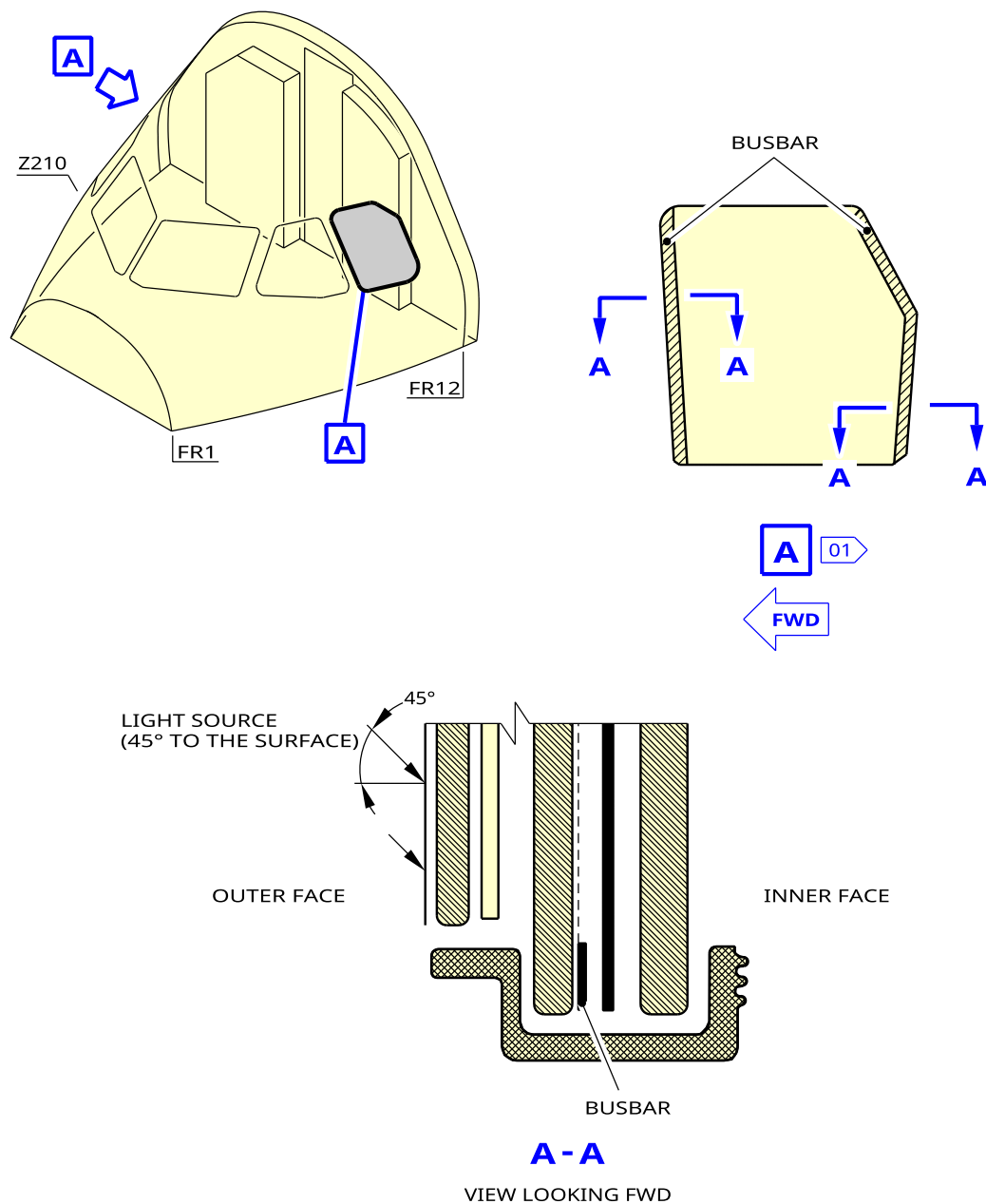
LH SHOWN
RH SYMMETRICAL

DESCRIPTION OF THE DEFECT	CORRECTIVE ACTION
DISTORTION AND OPTICAL DEVIATIONS THE FIXED WINDOW CAN HAVE DISTORTION AND OPTICAL DEVIATIONS IN ZONE 1 AND/OR ZONE 2 BECAUSE OF THE MANUFACTURING PROCESS OF THE WINDOW.	THIS DAMAGE CAN OCCUR IN ZONE 1 AND/OR ZONE 2. IF THERE IS IMPAIRED VISIBILITY BECAUSE OF DISTORTION AND OPTICAL DEVIATIONS IN ZONE 1, REPLACE THE WINDOW PANEL. NOTE: THE IMPAIRED VISIBILITY IS PERMITTED IN ZONE 2.

N_MM_561112_6_SHM2_01_00

Glass Fixed Window - Distortion and Optical Deviations
Figure 1 / Graphic 56-11-12 -991-08100

TF-N
AIRCRAFT MAINTENANCE MANUAL
WINDOW - FIXED - INSPECTION/CHECK



NOTE:

01 LH SHOWN, RH SYMMETRICAL

N_MM_561112_6_GMM0_01_01

Glass Fixed Window - Burn Spot Inspection Procedure
Figure 1 / Graphic 56-11-12 -991-02100

TF-N
AIRCRAFT MAINTENANCE MANUAL
SLIDING WINDOWS - INSPECTION/CHECK

TASK 56-12-11-200-001 -A

On A/C 001-999

Inspection of the Sliding Windows

FIN: 8DG1, 8DG2

1. Reason for the Job

NOTE : This procedure is applicable to:

- The acrylic sliding windows (GKN supplier, P/N NH...)
- The glass sliding windows (PPG supplier, P/N NP...)
- The glass sliding windows (SGS supplier, P/N SPSA320-3-X-X or P/N SPSA320-4-X-X and P/N SPSA320-7-X-X or P/N SPSA320-8-X-X).

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	1	ACCESS PLATFORM 5M (16 FT)- ADJUSTABLE
No specific	1	FLASHLIGHT
No specific	AR	LIGHT SOURCE - WHITE
No specific	1	MAGNIFYING GLASS POWER 10X
No specific	1	RULER 150 MM (6 IN) - METALLIC
No specific	AR	SAFETY CLIP - CIRCUIT BREAKER
SJ201	1	SURFACE TESTER
SJ210	1	SURFACE TESTER

B. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
210	CKPT,FWD COMPT BHD TO FLT COMPT BULKHEAD
211	COCKPIT
212	COCKPIT

C. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 56-10-00-110-003-A	Cleaning of the L/R Sliding Windows
Ref. AMM 56-10-00-200-004-A	Measurement of the Scratch Depth With Mitutoyo Surface Tester SJ201
Ref. AMM 56-10-00-200-004-A-01	Measurement of the Scratch Depth with Mitutoyo Surface Tester SJ210
Ref. AMM 56-12-11-000-001-A	Removal of the Sliding Windows
Ref. AMM 56-12-11-000-001-A-02	Removal of the Sliding Windows
Ref. AMM 56-12-11-000-003-A	Removal of the Sliding Window Seal
Ref. AMM 56-12-11-000-003-A-01	Removal of the Sliding Window Seal
Ref. AMM 56-12-11-010-001-A	Opening of the Cockpit Sliding Windows
Ref. AMM 56-12-11-400-001-A	Installation of the Sliding Windows
Ref. AMM 56-12-11-400-001-A-02	Installation of the Sliding Windows
Ref. AMM 56-12-11-400-003-A	Installation of the Sliding Window Seal
Ref. AMM 56-12-11-400-003-A-01	Installation of the Sliding Window Seal
Ref. AMM 56-12-11-410-001-A	Closing of the Cockpit Sliding Windows
Ref. Fig. (1 / 56-12-11-991-00200)	
Ref. Fig. (1 / 56-12-11-991-00300)	
Ref. Fig. (1 / 56-12-11-991-00400)	
Ref. Fig. (1 / 56-12-11-991-00500)	
Ref. Fig. (1 / 56-12-11-991-01200)	
Ref. Fig. (1 / 56-12-11-991-01300)	
Ref. Fig. (1 / 56-12-11-991-01700)	
Ref. Fig. (1 / 56-12-11-991-12000)	
Ref. Fig. (1 / 56-12-11-991-13100)	
Ref. Fig. (1 / 56-12-11-991-01800)	

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REFERENCE	DESIGNATION
Ref. Fig. (1 / 56-12-11-991-01900)	
Ref. Fig. (1 / 56-12-11-991-02100)	
Ref. Fig. (1 / 56-12-11-991-02200)	
Ref. Fig. (1 / 56-12-11-991-02300)	
Ref. Fig. (1 / 56-12-11-991-02400)	
Ref. Fig. (1 / 56-12-11-991-02500)	
Ref. Fig. (1 / 56-12-11-991-01400)	
Ref. Fig. (1 / 56-12-11-991-02600)	
Ref. Fig. (1 / 56-12-11-991-12100)	
Ref. Fig. (1 / 56-12-11-991-13200)	
Ref. Fig. (1 / 56-12-11-991-02700)	

3. Job Set-up

Subtask 56-12-11-941-057-A

On A/C 001-999

A. Get Access

- (1) Put the ACCESS PLATFORM 5M (16 FT)- ADJUSTABLE in position in front of the sliding windows at zone 210.

NOTE : This step is only applicable to the detailed inspection from the outside of the aircraft.

- (2) Clean the sliding windows [Ref. AMM 56-10-00-110-003-A](#).
- (3) Open the sliding windows until you see the sliding-window pressure seals
[Ref. AMM 56-12-11-010-001-A](#).

Subtask 56-12-11-865-068-A

On A/C 001-999

WARNING: MAKE SURE THAT THE CIRCUIT BREAKERS RELATED TO THE WEATHER RADAR ARE OPENED, SAFETIED AND TAGGED.

- B. Open, safety and tag the circuit breaker(s) that follow(s). Use the SAFETY CLIP - CIRCUIT BREAKER as necessary.

PANEL	DESIGNATION	FIN	LOCATION
121VU	COM NAV/RADAR/2	5SQ2	K14
121VU	COM NAV/RADAR/1	5SQ1	K13
122VU	ANTI ICE/WINDOWS/R	4DG2	W14
122VU	ANTI ICE/WHC/2	5DG2	W13
122VU	ANTI ICE/WINDOWS/L	4DG1	X14
122VU	ANTI ICE/WHC/1	5DG1	X13
123VU	ANTI ICE/L/WSHLD	1DG1	AF10
123VU	ANTI ICE/R/WSHLD	1DG2	AF03

4. Procedure

Subtask 56-12-11-210-057-A

On A/C 001-999

A. Inspection of the Sliding-Window Pressure Seal

NOTE : We recommend that you do the inspection of the sliding-window pressure seal from the outside of the aircraft when the window is open.

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NOTE : It is permitted to fly with a damaged pressure seal (all types of damage) with no time limit. This is only possible if there is no pressure leak which causes an unsatisfactory noise level for the crew in the cockpit.

- (1) Make sure that there are no cracks or blisters on the sliding-window pressure seal.
- (2) Make sure that the surface of the pressure seal that touches the sliding window is not damaged. If the sliding-window pressure seal is damaged, do the steps that follow:
 - (a) For the sliding window assemblies that have windows from the SGS supplier (Part Number (P/N) SPSA320-3-X-X and P/N SPSA320-4-X-X), replace the pressure seal [Ref. AMM 56-12-11-000-003-A](#) [Ref. AMM 56-12-11-000-003-A-01](#) and [Ref. AMM 56-12-11-400-003-A](#) [Ref. AMM 56-12-11-400-003-A-01](#).
 - (b) For the sliding window assemblies that have windows from the GKN, PPG and SGS suppliers (P/N SPSA320-7-X-X and P/N SPSA320-8-X-X), replace the sliding window [Ref. AMM 56-12-11-000-001-A](#) [Ref. AMM 56-12-11-000-001-A-02](#) and [Ref. AMM 56-12-11-400-001-A](#) [Ref. AMM 56-12-11-400-001-A-02](#).

Subtask 56-12-11-220-055-A

On A/C 001-999

B. Inspection of the Sliding Windows

- (1) Do an inspection of the acrylic sliding windows (GKN supplier, Part Number (P/N) NH...):

NOTE : In the figures about the permitted damage on the sliding windows, the word "impaired" refers to the visibility. The visibility is impaired when the defects have an unwanted effect on it.

If the visibility is impaired, it is necessary to replace the window panel.

- (a) Examine the acrylic sliding windows for:
 - 1 Cracks:
Ref. Fig. (1 / 56-12-11-991-00200)
 - 2 Scratches:
Ref. Fig. (1 / 56-12-11-991-00300)
 - 3 Chips:
Ref. Fig. (1 / 56-12-11-991-00400)
 - 4 Delamination:
Ref. Fig. (1 / 56-12-11-991-00500)
 - 5 Bubbles:
Ref. Fig. (1 / 56-12-11-991-01200)
 - 6 Discoloration or burning:
Ref. Fig. (1 / 56-12-11-991-01200)
 - 7 Transparency:
Ref. Fig. (1 / 56-12-11-991-01200)
 - 8 Crazing:
Ref. Fig. (1 / 56-12-11-991-01300)
 - 9 Burn spots:
Ref. Fig. (1 / 56-12-11-991-01700)
 - 10 Heating film cracks:
Ref. Fig. (1 / 56-12-11-991-12000)
 - 11 Distortion and optical deviations:

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Ref. Fig. (1 / 56-12-11-991-13100)

NOTE : Distortion and optical deviations have no effect on the structural strength of the window. The sliding window can have distortion and optical deviations in zone 1 and/or zone 2 because of the manufacturing process of the window.

- a Make sure that the visibility is not impaired because of distortion and optical deviations in zone 1. If the visibility is impaired, replace the sliding windows
[Ref. AMM 56-12-11-000-001-A](#) [Ref. AMM 56-12-11-000-001-A-02](#) and
[Ref. AMM 56-12-11-400-001-A](#) [Ref. AMM 56-12-11-400-001-A-02](#).

NOTE : An impaired visibility is permitted in zone 2.

NOTE : Zone 2 is the 50 mm (1.969 in.) wide shaded border from the daylight opening.

- (b) Examine the acrylic sliding windows for burn spots:

Ref. Fig. (1 / 56-12-11-991-01800)

- 1 Point the FLASHLIGHT to the outer surface of the window panel, at an angle of 45 degrees.
- 2 Examine the busbar for burn spots.

- (c) Examine the acrylic sliding windows for crazing:

Ref. Fig. (1 / 56-12-11-991-01900)

- 1 Put the RULER 150 MM (6 IN) - METALLIC perpendicular to the surface of the sliding window panel where you see the maximum depth of the crazing.
- 2 Look at the ruler reflection at an angle of 45 degrees to find the depth of the crazing.

NOTE : The correct depth is approximately two times the measured depth.

NOTE : You can use the MAGNIFYING GLASS POWER 10X as an optical aid where the depth of the crazing is at a minimum.

- (2) Do an inspection of the glass sliding windows (PPG supplier, P/N NP... and SGS supplier, P/N SPSA320-3-X-X or P/N SPSA320-4-X-X, P/N SPSA320-7-X-X or P/N SPSA320-8-X-X):

NOTE : In the figures about the permitted damage on the sliding windows, the word "impaired" refers to the visibility. The visibility is impaired when the defects have an unwanted effect on it.

If the visibility is impaired, it is necessary to replace the window panel.

- (a) Examine the glass sliding windows for:

- 1 Cracks:
Ref. Fig. (1 / 56-12-11-991-02100)
- 2 Scratches:
Ref. Fig. (1 / 56-12-11-991-02200)

NOTE : There are two procedures to measure the depth of the scratches. The fingernail procedure measures the approximate depth of the scratches. The MITUTOYO surface-tester procedure measures the accurate depth of the scratches. If a MITUTOYO surface tester is available, Airbus

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recommends that you use the MITUTOYO surface-tester procedure because it gives accurate results.

- a If you find scratches, do a check of the depth of the scratches:
 - With the MITUTOYO **SURFACE TESTER** (SJ201) (SJ201) or MITUTOYO **SURFACE TESTER** (SJ210) (SJ210) [Ref. AMM 56-10-00-200-004-A](#) [Ref. AMM 56-10-00-200-004-A-01](#), or
 - With your fingernail.
- 3 Chips:
Ref. Fig. (1 / 56-12-11-991-02300)
- 4 Delamination:
Ref. Fig. (1 / 56-12-11-991-02400)
- 5 Bubbles:
Ref. Fig. (1 / 56-12-11-991-02500)
- 6 Discoloration:
Ref. Fig. (1 / 56-12-11-991-02500)
- 7 Transparency:
Ref. Fig. (1 / 56-12-11-991-02500)
- 8 Interlayer microflakes:
Ref. Fig. (1 / 56-12-11-991-01400)
- 9 Burn spots:
Ref. Fig. (1 / 56-12-11-991-02600)
- 10 Heating film cracks:
Ref. Fig. (1 / 56-12-11-991-12100)
- 11 Distortion and optical deviations:
Ref. Fig. (1 / 56-12-11-991-13200)

NOTE : Distortion and optical deviations have no effect on the structural strength of the window. The sliding window can have distortion and optical deviations in zone 1 and/or zone 2 because of the manufacturing process of the window.

- a Make sure that the visibility is not impaired because of distortion and optical deviations in zone 1. If the visibility is impaired, replace the sliding windows [Ref. AMM 56-12-11-000-001-A](#) [Ref. AMM 56-12-11-000-001-A-02](#) and [Ref. AMM 56-12-11-400-001-A](#) [Ref. AMM 56-12-11-400-001-A-02](#).

NOTE : An impaired visibility is permitted in zone 2.

NOTE : Zone 2 is the 50 mm (1.969 in.) wide shaded border from the daylight opening.

- (b) Examine the glass sliding windows for burn spots:
Ref. Fig. (1 / 56-12-11-991-02700)
 - 1 Point the LIGHT SOURCE - WHITE to the outer surface of the window panel, at an angle of 45 degrees.
 - 2 Examine the busbar for burn spots.

5. Close-up

Subtask 56-12-11-865-069-A

[On A/C 001-999](#)

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- A. Remove the SAFETY CLIP - CIRCUIT BREAKER and the tag(s) and close this (these) circuit breaker(s):

PANEL	DESIGNATION	FIN	LOCATION
121VU	COM NAV/RADAR/2	5SQ2	K14
121VU	COM NAV/RADAR/1	5SQ1	K13
122VU	ANTI ICE/WINDOWS/R	4DG2	W14
122VU	ANTI ICE/WHC/2	5DG2	W13
122VU	ANTI ICE/WINDOWS/L	4DG1	X14
122VU	ANTI ICE/WHC/1	5DG1	X13
123VU	ANTI ICE/L/WSHLD	1DG1	AF10
123VU	ANTI ICE/R/WSHLD	1DG2	AF03

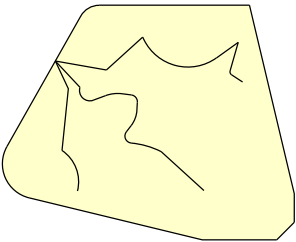
Subtask 56-12-11-942-055-A

On A/C 001-999

- B. Close Access

- (1) Close the sliding windows [Ref. AMM 56-12-11-410-001-A](#).
- (2) Remove the access platform(s).
- (3) Remove the ground support and maintenance equipment, the special and standard tools and all other items.

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DESCRIPTION OF THE DEFECT	CORRECTIVE ACTION
<u>CRACKS</u> DEFECT LINE TYPES IN THE DEPTH OF THE PLY OF THE ACRYLIC SLIDING WINDOW. 	<u>OUTER/INNER PLY</u> NOT PERMITTED DO NOT APPLY ELECTRICAL HEAT TO THE ACRYLIC SLIDING WINDOW WITH CRACKS IF THERE IS AN ARCING.

N_MM_561211_6_ACM0_01_01

Acrylic Sliding Window - Permitted Damage
Figure 1 / Graphic 56-12-11 -991-00200

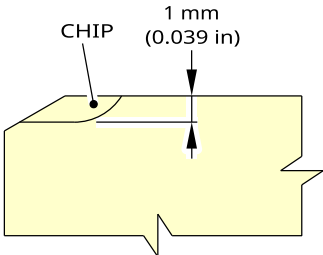
TF-N
AIRCRAFT MAINTENANCE MANUAL
SLIDING WINDOWS - INSPECTION/CHECK

DESCRIPTION OF THE DEFECT	CORRECTIVE ACTION
<u>SCRATCHES</u> DEFECT LINE TYPE ON THE EXTERNAL SURFACE OF THE ACRYLIC SLIDING WINDOW.	<u>OUTER/INNER PLY</u> MAXIMUM DEPTH: 1 mm (0.039 in) PERMITTED IF THE SCRATCHES ARE IN THE TOLERANCE.

N_MM_561211_6_AEM0_01_01

Acrylic Sliding Window - Permitted Damage
Figure 1 / Graphic 56-12-11 -991-00300

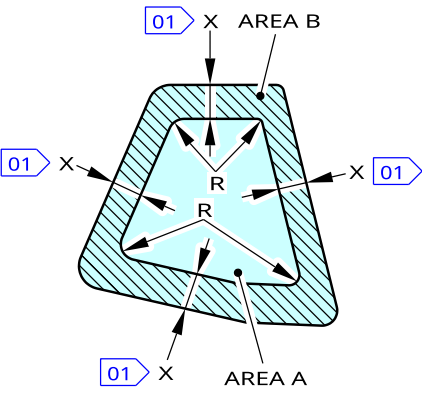
TF-N
AIRCRAFT MAINTENANCE MANUAL
SLIDING WINDOWS - INSPECTION/CHECK

DESCRIPTION OF THE DEFECT	CORRECTIVE ACTION
<u>CHIPS</u> FLAKES OF BROKEN GLASS FROM THE SURFACE AND THE EDGES OF THE ACRYLIC SLIDING WINDOW. <u>DESCRIPTION OF A CHIP</u> 	<u>OUTER/INNER PLY</u> MAXIMUM DEPTH: 1 mm (0.039 in) PERMITTED IF THE CHIPS ARE IN THE TOLERANCE.

N_MM_561211_6_AGM0_01_01

Acrylic Sliding Window - Permitted damage
Figure 1 / Graphic 56-12-11 -991-00400

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AIRCRAFT MAINTENANCE MANUAL
SLIDING WINDOWS - INSPECTION/CHECK

DESCRIPTION OF DEFECT	CORRECTIVE ACTION
<u>DELAMINATION</u> SEPARATION OF THE MATERIAL OCCURS FREQUENTLY AROUND THE EDGES, THE SENSORS AND IN AREAS WITH NO HEATING. THE AREA WITH DELAMINATION CAN BE TRANSPARENT. THIS AREA WILL NOT BE TRANSPARENT IF MOISTURE GOES INTO IT. <u>DELAMINATION LIMITS</u>  X = 76.20 mm (3.00 in) R = 76.20 mm (3.00 in) NOTE: 01 DIMENSION X IS MEASURED FROM THE EDGE OF THE DAYLIGHT OPENING (EDGE OF SEAL ON EXTERNAL SURFACE).	<u>OUTER/INNER PLY</u> AREA A: – SATISFACTORY IF EACH DELAMINATION AREA HAS A SURFACE OF NO MORE THAN 1290 mm² (2.0 in²). – THE TOTAL SURFACE OF THESE AREAS MUST NOT BE MORE THAN 3871 mm² (6.0 in²). – MAKE SURE THAT THE DELAMINATION AREAS DO NOT MAKE THE VISIBILITY UNSATISFACTORY. AREA B: DELAMINATION ON AREA B IS PERMITTED. IF THE DELAMINATION AREA IS ACROSS THE TWO ZONES (A AND B), YOU MUST SEE IT AS TWO DIFFERENT DELAMINATION CONDITIONS (ONE IN AREA A AND ONE IN AREA B). THUS, YOU MUST DO TWO ISOLATED CORRECTIONS: ONE FOR AREA A AND ONE FOR AREA B.

NOTE:

THERE ARE STRESS-RELEASE SEPARATORS ABOVE THE BUSBARS IN THE VISION AREA.
IT IS POSSIBLE TO THINK THAT THEY CAN BE DELAMINATION AREAS.

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Acrylic Sliding Window - Permitted Damage
Figure 1 / Graphic 56-12-11 -991-00500

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DESCRIPTION OF THE DEFECT	CORRECTIVE ACTION
<u>BUBBLES</u> WHEN THE ACRYLIC MATERIAL BECOMES TOO HOT, THE GAS CAUSES SMALL BUBBLES. THE BUBBLES DO NOT DECREASE THE STRUCTURAL INTEGRITY.	<u>OUTER/INNER PLY</u> ISOLATED BUBBLES ARE PERMITTED TO A MAXIMUM DIAMETER OF 6.4 mm (0.25 in) IF THE VISIBILITY IS NOT IMPAIRED.
<u>CAUSES OF THE DISCOLORATION:</u> - BURNING OF THE CONDUCTIVE FILM. - CARBON DISCOLORATION BECAUSE OF THE REFLECTION FROM THE BUSBAR AND THE CONDUCTIVE FILM. THIS REFLECTION CHANGES (RAINBOW COLORS) WHEN YOU LOOK AT IT FROM AN TILTED ANGLE. <u>CAUSES OF THE BURNING:</u> - FREQUENT HIGH TEMPERATURES WHICH MAKE THE INTERLAYER BECOME BROWN AT HOT POINTS. - BLACK OR BROWN DISCOLORATION OR BURNING ARE POSSIBLE DEFECTS. DISCOLORATION OR BURNING HAS NO EFFECT ON THE STRUCTURAL INTEGRITY.	PERMITTED IF THE VISIBILITY IS NOT IMPAIRED.
<u>TRANSPARENCY</u> HALOS ON THE SURFACE OF THE ACRYLIC SLIDING WINDOW CAN MAKE IT LESS TRANSPARENT.	PERMITTED IF THE VISIBILITY IS NOT IMPAIRED.

N_MM_561211_6_ALM0_01_02

Acrylic Sliding Window - Permitted Damage
Figure 1 / Graphic 56-12-11 -991-01200

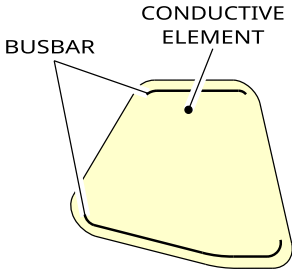
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DESCRIPTION OF THE DEFECT	CORRECTIVE ACTION
<u>CRAZING</u> - SERIES OF CRACKS PERPENDICULAR TO THE SURFACE BUT WHICH DO NOT EXTEND THROUGH THE PANEL. - YOU CAN SEE SOME SURFACE CRACKS ONLY FROM A SPECIFIED ANGLE (FOR EXAMPLE WHEN THERE IS A LIGHT REFLECTION FROM THE CRACK SURFACE). - TENSION AND CONTAMINATION CAUSE CRAZING.	<u>INNER/OUTER PLY</u> - CRAZING ON THE FULL SURFACE IS PERMITTED IF THE DEPTH IS NOT MORE THAN 0.5 mm (0.02 in). CRAZING ON THE OUTER SURFACE OF AN AREA OR A NUMBER OF ISOLATED AREAS IS PERMITTED: - WHERE THE TOTAL AREA IS NOT MORE THAN 500 mm² (0.78 in²) - IF THE DEPTH IS NOT MORE THAN 1.0 mm (0.039 in). - CRAZING ON ONE AREA OF THE OUTER SURFACE WHICH IS NOT MORE THAN 100 mm² (0.16 in²) IS PERMITTED IF THE DEPTH IS NOT MORE THAN 1.5 mm (0.06 in). - NOT PERMITTED IF THE CRAZING IS MORE THAN THE OUTER SURFACE OR NOT IN THE TOLERANCE.

N_MM_561211_6_AMM0_01_01

Acrylic Sliding Window - Permitted Damage
Figure 1 / Graphic 56-12-11 -991-01300

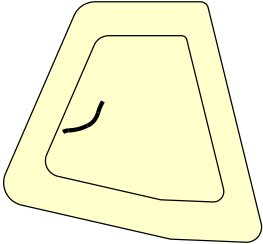
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SLIDING WINDOWS - INSPECTION/CHECK

DESCRIPTION OF THE DEFECTS	CORRECTIVE ACTION
<u>BURN MARK</u> - <u>BUSBAR AND CONDUCTIVE ELEMENT LOCATION</u> 	
- <u>BURN MARK ON THE BUSBAR</u> AN ARCING CAUSES BURN MARK ON THE BUSBAR AREA AFTER A BUSBAR DAMAGE.	PERMITTED IF THE DIAMETER OF THE MARK IS LESS THAN 10 mm (0.39 in). IF THE DIAMETER OF THE MARK IS MORE THAN 10 mm (0.39 in), STOP THE SIDE-SLIDING WINDOW HEATING AS FOLLOWS: - OPEN, SAFETY AND TAG THE CIRCUIT BREAKERS: - FOR THE LEFT SIDE: 4DG1 (122VU) - FOR THE RIGHT SIDE: 4DG2 (122VU).
- <u>BURN MARK ON THE CONDUCTIVE ELEMENT</u> CAUSE OF THE BURN MARK: - DAMAGE TO THE HEATING ELEMENT CAUSES BURN MARK ON THE CONDUCTIVE ELEMENT. - FREQUENT HIGH TEMPERATURES WHICH MAKE THE INTERLAYER BROWN AT HOT POINTS.	PERMITTED IF THE VISIBILITY DOES NOT DECREASE.

N_MM_561211_6_AQM0_01_03

Acrylic Sliding Window - Permitted Damage
Figure 1 / Graphic 56-12-11 -991-01700

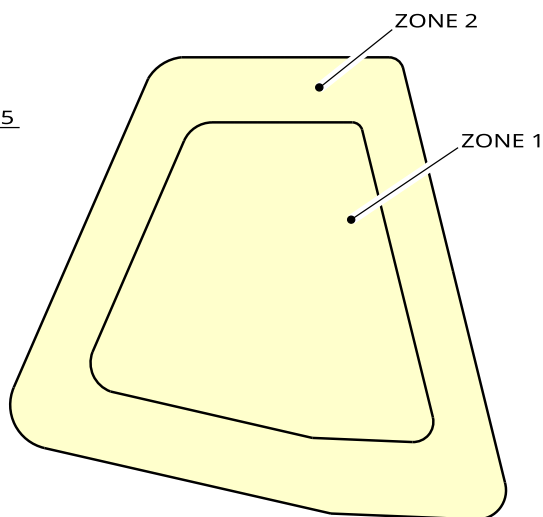
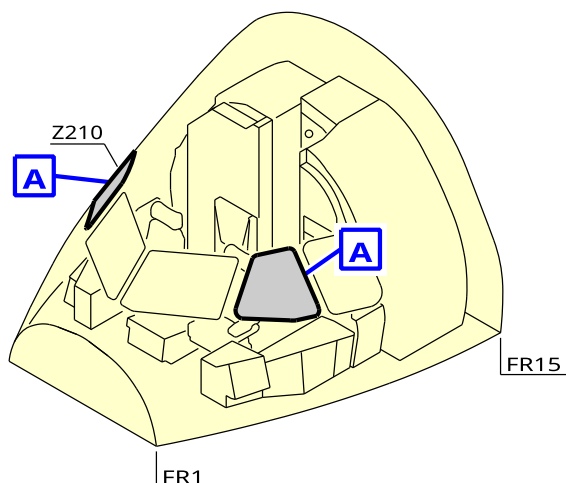
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AIRCRAFT MAINTENANCE MANUAL
SLIDING WINDOWS - INSPECTION/CHECK

DESCRIPTION OF THE DEFECT	DAMAGE LOCATION	PERMITTED/NOT PERMITTED DAMAGE
<u>HEATING FILM CRACKS</u> THE HEATING FILM CRACKS CAN CAUSE AN ELECTRICAL FAILURE OF THE HEATING SYSTEM. THIS TYPE OF DAMAGE HAS NO EFFECT ON THE STRUCTURAL INTEGRITY. NOTE: THE CRACKS OF THE HEATING FILM CAN LOOK THE SAME AS THE CRACKS OF THE MIDDLE PLY. <u>DESCRIPTION OF THE HEATING FILM CRACKS</u> 	BETWEEN THE OUTER PLY AND THE MIDDLE PLY OF THE ACRYLIC SLIDING WINDOW.	THE AIRCRAFT DISPATCH IS PERMITTED IF: -THE VISIBILITY DOES NOT DECREASE -THE CIRCUIT BREAKERS OF THE SLIDING WINDOW ARE OPEN BEFORE THE FLIGHT.

N_MM_561211_6_AAA0_01_01

Acrylic Sliding Window - Permitted Damage
Figure 1 / Graphic 56-12-11 -991-12000

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ZONE 2 = 50 mm (1.969 in) WIDE SHADED BORDER
FROM THE DAYLIGHT OPENING



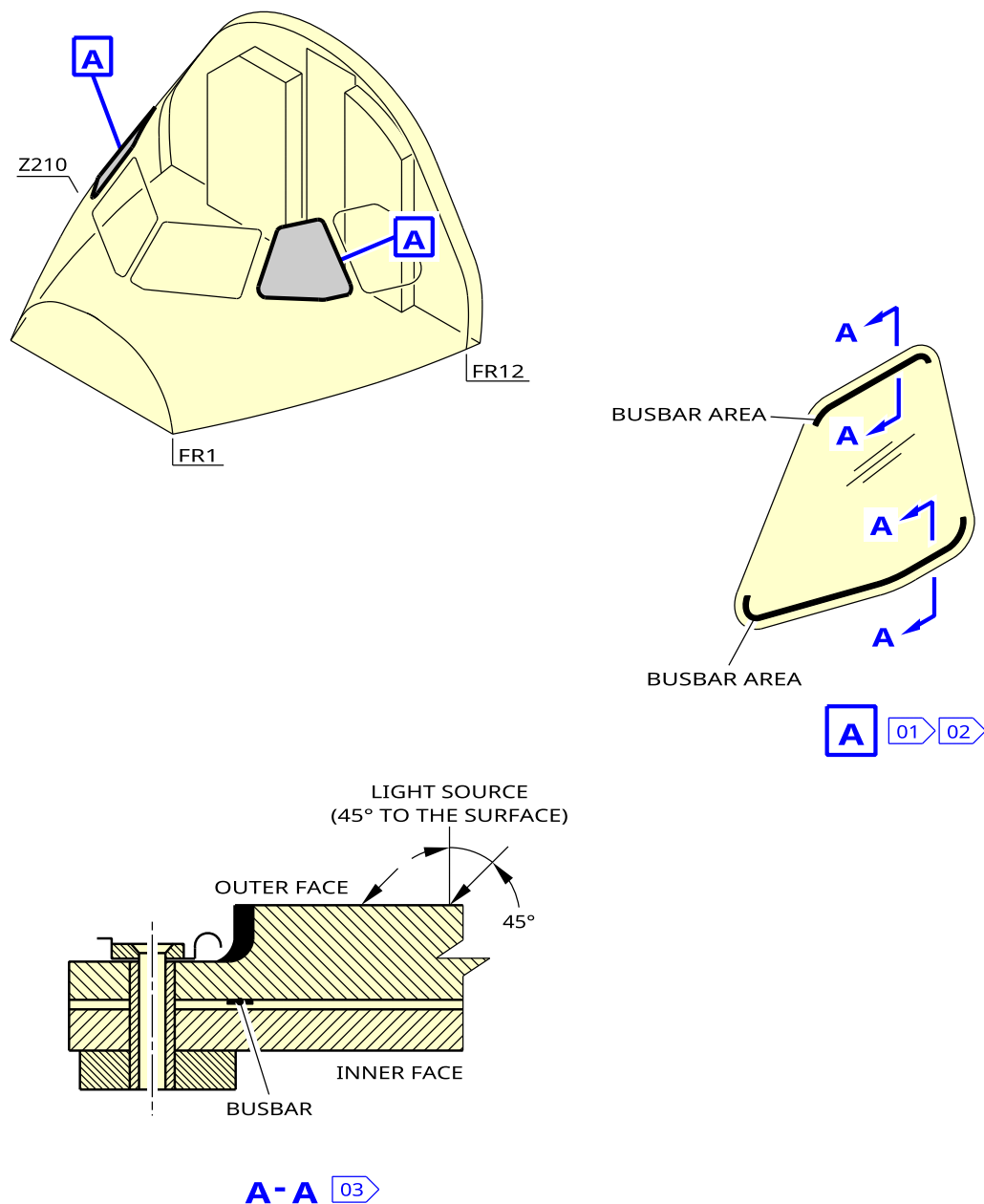
LH SHOWN
RH SYMMETRICAL

DESCRIPTION OF THE DEFECT	CORRECTIVE ACTION
DISTORTION AND OPTICAL DEVIATIONS THE SLIDING WINDOW CAN HAVE DISTORTION AND OPTICAL DEVIATIONS IN ZONE 1 AND/OR ZONE 2 BECAUSE OF THE MANUFACTURING PROCESS OF THE WINDOW.	THIS DAMAGE CAN OCCUR IN ZONE 1 AND/OR ZONE 2. IF THE VISIBILITY IS IMPAIRED BECAUSE OF DISTORTION AND OPTICAL DEVIATIONS IN ZONE 1, REPLACE THE WINDOW PANEL. NOTE: AN IMPAIRED VISIBILITY IS PERMITTED IN ZONE 2.

N_MM_561211_6_SHM1_01_01

Acrylic Sliding Window - Distortion and Optical Deviations
Figure 1 / Graphic 56-12-11 -991-13100

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SLIDING WINDOWS - INSPECTION/CHECK



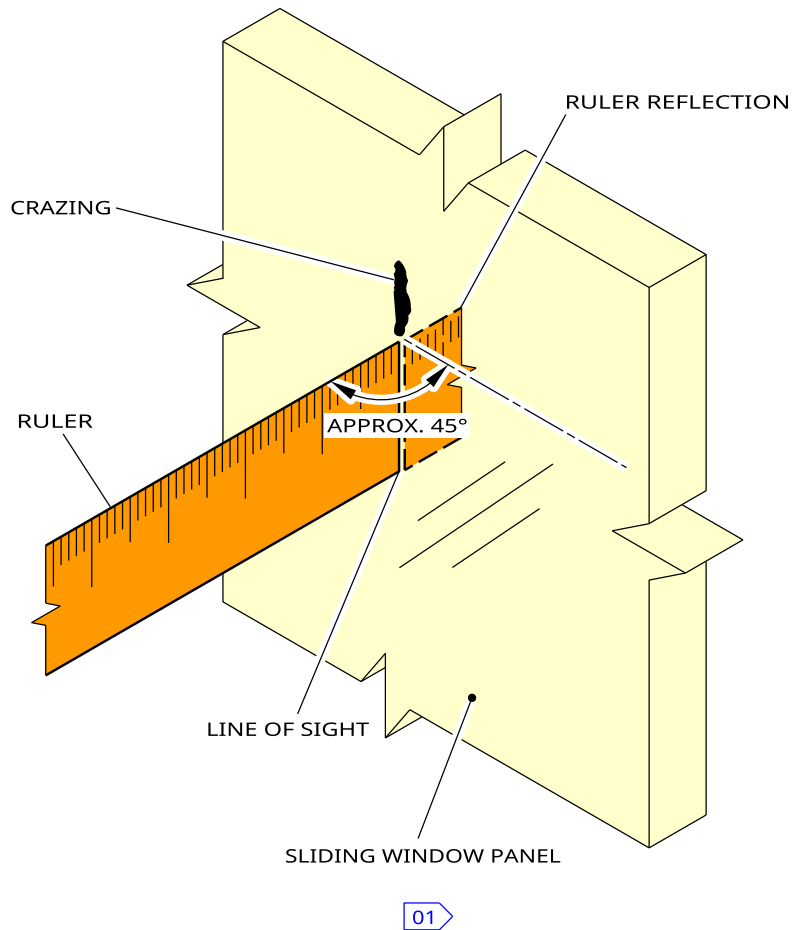
NOTE:

- 01 LH SHOWN, RH SYMMETRICAL
- 02 THE ACRYLIC SLIDING WINDOWS (GKN SUPPLIER, P/N NH...)
- 03 TYPICAL

N_MM_561211_6_FAM0_01_03

Acrylic Sliding Window - Burn Spot Inspection Procedure
Figure 1 / Graphic 56-12-11 -991-01800

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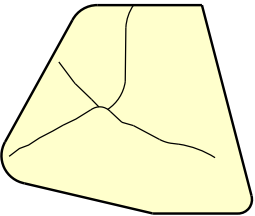
NOTE:

01 TYPICAL

N_MM_561211_6_GAM0_01_01

Acrylic Sliding Window - Crazing Inspection Procedure
Figure 1 / Graphic 56-12-11 -991-01900

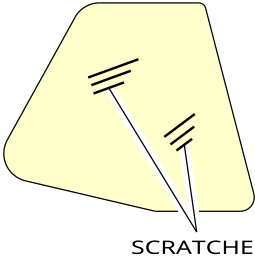
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SLIDING WINDOWS - INSPECTION/CHECK

DESCRIPTION OF THE DEFECT	DAMAGE LOCATION	PERMITTED/NOT PERMITTED DAMAGE
<u>CRACKS</u> DEFECT LINE TYPES IN THE DEPTH OF THE PLY OF THE GLASS SLIDING WINDOW. 	OUTER PLY	ONLY ONE FLIGHT IS PERMITTED IF: -THE VISIBILITY IS NOT IMPARED - THE RELATED FIXED LATERAL WINDOW HAS NO CRACKS.
	INNER/MIDDLE PLY	MIDDLE AND/OR INNER PLY(IES) WITH CRACKS IS(ARE) NOT PERMITTED.

N_MM_561211_6_GBM0_01_02

Glass Sliding Window - Permitted Damage
Figure 1 / Graphic 56-12-11 -991-02100

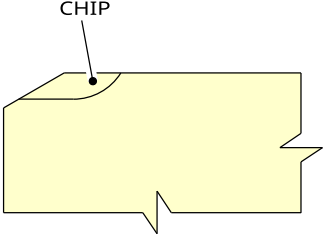
TF-N
AIRCRAFT MAINTENANCE MANUAL
SLIDING WINDOWS - INSPECTION/CHECK

DESCRIPTION OF DEFECT	CORRECTIVE ACTION
<u>SCRATCHES</u> LINE TYPE DEFECTS IN THE EXTERNAL SURFACE OF THE SLIDING WINDOW  SCRATCHES	<u>OUTER PLY</u> ACCEPTABLE IF THE VISIBILITY IS NOT IMPAIRED
	<u>INNER PLY</u> ACCEPTABLE: - IF YOU CANNOT FEEL THE SCRATCHES WHEN YOU TOUCH THE SLIDING WINDOW WITH YOUR FINGER NAIL OR - IF THE DEPTH OF THE SCRATCHES MEASURED WITH THE MITUTOYO SURFACE TESTER SJ201/SJ210 IS LESS THAN 0.05 mm (0.0019 in) (REF. TASK 561000-200-004). NOT ACCEPTABLE: - IF YOU CAN FEEL THE SCRATCHES WHEN YOU TOUCH THE SLIDING WINDOW WITH YOUR FINGER NAIL OR - IF THE DEPTH OF THE SCRATCHES MEASURED WITH THE MITUTOYO SURFACE TESTER SJ201/SJ210 IS MORE THAN 0.05 mm (0.0019 in) (REF. TASK 561000-200-004). NOTE: THERE ARE TWO PROCEDURES TO MEASURE THE DEPTH OF THE SCRATCHES. THE FINGERNAIL PROCEDURE HELPS YOU TO MEASURE AN APPROXIMATE DEPTH OF THE SCRATCHES. THE MITUTOYO SURFACE-TESTER PROCEDURE HELPS YOU TO MEASURE THE ACCURATE DEPTH OF THE SCRATCHES. IF THE MITUTOYO SURFACE TESTER IS AVAILABLE, THE MITUTOYO SURFACE-TESTER PROCEDURE IS RECOMMENDED BECAUSE IT GIVES ACCURATE MEASUREMENT RESULTS.

N_MM_561211_6_GEM0_01_03

Glass Sliding Window - Permitted Damage
Figure 1 / Graphic 56-12-11 -991-02200

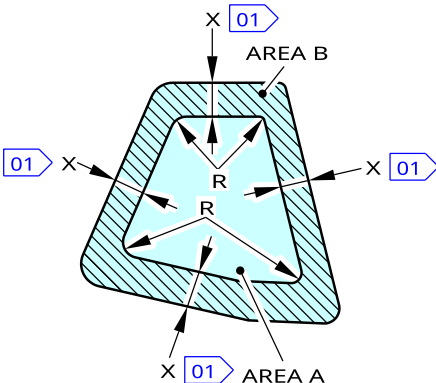
TF-N
AIRCRAFT MAINTENANCE MANUAL
SLIDING WINDOWS - INSPECTION/CHECK

DESCRIPTION OF THE DEFECT	CORRECTIVE ACTION
<u>CHIPS</u> FLAKES OF BROKEN GLASS FROM THE SURFACE AND THE EDGES OF THE GLASS SLIDING WINDOW. <u>DESCRIPTION OF A CHIP</u> 	<u>OUTER PLY</u> PERMITTED IF THE VISIBILITY IS NOT IMPAIRED.
	<u>INNER/MIDDLE PLY</u> NOT PERMITTED

N_MM_561211_6_GGM0_01_01

Glass Sliding Window - Permitted Damage
Figure 1 / Graphic 56-12-11 -991-02300

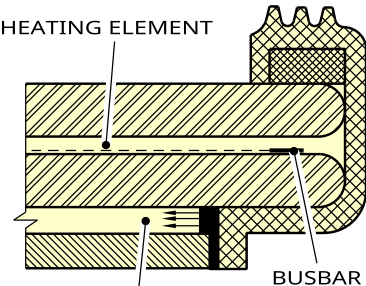
TF-N
AIRCRAFT MAINTENANCE MANUAL
SLIDING WINDOWS - INSPECTION/CHECK

DESCRIPTION OF DEFECT	CORRECTIVE ACTION
<u>DELAMINATION</u> THE SEPARATION OF THE GLASS FROM THE ADJACENT VINYL INTERLAYER OCCURS FREQUENTLY AROUND THE EDGES, THE SENSORS AND IN THE AREAS WITH NO HEATING. THE AREA WITH DELAMINATION CAN BE TRANSPARENT. THIS AREA WILL NOT BE TRANSPARENT IF THE MOISTURE GOES INTO IT. <u>DELAMINATION LIMITS:</u>  X = 76.20 mm (3.00 in) R = 76.20 mm (3.00 in) NOTE: 01 DIMENSION X IS MEASURED FROM THE EDGE OF THE DAYLIGHT OPENING (EDGE OF SEAL ON EXTERNAL SURFACE).	<u>MIDDLE, OUTER AND INNER PLY:</u> AREA A: NOT PERMITTED AREA B: DELAMINATION ON AREA B IS PERMITTED.

N_MM_561211_6_GJM0_01_06

Glass Sliding Window - Permitted Damage
Figure 1 / Graphic 56-12-11 -991-02400

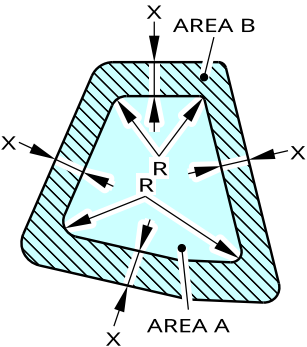
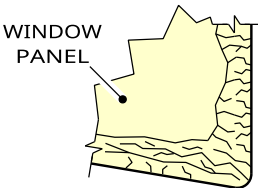
TF-N
AIRCRAFT MAINTENANCE MANUAL
SLIDING WINDOWS - INSPECTION/CHECK

DESCRIPTION OF THE DEFECTS	CORRECTIVE ACTION
BUBBLES WHEN THE VINYL MATERIAL BECOMES TOO HOT, THE GAS CAUSES SMALL BUBBLES. THE BUBBLES DO NOT DECREASE THE STRUCTURAL INTEGRITY.	INNER/MIDDLE/OUTER PLY PERMITTED IF THE VISIBILITY IS NOT IMPAIRED.
DISCOLORATION CAUSES OF THE DISCOLORATION: - CARBON DISCOLORATION BECAUSE OF THE REFLECTION FROM THE BUSBAR AND THE CONDUCTIVE FILM. THIS REFLECTION CHANGES (RAINBOW COLORS) WHEN YOU LOOK AT IT FROM AN TILTED ANGLE. - MIGRATION OF THE BROWN SEALANT TO THE PERIPHERY OF THE OUTER INTERLAYER. THIS INTERLAYER CAN EXTEND TO A WIDTH BETWEEN 10 mm (0.39 in) AND 20 mm (0.78 in) AT THE PERIPHERY OF THE WINDOW PANEL. SEALANT MIGRATION  HEATING ELEMENT BUSBAR SEALANT MIGRATION OUTER FACE	PERMITTED IF THE VISIBILITY IS NOT IMPAIRED.
TRANSPARENCY HALOS ON THE SURFACE OF THE GLASS SLIDING WINDOW CAN MAKE IT LESS TRANSPARENT.	PERMITTED IF THE VISIBILITY IS NOT IMPAIRED.

N_MM_561211_6_GLM0_01_02

Glass Sliding Window - Permitted Damage
Figure 1 / Graphic 56-12-11 -991-02500

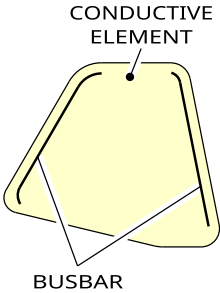
TF-N
AIRCRAFT MAINTENANCE MANUAL
SLIDING WINDOWS - INSPECTION/CHECK

DESCRIPTION OF THE DEFECT	CORRECTIVE ACTION
<u>INTERLAYER MICROFLAKES</u> MICROFLAKES ARE CHEMICAL REACTIONS IN THE PERIPHERY OF THE INTERLAYER THAT MOISTURE AND WINDOW AGEING CAUSE. <u>INTERLAYER MICROFLAKES LIMITS</u>  X = 76.20 mm (3.00 in) R = 76.20 mm (3.00 in) <u>DESCRIPTION OF THE INTERLAYER MICROFLAKES</u> 	AREA A: NOT PERMITTED AREA B: PERMITTED IF THE VISIBILITY IS NOT IMPAIRED. NOTE: - MICROFLAKES ONLY OCCUR IN THE INTERLAYERS - AN EQUIVALENT DAMAGE IN ONE OF THE GLASS PLIES CAN IMMEDIATELY BREAK THE PLY.

N_MM_561211_6_GPM0_01_01

Glass Sliding Window - Permitted Damage
Figure 1 / Graphic 56-12-11 -991-01400

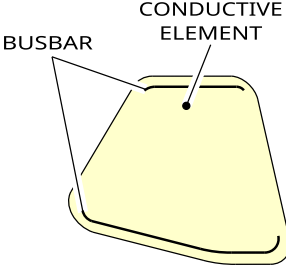
TF-N
AIRCRAFT MAINTENANCE MANUAL
SLIDING WINDOWS - INSPECTION/CHECK

DESCRIPTION OF THE DEFECTS	CORRECTIVE ACTION
<u>BURN MARK</u> - <u>BUSBAR AND CONDUCTIVE ELEMENT LOCATION</u>  - <u>BURN MARK ON THE BUSBAR</u> AN ARCING CAUSES BURN MARK ON THE BUSBAR AREA AFTER A BUSBAR DAMAGE.	PERMITTED IF THE DIAMETER OF THE BURN MARK IS LESS THAN 10 mm (0.39 in). PERMITTED IF THE VISIBILITY DOES NOT DECREASE. IF THE DIAMETER OF THE BURN MARK IS MORE THAN 10 mm (0.39 in), STOP THE SIDE-SLIDING WINDOW HEATING AS FOLLOWS: - OPEN, SAFETY AND TAG THE CIRCUIT BREAKERS: - FOR THE LEFT SIDE: 4DG1 (122VU) - FOR THE RIGHT SIDE: 4DG2 (122VU).
- <u>BURN MARK ON THE CONDUCTIVE ELEMENT</u> CAUSE OF THE BURN MARK: - DAMAGE TO THE HEATING ELEMENT CAUSES BURN MARK ON THE CONDUCTIVE ELEMENT. - FREQUENCY HIGH TEMPERATURES WHICH MAKE THE INTERLAYER BROWN AT HOT POINTS.	

N_MM_561211_6_GQM0_01_04

Glass Sliding Window - Permitted Damage
Figure 1 / Graphic 56-12-11 -991-02600

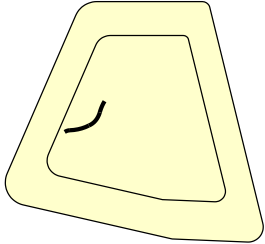
TF-N
AIRCRAFT MAINTENANCE MANUAL
SLIDING WINDOWS - INSPECTION/CHECK

DESCRIPTION OF THE DEFECTS	CORRECTIVE ACTION
<u>BURN MARK</u> - <u>BUSBAR AND CONDUCTIVE ELEMENT LOCATION</u>  - <u>BURN MARK ON THE BUSBAR</u> AN ARCING CAUSES BURN MARK ON THE BUSBAR AREA AFTER A BUSBAR DAMAGE.	PERMITTED IF THE DIAMETER OF THE BURN MARK IS LESS THAN 10 mm (0.39 in). PERMITTED IF THE VISIBILITY DOES NOT DECREASE. IF THE DIAMETER OF THE BURN MARK IS MORE THAN 10 mm (0.39 in), STOP THE SIDE-SLIDING WINDOW HEATING AS FOLLOWS: - OPEN, SAFETY AND TAG THE CIRCUIT BREAKERS: - FOR THE LEFT SIDE: 4DG1 (122VU) - FOR THE RIGHT SIDE: 4DG2 (122VU).
- <u>BURN MARK ON THE CONDUCTIVE ELEMENT</u> CAUSE OF THE BURN MARK: - DAMAGE TO THE HEATING ELEMENT CAUSES BURN MARK ON THE CONDUCTIVE ELEMENT. - FREQUENCY HIGH TEMPERATURES WHICH MAKE THE INTERLAYER BROWN AT HOT POINTS.	

N_MM_561211_6_GQM0_02_00

Glass Sliding Window - Permitted Damage
Figure 2 / Graphic 56-12-11 -991-02600

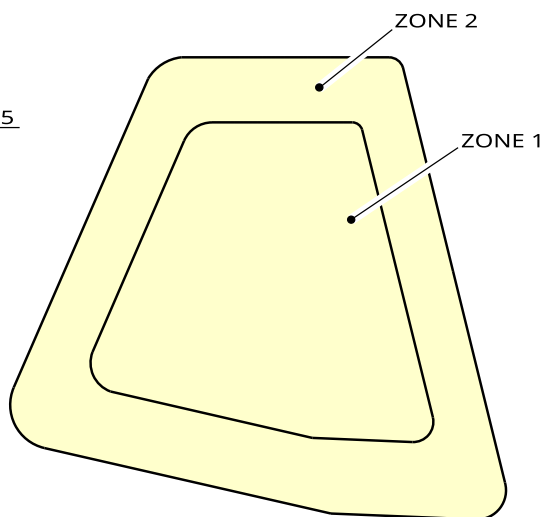
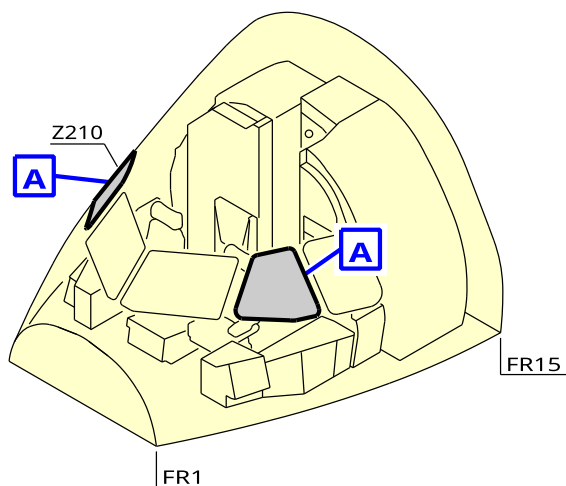
TF-N
AIRCRAFT MAINTENANCE MANUAL
SLIDING WINDOWS - INSPECTION/CHECK

DESCRIPTION OF THE DEFECT	DAMAGE LOCATION	PERMITTED/NOT PERMITTED DAMAGE
<u>HEATING FILM CRACKS</u> THE HEATING FILM CRACKS CAN CAUSE AN ELECTRICAL FAILURE OF THE HEATING SYSTEM. THIS TYPE OF DAMAGE HAS NO EFFECT ON THE STRUCTURAL INTEGRITY. NOTE: THE CRACKS OF THE HEATING FILM CAN LOOK THE SAME AS THE CRACKS OF THE MIDDLE PLY. <u>DESCRIPTION OF THE HEATING FILM CRACKS</u> 	BETWEEN THE OUTER PLY AND THE MIDDLE PLY OF THE GLASS SLIDING WINDOW.	THE AIRCRAFT DISPATCH IS PERMITTED IF: -THE VISIBILITY DOES NOT DECREASE. -THE CIRCUIT BREAKERS OF THE GLASS SLIDING WINDOW ARE OPEN BEFORE THE FLIGHT.

N_MM_561211_6_KTM0_01_01

Glass Sliding Window - Permitted Damage
Figure 1 / Graphic 56-12-11 -991-12100

TF-N
AIRCRAFT MAINTENANCE MANUAL
SLIDING WINDOWS - INSPECTION/CHECK



ZONE 2 = 50 mm (1.969 in) WIDE SHADED BORDER
FROM THE DAYLIGHT OPENING



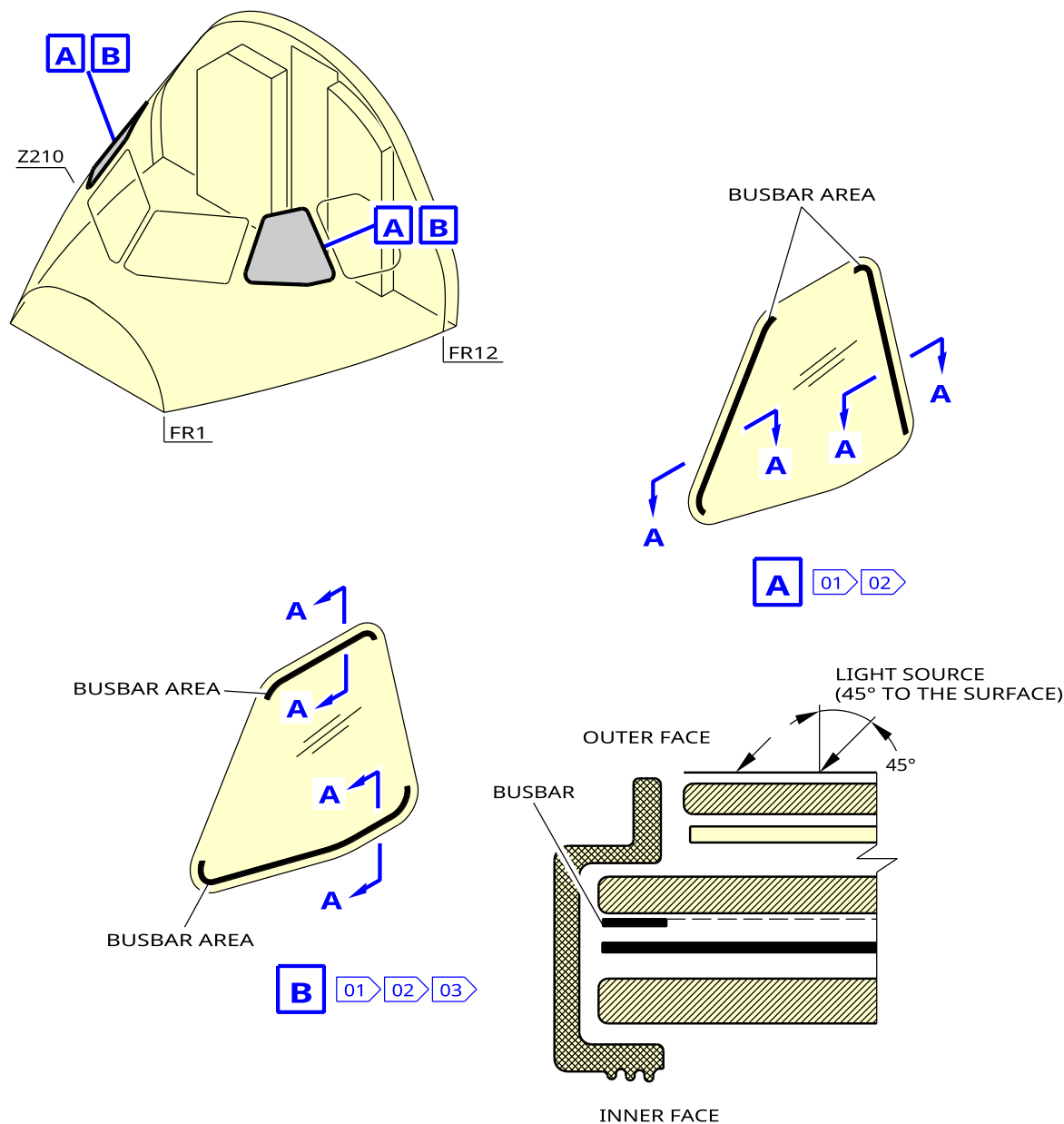
LH SHOWN
RH SYMMETRICAL

DESCRIPTION OF THE DEFECT	CORRECTIVE ACTION
DISTORTION AND OPTICAL DEVIATIONS THE SLIDING WINDOW CAN HAVE DISTORTION AND OPTICAL DEVIATIONS IN ZONE 1 AND/OR ZONE 2 BECAUSE OF THE MANUFACTURING PROCESS OF THE WINDOW.	THIS DAMAGE CAN OCCUR IN ZONE 1 AND/OR ZONE 2. IF THE VISIBILITY IS IMPAIRED BECAUSE OF DISTORTION AND OPTICAL DEVIATIONS IN ZONE 1, REPLACE THE WINDOW PANEL. NOTE: AN IMPAIRED VISIBILITY IS PERMITTED IN ZONE 2.

N_MM_561211_6_SHM2_01_01

Glass Sliding Window - Distortion and Optical Deviations
Figure 1 / Graphic 56-12-11 -991-13200

TF-N
AIRCRAFT MAINTENANCE MANUAL
SLIDING WINDOWS - INSPECTION/CHECK



NOTE:

- 01 LH SHOWN, RH SYMMETRICAL
- 02 THE GLASS SLIDING WINDOW (SGS SUPPLIER, P/N SPSA320-3-X-X OR P/N SPSA320-4-X-X, P/N SPSA320-7-X-X OR P/N SPSA320-8-X-X)
- 03 THE GLASS SLIDING WINDOW (PPG SUPPLIER, P/N NP...)
- 04 TYPICAL

A-A 04

N_MM_561211_6_GSM0_01_03

Glass Sliding Window - Burn Spot Inspection Procedure
Figure 1 / Graphic 56-12-11 -991-02700

 MET.25439539		Task Card Title: NO. 1 ENG FAN COWL LOCKING DEVICE INSP									
Task Card NO: 711300-C2-1-001		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP							
A/C Type: 321		Block Taskcard		Main Skill: PP,II,INSP,ACFT_TECH		PROD Final Acceptance:					
Project No: 01BASPC72608		Related Task Card:		A/C Area: 01							
Internal Check:		Check Type:		Work Order NO: 31909295		Tally Sequence NO: 92					
ENCLOSURE: ☐ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐											
Type Code: DET		Special Code: INSP		MSG Code: LUR, 8, MRB		Request: JOB-SETUP,INSPECTION,INSPECTION,JOEUP					
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRID OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.											
Zones: 430											
DOCUMENT REFERENCES:											
Doc Typ AMM		Reference 71-13-13-220-041-B									
ACCESS PANELS:											
437AL, 438AR											
MATERIAL REQUIREMENTS:											
Part Number No parts assigned to this Task Item		Description				Qty.					
TOOLING REQUIREMENTS:											
Part Number No tools assigned to this Task Item		Description				Qty.					
TASK PREPARATION											
Job Description											
A. PREPARE FOR THE TASK ACCOMPLISHMENT PER AMM TASK 71-13-13-220-041-B.											
STEP COMPLETION SIGN OFF:											
		Qualification / Skill		Number of employees		Duration		Number./Signature		Date	
Mechanic/s		PP/ACFT_TECH				HRS				Day Month Year	
Insp.						HRS					
TASK INSTRUCTION											
Job Description											
A. PERFORM A DETAILED VISUAL INSPECTION OF THE ADJUSTA KEEPERS OF THE RIGHT HANDFAN COWL AS PER AMM TASK 71-13-13-220-041-B.											



Task Card Title:

NO. 1 ENG FAN COWL LOCKING DEVICE INSP



Task Card NO:

711300-C2-1-001

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s			HRS				
Insp.	II/INSP		HRS				

TASK INSTRUCTION

Job Description

B. PERFORM A DETAILED VISUAL INSPECTION OF THE RESTRAINERS OF THE RIGHT HAND FAN COWL AS PER AMM TASK 71-13-13-220-041-B.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s			HRS				
Insp.	INSP/II		HRS				

TASK CLOSE UP

Job Description

A. PERFORM JOB CLOSE-UP PER AMM TASK 71-13-13-220-041-B.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PP/ACFT_TECH		HRS				
Insp.			HRS				

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below

Work order No: _____

Note: Make sure to creat the finding Work Order on task card level.

Remark's: _____



Task Card Title:

NO. 1 ENG FAN COWL LOCKING DEVICE INSP

Task Card NO:

711300-C2-1-001

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

(1) If Task card interrupted or performed by different personals updates the system and update the following according (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 2 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 3 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 4 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 5 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____

**Task Completion Close Up And
System Update**

Station**Date****Signature**

TF-N
AIRCRAFT MAINTENANCE MANUAL
LATCH - FAN COWL DOOR - INSPECTION/CHECK

TASK 71-13-13-220-041 -B

On A/C 003-250

Detailed Inspection of the Keeper Assy Fan Cowl Locking Device

1. Reason for the Job

Refer to the MPD TASK: 711300-C2

DETAILED INSPECTION OF THE KEEPER ASSY FAN COWL LOCKING DEVICE

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	AR	WARNING NOTICE(S)

B. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
FOR 1000EM1(ENGINE-1)	
437AL , 438AR	
FOR 1000EM2(ENGINE-2)	
447AL , 448AR	

C. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 71-13-00-010-040-B	Opening of the Fan Cowl Doors
Ref. AMM 71-13-00-410-040-B	Closing of the Fan Cowl Doors
Ref. AMM 71-13-13-000-041-A	Removal of the Fan Cowl Door Latches and Keepers
Ref. AMM 71-13-13-400-041-A	Installation of the Fan Cowl Door Latches and Keepers
Ref. Fig. (1 / 71-13-13-991-90300)	

3. Job Set-up

Subtask 71-13-13-941-055-A

On A/C 003-250

A. Safety Precautions

- (1) On the center pedestal, on the ENG panel 115VU:
 - (a) Put a WARNING NOTICE(S) to tell persons not to start the engine.
- (2) Make sure that the engine shutdown occurred five minutes or more before you do this procedure.
- (3) On the overhead maintenance panel 50VU:
 - (a) Make sure that the ON legend of the ENG/FADEC GND PWR/1(2) pushbutton switch is off.
 - (b) Put a WARNING NOTICE(S) to tell persons not to energize the FADEC 1(2).

Subtask 71-13-13-010-057-A

On A/C 003-250

B. Get Access

CAUTION: USE YOUR HANDS TO OPERATE THE FAN COWL LATCHES. IF YOU USE SOMETHING DIFFERENT FROM YOUR HANDS TO OPERATE THE LATCHES, YOU CAN CAUSE DAMAGE TO THE LATCHES.

- (1) Open the fan cowl doors: [Ref. AMM 71-13-00-010-040-B](#)
- (2) FOR 1000EM1 (ENGINE-1)
437AL 438AR
- (3) FOR 1000EM2 (ENGINE-2)
447AL 448AR

4. Procedure

Ref. Fig. 1 / 71-13-13-991-90300

TF-N
AIRCRAFT MAINTENANCE MANUAL
LATCH - FAN COWL DOOR - INSPECTION/CHECK

Subtask 71-13-13-220-053-A

On A/C 003-250

A. Visual Inspection of the Adjustable Keepers of the Right-Hand Fan Cowl

- (1) Do a visual inspection of adjustable keepers of the right-hand fan cowl as follows:
 - (a) Examine the adjusta-keepers for cracks. If any cracks are found, replace the adjustable keepers [Ref. AMM 71-13-13-000-041-A](#) and [Ref. AMM 71-13-13-400-041-A](#).
 - (b) Examine the clevises of the adjustable keepers for cracks. If any cracks are found, replace the clevises [Ref. AMM 71-13-13-000-041-A](#) and [Ref. AMM 71-13-13-400-041-A](#).
 - (c) Make sure that the adjusta-keepers are correctly attached. Tighten or replace the attachment hardware as necessary [Ref. AMM 71-13-13-000-041-A](#) and [Ref. AMM 71-13-13-400-041-A](#).
 - (d) Clevis use recommendation

CAUTION: DO NOT TURN THE CLEVISES. IF YOU TURN THE CLEVISES, YOU CAN CAUSE DAMAGE TO THE HOUSING GUIDE. IF THE GUIDE IS DAMAGED, THE CLEVIS CAN TURN. IF THE CLEVIS CAN TURN, THE CLEVIS CAN NOT ENGAGE CORRECTLY WITH THE LATCH AND CAN KEEP THE LATCH FROM CLOSING CORRECTLY. IF THE LATCHES ARE NOT CLOSED CORRECTLY, THE FAN COWLS CAN OPEN IN FLIGHT.

- 1 Make sure the guides of the adjustable keepers are not broken and that the clevises will not turn. If the guides are broken or you can turn the clevises, replace the adjustable keepers [Ref. AMM 71-13-13-000-041-A](#) and [Ref. AMM 71-13-13-400-041-A](#).

Subtask 71-13-13-220-055-A

On A/C 003-250

B. Visual Inspection of the Restrainers of the Right-Hand Fan Cowl

POST SB RA32071-117

- (1) Do a visual inspection of the restrainers of the right-hand fan cowl as follows:
 - (a) Examine the restrainer for cracks. If any cracks are found, replace the restrainers.
 - (b) Make sure that the restrainers are correctly attached. Tighten or replace the attachment hardware as necessary.

END OF POST SB RA32071-117

5. Close-up

Subtask 71-13-13-410-055-A

On A/C 003-250

A. Close Access

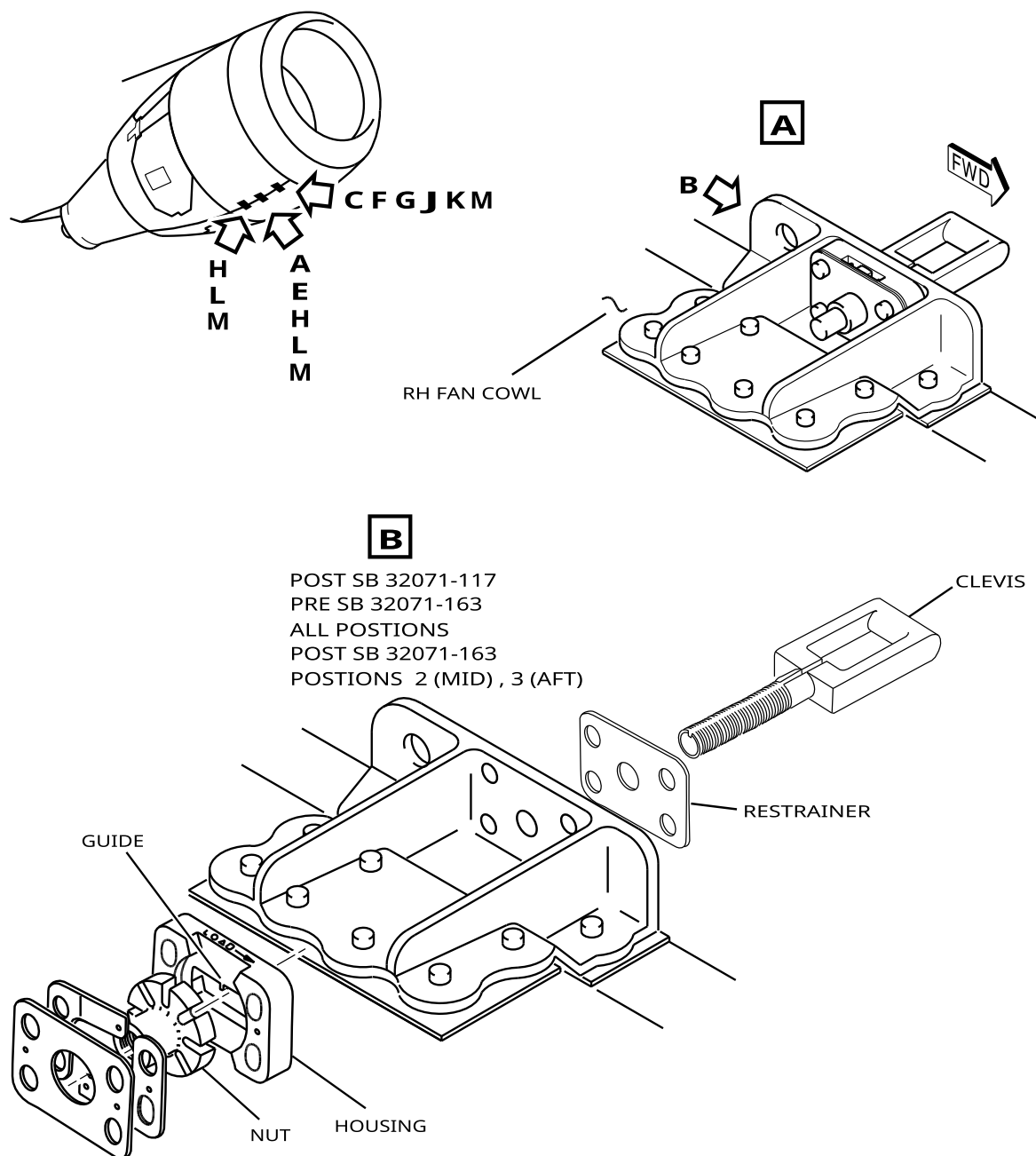
CAUTION: USE YOUR HANDS TO OPERATE THE FAN COWL LATCHES. IF YOU USE SOMETHING DIFFERENT FROM YOUR HANDS TO OPERATE THE LATCHES, YOU CAN CAUSE DAMAGE TO THE LATCHES.

- (1) Make sure that the work area is clean and clear of tools and other items.
- (2) Close the fan cowl doors: [Ref. AMM 71-13-00-410-040-B](#)
- (3) FOR 1000EM1 (ENGINE-1)

TF-N
AIRCRAFT MAINTENANCE MANUAL
LATCH - FAN COWL DOOR - INSPECTION/CHECK

- 437AL 438AR
(4) FOR 1000EM2 (ENGINE-2)
447AL 448AR

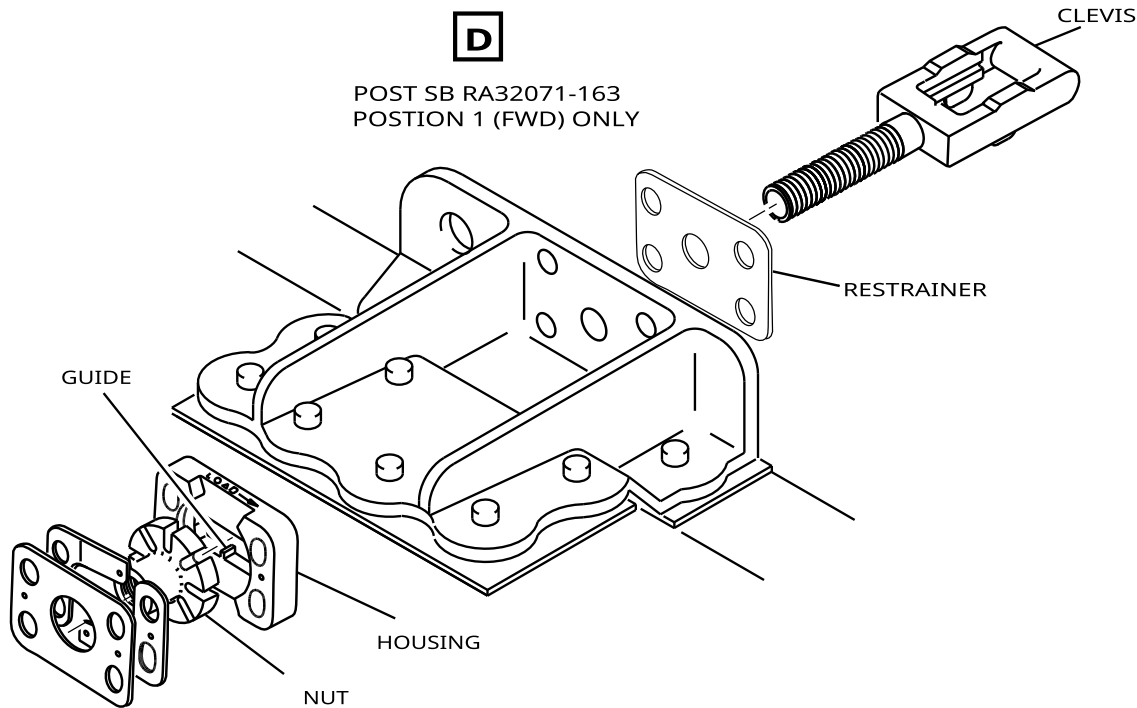
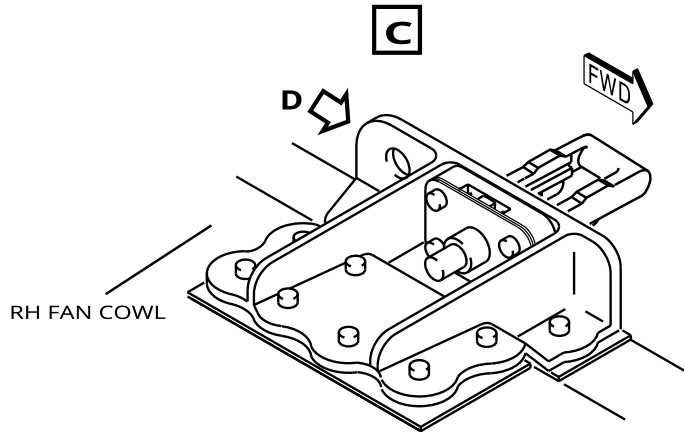
TF-N
AIRCRAFT MAINTENANCE MANUAL
LATCH - FAN COWL DOOR - INSPECTION/CHECK



N_MM_711313_6_RAM0_01_01

Right Fan Cowl Adjusta-Keepers and Restrainers
Figure 1 / Graphic 71-13-13 -991-90300

TF-N
AIRCRAFT MAINTENANCE MANUAL
LATCH - FAN COWL DOOR - INSPECTION/CHECK



CFM-521-01

NOTE:
ADJUSTA-KEEPER SHOWN
DISASSEMBLED FOR CLARITY
DO NOT DISASSEMBLE THE
KEEPER ASSEMBLY

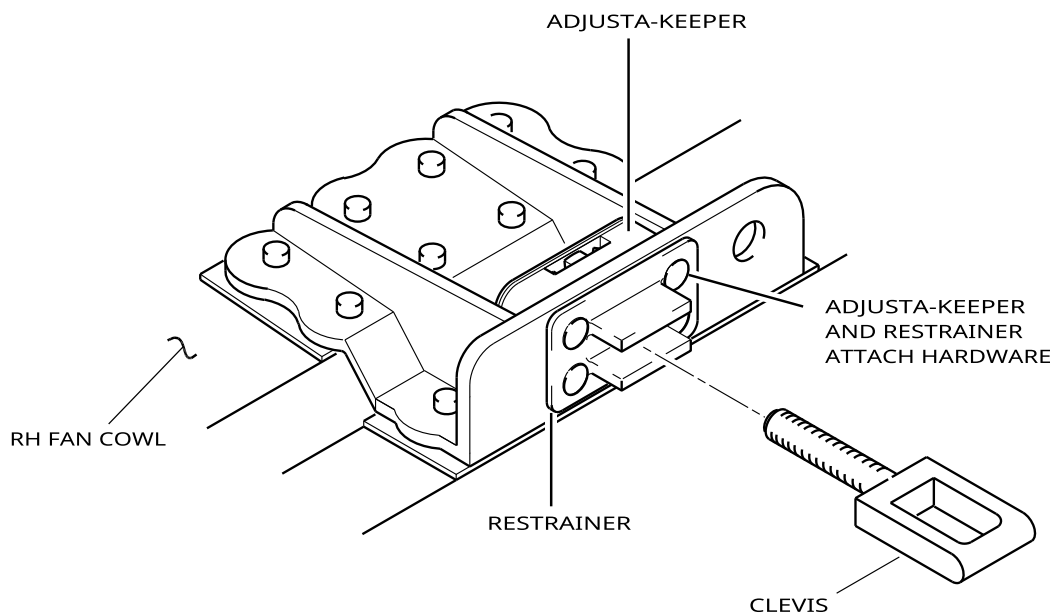
N_MM_711313_6_RAM0_02_01

Right Fan Cowl Adjusta-Keepers and Restrainers
Figure 2 / Graphic 71-13-13 -991-90300

TF-N
AIRCRAFT MAINTENANCE MANUAL
LATCH - FAN COWL DOOR - INSPECTION/CHECK

E

POST SB 32071-117
PRE SB 32071-163
ALL POSTIONS
POST SB 32071-163
POSTIONS 2 (MID) , 3 (AFT)



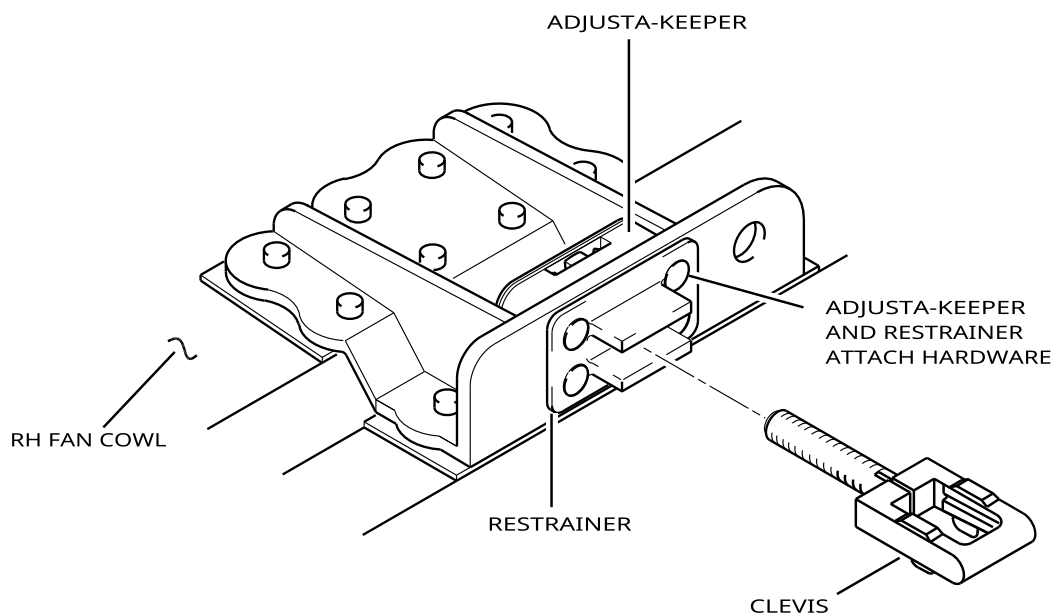
Right Fan Cowl Adjusta-Keepers and Restrainers
Figure 3 / Graphic 71-13-13 -991-90300

N_MM_711313_6_RAM0_03_00

TF-N
AIRCRAFT MAINTENANCE MANUAL
LATCH - FAN COWL DOOR - INSPECTION/CHECK

F

POST SB RA32071-163
POSITION 1 ONLY(FWD ONLY)
POST SB RA32071-117

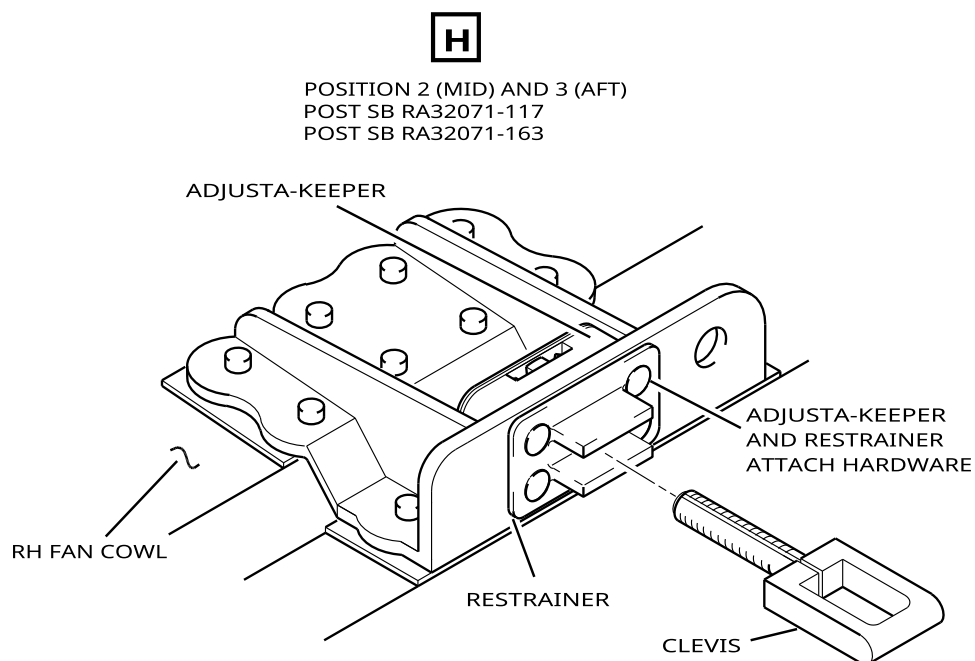
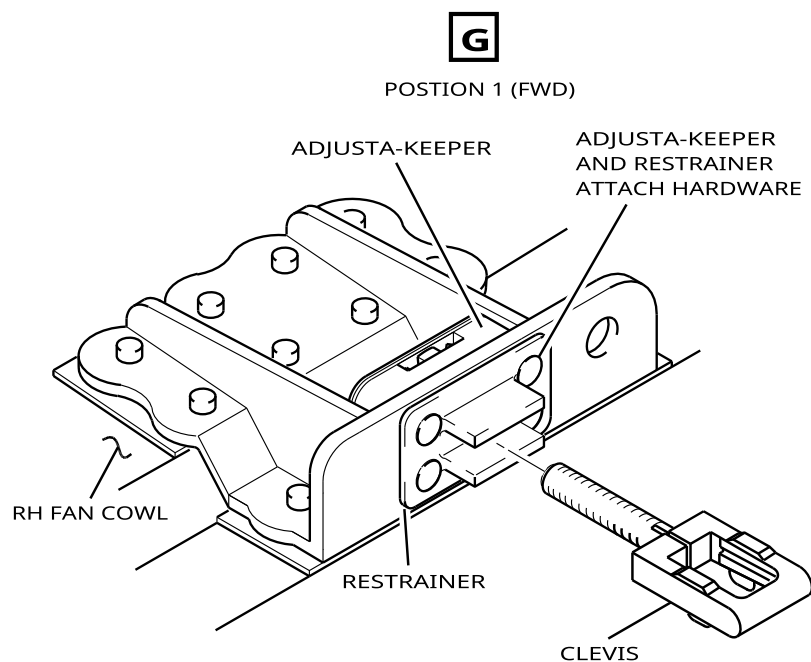


CFM-457-02

N_MM_711313_6_RAM0_04_00

Right Fan Cowl Adjusta-Keepers and Restrainers
Figure 4 / Graphic 71-13-13 -991-90300

TF-N
AIRCRAFT MAINTENANCE MANUAL
LATCH - FAN COWL DOOR - INSPECTION/CHECK



N_MM_711313_6_RAM0_05_00

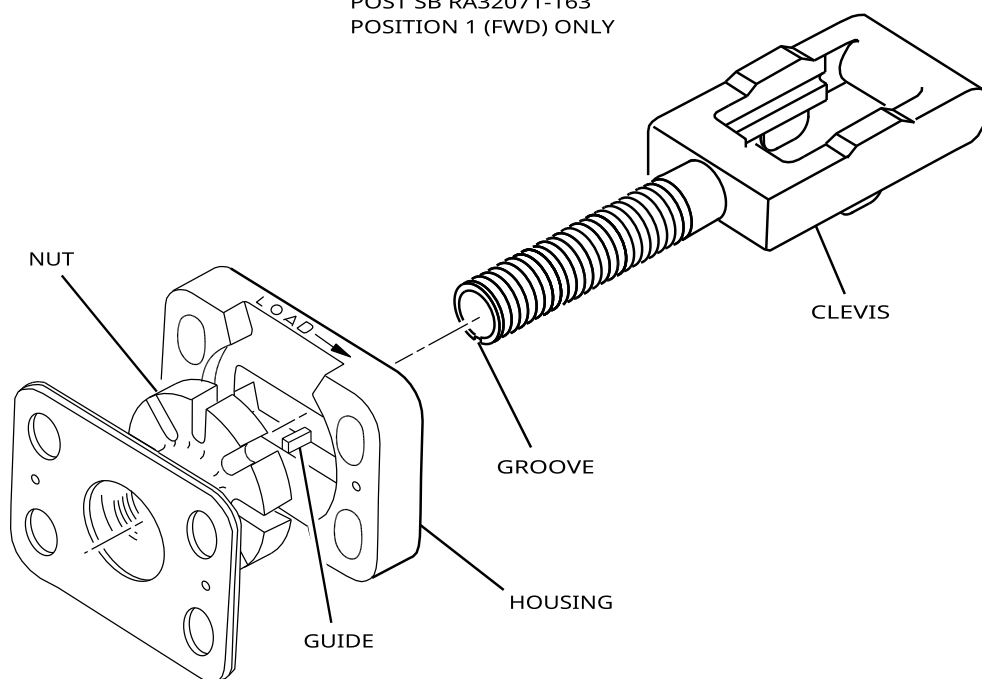
Right Fan Cowl Adjusta-Keepers and Restrainers
Figure 5 / Graphic 71-13-13 -991-90300

CFM-520-01

TF-N
AIRCRAFT MAINTENANCE MANUAL
LATCH - FAN COWL DOOR - INSPECTION/CHECK



POST SB RA32071-163
POSITION 1 (FWD) ONLY



NOTE:

ADJUSTA-KEEPER SHOWN
DISASSEMBLED FOR CLARITY
DO NOT DISASSEMBLE THE
KEEPER ASSEMBLY

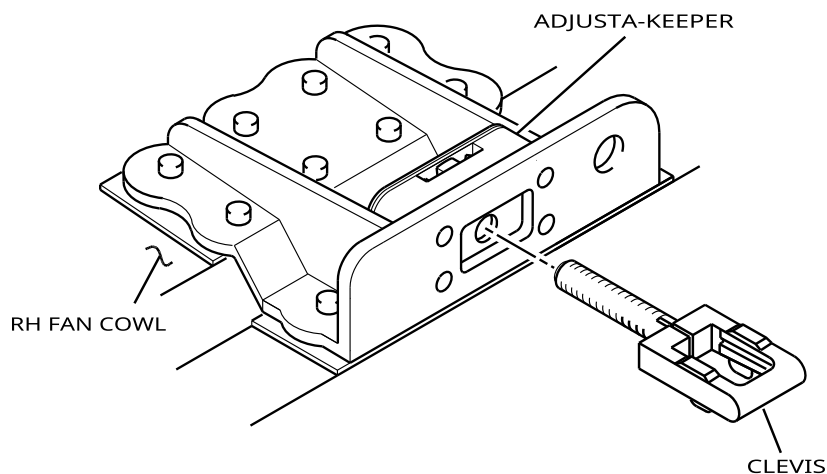
N_MM_711313_6_RAM0_06_00

Right Fan Cowl Adjusta-Keepers and Restrainers
Figure 6 / Graphic 71-13-13 -991-90300

TF-N
AIRCRAFT MAINTENANCE MANUAL
LATCH - FAN COWL DOOR - INSPECTION/CHECK

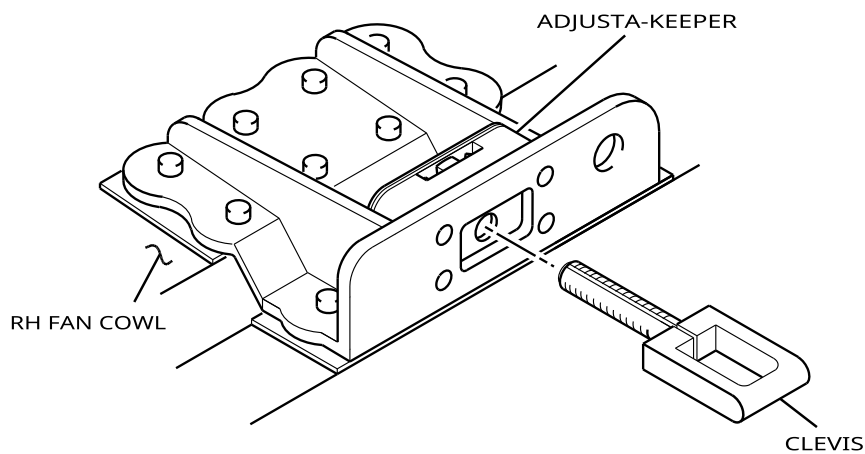
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POST SB RA32071-163
PRE SB RA32071-117
POSITION 1 (FWD)



L

POST SB RA320-71-163
PRE SB RA32071-117
POSITION 2 (CENTER) AND 3 (AFT)



N_MM_711313_6_RAM0_07_00

Right Fan Cowl Adjusta-Keepers and Restrainers
Figure 7 / Graphic 71-13-13 -991-90300

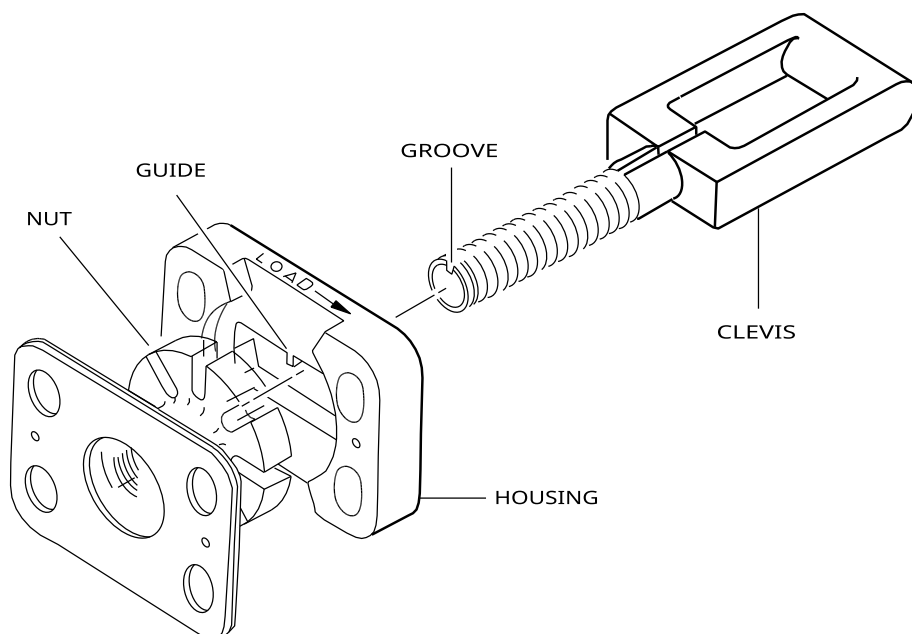
CFM-532-00

TF-N
AIRCRAFT MAINTENANCE MANUAL
LATCH - FAN COWL DOOR - INSPECTION/CHECK



PRE AND POST SB RA32071-117
POSITIONS 2 (MID) AND 3 (AFT)

PRE SB RA32071-163
POSITION 1 (FWD)



NOTE:
ADJUSTA-KEEPER SHOWN
DISASSEMBLED FOR CLARITY
DO NOT DISASSEMBLE THE
KEEPER ASSEMBLY

N_MM_711313_6_RAM0_08_00

Right Fan Cowl Adjusta-Keepers and Restraints
Figure 8 / Graphic 71-13-13 -991-90300

CFM-533-00

 MET.25495693		Task Card Title: NO. 2 ENG FAN COWL LOCKING DEVICE INSP							
Task Card NO: 711300-C2-1-002		Work Package: ASP/H-26 C7-CHK			A/C Registration: HZ-ASP				
A/C Type: 321		Block Taskcard		Main Skill: PP,INSP,II,ACFT_TECH		PROD Final Acceptance:			
Project No: 01BASPC72608		Related Task Card:		A/C Area: 02					
Internal Check:		Check Type:		Work Order NO: 31909295		Tally Sequence NO: 93			
ENCLOSURE: ☐ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐									
Type Code: DET		Special Code: INSP		MSG Code: LUR, 8, MRB		Request: JOB-SETUP,INSPECTION,INSPECTION,JOEUP			
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRID OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.									
Zones: 440									
DOCUMENT REFERENCES:									
Doc Typ		Reference							
AMM		71-13-13-220-041-B							
ACCESS PANELS:									
447AL, 448AR									
MATERIAL REQUIREMENTS:									
Part Number		Description				Qty.			
No parts assigned to this Task Item									
TOOLING REQUIREMENTS:									
Part Number		Description				Qty.			
No tools assigned to this Task Item									
TASK PREPARATION									
Job Description									
A. PREPARE FOR THE TASK ACCOMPLISHMENT PER AMM TASK 71-13-13-220-041-B.									
STEP COMPLETION SIGN OFF:									
		Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
							Day	Month	Year
Mechanic/s		PP/ACFT_TECH		HRS					
Insp.				HRS					
TASK INSTRUCTION									
Job Description									
A. PERFORM A DETAILED VISUAL INSPECTION OF THE ADJUSTA KEEPERS OF THE RIGHT HANDFAN COWL AS PER AMM TASK 71-13-13-220-041-B.									



Task Card Title:

NO. 2 ENG FAN COWL LOCKING DEVICE INSP



Task Card NO:

711300-C2-1-002

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s			HRS				
Insp.	INSP/II		HRS				

TASK INSTRUCTION

Job Description

B. PERFORM A DETAILED VISUAL INSPECTION OF THE RESTRAINERS OF THE RIGHT HAND FAN COWL AS PER AMM TASK 71-13-13-220-041-B.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s			HRS				
Insp.	INSP/II		HRS				

TASK CLOSE UP

Job Description

A. PERFORM JOB CLOSE-UP PER AMM TASK 71-13-13-220-041-B.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PP/ACFT_TECH		HRS				
Insp.			HRS				

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below

Work order No: _____

Note: Make sure to creat the finding Work Order on task card level.

Remark's: _____



MET.25495693

Task Card Title:

NO. 2 ENG FAN COWL LOCKING DEVICE INSP



Task Card NO:

711300-C2-1-002

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

(1) If Task card interrupted or performed by different personals updates the system and update the following according (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 2 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 3 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 4 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 5 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____

**Task Completion Close Up And
System Update**

Station

Date

Signature

TF-N
AIRCRAFT MAINTENANCE MANUAL
LATCH - FAN COWL DOOR - INSPECTION/CHECK

TASK 71-13-13-220-041 -B

On A/C 003-250

Detailed Inspection of the Keeper Assy Fan Cowl Locking Device

1. Reason for the Job

Refer to the MPD TASK: 711300-C2

DETAILED INSPECTION OF THE KEEPER ASSY FAN COWL LOCKING DEVICE

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	AR	WARNING NOTICE(S)

B. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
FOR 1000EM1(ENGINE-1)	
437AL , 438AR	
FOR 1000EM2(ENGINE-2)	
447AL , 448AR	

C. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 71-13-00-010-040-B	Opening of the Fan Cowl Doors
Ref. AMM 71-13-00-410-040-B	Closing of the Fan Cowl Doors
Ref. AMM 71-13-13-000-041-A	Removal of the Fan Cowl Door Latches and Keepers
Ref. AMM 71-13-13-400-041-A	Installation of the Fan Cowl Door Latches and Keepers
Ref. Fig. (1 / 71-13-13-991-90300)	

3. Job Set-up

Subtask 71-13-13-941-055-A

On A/C 003-250

A. Safety Precautions

- (1) On the center pedestal, on the ENG panel 115VU:
 - (a) Put a WARNING NOTICE(S) to tell persons not to start the engine.
- (2) Make sure that the engine shutdown occurred five minutes or more before you do this procedure.
- (3) On the overhead maintenance panel 50VU:
 - (a) Make sure that the ON legend of the ENG/FADEC GND PWR/1(2) pushbutton switch is off.
 - (b) Put a WARNING NOTICE(S) to tell persons not to energize the FADEC 1(2).

Subtask 71-13-13-010-057-A

On A/C 003-250

B. Get Access

CAUTION: USE YOUR HANDS TO OPERATE THE FAN COWL LATCHES. IF YOU USE SOMETHING DIFFERENT FROM YOUR HANDS TO OPERATE THE LATCHES, YOU CAN CAUSE DAMAGE TO THE LATCHES.

- (1) Open the fan cowl doors: [Ref. AMM 71-13-00-010-040-B](#)
- (2) FOR 1000EM1 (ENGINE-1)
437AL 438AR
- (3) FOR 1000EM2 (ENGINE-2)
447AL 448AR

4. Procedure

Ref. Fig. 1 / 71-13-13-991-90300

TF-N
AIRCRAFT MAINTENANCE MANUAL
LATCH - FAN COWL DOOR - INSPECTION/CHECK

Subtask 71-13-13-220-053-A

On A/C 003-250

A. Visual Inspection of the Adjustable Keepers of the Right-Hand Fan Cowl

- (1) Do a visual inspection of adjustable keepers of the right-hand fan cowl as follows:
 - (a) Examine the adjusta-keepers for cracks. If any cracks are found, replace the adjustable keepers [Ref. AMM 71-13-13-000-041-A](#) and [Ref. AMM 71-13-13-400-041-A](#).
 - (b) Examine the clevises of the adjustable keepers for cracks. If any cracks are found, replace the clevises [Ref. AMM 71-13-13-000-041-A](#) and [Ref. AMM 71-13-13-400-041-A](#).
 - (c) Make sure that the adjusta-keepers are correctly attached. Tighten or replace the attachment hardware as necessary [Ref. AMM 71-13-13-000-041-A](#) and [Ref. AMM 71-13-13-400-041-A](#).
 - (d) Clevis use recommendation

CAUTION: DO NOT TURN THE CLEVISES. IF YOU TURN THE CLEVISES, YOU CAN CAUSE DAMAGE TO THE HOUSING GUIDE. IF THE GUIDE IS DAMAGED, THE CLEVIS CAN TURN. IF THE CLEVIS CAN TURN, THE CLEVIS CAN NOT ENGAGE CORRECTLY WITH THE LATCH AND CAN KEEP THE LATCH FROM CLOSING CORRECTLY. IF THE LATCHES ARE NOT CLOSED CORRECTLY, THE FAN COWLS CAN OPEN IN FLIGHT.

- 1 Make sure the guides of the adjustable keepers are not broken and that the clevises will not turn. If the guides are broken or you can turn the clevises, replace the adjustable keepers [Ref. AMM 71-13-13-000-041-A](#) and [Ref. AMM 71-13-13-400-041-A](#).

Subtask 71-13-13-220-055-A

On A/C 003-250

B. Visual Inspection of the Restrainers of the Right-Hand Fan Cowl

POST SB RA32071-117

- (1) Do a visual inspection of the restrainers of the right-hand fan cowl as follows:
 - (a) Examine the restrainer for cracks. If any cracks are found, replace the restrainers.
 - (b) Make sure that the restrainers are correctly attached. Tighten or replace the attachment hardware as necessary.

END OF POST SB RA32071-117

5. Close-up

Subtask 71-13-13-410-055-A

On A/C 003-250

A. Close Access

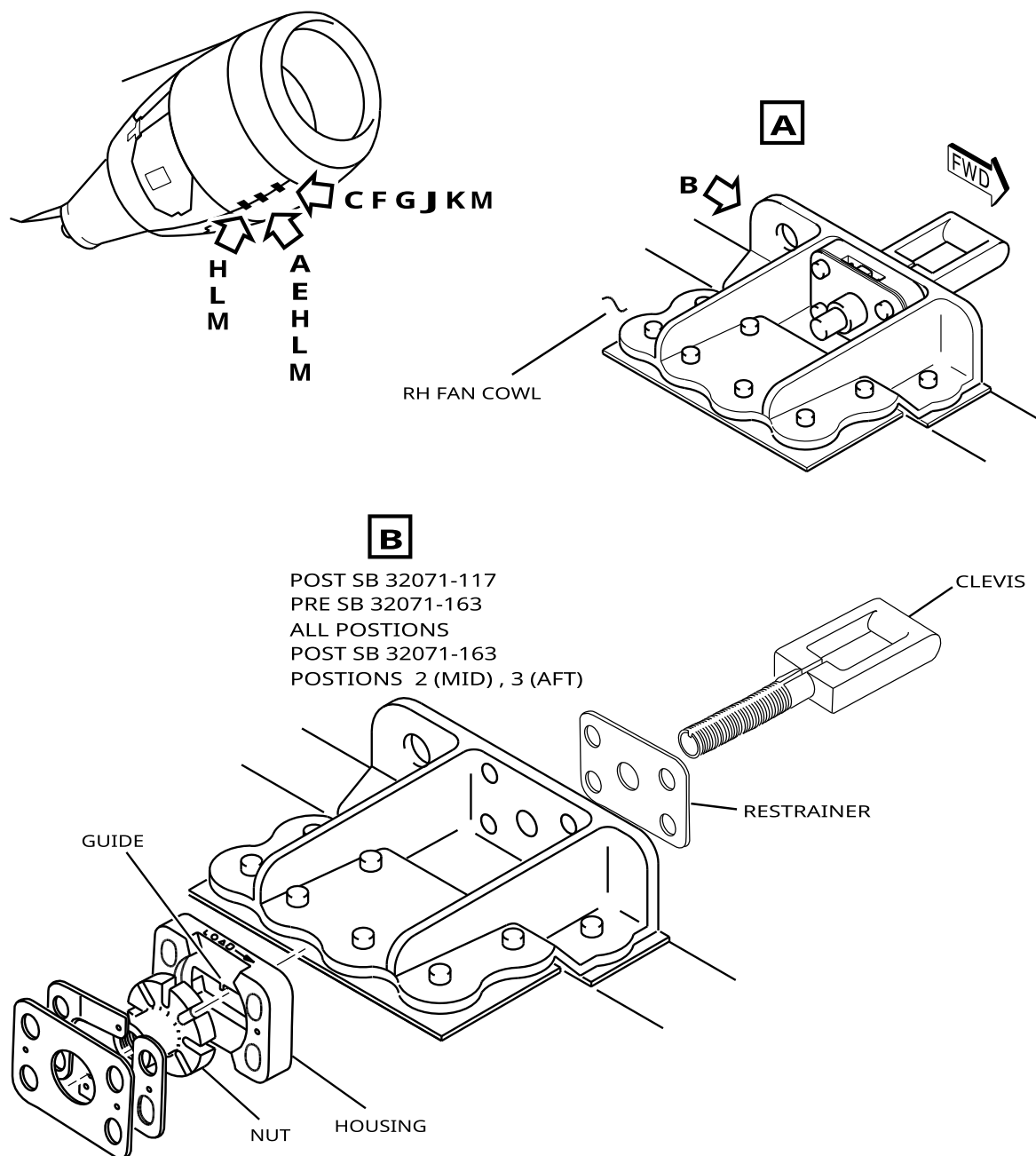
CAUTION: USE YOUR HANDS TO OPERATE THE FAN COWL LATCHES. IF YOU USE SOMETHING DIFFERENT FROM YOUR HANDS TO OPERATE THE LATCHES, YOU CAN CAUSE DAMAGE TO THE LATCHES.

- (1) Make sure that the work area is clean and clear of tools and other items.
- (2) Close the fan cowl doors: [Ref. AMM 71-13-00-410-040-B](#)
- (3) FOR 1000EM1 (ENGINE-1)

TF-N
AIRCRAFT MAINTENANCE MANUAL
LATCH - FAN COWL DOOR - INSPECTION/CHECK

- 437AL 438AR
(4) FOR 1000EM2 (ENGINE-2)
447AL 448AR

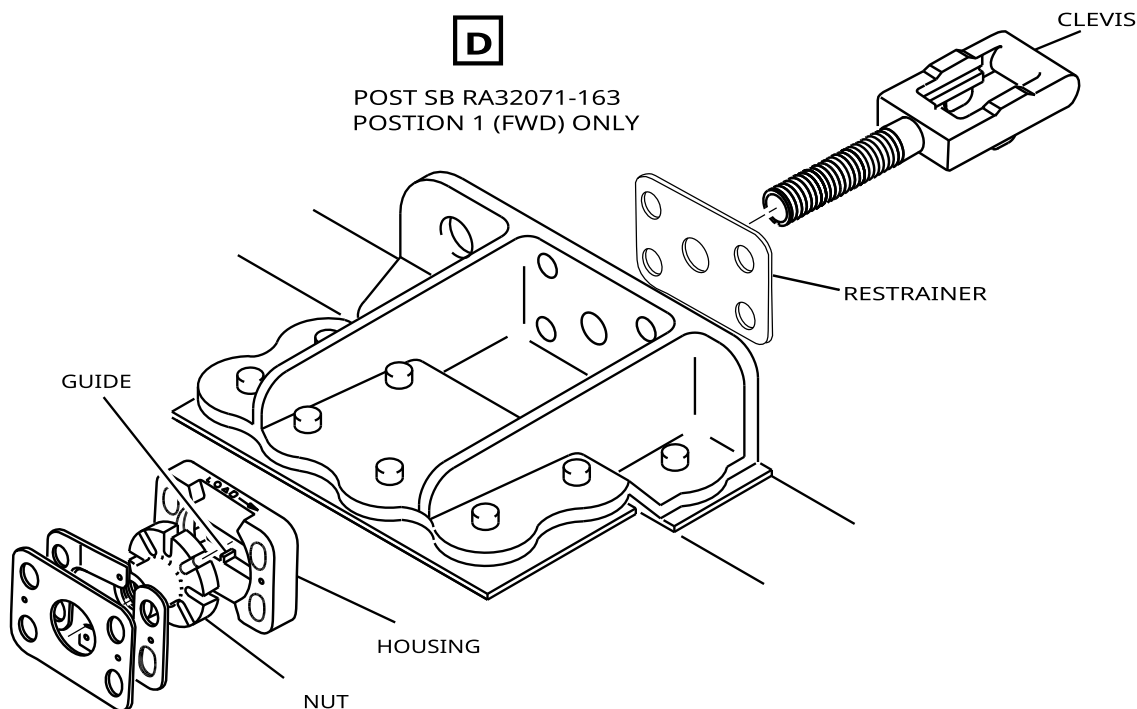
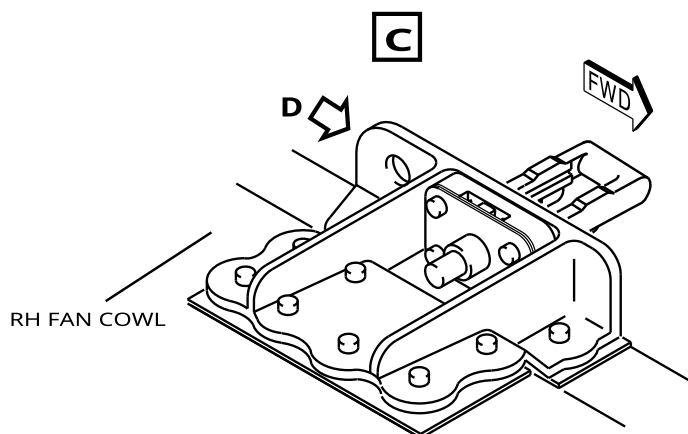
TF-N
AIRCRAFT MAINTENANCE MANUAL
LATCH - FAN COWL DOOR - INSPECTION/CHECK



N_MM_711313_6_RAM0_01_01

Right Fan Cowl Adjusta-Keepers and Restrainers
Figure 1 / Graphic 71-13-13 -991-90300

TF-N
AIRCRAFT MAINTENANCE MANUAL
LATCH - FAN COWL DOOR - INSPECTION/CHECK



CFM-521-01

NOTE:
ADJUSTA-KEEPER SHOWN
DISASSEMBLED FOR CLARITY
DO NOT DISASSEMBLE THE
KEEPER ASSEMBLY

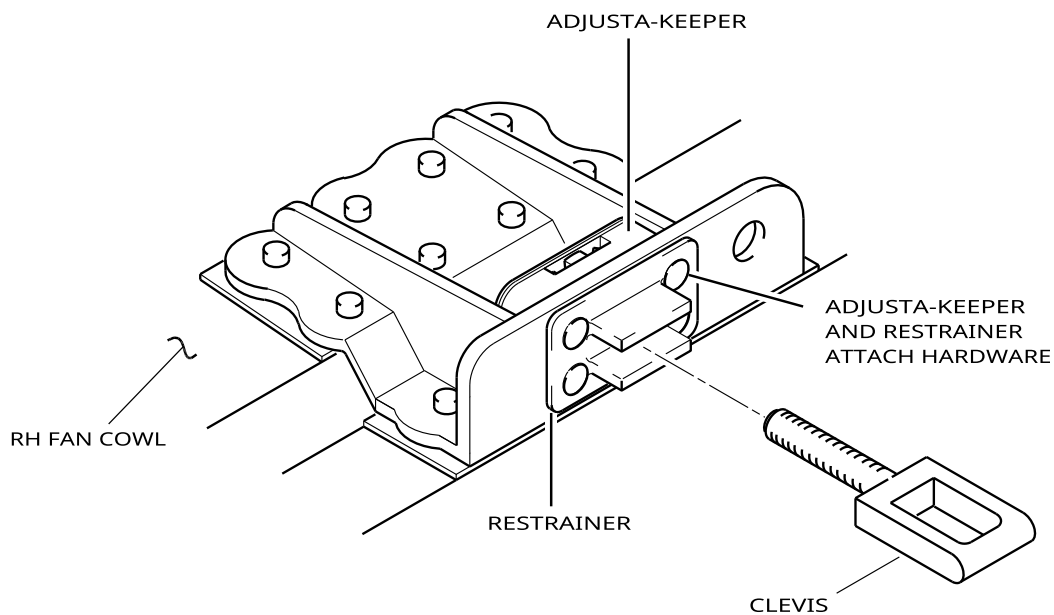
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Right Fan Cowl Adjusta-Keepers and Restrainers
Figure 2 / Graphic 71-13-13 -991-90300

TF-N
AIRCRAFT MAINTENANCE MANUAL
LATCH - FAN COWL DOOR - INSPECTION/CHECK

E

POST SB 32071-117
PRE SB 32071-163
ALL POSTIONS
POST SB 32071-163
POSTIONS 2 (MID) , 3 (AFT)



CFM-522-01

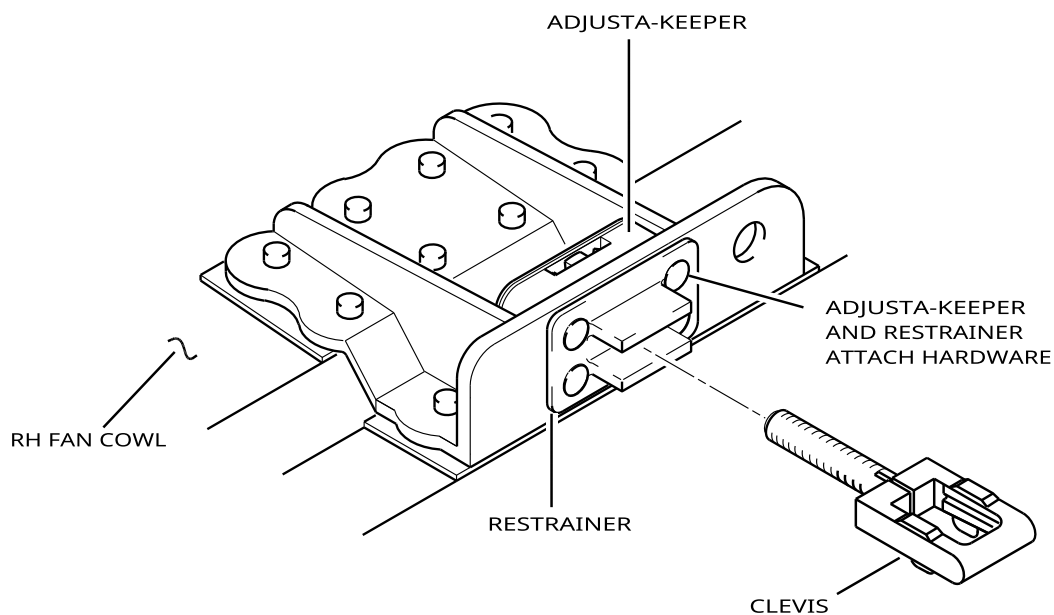
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Right Fan Cowl Adjusta-Keepers and Restrainers
Figure 3 / Graphic 71-13-13 -991-90300

TF-N
AIRCRAFT MAINTENANCE MANUAL
LATCH - FAN COWL DOOR - INSPECTION/CHECK

F

POST SB RA32071-163
POSITION 1 ONLY(FWD ONLY)
POST SB RA32071-117

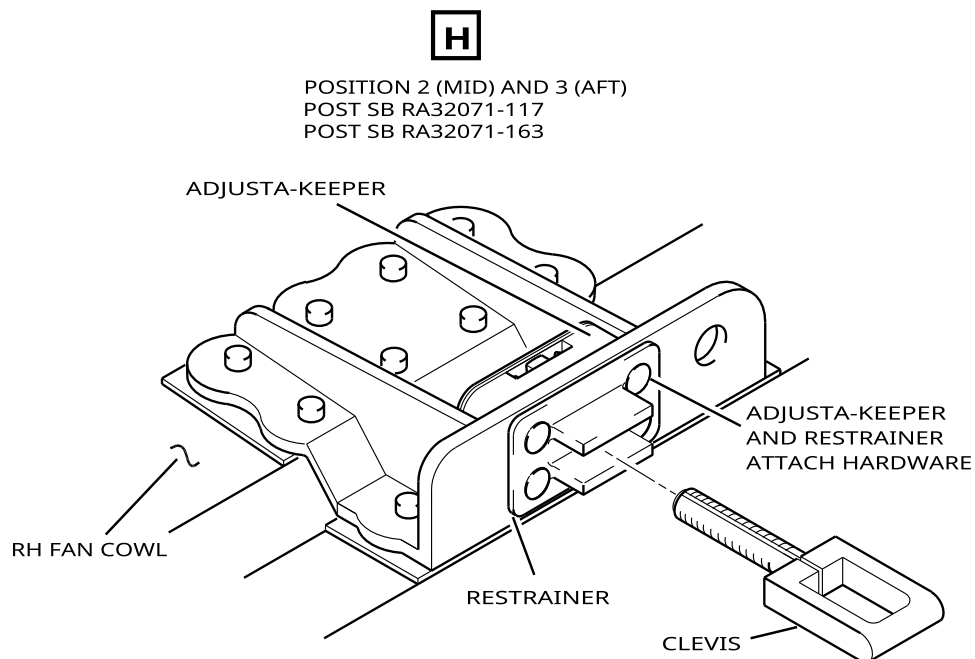
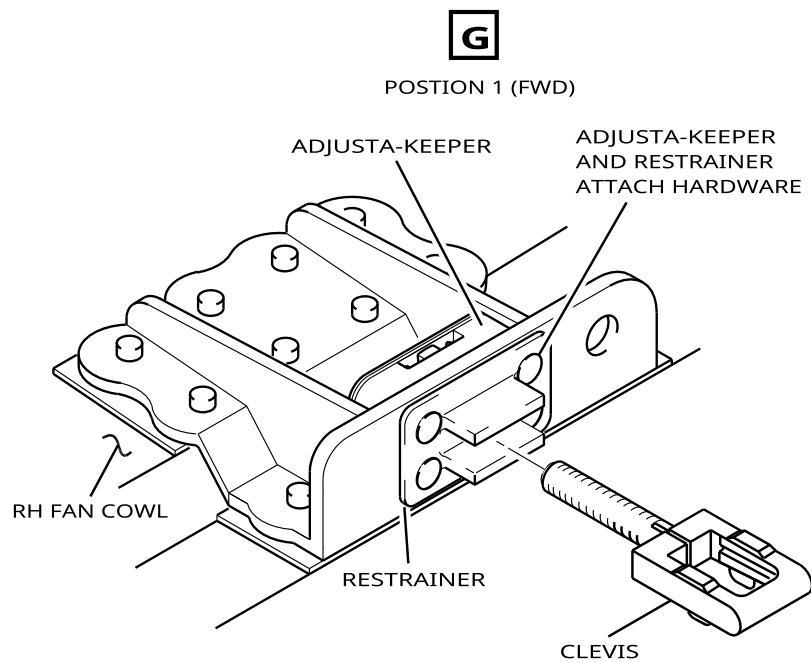


CFM-457-02

N_MM_711313_6_RAM0_04_00

Right Fan Cowl Adjusta-Keepers and Restrainers
Figure 4 / Graphic 71-13-13 -991-90300

TF-N
AIRCRAFT MAINTENANCE MANUAL
LATCH - FAN COWL DOOR - INSPECTION/CHECK



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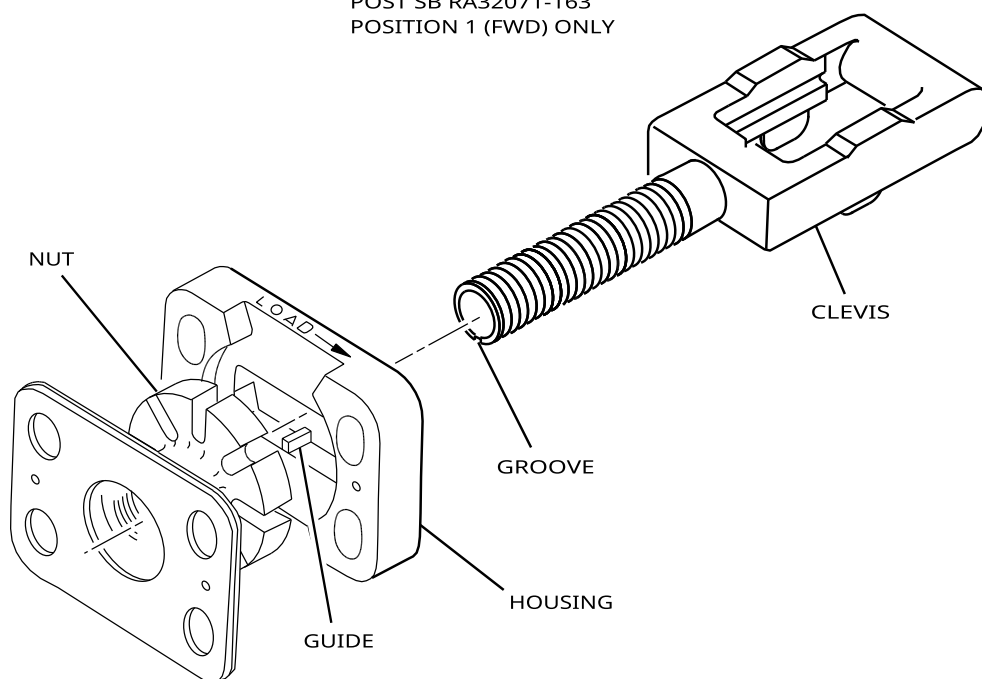
Right Fan Cowl Adjusta-Keepers and Restrainers
Figure 5 / Graphic 71-13-13 -991-90300

CFM-520-01

TF-N
AIRCRAFT MAINTENANCE MANUAL
LATCH - FAN COWL DOOR - INSPECTION/CHECK



POST SB RA32071-163
POSITION 1 (FWD) ONLY



NOTE:

ADJUSTA-KEEPER SHOWN
DISASSEMBLED FOR CLARITY
DO NOT DISASSEMBLE THE
KEEPER ASSEMBLY

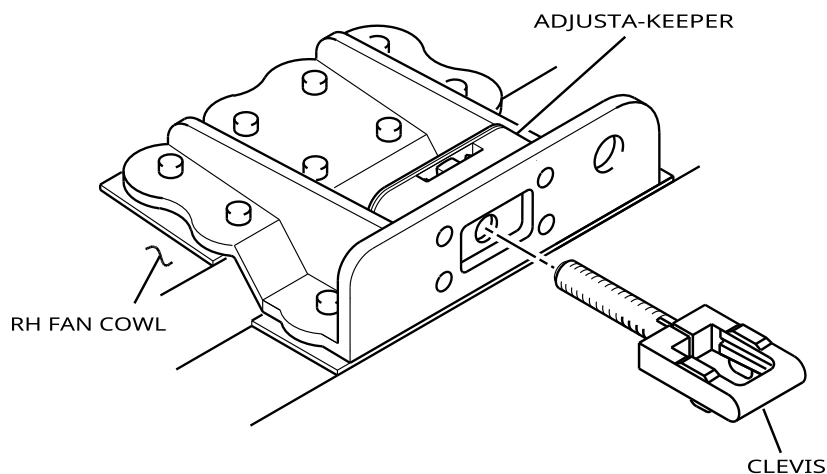
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Right Fan Cowl Adjusta-Keepers and Restrainers
Figure 6 / Graphic 71-13-13 -991-90300

TF-N
AIRCRAFT MAINTENANCE MANUAL
LATCH - FAN COWL DOOR - INSPECTION/CHECK

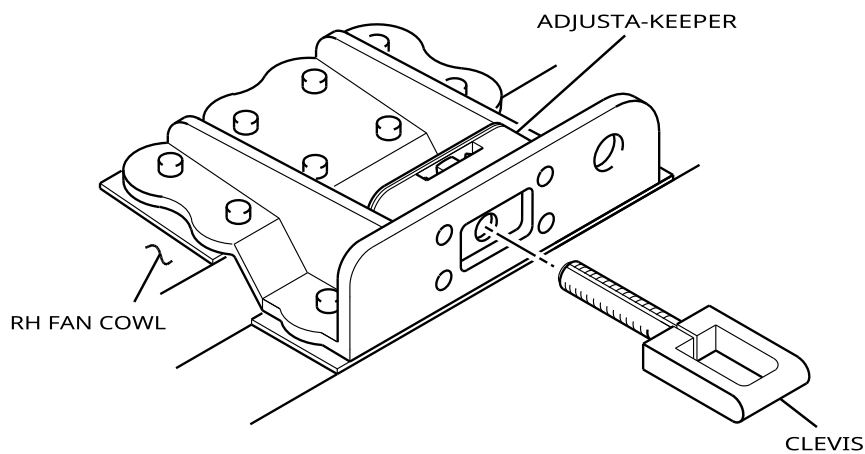
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POST SB RA32071-163
PRE SB RA32071-117
POSITION 1 (FWD)



L

POST SB RA320-71-163
PRE SB RA32071-117
POSITION 2 (CENTER) AND 3 (AFT)



N_MM_711313_6_RAM0_07_00

Right Fan Cowl Adjusta-Keepers and Restrainers
Figure 7 / Graphic 71-13-13 -991-90300

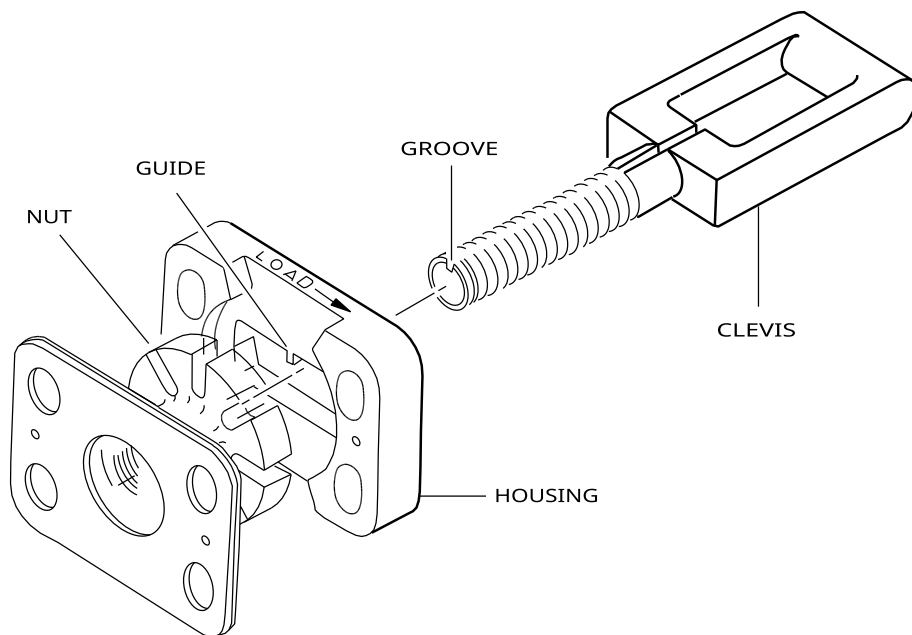
CFM-532-00

TF-N
AIRCRAFT MAINTENANCE MANUAL
LATCH - FAN COWL DOOR - INSPECTION/CHECK



PRE AND POST SB RA32071-117
POSITIONS 2 (MID) AND 3 (AFT)

PRE SB RA32071-163
POSITION 1 (FWD)



NOTE:
ADJUSTA-KEEPER SHOWN
DISASSEMBLED FOR CLARITY
DO NOT DISASSEMBLE THE
KEEPER ASSEMBLY

N_MM_711313_6_RAM0_08_00

Right Fan Cowl Adjusta-Keepers and Restraints
Figure 8 / Graphic 71-13-13 -991-90300

CFM-533-00

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Task Card Title:

NO. 1 ENGINE COMPRESSOR WASH

Task Card NO:

72-00-00-01-0-SVA-001

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

Insp.

HRS

Job Description**2. COMPRESSOR CLEANING:**

NOTE: WHILE ACCOMPLISHING THIS FUEL CONSERVATION CARDS, ENSURE THAT AUTO SPEED BRAKE SYSTEM **TOLERANCES** PROVIDED IN THE AMM ARE MAINTAINED IN THE **OPTIMUM** RANGE.

A. PERFORM ENGINE COMPRESSOR WASH USING ECOPOWER [P&W EA# 09IZ001] (PREFERRED) OR THE ALTERNATIVE EQUIPMENT PER AMM TASK 72-00-00-100-026-A.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PP/ACFT_TECH		HRS				
Insp.			HRS				

Job Description**3. EQUIPMENT REMOVAL:**

A. REMOVE THE ECOPOWER EQUIPMENT AND RECONNECT THE DICONNECTED ITEMS DURING PREPARATION PER AMM TASK 72-00-00-100-026-A.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PP/ACFT_TECH		HRS				
Insp.	INSP/RII		HRS				

Job Description**4. LINES INSTALLATION:**

A. INSTALL PS3 AND P25 LINES (IF INSTALLED) TO THE ECU AS FOLLOWS:

CAUTION: DO NOT BLOW COMPRESSED AIR OR NITROGEN INTO THE ECU AS DAMAGE TO THE ECU CAN OCCUR.

(1) REMOVE THE PLUGS FROM THE PS3 AND P25 (IF INSTALLED) LINES AND FROM THE ECU.

CAUTION: DURING INSTALLATION, MAKE SURE THAT THE PS3 LINE DOES NOT TOUCH THE LOWER STARTER DUCT OR OTHER PARTS IN THIS AREA. IF IT DOES, CHAFING OF THE PS3 HOSE CAN OCCUR.

(2) BLOW PS3 AND P25 LINES CLEAR OF WATER USING 20 PSI COMPRESSED AIR OR NITROGEN.

(3) CONNECT PS3 LINE TO ECU, TORQUE TO BETWEEN 270 AND 300 LBF.IN (3.05 AND 3.39 M.DAN) AND SAFETY WITH LOCKWIRE.

(4) CONNECT P25 (IF INSTALLED) LINE TO ECU, TORQUE TO BETWEEN 180 AND 200 LBF.IN (2.03 AND 2.26 M.DAN) AND SAFETY WITH LOCKWIRE.

(5) REMOVE THE STANDARD PROTECTIVE COVER FROM THE NACELLE ANTI-ICE LINE PORT



Task Card Title:

NO. 1 ENGINE COMPRESSOR WASH

Task Card NO:

72-00-00-01-0-SVA-001

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

OF THE NAI VALVE.

- (6) CONNECT THE RIGID ANTI-ICE SERVO PRESSURE LINE (25) TO NACELLE ANTI-ICE VALVE (4000 DN). TORQUE THE FORWARD HOSE NUT TO BETWEEN 135 AND 150 LBF.IN (1.52 AND 1.69 M.DAN).
- (7) MAKE SURE THAT THE WORK AREA IS CLEAN AND CLEAR OF TOOLS.
- (8) CLOSE THE FAN COWLS 437AL, 438AR (REF. AMM TASK 71-13-00-410-040).

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PP/ACFT_TECH		HRS				
Insp.	INSP/RII		HRS				

Job Description

5. POST ENGINE WASH TEST:
- A. FOR AMM TASK 72-00-00-100-026, PERFORM POST ENGINE WASH TEST USING PURE WATER.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PP/ACFT_TECH		HRS				
Insp.			HRS				

Job Description

5. POST ENGINE WASH TEST:
- B. FOR ECOPOWER PERFORM POST ENGINE WASH TEST AS FOLLOWS:
- (1) ON THE AIR COND PANEL 30VU, MAKE SURE THAT THE CROSSBLEED VALVE SELECTOR SWITCH IS IN THE OPEN POSITION.
- (2) START THE ENGINE WITHIN 2 HOURS (OR 30 MINUTES IN ICING CONDITION) OF THE LAST WASH CYCLE TO PURGE THE LUBE/SUMP SYSTEM OF ANY WATER INGESTION.
(REF. AMM TASK 71-00-00-710-003)
- NOTE: APU BLEED VALVE MUST BE CLOSED (THE ON LEGEND OF APU PUSHBUTTON SWITCH IS OFF) AFTER ENGINE START, TO AVOID THE SUPPLY OF THE AIR TO THE WING-ANTI-ICE SYSTEMS FROM APU.
- THIS ENGINE RUN IS DONE TO ENSURE THAT THE BLEED SYSTEM IS FREE FROM CONTAMINANT. SMOKE/SMELL IN CABIN CAN RESULT IF CONTAMINANT ENTERS IN THE BLEED SYSTEM.
- CORROSION PREVENTIVE PRODUCTS SHOULD BE USED. IF NONE OF THE CORROSION



MET.29291014

Task Card Title:

NO. 1 ENGINE COMPRESSOR WASH

Task Card NO:

72-00-00-01-0-SVA-001

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

PREVENTIVE PRODUCTS SPECIFIED ABOVE IS AVAILABLE, THE AIRCRAFT MUST RESUME REVENUE SERVICE OPERATION WITHIN 24 HOURS FOLLOWING THIS POST WASH TEST. DO NOT ACTUATE WAI SYSTEMS BEFORE THE 5-MINUTE PERIOD AT IDLE IS OVER IN ORDER TO AVOID WAI SYSTEMS OVERHEAT.

(3) LET THE ENGINE RUN AT IDLE FOR 5 MINUTES.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PP/ACFT_TECH		HRS				
Insp.			HRS				

Job Description**5. POST ENGINE WASH TEST:****B. CONTINUE POST ENGINE WASH TEST FOR ECOPOWER AS FOLLOWS:****(4) AFTER 5 MINUTES FROM ENGINE START, PERFORM THE FOLLOWING:**

- (a) PRESS THE ENG 1 BLEED PUSHBUTTON SWITCH TO OPERATE THE ENGINE BLEED SYSTEM.
- (b) MAKE SURE THAT THE OFF LEGEND OF THE ENG1 BLEED PUSHBUTTON SWITCH GOES OFF WHEN THE ENGINE BLEED SYSTEM OPERATES.
- (c) PRESS THE ANTL-ICE/WING PUSHBUTTON SWITCH TO OPERATE THE WING ANTI-ICE SYSTEM.
NOTE: THE AIM IS TO PURGE WATER POTENTIALLY TRAPPED IN HIGH PRESSURE (HP) BLEED DUCTS AND HP ENGINE BLEED CHAMBERS.

- (d) MAKE SURE THAT THE ON LEGEND OF THE ANTI ICE/WING PUSHBUTTON SWITCH GOES ON WHEN THE WING ANTI-ICE SYSTEM OPERATES.

NOTE: THE WING ANTI-ICE (WAI) VALVES WILL AUTOMATICALLY CLOSE 30 SECONDS AFTER THE WING ANTI-ICE SYSTEM IS SELECTED ON. THIS IS TO PREVENT WING SLATS OVERHEAT. HOWEVER, THE ON LEGEND REMAINS ILLUMINATED UNTIL THE SWITCH IS DESELECTED.

- (e) MAKE SURE THAT PRESSURE REGULATING VALVE (PRV) AND HIGH PRESSURE VALVE (HPV) ARE OPEN ON ECAM BLEED PAGE WITH ENGINE RUNNING AT THIS SPEED WHILE WAI VALVE IS OPEN.
- (f) ONCE WAI VALVE IS CLOSED, LET THE ENGINE RUN AT IDLE FOR 5 MINUTES WHILE DOING THE FOLLOWING STEPS:

NOTE: DO NOT ACTUATE WAI SYSTEMS BEFORE THE 5-MINUTE PERIOD AT IDLE IS OVER IN ORDER TO AVOID WAI SYSTEMS OVERHEAT.

- 1) PRESS THE ANTI ICE/WING PUSHBUTTON SWITCH AND MAKE SURE THAT THE ON LEGEND OF THE ANTI ICE/WING PUSHBUTTON SWITCH GOES OFF.
- 2) TURN THE CROSSBLEED VALVE SELECTOR SWITCH TO THE AUTO POSITION.
- 3) PRESS THE PACK 1 PUSHBUTTON SWITCH TO OPERATE THE AIR CONDITIONING PACK.
- 4) MAKE SURE THAT THE OFF LEGEND OF THE PACK 1 PUSHBUTTON GOES OFF.



Task Card Title:

NO. 1 ENGINE COMPRESSOR WASH

Task Card NO:

72-00-00-01-0-SVA-001

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP**STEP COMPLETION SIGN OFF:**

	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
						Day	Month	Year
Mechanic/s	PP/ACFT_TECH			HRS				
Insp.				HRS				

Job Description

5. POST ENGINE WASH TEST:

B. CONTINUE POST ENGINE WASH TEST FOR ECOPOWER AS FOLLOWS:

(5) AFTER 10 MINUTES OF ENGINE RUNNING, CHECK FOR EVIDENCE OF ODOR/SMOKE IN THE CABIN.

(a) IF NO EVIDENCE IS FOUND, SHUT DOWN THE ENGINE. (REF. AMM TASK 71-00-00-710-028)

(b) IF EVIDENCE IS FOUND, CONTINUE TO OPERATE THE ENGINE AT IDLE UNTIL NO ODOR/SMOKE REMAINS IN THE CABIN.

(c) AFTER 10 MINUTES ENGINE RUNNING, IF CONTINUOUS ODOR/SMOKE REMAINS IN THE CABIN, SHUT DOWN THE ENGINE (REF. AMM TASK 71-00-00-710-028) AND DECONTAMINATE THE ENVIRONMENTAL CONTROL SYSTEM (ECS). NOTE THAT THE APU BLEED VALVE MUST BE OPEN BEFORE ENGINE SHUTDOWN.

1) WHEN AMBIENT AIR TEMPERATURE IS BELOW 75.2°F (24°C).

(REF. AMM TASK 21-00-00-615-003)

2) WHEN AMBIENT AIR TEMPERATURE IS ABOVE 75.2°F (24°C).

(REF. AMM TASK 21-00-00-615-004)

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
						Day	Month	Year
Mechanic/s	PP/ACFT_TECH			HRS				
Insp.				HRS				

Job Description

5. POST ENGINE WASH TEST:

B. CONTINUE POST ENGINE WASH TEST FOR ECOPOWER AS FOLLOWS:

WARNING: OIL WILL BE HOT. PERSONNEL MUST WEAR PROPER PERSONAL PROTECTIVE EQUIPMENT WHILE TAKING THE OIL SAMPLE. WEAR NYTRILE OR RUBBER GLOVES TO PROTECT HANDS FROM EXPOSURE TO ENGINE OILS.

(6) PERFORM AN OIL WATER CONTENT TEST AS FOLLOWS:

(a) OBTAIN AN ENGINE OIL SAMPLE OF APPROXIMATELY 2 OUNCES (60 ML).

(b) REINSTALL THE MAIN OIL TANK CAP AND ENSURE IT IS PROPERLY LOCKED IN POSITION.

NOTE: IT IS PERMISSIBLE TO USE THE THIN FILM CAPACITIVE POLYMER (TFCP) TESTER, PALL METER (WS04 WATER SENSOR) TESTER OR EQUIVALENT.

(c) FOR THE TFCP OIL WATER CONTENT TESTER, PERFORM THE FOLLOWING:

NOTE: IT IS PERMISSIBLE TO INSERT THE TFCP TESTER DIRECTLY INTO THE MAIN OIL TANK FILL FOR THE TFCP OIL WATER CONTENT TESTER (TFCP) TEST.

1) SET OIL WATER CONTENT TESTER TO MIL-L-23699 OIL, OR EQUIVALENT.



MET.29291014

Task Card Title:

NO. 1 ENGINE COMPRESSOR WASH

Task Card NO:

72-00-00-01-0-SVA-001

Work Package:

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A/C Registration:

HZ-ASP

2) STIR THE PROBE IN THE OIL UNTIL THE UNIT HAS REACHED THE STEADY STATE LEVEL OF WATER CONCENTRATION FOR 20 SECONDS. THIS NORMALLY TAKES AT LEAST TWO MINUTES OF STIRRING TO COMPLETE.

- (d) FOR PALL METER (WS04 WATER SENSOR) TESTER, THE PROBE SHOULD BE INSERT INTO THE TEST FLUID SO THAT THE PROBE TIP IS COMPLETELY IMMersed IN THE FLUID. ALLOW ABOUT THREE MINUTES FOR THE READING TO STABILIZE BEFORE TAKING THE MEASUREMENT.

NOTE: PLEASE NOTE THAT THE SENSOR IN THE PROBE IS SENSITIVE TO LIGHT. DO NOT USE WHERE LIGHT MAY SHINE DIRECTLY ON THE SENSOR.

- (e) ENGINE OIL/WATER CONTENT LIMITS:

- 1) THE WATER CONTENT IN THE OIL MUST BE BELOW 1000 PARTS PER MILLION (PPM) BY WEIGHT OR BY VOLUME.
- 2) IF THE OIL EXCEEDS 1000 PPM BY WEIGHT OR BY VOLUME, PERFORM ONE OF THE FOLLOWING:
 - a) DRAIN AND SERVICE OIL TANK WITH FRESH OIL.
(REF. AMM, 72-00-00, SERVICING).
 - b) RE-RUN AT IDLE AND TEST AGAIN FOR OIL/WATER CONTENT.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PP/ACFT_TECH		HRS				
Insp.			HRS				

Job Description

5. ENGINE WASH DATA SHEET:

C. FILL THE ENGINE WASH DATA TABLE.

A/C Registration	
A/C Type	
Engine Position	
Engine Serial #	
Date	
Station	
TDS (PPM) Cycle 1	
TDS (PPM) Cycle 2	
TDS (PPM) Cycle 3	
TDS (PPM) Cycle 4	
TDS (PPM) Cycle 5	
Oil PPM	



MET.29291014

Task Card Title:

NO. 1 ENGINE COMPRESSOR WASH

Task Card NO:

72-00-00-01-0-SVA-001

Work Package:

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A/C Registration:

HZ-ASP

Oil Temp

Oil % Saturation

OAT (° F)

IPA %

Note:**STEP COMPLETION SIGN OFF:**

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PP/ACFT_TECH		HRS				
Insp.			HRS				

Job Description

6. ENGINE WASH DATA SHEET DELIVERY:

D. DELIVER OR E-MAIL A COPY OF THE ENGINE WASH DATA TO SV AIRCRAFT MAINTENANCE PLANNING AND CONTROL DEPARTMENT CC# 873
E-MAIL: svamp@saudia.com.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PROD/PLANNER		HRS				
Insp.			HRS				

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below
Work order No: _____

Note: Make sure to creat the finding Work Order on task card level.

Remark's: _____



MET.29291014

Task Card Title:

NO. 1 ENGINE COMPRESSOR WASH

Task Card NO:

72-00-00-01-0-SVA-001

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

(1) If Task card interrupted or performed by different personals updates the system and update the following according (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 2 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 3 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 4 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 5 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____

**Task Completion Close Up And
System Update**

Station**Date****Signature**

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AIRCRAFT MAINTENANCE MANUAL
ENGINE - GENERAL - CLEANING/PAINTING

TASK 72-00-00-100-026 -A

On A/C 003-250

Washing of the Engine Gas Path With Pure Water

WARNING: BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER ENGINE SHUTDOWN. USE APPLICABLE THERMAL GLOVES. THE ENGINE PARTS AND THE ENGINE OIL CAN STAY HOT FOR ONE HOUR AFTER SHUTDOWN AND CAN BURN YOU.

WARNING: BE CAREFUL WHEN YOU USE CONSUMABLE MATERIALS. OBEY THE MATERIAL MANUFACTURER'S INSTRUCTIONS AND YOUR LOCAL REGULATIONS.

CAUTION: IF CLEANING AGENT IS USED, PROPER AIRCRAFT MAINTENANCE CONFIGURATION, ENGINE RINSING AND POST ENGINE WASH TEST MUST BE COMPLETED TO PREVENT THE POSSIBILITY OF FUMES IN THE CABIN WHEN AIRCRAFT RESUMES REVENUE FLIGHT OPERATION.

FIN: 1000EM1, 1000EM2

1. Reason for the Job

For Exhaust Gas Temperature (EGT) margin recovery, it is recommended to perform the engine gas path water wash with pure water only.

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	AR	COVER - PROTECTION
No specific	1	TDS TOTAL DISSOLVED SOLID METER OR EQUIVALENT
No specific	AR	WARNING NOTICE(S)
No specific		Torque wrench: range to between 150 and 160 lbf.in (1.70 and 1.81 m.daN)
No specific		Torque wrench: range to between 180 and 200 lbf.in (2.03 and 2.26 m.daN)
No specific		Torque wrench: range to between 270 and 300 lbf.in (3.05 and 3.39 m.daN)

B. Consumable Materials

REFERENCE	DESIGNATION
CP8001	lockwire 0.032 in. (0.8 mm) dia.
CP8002	lockwire 0.032 in. (0.8 mm) dia

C. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
210	COCKPIT FWD COMPARTMENT BULKHEAD TO FLIGHT COMPARTMENT BULKHEAD
400	POWER PLANT, NACELLES AND PYLONS
FOR 1000EM1(ENGINE-1)	
437AL , 438AR	
FOR 1000EM2(ENGINE-2)	
447AL , 448AR	

D. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 12-13-79-610-002-A	Check oil level and replenish Check Pop Out Indicator of EMCD (Electrical Master Chip Detector)
Ref. AMM 12-13-79-610-002-A-01	Check oil level and replenish (Pressure Filling)
Ref. AMM 21-00-00-615-803-A	Level B Removal of the Contamination of the Environmental Control System (ECS)
Ref. AMM 21-00-00-615-803-B	Level B Removal of the Contamination of the Environmental Control System (ECS) With TCT Part Number (P/N) SBE2272

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REFERENCE	DESIGNATION
Ref. AMM 24-41-00-861-002-A	Energize the Aircraft Electrical Circuits from the External Power
Ref. AMM 24-41-00-861-002-A-01	Energize the Aircraft Electrical Circuits from the APU
Ref. AMM 24-41-00-861-002-A-02	Energize the Aircraft Electrical Circuits from Engine 1(2)
Ref. AMM 24-41-00-862-002-A	De-energize the Aircraft Electrical Circuits Supplied from the External Power
Ref. AMM 24-41-00-862-002-A-01	De-energize the Aircraft Electrical Circuits Supplied from the APU
Ref. AMM 24-41-00-862-002-A-02	De-energize the Aircraft Electrical Circuits Supplied from the Engine 1(2)
Ref. AMM 71-00-00-710-001-B	Dry Motoring Check
Ref. AMM 71-00-00-710-003-B	Engine Automatic Start
Ref. AMM 71-00-00-710-028-B	Engine Shutdown
Ref. AMM 71-00-00-710-802-A	Water Wash Crank
Ref. AMM 71-13-00-010-040-B	Opening of the Fan Cowl Doors
Ref. AMM 71-13-00-410-040-B	Closing of the Fan Cowl Doors
Ref. AMM 72-00-00-100-008-A	Engine Washing Requirements
Ref. Fig. (1 / 72-00-00-991-32500)	
Ref. Fig. (1 / 72-00-00-991-30400)	

3. Job Set-up

Subtask 72-00-00-941-065-A

On A/C 003-250

A. Safety Precautions

- (1) On the center pedestal, on the ENG panel 115VU:
 - (a) Put a WARNING NOTICE(S) to tell persons not to start the engine.
- (2) Make sure that the engine shutdown occurred five minutes or more before you do this procedure.
- (3) On the overhead maintenance panel 50VU:
 - (a) Make sure that the ENG/FADEC GND PWR/1(2) pushbutton switch is OFF (the ON legend of the ENG/FADEC GND PWR/1(2) pushbutton switch is OFF).
 - (b) Put a WARNING NOTICE(S) to tell persons not to energize the FADEC 1(2).

Subtask 72-00-00-010-072-A

On A/C 003-250

B. Get Access

- (1) Open the fan cowl doors [Ref. AMM 71-13-00-010-040-B](#):
 - (a) FOR 1000EM1 (ENGINE-1)
437AL 438AR
 - (b) FOR 1000EM2 (ENGINE-2)
447AL 448AR

4. Procedure

Subtask 72-00-00-110-057-A

On A/C 003-250

A. Preparation of the Engine for Washing

- (1) For the engine washing requirements [Ref. AMM 72-00-00-100-008-A](#).

NOTE : The disconnection of the P25 line is only available for the engines equipped with the PMUX option.

CAUTION: USE TWO WRENCHES TO REMOVE OR INSTALL THE TUBE COUPLING NUTS:

- ONE WRENCH TO HOLD THE UNION, AND
- ONE TO LOOSEN OR TIGHTEN THE COUPLING NUT.

THIS PREVENTS DAMAGE TO THE PARTS.

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- (2) Disconnect the PS3 line and the P25 line (if installed) from the Electronic Control Unit (ECU).
- (3) Plug the open tubes and ports of the ECU with the COVER - PROTECTION.
- (4) Disconnect the flexible hose (25) from the Nacelle Anti-Icing (NAI) valve:
Ref. Fig. (1 / 72-00-00-991-32500)
 - (a) Let the flexible hose free (not plugged) during the washing of the engine.
- (5) Install the COVER - PROTECTION on the anti-ice servo-pressure-line port of the NAI valve.
- (6) Prepare an anti-freeze mixture, if necessary, as defined in the engine washing requirements
Ref. AMM 72-00-00-100-008-A.
- (7) Service the oil tank:
 - (a) Do a check of oil level of the engine Ref. AMM 12-13-79-610-002-A
Ref. AMM 12-13-79-610-002-A-01.

Subtask 72-00-00-861-051-A

On A/C 003-250

- B. Energize the aircraft electrical circuits Ref. AMM 24-41-00-861-002-A
Ref. AMM 24-41-00-861-002-A-01 Ref. AMM 24-41-00-861-002-A-02.

Subtask 72-00-00-860-051-A

On A/C 003-250

- C. Aircraft Maintenance Configuration
- (1) On the ANTI-ICE section of panel 25VU, make sure that the ANTI ICE/ENG 1(2) pushbutton switch is OFF (the ON legend of the ANTI ICE/ENG 1(2) pushbutton switch is OFF).
 - (2) On the AIR COND panel 30VU, make sure that ENG 1(2) BLEED pushbutton switch is OFF (the OFF legend of the ENG1(2) BLEED is ON).
 - (3) On the AIR COND panel 30VU, make sure that PACK 1 and PACK 2 pushbutton switches are OFF (the OFF legend of the PACK 1 and the PACK 2 pushbutton switches are ON).

Subtask 72-00-00-110-058-B

On A/C 003-250

EMB SB 73-1100 for A/C 003-004; 051-054; 101-109

- D. Procedure to Wash the Engine

CAUTION: DO NOT WASH AN ENGINE WHICH HAS BEEN EXPOSED TO DRY
CHEMICAL FIRE EXTINGUISHING AGENTS.

CAUTION: DO NOT WASH AN ENGINE UNLESS THE EGT IS BELOW 150 DEG.F (66
DEG.C).

CAUTION: DO NOT WASH AN ENGINE AT TEMPERATURE BELOW 14 DEG.F (-10
DEG.C).

CAUTION: THE FUEL PUMP AND HYDRO-MECHANICAL UNIT (HMU) ARE FUEL
LUBRICATED. DO NOT MOTOR, START, OR OPERATE ENGINE UNLESS A
POSITIVE FUEL INLET PRESSURE IS INDICATED ON THE LOWER ECAM
(FUEL SYSTEM PAGE). DAMAGE TO COMPONENTS MAY RESULT.

CAUTION: DO NOT EXCEED THE STARTER DUTY CYCLE. DAMAGE TO THE STARTER
CAN OCCUR.

CAUTION: MAKE SURE THE N1 ROTOR TURNS FREELY THROUGHOUT WASHING TO
PREVENT OIL SYSTEM CONTAMINATION.

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CAUTION: DO NOT APPLY WATER AT MORE THAN 6 US GAL (23 LITERS) PER MINUTE. IF YOU DO, ENGINE DAMAGE CAN OCCUR.

CAUTION: DRY MOTOR CYCLE DURATIONS MUST NOT EXCEED 2 MINUTES (REFER TO AMM TASK 71-00-00-710-001).

- (1) Dry motor the engine for two minutes [Ref. AMM 71-00-00-710-001-B](#) or do the water wash crank [Ref. AMM 71-00-00-710-802-A](#).

NOTE : The water wash crank is an alternate procedure available for the FADEC POST 5BS2 software. This task is to optimize the washing procedure by positioning the valves in the better way.

- (2) After 10 seconds of motoring with a positive fan rotation, inject water at one o'clock position (aft view) on the Low Pressure Compressor (LPC) inlet through the fan blades (or at the LPC inlet if a specific rig is used).
- (3) Stop the water injection before the end of the engine motoring.
- (4) Let the engine soak for five minutes.
- (5) Repeat paragraph 4.D.(1) to paragraph 4.D.(4).
- (6) Let the engine soak for five minutes.
- (7) Dry motor the engine again [Ref. AMM 71-00-00-710-001-B](#).

NOTE : Water can potentially drain out of the disconnected anti-ice servo pressure-line. If water drains out, continue to dry motor the engine until there is no more water draining out of the line.

Subtask 72-00-00-110-087-A

On A/C 003-250

- E. Procedure to Wash the Engine With the Total Dissolved-Solid Meter or Equivalent - Alternative Procedure Available

CAUTION: DO NOT WASH AN ENGINE WHICH HAS BEEN EXPOSED TO DRY CHEMICAL FIRE EXTINGUISHING AGENTS.

CAUTION: DO NOT WASH AN ENGINE UNLESS THE EGT IS BELOW 150 DEG.F (66 DEG.C).

CAUTION: DO NOT WASH AN ENGINE AT TEMPERATURE BELOW 14 DEG.F (-10 DEG.C).

CAUTION: THE FUEL PUMP AND HYDRO-MECHANICAL UNIT (HMU) ARE FUEL LUBRICATED. DO NOT MOTOR, START, OR OPERATE ENGINE UNLESS A POSITIVE FUEL INLET PRESSURE IS INDICATED ON THE LOWER ECAM (FUEL SYSTEM PAGE). DAMAGE TO COMPONENTS MAY RESULT.

CAUTION: DO NOT EXCEED THE STARTER DUTY CYCLE. DAMAGE TO THE STARTER CAN OCCUR.

CAUTION: MAKE SURE THE N1 ROTOR TURNS FREELY THROUGHOUT WASHING TO PREVENT OIL SYSTEM CONTAMINATION.

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ENGINE - GENERAL - CLEANING/PAINTING

CAUTION: DO NOT APPLY WATER AT MORE THAN 6 US GAL (23 LITERS) PER MINUTE. IF YOU DO, ENGINE DAMAGE CAN OCCUR.

CAUTION: DRY MOTOR CYCLE DURATIONS MUST NOT EXCEED 2 MINUTES (REFER TO AMM TASK 71-00-00-710-001).

- (1) Optional - collect a sample of wash supply water and measure the total dissolved solid with the TDS TOTAL DISSOLVED SOLID METER OR EQUIVALENT (supply-water total dissolved-solid). Make a record of the initial measurement.

NOTE : Use a TDS TOTAL DISSOLVED SOLID METER OR EQUIVALENT with properties that follow:

- Measurement range of 0 to 9990 ppm total dissolved solids
- Resolution of 1 ppm
- Readout accuracy of plus or minus two percent.

- (2) Dry motor the engine for two minutes [Ref. AMM 71-00-00-710-001-B](#) or do the water wash crank [Ref. AMM 71-00-00-710-802-A](#).

NOTE : The water wash crank is an alternative procedure available for the FADEC POST 5BS2 software. This task is to optimize the washing procedure by positioning the valves in the better way.

- (3) After 10 seconds of motoring with a positive fan rotation, inject water at the one o'clock position (aft view) on the Low Pressure Compressor (LPC) inlet through the fan blades (or at the LPC inlet if a specific rig is used).
- (4) Stop the water injection before the end of the engine motoring and if necessary, do these steps:
- (a) During the engine coast-down, collect a sample of the wash water that drains from the exhaust area of the engine.
 - (b) Measure and make a record of the total dissolved solid with the TDS TOTAL DISSOLVED SOLID METER OR EQUIVALENT.
 - (c) Subtract initial-supply-water total-dissolved-solid value, measured in paragraph 4.E.(1) above, from the rinse effluent value.
 - 1 If the net total-dissolved-solid value is less than 150 ppm, no additional rinse cycles are necessary.
 - 2 If this value is more than 150 ppm, do additional wash/soak cycles until the net total-dissolved-solid falls below 150 ppm, up to a maximum of four wash cycles.
- (5) Let the engine soak for five minutes.
- (6) Repeat paragraph 4.E.(1) to paragraph 4.E.(4).
- (7) Let the engine soak for five minutes.
- (8) Dry motor the engine [Ref. AMM 71-00-00-710-001-B](#).

NOTE : Water can potentially drain out of the disconnected anti-ice servo pressure-line. If water drains out, continue to dry motor the engine until there is no more water draining out from the line.

Subtask 72-00-00-420-055-A

On A/C 003-250

F. Installation of PS3 Line and P25 Line (if installed)

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AIRCRAFT MAINTENANCE MANUAL
ENGINE - GENERAL - CLEANING/PAINTING

WARNING: USE PROTECTIVE GOGGLES AND GLOVES EACH TIME YOU CUT LOCKWIRE. DISCARD LOOSE OR USED LOCKWIRE IMMEDIATELY. LOOSE OR USED LOCKWIRE CAN CUT YOU OR MAKE YOU BLIND.

CAUTION: DO NOT BLOW COMPRESSED AIR OR NITROGEN INTO THE ECU AS DAMAGE TO THE ECU CAN OCCUR.

CAUTION: DURING INSTALLATION, MAKE SURE THAT THE PS3 LINE DOES NOT TOUCH THE LOWER STARTER DUCT OR OTHER PARTS IN THIS AREA. IF IT DOES, CHAFING OF THE PS3 HOSE CAN OCCUR.

- (1) Remove the plugs from the PS3 and P25 lines and from the ECU.
- (2) Blow PS3 and P25 lines clear of water using compressed air or nitrogen.
 - (a) Do a visual inspection to make sure that the PS3 line weep-hole is clear and there is no blockage.
- (3) Connect PS3 line to ECU, TORQUE **to between 270 and 300 lbf.in (3.05 and 3.39 m.daN)** and safety with lockwire 0.032 in. (0.8 mm) dia. (Material No. **CP8001**) or lockwire 0.032 in. (0.8 mm) dia (Material No. **CP8002**) .
- (4) Connect P25 line to ECU, TORQUE **to between 180 and 200 lbf.in (2.03 and 2.26 m.daN)** and safety with lockwire 0.032 in. (0.8 mm) dia. (Material No. **CP8001**) or lockwire 0.032 in. (0.8 mm) dia (Material No. **CP8002**) .
- (5) Remove the COVER - PROTECTION from the nacelle anti-ice line port of the NAI valve.
- (6) Connect the flexible hose (25) to the nacelle anti-ice valve (4000 DN). TORQUE the flexible hose coupling **to between 150 and 160 lbf.in (1.70 and 1.81 m.daN)**.

Subtask 72-00-00-410-085-A

On A/C 003-250

G. Close Access

- (1) Make sure that the work area is clean and clear of tools and other items.
- (2) Close the following engine fan cowl doors [Ref. AMM 71-13-00-410-040-B](#):
 - (a) FOR 1000EM1 (ENGINE-1)
437AL 438AR
 - (b) FOR 1000EM2 (ENGINE-2)
447AL 448AR

Subtask 72-00-00-620-063-A

On A/C 003-250

H. Post-Engine Wash Test - Alternative Procedure Available

Ref. Fig. (1 / 72-00-00-991-30400)

NOTE : The aim of this post-engine wash test is:

- To purge water from the oil system in order to prevent potential bearing/sump corrosion/deterioration
- To prevent fumes in the cockpit and in the cabin when the airplane is put back into service.

- (1) On AIR COND panel 30VU, make sure that:
 - The crossbleed-valve selector switch is in the OPEN position.
- (2) Start the engine within two hours (or 30 minutes in icing condition) of the last wash cycle to purge the lube/sump system of any water ingestion [Ref. AMM 71-00-00-710-003-B](#).

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NOTE : The APU bleed valve must be closed (the ON legend of the APU bleed pushbutton switch off) after the engine start, to avoid the supply of air to the Wing Anti-Ice systems from the APU.

NOTE : This engine run is done to ensure that the bleed system is free from contaminants. Smoke/smell in the cabin and Starter Air Valve (SAV) contamination can result if contaminants go into the bleed system.

(a) Let the engine run at idle for five minutes and after, do the following:

- 1 Press the ANTI-ICE/ENG pushbutton switch to operate the engine anti-ice system.
- 2 After five minutes, set the ANTI-ICE/ENG pushbutton switch to OFF.
- 3 Press the ENG 1(2) BLEED pushbutton switch to operate the engine bleed system.
- 4 Make sure that the OFF legend of the ENG1(2) BLEED pushbutton switch goes off when the engine bleed system operates.
- 5 Press the ANTI ICE/WING pushbutton switch to operate the WAI system.

NOTE : The aim is to purge water potentially trapped in the High Pressure (HP) bleed ducts and the HP engine bleed chambers.

- 6 Make sure that the ON legend of the ANTI ICE/WING pushbutton switch comes on when the WAI system operates.

NOTE : 30 seconds after you push the ANTI ICE/WING pushbutton switch, the WAI valves will automatically close to prevent wing slat overheat. But the ON legend of the pushbutton switch will stay on until you release the pushbutton switch.

- 7 Make sure that the Pressure Regulating Valve (PRV) and the High Pressure Valve (HPV) are open on the ECAM BLEED page with the engine running at this speed while the WAI valve is open.
- 8 Once the WAI valve is closed, let the engine run at idle for five minutes.

NOTE : Do not actuate WAI systems before the 5-minute period at idle is over in order to avoid the WAI system overheat.

- 9 Increase and stabilize the engine thrust to at least 60 percent Fan (N1) for ten minutes.
- 10 Press the ANTI ICE/WING pushbutton switch two times to operate the WAI system once more.

NOTE : The aim is to purge water potentially trapped in the IP bleed ducts and the IP engine bleed chambers.

NOTE : If the Wing Anti-Ice ON legend is still on, pressing twice the WAI pushbutton switch enables a second WAI cycle of 30 seconds to be started.

NOTE : Do not perform more than two attempts with a minimum of five minutes between the two attempts, in order to avoid the WAI system overheat.

- 11 Make sure that the ON legend of the ANTI ICE/WING pushbutton switch comes on when the WAI system operates.

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- 12 Make sure that the PRV is open and the HPV is closed on the ECAM BLEED page with the engine running at this speed while the WAI valve is open.
- (b) Once the WAI valve is closed, pull the throttle to stabilize the engine thrust at idle for five minutes and after, do the following:
 - 1 Press the ANTI ICE/WING pushbutton switch and make sure that the ON legend of the ANTI ICE/WING pushbutton switch goes OFF.
 - 2 Turn the crossbleed-valve selector switch to the AUTO position.
 - 3 Press the PACK 1(2) pushbutton switch to operate the air conditioning pack.
 - 4 Make sure that the OFF legend of the PACK 1(2) pushbutton switch goes off.
 - 5 Check for evidence of anti-freeze mixture odor/smoke in the cabin.
 - If evidence is found, continue to operate the engine at idle until no odor/smoke remains in the cabin.
 - After ten minutes of engine running, if continuous odor/smoke remains in the cabin, shut down the engine [Ref. AMM 71-00-00-710-028-B](#) and decontaminate the Environmental Control System (ECS) [Ref. AMM 21-00-00-615-803-A](#) [Ref. AMM 21-00-00-615-803-B](#).
Note that the APU Bleed valve must be open before the engine shutdown.
 - If no evidence is found, shut the engine down [Ref. AMM 71-00-00-710-028-B](#).

Subtask 72-00-00-620-070-A

On A/C 003-250

CAUTION: DO NOT USE THIS ALTERNATIVE PROCEDURE IF THE TEMPERATURE IS LESS THAN +5 DEG. C.

J. Post-Engine Wash Test - Alternative Procedure

NOTE : The aim of this post-engine wash test is:

- To purge water from oil system in order to prevent potential bearing/sump corrosion/deterioration
 - To prevent fumes in the cockpit and in the cabin when the airplane is put back into service.
- (1) On the AIR COND panel 30VU, make sure that:
 - PACK 1 and PACK 2 are OFF prior to the start of the procedure that follows.
 - The crossbleed-valve selector switch is in the OPEN position.
 - ENG 1 BLEED and ENG 2 BLEED are OFF.
 - (2) On the VENTILATION panel 22VU, make sure that:
 - CAB FANS pushbutton switch is in the OFF position.
 - (3) Start the engine within 2 hours (or 30 minutes in icing condition) of the last wash cycle to purge the lube/sump system of any water ingestion [Ref. AMM 71-00-00-710-003-B](#).

NOTE : This engine run is done to ensure that the bleed system is free from contaminant. Smoke/Smell in the cabin and Starter Air Valve (SAV) contamination can result if contaminant enters the bleed system.

- (4) After engine start, push the APU BLEED pushbutton switch to the OFF position. The APU bleed valve must be closed, to avoid the supply of air to the wing anti-ice systems from the APU.
- (5) Let the engine run at idle for 5 minutes.
- (6) Press the ANTI ICE/ENG pushbutton switch to operate the engine anti-ice system.

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ENGINE - GENERAL - CLEANING/PAINTING

- (7) After 5 minutes put the ANTI ICE/ENG pushbutton switch to OFF.
- (8) Do the following:
 - (a) Press the ENG 1(2) BLEED pushbutton switch to ON. Make sure that the OFF legend goes off when the engine bleed system operates.
 - (b) Press the ANTI ICE/WING pushbutton switch to ON. Make sure that the ON legend goes on when the wing anti ice system operates.

NOTE : The aim is to purge water potentially trapped in High Pressure (HP) bleed ducts and HP engine bleed chambers.

NOTE : 30 seconds after you push the ANTI ICE/WING pushbutton switch, the WAI valves will automatically close to prevent wing slat overheat. But the ON legend of the pushbutton switch will stay on until you release the pushbutton switch.

- (c) Make sure that Pressure Regulating Valve (PRV) and High Pressure Valve (HPV) are open on ECAM BLEED page with engine running at this speed while WAI valves are open.
- (9) Once WAI valves are closed, let the engine run at idle for 5 minutes.

NOTE : Do not actuate WAI system before the 5-minute period at idle is over in order to avoid WAI system overheat.

- (10) Press twice the ANTI ICE/WING pushbutton switch to operate the WAI system once more.

NOTE : If the Wing Anti-Ice ON legend is still on, pressing twice the WAI pushbutton switch enables a second WAI cycle of 30 seconds to be started.

NOTE : Do not perform more than two attempts with a minimum of 5 minutes between the two attempts, in order to avoid WAI systems overheat.

- (a) Make sure that the ON legend goes on when the wing anti ice system operates.
 - (b) Make sure that Pressure Regulating Valve (PRV) and High Pressure Valve (HPV) are open on ECAM BLEED page with engine running at this speed while WAI valves are open.
- (11) Once the WAI valves are closed, stabilize the engine thrust at idle for 5 minutes.
- (12) Press once the ANTI ICE/WING pushbutton switch and make sure that the ON legend goes off.

NOTE : Do not press the WAI pushbutton switch twice. If the WAI pushbutton switch ON legend is still on, pressing twice the WAI pushbutton switch enables a second WAI cycle of 30 seconds to be started.

- (13) Push the APU BLEED pushbutton switch to the ON position.
- (14) Turn the crossbleed-valve selector switch to the AUTO position.
- (15) Press the PACK 1(2) pushbutton switch to operate the air conditioning pack.
- (16) Make sure that the OFF legend of the PACK 1(2) pushbutton switch goes off.
- (17) Press the CAB FANS pushbutton switch.
- (18) Make sure that the OFF legend of the CAB FANS pushbutton switch goes off.
- (19) Shut down the engine [Ref. AMM 71-00-00-710-028-B](#).

5. Close-up

Subtask 72-00-00-869-065-A

On A/C 003-250

TF-N
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ENGINE - GENERAL - CLEANING/PAINTING

A. Make sure that the work area is clean and clear of tools and other items.

Subtask 72-00-00-862-051-A

On A/C 003-250

B. De-energize the aircraft electrical circuits [Ref. AMM 24-41-00-862-002-A](#)

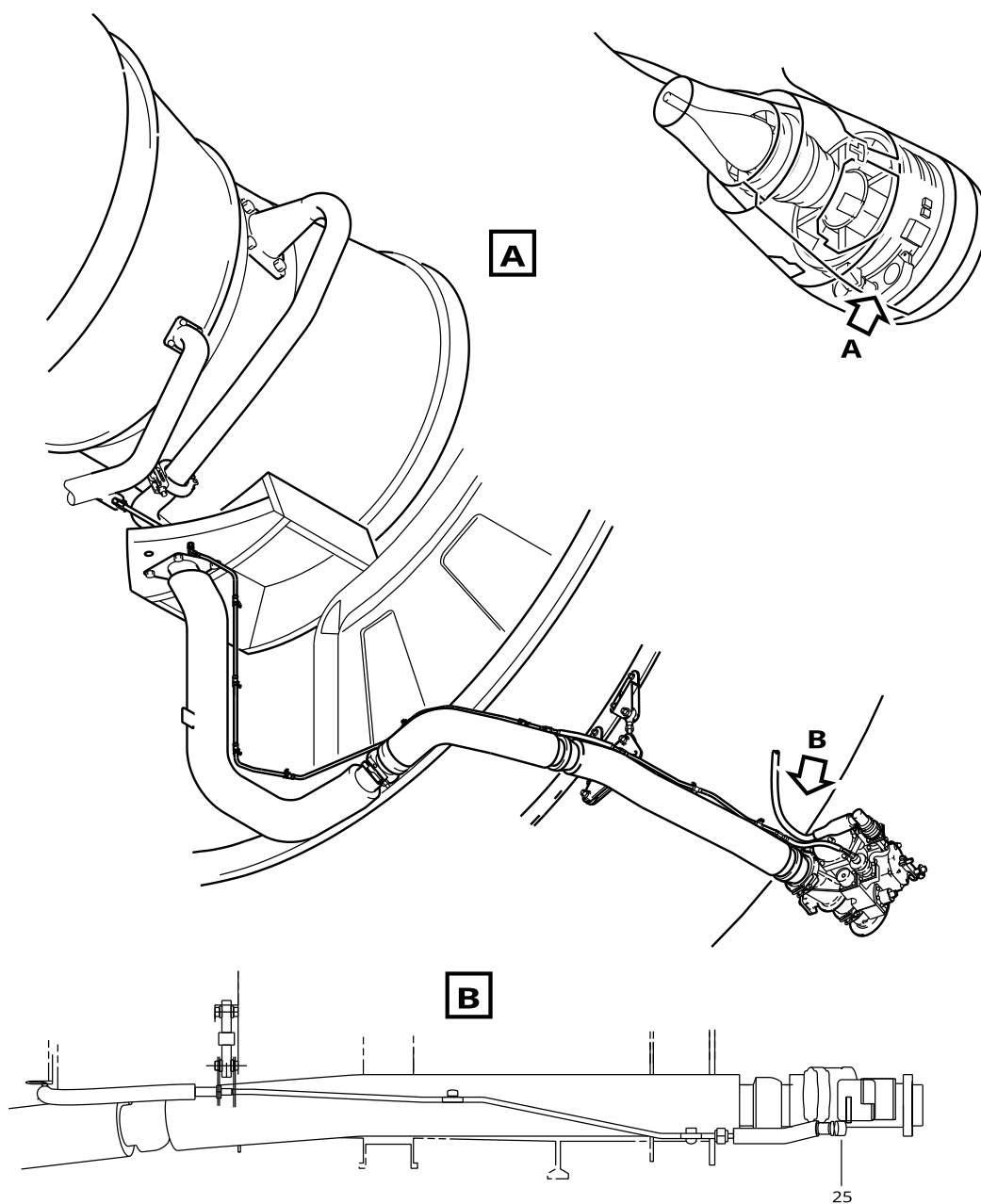
[Ref. AMM 24-41-00-862-002-A-01](#) [Ref. AMM 24-41-00-862-002-A-02](#).

Subtask 72-00-00-942-053-A

On A/C 003-250

C. Remove the WARNING NOTICE(S).

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ENGINE - GENERAL - CLEANING/PAINTING



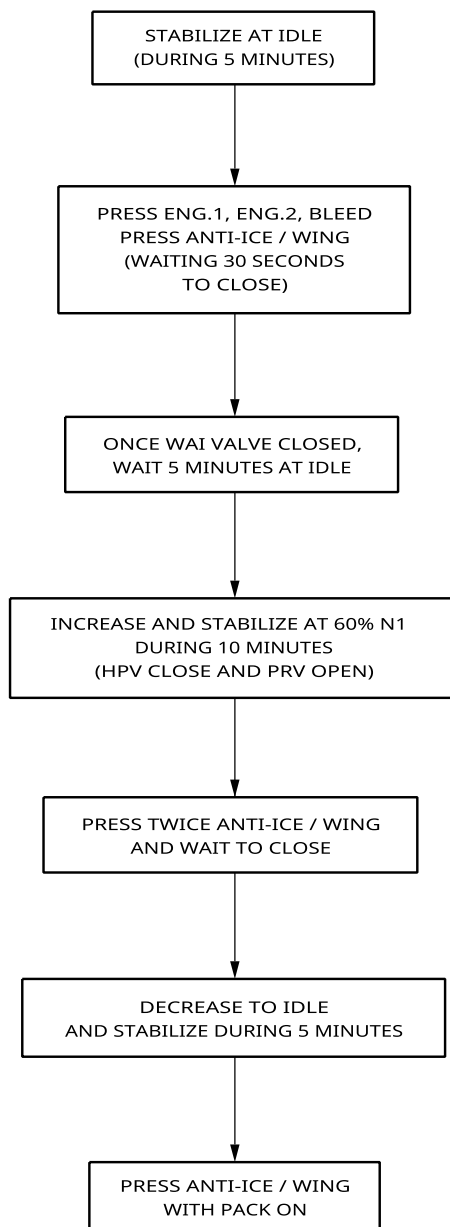
C1M-444-00

N_MM_720000_7_RHM0_01_01

Anti-Ice Muscle Line
Figure 1 / Graphic 72-00-00 -991-32500

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ENGINE - GENERAL - CLEANING/PAINTING

STARTING CONFIGURATION



N_MM_720000_7_RFM0_01_01

Engine Cleaning (Washing with Pure Water)
Figure 1 / Graphic 72-00-00 -991-30400

S-M156-MM-04706-04-B

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MET.29291013

Task Card Title:

NO. 2 ENGINE COMPRESSOR WASH

Task Card NO:

72-00-00-01-0-SVA-002

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

Insp.

HRS

Job Description**2. COMPRESSOR CLEANING:**

NOTE: WHILE ACCOMPLISHING THIS FUEL CONSERVATION CARDS, ENSURE THAT AUTO SPEED BRAKE SYSTEM **TOLERANCES** PROVIDED IN THE AMM ARE MAINTAINED IN THE **OPTIMUM** RANGE.

- A. PERFORM ENGINE COMPRESSOR WASH USING ECOPOWER [P&W EA# 09IZ001] (PREFERRED) OR THE ALTERNATIVE EQUIPMENT PER AMM TASK 72-00-00-100-026-A.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PP/ACFT_TECH		HRS				
Insp.			HRS				

Job Description**3. EQUIPMENT REMOVAL:**

- A. REMOVE THE ECOPOWER EQUIPMENT AND RECONNECT THE DICONNECTED ITEMS DURING PREPARATION PER AMM TASK 72-00-00-100-026-A.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PP/ACFT_TECH		HRS				
Insp.	INSP/RII		HRS				

Job Description**4. LINES INSTALLATION:**

- A. INSTALL PS3 AND P25 LINES (IF INSTALLED) TO THE ECU AS FOLLOWS:

CAUTION: DO NOT BLOW COMPRESSED AIR OR NITROGEN INTO THE ECU AS DAMAGE TO THE ECU CAN OCCUR.

- (1) REMOVE THE PLUGS FROM THE PS3 AND P25 (IF INSTALLED) LINES AND FROM THE ECU.

CAUTION: DURING INSTALLATION, MAKE SURE THAT THE PS3 LINE DOES NOT TOUCH THE LOWER STARTER DUCT OR OTHER PARTS IN THIS AREA. IF IT DOES, CHAFING OF THE PS3 HOSE CAN OCCUR.

- (2) BLOW PS3 AND P25 LINES CLEAR OF WATER USING 20 PSI COMPRESSED AIR OR NITROGEN.

- (3) CONNECT PS3 LINE TO ECU, TORQUE TO BETWEEN 270 AND 300 LBF.IN (3.05 AND 3.39 M.DAN) AND SAFETY WITH LOCKWIRE.

- (4) CONNECT P25 (IF INSTALLED) LINE TO ECU, TORQUE TO BETWEEN 180 AND 200 LBF.IN (2.03 AND 2.26 M.DAN) AND SAFETY WITH LOCKWIRE.

- (5) REMOVE THE STANDARD PROTECTIVE COVER FROM THE NACELLE ANTI-ICE LINE PORT



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OF THE NAI VALVE.

- (6) CONNECT THE RIGID ANTI-ICE SERVO PRESSURE LINE (25) TO NACELLE ANTI-ICE VALVE (4000 DN). TORQUE THE FORWARD HOSE NUT TO BETWEEN 135 AND 150 LBF.IN (1.52 AND 1.69 M.DAN).
- (7) MAKE SURE THAT THE WORK AREA IS CLEAN AND CLEAR OF TOOLS.
- (8) CLOSE THE FAN COWLS 437AL, 438AR (REF. AMM TASK 71-13-00-410-040).

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PP/ACFT_TECH		HRS				
Insp.	INSP/RII		HRS				

Job Description

5. POST ENGINE WASH TEST:
- A. FOR AMM TASK 72-00-00-100-026, PERFORM POST ENGINE WASH TEST USING PURE WATER.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PP/ACFT_TECH		HRS				
Insp.			HRS				

Job Description

5. POST ENGINE WASH TEST:
- B. FOR ECOPOWER PERFORM POST ENGINE WASH TEST AS FOLLOWS:
- (1) ON THE AIR COND PANEL 30VU, MAKE SURE THAT THE CROSSBLEED VALVE SELECTOR SWITCH IS IN THE OPEN POSITION.
- (2) START THE ENGINE WITHIN 2 HOURS (OR 30 MINUTES IN ICING CONDITION) OF THE LAST WASH CYCLE TO PURGE THE LUBE/SUMP SYSTEM OF ANY WATER INGESTION.
(REF. AMM TASK 71-00-00-710-003)
- NOTE: APU BLEED VALVE MUST BE CLOSED (THE ON LEGEND OF APU PUSHBUTTON SWITCH IS OFF) AFTER ENGINE START, TO AVOID THE SUPPLY OF THE AIR TO THE WING-ANTI-ICE SYSTEMS FROM APU.
- THIS ENGINE RUN IS DONE TO ENSURE THAT THE BLEED SYSTEM IS FREE FROM CONTAMINANT. SMOKE/SMELL IN CABIN CAN RESULT IF CONTAMINANT ENTERS IN THE BLEED SYSTEM.
- CORROSION PREVENTIVE PRODUCTS SHOULD BE USED. IF NONE OF THE CORROSION



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PREVENTIVE PRODUCTS SPECIFIED ABOVE IS AVAILABLE, THE AIRCRAFT MUST RESUME REVENUE SERVICE OPERATION WITHIN 24 HOURS FOLLOWING THIS POST WASH TEST. DO NOT ACTUATE WAI SYSTEMS BEFORE THE 5-MINUTE PERIOD AT IDLE IS OVER IN ORDER TO AVOID WAI SYSTEMS OVERHEAT.

(3) LET THE ENGINE RUN AT IDLE FOR 5 MINUTES.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PP/ACFT_TECH		HRS				
Insp.			HRS				

Job Description**5. POST ENGINE WASH TEST:****B. CONTINUE POST ENGINE WASH TEST FOR ECOPOWER AS FOLLOWS:****(4) AFTER 5 MINUTES FROM ENGINE START, PERFORM THE FOLLOWING:**

- (a) PRESS THE ENG 1 BLEED PUSHBUTTON SWITCH TO OPERATE THE ENGINE BLEED SYSTEM.
- (b) MAKE SURE THAT THE OFF LEGEND OF THE ENG1 BLEED PUSHBUTTON SWITCH GOES OFF WHEN THE ENGINE BLEED SYSTEM OPERATES.
- (c) PRESS THE ANTL-ICE/WING PUSHBUTTON SWITCH TO OPERATE THE WING ANTI-ICE SYSTEM.
NOTE: THE AIM IS TO PURGE WATER POTENTIALLY TRAPPED IN HIGH PRESSURE (HP) BLEED DUCTS AND HP ENGINE BLEED CHAMBERS.

- (d) MAKE SURE THAT THE ON LEGEND OF THE ANTI ICE/WING PUSHBUTTON SWITCH GOES ON WHEN THE WING ANTI-ICE SYSTEM OPERATES.

NOTE: THE WING ANTI-ICE (WAI) VALVES WILL AUTOMATICALLY CLOSE 30 SECONDS AFTER THE WING ANTI-ICE SYSTEM IS SELECTED ON. THIS IS TO PREVENT WING SLATS OVERHEAT. HOWEVER, THE ON LEGEND REMAINS ILLUMINATED UNTIL THE SWITCH IS DESELECTED.

- (e) MAKE SURE THAT PRESSURE REGULATING VALVE (PRV) AND HIGH PRESSURE VALVE (HPV) ARE OPEN ON ECAM BLEED PAGE WITH ENGINE RUNNING AT THIS SPEED WHILE WAI VALVE IS OPEN.
- (f) ONCE WAI VALVE IS CLOSED, LET THE ENGINE RUN AT IDLE FOR 5 MINUTES WHILE DOING THE FOLLOWING STEPS:

NOTE: DO NOT ACTUATE WAI SYSTEMS BEFORE THE 5-MINUTE PERIOD AT IDLE IS OVER IN ORDER TO AVOID WAI SYSTEMS OVERHEAT.

- 1) PRESS THE ANTI ICE/WING PUSHBUTTON SWITCH AND MAKE SURE THAT THE ON LEGEND OF THE ANTI ICE/WING PUSHBUTTON SWITCH GOES OFF.
- 2) TURN THE CROSSBLEED VALVE SELECTOR SWITCH TO THE AUTO POSITION.
- 3) PRESS THE PACK 1 PUSHBUTTON SWITCH TO OPERATE THE AIR CONDITIONING PACK.
- 4) MAKE SURE THAT THE OFF LEGEND OF THE PACK 1 PUSHBUTTON GOES OFF.



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Task Card Title:

NO. 2 ENGINE COMPRESSOR WASH

Task Card NO:

72-00-00-01-0-SVA-002

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP**STEP COMPLETION SIGN OFF:**

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PP/ACFT_TECH		HRS				
Insp.			HRS				

Job Description

5. POST ENGINE WASH TEST:

B. CONTINUE POST ENGINE WASH TEST FOR ECOPOWER AS FOLLOWS:

(5) AFTER 10 MINUTES OF ENGINE RUNNING, CHECK FOR EVIDENCE OF ODOR/SMOKE IN THE CABIN.

(a) IF NO EVIDENCE IS FOUND, SHUT DOWN THE ENGINE. (REF. AMM TASK 71-00-00-710-028)

(b) IF EVIDENCE IS FOUND, CONTINUE TO OPERATE THE ENGINE AT IDLE UNTIL NO ODOR/SMOKE REMAINS IN THE CABIN.

(c) AFTER 10 MINUTES ENGINE RUNNING, IF CONTINUOUS ODOR/SMOKE REMAINS IN THE CABIN, SHUT DOWN THE ENGINE (REF. AMM TASK 71-00-00-710-028) AND DECONTAMINATE THE ENVIRONMENTAL CONTROL SYSTEM (ECS). NOTE THAT THE APU BLEED VALVE MUST BE OPEN BEFORE ENGINE SHUTDOWN.

1) WHEN AMBIENT AIR TEMPERATURE IS BELOW 75.2°F (24°C).

(REF. AMM TASK 21-00-00-615-003)

2) WHEN AMBIENT AIR TEMPERATURE IS ABOVE 75.2°F (24°C).

(REF. AMM TASK 21-00-00-615-004)

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PP/ACFT_TECH		HRS				
Insp.			HRS				

Job Description

5. POST ENGINE WASH TEST:

B. CONTINUE POST ENGINE WASH TEST FOR ECOPOWER AS FOLLOWS:

WARNING: OIL WILL BE HOT. PERSONNEL MUST WEAR PROPER PERSONAL PROTECTIVE EQUIPMENT WHILE TAKING THE OIL SAMPLE. WEAR NYTRILE OR RUBBER GLOVES TO PROTECT HANDS FROM EXPOSURE TO ENGINE OILS.

(6) PERFORM AN OIL WATER CONTENT TEST AS FOLLOWS:

(a) OBTAIN AN ENGINE OIL SAMPLE OF APPROXIMATELY 2 OUNCES (60 ML).

(b) REINSTALL THE MAIN OIL TANK CAP AND ENSURE IT IS PROPERLY LOCKED IN POSITION.

NOTE: IT IS PERMISSIBLE TO USE THE THIN FILM CAPACITIVE POLYMER (TFCP) TESTER, PALL METER (WS04 WATER SENSOR) TESTER OR EQUIVALENT.

(c) FOR THE TFCP OIL WATER CONTENT TESTER, PERFORM THE FOLLOWING:

NOTE: IT IS PERMISSIBLE TO INSERT THE TFCP TESTER DIRECTLY INTO THE MAIN OIL TANK FILL FOR THE TFCP OIL WATER CONTENT TESTER (TFCP) TEST.

1) SET OIL WATER CONTENT TESTER TO MIL-L-23699 OIL, OR EQUIVALENT.



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Work Package:

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A/C Registration:

HZ-ASP

2) STIR THE PROBE IN THE OIL UNTIL THE UNIT HAS REACHED THE STEADY STATE LEVEL OF WATER CONCENTRATION FOR 20 SECONDS. THIS NORMALLY TAKES AT LEAST TWO MINUTES OF STIRRING TO COMPLETE.

- (d) FOR PALL METER (WS04 WATER SENSOR) TESTER, THE PROBE SHOULD BE INSERT INTO THE TEST FLUID SO THAT THE PROBE TIP IS COMPLETELY IMMersed IN THE FLUID. ALLOW ABOUT THREE MINUTES FOR THE READING TO STABILIZE BEFORE TAKING THE MEASUREMENT.

NOTE: PLEASE NOTE THAT THE SENSOR IN THE PROBE IS SENSITIVE TO LIGHT. DO NOT USE WHERE LIGHT MAY SHINE DIRECTLY ON THE SENSOR.

- (e) ENGINE OIL/WATER CONTENT LIMITS:

- 1) THE WATER CONTENT IN THE OIL MUST BE BELOW 1000 PARTS PER MILLION (PPM) BY WEIGHT OR BY VOLUME.
- 2) IF THE OIL EXCEEDS 1000 PPM BY WEIGHT OR BY VOLUME, PERFORM ONE OF THE FOLLOWING:
 - a) DRAIN AND SERVICE OIL TANK WITH FRESH OIL.
(REF. AMM, 72-00-00, SERVICING).
 - b) RE-RUN AT IDLE AND TEST AGAIN FOR OIL/WATER CONTENT.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PP/ACFT_TECH		HRS				
Insp.			HRS				

Job Description

5. ENGINE WASH DATA SHEET:

C. FILL THE ENGINE WASH DATA TABLE.

A/C Registration	
A/C Type	
Engine Position	
Engine Serial #	
Date	
Station	
TDS (PPM) Cycle 1	
TDS (PPM) Cycle 2	
TDS (PPM) Cycle 3	
TDS (PPM) Cycle 4	
TDS (PPM) Cycle 5	
Oil PPM	



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Task Card Title:

NO. 2 ENGINE COMPRESSOR WASH



Task Card NO:

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Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

Oil Temp

Oil % Saturation

OAT (° F)

IPA %

Note:

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
						Day	Month	Year
Mechanic/s	PP/ACFT_TECH			HRS				
Insp.				HRS				

Job Description

6. ENGINE WASH DATA SHEET DELIVERY:

D. DELIVER OR E-MAIL A COPY OF THE ENGINE WASH DATA TO SV AIRCRAFT MAINTENANCE PLANNING AND CONTROL DEPARTMENT CC# 873
E-MAIL: svamp@saudia.com.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration		Number./Signature	Date		
						Day	Month	Year
Mechanic/s	PROD/PLANNER			HRS				
Insp.				HRS				

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below
Work order No: _____

Note: Make sure to creat the finding Work Order on task card level.

Remark's: _____



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Task Card Title:

NO. 2 ENGINE COMPRESSOR WASH

Task Card NO:

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Work Package:

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A/C Registration:

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(1) If Task card interrupted or performed by different personals updates the system and update the following according (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 2 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 3 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 4 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 5 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____

**Task Completion Close Up And
System Update**

Station**Date****Signature**

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ENGINE - GENERAL - CLEANING/PAINTING

TASK 72-00-00-100-026 -A

On A/C 003-250

Washing of the Engine Gas Path With Pure Water

WARNING: BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER ENGINE SHUTDOWN. USE APPLICABLE THERMAL GLOVES. THE ENGINE PARTS AND THE ENGINE OIL CAN STAY HOT FOR ONE HOUR AFTER SHUTDOWN AND CAN BURN YOU.

WARNING: BE CAREFUL WHEN YOU USE CONSUMABLE MATERIALS. OBEY THE MATERIAL MANUFACTURER'S INSTRUCTIONS AND YOUR LOCAL REGULATIONS.

CAUTION: IF CLEANING AGENT IS USED, PROPER AIRCRAFT MAINTENANCE CONFIGURATION, ENGINE RINSING AND POST ENGINE WASH TEST MUST BE COMPLETED TO PREVENT THE POSSIBILITY OF FUMES IN THE CABIN WHEN AIRCRAFT RESUMES REVENUE FLIGHT OPERATION.

FIN: 1000EM1, 1000EM2

1. Reason for the Job

For Exhaust Gas Temperature (EGT) margin recovery, it is recommended to perform the engine gas path water wash with pure water only.

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	AR	COVER - PROTECTION
No specific	1	TDS TOTAL DISSOLVED SOLID METER OR EQUIVALENT
No specific	AR	WARNING NOTICE(S)
No specific		Torque wrench: range to between 150 and 160 lbf.in (1.70 and 1.81 m.daN)
No specific		Torque wrench: range to between 180 and 200 lbf.in (2.03 and 2.26 m.daN)
No specific		Torque wrench: range to between 270 and 300 lbf.in (3.05 and 3.39 m.daN)

B. Consumable Materials

REFERENCE	DESIGNATION
CP8001	lockwire 0.032 in. (0.8 mm) dia.
CP8002	lockwire 0.032 in. (0.8 mm) dia

C. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
210	COCKPIT FWD COMPARTMENT BULKHEAD TO FLIGHT COMPARTMENT BULKHEAD
400	POWER PLANT, NACELLES AND PYLONS
FOR 1000EM1(ENGINE-1)	
437AL , 438AR	
FOR 1000EM2(ENGINE-2)	
447AL , 448AR	

D. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 12-13-79-610-002-A	Check oil level and replenish Check Pop Out Indicator of EMCD (Electrical Master Chip Detector)
Ref. AMM 12-13-79-610-002-A-01	Check oil level and replenish (Pressure Filling)
Ref. AMM 21-00-00-615-803-A	Level B Removal of the Contamination of the Environmental Control System (ECS)
Ref. AMM 21-00-00-615-803-B	Level B Removal of the Contamination of the Environmental Control System (ECS) With TCT Part Number (P/N) SBE2272

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REFERENCE	DESIGNATION
Ref. AMM 24-41-00-861-002-A	Energize the Aircraft Electrical Circuits from the External Power
Ref. AMM 24-41-00-861-002-A-01	Energize the Aircraft Electrical Circuits from the APU
Ref. AMM 24-41-00-861-002-A-02	Energize the Aircraft Electrical Circuits from Engine 1(2)
Ref. AMM 24-41-00-862-002-A	De-energize the Aircraft Electrical Circuits Supplied from the External Power
Ref. AMM 24-41-00-862-002-A-01	De-energize the Aircraft Electrical Circuits Supplied from the APU
Ref. AMM 24-41-00-862-002-A-02	De-energize the Aircraft Electrical Circuits Supplied from the Engine 1(2)
Ref. AMM 71-00-00-710-001-B	Dry Motoring Check
Ref. AMM 71-00-00-710-003-B	Engine Automatic Start
Ref. AMM 71-00-00-710-028-B	Engine Shutdown
Ref. AMM 71-00-00-710-802-A	Water Wash Crank
Ref. AMM 71-13-00-010-040-B	Opening of the Fan Cowl Doors
Ref. AMM 71-13-00-410-040-B	Closing of the Fan Cowl Doors
Ref. AMM 72-00-00-100-008-A	Engine Washing Requirements
Ref. Fig. (1 / 72-00-00-991-32500)	
Ref. Fig. (1 / 72-00-00-991-30400)	

3. Job Set-up

Subtask 72-00-00-941-065-A

On A/C 003-250

A. Safety Precautions

- (1) On the center pedestal, on the ENG panel 115VU:
 - (a) Put a WARNING NOTICE(S) to tell persons not to start the engine.
- (2) Make sure that the engine shutdown occurred five minutes or more before you do this procedure.
- (3) On the overhead maintenance panel 50VU:
 - (a) Make sure that the ENG/FADEC GND PWR/1(2) pushbutton switch is OFF (the ON legend of the ENG/FADEC GND PWR/1(2) pushbutton switch is OFF).
 - (b) Put a WARNING NOTICE(S) to tell persons not to energize the FADEC 1(2).

Subtask 72-00-00-010-072-A

On A/C 003-250

B. Get Access

- (1) Open the fan cowl doors [Ref. AMM 71-13-00-010-040-B](#):
 - (a) FOR 1000EM1 (ENGINE-1)
437AL 438AR
 - (b) FOR 1000EM2 (ENGINE-2)
447AL 448AR

4. Procedure

Subtask 72-00-00-110-057-A

On A/C 003-250

A. Preparation of the Engine for Washing

- (1) For the engine washing requirements [Ref. AMM 72-00-00-100-008-A](#).

NOTE : The disconnection of the P25 line is only available for the engines equipped with the PMUX option.

CAUTION: USE TWO WRENCHES TO REMOVE OR INSTALL THE TUBE COUPLING NUTS:

- ONE WRENCH TO HOLD THE UNION, AND
- ONE TO LOOSEN OR TIGHTEN THE COUPLING NUT.

THIS PREVENTS DAMAGE TO THE PARTS.

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- (2) Disconnect the PS3 line and the P25 line (if installed) from the Electronic Control Unit (ECU).
- (3) Plug the open tubes and ports of the ECU with the COVER - PROTECTION.
- (4) Disconnect the flexible hose (25) from the Nacelle Anti-Icing (NAI) valve:
Ref. Fig. (1 / 72-00-00-991-32500)
 - (a) Let the flexible hose free (not plugged) during the washing of the engine.
- (5) Install the COVER - PROTECTION on the anti-ice servo-pressure-line port of the NAI valve.
- (6) Prepare an anti-freeze mixture, if necessary, as defined in the engine washing requirements
Ref. AMM 72-00-00-100-008-A.
- (7) Service the oil tank:
 - (a) Do a check of oil level of the engine Ref. AMM 12-13-79-610-002-A
Ref. AMM 12-13-79-610-002-A-01.

Subtask 72-00-00-861-051-A

On A/C 003-250

- B. Energize the aircraft electrical circuits Ref. AMM 24-41-00-861-002-A
Ref. AMM 24-41-00-861-002-A-01 Ref. AMM 24-41-00-861-002-A-02.

Subtask 72-00-00-860-051-A

On A/C 003-250

- C. Aircraft Maintenance Configuration
- (1) On the ANTI-ICE section of panel 25VU, make sure that the ANTI ICE/ENG 1(2) pushbutton switch is OFF (the ON legend of the ANTI ICE/ENG 1(2) pushbutton switch is OFF).
 - (2) On the AIR COND panel 30VU, make sure that ENG 1(2) BLEED pushbutton switch is OFF (the OFF legend of the ENG1(2) BLEED is ON).
 - (3) On the AIR COND panel 30VU, make sure that PACK 1 and PACK 2 pushbutton switches are OFF (the OFF legend of the PACK 1 and the PACK 2 pushbutton switches are ON).

Subtask 72-00-00-110-058-B

On A/C 003-250

EMB SB 73-1100 for A/C 003-004; 051-054; 101-109

- D. Procedure to Wash the Engine

CAUTION: DO NOT WASH AN ENGINE WHICH HAS BEEN EXPOSED TO DRY
CHEMICAL FIRE EXTINGUISHING AGENTS.

CAUTION: DO NOT WASH AN ENGINE UNLESS THE EGT IS BELOW 150 DEG.F (66
DEG.C).

CAUTION: DO NOT WASH AN ENGINE AT TEMPERATURE BELOW 14 DEG.F (-10
DEG.C).

CAUTION: THE FUEL PUMP AND HYDRO-MECHANICAL UNIT (HMU) ARE FUEL
LUBRICATED. DO NOT MOTOR, START, OR OPERATE ENGINE UNLESS A
POSITIVE FUEL INLET PRESSURE IS INDICATED ON THE LOWER ECAM
(FUEL SYSTEM PAGE). DAMAGE TO COMPONENTS MAY RESULT.

CAUTION: DO NOT EXCEED THE STARTER DUTY CYCLE. DAMAGE TO THE STARTER
CAN OCCUR.

CAUTION: MAKE SURE THE N1 ROTOR TURNS FREELY THROUGHOUT WASHING TO
PREVENT OIL SYSTEM CONTAMINATION.

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CAUTION: DO NOT APPLY WATER AT MORE THAN 6 US GAL (23 LITERS) PER MINUTE. IF YOU DO, ENGINE DAMAGE CAN OCCUR.

CAUTION: DRY MOTOR CYCLE DURATIONS MUST NOT EXCEED 2 MINUTES (REFER TO AMM TASK 71-00-00-710-001).

- (1) Dry motor the engine for two minutes [Ref. AMM 71-00-00-710-001-B](#) or do the water wash crank [Ref. AMM 71-00-00-710-802-A](#).

NOTE : The water wash crank is an alternate procedure available for the FADEC POST 5BS2 software. This task is to optimize the washing procedure by positioning the valves in the better way.

- (2) After 10 seconds of motoring with a positive fan rotation, inject water at one o'clock position (aft view) on the Low Pressure Compressor (LPC) inlet through the fan blades (or at the LPC inlet if a specific rig is used).
- (3) Stop the water injection before the end of the engine motoring.
- (4) Let the engine soak for five minutes.
- (5) Repeat paragraph 4.D.(1) to paragraph 4.D.(4).
- (6) Let the engine soak for five minutes.
- (7) Dry motor the engine again [Ref. AMM 71-00-00-710-001-B](#).

NOTE : Water can potentially drain out of the disconnected anti-ice servo pressure-line. If water drains out, continue to dry motor the engine until there is no more water draining out of the line.

Subtask 72-00-00-110-087-A

On A/C 003-250

- E. Procedure to Wash the Engine With the Total Dissolved-Solid Meter or Equivalent - Alternative Procedure Available

CAUTION: DO NOT WASH AN ENGINE WHICH HAS BEEN EXPOSED TO DRY CHEMICAL FIRE EXTINGUISHING AGENTS.

CAUTION: DO NOT WASH AN ENGINE UNLESS THE EGT IS BELOW 150 DEG.F (66 DEG.C).

CAUTION: DO NOT WASH AN ENGINE AT TEMPERATURE BELOW 14 DEG.F (-10 DEG.C).

CAUTION: THE FUEL PUMP AND HYDRO-MECHANICAL UNIT (HMU) ARE FUEL LUBRICATED. DO NOT MOTOR, START, OR OPERATE ENGINE UNLESS A POSITIVE FUEL INLET PRESSURE IS INDICATED ON THE LOWER ECAM (FUEL SYSTEM PAGE). DAMAGE TO COMPONENTS MAY RESULT.

CAUTION: DO NOT EXCEED THE STARTER DUTY CYCLE. DAMAGE TO THE STARTER CAN OCCUR.

CAUTION: MAKE SURE THE N1 ROTOR TURNS FREELY THROUGHOUT WASHING TO PREVENT OIL SYSTEM CONTAMINATION.

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CAUTION: DO NOT APPLY WATER AT MORE THAN 6 US GAL (23 LITERS) PER MINUTE. IF YOU DO, ENGINE DAMAGE CAN OCCUR.

CAUTION: DRY MOTOR CYCLE DURATIONS MUST NOT EXCEED 2 MINUTES (REFER TO AMM TASK 71-00-00-710-001).

- (1) Optional - collect a sample of wash supply water and measure the total dissolved solid with the TDS TOTAL DISSOLVED SOLID METER OR EQUIVALENT (supply-water total dissolved-solid). Make a record of the initial measurement.

NOTE : Use a TDS TOTAL DISSOLVED SOLID METER OR EQUIVALENT with properties that follow:

- Measurement range of 0 to 9990 ppm total dissolved solids
- Resolution of 1 ppm
- Readout accuracy of plus or minus two percent.

- (2) Dry motor the engine for two minutes [Ref. AMM 71-00-00-710-001-B](#) or do the water wash crank [Ref. AMM 71-00-00-710-802-A](#).

NOTE : The water wash crank is an alternative procedure available for the FADEC POST 5BS2 software. This task is to optimize the washing procedure by positioning the valves in the better way.

- (3) After 10 seconds of motoring with a positive fan rotation, inject water at the one o'clock position (aft view) on the Low Pressure Compressor (LPC) inlet through the fan blades (or at the LPC inlet if a specific rig is used).
- (4) Stop the water injection before the end of the engine motoring and if necessary, do these steps:
- (a) During the engine coast-down, collect a sample of the wash water that drains from the exhaust area of the engine.
 - (b) Measure and make a record of the total dissolved solid with the TDS TOTAL DISSOLVED SOLID METER OR EQUIVALENT.
 - (c) Subtract initial-supply-water total-dissolved-solid value, measured in paragraph 4.E.(1) above, from the rinse effluent value.
 - 1 If the net total-dissolved-solid value is less than 150 ppm, no additional rinse cycles are necessary.
 - 2 If this value is more than 150 ppm, do additional wash/soak cycles until the net total-dissolved-solid falls below 150 ppm, up to a maximum of four wash cycles.
- (5) Let the engine soak for five minutes.
- (6) Repeat paragraph 4.E.(1) to paragraph 4.E.(4).
- (7) Let the engine soak for five minutes.
- (8) Dry motor the engine [Ref. AMM 71-00-00-710-001-B](#).

NOTE : Water can potentially drain out of the disconnected anti-ice servo pressure-line. If water drains out, continue to dry motor the engine until there is no more water draining out from the line.

Subtask 72-00-00-420-055-A

On A/C 003-250

F. Installation of PS3 Line and P25 Line (if installed)

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WARNING: USE PROTECTIVE GOGGLES AND GLOVES EACH TIME YOU CUT LOCKWIRE. DISCARD LOOSE OR USED LOCKWIRE IMMEDIATELY. LOOSE OR USED LOCKWIRE CAN CUT YOU OR MAKE YOU BLIND.

CAUTION: DO NOT BLOW COMPRESSED AIR OR NITROGEN INTO THE ECU AS DAMAGE TO THE ECU CAN OCCUR.

CAUTION: DURING INSTALLATION, MAKE SURE THAT THE PS3 LINE DOES NOT TOUCH THE LOWER STARTER DUCT OR OTHER PARTS IN THIS AREA. IF IT DOES, CHAFING OF THE PS3 HOSE CAN OCCUR.

- (1) Remove the plugs from the PS3 and P25 lines and from the ECU.
- (2) Blow PS3 and P25 lines clear of water using compressed air or nitrogen.
 - (a) Do a visual inspection to make sure that the PS3 line weep-hole is clear and there is no blockage.
- (3) Connect PS3 line to ECU, TORQUE **to between 270 and 300 lbf.in (3.05 and 3.39 m.daN)** and safety with lockwire 0.032 in. (0.8 mm) dia. (Material No. **CP8001**) or lockwire 0.032 in. (0.8 mm) dia (Material No. **CP8002**) .
- (4) Connect P25 line to ECU, TORQUE **to between 180 and 200 lbf.in (2.03 and 2.26 m.daN)** and safety with lockwire 0.032 in. (0.8 mm) dia. (Material No. **CP8001**) or lockwire 0.032 in. (0.8 mm) dia (Material No. **CP8002**) .
- (5) Remove the COVER - PROTECTION from the nacelle anti-ice line port of the NAI valve.
- (6) Connect the flexible hose (25) to the nacelle anti-ice valve (4000 DN). TORQUE the flexible hose coupling **to between 150 and 160 lbf.in (1.70 and 1.81 m.daN)**.

Subtask 72-00-00-410-085-A

On A/C 003-250

G. Close Access

- (1) Make sure that the work area is clean and clear of tools and other items.
- (2) Close the following engine fan cowl doors [Ref. AMM 71-13-00-410-040-B](#):
 - (a) FOR 1000EM1 (ENGINE-1)
437AL 438AR
 - (b) FOR 1000EM2 (ENGINE-2)
447AL 448AR

Subtask 72-00-00-620-063-A

On A/C 003-250

H. Post-Engine Wash Test - Alternative Procedure Available

Ref. Fig. (1 / 72-00-00-991-30400)

NOTE : The aim of this post-engine wash test is:

- To purge water from the oil system in order to prevent potential bearing/sump corrosion/deterioration
- To prevent fumes in the cockpit and in the cabin when the airplane is put back into service.

- (1) On AIR COND panel 30VU, make sure that:
 - The crossbleed-valve selector switch is in the OPEN position.
- (2) Start the engine within two hours (or 30 minutes in icing condition) of the last wash cycle to purge the lube/sump system of any water ingestion [Ref. AMM 71-00-00-710-003-B](#).

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NOTE : The APU bleed valve must be closed (the ON legend of the APU bleed pushbutton switch off) after the engine start, to avoid the supply of air to the Wing Anti-Ice systems from the APU.

NOTE : This engine run is done to ensure that the bleed system is free from contaminants. Smoke/smell in the cabin and Starter Air Valve (SAV) contamination can result if contaminants go into the bleed system.

- (a) Let the engine run at idle for five minutes and after, do the following:
- 1 Press the ANTI-ICE/ENG pushbutton switch to operate the engine anti-ice system.
 - 2 After five minutes, set the ANTI-ICE/ENG pushbutton switch to OFF.
 - 3 Press the ENG 1(2) BLEED pushbutton switch to operate the engine bleed system.
 - 4 Make sure that the OFF legend of the ENG1(2) BLEED pushbutton switch goes off when the engine bleed system operates.
 - 5 Press the ANTI ICE/WING pushbutton switch to operate the WAI system.

NOTE : The aim is to purge water potentially trapped in the High Pressure (HP) bleed ducts and the HP engine bleed chambers.

- 6 Make sure that the ON legend of the ANTI ICE/WING pushbutton switch comes on when the WAI system operates.

NOTE : 30 seconds after you push the ANTI ICE/WING pushbutton switch, the WAI valves will automatically close to prevent wing slat overheat. But the ON legend of the pushbutton switch will stay on until you release the pushbutton switch.

- 7 Make sure that the Pressure Regulating Valve (PRV) and the High Pressure Valve (HPV) are open on the ECAM BLEED page with the engine running at this speed while the WAI valve is open.
- 8 Once the WAI valve is closed, let the engine run at idle for five minutes.

NOTE : Do not actuate WAI systems before the 5-minute period at idle is over in order to avoid the WAI system overheat.

- 9 Increase and stabilize the engine thrust to at least 60 percent Fan (N1) for ten minutes.
- 10 Press the ANTI ICE/WING pushbutton switch two times to operate the WAI system once more.

NOTE : The aim is to purge water potentially trapped in the IP bleed ducts and the IP engine bleed chambers.

NOTE : If the Wing Anti-Ice ON legend is still on, pressing twice the WAI pushbutton switch enables a second WAI cycle of 30 seconds to be started.

NOTE : Do not perform more than two attempts with a minimum of five minutes between the two attempts, in order to avoid the WAI system overheat.

- 11 Make sure that the ON legend of the ANTI ICE/WING pushbutton switch comes on when the WAI system operates.

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- 12 Make sure that the PRV is open and the HPV is closed on the ECAM BLEED page with the engine running at this speed while the WAI valve is open.
- (b) Once the WAI valve is closed, pull the throttle to stabilize the engine thrust at idle for five minutes and after, do the following:
 - 1 Press the ANTI ICE/WING pushbutton switch and make sure that the ON legend of the ANTI ICE/WING pushbutton switch goes OFF.
 - 2 Turn the crossbleed-valve selector switch to the AUTO position.
 - 3 Press the PACK 1(2) pushbutton switch to operate the air conditioning pack.
 - 4 Make sure that the OFF legend of the PACK 1(2) pushbutton switch goes off.
 - 5 Check for evidence of anti-freeze mixture odor/smoke in the cabin.
 - If evidence is found, continue to operate the engine at idle until no odor/smoke remains in the cabin.
 - After ten minutes of engine running, if continuous odor/smoke remains in the cabin, shut down the engine [Ref. AMM 71-00-00-710-028-B](#) and decontaminate the Environmental Control System (ECS) [Ref. AMM 21-00-00-615-803-A](#) [Ref. AMM 21-00-00-615-803-B](#).
Note that the APU Bleed valve must be open before the engine shutdown.
 - If no evidence is found, shut the engine down [Ref. AMM 71-00-00-710-028-B](#).

Subtask 72-00-00-620-070-A

On A/C 003-250

CAUTION: DO NOT USE THIS ALTERNATIVE PROCEDURE IF THE TEMPERATURE IS LESS THAN +5 DEG. C.

J. Post-Engine Wash Test - Alternative Procedure

NOTE : The aim of this post-engine wash test is:

- To purge water from oil system in order to prevent potential bearing/sump corrosion/deterioration
 - To prevent fumes in the cockpit and in the cabin when the airplane is put back into service.
- (1) On the AIR COND panel 30VU, make sure that:
 - PACK 1 and PACK 2 are OFF prior to the start of the procedure that follows.
 - The crossbleed-valve selector switch is in the OPEN position.
 - ENG 1 BLEED and ENG 2 BLEED are OFF.
 - (2) On the VENTILATION panel 22VU, make sure that:
 - CAB FANS pushbutton switch is in the OFF position.
 - (3) Start the engine within 2 hours (or 30 minutes in icing condition) of the last wash cycle to purge the lube/sump system of any water ingestion [Ref. AMM 71-00-00-710-003-B](#).

NOTE : This engine run is done to ensure that the bleed system is free from contaminant. Smoke/Smell in the cabin and Starter Air Valve (SAV) contamination can result if contaminant enters the bleed system.

- (4) After engine start, push the APU BLEED pushbutton switch to the OFF position. The APU bleed valve must be closed, to avoid the supply of air to the wing anti-ice systems from the APU.
- (5) Let the engine run at idle for 5 minutes.
- (6) Press the ANTI ICE/ENG pushbutton switch to operate the engine anti-ice system.

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- (7) After 5 minutes put the ANTI ICE/ENG pushbutton switch to OFF.
- (8) Do the following:
 - (a) Press the ENG 1(2) BLEED pushbutton switch to ON. Make sure that the OFF legend goes off when the engine bleed system operates.
 - (b) Press the ANTI ICE/WING pushbutton switch to ON. Make sure that the ON legend goes on when the wing anti ice system operates.

NOTE : The aim is to purge water potentially trapped in High Pressure (HP) bleed ducts and HP engine bleed chambers.

NOTE : 30 seconds after you push the ANTI ICE/WING pushbutton switch, the WAI valves will automatically close to prevent wing slat overheat. But the ON legend of the pushbutton switch will stay on until you release the pushbutton switch.

- (c) Make sure that Pressure Regulating Valve (PRV) and High Pressure Valve (HPV) are open on ECAM BLEED page with engine running at this speed while WAI valves are open.
- (9) Once WAI valves are closed, let the engine run at idle for 5 minutes.

NOTE : Do not actuate WAI system before the 5-minute period at idle is over in order to avoid WAI system overheat.

- (10) Press twice the ANTI ICE/WING pushbutton switch to operate the WAI system once more.

NOTE : If the Wing Anti-Ice ON legend is still on, pressing twice the WAI pushbutton switch enables a second WAI cycle of 30 seconds to be started.

NOTE : Do not perform more than two attempts with a minimum of 5 minutes between the two attempts, in order to avoid WAI systems overheat.

- (a) Make sure that the ON legend goes on when the wing anti ice system operates.
 - (b) Make sure that Pressure Regulating Valve (PRV) and High Pressure Valve (HPV) are open on ECAM BLEED page with engine running at this speed while WAI valves are open.
- (11) Once the WAI valves are closed, stabilize the engine thrust at idle for 5 minutes.
- (12) Press once the ANTI ICE/WING pushbutton switch and make sure that the ON legend goes off.

NOTE : Do not press the WAI pushbutton switch twice. If the WAI pushbutton switch ON legend is still on, pressing twice the WAI pushbutton switch enables a second WAI cycle of 30 seconds to be started.

- (13) Push the APU BLEED pushbutton switch to the ON position.
- (14) Turn the crossbleed-valve selector switch to the AUTO position.
- (15) Press the PACK 1(2) pushbutton switch to operate the air conditioning pack.
- (16) Make sure that the OFF legend of the PACK 1(2) pushbutton switch goes off.
- (17) Press the CAB FANS pushbutton switch.
- (18) Make sure that the OFF legend of the CAB FANS pushbutton switch goes off.
- (19) Shut down the engine [Ref. AMM 71-00-00-710-028-B](#).

5. Close-up

Subtask 72-00-00-869-065-A

On A/C 003-250

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A. Make sure that the work area is clean and clear of tools and other items.

Subtask 72-00-00-862-051-A

On A/C 003-250

B. De-energize the aircraft electrical circuits [Ref. AMM 24-41-00-862-002-A](#)

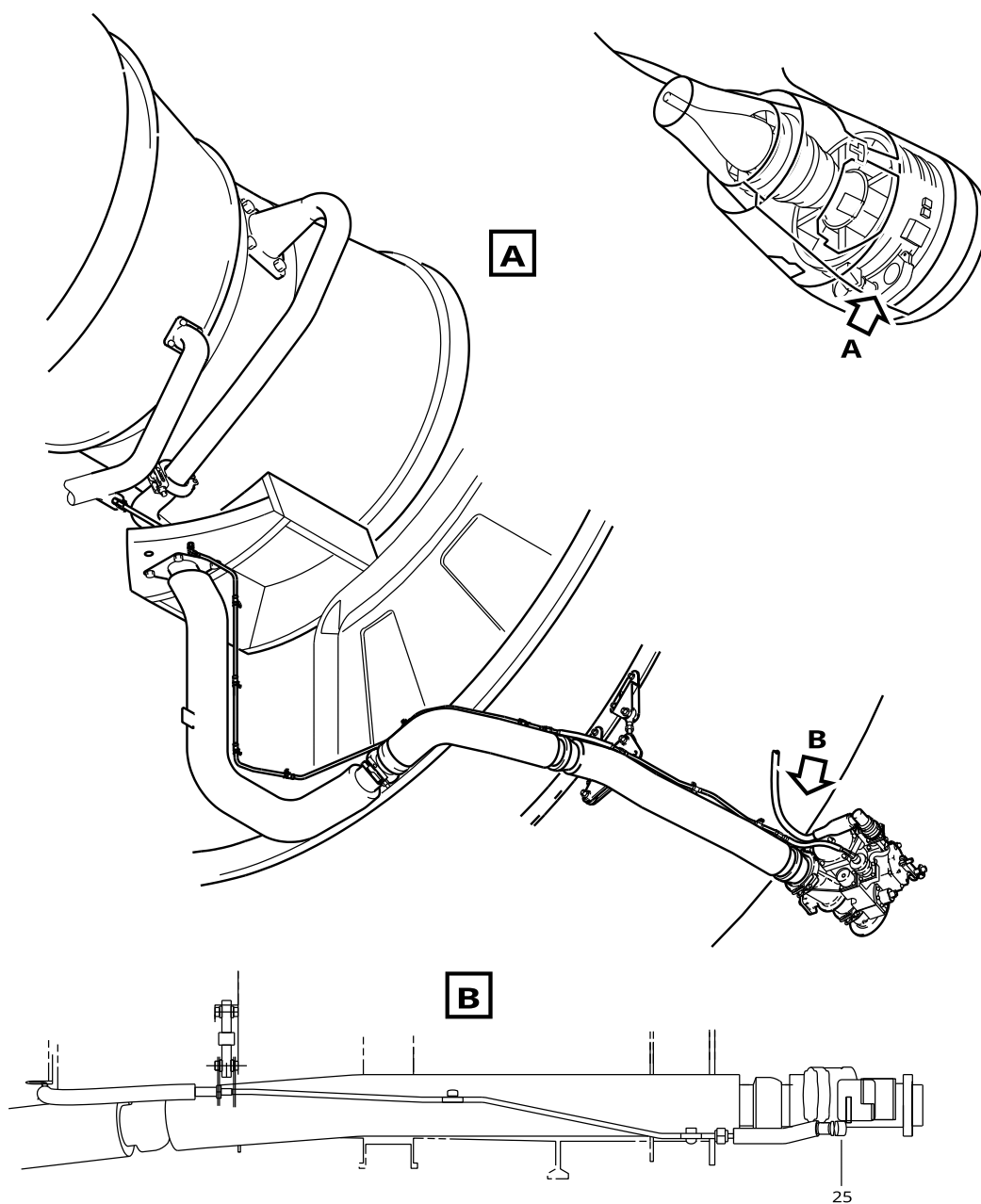
[Ref. AMM 24-41-00-862-002-A-01](#) [Ref. AMM 24-41-00-862-002-A-02](#).

Subtask 72-00-00-942-053-A

On A/C 003-250

C. Remove the WARNING NOTICE(S).

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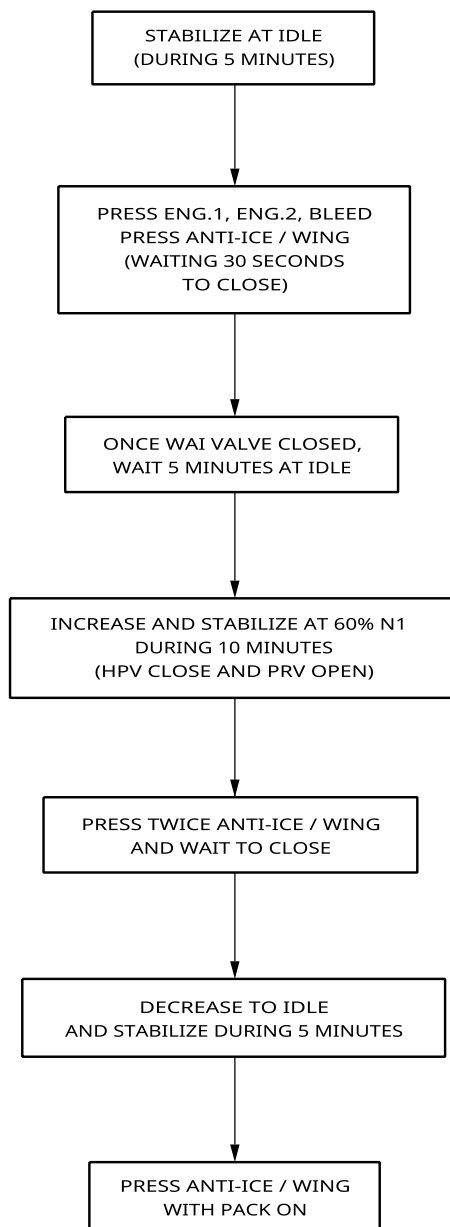


N_MM_720000_7_RHM0_01_01

Anti-Ice Muscle Line
Figure 1 / Graphic 72-00-00 -991-32500

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STARTING CONFIGURATION



N_MM_720000_7_RFM0_01_01

Engine Cleaning (Washing with Pure Water)
Figure 1 / Graphic 72-00-00 -991-30400

S-M156-MM-04706-04-B

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Task Card Title:

NO. 1 ENGINE INLET FAN SECTION INSP

Task Card NO:

722000-C1-1-001

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP**STEP COMPLETION SIGN OFF:**

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s				HRS			
Insp.	II/INSP			HRS			

CLOSE-UP**Job Description**

C. PERFORM JOB CLOSE-UP PER AMM TASK 72-23-00-210-003-C.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PP/ACFT_TECH			HRS			
Insp.				HRS			

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below

Work order No: _____

Note: Make sure to create the finding Work Order on task card level.

Remark's: _____



Task Card Title:

NO. 1 ENGINE INLET FAN SECTION INSP

Task Card NO:

722000-C1-1-001

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

(1) If Task card interrupted or performed by different personals updates the system and update the following according (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 2 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 3 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 4 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 5 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____

**Task Completion Close Up And
System Update**

Station**Date****Signature**

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FAN FRAME ASSEMBLY - INSPECTION/CHECK

TASK 72-23-00-210-003 -C

On A/C 003-250

Detailed Inspection of Spinner, Engine Inlet, Fan Blades, Fan Outlet and Abradable Material

WARNING: BE CAREFUL WHEN YOU USE CONSUMABLE MATERIALS. OBEY THE MATERIAL MANUFACTURER'S INSTRUCTIONS AND YOUR LOCAL REGULATIONS.

WARNING: BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER ENGINE SHUTDOWN. USE APPLICABLE THERMAL GLOVES. THE ENGINE PARTS AND THE ENGINE OIL CAN STAY HOT FOR ONE HOUR AFTER SHUTDOWN AND CAN BURN YOU.

FIN: 1000EM1, 1000EM2

1. Reason for the Job

Refer to the MPD TASK: 722000-C1

DETAILED INSPECTION OF SPINNER, ENGINE INLET, FAN BLADES, FAN OUTLET AND ABRADABLE MATERIAL.

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	1	ACCESS PLATFORM 1M (3 FT)-ADJUSTABLE
No specific	1	MAT - PROTECTIVE
No specific	AR	WARNING NOTICE(S)

B. Consumable Materials

REFERENCE	DESIGNATION
CP2216	Loctite 496
CP2516	(Speed tape)

C. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
400	POWER PLANT, NACELLES AND PYLONS
FOR 1000EM1(ENGINE-1)	
437AL , 438AR , 451AL , 452AR	
FOR 1000EM2(ENGINE-2)	
447AL , 448AR , 461AL , 462AR	

D. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 71-13-00-010-040-B	Opening of the Fan Cowl Doors
Ref. AMM 71-13-00-410-040-B	Closing of the Fan Cowl Doors
Ref. AMM 72-00-00-200-025-A	Over-Serviceable-Limit Extensions
Ref. AMM 72-00-00-200-025-A-01	Over-Serviceable-Limit Extensions
Ref. AMM 72-21-00-000-005-A	Removal of the Spinner Front Cone
Ref. AMM 72-21-00-000-006-A	Removal of the Spinner Rear Cone
Ref. AMM 72-21-00-210-009-A	Inspection of Fan Rotor Blades Installed on the Fan Disk
Ref. AMM 72-21-00-300-001-A	Replacement of the Crimped Self-Locking Nuts on the Spinner Rear Cone
Ref. AMM 72-21-00-400-005-A	Installation of the Spinner Front Cone
Ref. AMM 72-21-00-400-006-A	Installation of the Spinner Rear Cone
Ref. AMM 72-23-00-000-008-A	Removal of the Fan Outlet Guide Vane (OGV) Doublets
Ref. AMM 72-23-00-300-040-A	Local Reconditioning of the Abradable Material- Defects less than 0.12 in. (3 mm) deep and 0.50 sq in (3.2 sq cm)
Ref. AMM 72-23-00-300-043-A	Repair of the Abradable (material) Shroud with Circular Patches

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REFERENCE	DESIGNATION
Ref. AMM 72-23-00-300-805-A	Fan Frame Assembly - Fan Inlet Case Repair - Local Reconditioning of the Abradable Silicone Liner (Large Depth)
Ref. AMM 72-23-00-300-806-A	Partial Replacement of the Abradable Liner with Square/Rectangular Patches
Ref. AMM 72-23-00-300-807-A	Partial Replacement of the Abradable Liner with Large Circular Patches
Ref. AMM 72-23-00-300-812-A	Repair of Fan Composite Outlet Guide Vane (OGV)
Ref. AMM 72-23-00-400-008-A	Installation of the Fan Outlet Guide Vane (OGV) Doublets
Ref. AMM 72-55-00-960-002-A	Replacement of LPT Shaft Plug O-rings
Ref. AMM 78-30-00-081-041-B	Make the Thrust Reverser System Serviceable after Maintenance
Ref. AMM 78-30-00-481-041-B	Make the Thrust Reverser System Unserviceable for Maintenance
Ref. AMM 78-36-00-010-040-B	Opening of the Thrust Reverser Doors
Ref. AMM 78-36-00-410-040-B	Closing of the Thrust Reverser Doors
Ref. Fig. (1 / 72-23-00-991-30800)	
Ref. Fig. (1 / 72-23-00-991-31000)	
Ref. Fig. (1 / 72-23-00-991-33300)	
Ref. Fig. (1 / 72-23-00-991-33400)	

3. Job Set-up

Subtask 72-23-00-941-080-A

On A/C 003-250

A. Safety Precautions

- (1) On the center pedestal, on ENG panel 115VU:
 - (a) Put the WARNING NOTICE(S) in position to tell persons not to start the engine.
- (2) Make sure that the engine shutdown occurred five minutes or more before you do this procedure.
- (3) On overhead maintenance panel 50VU:
 - (a) Make sure that the ON legend of the ENG/FADEC GND PWR/1(2) pushbutton switch is off.
 - (b) Put the WARNING NOTICE(S) in position to tell persons not to energize the FADEC 1(2).
- (4) Put the ACCESS PLATFORM 1M (3 FT)-ADJUSTABLE in position.
- (5) Install the MAT - PROTECTIVE in the lower half of the inlet cowl.

Subtask 72-23-00-010-090-A

On A/C 003-250

B. Get Access

- (1) Open the fan cowl doors [Ref. AMM 71-13-00-010-040-B](#):
 - (a) FOR 1000EM1 (ENGINE-1)
437AL 438AR
 - (b) FOR 1000EM2 (ENGINE-2)
447AL 448AR

Subtask 72-23-00-040-067-A

On A/C 003-250

C. Make the thrust reverser unserviceable [Ref. AMM 78-30-00-481-041-B](#).

Subtask 72-23-00-010-102-A

On A/C 003-250

D. Get Access

- (1) Open the thrust-reverser cowl doors [Ref. AMM 78-36-00-010-040-B](#):
 - (a) FOR 1000EM1 (ENGINE-1)
451AL 452AR
 - (b) FOR 1000EM2 (ENGINE-2)
461AL 462AR

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Subtask 72-23-00-020-077-A

On A/C 003-250

E. Removal of the Spinner Cone

(1) Remove the spinner front cone [Ref. AMM 72-21-00-000-005-A](#).

(2) Remove the spinner rear cone [Ref. AMM 72-21-00-000-006-A](#).

4. Procedure

Ref. Fig. 1 / 72-23-00-991-30800

Ref. Fig. 1 / 72-23-00-991-31000

Ref. Fig. 1 / 72-23-00-991-33300

Ref. Fig. 1 / 72-23-00-991-33400

Subtask 72-23-00-210-058-A

On A/C 003-250

A. Detailed Visual Inspection of the Spinner, Engine Inlet, Fan Blades, Fan Outlet and Abradable Material

INSPECT/CHECK	MAXIMUM SERVICEABLE LIMITS	REMARKS
1.Spinner front cone for:		
A.Cracks	Not serviceable	Install a new or serviceable spinner front cone Ref. AMM 72-21-00-400-005-A .
B.Nicks, dents and scratches	No limited quantity	
C.Loose or missing bonded washer	Not serviceable	Install the washer again with Loctite 496 (Material No. CP2216) and hold it in pressure.
D.Tip erosion	No limit if radius is not greater than 0.28 in. (7.11 mm) after you blend	Do the blending procedure to the maximum serviceable limits or install a new serviceable spinner front cone Ref. AMM 72-21-00-400-005-A .
E.Corrosion marks (white mushroom pattern) before part removal	These corrosion marks are not serviceable	Refer to over-serviceable-limit extensions Ref. AMM 72-00-00-200-025-A Ref. AMM 72-00-00-200-025-A-01 .
F.Swirl pattern faded or missing paint	Serviceable. Any amount of faded or missing paint permitted	
2.Spinner rear cone for:		
A.Cracks	Not serviceable	Install a new or serviceable spinner rear cone Ref. AMM 72-21-00-400-006-A .
B.Dents	No limited quantity	
C.Damaged surface protection	No limit if not more than ten percent of the total surface	If more than ten percent of the total surface, install a new or serviceable spinner rear cone Ref. AMM 72-21-00-400-006-A .
D.Missing self-locking nuts	Not serviceable	Replace the nuts Ref. AMM 72-21-00-300-001-A or install a new or serviceable spinner rear cone Ref. AMM 72-21-00-400-006-A .
E.Self-locking nuts with loss of self-locking quality	Not serviceable if running torque is less than 4 lbf.in (0.05 m.daN)	Replace the nuts Ref. AMM 72-21-00-300-001-A or install a new or serviceable spinner rear cone Ref. AMM 72-21-00-400-006-A .
F.Self-locking nuts with axial looseness	Not more than 0.008 in. (0.20 mm)	Crimp again (no crack permitted) or replace the nuts Ref. AMM 72-21-00-300-001-A .
G.Self-locking nuts with lateral looseness	Not more than 0.008 in. (0.20 mm)	Replace the nuts Ref. AMM 72-21-00-300-001-A or install a new or serviceable spinner rear cone Ref. AMM 72-21-00-400-006-A .

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INSPECT/CHECK	MAXIMUM SERVICEABLE LIMITS	REMARKS
H.Oil wetting at inner of spinner cone or fan disk or forward fan shaft	None permitted	Refer to the over-serviceable-limit extensions Ref. AMM 72-00-00-200-025-A Ref. AMM 72-00-00-200-025-A-01 . Replace the O-rings on the Low Pressure Turbine (LPT) shaft plug Ref. AMM 72-55-00-960-002-A .
J.Seal ring tears	Maximum cumulated length 0.4 in. (10.16 mm)	Refer to the serviceable-limit extensions Ref. AMM 72-00-00-200-025-A Ref. AMM 72-00-00-200-025-A-01 . Install a new or serviceable spinner rear cone Ref. AMM 72-21-00-400-006-A . NOTE : When the serviceable limits for the steps that follow are more than specified, refer to the over-serviceable-limit extensions if necessary Ref. AMM 72-00-00-200-025-A Ref. AMM 72-00-00-200-025-A-01 .
3.Inlet case for: Ref. Fig. (1 / 72-23-00-991-30800)		
A.Cracks	Not permitted	
B.Cracks or crazing in the rubber of AGB mount	Any amount	
C.Nicks	On area A, if 0.012 in. (0.30 mm) max. depth and 0.046 in.2 (29.68 mm2) max. surface. On area B, if 0.008 in. (0.20 mm) max. depth and 0.046 in.2 (29.68 mm2) max. surface. On area C, if 0.012 in. (0.30 mm) max. depth and 0.077 in.2 (49.68 mm2) max. surface.	50-cycle serviceable extension is allowed.
D.Scratches	0.008 in. (0.20 mm) max. depth and 1.968 in. (49.99 mm) max. length	50-cycle serviceable extension is allowed.
E.Dents	On area A, if 0.002 in. (0.05 mm) max. depth and 0.124 in.2 (80.00 mm2) max. surface. On area B, if 0.008 in. (0.20 mm) max. depth and 0.124 in.2 (80.00 mm2) max. surface. On area C, if 0.02 in. (0.51 mm) max. depth and 0.155 in.2 (100.00 mm2) max. surface.	50-cycle serviceable extension is allowed.
F.Wear	On areas A and B, if 0.004 in. (0.10 mm) max. depth and 0.465 in.2 (300.00 mm2) max. surface. On area C, if 0.004 in. (0.10 mm) max. depth and 0.62 in.2 (400.00 mm2) max. surface.	50-cycle serviceable extension is allowed.
4.Fan rotor blades: Ref. AMM 72-21-00-210-009-A		
5.Abradable for: Ref. Fig. (1 / 72-23-00-991-30800)		
A.Cracks	Any amount	
B.Rubs	Any amount	

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INSPECT/CHECK	MAXIMUM SERVICEABLE LIMITS	REMARKS
C.Crater (length<5x width) and damage other than crack rub and circumferential defect	Cumulated damage is less than 160 in.2 (103225.6 mm2) with no area larger than 30 in.2 (19354.8 mm2) and: a) The maximum axial width (parallel to engine axis) is less than 2.5 in. (63.50 mm) b) The minimum distance between the areas is equal to the maximum dimension of the nearest damaged area c) The fan case parent metal does not show through	For defect above those limits: Local reconditioning: Ref. AMM 72-23-00-300-040-A Circular patch: Ref. AMM 72-23-00-300-043-A Large circular patch: Ref. AMM 72-23-00-300-807-A Square/rectangular patch: Ref. AMM 72-23-00-300-806-A Large depth: Ref. AMM 72-23-00-300-805-A Refer to the over-serviceable-limits extensions: Ref. AMM 72-00-00-200-025-A Ref. AMM 72-00-00-200-025-A-01 NOTE : It is permitted to do a circular patch and local repairs at the same time.
D.Grooves (orientated within 0 deg. and 80 deg. from the engine axis and length>5x width)	All damage is permitted if: a) The maximum width of one groove is not more than 0.32 in. (8.13 mm) b) A maximum of total width of 1.25 in. (31.75 mm) of groove is with a minimum separation of 0.1 in. (2.54 mm) c) The fan case parent metal does not show through	For defect above those limits: Local reconditioning: Ref. AMM 72-23-00-300-040-A Large circular patch: Ref. AMM 72-23-00-300-807-A Square/rectangular patch: Ref. AMM 72-23-00-300-806-A Large depth: Ref. AMM 72-23-00-300-805-A Refer to the over-serviceable-limits extensions: Ref. AMM 72-00-00-200-025-A Ref. AMM 72-00-00-200-025-A-01 .
E.Circumferential defect (360 deg. or orientated within 80 deg. to 90 deg. from the engine axis)	All damage is permitted if: a) The maximum axial width (parallel to the engine axis) of one or cumulated defect is less than 1.5 in. (38.10 mm) b) The fan case parent metal does not show through	For defect above those limits: Local reconditioning: Ref. AMM 72-23-00-300-040-A Large circular patch: Ref. AMM 72-23-00-300-807-A Square/rectangular patch: Ref. AMM 72-23-00-300-806-A Large depth: Ref. AMM 72-23-00-300-805-A Refer to the over-serviceable-limits extensions: Ref. AMM 72-00-00-200-025-A Ref. AMM 72-00-00-200-025-A-01 .
6.Outlet Guide Vanes (OGV) for: Ref. Fig. (1 / 72-23-00-991-31000)		
A.Broken vane	Not serviceable	Replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A or remove the vane doublet to be in 6. B. Missing Vane condition.
B.Missing vane	One doublet missing	10-cycle extension allowed with one doublet missing. Complete set inspection required when new doublet fitted. Replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .
C.Cracks		

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INSPECT/CHECK	MAXIMUM SERVICEABLE LIMITS	REMARKS
Airfoil	Not serviceable	Replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .
Outer platform	Cracks from or along the airfoil are defined on the figure. See sheet 3 of the figure Ref. Fig. (1 / 72-23-00-991-31000)	50-cycle extension allowed if cracks are found out of tolerance. Replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .
Inner platform Ref. Fig. (1 / 72-23-00-991-33300)	Any amount of longitudinal cracks in Area F	
	Transversal cracks are not serviceable	1500-cycle extension is allowed if the inner platform has not more than:
		- 30 transversal cracks
		- 15 transversal cracks included 5 opened transversal cracks maximum in area E
		- 15 transversal cracks included 5 opened transversal cracks maximum in area I
		- 15 transversal cracks included 5 opened transversal cracks maximum in area M.
		This over-serviceable extension can be renewed after inspection.
		Do not change the place of a cracked OGV on the engine.
		If the extension is not applicable, replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .
D.Missing material on the inner platform. See Sheet 1 of Ref. Fig. (1 / 72-23-00-991-33400)		
Extrados or front edge	Not Serviceable	10-cycle extension is allowed for missing material, only on one side and less than 0.16 in.2 (103.23 mm2) on two platforms maximum.
		No missing OGV is allowed on the set. Do not change the place of an OGV with missing material on the engine.
		If the extension is not applicable, replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .
Intrados or aft edge	Not serviceable	See Sheet 2 of Ref. Fig. (1 / 72-23-00-991-33400) for over-serviceable-limit extensions, if material is missing on only one side of the inner platform.
		No missing OGV is allowed on the set. Do not change the place of an OGV with missing material on the engine.
		If the extension is not applicable, replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .
E.Delamination	Not serviceable	Replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .

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INSPECT/CHECK	MAXIMUM SERVICEABLE LIMITS	REMARKS
F.Eroded surface (glass film)	50 percent of airfoil	Replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .
G.Unbonding of metal leading edge	Up to 0.31 in.2 (200.00 mm2) serviceable if no parent composite material.	Replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .
	Over 0.31 in.2 (200.00 mm2).	Remove the unbonding area. Refer to paragraph H. Metal leading edge missing.
H.Metal leading edge missing	Up to 0.78 in. (19.81 mm) height and 0.39 in. (9.91 mm) wide.	100-cycle extension allowed if missing material is over the maximum serviceable limits.(Speed tape) (Material No. CP2516) usage is recommended when missing metal on leading edge to preserve repair possibility, avoiding any damage on composite material Ref. AMM 72-23-00-300-812-A .
J.Outer and inner platform debonding	Not serviceable	Replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .
K.Nicks and/or dents, on the leading edge (metallic leading edge)	Any amount of damage less than 0.12 in. (3.05 mm) deep and metal not torn: serviceable. If damage less than 0.12 in. (3.05 mm) deep and metal torn, refer to paragraph G.	A maximum service extension of 100 cycles is allowed if damage is between 0.12 in. (3.05 mm) and 0.19 in. (4.83 mm) deep.
L.Nicks and/or dents, on the trailing edge	No limited number if less than 0.01 in. (0.25 mm) depth.	Replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .
M.Nicks and/or dents on the concave or convex side	No limited number if less than 0.02 in. (0.51 mm) depth.	Replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .
N.Nicks and/or dents on the inner or outer platform	No limited number if less than 0.02 in. (0.51 mm) depth.	Replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .

5. Close-up

Subtask 72-23-00-420-087-A

[On A/C 003-250](#)

A. Installation of the Spinner Cone:

- (1) Install the spinner rear cone [Ref. AMM 72-21-00-400-006-A](#).
- (2) Install the spinner front cone [Ref. AMM 72-21-00-400-005-A](#).

Subtask 72-23-00-410-084-A

[On A/C 003-250](#)

B. Close Access

- (1) Make sure that the work area is clean and clear of tools and other items.
- (2) Close the thrust reverser doors [Ref. AMM 78-36-00-410-040-B](#):
 - (a) FOR 1000EM1 (ENGINE-1)
451AL 452AR
 - (b) FOR 1000EM2 (ENGINE-2)
461AL 462AR

Subtask 72-23-00-440-063-A

[On A/C 003-250](#)

C. Make the thrust reverser serviceable [Ref. AMM 78-30-00-081-041-B](#).

Subtask 72-23-00-410-094-A

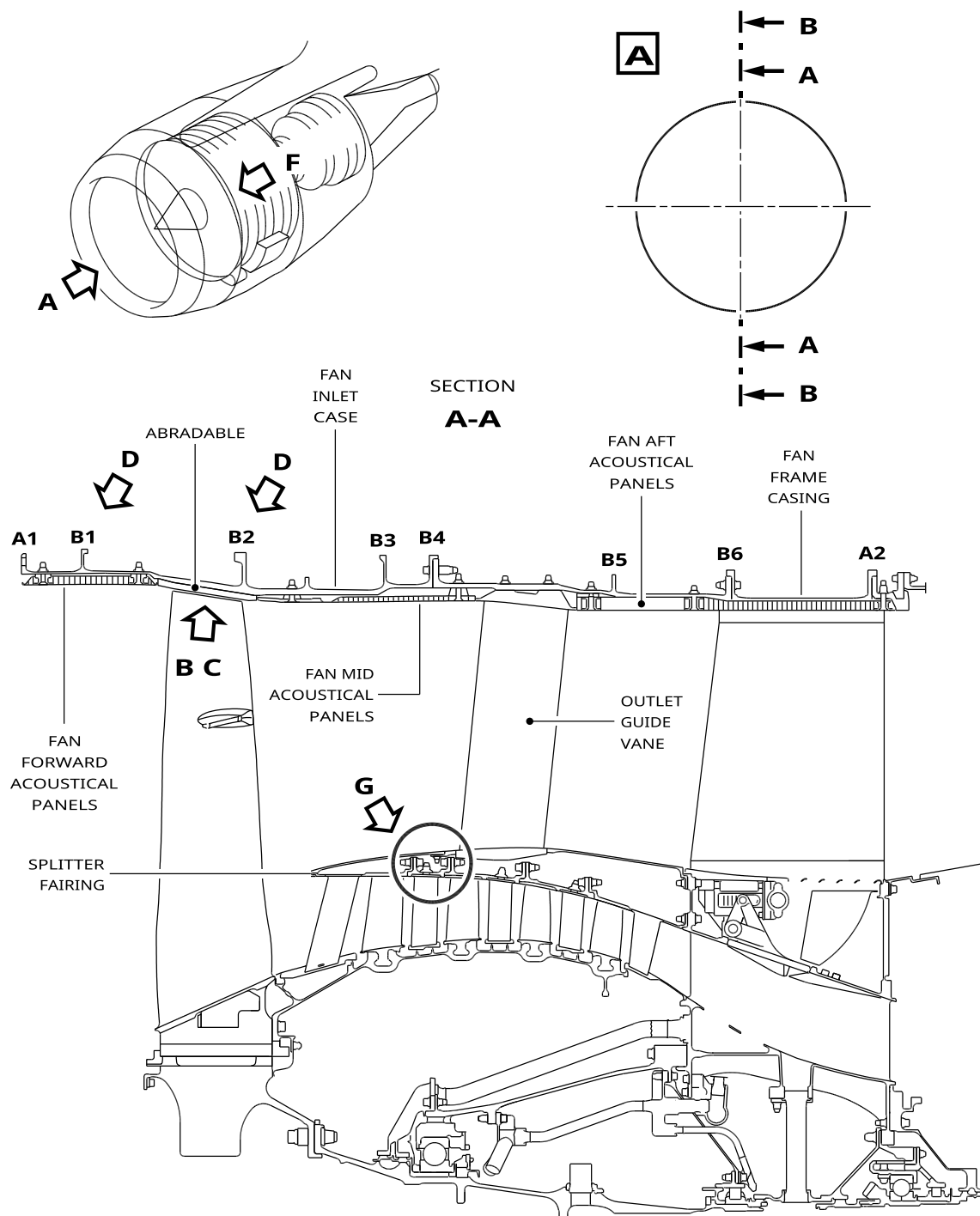
[On A/C 003-250](#)

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D. Close Access

- (1) Close the fan cowl doors [Ref. AMM 71-13-00-410-040-B](#):
 - (a) FOR 1000EM1 (ENGINE-1)
437AL 438AR
 - (b) FOR 1000EM2 (ENGINE-2)
447AL 448AR
- (2) Remove the MAT - PROTECTIVE from the inlet cowl.
- (3) Remove the access platform(s).
- (4) Remove the WARNING NOTICE(S).

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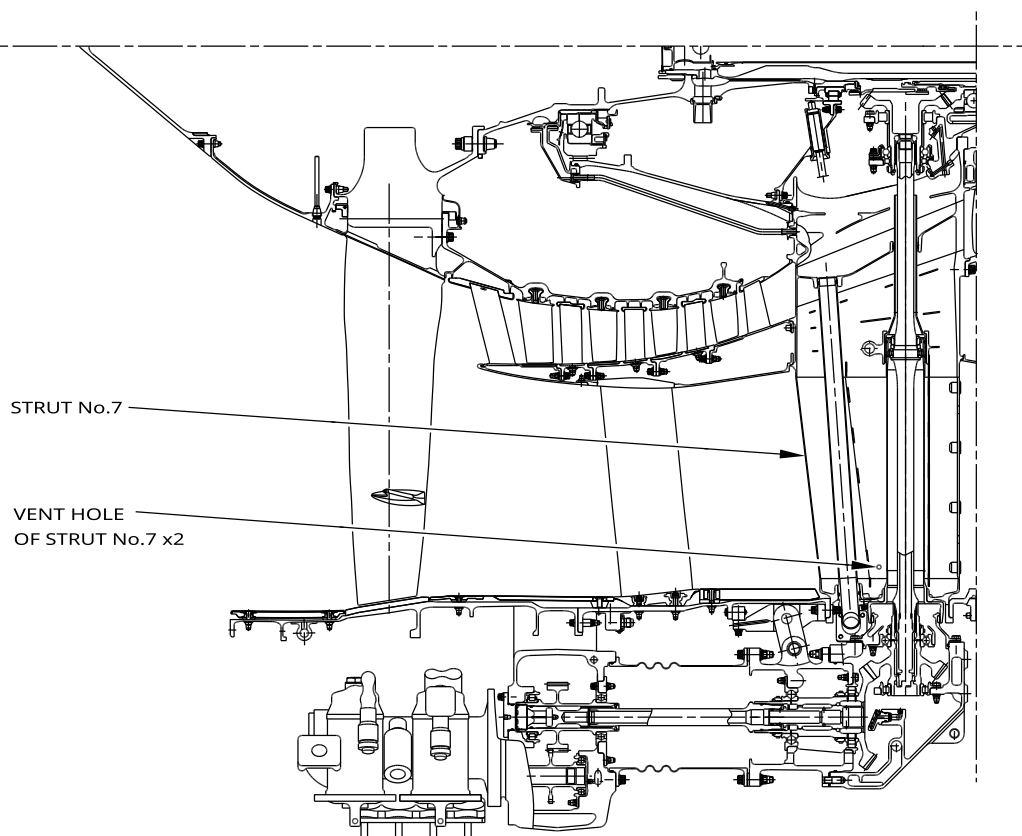
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Fan Frame Casing, Fan Inlet Case and Abradable Cartridge - Inspection/Check
Figure 1 / Graphic 72-23-00 -991-30800

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FAN FRAME ASSEMBLY - INSPECTION/CHECK

SECTION

B-B



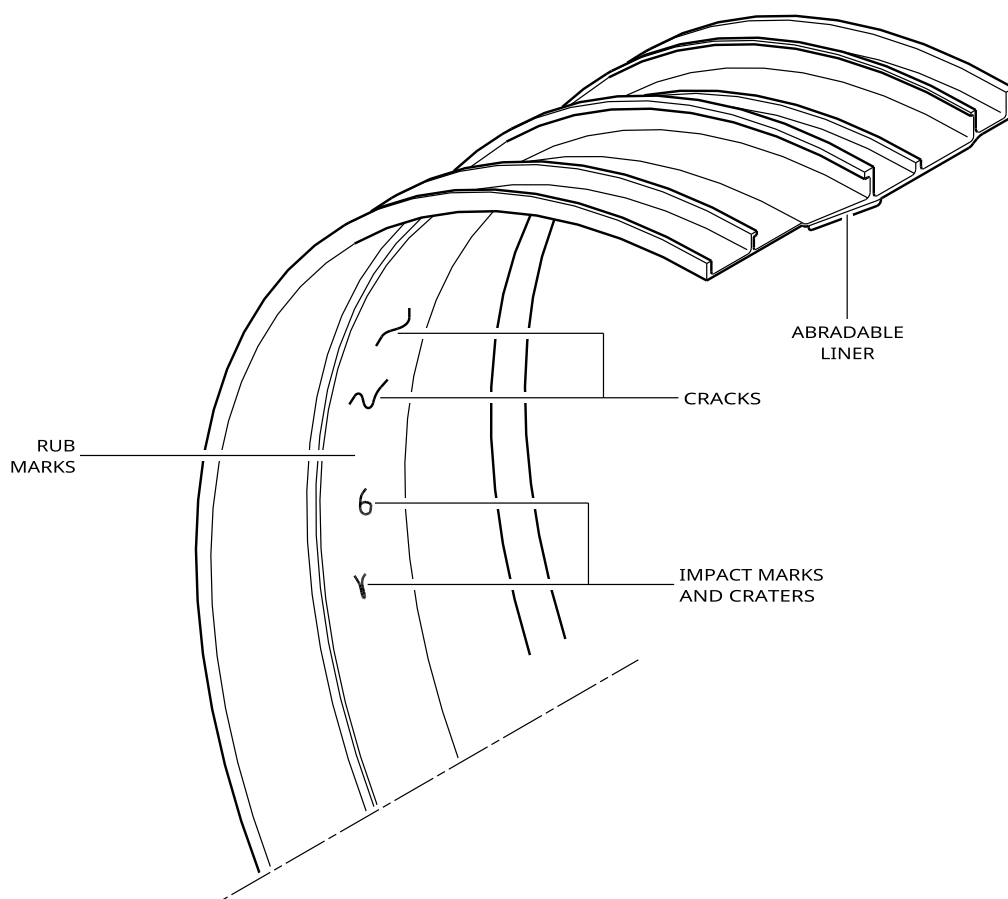
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Fan Frame Casing, Fan Inlet Case and Abradable Cartridge - Inspection/Check
Figure 2 / Graphic 72-23-00 -991-30800

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B

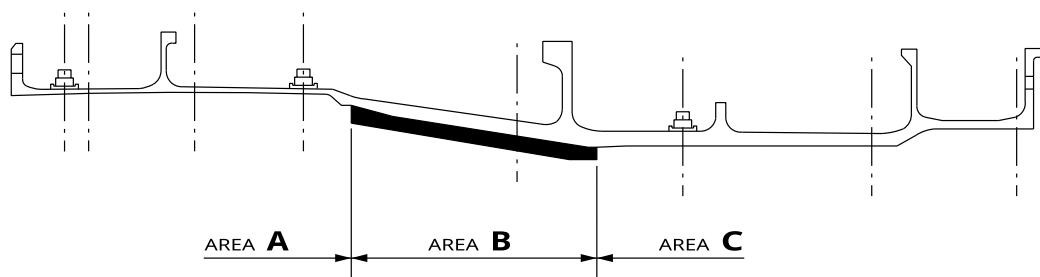
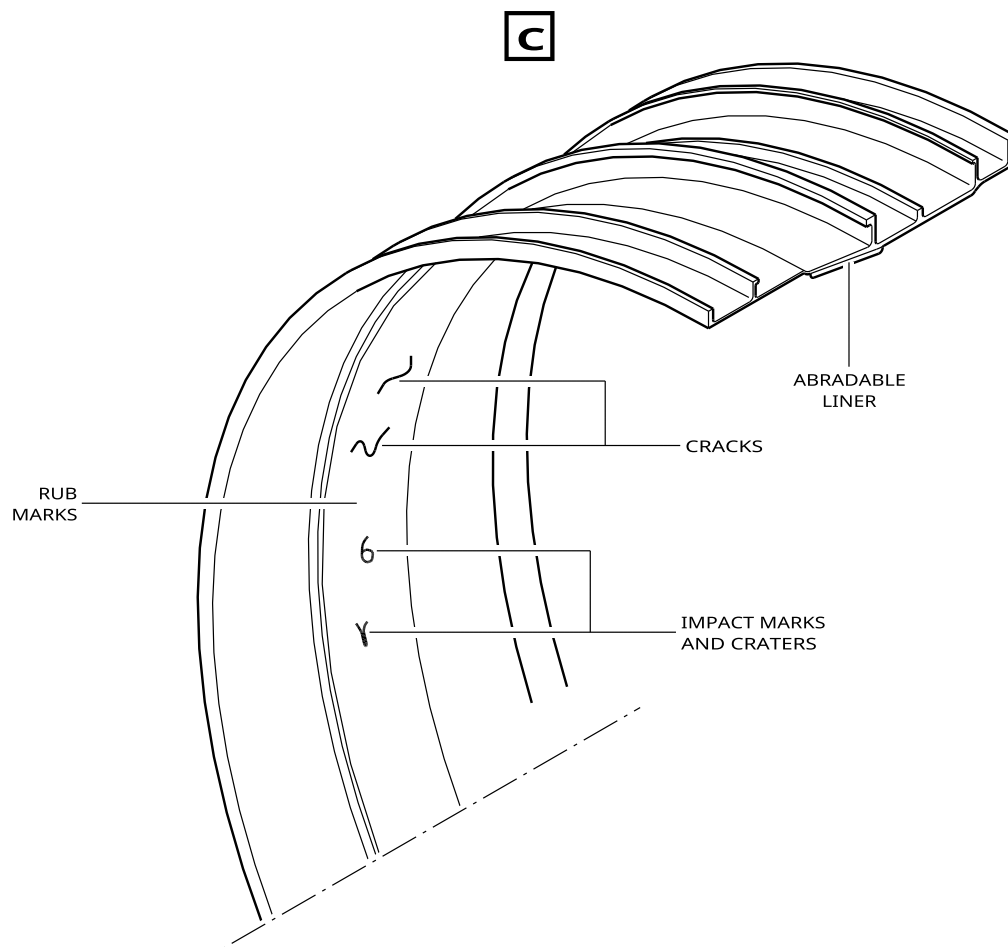


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Fan Frame Casing, Fan Inlet Case and Abradable Cartridge - Inspection/Check
Figure 3 / Graphic 72-23-00 -991-30800

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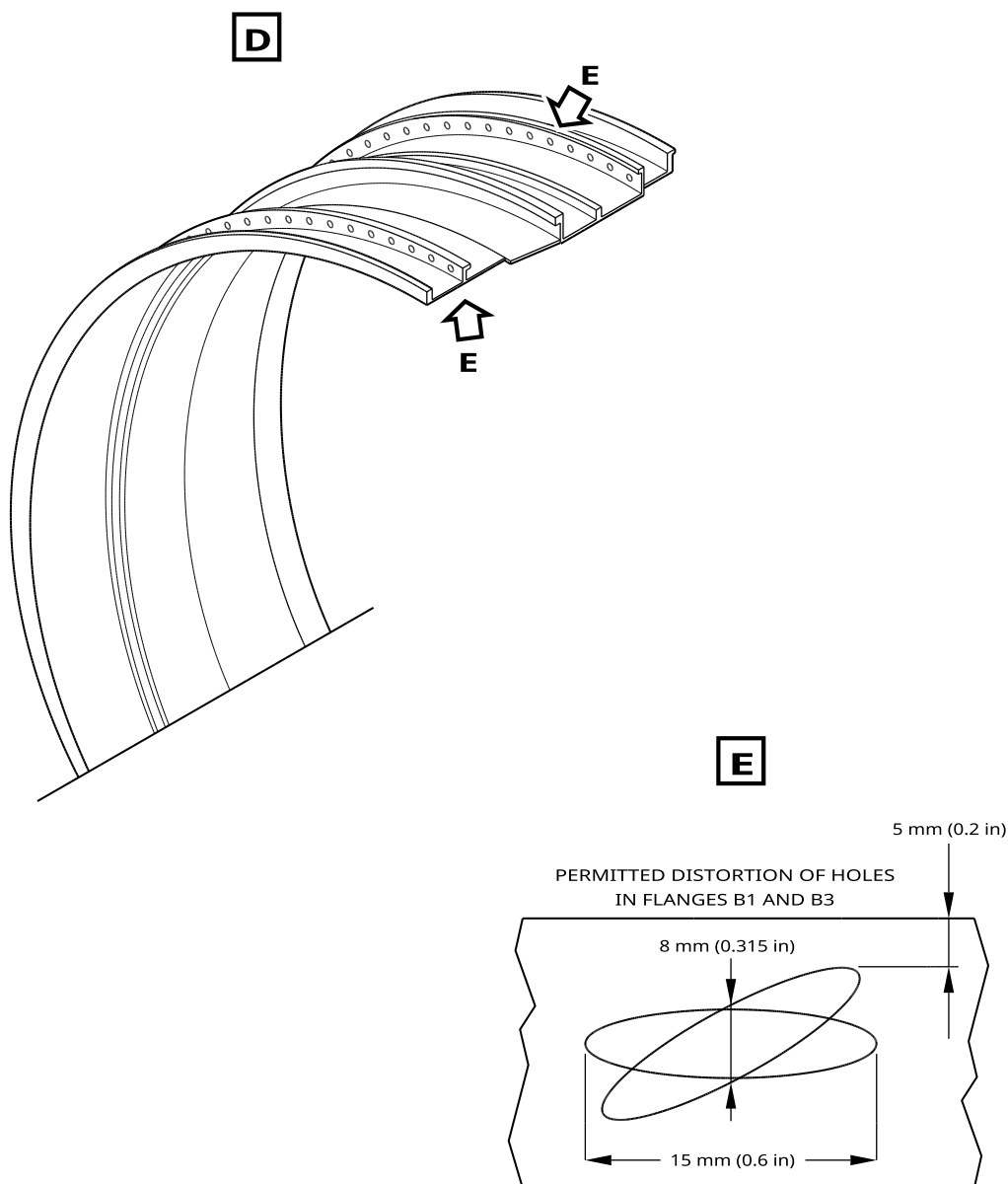


N_MM_722300_6_RAM0_04_02

Fan Frame Casing, Fan Inlet Case and Abradable Cartridge - Inspection/Check
Figure 4 / Graphic 72-23-00 -991-30800

S-M56-MM-04447-00-B

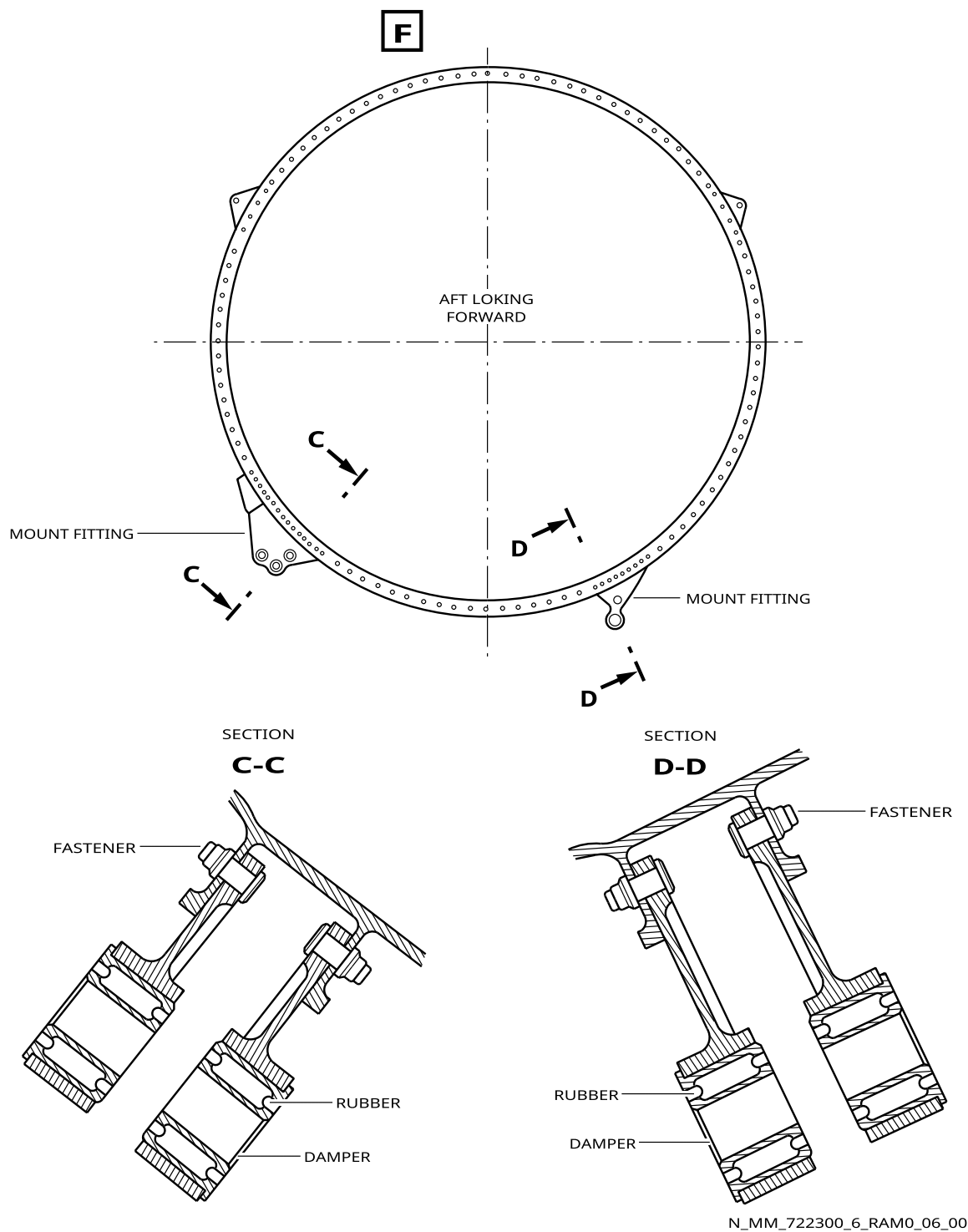
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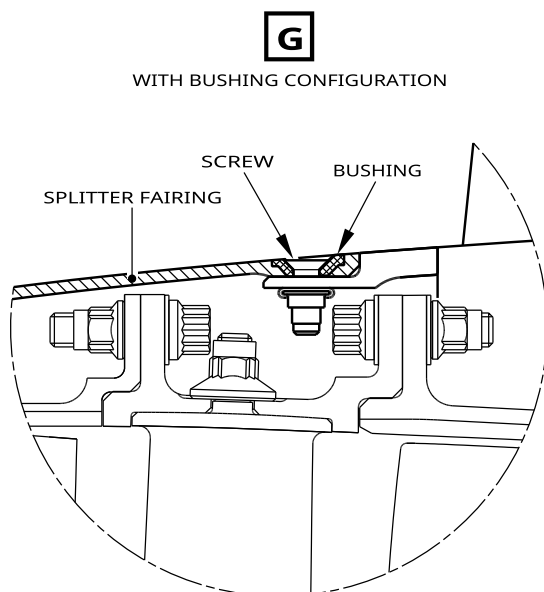
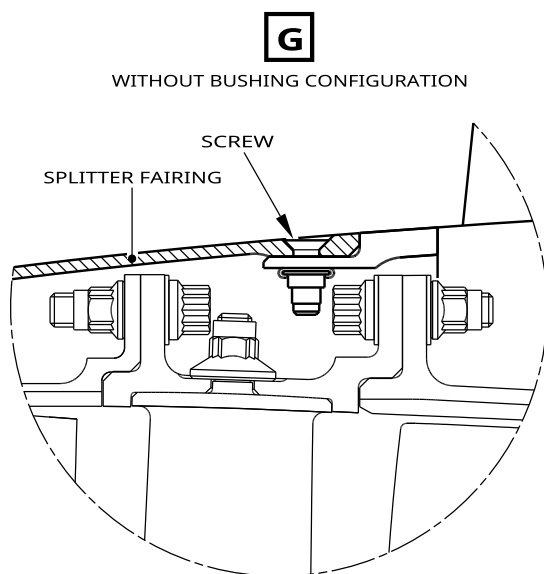
Fan Frame Casing, Fan Inlet Case and Abradable Cartridge - Inspection/Check
Figure 5 / Graphic 72-23-00 -991-30800

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FAN FRAME ASSEMBLY - INSPECTION/CHECK



Fan Frame Casing, Fan Inlet Case and Abradable Cartridge - Inspection/Check
Figure 6 / Graphic 72-23-00 -991-30800

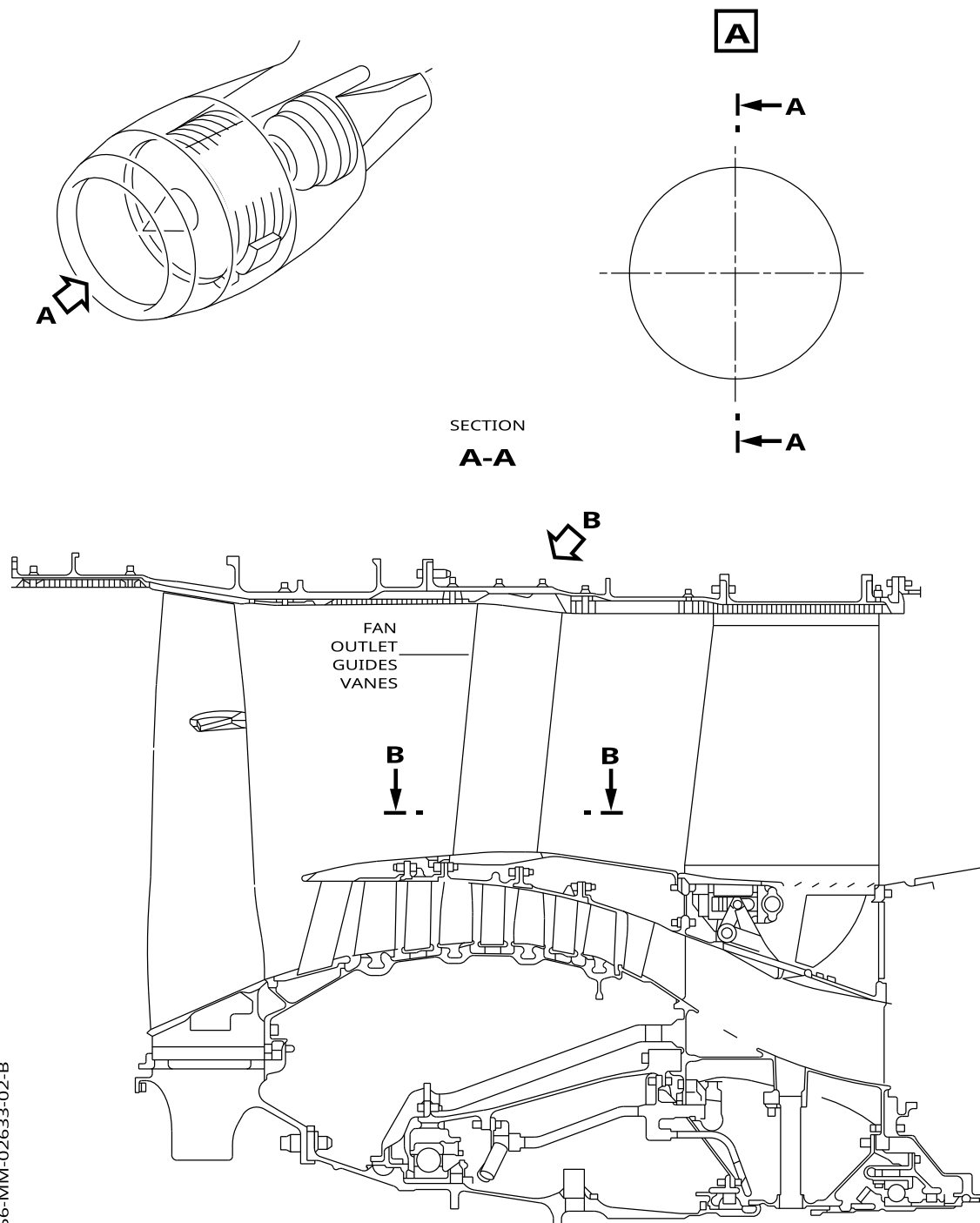
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S-M56-MM-05975-01
N_MM_722300_6_RAM0_07_00

Fan Frame Casing, Fan Inlet Case and Abradable Cartridge - Inspection/Check
Figure 7 / Graphic 72-23-00 -991-30800

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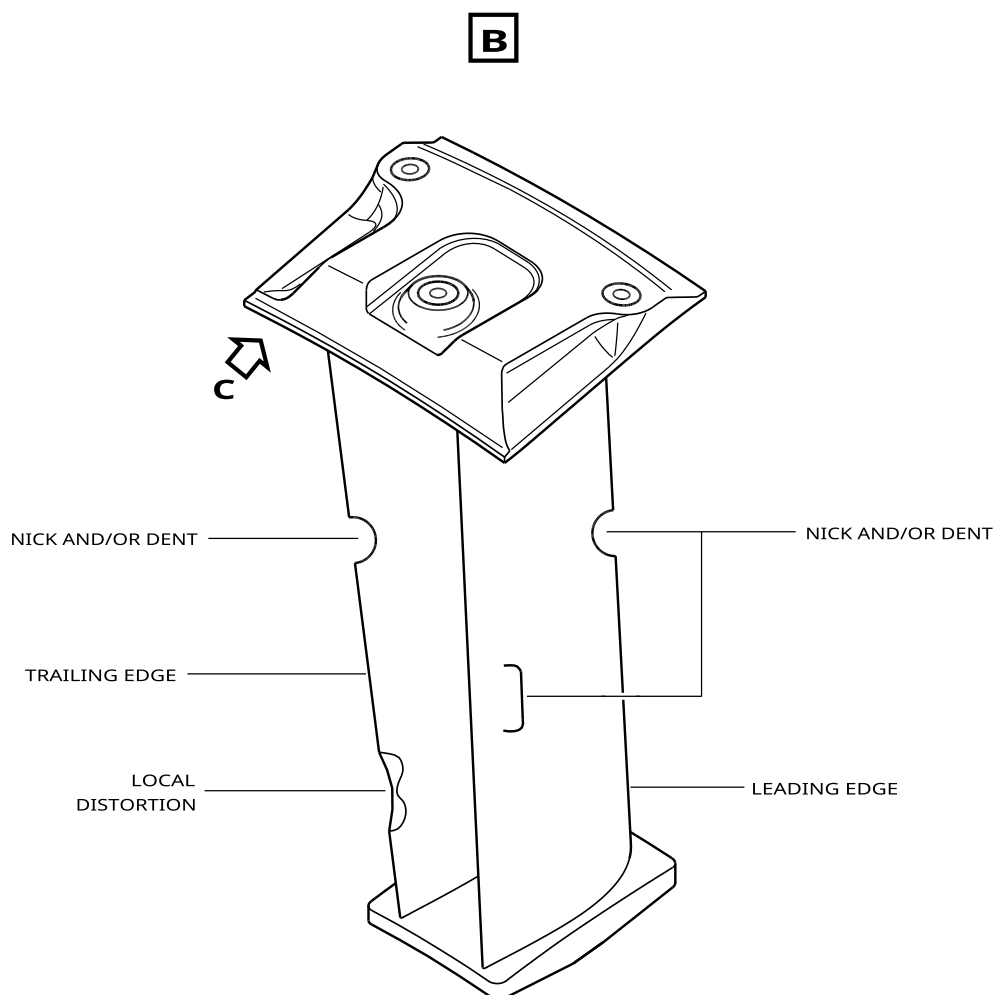


S-M56-MM-02633-02-B

N_MM_722300_6_RCM0_01_02

Inspection/Check of the Fan-Outlet Guide Vanes (OGV)
Figure 1 / Graphic 72-23-00 -991-31000

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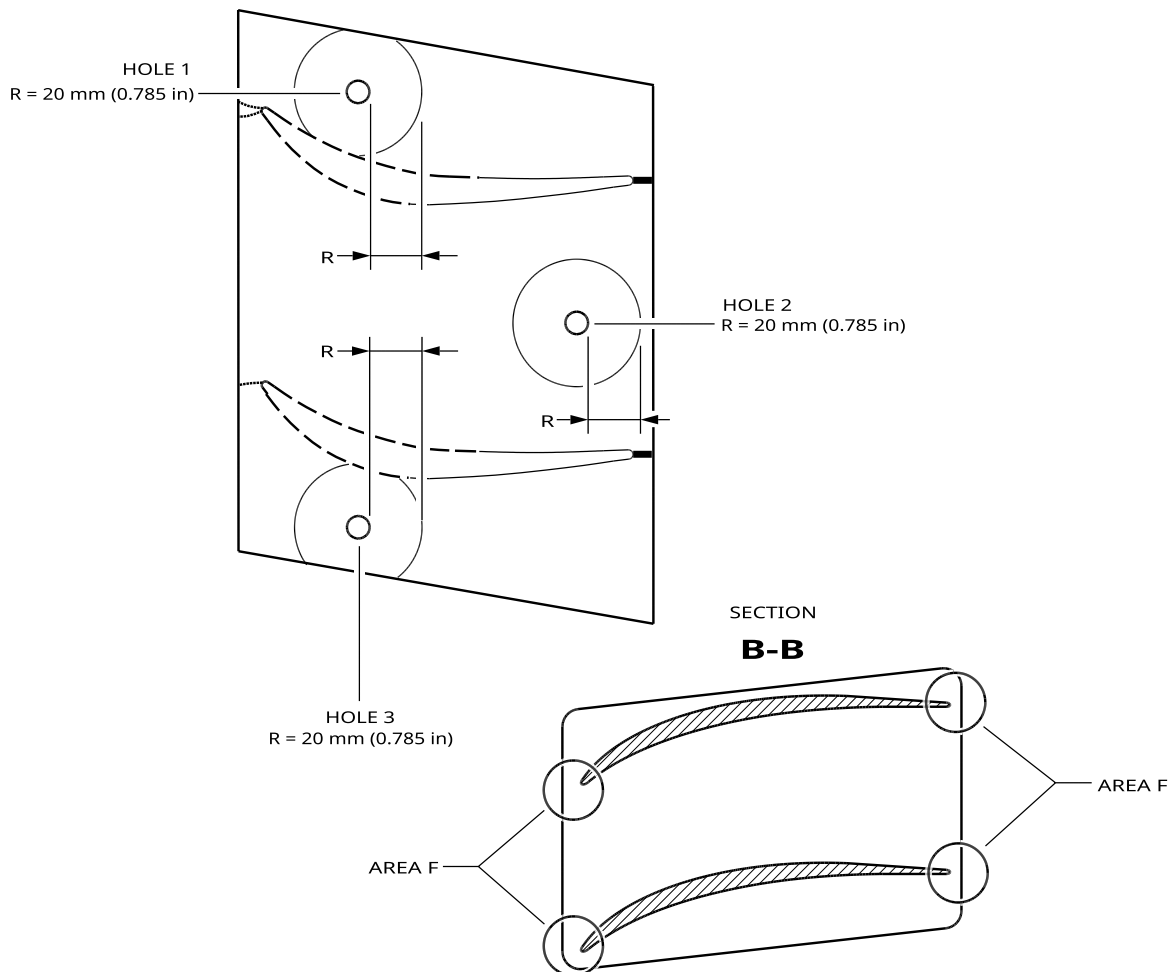


N_MM_722300_6_RCM0_02_02

Inspection/Check of the Fan-Outlet Guide Vanes (OGV)
Figure 2 / Graphic 72-23-00 -991-31000

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FAN FRAME ASSEMBLY - INSPECTION/CHECK

C



NOTE:



CRACKS IN TO RADIUS "R" ARE PERMITTED



CRACKS FROM THE AIRFOIL LEADING EDGE TO THE PLATFORM FRONT SIDE ARE PERMITTED. PIECE OF PU. PLATFORM MISSING IS NOT PERMITTED.



CRACKS FROM THE AIRFOIL TRAILING EDGE TO THE REAR SIDE PLATFORM ARE NOT PERMITTED.



CRACKS ALONG THE AIRFOIL ARE ALLOWED WITH A REQUIRED INSPECTION EVERY "A 600 F.H. CHECK" TO CONTROL THAT CRACKS ARE NOT GOING UP TO THE TRAILING EDGE. (IF CRACKS GO UP TO THE TRAILING EDGE THE PART IS NOT SERVICEABLE)

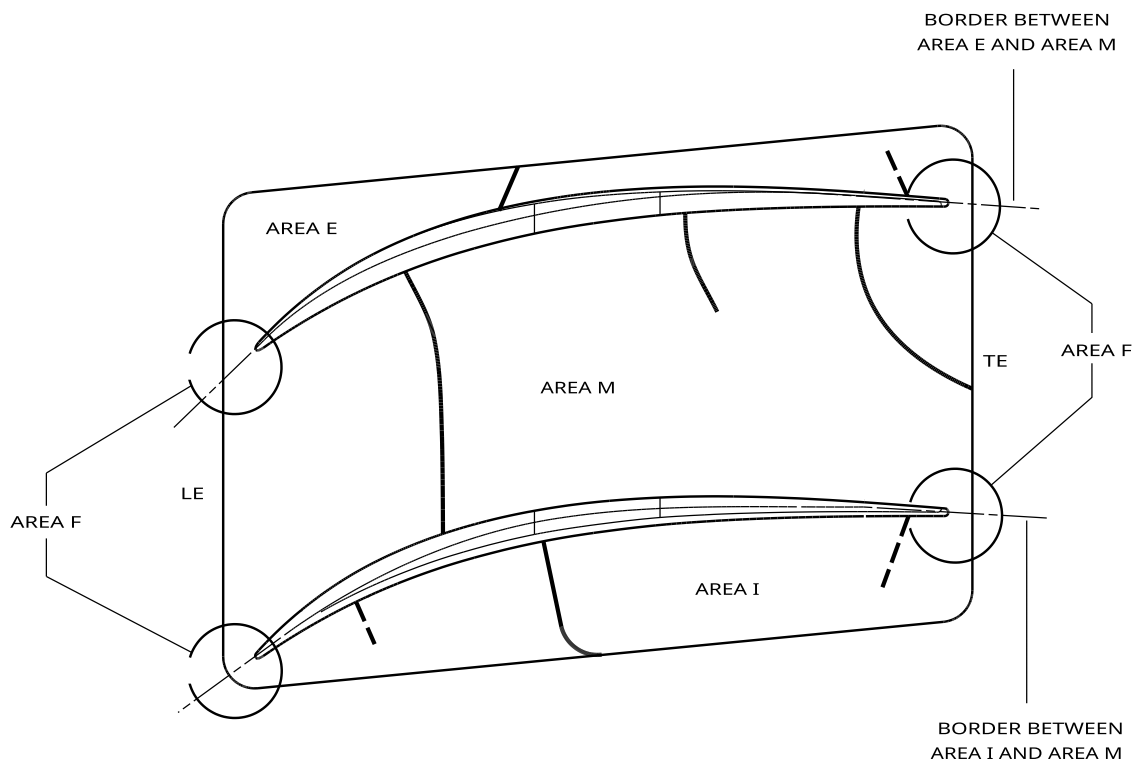
N_MM_722300_6_RCM0_03_02

Inspection/Check of the Fan-Outlet Guide Vanes (OGV)
Figure 3 / Graphic 72-23-00 -991-31000

S-M56-MM-04249-06-B

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OGV INNER PLATFORM FOR CRACKS



NOTE:

— OPENED TRANSVERSAL CRACK

- - - TRANSVERSAL CRACK

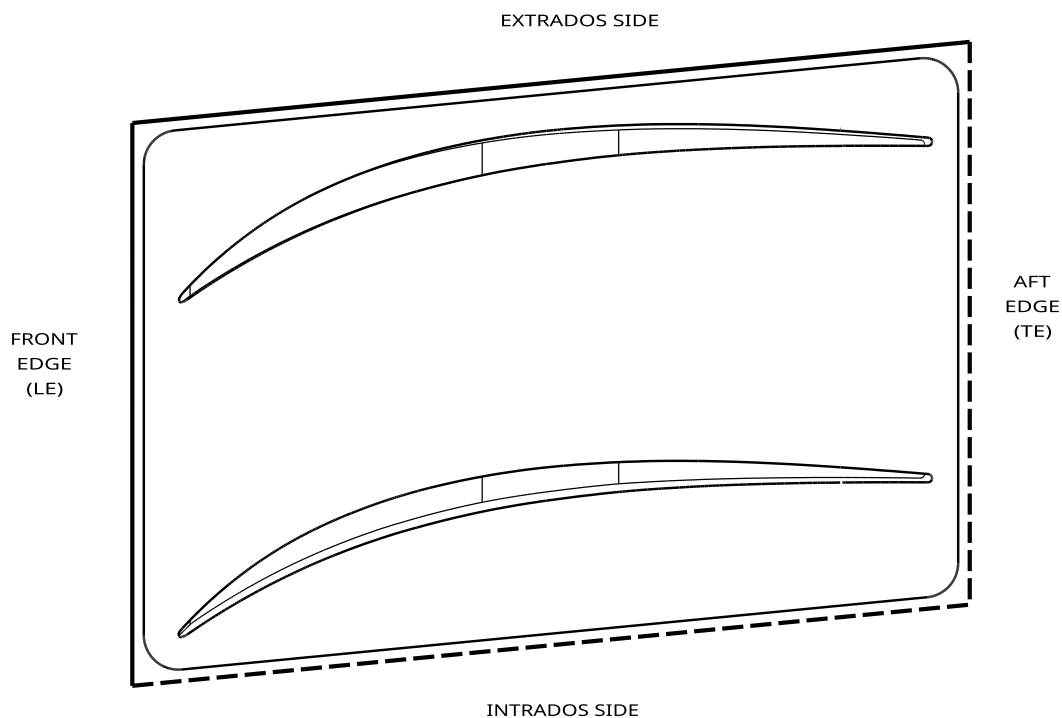
DO NOT CHANGE THE LOCATION OF THE OGV CRACKED ON THE ENGINE.

N_MM_722300_6_RCN0_01_00

Outlet Guide Vanes (OGV) Inner Platform for Cracks
Figure 1 / Graphic 72-23-00 -991-33300

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OGV INNER PLATFORM FOR MISSING MATERIAL



NOTE:

— REFER TO 6.D MISSING MATERIAL ON INNER PLATFORM FOR EXTRADOS OR FRONT EDGE.

- - - REFER TO 6.D MISSING MATERIAL ON INNER PLATFORM FOR INTRADOS OR AFT EDGE.

DO NOT CHANGE THE LOCATION OF THE OGV WITH MISSING MATERIAL ON THE ENGINE

S-M56-MM-04962-02

N_MM_722300_6_RCP0_01_01

Outlet Guide Vanes (OGV) Inner Platform for missing material

Figure 1 / Graphic 72-23-00 -991-33400

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MISSING MATERIAL ON INNER PLATFORM
FOR INTRADOS OR AFT EDGE

CONDITION	OVER SERVICEABLE LIMIT EXTENSION
IF MISSING MATERIAL IS LESS THAN 0.08 SQUARE in (50 SQUARE mm).	300 CYCLES EXTENSION ALLOWED
IF MISSING MATERIAL IS LESS THAN 0.16 SQUARE in (100 SQUARE mm).	100 CYCLES EXTENSION ALLOWED
IF MISSING MATERIAL IS LESS THAN 0.47 SQUARE in (300 SQUARE mm).	10 CYCLES EXTENSION ALLOWED

NOTE:

DO NOT CHANGE THE PLACE OF OGV WITH MISSING MATERIAL ON THE ENGINE

N_MM_722300_6_RCP0_02_00

Outlet Guide Vanes (OGV) Inner Platform for missing material
Figure 2 / Graphic 72-23-00 -991-33400

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MET.25133140

Task Card Title:

NO. 2 ENGINE INLET FAN SECTION INSP

Task Card NO:

722000-C1-1-002

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP**STEP COMPLETION SIGN OFF:**

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s				HRS			
Insp.	INSP/II			HRS			

CLOSE-UP**Job Description**

C. PERFORM JOB CLOSE-UP PER AMM TASK 72-23-00-210-003-C.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PP/ACFT_TECH			HRS			
Insp.				HRS			

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below

Work order No: _____

Note: Make sure to creat the finding Work Order on task card level.

Remark's: _____



Task Card Title:

NO. 2 ENGINE INLET FAN SECTION INSP

Task Card NO:

722000-C1-1-002

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

(1) If Task card interrupted or performed by different personals updates the system and update the following according (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 2 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 3 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 4 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 5 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____

**Task Completion Close Up And
System Update**

Station**Date****Signature**

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FAN FRAME ASSEMBLY - INSPECTION/CHECK

TASK 72-23-00-210-003 -C

On A/C 003-250

Detailed Inspection of Spinner, Engine Inlet, Fan Blades, Fan Outlet and Abradable Material

WARNING: BE CAREFUL WHEN YOU USE CONSUMABLE MATERIALS. OBEY THE MATERIAL MANUFACTURER'S INSTRUCTIONS AND YOUR LOCAL REGULATIONS.

WARNING: BE CAREFUL WHEN YOU DO WORK ON THE ENGINE PARTS AFTER ENGINE SHUTDOWN. USE APPLICABLE THERMAL GLOVES. THE ENGINE PARTS AND THE ENGINE OIL CAN STAY HOT FOR ONE HOUR AFTER SHUTDOWN AND CAN BURN YOU.

FIN: 1000EM1, 1000EM2

1. Reason for the Job

Refer to the MPD TASK: 722000-C1

DETAILED INSPECTION OF SPINNER, ENGINE INLET, FAN BLADES, FAN OUTLET AND ABRADABLE MATERIAL.

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	1	ACCESS PLATFORM 1M (3 FT)-ADJUSTABLE
No specific	1	MAT - PROTECTIVE
No specific	AR	WARNING NOTICE(S)

B. Consumable Materials

REFERENCE	DESIGNATION
CP2216	Loctite 496
CP2516	(Speed tape)

C. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
400	POWER PLANT, NACELLES AND PYLONS
FOR 1000EM1(ENGINE-1)	
437AL , 438AR , 451AL , 452AR	
FOR 1000EM2(ENGINE-2)	
447AL , 448AR , 461AL , 462AR	

D. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 71-13-00-010-040-B	Opening of the Fan Cowl Doors
Ref. AMM 71-13-00-410-040-B	Closing of the Fan Cowl Doors
Ref. AMM 72-00-00-200-025-A	Over-Serviceable-Limit Extensions
Ref. AMM 72-00-00-200-025-A-01	Over-Serviceable-Limit Extensions
Ref. AMM 72-21-00-000-005-A	Removal of the Spinner Front Cone
Ref. AMM 72-21-00-000-006-A	Removal of the Spinner Rear Cone
Ref. AMM 72-21-00-210-009-A	Inspection of Fan Rotor Blades Installed on the Fan Disk
Ref. AMM 72-21-00-300-001-A	Replacement of the Crimped Self-Locking Nuts on the Spinner Rear Cone
Ref. AMM 72-21-00-400-005-A	Installation of the Spinner Front Cone
Ref. AMM 72-21-00-400-006-A	Installation of the Spinner Rear Cone
Ref. AMM 72-23-00-000-008-A	Removal of the Fan Outlet Guide Vane (OGV) Doublets
Ref. AMM 72-23-00-300-040-A	Local Reconditioning of the Abradable Material- Defects less than 0.12 in. (3 mm) deep and 0.50 sq in (3.2 sq cm)
Ref. AMM 72-23-00-300-043-A	Repair of the Abradable (material) Shroud with Circular Patches

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REFERENCE	DESIGNATION
Ref. AMM 72-23-00-300-805-A	Fan Frame Assembly - Fan Inlet Case Repair - Local Reconditioning of the Abradable Silicone Liner (Large Depth)
Ref. AMM 72-23-00-300-806-A	Partial Replacement of the Abradable Liner with Square/Rectangular Patches
Ref. AMM 72-23-00-300-807-A	Partial Replacement of the Abradable Liner with Large Circular Patches
Ref. AMM 72-23-00-300-812-A	Repair of Fan Composite Outlet Guide Vane (OGV)
Ref. AMM 72-23-00-400-008-A	Installation of the Fan Outlet Guide Vane (OGV) Doublets
Ref. AMM 72-55-00-960-002-A	Replacement of LPT Shaft Plug O-rings
Ref. AMM 78-30-00-081-041-B	Make the Thrust Reverser System Serviceable after Maintenance
Ref. AMM 78-30-00-481-041-B	Make the Thrust Reverser System Unserviceable for Maintenance
Ref. AMM 78-36-00-010-040-B	Opening of the Thrust Reverser Doors
Ref. AMM 78-36-00-410-040-B	Closing of the Thrust Reverser Doors
Ref. Fig. (1 / 72-23-00-991-30800)	
Ref. Fig. (1 / 72-23-00-991-31000)	
Ref. Fig. (1 / 72-23-00-991-33300)	
Ref. Fig. (1 / 72-23-00-991-33400)	

3. Job Set-up

Subtask 72-23-00-941-080-A

On A/C 003-250

A. Safety Precautions

- (1) On the center pedestal, on ENG panel 115VU:
 - (a) Put the WARNING NOTICE(S) in position to tell persons not to start the engine.
- (2) Make sure that the engine shutdown occurred five minutes or more before you do this procedure.
- (3) On overhead maintenance panel 50VU:
 - (a) Make sure that the ON legend of the ENG/FADEC GND PWR/1(2) pushbutton switch is off.
 - (b) Put the WARNING NOTICE(S) in position to tell persons not to energize the FADEC 1(2).
- (4) Put the ACCESS PLATFORM 1M (3 FT)-ADJUSTABLE in position.
- (5) Install the MAT - PROTECTIVE in the lower half of the inlet cowl.

Subtask 72-23-00-010-090-A

On A/C 003-250

B. Get Access

- (1) Open the fan cowl doors [Ref. AMM 71-13-00-010-040-B](#):
 - (a) FOR 1000EM1 (ENGINE-1)
437AL 438AR
 - (b) FOR 1000EM2 (ENGINE-2)
447AL 448AR

Subtask 72-23-00-040-067-A

On A/C 003-250

C. Make the thrust reverser unserviceable [Ref. AMM 78-30-00-481-041-B](#).

Subtask 72-23-00-010-102-A

On A/C 003-250

D. Get Access

- (1) Open the thrust-reverser cowl doors [Ref. AMM 78-36-00-010-040-B](#):
 - (a) FOR 1000EM1 (ENGINE-1)
451AL 452AR
 - (b) FOR 1000EM2 (ENGINE-2)
461AL 462AR

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Subtask 72-23-00-020-077-A

On A/C 003-250

E. Removal of the Spinner Cone

(1) Remove the spinner front cone [Ref. AMM 72-21-00-000-005-A](#).

(2) Remove the spinner rear cone [Ref. AMM 72-21-00-000-006-A](#).

4. Procedure

Ref. Fig. 1 / 72-23-00-991-30800

Ref. Fig. 1 / 72-23-00-991-31000

Ref. Fig. 1 / 72-23-00-991-33300

Ref. Fig. 1 / 72-23-00-991-33400

Subtask 72-23-00-210-058-A

On A/C 003-250

A. Detailed Visual Inspection of the Spinner, Engine Inlet, Fan Blades, Fan Outlet and Abradable Material

INSPECT/CHECK	MAXIMUM SERVICEABLE LIMITS	REMARKS
1.Spinner front cone for:		
A.Cracks	Not serviceable	Install a new or serviceable spinner front cone Ref. AMM 72-21-00-400-005-A .
B.Nicks, dents and scratches	No limited quantity	
C.Loose or missing bonded washer	Not serviceable	Install the washer again with Loctite 496 (Material No. CP2216) and hold it in pressure.
D.Tip erosion	No limit if radius is not greater than 0.28 in. (7.11 mm) after you blend	Do the blending procedure to the maximum serviceable limits or install a new serviceable spinner front cone Ref. AMM 72-21-00-400-005-A .
E.Corrosion marks (white mushroom pattern) before part removal	These corrosion marks are not serviceable	Refer to over-serviceable-limit extensions Ref. AMM 72-00-00-200-025-A Ref. AMM 72-00-00-200-025-A-01 .
F.Swirl pattern faded or missing paint	Serviceable. Any amount of faded or missing paint permitted	
2.Spinner rear cone for:		
A.Cracks	Not serviceable	Install a new or serviceable spinner rear cone Ref. AMM 72-21-00-400-006-A .
B.Dents	No limited quantity	
C.Damaged surface protection	No limit if not more than ten percent of the total surface	If more than ten percent of the total surface, install a new or serviceable spinner rear cone Ref. AMM 72-21-00-400-006-A .
D.Missing self-locking nuts	Not serviceable	Replace the nuts Ref. AMM 72-21-00-300-001-A or install a new or serviceable spinner rear cone Ref. AMM 72-21-00-400-006-A .
E.Self-locking nuts with loss of self-locking quality	Not serviceable if running torque is less than 4 lbf.in (0.05 m.daN)	Replace the nuts Ref. AMM 72-21-00-300-001-A or install a new or serviceable spinner rear cone Ref. AMM 72-21-00-400-006-A .
F.Self-locking nuts with axial looseness	Not more than 0.008 in. (0.20 mm)	Crimp again (no crack permitted) or replace the nuts Ref. AMM 72-21-00-300-001-A .
G.Self-locking nuts with lateral looseness	Not more than 0.008 in. (0.20 mm)	Replace the nuts Ref. AMM 72-21-00-300-001-A or install a new or serviceable spinner rear cone Ref. AMM 72-21-00-400-006-A .

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INSPECT/CHECK	MAXIMUM SERVICEABLE LIMITS	REMARKS
H.Oil wetting at inner of spinner cone or fan disk or forward fan shaft	None permitted	Refer to the over-serviceable-limit extensions Ref. AMM 72-00-00-200-025-A Ref. AMM 72-00-00-200-025-A-01 . Replace the O-rings on the Low Pressure Turbine (LPT) shaft plug Ref. AMM 72-55-00-960-002-A .
J.Seal ring tears	Maximum cumulated length 0.4 in. (10.16 mm)	Refer to the serviceable-limit extensions Ref. AMM 72-00-00-200-025-A Ref. AMM 72-00-00-200-025-A-01 . Install a new or serviceable spinner rear cone Ref. AMM 72-21-00-400-006-A . NOTE : When the serviceable limits for the steps that follow are more than specified, refer to the over-serviceable-limit extensions if necessary Ref. AMM 72-00-00-200-025-A Ref. AMM 72-00-00-200-025-A-01 .
3.Inlet case for: Ref. Fig. (1 / 72-23-00-991-30800)		
A.Cracks	Not permitted	
B.Cracks or crazing in the rubber of AGB mount	Any amount	
C.Nicks	On area A, if 0.012 in. (0.30 mm) max. depth and 0.046 in.2 (29.68 mm2) max. surface. On area B, if 0.008 in. (0.20 mm) max. depth and 0.046 in.2 (29.68 mm2) max. surface. On area C, if 0.012 in. (0.30 mm) max. depth and 0.077 in.2 (49.68 mm2) max. surface.	50-cycle serviceable extension is allowed.
D.Scratches	0.008 in. (0.20 mm) max. depth and 1.968 in. (49.99 mm) max. length	50-cycle serviceable extension is allowed.
E.Dents	On area A, if 0.002 in. (0.05 mm) max. depth and 0.124 in.2 (80.00 mm2) max. surface. On area B, if 0.008 in. (0.20 mm) max. depth and 0.124 in.2 (80.00 mm2) max. surface. On area C, if 0.02 in. (0.51 mm) max. depth and 0.155 in.2 (100.00 mm2) max. surface.	50-cycle serviceable extension is allowed.
F.Wear	On areas A and B, if 0.004 in. (0.10 mm) max. depth and 0.465 in.2 (300.00 mm2) max. surface. On area C, if 0.004 in. (0.10 mm) max. depth and 0.62 in.2 (400.00 mm2) max. surface.	50-cycle serviceable extension is allowed.
4.Fan rotor blades: Ref. AMM 72-21-00-210-009-A		
5.Abradable for: Ref. Fig. (1 / 72-23-00-991-30800)		
A.Cracks	Any amount	
B.Rubs	Any amount	

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INSPECT/CHECK	MAXIMUM SERVICEABLE LIMITS	REMARKS
C.Crater (length<5x width) and damage other than crack rub and circumferential defect	Cumulated damage is less than 160 in.2 (103225.6 mm2) with no area larger than 30 in.2 (19354.8 mm2) and: a) The maximum axial width (parallel to engine axis) is less than 2.5 in. (63.50 mm) b) The minimum distance between the areas is equal to the maximum dimension of the nearest damaged area c) The fan case parent metal does not show through	For defect above those limits: Local reconditioning: Ref. AMM 72-23-00-300-040-A Circular patch: Ref. AMM 72-23-00-300-043-A Large circular patch: Ref. AMM 72-23-00-300-807-A Square/rectangular patch: Ref. AMM 72-23-00-300-806-A Large depth: Ref. AMM 72-23-00-300-805-A Refer to the over-serviceable-limits extensions: Ref. AMM 72-00-00-200-025-A Ref. AMM 72-00-00-200-025-A-01 NOTE : It is permitted to do a circular patch and local repairs at the same time.
D.Grooves (orientated within 0 deg. and 80 deg. from the engine axis and length>5x width)	All damage is permitted if: a) The maximum width of one groove is not more than 0.32 in. (8.13 mm) b) A maximum of total width of 1.25 in. (31.75 mm) of groove is with a minimum separation of 0.1 in. (2.54 mm) c) The fan case parent metal does not show through	For defect above those limits: Local reconditioning: Ref. AMM 72-23-00-300-040-A Large circular patch: Ref. AMM 72-23-00-300-807-A Square/rectangular patch: Ref. AMM 72-23-00-300-806-A Large depth: Ref. AMM 72-23-00-300-805-A Refer to the over-serviceable-limits extensions: Ref. AMM 72-00-00-200-025-A Ref. AMM 72-00-00-200-025-A-01 .
E.Circumferential defect (360 deg. or orientated within 80 deg. to 90 deg. from the engine axis)	All damage is permitted if: a) The maximum axial width (parallel to the engine axis) of one or cumulated defect is less than 1.5 in. (38.10 mm) b) The fan case parent metal does not show through	For defect above those limits: Local reconditioning: Ref. AMM 72-23-00-300-040-A Large circular patch: Ref. AMM 72-23-00-300-807-A Square/rectangular patch: Ref. AMM 72-23-00-300-806-A Large depth: Ref. AMM 72-23-00-300-805-A Refer to the over-serviceable-limits extensions: Ref. AMM 72-00-00-200-025-A Ref. AMM 72-00-00-200-025-A-01 .
6.Outlet Guide Vanes (OGV) for: Ref. Fig. (1 / 72-23-00-991-31000)		
A.Broken vane	Not serviceable	Replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A or remove the vane doublet to be in 6. B. Missing Vane condition.
B.Missing vane	One doublet missing	10-cycle extension allowed with one doublet missing. Complete set inspection required when new doublet fitted. Replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .
C.Cracks		

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INSPECT/CHECK	MAXIMUM SERVICEABLE LIMITS	REMARKS
Airfoil	Not serviceable	Replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .
Outer platform	Cracks from or along the airfoil are defined on the figure. See sheet 3 of the figure Ref. Fig. (1 / 72-23-00-991-31000)	50-cycle extension allowed if cracks are found out of tolerance. Replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .
Inner platform Ref. Fig. (1 / 72-23-00-991-33300)	Any amount of longitudinal cracks in Area F	
	Transversal cracks are not serviceable	1500-cycle extension is allowed if the inner platform has not more than:
		- 30 transversal cracks
		- 15 transversal cracks included 5 opened transversal cracks maximum in area E
		- 15 transversal cracks included 5 opened transversal cracks maximum in area I
		- 15 transversal cracks included 5 opened transversal cracks maximum in area M.
		This over-serviceable extension can be renewed after inspection.
		Do not change the place of a cracked OGV on the engine.
		If the extension is not applicable, replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .
D.Missing material on the inner platform. See Sheet 1 of Ref. Fig. (1 / 72-23-00-991-33400)		
Extrados or front edge	Not Serviceable	10-cycle extension is allowed for missing material, only on one side and less than 0.16 in.2 (103.23 mm2) on two platforms maximum.
		No missing OGV is allowed on the set. Do not change the place of an OGV with missing material on the engine.
		If the extension is not applicable, replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .
Intrados or aft edge	Not serviceable	See Sheet 2 of Ref. Fig. (1 / 72-23-00-991-33400) for over-serviceable-limit extensions, if material is missing on only one side of the inner platform.
		No missing OGV is allowed on the set. Do not change the place of an OGV with missing material on the engine.
		If the extension is not applicable, replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .
E.Delamination	Not serviceable	Replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .

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INSPECT/CHECK	MAXIMUM SERVICEABLE LIMITS	REMARKS
F.Eroded surface (glass film)	50 percent of airfoil	Replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .
G.Unbonding of metal leading edge	Up to 0.31 in.2 (200.00 mm2) serviceable if no parent composite material.	Replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .
	Over 0.31 in.2 (200.00 mm2).	Remove the unbonding area. Refer to paragraph H. Metal leading edge missing.
H.Metal leading edge missing	Up to 0.78 in. (19.81 mm) height and 0.39 in. (9.91 mm) wide.	100-cycle extension allowed if missing material is over the maximum serviceable limits.(Speed tape) (Material No. CP2516) usage is recommended when missing metal on leading edge to preserve repair possibility, avoiding any damage on composite material Ref. AMM 72-23-00-300-812-A .
J.Outer and inner platform debonding	Not serviceable	Replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .
K.Nicks and/or dents, on the leading edge (metallic leading edge)	Any amount of damage less than 0.12 in. (3.05 mm) deep and metal not torn: serviceable. If damage less than 0.12 in. (3.05 mm) deep and metal torn, refer to paragraph G.	A maximum service extension of 100 cycles is allowed if damage is between 0.12 in. (3.05 mm) and 0.19 in. (4.83 mm) deep.
L.Nicks and/or dents, on the trailing edge	No limited number if less than 0.01 in. (0.25 mm) depth.	Replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .
M.Nicks and/or dents on the concave or convex side	No limited number if less than 0.02 in. (0.51 mm) depth.	Replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .
N.Nicks and/or dents on the inner or outer platform	No limited number if less than 0.02 in. (0.51 mm) depth.	Replace the vane doublet Ref. AMM 72-23-00-000-008-A and Ref. AMM 72-23-00-400-008-A .

5. Close-up

Subtask 72-23-00-420-087-A

[On A/C 003-250](#)

A. Installation of the Spinner Cone:

- (1) Install the spinner rear cone [Ref. AMM 72-21-00-400-006-A](#).
- (2) Install the spinner front cone [Ref. AMM 72-21-00-400-005-A](#).

Subtask 72-23-00-410-084-A

[On A/C 003-250](#)

B. Close Access

- (1) Make sure that the work area is clean and clear of tools and other items.
- (2) Close the thrust reverser doors [Ref. AMM 78-36-00-410-040-B](#):
 - (a) FOR 1000EM1 (ENGINE-1)
451AL 452AR
 - (b) FOR 1000EM2 (ENGINE-2)
461AL 462AR

Subtask 72-23-00-440-063-A

[On A/C 003-250](#)

C. Make the thrust reverser serviceable [Ref. AMM 78-30-00-081-041-B](#).

Subtask 72-23-00-410-094-A

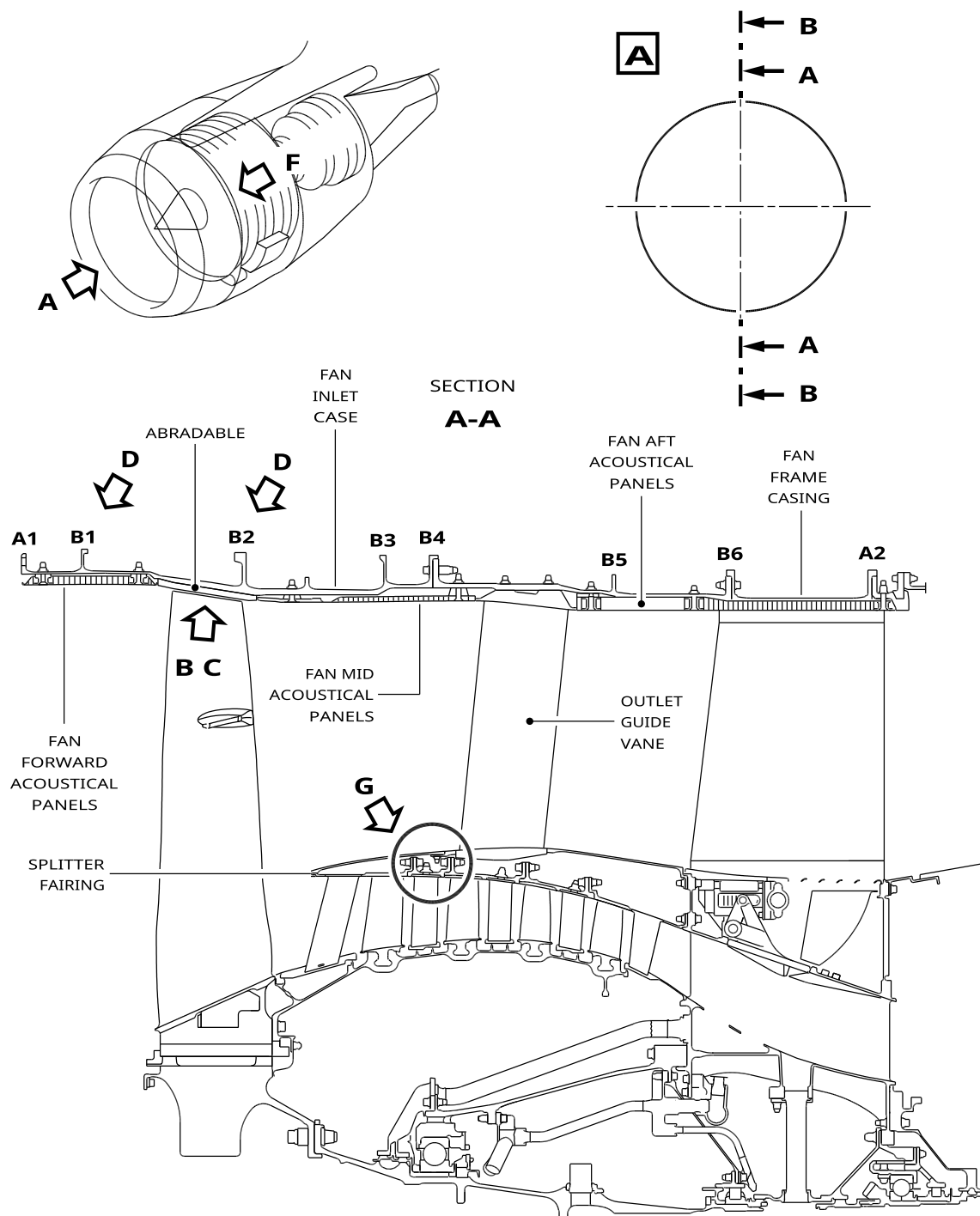
[On A/C 003-250](#)

TF-N
AIRCRAFT MAINTENANCE MANUAL
FAN FRAME ASSEMBLY - INSPECTION/CHECK

D. Close Access

- (1) Close the fan cowl doors [Ref. AMM 71-13-00-410-040-B](#):
 - (a) FOR 1000EM1 (ENGINE-1)
437AL 438AR
 - (b) FOR 1000EM2 (ENGINE-2)
447AL 448AR
- (2) Remove the MAT - PROTECTIVE from the inlet cowl.
- (3) Remove the access platform(s).
- (4) Remove the WARNING NOTICE(S).

TF-N
AIRCRAFT MAINTENANCE MANUAL
FAN FRAME ASSEMBLY - INSPECTION/CHECK



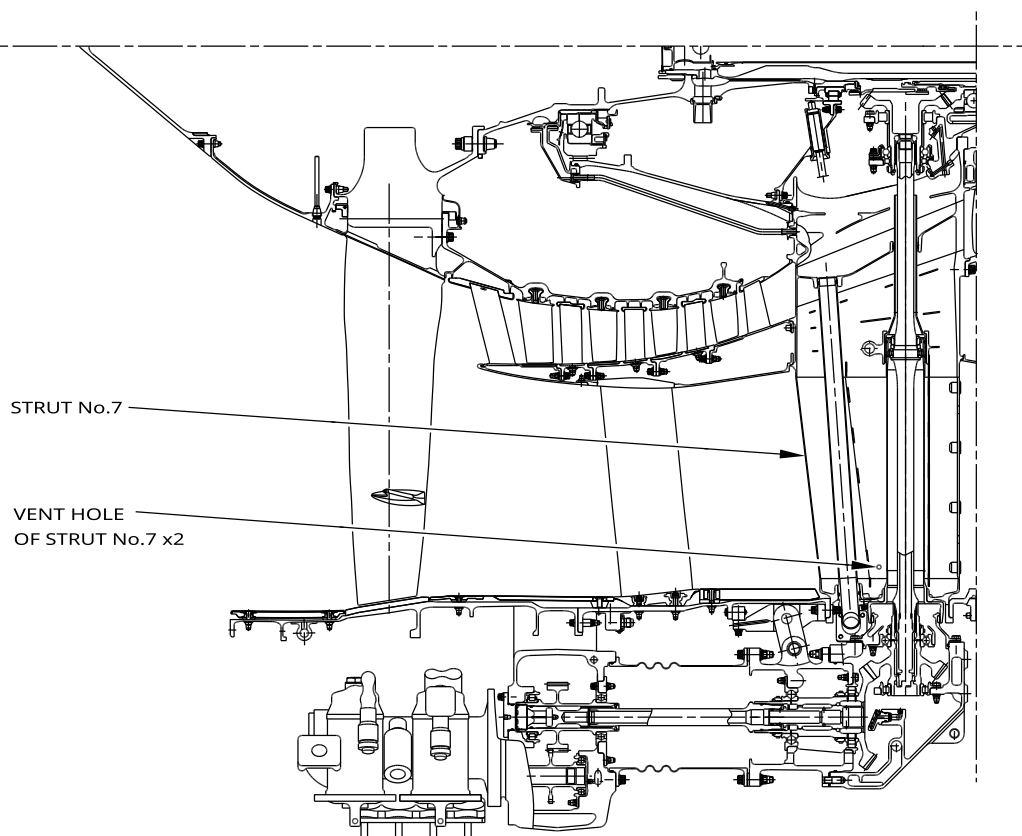
S-M56-MM-02631-03
N_MM_722300_6_RAM0_01_04

Fan Frame Casing, Fan Inlet Case and Abradable Cartridge - Inspection/Check
Figure 1 / Graphic 72-23-00 -991-30800

TF-N
AIRCRAFT MAINTENANCE MANUAL
FAN FRAME ASSEMBLY - INSPECTION/CHECK

SECTION

B-B



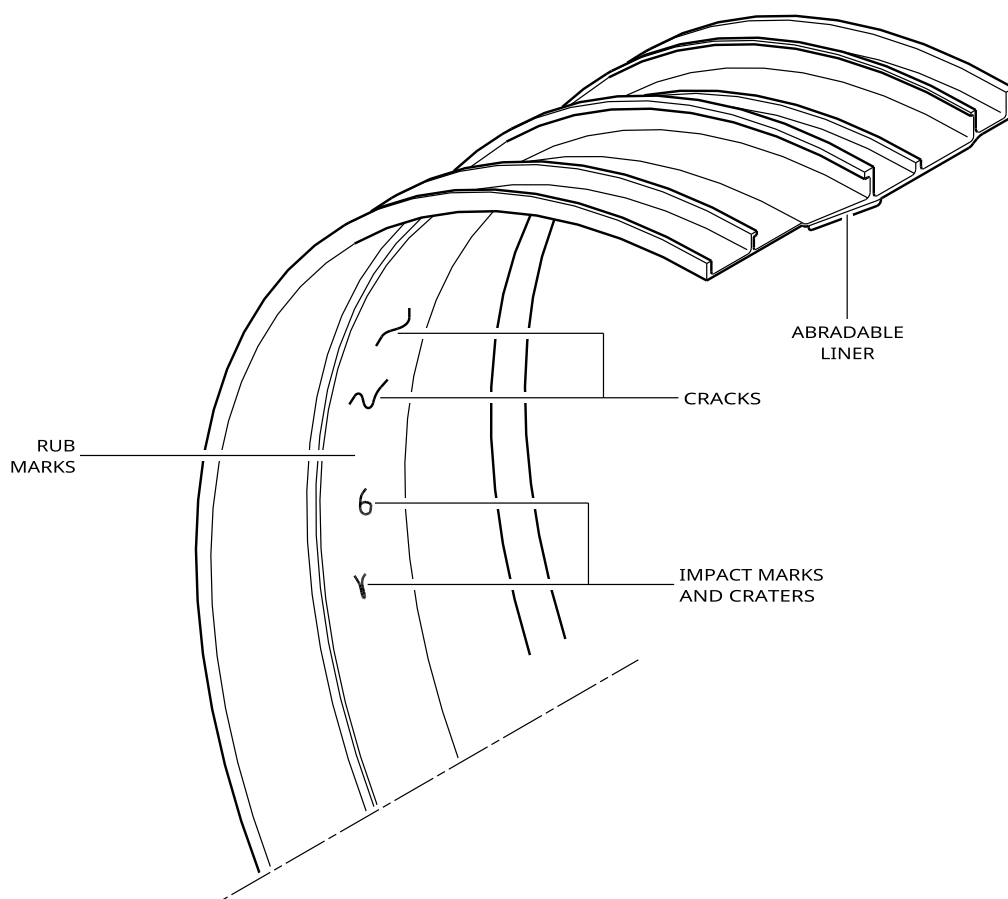
N_MM_722300_6_RAM0_02_01

Fan Frame Casing, Fan Inlet Case and Abradable Cartridge - Inspection/Check
Figure 2 / Graphic 72-23-00 -991-30800

S-M56-MM-05323-01

TF-N
AIRCRAFT MAINTENANCE MANUAL
FAN FRAME ASSEMBLY - INSPECTION/CHECK

B

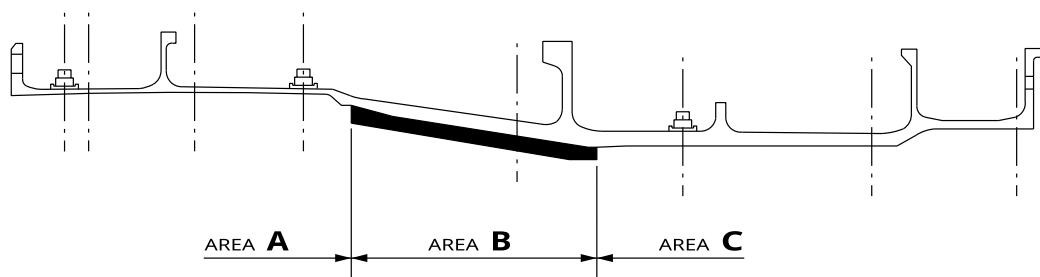
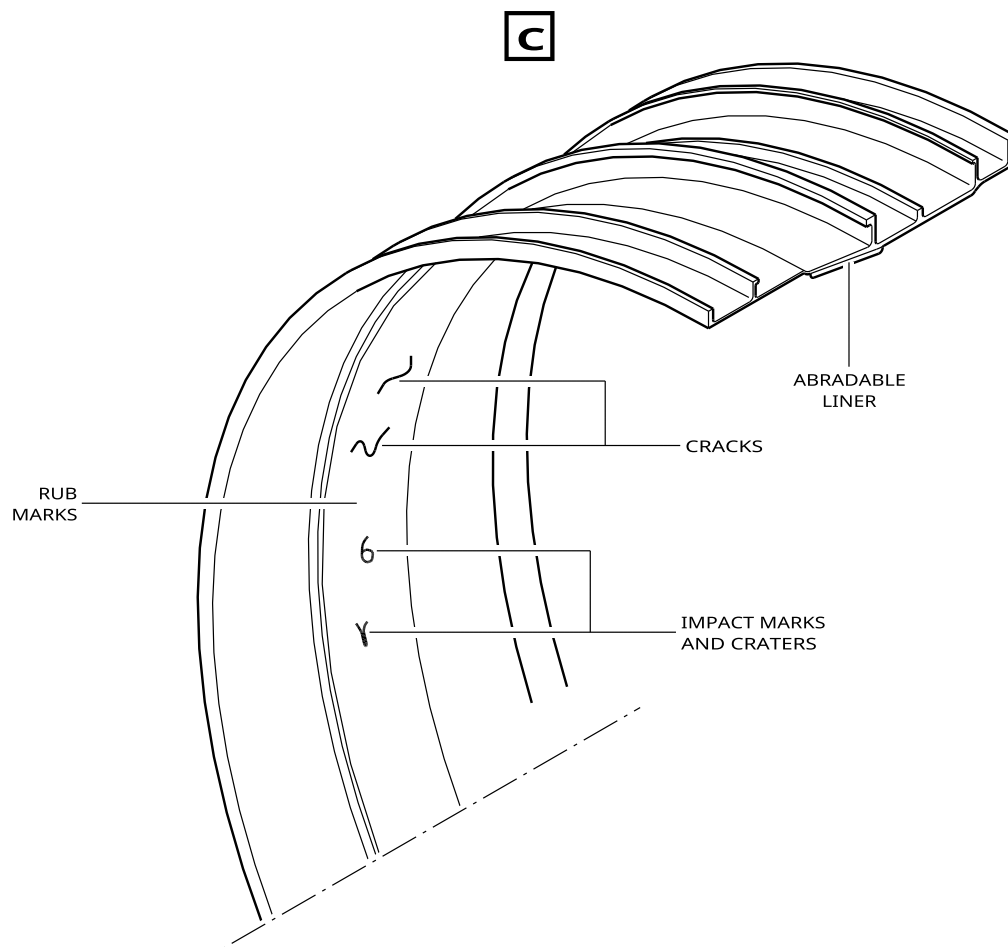


S-M56-MM-03486-00-B

N_MM_722300_6_RAM0_03_01

Fan Frame Casing, Fan Inlet Case and Abradable Cartridge - Inspection/Check
Figure 3 / Graphic 72-23-00 -991-30800

TF-N
AIRCRAFT MAINTENANCE MANUAL
FAN FRAME ASSEMBLY - INSPECTION/CHECK

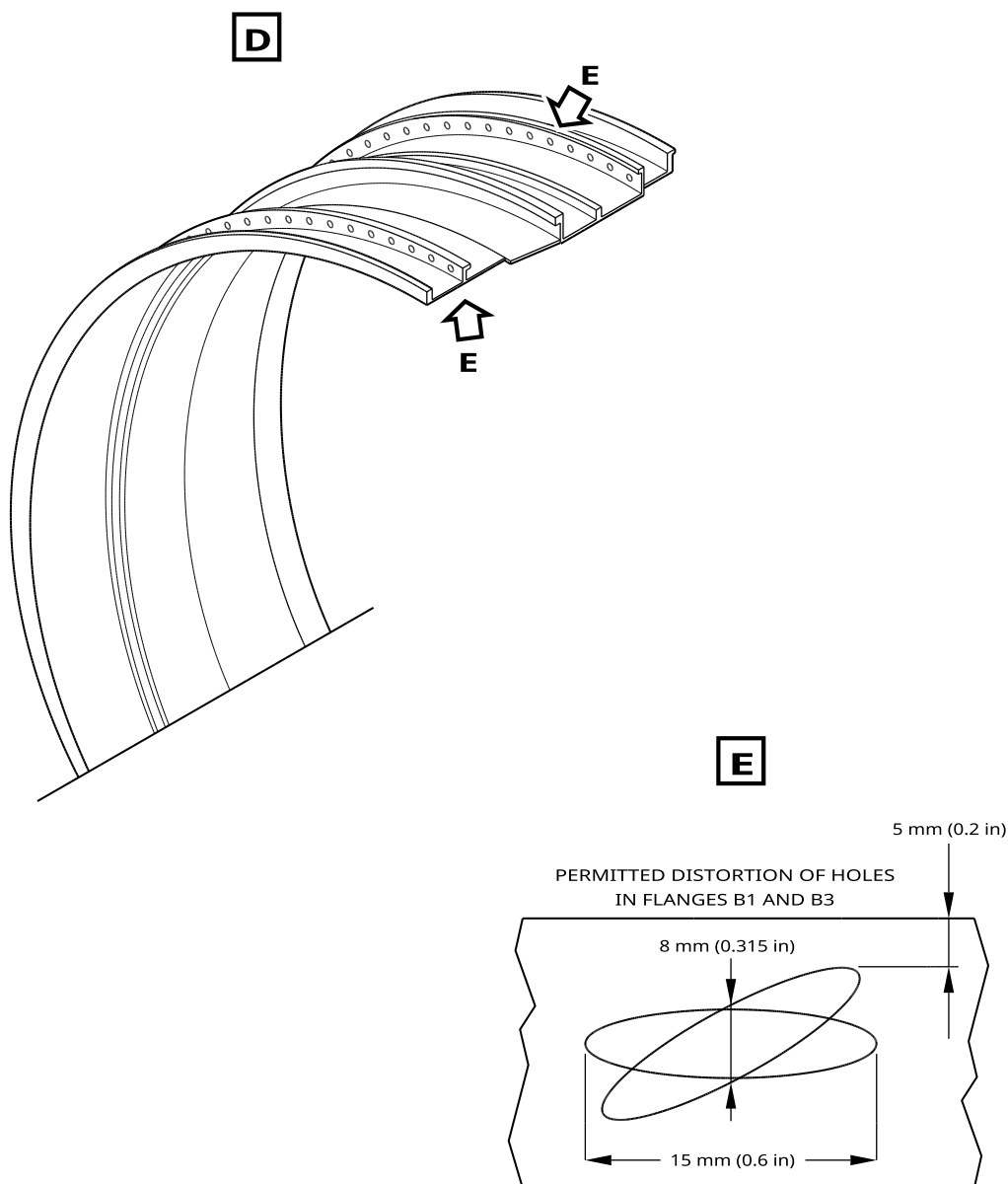


N_MM_722300_6_RAM0_04_02

Fan Frame Casing, Fan Inlet Case and Abradable Cartridge - Inspection/Check
Figure 4 / Graphic 72-23-00 -991-30800

S-M56-MM-04447-00-B

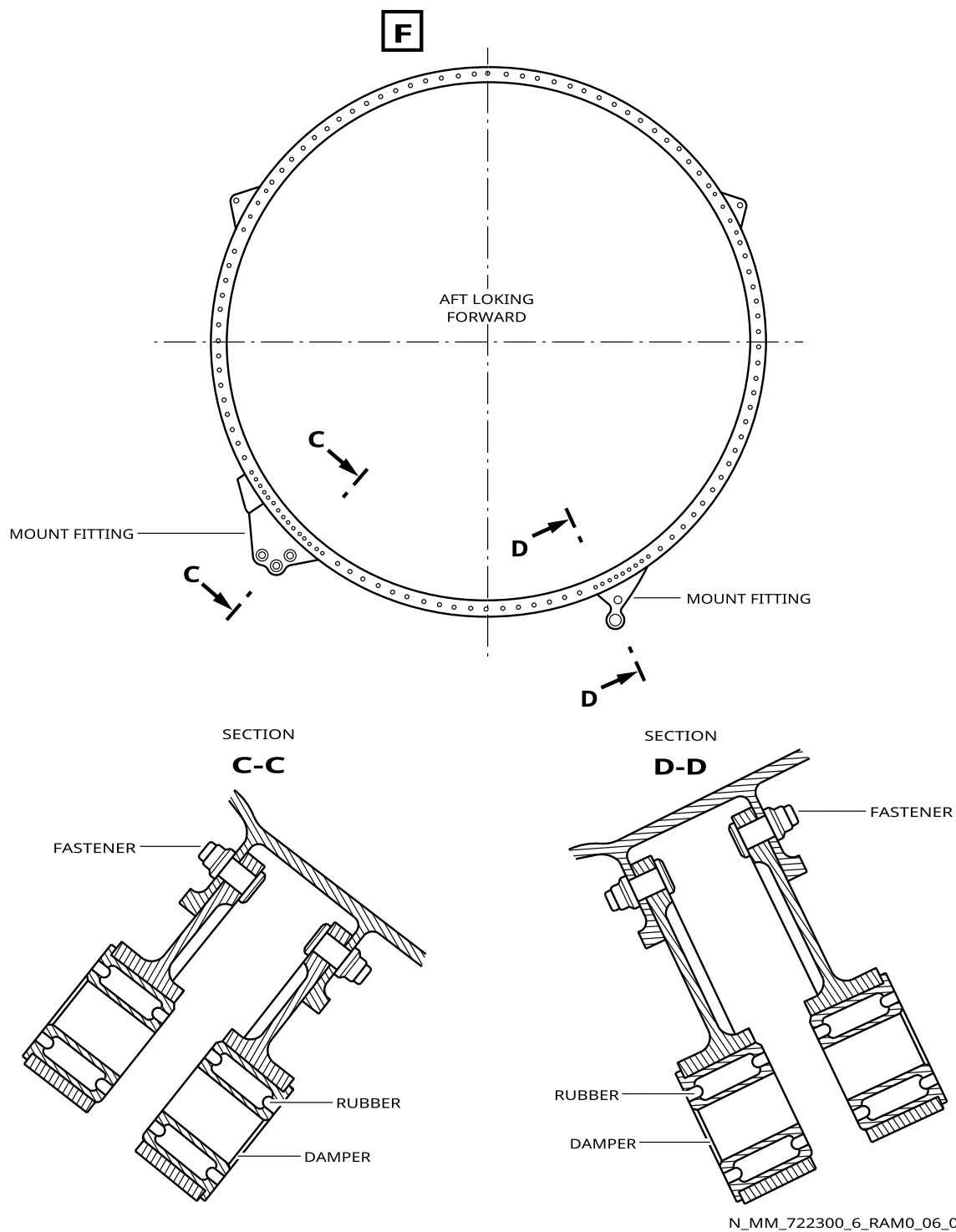
TF-N
AIRCRAFT MAINTENANCE MANUAL
FAN FRAME ASSEMBLY - INSPECTION/CHECK



N_MM_722300_6_RAM0_05_01

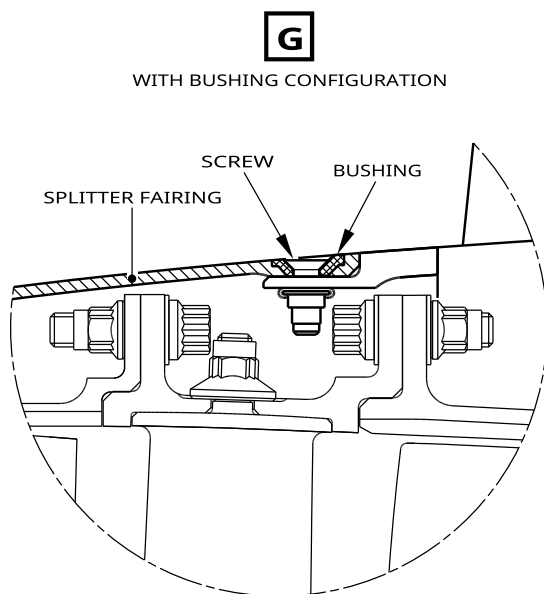
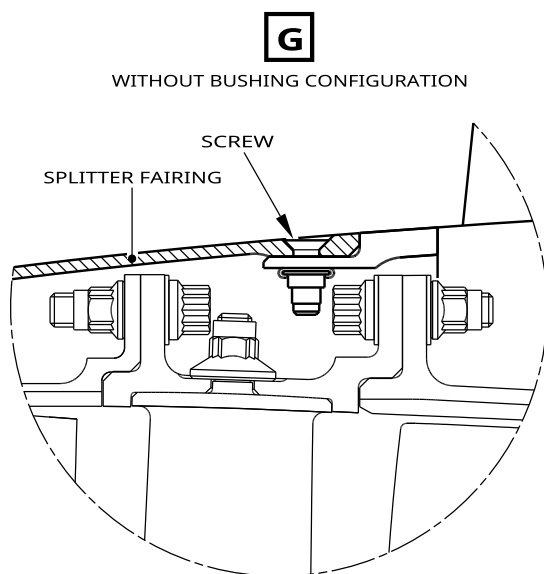
Fan Frame Casing, Fan Inlet Case and Abradable Cartridge - Inspection/Check
Figure 5 / Graphic 72-23-00 -991-30800

TF-N
AIRCRAFT MAINTENANCE MANUAL
FAN FRAME ASSEMBLY - INSPECTION/CHECK



Fan Frame Casing, Fan Inlet Case and Abradable Cartridge - Inspection/Check
Figure 6 / Graphic 72-23-00 -991-30800

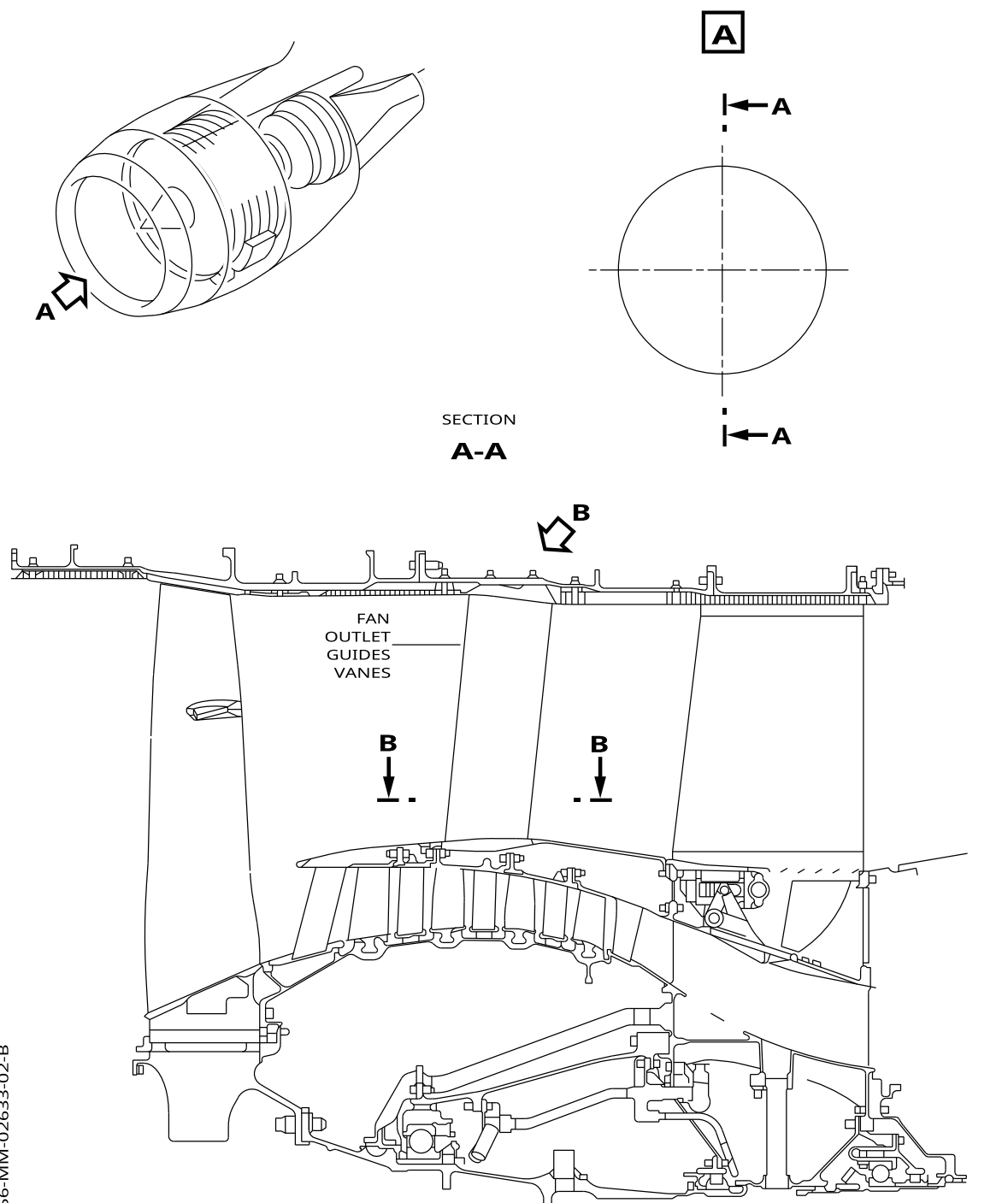
TF-N
AIRCRAFT MAINTENANCE MANUAL
FAN FRAME ASSEMBLY - INSPECTION/CHECK



S-M56-MM-05975-01
N_MM_722300_6_RAM0_07_00

Fan Frame Casing, Fan Inlet Case and Abradable Cartridge - Inspection/Check
Figure 7 / Graphic 72-23-00 -991-30800

TF-N
AIRCRAFT MAINTENANCE MANUAL
FAN FRAME ASSEMBLY - INSPECTION/CHECK

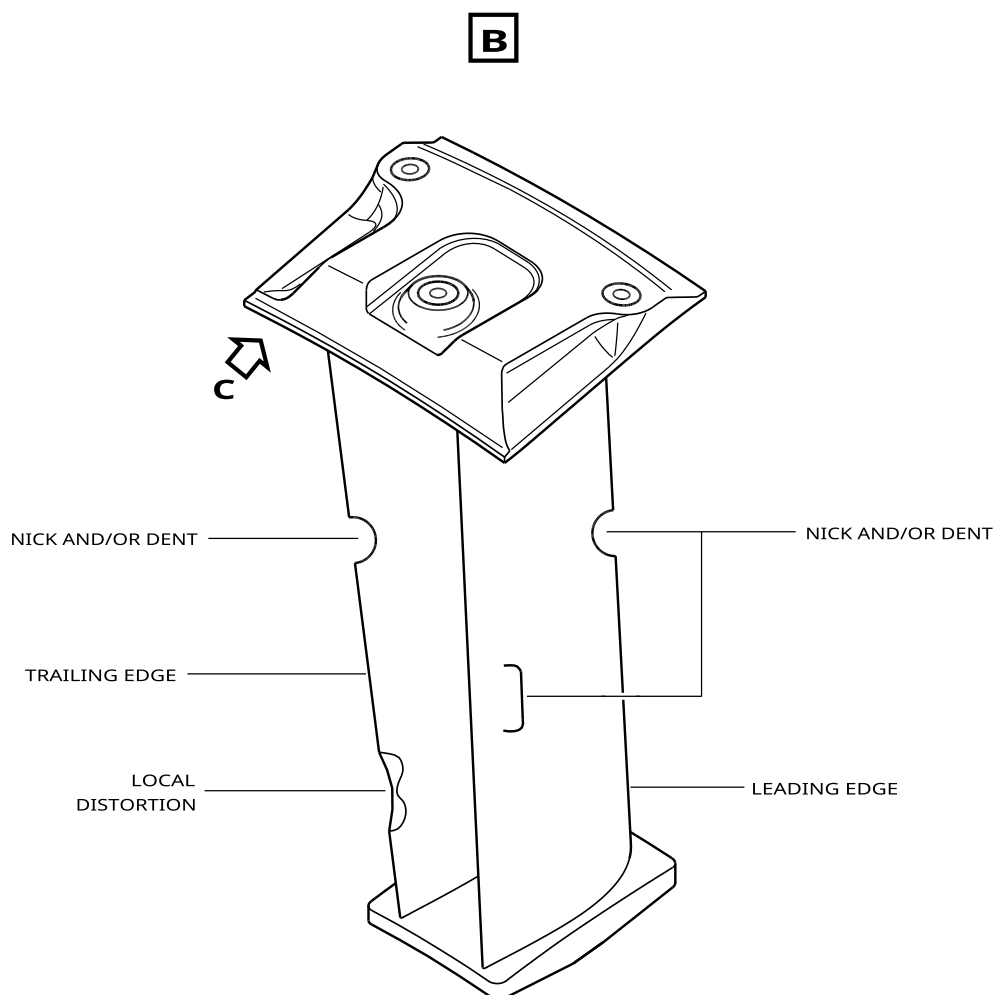


S-M56-MM-02633-02-B

N_MM_722300_6_RCM0_01_02

Inspection/Check of the Fan-Outlet Guide Vanes (OGV)
Figure 1 / Graphic 72-23-00 -991-31000

TF-N
AIRCRAFT MAINTENANCE MANUAL
FAN FRAME ASSEMBLY - INSPECTION/CHECK

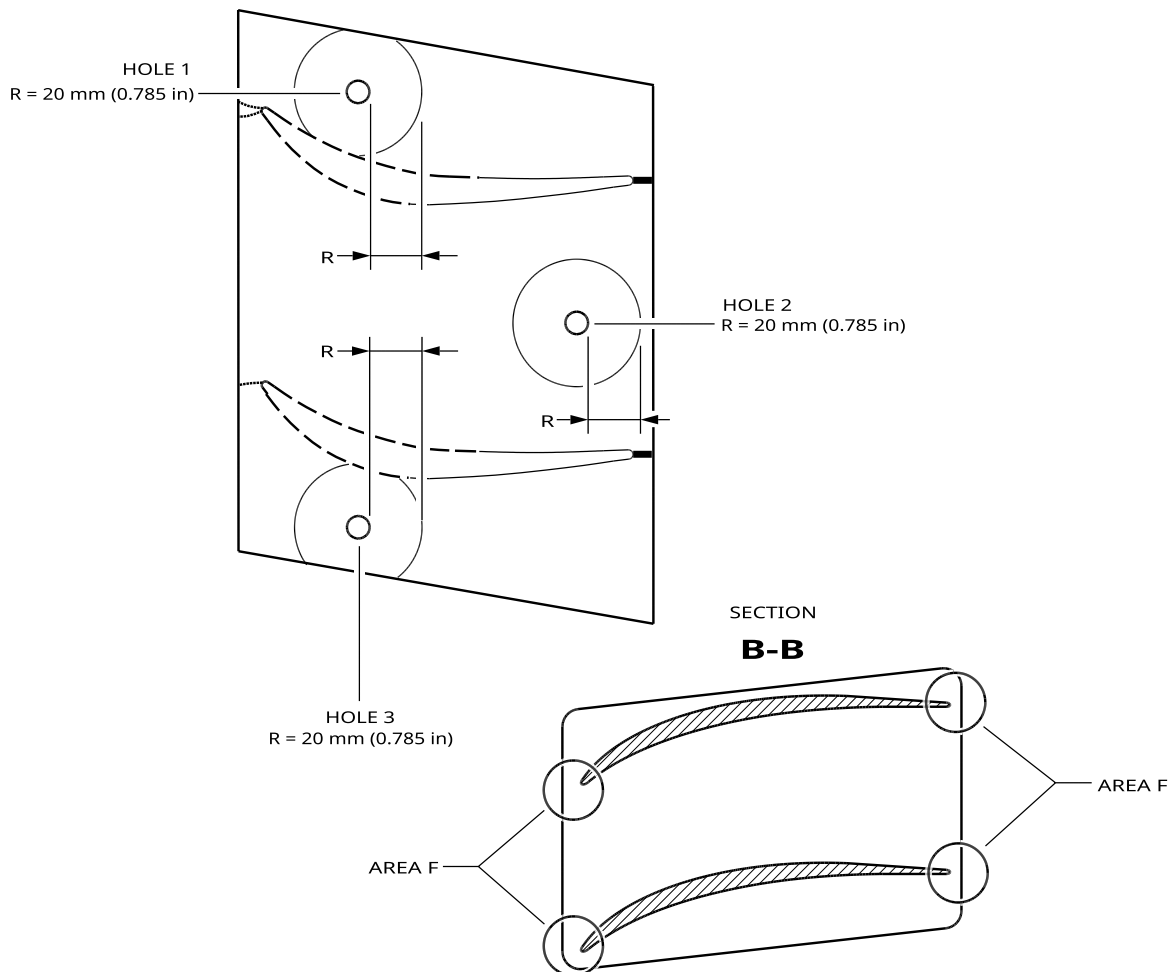


N_MM_722300_6_RCM0_02_02

Inspection/Check of the Fan-Outlet Guide Vanes (OGV)
Figure 2 / Graphic 72-23-00 -991-31000

TF-N
AIRCRAFT MAINTENANCE MANUAL
FAN FRAME ASSEMBLY - INSPECTION/CHECK

C



NOTE:



CRACKS IN TO RADIUS "R" ARE PERMITTED



CRACKS FROM THE AIRFOIL LEADING EDGE TO THE PLATFORM FRONT SIDE ARE PERMITTED. PIECE OF PU. PLATFORM MISSING IS NOT PERMITTED.



CRACKS FROM THE AIRFOIL TRAILING EDGE TO THE REAR SIDE PLATFORM ARE NOT PERMITTED.



CRACKS ALONG THE AIRFOIL ARE ALLOWED WITH A REQUIRED INSPECTION EVERY "A 600 F.H. CHECK" TO CONTROL THAT CRACKS ARE NOT GOING UP TO THE TRAILING EDGE. (IF CRACKS GO UP TO THE TRAILING EDGE THE PART IS NOT SERVICEABLE)

N_MM_722300_6_RCM0_03_02

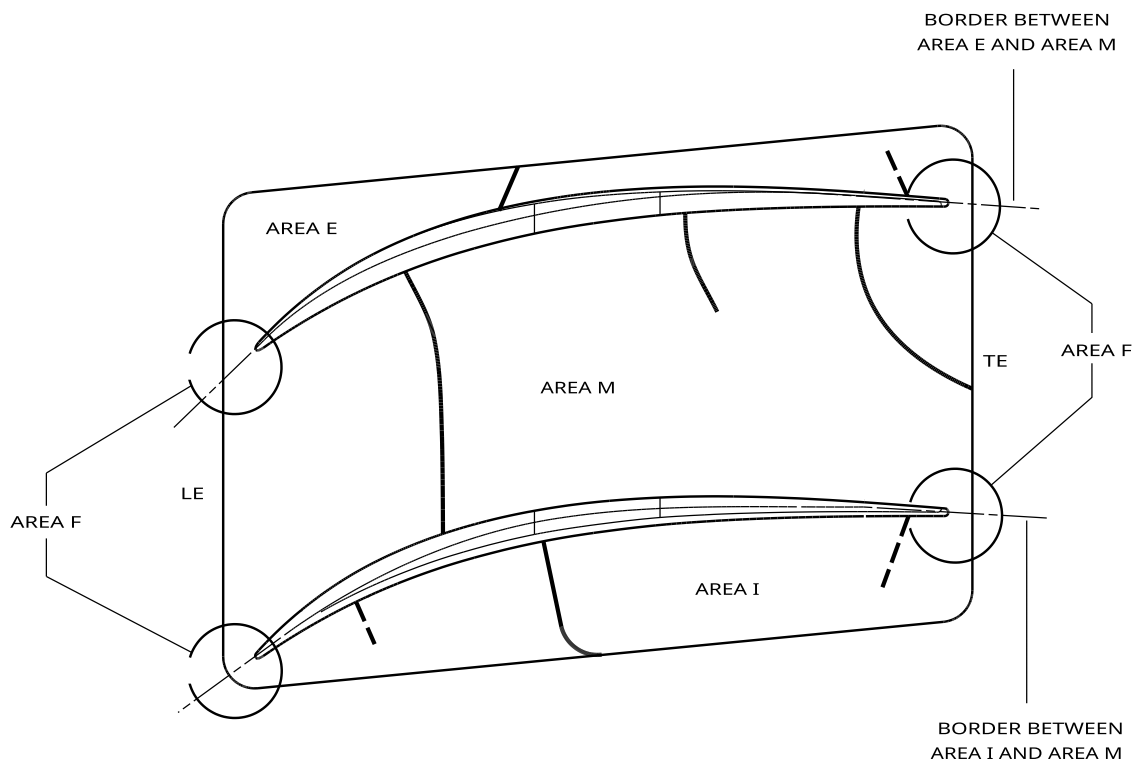
Inspection/Check of the Fan-Outlet Guide Vanes (OGV)

Figure 3 / Graphic 72-23-00 -991-31000

S-M56-MM-04249-06-B

TF-N
AIRCRAFT MAINTENANCE MANUAL
FAN FRAME ASSEMBLY - INSPECTION/CHECK

OGV INNER PLATFORM FOR CRACKS



NOTE:

—— OPENED TRANSVERSAL CRACK

--- TRANSVERSAL CRACK

DO NOT CHANGE THE LOCATION OF THE OGV CRACKED ON THE ENGINE.

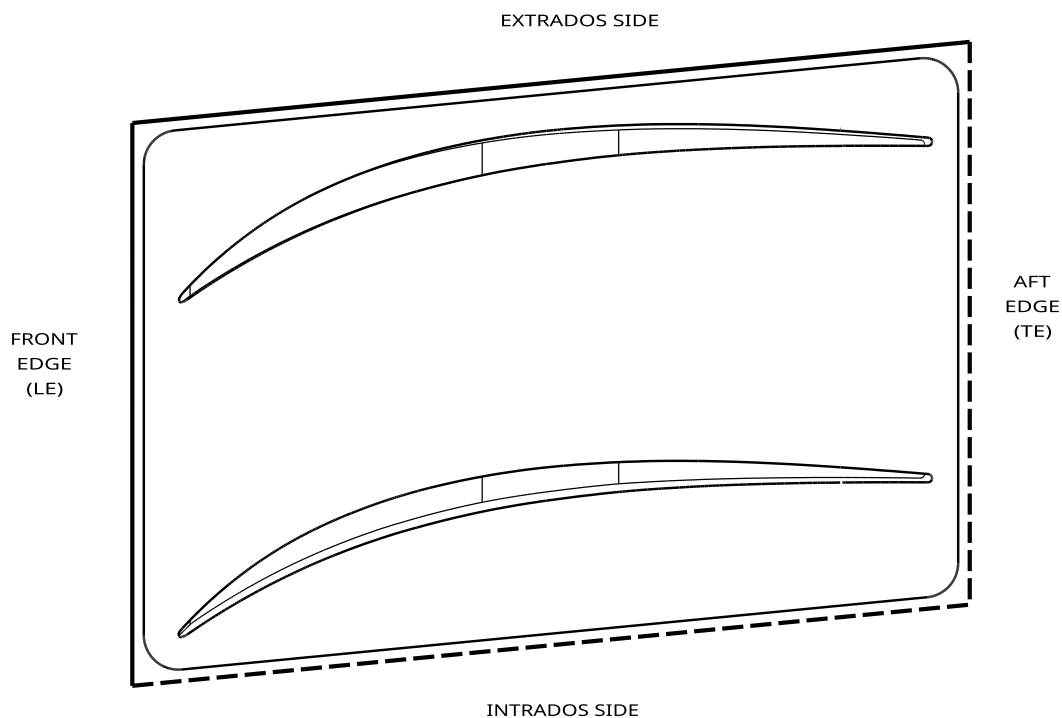
N_MM_722300_6_RCN0_01_00

Outlet Guide Vanes (OGV) Inner Platform for Cracks
Figure 1 / Graphic 72-23-00 -991-33300

S-M56-MM-04961-01

TF-N
AIRCRAFT MAINTENANCE MANUAL
FAN FRAME ASSEMBLY - INSPECTION/CHECK

OGV INNER PLATFORM FOR MISSING MATERIAL



NOTE:

— REFER TO 6.D MISSING MATERIAL ON INNER PLATFORM FOR EXTRADOS OR FRONT EDGE.

- - - REFER TO 6.D MISSING MATERIAL ON INNER PLATFORM FOR INTRADOS OR AFT EDGE.

DO NOT CHANGE THE LOCATION OF THE OGV WITH MISSING MATERIAL ON THE ENGINE

S-M56-MM-04962-02

N_MM_722300_6_RCP0_01_01

Outlet Guide Vanes (OGV) Inner Platform for missing material

Figure 1 / Graphic 72-23-00 -991-33400

TF-N
AIRCRAFT MAINTENANCE MANUAL
FAN FRAME ASSEMBLY - INSPECTION/CHECK

MISSING MATERIAL ON INNER PLATFORM
FOR INTRADOS OR AFT EDGE

CONDITION	OVER SERVICEABLE LIMIT EXTENSION
IF MISSING MATERIAL IS LESS THAN 0.08 SQUARE in (50 SQUARE mm).	300 CYCLES EXTENSION ALLOWED
IF MISSING MATERIAL IS LESS THAN 0.16 SQUARE in (100 SQUARE mm).	100 CYCLES EXTENSION ALLOWED
IF MISSING MATERIAL IS LESS THAN 0.47 SQUARE in (300 SQUARE mm).	10 CYCLES EXTENSION ALLOWED

NOTE:

DO NOT CHANGE THE PLACE OF OGV WITH MISSING MATERIAL ON THE ENGINE

N_MM_722300_6_RCP0_02_00

Outlet Guide Vanes (OGV) Inner Platform for missing material
Figure 2 / Graphic 72-23-00 -991-33400

 MET.28394722		Task Card Title: NO. 1 ENGINE EMCD CHECK			
Task Card NO: 792000-C8-1-001		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP	
A/C Type: 321		Block Taskcard		Main Skill: PP,ACFT_TECH	
Project No: 01BASPC72608		Related Task Card:		A/C Area: 01	
Internal Check:		Check Type:		Work Order NO: 31909295	
				Tally Sequence NO: 98	
ENCLOSURE: ☐ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐					
Type Code: DET		Special Code: PP		MSG Code: SOEE, 6, MRB, LUR	
				Request: Phase: OPS/LK	
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRID OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.					
Zones: 435					
DOCUMENT REFERENCES:					
Doc Typ		Reference			
AMM		79-00-00-281-002-A			
TR		SA00ME2302736			
ACCESS PANELS:					
437AL, 438AR					
MATERIAL REQUIREMENTS:					
Part Number		Description			Qty.
9200-116-0		PACKING-PREFORMED O-RING U/O CFM56-5			0.0
M83248-1-016		PACKING			1.0
TOOLING REQUIREMENTS:					
Part Number		Description			Qty.
856A1364G02		PRINT SET (CFM-CD)			1.0
856A2683G01		CHIP DETECTOR MAT. IDENTIFICATION TO			0.0
TASK INSTRUCTION					
Job Description					
A. CHECK THE ENGINE ELECTRICAL MASTER CHIP DETECTOR (EMCD) PER AMM TASK 79-00-00-281-002-A. B. COLLECT ANY DEBRIS FOUND ON THE EMCD IN A BAG AND WRITE: 1. AIRCRAFT REGISTRATION 2. ENGINE POSITION 3. ENGINE SERIAL NUMBER 4. PERFORMED DATE C. SEND THE DEBRIS FOUND ON THE EMCD TO FLUID ANALYSIS LAB (FAL) 1. NOTE: THE DEBRIS TO BE SENT THROUGH:					



Task Card Title:

NO. 1 ENGINE EMCD CHECK

Task Card NO:

792000-C8-1-001

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

a. IF AIRCRAFT IN ST MRO (LOCAL): ROUTE TO FLUID LAB

b. IF AIRCRAFT AT OOK MRO: ON-SITE SV PLANNER

2. NOTE: IF ANY UNACCEPTABLE FINDING NOTICED, REPORT TO MCC AIRBUS VIA TECHNICAL QUERY GROUP: **MCC-AB-AP** and TO POWERPLANT ENGINEERING VIA TECHNICAL QUERY GROUP: **ENG-CON-PP**.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PP/ACFT_TECH		HRS				
Insp.			HRS				

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below

 Work order No: _____

Note: Make sure to creat the finding Work Order on task card level.

 Remark's: _____



Task Card Title:

NO. 1 ENGINE EMCD CHECK

Task Card NO:

792000-C8-1-001

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

(1) If Task card interrupted or performed by different personals updates the system and update the following according (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 2 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 3 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 4 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 5 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____

**Task Completion Close Up And
System Update**

Station**Date****Signature**

TF-N
AIRCRAFT MAINTENANCE MANUAL
OIL - GENERAL - INSPECTION/CHECK

TASK 79-00-00-281-002 -A

On A/C 003-250

Check of the Electrical Master Chip Detector (EMCD) for Particles

FIN: 1000EM1, 1000EM2

1. Reason for the Job

Refer to the MPD TASK: 792000-C8

CHECK MASTER MAGNETIC CHIP DETECTOR FOR PARTICLES

(Inspection of the Electrical Master Chip Detector at scheduled intervals).

This inspection is required when the pop-out on the visual indicator is extended during operation.

This inspection is required when performing TSM Task 79-00-00-810-820/821 (Oil filter clogged).

detection of particles and gives recommendations for Line Maintenance.

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	AR	WARNING NOTICE(S)

B. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
FOR 1000EM1(ENGINE-1)	
437AL , 438AR	
FOR 1000EM2(ENGINE-2)	
447AL , 448AR	

C. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 71-13-00-010-040-B	Opening of the Fan Cowl Doors
Ref. AMM 71-13-00-410-040-B	Closing of the Fan Cowl Doors
Ref. AMM 79-00-00-281-003-A	Check of FWD Sump, AFT Sump, AGB and TGB Scavenge Screens
Ref. AMM 79-00-00-281-004-A	Preparation of the Particles from the Electrical Master Chip Detector (EMCD)
Ref. AMM 79-00-00-281-006-A	Ground Run for Additional Particle Collection
Ref. Fig. (1 / 79-00-00-991-30100)	
Ref. Fig. (1 / 79-00-00-991-52000)	

3. Job Set-up

Subtask 79-00-00-941-055-A

On A/C 003-250

A. Safety Precautions

- (1) On the center pedestal, on the ENG panel 115VU:
 - (a) Put a WARNING NOTICE(S) to tell persons not to start the engine.
- (2) Make sure that the engine shutdown occurred five minutes or more before you do this procedure.
- (3) On the overhead maintenance panel 50VU:
 - (a) Make sure that the ON legend of the ENG/FADEC GND PWR/1(2) pushbutton switch is off.
 - (b) Put a WARNING NOTICE(S) to tell persons not to energize the FADEC 1(2).

Subtask 79-00-00-010-055-A

On A/C 003-250

B. Open the fan cowls doors [Ref. AMM 71-13-00-010-040-B](#):

- (1) FOR 1000EM1 (ENGINE-1)
437AL 438AR
- (2) FOR 1000EM2 (ENGINE-2)
447AL 448AR .

4. Procedure

Subtask 79-00-00-281-050-A

On A/C 003-250

A. Inspection of the Electrical Master Chip Detector (EMCD)

Ref. Fig. (1 / 79-00-00-991-30100)

Ref. Fig. (1 / 79-00-00-991-52000)

(1) Inspect the EMCD [Ref. AMM 79-00-00-281-004-A](#).

(a) If any findings on the EMCD, do a check of Fwd sump, Aft sump, AGB and TGB scavenge screens [Ref. AMM 79-00-00-281-003-A](#) for further troubleshooting.

(b) If no findings in the scavenge screens, perform ground run for additional particle collection [Ref. AMM 79-00-00-281-006-A](#) for further troubleshooting.

(c) If no findings after step (b), no further action.

5. Close-up

Subtask 79-00-00-410-055-A

On A/C 003-250

A. Close Access

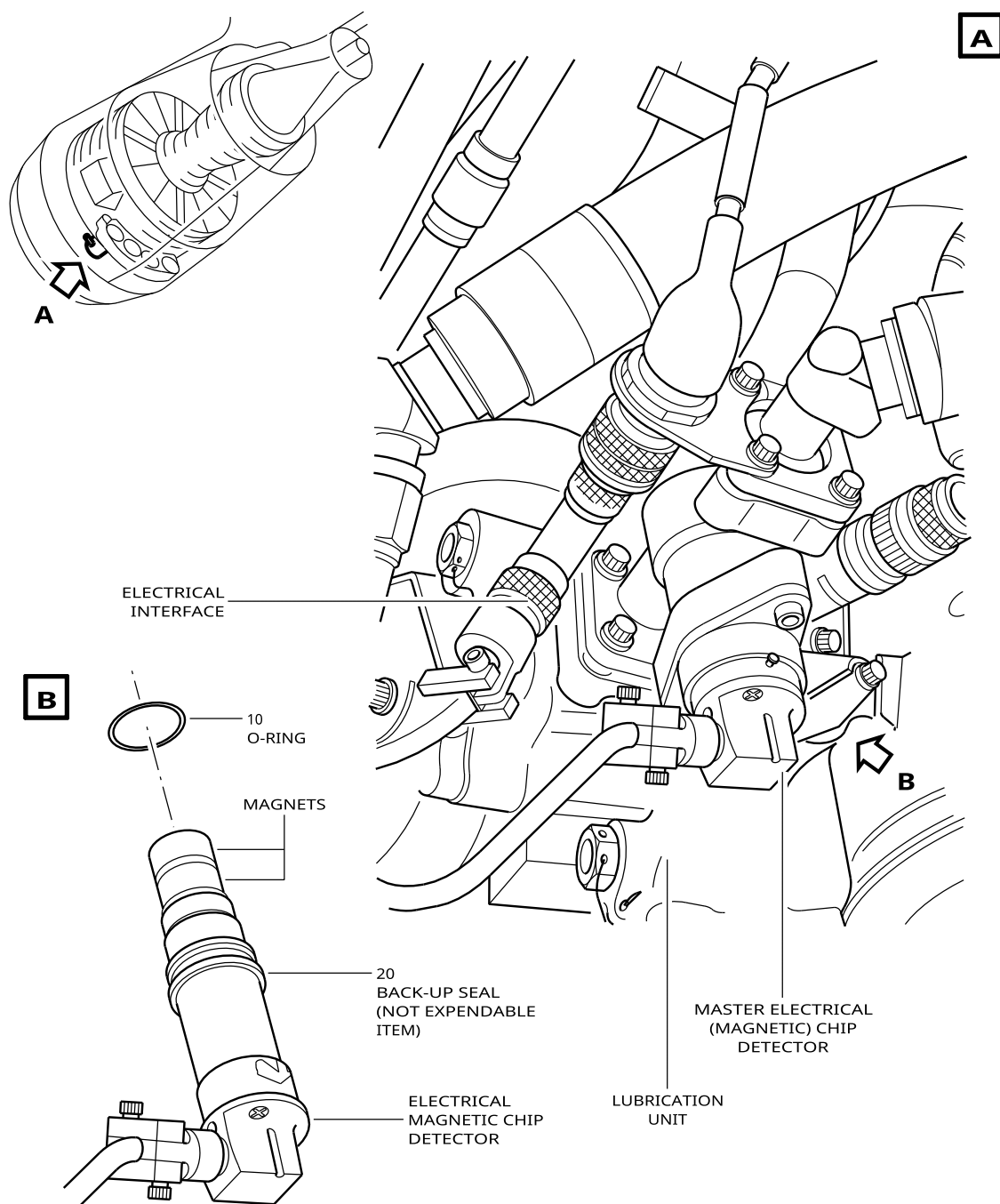
(1) Make sure that the work area is clean and clear of tools and other items.

(2) Close the fan cowl doors [Ref. AMM 71-13-00-410-040-B](#):

(a) FOR 1000EM1 (ENGINE-1)
437AL 438AR

(b) FOR 1000EM2 (ENGINE-2)
447AL 448AR

(3) Remove the WARNING NOTICE(S).

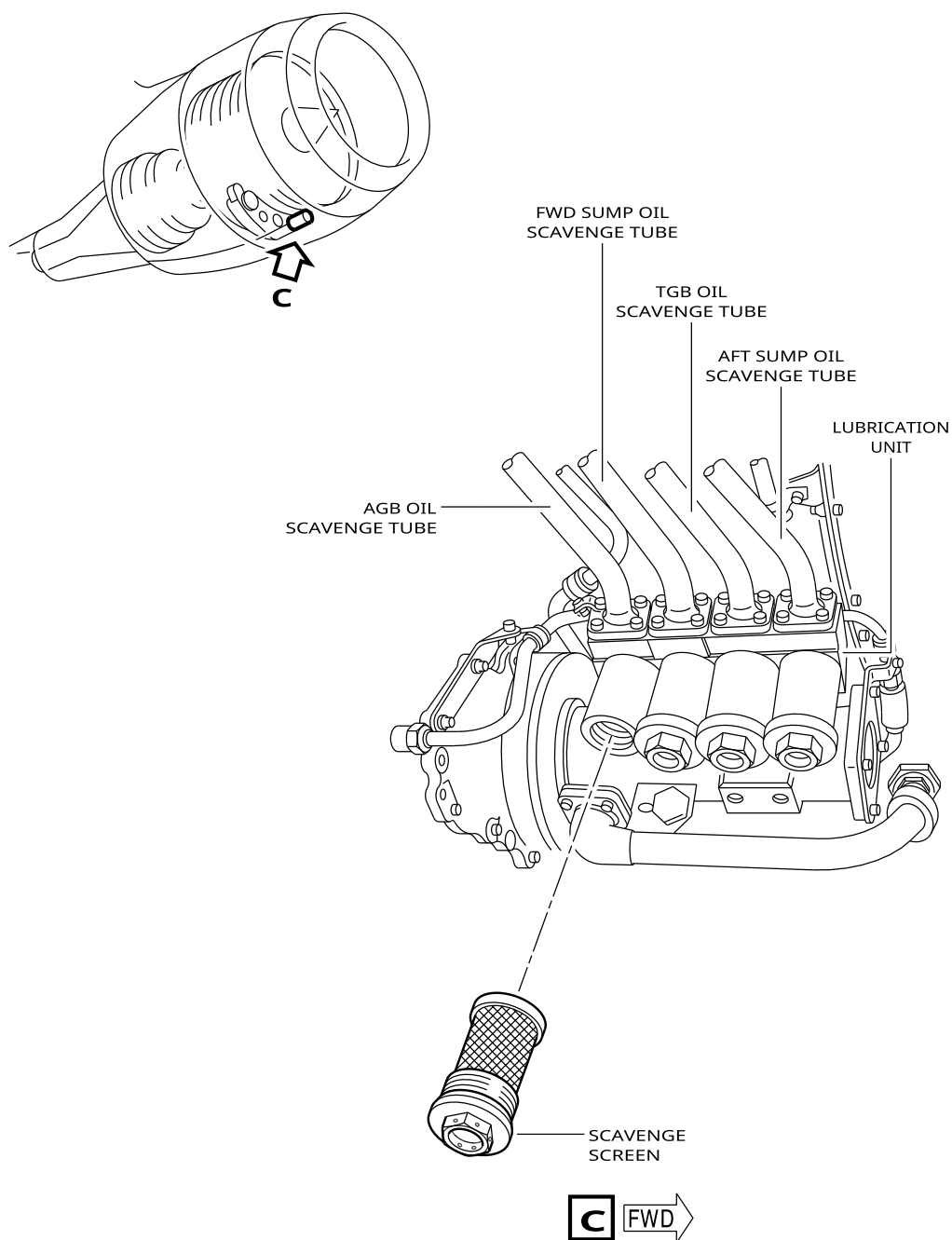


S-M56-MM-04252-00-B

N_MM_790000_6_RBM0_01_00

Position of the Electrical Master Magnetic Chip Detector for Particles
Figure 1 / Graphic 79-00-00 -991-30100

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AIRCRAFT MAINTENANCE MANUAL
OIL - GENERAL - INSPECTION/CHECK



S-M56-MM-02699-03

N_MM_790000_6_RBQ0_01_00

Position of the AGB and TGB Scavenge Screens
Figure 1 / Graphic 79-00-00 -991-52000

 MET.28358178		Task Card Title: NO. 2 ENGINE EMCD CHECK			
Task Card NO: 792000-C8-1-002		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP	
A/C Type: 321		Block Taskcard		Main Skill: PP,ACFT_TECH	
Project No: 01BASPC72608		Related Task Card:		A/C Area: 02	
Internal Check:		Check Type:		Work Order NO: 31909295	
				Tally Sequence NO: 99	
ENCLOSURE: ☐ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐					
Type Code: DET		Special Code: PP		MSG Code: SOEE, 6, MRB, LUR	
				Request: Phase: OPS/LK	
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRID OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.					
Zones: 445					
DOCUMENT REFERENCES:					
Doc Typ		Reference			
AMM		79-00-00-281-002-A			
TR		SA00ME2302736			
ACCESS PANELS:					
447AL, 448AR					
MATERIAL REQUIREMENTS:					
Part Number		Description			Qty.
9200-116-0		PACKING-PREFORMED O-RING U/O CFM56-5			0.0
M83248-1-016		PACKING			1.0
TOOLING REQUIREMENTS:					
Part Number		Description			Qty.
856A1364G02		PRINT SET (CFM-CD)			1.0
856A2683G01		CHIP DETECTOR MAT. IDENTIFICATION TO			0.0
TASK INSTRUCTION					
Job Description					
A. CHECK THE ENGINE ELECTRICAL MASTER CHIP DETECTOR (EMCD) PER AMM TASK 79-00-00-281-002-A. B. COLLECT ANY DEBRIS FOUND ON THE EMCD IN A BAG AND WRITE: 1. AIRCRAFT REGISTRATION 2. ENGINE POSITION 3. ENGINE SERIAL NUMBER 4. PERFORMED DATE C. SEND THE DEBRIS FOUND ON THE EMCD TO FLUID ANALYSIS LAB (FAL) 1. NOTE: THE DEBRIS TO BE SENT THROUGH:					



MET.28358178

Task Card Title:

NO. 2 ENGINE EMCD CHECK

Task Card NO:

792000-C8-1-002

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

a. IF AIRCRAFT IN ST MRO (LOCAL): ROUTE TO FLUID LAB

b. IF AIRCRAFT AT OOK MRO: ON-SITE SV PLANNER

2. NOTE: IF ANY UNACCEPTABLE FINDING NOTICED, REPORT TO MCC AIRBUS VIA TECHNICAL QUERY GROUP: **MCC-AB-AP** and TO POWERPLANT ENGINEERING VIA TECHNICAL QUERY GROUP: **ENG-CON-PP**.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	PP/ACFT_TECH		HRS				
Insp.			HRS				

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below

Work order No: _____

Note: Make sure to creat the finding Work Order on task card level.

Remark's: _____



Task Card Title:

NO. 2 ENGINE EMCD CHECK

Task Card NO:

792000-C8-1-002

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

(1) If Task card interrupted or performed by different personals updates the system and update the following according (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 2 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 3 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 4 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 5 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____

**Task Completion Close Up And
System Update**

Station**Date****Signature**

TF-N
AIRCRAFT MAINTENANCE MANUAL
OIL - GENERAL - INSPECTION/CHECK

TASK 79-00-00-281-002 -A

On A/C 003-250

Check of the Electrical Master Chip Detector (EMCD) for Particles

FIN: 1000EM1, 1000EM2

1. Reason for the Job

Refer to the MPD TASK: 792000-C8

CHECK MASTER MAGNETIC CHIP DETECTOR FOR PARTICLES

(Inspection of the Electrical Master Chip Detector at scheduled intervals).

This inspection is required when the pop-out on the visual indicator is extended during operation.

This inspection is required when performing TSM Task 79-00-00-810-820/821 (Oil filter clogged).

detection of particles and gives recommendations for Line Maintenance.

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	AR	WARNING NOTICE(S)

B. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
FOR 1000EM1(ENGINE-1)	
437AL , 438AR	
FOR 1000EM2(ENGINE-2)	
447AL , 448AR	

C. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 71-13-00-010-040-B	Opening of the Fan Cowl Doors
Ref. AMM 71-13-00-410-040-B	Closing of the Fan Cowl Doors
Ref. AMM 79-00-00-281-003-A	Check of FWD Sump, AFT Sump, AGB and TGB Scavenge Screens
Ref. AMM 79-00-00-281-004-A	Preparation of the Particles from the Electrical Master Chip Detector (EMCD)
Ref. AMM 79-00-00-281-006-A	Ground Run for Additional Particle Collection
Ref. Fig. (1 / 79-00-00-991-30100)	
Ref. Fig. (1 / 79-00-00-991-52000)	

3. Job Set-up

Subtask 79-00-00-941-055-A

On A/C 003-250

A. Safety Precautions

- (1) On the center pedestal, on the ENG panel 115VU:
 - (a) Put a WARNING NOTICE(S) to tell persons not to start the engine.
- (2) Make sure that the engine shutdown occurred five minutes or more before you do this procedure.
- (3) On the overhead maintenance panel 50VU:
 - (a) Make sure that the ON legend of the ENG/FADEC GND PWR/1(2) pushbutton switch is off.
 - (b) Put a WARNING NOTICE(S) to tell persons not to energize the FADEC 1(2).

Subtask 79-00-00-010-055-A

On A/C 003-250

B. Open the fan cowls doors [Ref. AMM 71-13-00-010-040-B](#):

- (1) FOR 1000EM1 (ENGINE-1)
437AL 438AR
- (2) FOR 1000EM2 (ENGINE-2)
447AL 448AR .

4. Procedure

Subtask 79-00-00-281-050-A

On A/C 003-250

A. Inspection of the Electrical Master Chip Detector (EMCD)

Ref. Fig. (1 / 79-00-00-991-30100)

Ref. Fig. (1 / 79-00-00-991-52000)

(1) Inspect the EMCD [Ref. AMM 79-00-00-281-004-A](#).

(a) If any findings on the EMCD, do a check of Fwd sump, Aft sump, AGB and TGB scavenge screens [Ref. AMM 79-00-00-281-003-A](#) for further troubleshooting.

(b) If no findings in the scavenge screens, perform ground run for additional particle collection [Ref. AMM 79-00-00-281-006-A](#) for further troubleshooting.

(c) If no findings after step (b), no further action.

5. Close-up

Subtask 79-00-00-410-055-A

On A/C 003-250

A. Close Access

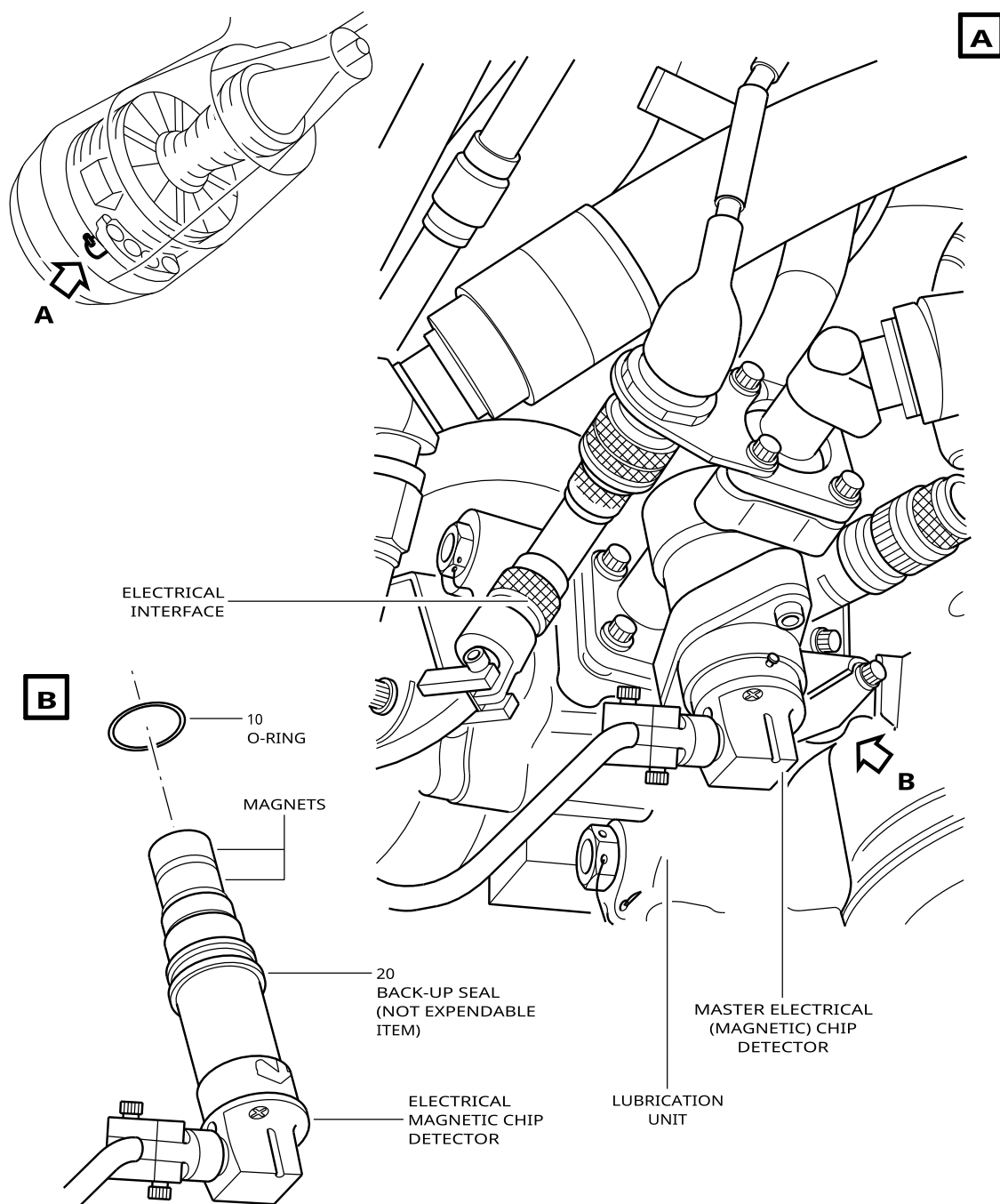
(1) Make sure that the work area is clean and clear of tools and other items.

(2) Close the fan cowl doors [Ref. AMM 71-13-00-410-040-B](#):

(a) FOR 1000EM1 (ENGINE-1)
437AL 438AR

(b) FOR 1000EM2 (ENGINE-2)
447AL 448AR

(3) Remove the WARNING NOTICE(S).

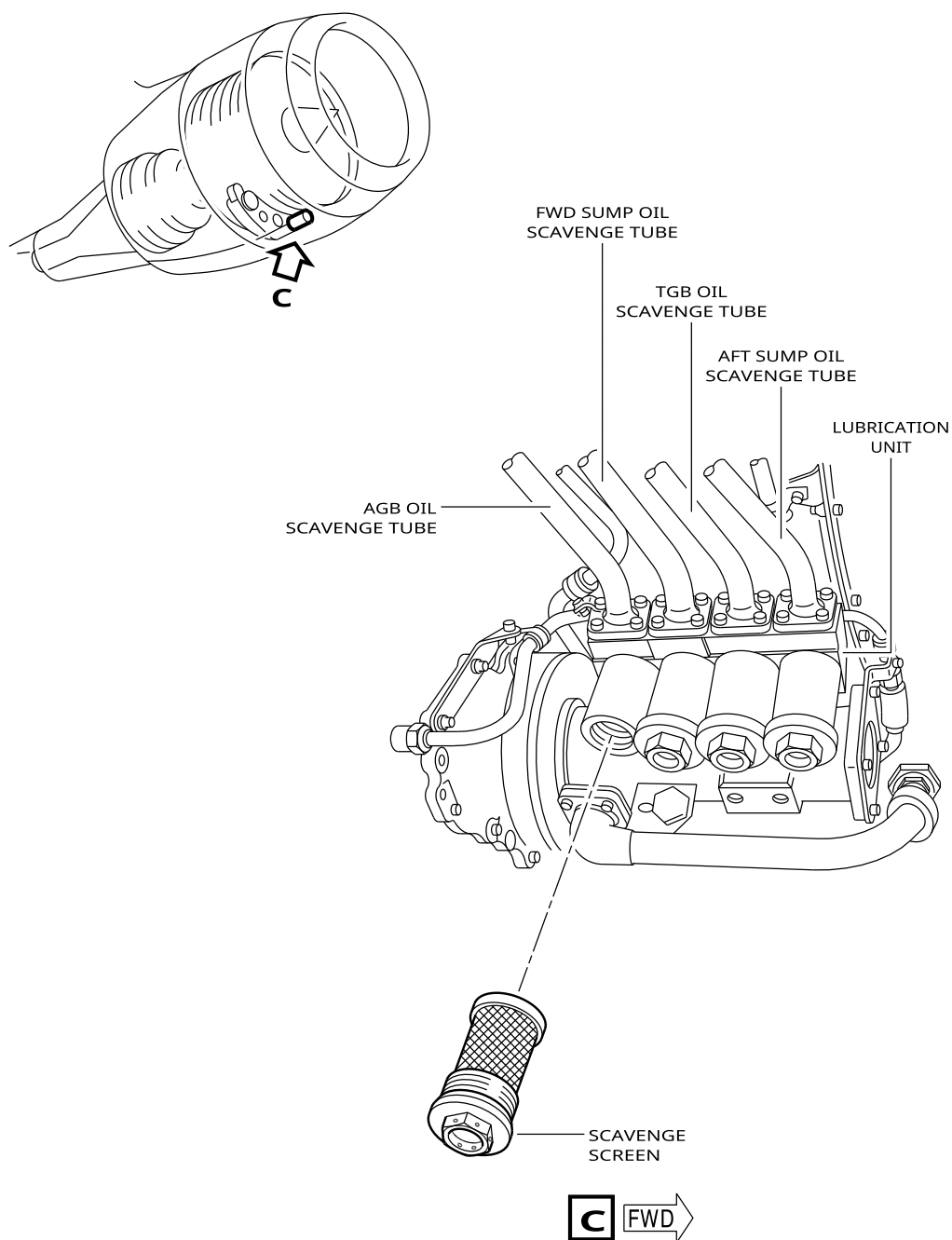


S-M56-MM-04252-00-B

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Position of the Electrical Master Magnetic Chip Detector for Particles
Figure 1 / Graphic 79-00-00 -991-30100

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OIL - GENERAL - INSPECTION/CHECK



S-M56-MM-02699-03

N_MM_790000_6_RBQ0_01_00

Position of the AGB and TGB Scavenge Screens
Figure 1 / Graphic 79-00-00 -991-52000

 MET.28358433		Task Card Title: NO. 1 ENGINE STARTER CHIP DETECTOR					
Task Card NO: 801110-C1-1-001		Work Package: ASP/H-26 C7-CHK		A/C Registration: HZ-ASP			
A/C Type: 321		Block Taskcard		Main Skill: ACFT_TECH,AP		PROD Final Acceptance:	
Project No: 01BASPC72608		Related Task Card:		A/C Area: 01			
Internal Check:		Check Type:		Work Order NO: 31909295		Tally Sequence NO: 100	
ENCLOSURE: ☐ CMR: ☐ CPCP: ☐ ETOPS: ☐ CDCCL: ☐ AD: ☐ ALI: ☐							
Type Code: CHK		Special Code: AP		MSG Code: LUR, 6, 9, MRB, SOEE		Request: Phase: JOB SETUP, TC/P, JOB-C LOSEUP	
I HEREBY CERTIFY THAT ALL WARNINGS AND CAUTIONS HAVE BEEN READ AND THE WORK HAS BEEN CARRIED OUT IN ACCORDING WITH ALL LAID DOWN PROCEDURES, ATTACHED NOTES AND INFORMATION.							
Zones: 436							
DOCUMENT REFERENCES:							
Doc Typ		Reference					
AMM		80-11-10-210-001-C					
AMM		80-11-10-210-001-C					
TR		SA00ME2302736					
ACCESS PANELS:							
437AL, 438AR							
MATERIAL REQUIREMENTS:							
Part Number		Description				Qty.	
AEROSHELLOIL560		JET ENGINE OIL (TYPE II HTS)				0.0	
CP2442		ENGINE OIL FOR LEASED AIRCRAFT				1.0	
TOOLING REQUIREMENTS:							
Part Number		Description				Qty.	
No tools assigned to this Task Item							
1. JOB-SETUP							
Job Description							
A.PER FORM JOB SETUP FOR ENG STARTER CHIP DETECTOR CHK AS PER AMM 80-11-10-210-001-C.							
STEP COMPLETION SIGN OFF:							
	Qualification / Skill	Number of employees	Duration		Number./Signature	Date	
				HRS		Day	Month
Mechanic/s	ACFT_TECH/AP			HRS			
Insp.				HRS			
2. WORK INSTRUCTIONS							
Job Description							
2.A. PERFORM ENGINE STARTER CHIP DETECTOR CHECK PER AMM TASK 80-11-10-210-001-C.							



MET.28358433

Task Card Title:

NO. 1 ENGINE STARTER CHIP DETECTOR

Task Card NO:

801110-C1-1-001

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

- a) TAKE PICTURES OF THE MAGNETIC PLUG PROBE, AND PARTICLES.
- b) CAREFULLY EXAMINE AND RECORD THE MAXIMUM SIZE OF THE PARTICLES AND DECIDE THE SERVICEABILITY OF THE STARTER ACCORDINGLY.
- c) RECORD THE ADDED OIL QUANTITY.
- d) ISSUE WORK ORDER TO:
- ATTACH THE PICTURES OF THE MAGNETIC PLUG PROBE.
 - THE RECORD OF PARTICLES MAXIMUM SIZE.
 - THE RECORD OF ADDED OIL QUANTITY.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	ACFT_TECH/AP		HRS				
Insp.			HRS				

3.JOB-CLOSEUP**Job Description**

A.PERFORM JOB-CLOSEUP AS PER AMM 80-11-10-210-001-C.

STEP COMPLETION SIGN OFF:

	Qualification / Skill	Number of employees	Duration	Number./Signature	Date		
					Day	Month	Year
Mechanic/s	ACFT_TECH/AP		HRS				
Insp.			HRS				

-Aircraft Maintenance Manual used: Revision No. _____ Dated. _____.

Task card finding (Tick Applicable Box) [] Yes [] No. If yes update the system and list Task card finding WO below

Work order No: _____

Note: Make sure to creat the finding Work Order on task card level.

Remark's: _____



MET.28358433

Task Card Title:

NO. 1 ENGINE STARTER CHIP DETECTOR

Task Card NO:

801110-C1-1-001

Work Package:

ASP/H-26 C7-CHK

A/C Registration:

HZ-ASP

(1) If Task card interrupted or performed by different personals updates the system and update the following according (Tick Applicable Box)

- Interruption No. 1 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 2 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 3 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 4 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____
- Interruption No. 5 [] Task card [] AMM [] Other specify. _____
 Performed by: _____ Page No. _____ Step From _____ To _____

(2) Record any dimension or torque through the performed steps and sign-off as applicable:

- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____
- Dimension/torque: _____ Mech _____ INSP _____ Page No. _____ Step _____
 Remark's: _____

**Task Completion Close Up And
System Update**

Station**Date****Signature**

TF-N
AIRCRAFT MAINTENANCE MANUAL
STARTER - INSPECTION/CHECK

TASK 80-11-10-210-001 -C

On A/C 003-250

Check Chip Detector

1. Reason for the Job

Refer to the MPD TASK: 801110-C1

CHECK CHIP DETECTOR

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	AR	MAGNIFYING GLASS POWER 10X
No specific	AR	TISSUE PAPER
No specific	AR	WARNING NOTICE(S)

B. Consumable Materials

REFERENCE	DESIGNATION
CP2442	engine oil

C. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
FOR 1000EM1(ENGINE-1)	
437AL , 438AR	
FOR 1000EM2(ENGINE-2)	
447AL , 448AR	

D. Expendable Parts

FIG.ITEM	DESIGNATION	IPC-CSN
100	O-RING	IPC-CSN(80-11-10-01 ITEM 100)

E. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 12-13-80-610-001-C	Drain and Replenish Oil System
Ref. AMM 71-13-00-010-040-B	Opening of the Fan Cowl Doors
Ref. AMM 71-13-00-410-040-B	Closing of the Fan Cowl Doors
Ref. AMM 80-11-10-000-002-A	Removal of the Pneumatic Starter
Ref. AMM 80-11-10-400-002-A	Installation of the Pneumatic Starter
Ref. Fig. (1 / 80-11-10-991-30000)	
Ref. Fig. (1 / 80-11-10-991-30100)	

3. Job Set-up

Subtask 80-11-10-941-052-A

On A/C 003-250

A. Safety Precautions

- (1) On the center pedestal, on the ENG panel 115VU:
 - (a) Put a WARNING NOTICE(S) to tell persons not to start the engine.
- (2) Make sure that the engine shutdown occurred five minutes or more before you do this procedure.
- (3) On the overhead maintenance panel 50VU:
 - (a) Make sure that the ON legend of the ENG/FADEC GND PWR/1(2) pushbutton switch is off.
 - (b) Put a WARNING NOTICE(S) to tell persons not to energize the FADEC 1(2).

Subtask 80-11-10-010-053-A

On A/C 003-250

B. Open the fan cowl doors: [Ref. AMM 71-13-00-010-040-B](#)

- (1) FOR 1000EM1 (ENGINE-1)
437AL 438AR

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AIRCRAFT MAINTENANCE MANUAL
STARTER - INSPECTION/CHECK

- (2) FOR 1000EM2 (ENGINE-2)
447AL 448AR

4. Procedure

Ref. Fig. 1 / 80-11-10-991-30000

Ref. Fig. 1 / 80-11-10-991-30100

Subtask 80-11-10-020-052-A

On A/C 003-250

A. Removal of Starter Magnetic Plug for Inspection.

- (1) Remove the knurled magnetic plug (90) from oil drain plug (110):
(a) Push and turn counter-clockwise to release the plug (90).

NOTE : The oil drain plug (110) has a built-in check valve to permit removal of magnetic plug (90) without draining oil from the starter.

Subtask 80-11-10-210-052-A

On A/C 003-250

B. Inspection of the Magnetic Plug Probe (90).

NOTE : Before examination for metal particles, carefully prevent a further contamination of the magnetic plug probe by metal particles from the tool box or work area.

- (1) Examine the magnetic plug probe for the quantity and type of the contaminants. If the contamination you see on the probe looks important, do as follows:

Ref. Fig. (1 / 80-11-10-991-30100)

NOTE : Carbon and dirt may form sludge on the probe along with metal particles, and cause misleading appearance in the quantity of contamination.

- (a) Clean the probe with TISSUE PAPER and retain the removed particles.
(b) Examine the metallic contaminants under a MAGNIFYING GLASS POWER 10X. Use the inspection criteria which follow :
- 1 A small accumulation of fine slivers and relatively flat flakes of metal shows a normal wear.
 - 2 A larger accumulation of particles, or particles larger than 0.1 in. (2.54 mm) in any direction, show there is some internal damage and that a starter replacement is necessary. Ref. AMM 80-11-10-000-002-A and Ref. AMM 80-11-10-400-002-A.

Subtask 80-11-10-420-052-A

On A/C 003-250

C. Installation of Magnetic Plug after Inspection.

- (1) Remove and discard the two O-rings (100).
(2) Lubricate two new IPC-CSN(80-11-10-01 ITEM 100) O-RING (100) with engine oil (Material No. **CP2442**) and put them on the knurled magnetic plug (90).
(3) Install the knurled magnetic plug (90) in oil drain plug (110):
(a) Push inward and turn clockwise to engage locking device.

Subtask 80-11-10-613-050-A

On A/C 003-250

D. Do the servicing of the starter, if necessary Ref. AMM 12-13-80-610-001-C

5. Close-up

Subtask 80-11-10-410-052-A

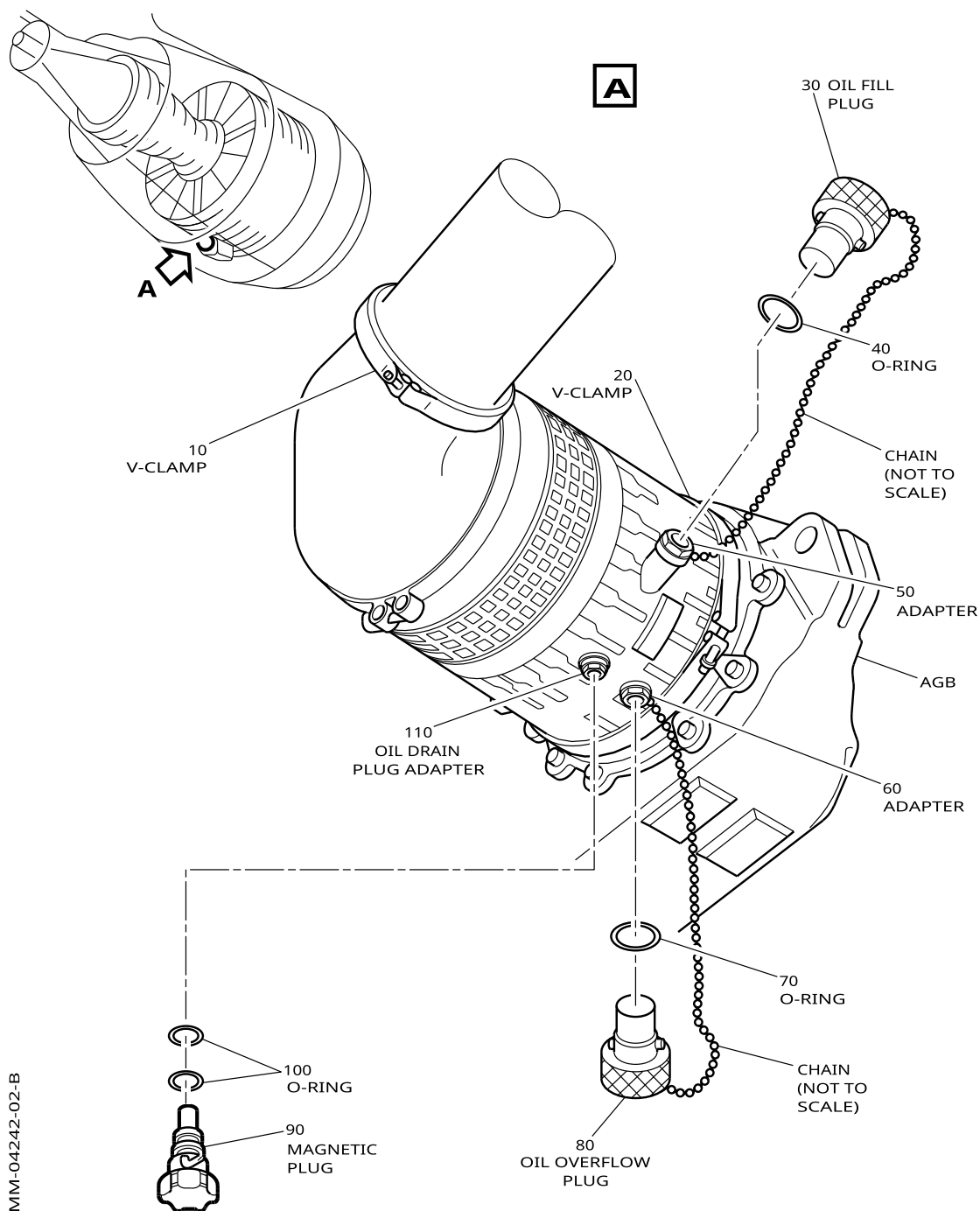
On A/C 003-250

TF-N
AIRCRAFT MAINTENANCE MANUAL
STARTER - INSPECTION/CHECK

A. Close Access

- (1) Make sure that the work area is clean and clear of tools and other items.
- (2) Close the fan cowl door: [Ref. AMM 71-13-00-410-040-B](#)
 - (a) FOR 1000EM1 (ENGINE-1)
437AL 438AR
 - (b) FOR 1000EM2 (ENGINE-2)
447AL 448AR
- (3) Remove the WARNING NOTICE(S).

TF-N
AIRCRAFT MAINTENANCE MANUAL
STARTER - INSPECTION/CHECK



S-M56-MM-04242-02-B

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Engine Starter Inspection
Figure 1 / Graphic 80-11-10 -991-30000

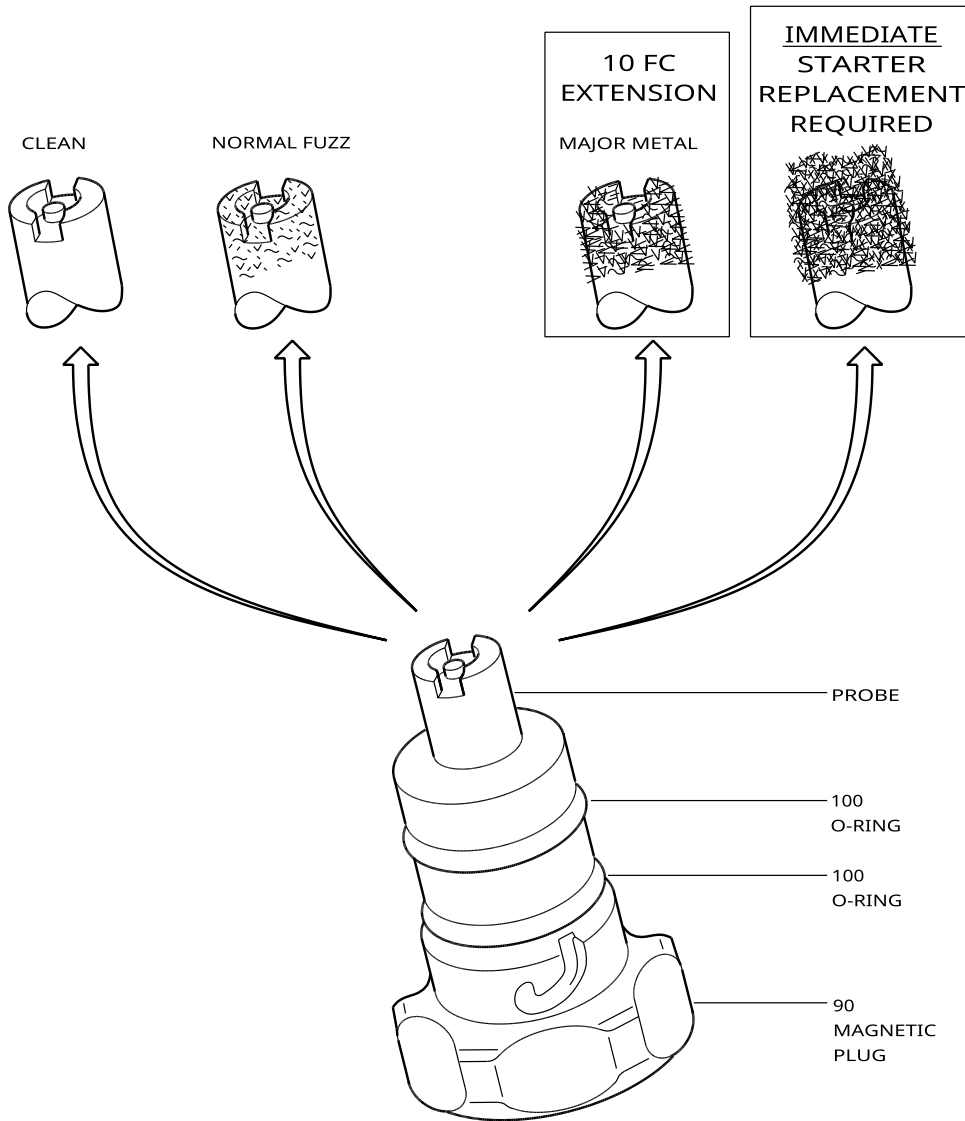
TF-N
AIRCRAFT MAINTENANCE MANUAL
STARTER - INSPECTION/CHECK

NORMAL :

A SMALL ACCUMULATION OF FINE SLIVERS AND RELATIVELY FLAT FLAKES OF METAL SHOWS A NORMAL WEAR.

ABNORMAL :

A LARGE ACCUMULATION OF PARTICULES OR PARTICULES LARGER THAN 0.1 in (2.54 mm) IN ANY DIRECTION, SHOWS THERE IS SOME INTERNAL DAMAGE AND THAT A STARTER REPLACEMENT IS NECESSARY.



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Engine Starter Inspection
Figure 1 / Graphic 80-11-10 -991-30100

S-M56-MM-04244-02

TF-N
AIRCRAFT MAINTENANCE MANUAL
STARTER - INSPECTION/CHECK

TASK 80-11-10-210-001 -C

On A/C 003-250

Check Chip Detector

1. Reason for the Job

Refer to the MPD TASK: 801110-C1

CHECK CHIP DETECTOR

2. Job Set-up Information

A. Fixtures, Tools, Test and Support Equipment

REFERENCE	QTY	DESIGNATION
No specific	AR	MAGNIFYING GLASS POWER 10X
No specific	AR	TISSUE PAPER
No specific	AR	WARNING NOTICE(S)

B. Consumable Materials

REFERENCE	DESIGNATION
CP2442	engine oil

C. Work Zones and Access Panels

ZONE/ACCESS	ZONE DESCRIPTION
FOR 1000EM1(ENGINE-1)	
437AL , 438AR	
FOR 1000EM2(ENGINE-2)	
447AL , 448AR	

D. Expendable Parts

FIG.ITEM	DESIGNATION	IPC-CSN
100	O-RING	IPC-CSN(80-11-10-01 ITEM 100)

E. Referenced Information

REFERENCE	DESIGNATION
Ref. AMM 12-13-80-610-001-C	Drain and Replenish Oil System
Ref. AMM 71-13-00-010-040-B	Opening of the Fan Cowl Doors
Ref. AMM 71-13-00-410-040-B	Closing of the Fan Cowl Doors
Ref. AMM 80-11-10-000-002-A	Removal of the Pneumatic Starter
Ref. AMM 80-11-10-400-002-A	Installation of the Pneumatic Starter
Ref. Fig. (1 / 80-11-10-991-30000)	
Ref. Fig. (1 / 80-11-10-991-30100)	

3. Job Set-up

Subtask 80-11-10-941-052-A

On A/C 003-250

A. Safety Precautions

- (1) On the center pedestal, on the ENG panel 115VU:
 - (a) Put a WARNING NOTICE(S) to tell persons not to start the engine.
- (2) Make sure that the engine shutdown occurred five minutes or more before you do this procedure.
- (3) On the overhead maintenance panel 50VU:
 - (a) Make sure that the ON legend of the ENG/FADEC GND PWR/1(2) pushbutton switch is off.
 - (b) Put a WARNING NOTICE(S) to tell persons not to energize the FADEC 1(2).

Subtask 80-11-10-010-053-A

On A/C 003-250

B. Open the fan cowl doors: [Ref. AMM 71-13-00-010-040-B](#)

- (1) FOR 1000EM1 (ENGINE-1)
437AL 438AR

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AIRCRAFT MAINTENANCE MANUAL
STARTER - INSPECTION/CHECK

- (2) FOR 1000EM2 (ENGINE-2)
447AL 448AR

4. Procedure

Ref. Fig. 1 / 80-11-10-991-30000

Ref. Fig. 1 / 80-11-10-991-30100

Subtask 80-11-10-020-052-A

On A/C 003-250

A. Removal of Starter Magnetic Plug for Inspection.

- (1) Remove the knurled magnetic plug (90) from oil drain plug (110):
(a) Push and turn counter-clockwise to release the plug (90).

NOTE : The oil drain plug (110) has a built-in check valve to permit removal of magnetic plug (90) without draining oil from the starter.

Subtask 80-11-10-210-052-A

On A/C 003-250

B. Inspection of the Magnetic Plug Probe (90).

NOTE : Before examination for metal particles, carefully prevent a further contamination of the magnetic plug probe by metal particles from the tool box or work area.

- (1) Examine the magnetic plug probe for the quantity and type of the contaminants. If the contamination you see on the probe looks important, do as follows:

Ref. Fig. (1 / 80-11-10-991-30100)

NOTE : Carbon and dirt may form sludge on the probe along with metal particles, and cause misleading appearance in the quantity of contamination.

- (a) Clean the probe with TISSUE PAPER and retain the removed particles.
(b) Examine the metallic contaminants under a MAGNIFYING GLASS POWER 10X. Use the inspection criteria which follow :
- 1 A small accumulation of fine slivers and relatively flat flakes of metal shows a normal wear.
 - 2 A larger accumulation of particles, or particles larger than 0.1 in. (2.54 mm) in any direction, show there is some internal damage and that a starter replacement is necessary. Ref. AMM 80-11-10-000-002-A and Ref. AMM 80-11-10-400-002-A.

Subtask 80-11-10-420-052-A

On A/C 003-250

C. Installation of Magnetic Plug after Inspection.

- (1) Remove and discard the two O-rings (100).
(2) Lubricate two new IPC-CSN(80-11-10-01 ITEM 100) O-RING (100) with engine oil (Material No. **CP2442**) and put them on the knurled magnetic plug (90).
(3) Install the knurled magnetic plug (90) in oil drain plug (110):
(a) Push inward and turn clockwise to engage locking device.

Subtask 80-11-10-613-050-A

On A/C 003-250

D. Do the servicing of the starter, if necessary Ref. AMM 12-13-80-610-001-C

5. Close-up

Subtask 80-11-10-410-052-A

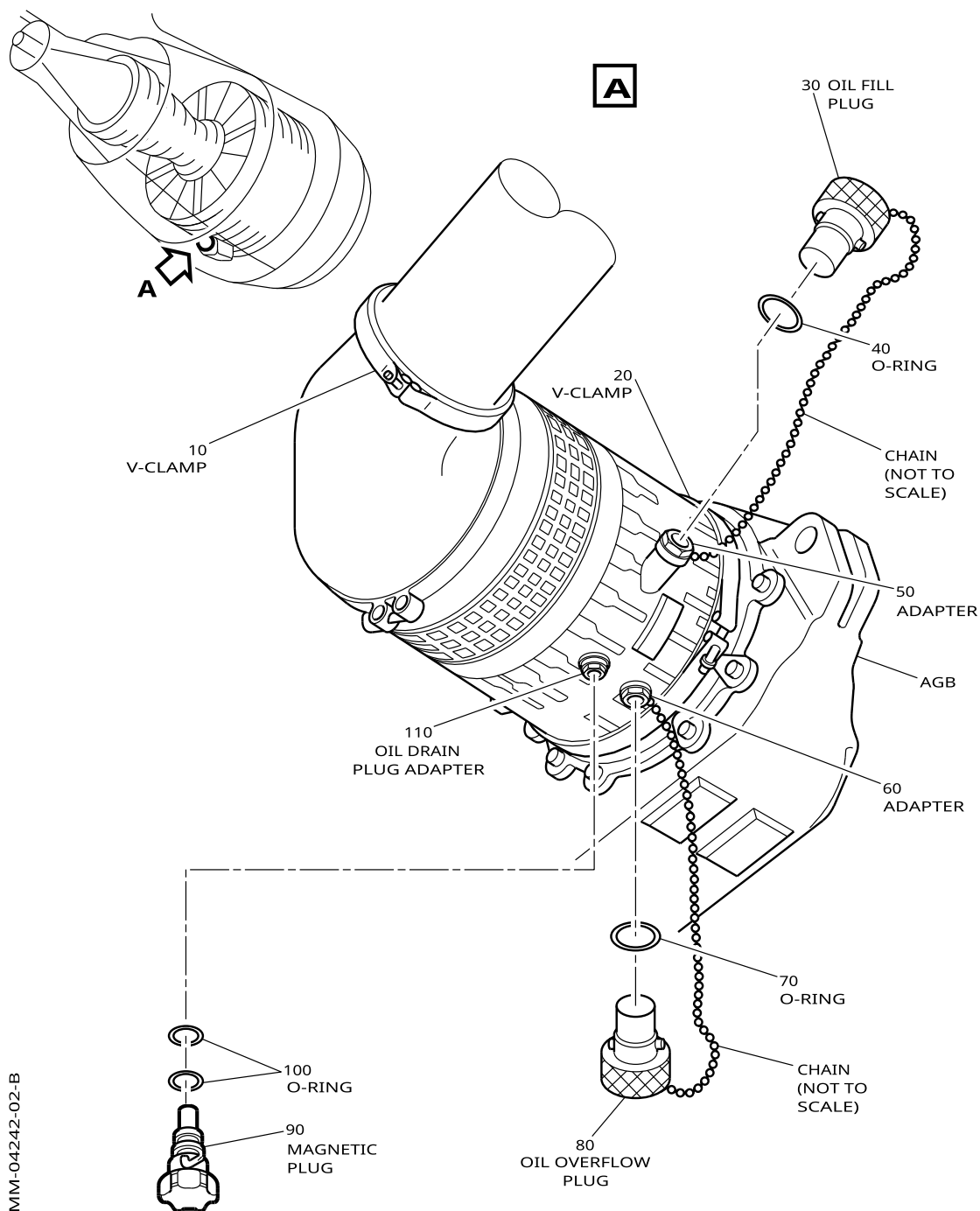
On A/C 003-250

TF-N
AIRCRAFT MAINTENANCE MANUAL
STARTER - INSPECTION/CHECK

A. Close Access

- (1) Make sure that the work area is clean and clear of tools and other items.
- (2) Close the fan cowl door: [Ref. AMM 71-13-00-410-040-B](#)
 - (a) FOR 1000EM1 (ENGINE-1)
437AL 438AR
 - (b) FOR 1000EM2 (ENGINE-2)
447AL 448AR
- (3) Remove the WARNING NOTICE(S).

TF-N
AIRCRAFT MAINTENANCE MANUAL
STARTER - INSPECTION/CHECK



S-M56-MM-04242-02-B

N_MM_801110_6_RAM0_01_01

Engine Starter Inspection
Figure 1 / Graphic 80-11-10 -991-30000

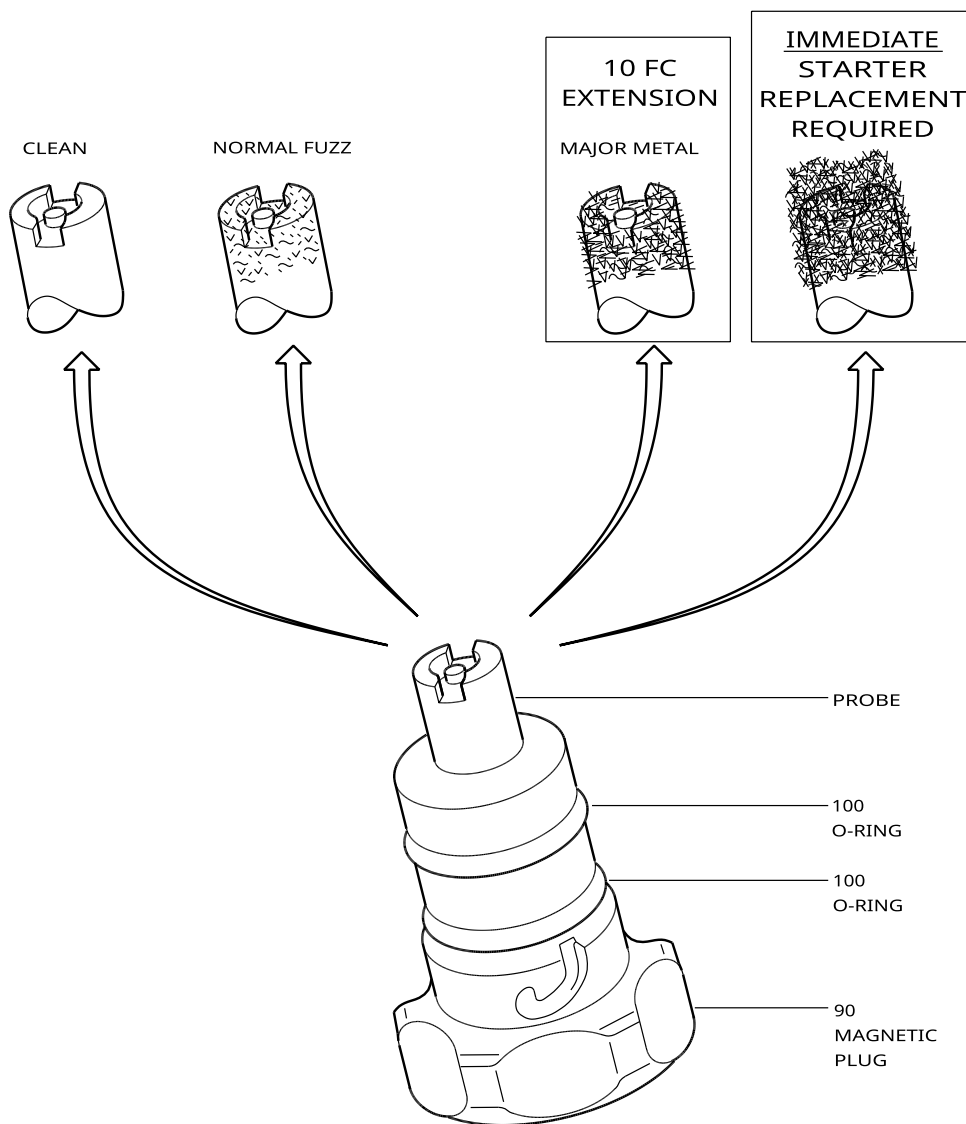
TF-N
AIRCRAFT MAINTENANCE MANUAL
STARTER - INSPECTION/CHECK

NORMAL :

A SMALL ACCUMULATION OF FINE SLIVERS AND RELATIVELY FLAT FLAKES OF METAL SHOWS A NORMAL WEAR.

ABNORMAL :

A LARGE ACCUMULATION OF PARTICULES OR PARTICULES LARGER THAN 0.1 in (2.54 mm) IN ANY DIRECTION, SHOWS THERE IS SOME INTERNAL DAMAGE AND THAT A STARTER REPLACEMENT IS NECESSARY.



N_MM_801110_6_RBM0_01_01

Engine Starter Inspection
Figure 1 / Graphic 80-11-10 -991-30100

S-M56-MM-04244-02